

Distinction Theory Unified Collection

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Active Finite Distinction Systems

*A Formal Core for Boundary Maintenance under Finite Capacity
Capacity Deficit, Dissipation, Externalization, Collapse, and Invariant
Persistence*

FORMAL CORE EDITION

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*A formal core for what active finite systems must pay
in order to maintain a boundary under limited capacity.*

Companion formal core to *Distinction Theory: A General Theory of Finite Systems*, Zenodo,
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EPISTEMIC NOTICE

This manuscript does not present Distinction Theory as an undifferentiated theory of everything. It presents a restricted mathematical theory of *active finite distinction systems*: systems that maintain a boundary through state-dependent updates under finite representational capacity and finite resource budgets.

The core chain is not asserted as rhetoric. It is decomposed into formal definitions, conditional theorems, physical bridge assumptions, normal-form dynamics, and quarantined applications:

distinction $\rightarrow$ boundary $\rightarrow$ finite capacity $\rightarrow$ capacity deficit
 $\rightarrow$ approximation $\rightarrow$ irreversible update $\rightarrow$ dissipation
 $\rightarrow$ prune/externalize/collapse $\rightarrow$ invariant persistence.

The mathematical core can fail only by mathematical counterexample or by failure of stated hypotheses. The physical bridge can fail by violation of its thermodynamic assumptions. Domain applications can fail without propagating failure back to the core.

A failed claim is not reinterpreted into survival. Each claim carries an explicit proof type, dependency class, and failure consequence.

Abstract

This document defines active finite distinction systems as finite-capacity systems that maintain boundaries through state-dependent updates under resource constraints. It develops the formal core of Distinction Theory, including representational capacity, rate-distortion capacity deficit, conditional approximation proliferation, Landauer dissipation floor under physical bridge assumptions, prune–externalize–collapse trichotomy, and invariant-supported persistence. Domain applications are quarantined behind explicit bridge assumptions and operational tests.

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Version relationship:

This document is the formal FDS Core companion to the full claim-space edition: *Distinction Theory: A General Theory of Finite Systems*, Zenodo, doi:10.5281/zenodo.20130174.

Reader Orientation

This document extracts the formal FDS core from the broader Distinction Theory claim-space archive. It is organized as a theory with three layers.

Layer 1: Formal FDS Core.

Definitions and conditional mathematical theorems about finite distinction systems. This layer uses partitions, information capacity, rate-distortion demand, conditional entropy, update maps, perturbation families, and invariant quotients.

Layer 2: Physical Bridge.

Conditions under which formal updates are realized by physical substrates coupled to thermal environments. This layer introduces Landauer costs, physical dissipation, free-energy budgets, and maintenance failure.

Layer 3: Quarantined Applications.

Biology, cognition, artificial intelligence, civilization, geometry, black holes, and other domains are treated only through explicit domain bridge assumptions and operational tests.

The purpose of this reconstruction is not to make the theory immune to criticism. It is to make every criticism land in the correct place.

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Part I.

Formal Kernel: Active Finite Distinction Systems

1. The Primitive and the Scope of the Theory

1.1. The Distinction Primitive

Definition 1.1 (Distinction). A *distinction* is an operation or relation that separates at least two alternatives within a possibility space. Formally, at the most elementary level, a distinction may be represented by a nontrivial partition

$$\mathcal{D} = \{A_i\}_{i=1}^n, \quad A_i \cap A_j = \emptyset \ (i \neq j), \quad \bigcup_i A_i = X, \quad n \geq 2.$$

The primitive is not the claim that every distinction is physical, costly, or thermodynamic. A purely formal distinction is a logical separation. A realized distinction in an active finite system is an operational state maintained under constraints.

Proposition 1.2 (Denial performs distinction). *Any coherent denial of distinction distinguishes the denial from the proposition denied. Thus the act of denial presupposes the operation it rejects.*

Proof. Let P be the proposition “distinction is required for coherent identity”. A denial asserts $\neg P$ rather than P . The assertion is meaningful only if $\neg P$ is distinguished from P . Hence the denial performs a distinction. This is a logical observation, not a physical theorem. $\square$

1.2. What the Primitive Does Not Prove

Not claimed. The distinction primitive alone does not derive thermodynamics, biology, cognition, artificial agency, black holes, spacetime geometry, or civilization. Those claims require finite-system restrictions, active-boundary hypotheses, physical bridge assumptions, and domain-specific mappings.

1.3. Finite Systems as the Domain of Application

The formal theory begins only when the primitive is restricted to realized bounded systems with finite representational and energetic resources. The central object is not an abstract distinction, but an active finite distinction system.

Definition 1.3 (Proof status labels). Each claim in this document carries one of four proof statuses.

1. The Primitive and the Scope of the Theory

Formal.	A mathematical statement derivable from definitions and stated hypotheses.
Conditional.	A mathematical or physical statement derivable only under explicitly stated bridge assumptions.
Normal-form.	A dynamical model describing a generic regime, not a universal law.
Domain-bridge.	An applied claim requiring a mapping from the FDS formalism to a specific empirical domain.

1.3.1. From the Distinction Primitive to the FDS Object

The distinction primitive by itself is not a complete physical or computational theory. A bare distinction only asserts a separation: some difference is drawn between what is counted as one side and what is counted as another. To obtain a theory of active finite systems, the primitive must be realized under additional operational constraints: the distinction must be finite, maintained, updated, exposed to perturbation, and evaluated over time.

The twelve-component FDS object is therefore not introduced as an arbitrary list of variables. It is an explicit operational decomposition of what is otherwise often smuggled into informal uses of the word “distinction”. Each component records one functional role required for a finite distinction to be actively maintained rather than merely declared.

First, a maintained distinction requires a partitioned domain: an internal state space X , an environment E , and a boundary B across which the self-environment distinction is operationally defined. Without this partition, there is no object whose boundary can be maintained.

Second, a finite system cannot preserve all distinctions available in its environment. It therefore requires an internal memory or model space M , a coarse-graining or projection map π , and a finite resource budget Φ . These terms make explicit the finiteness that is otherwise hidden in informal talk of an observer, agent, organism, or record-bearing system.

Third, an active distinction is not static. The system must receive observations through a channel Y , update its internal state through an update rule U , and possess an action or intervention space A through which its updates can matter for future boundary-maintenance loss. Without update and intervention, the system is at most a passive record or static partition.

Fourth, maintenance requires vulnerability. There must be a loss function ℓ specifying what counts as boundary-maintenance failure, a perturbation or pruning family $\mathcal{P}$ specifying the admissible transformations under which persistence is assessed, and an update timescale τ converting static information quantities into rates, costs, and dynamical constraints.

1. The Primitive and the Scope of the Theory

Thus the FDS tuple

$$S = (X, E, B, M, Y, A, U, \pi, \ell, \Phi, \mathcal{P}, \tau)$$

should be read as a minimal operational anatomy of an active finite distinction. Individual components may be compressed or represented differently in particular formalisms, but the functional roles they mark cannot be removed without either changing the object of study or reintroducing hidden assumptions.

2. The Finite Distinction System

2.1. The Twelve-Component Object

Definition 2.1 (Finite Distinction System). A *finite distinction system* is a tuple

$$\mathcal{S} = (X, E, B, M, Y, A, U, \pi, \ell, \Phi, \mathcal{P}, \tau),$$

where:

X	is the internal state space of the system.
E	is the environmental state space.
B	is the boundary or interface variable separating system from environment.
M	is the internal memory, model, or representational state space.
Y	is the observation channel available at the boundary.
A	is the action or boundary-update space.
U	is the internal update map, typically $U : M \times Y \rightarrow M$ or a stochastic kernel $\mathbf{P}(M_{t+1} \mid M_t, Y_t)$.
π	is the finite distinction projection, coarse-graining, or partition map.
ℓ	is the boundary-maintenance loss function.
Φ	is the finite resource or free-energy budget functional.
$\mathcal{P}$	is the admissible family of pruning, coarse-graining, perturbation, or degradation operators.
τ	is the update timescale converting per-step information loss into rates or powers.

Remark 2.2. The twelfth component τ is not cosmetic. Without a timescale, information-theoretic quantities remain dimensionless counts. With τ , they become rates and can be connected to thermodynamic power bounds.

2.2. Active and Passive Boundaries

Definition 2.3 (Active boundary). An FDS has an *active boundary* if its update rule is nontrivial and participates in future boundary maintenance:

$$\mathbb{P}(U(M_t, Y_t) \neq M_t) > 0$$

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and there exists $k > 0$ such that

$$I(M_{t+1}; \ell_{t+k}) > 0.$$

The second condition excludes irrelevant state noise. A state change counts as active-boundary-relevant only if it is statistically related to future boundary-maintenance loss; empirical applications require the stronger causal qualification in Remark 2.8.

Remark 2.4 (Intervention criterion — cross-reference). The mutual-information condition $I(M_{t+1}; \ell_{t+k}) > 0$ is a relevance screen; for the empirical causal strengthening, see Remark 2.8.

Definition 2.5 (Passive boundary). A system has a *passive boundary* when its persistence is supported by static physical constraints or stable configurations without task-directed internal updating. Passive persistence may be real and stable, but it does not enter the deficit-dissipation cascade unless an active update channel is introduced.

Domain Restriction 0

The deficit-dissipation-pruning cascade applies only to active-boundary finite distinction systems. Passive crystals, inert rocks, stable nuclei, and purely formal mathematical objects are not governed by the FDS cascade merely because they possess boundaries.

Criterion 2.6 (Active Boundary Qualification). An FDS enters the core Distinction Theory cascade only if its boundary-maintenance loss depends on nontrivial internal updates. Formally, the necessary qualification is

$$\mathbb{P}(U(M_t, Y_t) \neq M_t) > 0 \quad \text{and} \quad I(M_{t+1}; \ell_{t+k}) > 0$$

for some $k > 0$.

Remark 2.7 (Justification). If $U(M_t, Y_t) = M_t$ almost surely, no internal update is performed and no internal correction channel exists. If U changes M_t but $I(M_{t+1}; \ell_{t+k}) = 0$ for all $k > 0$, the update is not statistically relevant to boundary maintenance under the chosen variables. In either case the system lacks active-boundary dynamics in the sense required by the cascade. Conversely, if both conditions hold, the system possesses a nontrivial internal update relevant to future boundary-maintenance loss. This is precisely the domain condition.

Remark 2.8 (Causal strengthening for empirical systems). The mutual-information condition $I(M_{t+1}; \ell_{t+k}) > 0$ is a minimal formal relevance condition, not by itself a causal intervention test. In empirical applications, active boundary maintenance should be operationalized by an intervention or ablation criterion: there must exist an admissible null update channel $U_\emptyset$ (such as freezing, randomizing, or

2. The Finite Distinction System

identity-updating the internal update) and some $k > 0$ such that

$$\mathbb{E}[\ell_{t+k} \mid do(U)] \neq \mathbb{E}[\ell_{t+k} \mid do(U_\emptyset)].$$

This excludes passive observers whose internal states merely correlate with future boundary loss through a common environmental cause.

2.3. Non-Domain Cases

The following are not core FDS cascade cases unless extra active-boundary structure is specified:

- inert passive solids;
- equilibrium crystals without active repair;
- closed unitary quantum evolution without coarse-grained irreversible update;
- static mathematical structures;
- read-only storage without overwrite, erasure, or maintenance update;
- systems whose internal changes do not affect boundary-maintenance loss.

Part II.

Information Theorems: Capacity, Approximation, and Complexity

3. Finite Distinction Representation

3.1. Representational Capacity

Definition 3.1 (Internal representational capacity). For a finite memory/model space M , define the internal representational capacity

$$C_S = \log_2 |M|.$$

When M is continuous or effectively continuous, C_S is replaced by the appropriate operational capacity under finite resolution, noise, or coding constraints.

Definition 3.2 (Task-relevant environmental variable). Let

$$Z_t = \psi(E_t, B_t)$$

be a task-relevant environmental statistic relevant to maintaining the boundary up to a specified distortion or loss tolerance. The theory does not require full representation of E_t ; it requires only the representation needed for the boundary-maintenance task.

Theorem 3.3 (Finite Distinction Representation). *For any FDS with finite internal model capacity C_S , the mutual information between the environment and the internal model satisfies*

$$I(E_t; M_t) \leq C_S.$$

Proof. By the data-processing inequality and the entropy bound on a finite memory state,

$$I(E_t; M_t) \leq H(M_t) \leq \log_2 |M| = C_S.$$

□

3.2. Operational Meaning

This theorem does not say that the environment is finite. It says that the internally maintained distinction structure is finite. The system may face an environment whose task-relevant complexity exceeds what the system can encode internally.

4. Capacity Deficit

4.1. Rate-Distortion Demand

Definition 4.1 (Admissible task statistics). Let Ψ be a family of admissible statistics $\psi : (E, B) \rightarrow Z$ that are causally accessible through the observation channel Y . A statistic $Z_t = \psi(E_t, B_t)$ is task-relevant if it captures information needed to maintain the boundary within tolerated distortion. Admissible statistics must preserve sufficient information for the stated boundary-maintenance criterion; trivial constant maps are excluded unless the task itself is trivial. The family Ψ is fixed by the task specification and sensor–action architecture, not chosen post hoc after observing failure.

Definition 4.2 (Pre-registered admissible task families). To prevent post-hoc selection of task statistics, the admissible family Ψ must be fixed before evaluating capacity deficit. Typical admissible families include:

- (i) **Control or homeostasis tasks.** Ψ_{ctrl} contains statistics of (E_t, B_t) sufficient to select actions that keep the boundary variable B_t inside a specified viable set $\mathcal{V}$ up to tolerance ε .
- (ii) **Predictive tracking tasks.** Ψ_{pred} contains statistics sufficient to predict a pre-specified future observable Y_{t+k} or target variable Z_{t+k} under a fixed loss function.
- (iii) **Information-bottleneck tasks.** Ψ_{IB} contains compressed statistics that preserve task-relevant information about a pre-specified target Z under a stated rate-distortion or information-bottleneck objective.

The class Ψ is part of the task specification, not a variable chosen after observing system failure.

Definition 4.3 (Rate-distortion demand). For an admissible statistic ψ , update window τ , and tolerated distortion ε , let

$$R_{\psi(E,B)}^{(\tau)}(\varepsilon)$$

be the minimum number of bits per update window τ required to encode $\psi(E, B)$ so that the expected boundary-maintenance distortion remains below ε . The minimal rate-distortion demand over the admissible class is

$$R_{\min}^{(\tau)}(\varepsilon) = \inf_{\psi \in \Psi} R_{\psi(E,B)}^{(\tau)}(\varepsilon).$$

Definition 4.4 (Capacity deficit). The *capacity deficit* of an FDS at tolerance ε

4. Capacity Deficit

over window τ is

$$\Delta_\varepsilon(\tau) = R_{\min}^{(\tau)}(\varepsilon) - C_S.$$

If $\Delta_\varepsilon(\tau) > 0$, the system is in predictive capacity deficit at tolerance ε .

Theorem 4.5 (Capacity Deficit Theorem). *If*

$$R_{\min}^{(\tau)}(\varepsilon) > C_S,$$

then no purely internal model M_t of capacity C_S can encode even the least demanding admissible task statistic to distortion ε over window τ . The system must approximate, externalize, relax the task, or suffer boundary-maintenance failure.

Proof. For any admissible ψ , any representation of $\psi(E, B)$ achieving distortion at most ε over window τ requires at least $R_{\psi(E, B)}^{(\tau)}(\varepsilon)$ bits by the rate-distortion theorem. Taking the infimum, $R_{\min}^{(\tau)}(\varepsilon)$ is the minimal encoding demand across the admissible class. By Theorem 3.3, the system can internally maintain at most C_S bits. If $R_{\min}^{(\tau)}(\varepsilon) > C_S$, no admissible statistic can be encoded internally to the required distortion tolerance ε . Therefore at least one of the following must occur: representation is lossy beyond ε , representation is externalized outside M , the task tolerance is relaxed, or boundary-maintenance performance fails. $\square$

Corollary 4.6 (Approximation Necessity). *For an active-boundary FDS with $\Delta_\varepsilon(\tau) > 0$, approximation is not optional. It is forced by the information-theoretic mismatch between admissible task demand and internal capacity.*

Remark 4.7. The capacity deficit is not a vague cognitive gap. It is a computable gap between a task-relevant rate-distortion demand and an internal representational capacity.

5. Approximation Proliferation (Conditional)

5.1. Minimal Sufficient Complexity

Definition 5.1 (Minimal sufficient model complexity). Let $\mathfrak{C}$ be a fixed coding family (representation language, model architecture, or hypothesis class) that does not change during the considered interval. Let $C(M)$ be a model-complexity functional, such as description length, state count, algorithmic cost, parameter count under a scheme in $\mathfrak{C}$, or thermodynamic maintenance load. Define

$$C_t^* = \min\{C(M) : \mathbb{E}[\ell_t(M)] \leq \ell_c\},$$

where ℓ_c is an admissible boundary-maintenance threshold, and the minimum is taken only over models inside $\mathfrak{C}$, not over future inventions of new representation languages.

Definition 5.2 (Persistent task novelty). Fix a coding family $\mathfrak{C}$, a loss threshold ℓ_c , and the filtration $\mathcal{F}_t$ generated by the system's observations and internal states up to time t . The task process has persistent novelty at rate $\eta > 0$ if, for infinitely many update windows,

$$\mathbb{E} \left[\inf_{M \in \mathfrak{C}: C(M) \leq C_t^*} \mathbb{E}[\ell_{t+1}(M) \mid \mathcal{F}_t] - \ell_c \mid \mathcal{F}_t \right]_+ \geq \eta.$$

Equivalently, the current minimum sufficient model fails to absorb future task-relevant variation without refinement, pruning, externalization, task relaxation, or compression improvement. The novelty rate η is relative to the fixed coding family $\mathfrak{C}$: even if the environment has positive entropy rate, a sufficiently expressive $\mathfrak{C}$ may absorb the variation without complexity growth.

Remark 5.3 (Entropy-rate sufficient condition). A positive entropy rate $h(Z) > 0$ of the task-relevant process Z_t does not by itself imply model-complexity proliferation. It does so only when the induced innovations are not compressible within the fixed coding family $\mathfrak{C}$ at distortion ε , i.e. when the conditional rate-distortion residual remains bounded away from zero:

$$R_{Z_{t+1}|\mathcal{F}_t, \mathfrak{C}}^{(\tau)}(\varepsilon) \geq \eta > 0$$

over a positive-density set of update windows.

Theorem 5.4 (Conditional Approximation Proliferation under Fixed Coding Fam-

5. Approximation Proliferation (Conditional)

ily). Assume $\Delta_\varepsilon(\tau) > 0$ over a relevant window, persistent task novelty, no active pruning, no externalization, no task relaxation, and no improvement in compression scheme. Then the minimal sufficient model complexity is a submartingale:

$$\mathbb{E}[C_{t+1}^* \mid \mathcal{F}_t] \geq C_t^*.$$

Remark 5.5 (Proof sketch). Under capacity deficit, the system cannot internally encode all task-relevant distinctions to tolerance ε over the update window. Persistent novelty generates residual cases not captured by the current sufficient model. If the system must keep expected loss below ℓ_c and cannot prune, externalize, relax the task, or improve compression, the only remaining operation is to add or refine internal distinctions. Thus the minimum sufficient complexity cannot decrease in conditional expectation, giving the submartingale inequality. A fully formal proof would require a specified filtration $\mathcal{F}_t$, a measurable novelty process with a positive-rate definition, and a submartingale convergence argument. The present form is a conditional formal theorem.

Remark 5.6. This theorem does not say that observed representation length always increases. Better compression, abstraction, external memory, task relaxation, or pruning may reduce observed complexity. The theorem concerns minimum sufficient complexity under explicitly restricted conditions.

Remark 5.7. This result does not exclude discontinuous decreases in maintained complexity when a superior abstraction, compression code, pruning operation, or externalization channel becomes available. Historical examples include the transition from geocentric epicycles to Keplerian orbits, or the replacement of hand-crafted features by learned representations in machine learning. The theorem states only that in the absence of such improvements, persistent novelty forces the minimum sufficient complexity to be a submartingale.

Part III.

Physical Bridge: Irreversibility, Dissipation, and Free-Energy Budgets

6. Logical Irreversibility and Landauer Cost

6.1. Preimage Conditional Entropy

Definition 6.1 (Logical erasure per update). For an update $M_{t+1} = U(M_t, Y_t)$, define the logical erasure in bits by

$$b_t = H(M_t \mid M_{t+1}, Y_t).$$

This quantity measures how much uncertainty remains about the prior memory state after the new memory state and current input are known. It is the preimage information lost by the update.

Remark 6.2. A forward divergence such as $D_{\text{KL}}(\mathbb{P}(M_{t+1}) \parallel \mathbb{P}(U(M_t, Y_t)))$ does not measure erasure when M_{t+1} is defined by $U(M_t, Y_t)$; in that case the two distributions coincide. Logical erasure is a property of non-invertible preimages, not of forward equality.

Assumption 6.3 (Physical Landauer bridge). A logically irreversible update implemented by a physical substrate coupled to a thermal environment at temperature $T > 0$ dissipates at least $k_B T \ln 2$ per erased bit, under the standard physical conditions of Landauer's principle.

Theorem 6.4 (Landauer Dissipation Floor). *For an active-boundary FDS physically implementing irreversible memory updates at timescale τ , the informational heat dissipation rate obeys*

$$\dot{Q}_{\text{info}} \geq \frac{k_B T \ln 2}{\tau} H(M_t \mid M_{t+1}, Y_t).$$

The total maintenance cost decomposes as

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{phys}} + \dot{Q}_{\text{info}},$$

where $\dot{Q}_{\text{phys}}$ includes non-informational physical losses such as control, coupling, leakage, friction, error correction, clocking, isolation, and transport.

Proof. By definition, $H(M_t \mid M_{t+1}, Y_t)$ is the number of erased bits per update. Under the physical Landauer bridge, each erased bit costs at least $k_B T \ln 2$ of dissipated heat. Dividing by update timescale τ gives the rate bound. Physical losses independent of logical erasure add to the informational term. $\square$

6. Logical Irreversibility and Landauer Cost

Remark 6.5. If $H(M_t \mid M_{t+1}, Y_t) = 0$, the update is logically reversible relative to Y_t and no Landauer erasure cost follows from this term. It does not follow that the system is passive or cost-free. Reversible active computation may still require nonzero $\dot{Q}_{\text{phys}}$.

6.2. Failure Condition for the Physical Bridge

Failure Condition 6.6 (Landauer bridge failure). If a physical system satisfying the stated conditions of the Landauer bridge performs logically irreversible many-to-one overwrite or erasure at a reliable average cost below $k_{\text{B}}T \ln 2$ per erased bit, the physical bridge fails. Formal FDS theorems remain intact; thermodynamic conclusions depending on the bridge are demoted or abandoned.

7. Free-Energy Budgets and Maintenance Failure

Definition 7.1 (Available free-energy input). Let $\dot{F}_{\text{in}}(t)$ denote the free-energy input rate available to support boundary maintenance, internal updating, pruning, externalization, repair, and control.

Definition 7.2 (Maintenance failure region). An FDS enters the maintenance failure region when, over a relevant interval $[t_0, t_1]$,

$$\int_{t_0}^{t_1} \dot{Q}_{\text{total}}(t) dt > \int_{t_0}^{t_1} \dot{F}_{\text{in}}(t) dt$$

without compensating resource reserves, external subsidies, task relaxation, pruning, or externalization.

Assumption 7.3 (Loss–resource coupling). In a physically realized FDS, boundary-maintenance loss may feed back into the available resource budget through a nonnegative damage functional $D_\ell(\ell_t) \geq 0$. Over one update window τ ,

$$\Phi_{t+1} \leq \Phi_t + \int_t^{t+\tau} \dot{F}_{\text{in}}(s) ds - \int_t^{t+\tau} \dot{Q}_{\text{total}}(s) ds - D_\ell(\ell_t).$$

The damage functional may take various forms: linear ($\kappa \ell_t$), threshold-based (0 for $\ell \leq \ell_c$, positive above), or catastrophic (∞ for $\ell > \ell_{\text{death}}$). The formal FDS core does not require this coupling; it becomes active only in physical or domain-bridge models where boundary failure consumes repair capacity, destroys structure, or reduces future free-energy access.

Remark 7.4 (Model families for loss–resource coupling). The damage functional D_ℓ is domain-specific. Common choices include linear damage $D_\ell(\ell) = \kappa \ell$, threshold damage $D_\ell(\ell) = \kappa(\ell - \ell_c)_+$, accelerated damage $D_\ell(\ell) = \kappa(e^{\alpha(\ell - \ell_c)} - 1)_+$, and catastrophic damage $D_\ell(\ell) = \infty$ for $\ell > \ell_{\text{death}}$. The FDS core does not select among these forms; empirical domain bridges must specify which coupling applies.

Theorem 7.5 (Prune–Externalize–Collapse Trichotomy). *Let an active-boundary FDS satisfy $\Delta_\varepsilon(\tau) > 0$ over a relevant window and enter the maintenance failure region, with bounded $\dot{F}_{\text{in}}$ and no compensating external subsidy. Then the system cannot maintain the same active-boundary regime indefinitely; its trajectory must enter at least one of three outcome classes:*

- (i) pruning: reducing internal complexity or maintenance load;

7. Free-Energy Budgets and Maintenance Failure

- (ii) externalization: *shifting representational, energetic, or control load outside the internal memory budget M ;*
- (iii) collapse: *loss of boundary-maintenance capacity, task failure, or transition to a lower-complexity attractor.*

Proof. Under $\Delta_\varepsilon(\tau) > 0$ over the relevant window, the system faces a representational shortfall within its update budget. If continued internal updating and maintenance exceed the available free-energy input, the system cannot indefinitely maintain the same internal complexity and task burden. The admissible exits are: reduce the maintained internal load, move part of the load outside the internal budget, or fail to maintain the boundary/task. These are pruning, externalization, and collapse respectively. $\square$

Remark 7.6. The trichotomy replaces the older binary phrase “prune or collapse”. Many systems persist by externalizing distinctions into bodies, tools, texts, institutions, niches, databases, or environments. Externalization is not a loophole; it is a relocation of maintenance cost and boundary responsibility.

Part IV.

Persistence: Quotients, Perturbations, and Invariant-Supported Identity

8. Invariant-Supported Persistence

8.1. Identity Predicates and Quotient Maps

Definition 8.1 (Structural identity predicate). Let $R_A : X \rightarrow \{0, 1\}$ be an identity predicate for a structure A in state space X . The statement $R_A(x) = 1$ means that state x realizes structure A under the chosen resolution.

Definition 8.2 (Invariant quotient). Let $q : X \rightarrow T$ be a quotient or feature map into a reduced space T . The quotient is invariant under perturbation family $\mathcal{P}$ if

$$q \circ P_i = q$$

for all $P_i \in \mathcal{P}$.

Definition 8.3 (Invariant-supported identity). A structural identity predicate R_A is invariant-supported under $\mathcal{P}$ if there exist $q : X \rightarrow T$ and $\bar{R}_A : T \rightarrow \{0, 1\}$ such that

$$R_A = \bar{R}_A \circ q$$

and

$$q \circ P_i = q \quad \forall P_i \in \mathcal{P}.$$

Theorem 8.4 (Invariant Persistence). *If R_A is invariant-supported under $\mathcal{P}$, then identity is preserved under every admissible perturbation:*

$$R_A \circ P_i = R_A \quad \forall P_i \in \mathcal{P}.$$

Proof. Since $R_A = \bar{R}_A \circ q$ and $q \circ P_i = q$,

$$R_A \circ P_i = (\bar{R}_A \circ q) \circ P_i = \bar{R}_A \circ (q \circ P_i) = \bar{R}_A \circ q = R_A.$$

Thus the identity predicate is unchanged by all admissible perturbations. $\square$

Corollary 8.5 (Unprotected identity). *If no such invariant quotient exists, then the identity predicate is not algebraically protected by the perturbation family. The corollary does not imply that non-invariant identities must fail: active maintenance, repair, isolation, stochastic stability, or energy-consuming feedback may preserve them. It states only that algebraic protection through an invariant quotient is absent, so survival depends on ongoing compensatory mechanisms rather than structural persistence.*

Remark 8.6. The theory does not claim that every non-invariant identity immediately dies. It claims that non-invariant identities lack algebraic protection against

the allowed perturbation family and therefore require active maintenance, isolation, repair, or external support.

8.2. Illustrative Examples

Example 8.7 (Error-correcting codes). Let X be the space of physical code states and admissible corrupted states. Let P_i be an admissible error from a correctable error set. Let $q : X \rightarrow T$ map a physical state to its recovered logical equivalence class, i.e. the logical state obtained after syndrome extraction and recovery. Then for all correctable errors P_i ,

$$q \circ P_i = q.$$

The physical microstate and syndrome may change, but the recovered logical identity is invariant under the admissible error family.

Example 8.8 (Software API contracts). Let X be the full internal memory and implementation state of a service. Let P_i include garbage collection, memory reallocation, thread scheduling, or implementation refactoring. For a fixed admissible request class, let q map internal implementation states to contract-level input-output behavior. If the same valid request receives the same contract-level response before and after P_i , then the service identity is supported by the invariant quotient rather than by its microstate.

Example 8.9 (Topological winding numbers). For a gapped system with a topological winding number, let q map a microscopic configuration to its winding class. Perturbations that preserve the gap and remain within the admissible locality and symmetry class satisfy $q \circ P_i = q$. The protected identity is therefore invariant under that restricted perturbation family, not under arbitrary destructive perturbations.

9. Coarse-Grained Anti-Recurrence

Definition 9.1 (Effective truncation map). An effective truncation map is a non-injective coarse-graining or deletion operator

$$T : X \rightarrow X'$$

that identifies at least two distinct states: $x \neq x'$ but $T(x) = T(x')$.

Lemma 9.2 (Coarse-Grained Anti-Recurrence). *In an effective representation containing a non-injective truncation map, recurrence to an erased micro-distinction is impossible within that representation without external records or additional inverse information.*

Proof. If $T(x) = T(x')$ for $x \neq x'$, then the image state does not contain enough information to determine whether the preimage was x or x' . Any inverse selection requires information not present in the truncated representation. Therefore recurrence to the specific erased micro-distinction is impossible inside the effective state description alone. $\square$

Remark 9.3. This lemma is not a denial of microscopic reversibility, Hamiltonian recurrence, or unitary quantum evolution under their own hypotheses. It applies to coarse-grained effective representations containing non-injective truncation.

Proposition 9.4 (Coarse-graining and lumpability). *Let (X_t) be a Markov process on a microscopic state space X , and let $T : X \rightarrow X'$ be a non-injective coarse-graining map. The projected process $X'_t = T(X_t)$ is Markovian only under the usual lumpability condition [14]: transition probabilities out of any two microstates in the same fiber $T^{-1}(x')$ agree after projection. If lumpability fails, the projected dynamics carries hidden-state memory; recurrence to a specific erased micro-distinction requires information not contained in X'_t alone.*

Remark 9.5. Thus Lemma 9.2 is not a claim about microscopic irreversibility. It is a claim about loss of inverse information in the effective representation. Lumpability identifies the exceptional cases in which coarse-grained dynamics remains Markovian; non-lumpable projections generically introduce memory, apparent irreversibility, or entropy production at the effective level.

Part V.

Normal-Form Dynamics: Phase A, Phase B, and Failure Modes

10. The Minimal Dynamical Schema

10.1. Regime Variables

Let C_t denote maintained internal complexity, Φ_t available free-energy reserve, S_t pruning or simplification effort, and E_t^{ext} externalization load. A generic continuous-time normal form may be written as

$$\frac{dC}{dt} = G(\Delta_\varepsilon(\tau), C) - \text{Prune}(S, C) - X(E^{\text{ext}}, C),$$

$$\frac{d\Phi}{dt} = \dot{F}_{\text{in}} - \dot{Q}_{\text{phys}}(C, B) - \dot{Q}_{\text{info}}(U, M, Y) - \Gamma(S) - \Xi(E^{\text{ext}}),$$

where G is deficit-driven complexity growth, Prune pruning, X externalization relief, Γ pruning cost, and Ξ externalization cost.

Normal-Form Status

This chapter does not assert a universal oscillator theorem. It defines a normal-form regime structure. Depending on parameters and reset maps, an FDS may exhibit oscillation, monotone collapse, metastable maintenance, one-time failure, or externalized persistence.

10.2. Phase A and Phase B

Definition 10.1 (Phase A). Phase A is the growth-dominant regime in which capacity deficit and novelty drive model proliferation, exploration, expansion, or complexity accumulation:

$$G(\Delta_\varepsilon(\tau), C) > \text{Prune}(S, C) + X(E^{\text{ext}}, C).$$

Definition 10.2 (Phase B). Phase B is the maintenance, pruning, externalization, or collapse-dominant regime in which accumulated complexity cannot continue unchecked:

$$\text{Prune}(S, C) + X(E^{\text{ext}}, C) \geq G(\Delta_\varepsilon(\tau), C)$$

or boundary maintenance fails.

Proposition 10.3 (Normal-Form Regime). *Under finite capacity, finite free-energy input, positive deficit-driven complexity pressure, and nonzero maintenance cost, an active-boundary FDS cannot remain indefinitely in unbounded Phase-A complexity growth without pruning, externalization, task relaxation, or collapse.*

10. The Minimal Dynamical Schema

Proof. If C grows without bound while maintenance cost remains positive and nondecreasing in maintained complexity, then $\dot{Q}_{\text{total}}$ eventually exceeds bounded $\dot{F}_{\text{in}}$ unless pruning, externalization, task relaxation, or compression improvement offsets the growth. By Theorem 7.5, the system must exit unbounded Phase-A growth through one of the admissible channels. $\square$

10.3. Oscillation Is Conditional

Proposition 10.4 (Conditional oscillation). *If a system has replenishing free-energy input, threshold-triggered pruning or reset, and a return map whose fixed point is stable, then Phase-A/Phase-B oscillations may occur. Such oscillations are model-dependent and not universal consequences of the FDS core.*

Proof. Threshold-triggered reset maps can define a Poincare return map on a reduced state surface. If that map has a stable fixed point, oscillatory cycling follows. The result depends on the reset rule and stability hypotheses, and therefore belongs to normal-form dynamics rather than the formal core. $\square$

Part VI.

Quarantined Applications: Domain Bridges and Operational Tests

11. Application Protocol

11.1. The Three-Part Application Form

Protocol 11.1 (Domain Bridge Protocol). Every application of Distinction Theory outside the formal FDS core must be written in the following form:

FDS theorem+domain bridge assumptions+operational test $\Rightarrow$ domain conclusion.

Failure of a domain conclusion may invalidate the bridge assumptions or the operational mapping without invalidating the formal FDS theorem.

11.2. Application Claim Template

Each application must specify:

- (1) the system boundary B ;
- (2) the internal memory/model space M ;
- (3) the update operator U ;
- (4) whether the boundary is active under Criterion 2.6;
- (5) the task-relevant variable Z_t ;
- (6) the rate-distortion demand $R_{\min}^{(\tau)}(\varepsilon)$ over the admissible class;
- (7) the capacity estimate C_S ;
- (8) the erasure term $H(M_t \mid M_{t+1}, Y_t)$, if any;
- (9) the free-energy or resource budget Φ ;
- (10) the perturbation family $\mathcal{P}$;
- (11) the proposed invariant quotient q , if persistence is claimed;
- (12) the falsification condition.

12. Example: Artificial Agents

12.1. Non-Claim

Not claimed. The FDS core does not prove that any named model class can or cannot realize artificial general intelligence. It only states conditional constraints on active finite systems.

12.2. Conditional AI Bridge

Let an artificial system have state

$$M_t = (\theta, c_t, m_t^{\text{ext}}),$$

where θ are parameters, c_t is transient context or working state, and m_t^{ext} is durable external memory or environmental state.

Proposition 12.1 (Passive-mapper limitation, conditional). *If an artificial system has fixed parameters, no durable self-updating memory, no active pruning, no causal boundary control, and no externalization channel capable of reducing task-relevant capacity deficit, then it remains outside the strong FDS-agency class. Its apparent agency must be supplied by prompt structure, external operators, tools, environment, or wrapper systems.*

Proof. Under the stated conditions, the system lacks durable updates participating in future boundary-maintenance loss and lacks internal mechanisms for pruning or externalizing accumulated task-relevant distinctions. By Criterion 2.6, it does not qualify as a strong active-boundary agent. Any persistent agency must therefore be attributed to the larger coupled system that includes tools, memory, environment, or human operators. $\square$

Remark 12.2. A tool-using agent with writable memory, causal interventions, active pruning, and durable boundary control may satisfy the FDS conditions even if its base model has fixed parameters. The relevant system is the whole coupled architecture, not the neural network weights alone.

13. Other Domain Examples (Minimal)

Further domain mappings follow the same protocol: a biological organism maps membrane, regulatory memory, metabolism, repair, and ecological niche to B, M, Φ , pruning, and externalization; a disease process can be studied as a competing active-boundary subsystem if it exhibits boundary maintenance, resource acquisition, and adaptive updating; a civilization maps institutions, infrastructure, records, energy throughput, and governance to the FDS components. All such claims are domain-bridge theses, not theorems of the formal core, and require the full Application Protocol for operational testing.

Part VII.

Appendices

A. Notation Summary

Symbol	Meaning
X	Internal state space
E	Environmental state space
B	Boundary/interface variable
M	Internal memory/model space
Y	Observation channel
A	Action or boundary-update space
U	Internal update operator
π	Finite distinction projection / coarse-graining
ℓ	Boundary-maintenance loss
Φ	Resource or free-energy budget functional
$\mathcal{P}$	Perturbation/pruning/coarse-graining family
τ	Update timescale
C_S	Internal capacity, typically $\log_2 M $
Ψ	Admissible class of statistics $\psi : (E, B) \rightarrow Z$
$R_{\min}^{(\tau)}(\varepsilon)$	Minimal rate-distortion demand, $\inf_{\psi \in \Psi} R_{\psi(E, B)}^{(\tau)}(\varepsilon)$
$\Delta_\varepsilon(\tau)$	Capacity deficit over window τ , $R_{\min}^{(\tau)}(\varepsilon) - C_S$
b_t	Logical erasure, $H(M_t M_{t+1}, Y_t)$
q	Quotient or invariant feature map
R_A	Structural identity predicate

B. Theorem Dependency Table

Claim		Proof status	Dependencies and failure consequence
Active Qualification	Boundary	Definitional / Domain criterion	Fails only if active-boundary definition is replaced or shown incoherent.
Finite Representation	Distinction	Formal	Depends on finite internal capacity and information inequalities.
Capacity Deficit Theorem		Formal information-theoretic /	Depends on rate-distortion formalization and capacity estimate.
Approximation Proliferation		Conditional formal	Depends on persistent novelty and exclusion of pruning, externalization, task relaxation, and compression improvement.
Dissipation Floor		Physical bridge	Depends on Landauer conditions for physical irreversible erasure.
Trichotomy		Conditional physical	Depends on finite resource budget and maintenance cost exceeding input.
Invariant Persistence		Formal	Depends on existence of invariant quotient and perturbation family.
Normal-Form Regime		Normal-form	Describes regimes; does not claim universal oscillation.
Domain applications		Domain-bridge	Fail locally if mappings or operational tests fail.

B.1. Layered Defense Matrix

The three-layer architecture ensures that failure never propagates upward:

B. Theorem Dependency Table

Layer	Content	Failure Consequence
L0 (Formal Core)	The FDS definitions and core theorems	Can only be killed by a mathematical counterexample or logical inconsistency.
L1 (Physical Bridge)	Landauer bridge, dissipation floor, free-energy budgets	Failure suspends only the thermodynamic corollaries; L0 survives.
L2 (Quarantined Applications)	Life, intelligence, AI, civilization	Failure archives the specific domain bridge; no upstream damage.

C. Failure Registry

C.1. Formal Core Failures

A formal core claim fails if its definitions are inconsistent or a counterexample satisfies the hypotheses while violating the conclusion.

C.2. Physical Bridge Failures

The physical bridge fails if physical irreversible erasure under stated thermodynamic conditions reliably violates the Landauer lower bound, or if the assumed thermal, locality, or coupling conditions do not apply.

C.3. Normal-Form Failures

A normal-form claim fails only as a universal claim if treated as universal. In this document it is not universal; it is a regime model. Specific normal forms may fail to fit specific systems.

C.4. Application Failures

An application fails if its domain bridge assumptions are false, its operational variables are misidentified, or its predicted test outcome fails. Such failure does not propagate to the formal FDS core unless the application was the only support for a formal premise, which is explicitly disallowed.

D. Bibliographic Orientation

The formal core is naturally adjacent to several existing bodies of mathematics and physics:

- Shannon information theory and rate-distortion theory [2, 3, 5];
- stochastic thermodynamics and Landauer’s principle [7, 8, 10, 11];
- coarse-graining, renormalization group, and effective state descriptions;
- quotient spaces, invariants, fixed points, and algebraic topology;
- Markov blankets and active-boundary systems [12, 13];
- dynamical systems with resource constraints.

Key references for the formal framework include the core exposition [1] and a companion physical claim registry [16], to be archived separately. The rate-distortion bound follows the standard source-coding theorem; the Landauer bridge follows the established experimental and theoretical literature [9]; the collapse and persistence trichotomy connects to resource-constrained dynamical systems and stochastic thermodynamics.

These connections do not make Distinction Theory true by association. They specify where the theory must be formalized, tested, and, if necessary, broken.

E. Independent Criteria for Invariant-Supported Classification

A structure may be classified as invariant-supported under a perturbation family if it satisfies at least one of the following criteria:

- (I) **Global vs. local character.** The identity condition R_A depends on quantized, global features (topological winding numbers, Chern numbers, symmetry charges) rather than continuous, local micro-coordinates.
- (II) **Gap protection.** Destruction of the structure requires closing a spectral gap or crossing a non-perturbative phase boundary.
- (III) **Coarse-graining stability.** Under iterative renormalization-group flow, the defining features of R_A flow to a non-trivial fixed point.

A structure satisfying none of these criteria is identifier-dependent: its persistence relies on active maintenance, repair, isolation, or stochastic stability rather than on algebraic protection through an invariant quotient.

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Finite Distinguishability Budgets and Maintenance Bounds for Physical Observers

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A finite physical observer cannot register, preserve, update, and operationally use an unlimited number of distinctions. This paper defines an observer-relative distinguishability budget $N_{\mathcal{O}} = |\text{Im}(\pi_{\mathcal{O}})|$ and bit capacity $C_{\mathcal{O}} = \log_2 N_{\mathcal{O}}$ for finite record-bearing systems. Using established Bekenstein, holographic, channel, and Landauer-style update constraints as physical bridge inputs, we formulate accessible distinguishability as a bottleneck over internal memory, boundary access, communication channel, causal reach, and irreversible update throughput. We then define boundary-relative capacity deficit by comparing task-relevant rate-distortion demand $R_{\min}^{(\tau)}(\varepsilon; \Psi)$ with accessible capacity. When demand persistently exceeds accessible budget, a finite observer must coarse-grain, compress, externalize, prune, relax the task, or fail. The paper does not derive gravitational entropy bounds, quantum gravity, the Standard Model, the Einstein equations, or the numerical coefficient in the Bekenstein-Hawking entropy formula. Its purpose is narrower: to make finite distinguishability a precise, auditable, and testable bridge between finite-system theory, information bounds, and physical observation.

Reader Contract. This paper does not claim to derive the Bekenstein bound, holography, Landauer’s principle, the Einstein equations, quantum gravity, the Standard Model, or the numerical coefficient $1/4$ in the Bekenstein-Hawking entropy formula. It uses established entropy and thermodynamic constraints as physical bridge inputs. Its contribution is to express them as observer-relative distinguishability budgets for finite record-bearing systems and to derive maintenance consequences when task demand exceeds accessible capacity.

Claim-status summary

Keywords: finite distinguishability; holographic bound; Bekenstein bound; observer-relative entropy; active finite distinction systems; boundary maintenance; information bounds; causal horizons; Landauer principle; finite observers.

INTRODUCTION

The finite-observer problem

Physical theories often describe states as if all degrees of freedom in a region were simultaneously available to an ideal observer. Operational physics does not work that way. A real observer is a finite physical system. It has finite memory, finite energy, finite update rate, finite causal access, finite error tolerance, finite record stability, and a finite boundary through which information must pass. Even when the underlying mathematical

state space is large, the number of distinctions that can be registered and used by a finite system is limited.

Main concept

The central object of this paper is an observer-relative distinguishability budget. A finite system does not access the full physical state space directly. It resolves a space of possibilities into operationally distinguishable equivalence classes via a finite distinction projection $\pi_{\mathcal{O}}$:

$$N_{\mathcal{O}} = |\text{Im}(\pi_{\mathcal{O}})|, \quad C_{\mathcal{O}} = \log_2 N_{\mathcal{O}}, \quad (1)$$

where $\pi_{\mathcal{O}}$ is the projection by which $\mathcal{O}$, a finite record-bearing system, resolves a physical region into operationally distinguishable alternatives.

Why existing bounds matter

The Bekenstein bound limits the information that can be contained in a finite region of given energy and size [10]. The holographic principle asserts that the maximum entropy associated with a region scales with boundary area rather than volume [12–14]. Black-hole thermodynamics gives the limiting entropy

$$S_{\text{BH}} = \frac{k_{\text{B}} A}{4\ell_{\text{P}}^2}, \quad (2)$$

where A is horizon area and $\ell_{\text{P}} = \sqrt{\hbar G/c^3}$ is the Planck length [8, 9]. The present paper does not re-derive or modify these bounds. It treats them as physical bridge inputs that constrain the distinguishability of any finite record-bearing system.

TABLE I. Summary of central FDS-T1 claims, their epistemic status, and what would constitute falsification or demotion.

Claim	Status	What would fail
Finite record carrier $\rightarrow$ finite operational distinguishability	Operational / formal bridge	Bounded physical observer with unlimited reliable distinctions
Bekenstein ceiling on distinguishability	Physical bridge	Applicable bounded system exceeds Bekenstein information limit
Holographic ceiling	Physical bridge	Horizon-bounded system exceeds area information accounting
Accessible capacity bottleneck	Conditional theorem	Task uses distinctions that bypass all bottlenecks
Boundary-relative capacity deficit	Definition	Not falsified directly; only usefulness can fail
Budget-exit theorem	Conditional theorem	Full-fidelity maintenance under sustained deficit without exit
Maintenance inequality	Conditional physical bridge	Reliable irreversible erasure below Landauer cost under stated conditions
Budget crossing signature	Testable prediction	No transition in error, latency, compression, externalization, pruning, reset/erasure, heat, task relaxation, or failure under controlled crossing

Contributions

This paper makes nine contributions.

1. It defines an observer-relative finite distinguishability budget $N_{\mathcal{O}} = |\text{Im}(\pi_{\mathcal{O}})|$ and bit capacity $C_{\mathcal{O}} = \log_2 N_{\mathcal{O}}$.
2. It states conservative physical bridges from finite record carriers to Bekenstein and holographic information ceilings.
3. It distinguishes stock capacity (storage/boundary/channel/memory) from update throughput (Landauer-limited irreversible operations).
4. It formulates accessible capacity as the minimum of stock and throughput.
5. It defines boundary-relative capacity deficit by comparing rate-distortion task demand against accessible capacity.
6. It derives a budget-exit classification: coarse-graining or compression, externalization, pruning, task relaxation, or failure when demand exceeds accessible budget.
7. It provides a maintenance inequality connecting positive deficit to Landauer-style thermodynamic lower bounds.
8. It makes budget crossing the central testable prediction, with explicit observables and test protocols.
9. It gives falsification conditions and a reproducibility checklist.

What is not claimed

This paper does not derive the Bekenstein bound, holography, Landauer’s principle, the Einstein equations, quantum gravity, the Standard Model, or the numerical coefficient $1/4$ in the Bekenstein-Hawking entropy formula. It uses established entropy and thermodynamic constraints as physical bridge inputs. Its contribution is to express them as observer-relative distinguishability budgets for finite record-bearing systems and to derive maintenance consequences when task demand exceeds accessible capacity.

Relation to companion papers

This paper is the physical finite-observer bridge in the FDS sequence. It depends on the formal FDS core [1] and the DT archive [2] for foundational definitions. It does not duplicate the downstream applications already developed in companion papers. Artificial agency is treated in the FDS-AI Agency paper [3]; active pruning in protocell-like systems is treated in FDS-L1 [4]; reportable cognitive access is treated in FDS-C1 [6]; and life/cognitive bridge claims are registered in FDS-LC0 [7]. The present paper supplies the shared finite-observer budget language used by those papers: observer-relative finite distinction projections, accessible capacity, boundary-relative capacity deficit, maintenance cost, and budget-exit classification.

RELATED WORK

Bekenstein bounds and holography

Bekenstein proposed a universal upper bound on the entropy-to-energy ratio for bounded systems [10]. In one common form, a system of energy E confined to a sphere of radius R satisfies

$$S \leq \frac{2\pi k_B ER}{\hbar c}. \quad (3)$$

Measured in bits, this becomes

$$I \leq \frac{2\pi ER}{\hbar c \ln 2}. \quad (4)$$

The holographic principle strengthens the geometric intuition: the maximum entropy associated with a spatial region is controlled by the area of an appropriate boundary, not by the enclosed volume [12–14]. Bousso’s covariant entropy bound generalizes the area law to light-sheets and avoids some limitations of simple spatial-volume statements [14].

The present paper does not re-prove these bounds. It treats them as physical constraints that any finite distinguishability theory must respect. Its novelty is to place them inside the FDS architecture: entropy bounds become upper limits on the number of distinctions that a finite boundary-maintaining system can operationally resolve, preserve, or use.

Black-hole thermodynamics and horizon entropy

Black-hole thermodynamics established that horizons carry entropy and temperature [8, 9]. The entropy formula (2) is the paradigmatic area law. Jacobson’s thermodynamic derivation of the Einstein equation from local horizon thermodynamics suggests a deep connection between area, entropy, energy flux, and spacetime geometry [15]. Verlinde and related emergent-gravity approaches further explore the possibility that gravity is connected to information, entropy, and coarse-graining [16].

FDS-T1 remains upstream of a full gravity derivation. It does not infer Einstein equations. It supplies a finite-system reading of horizon entropy: a horizon is not only a geometric surface but also a boundary of finite distinguishability access for a class of observers.

Landauer erasure and information thermodynamics

Landauer’s principle links logically irreversible information erasure to a thermodynamic heat cost [17]. Bennett clarified how reversible computation can avoid erasure costs locally while still requiring attention to garbage, memory reuse, and thermodynamic accounting

[18]. Experiments have verified Landauer-type erasure costs in controlled systems [19–21]. Stochastic thermodynamics and feedback thermodynamics generalize these links in nonequilibrium systems [22–24].

FDS uses Landauer conservatively. It does not assert that every representation or inference step dissipates $k_B T \ln 2$ per bit. It applies the bridge to physically implemented logically irreversible operations: erasure, reset, many-to-one overwrite, and irreversible compression. In this paper, Landauer enters as an update-throughput constraint on finite distinguishability, not as a replacement for the Bekenstein or holographic bounds.

Observer-dependence and finite records

Operational and relational views of physics emphasize that measurements are records available to physical systems rather than disembodied facts [25, 26]. Quantum Darwinism studies how information about a system becomes redundantly recorded in an environment [27]. Decoherence explains why stable records emerge in open quantum systems [27, 28]. These traditions motivate the idea that observerhood is a physical record-bearing role.

The FDS observer is not defined by consciousness or biological subjectivity. It is a finite distinction-register: a physical system capable of registering, preserving, updating, and ordering distinctions under finite capacity. That definition makes information bounds directly relevant to observation.

Rate-distortion and bounded representation

Rate-distortion theory asks how many bits are required to represent a source under a specified distortion level [29–32]. FDS uses this language to define capacity deficit: a task can demand more distinctions than a system can maintain. In physical settings, the available capacity is not only computational but also bounded by energy, causal access, and horizon area.

Relation to standard information thermodynamics

Each ingredient in FDS-T1—Bekenstein bounds, holographic ceilings, Landauer erasure, rate-distortion theory—is individually well established. The novelty is not any one ingredient taken separately. It is the finite-observer maintenance accounting obtained when task-relevant rate-distortion demand is compared against a bottlenecked accessible capacity (internal memory, boundary ceiling, channel, update rate, and thermodynamic budget) and then converted into compulsory exits or irreversible update cost. Bekenstein and holographic bounds set the storage or access ceiling; Landauer sets

the per-irreversible-operation heat floor; rate-distortion sets the task demand. FDS-T1's contribution is the deficit-driven classification that connects them: when demand exceeds the bottlenecked accessible capacity, the system must coarse-grain, externalize, prune, relax the task, or fail; and the surplus distinctions that are irreversibly erased carry a Landauer-accounted thermodynamic cost.

FDS PRELIMINARIES

Distinctions and finite projections

Definition 1 (Distinction). *A distinction is an operation or relation that separates at least two alternatives within a possibility space. At the elementary formal level, a distinction may be represented by a nontrivial partition*

$$D = \{A_i\}_{i=1}^n, \quad A_i \cap A_j = \emptyset \ (i \neq j), \quad \bigcup_i A_i = X, \quad (5)$$

with $n \geq 2$.

A physical distinction is not merely a formal partition. It must be instantiated by a record, boundary, coupling, memory state, detector state, or other physical carrier. A finite system implements distinctions through a finite projection.

Definition 2 (Finite distinction projection). *Let Ω be a physical possibility space for a region or task. A finite observer $\mathcal{O}$ implements a finite distinction projection*

$$\pi_{\mathcal{O}} : \Omega \rightarrow \mathcal{Z}_{\mathcal{O}}, \quad (6)$$

where $\mathcal{Z}_{\mathcal{O}}$ is the finite set of distinction classes accessible to $\mathcal{O}$.

Definition 3 (Distinguishability budget). *The instantaneous distinguishability budget of $\mathcal{O}$ is*

$$N_{\mathcal{O}} = |\mathcal{Z}_{\mathcal{O}}| = |\text{Im}(\pi_{\mathcal{O}})|. \quad (7)$$

The bit capacity is

$$C_{\mathcal{O}} = \log_2 N_{\mathcal{O}}. \quad (8)$$

When $\mathcal{Z}_{\mathcal{O}}$ is continuous or effectively continuous, $C_{\mathcal{O}}$ denotes the operational number of reliably distinguishable states under the observer's noise, energy, resolution, and error-tolerance constraints.

Record-bearing observers

Definition 4 (Finite distinction-register). *A finite distinction-register is a physical system that can register, preserve, update, and order distinctions using finite*

resources. It must have a record carrier, a boundary or interface, a stability condition, and a finite update capacity.

This definition intentionally excludes two extremes. A disembodied mathematical observer is not a finite distinction-register because it has no physical record carrier. A purely passive boundary that never records or updates distinctions is also not an observer in this sense. A detector, memory device, organism, autonomous system, or laboratory apparatus can be a finite distinction-register when its boundary and records satisfy the relevant stability and update conditions.

Stock capacity and update throughput

The word ‘‘capacity’’ hides two distinct quantities: storage-like capacity and flow-like update throughput.

Definition 5 (Stock / accessible-state capacity). *The stock capacity of observer $\mathcal{O}$ in region $\mathcal{D}$ at time t is*

$$C_{\text{stock}}(\mathcal{O}, \mathcal{D}, t) = \min\{C_{\text{mem}}(\mathcal{O}, t), C_{\text{bd}}(\mathcal{O}, \mathcal{D}, t), C_{\text{chan}}(\mathcal{O}, \mathcal{D}, t), C_{\text{ext}}^{\text{eff}}(\mathcal{O}, t)\}, \quad (9)$$

where C_{mem} is internal memory capacity, C_{bd} is boundary-limited capacity, C_{chan} is channel-limited capacity, and $C_{\text{ext}}^{\text{eff}}$ is effective externalized capacity after indexing, verification, synchronization, latency, and repair costs.

Definition 6 (Update / irreversible-throughput capacity). *The maximum number of irreversible distinction updates over interval $[t, t + \tau]$ is bounded by the available free energy or allowable dissipation budget:*

$$I_{\text{update}}^{\text{max}}(t, t + \tau) \leq \int_t^{t+\tau} \frac{\dot{F}_{\text{avail}}(s)}{k_{\text{B}}T(s) \ln 2} ds, \quad (10)$$

where $\dot{F}_{\text{avail}}$ is the rate of available free-energy input or allowable dissipation for irreversible operations. This is not a holographic bound. It is an update-throughput ceiling derived from the Landauer bridge: the actual heat cost of irreversible updates satisfies $\dot{Q}_{\text{info}}(t) \geq k_{\text{B}}T(t) \ln 2 \dot{I}_{\text{irr}}(t)$, so the available budget $\dot{F}_{\text{avail}}$ upper-bounds the rate at which irreversible updates can be physically sustained.

The effective task-available capacity is the minimum of stock and throughput:

$$C_{\text{acc}}(\mathcal{O}, \mathcal{D}, t, \tau) = \min\{C_{\text{stock}}(\mathcal{O}, \mathcal{D}, t), I_{\text{update}}^{\text{max}}(t, t + \tau)\}. \quad (11)$$

Unless otherwise stated, all capacity terms entering C_{acc} are converted into usable task-relevant bits over the same update window $[t, t + \tau]$. Stock capacities are interpreted as the maximum number of simultaneously maintained distinctions that can be reliably preserved during the window, while throughput capacities are interpreted as the maximum number of irreversible updates

that can be executed during the window. This conversion ensures that the minimum in (11) compares quantities with compatible units. When no ambiguity arises, we write $C_{\text{acc}}(\mathcal{O}, \mathcal{D})$ as shorthand for $C_{\text{acc}}(\mathcal{O}, \mathcal{D}, t, \tau)$.

Definition 7 (Boundary-relative capacity deficit).

$$\Delta_{\text{FDS}}(\mathcal{O}, \mathcal{D}, \Psi, \varepsilon, \tau) = R_{\min}^{(\tau)}(\varepsilon; \Psi) - C_{\text{acc}}(\mathcal{O}, \mathcal{D}, t, \tau). \quad (12)$$

PHYSICAL BRIDGE ASSUMPTIONS

Finite distinguishability

Assumption 1 (Finite distinguishability). *A physically instantiated system that maintains identity has a finite distinguishability budget over any finite operational interval. Its accessible records, memory states, update channels, and boundary-maintenance operations are bounded by finite physical resources.*

This assumption is deliberately weak. It does not require that all physical systems saturate a black-hole bound. It requires only that physical systems do not possess infinite usable distinguishability. It is compatible with finite entropy bounds, finite channel capacities, finite memory, finite energy, finite precision, and finite causal access.

Bekenstein bridge

Assumption 2 (Bekenstein bridge). *For a weakly gravitating physical system of total energy E confined to radius R , the entropy and information accessible within that region obey the Bekenstein bound*

$$S \leq \frac{2\pi k_{\text{B}} E R}{\hbar c}, \quad I \leq \frac{2\pi E R}{\hbar c \ln 2}. \quad (13)$$

FDS does not derive this bound in the present paper. It uses it as an established physical bridge constraint.

Holographic bridge

Assumption 3 (Holographic ceiling). *For a horizon-bounded or gravitationally saturated region whose appropriate boundary area is A , the maximum accessible entropy satisfies*

$$S \leq \frac{k_{\text{B}} A}{4\ell_{\text{P}}^2}, \quad (14)$$

and therefore the maximum number of accessible bits satisfies

$$I \leq \frac{A}{4\ell_{\text{P}}^2 \ln 2}. \quad (15)$$

The qualifier “appropriate boundary” matters. In covariant settings the relevant area may be defined by a light-sheet or causal horizon rather than an arbitrary spatial surface [14].

Landauer update bridge

Assumption 4 (Landauer update bridge). *When a physical system performs a logically irreversible operation—erasure, many-to-one overwrite, reset, irreversible compression, or pruning of previously distinguishable alternatives—the operation incurs a thermodynamic entropy cost under standard Landauer conditions.*

This bridge is used only for update and erasure accounting. Stable storage and reversible evolution are not automatically charged with $k_{\text{B}} T \ln 2$ per bit.

Finite causal access

Assumption 5 (Finite causal access). *A finite physical observer cannot instantaneously access, register, update, and causally use arbitrary distant distinctions without unbounded update capacity. Distinguishability is therefore restricted by causal structure and finite update rate.*

This assumption is weaker than a derivation of relativity. It says only that finite observers require finite causal access. The numerical value of c is not derived here.

FINITE DISTINGUISHABILITY BUDGETS

The finite-budget proposition

Proposition 1 (Finite record carriers imply finite distinguishability). *Let $\mathcal{O}$ be a physical record-bearing system with finite internal record capacity, finite error tolerance, and finite update resources over interval $[t_0, t_1]$. Then $\mathcal{O}$ has a finite operational distinguishability budget over that interval.*

Proof. A distinction available to $\mathcal{O}$ must be encoded in a record or state difference that can be reliably discriminated under the observer’s noise and error tolerance. If the record substrate has finite capacity, only finitely many such distinguishable record states can be reliably preserved and accessed over the interval. If update resources are also finite, the number of new distinctions that can be acquired, overwritten, corrected, or stabilized is finite. Therefore the operational image of $\pi_{\mathcal{O}}$ is finite, and $N_{\mathcal{O}} < \infty$. $\square$

Remark 1. *This proposition is not a statement that the underlying mathematical Hilbert space is finite-*

dimensional. It is a statement about operationally accessible distinctions for a finite physical record carrier.

Distinguishability throughput

A finite observer may have both an instantaneous storage budget and a lifetime throughput budget. Let $\dot{I}_{\text{irr}}(t)$ be the rate at which logically irreversible distinction updates occur. Under the Landauer bridge,

$$\dot{Q}_{\text{info}}(t) \geq k_{\text{B}}T(t) \ln 2 \dot{I}_{\text{irr}}(t). \quad (16)$$

If the observer has finite free-energy input or dissipation allowance over $[t_0, t_1]$, then

$$\int_{t_0}^{t_1} \dot{I}_{\text{irr}}(t) dt \leq \int_{t_0}^{t_1} \frac{\dot{Q}_{\text{info}}(t)}{k_{\text{B}}T(t) \ln 2} dt, \quad (17)$$

with the inequality interpreted carefully when temperature varies or non-thermal reservoirs are involved. This is not a holographic bound. It is an update-throughput bound. It says that a finite observer cannot indefinitely acquire and erase distinctions for free.

Internal versus accessible budgets

Internal memory can be far below the physical entropy ceiling of the region. Conversely, a large physical region can contain many degrees of freedom that are inaccessible to a small observer. FDS therefore uses the bottleneck form (11). For observation and control tasks, what matters is not the total number of microstates in a region but the number of task-relevant distinctions that can be reliably acquired, represented, and used by the observer.

Theorem 1 (Accessible budget bottleneck). *For any finite observer $\mathcal{O}$ performing a task in region $\mathcal{D}$ over window τ , the operational capacity available to that task is upper-bounded by the minimum of stock capacity and update throughput:*

$$C_{\text{task}}(\mathcal{O}, \mathcal{D}, t, \tau) \leq C_{\text{acc}}(\mathcal{O}, \mathcal{D}, t, \tau). \quad (18)$$

Proof. A task-relevant distinction must pass multiple filters: it must be storable in memory or externalized records, physically available through the boundary, transmissible through a finite channel, and updatable within the relevant timescale. The effective usable capacity is the minimum of the stock bottleneck (memory, boundary, channel, externalization) and the update-throughput bottleneck. Therefore C_{task} cannot exceed C_{acc} . $\square$

HOLOGRAPHIC INFORMATION BOUNDS AS DISTINGUISHABILITY CEILINGS

Weak-gravity regime: Bekenstein ceiling

In weakly gravitating bounded systems, the Bekenstein bound gives

$$C_{\text{bd}}(\mathcal{O}, \mathcal{D}) \leq \frac{2\pi ER}{\hbar c \ln 2}, \quad (19)$$

where E and R characterize the bounded region relative to the operational setup.

Theorem 2 (Bekenstein distinguishability ceiling). *Assume the Bekenstein bridge applies to a bounded physical region $\mathcal{D}$ of energy E and radius R . Then no finite observer can operationally distinguish more than*

$$N_{\mathcal{O}} \leq \exp\left(\frac{2\pi ER}{\hbar c}\right) \quad (20)$$

region-states as mutually accessible alternatives through that region without violating the entropy bound.

Proof. The Bekenstein bridge bounds the entropy by $S/k_{\text{B}} \leq 2\pi ER/\hbar c$. The number of equiprobable distinguishable alternatives compatible with entropy S is $N = \exp(S/k_{\text{B}})$. Therefore $N \leq \exp(2\pi ER/\hbar c)$. In bits this is equivalent to (19). $\square$

Remark 2. *This theorem is conditional on the physical bridge. It does not derive the Bekenstein bound from the distinction primitive alone.*

Remark 3. *Here $N_{\mathcal{O}}$ should be read as the maximum effective number of reliably distinguishable, thermodynamically admissible alternatives, or equivalently the equiprobable support size associated with the entropy ceiling. For nonuniform ensembles, the corresponding quantity is the effective entropy number rather than a raw set cardinality.*

Horizon-saturating regime: area ceiling

When gravitational collapse or horizon saturation becomes relevant, the area law dominates the naive volume count.

Theorem 3 (Holographic distinguishability ceiling). *Assume the holographic ceiling applies to a causal or horizon-bounded region $\mathcal{D}$ with boundary area A . Then the accessible distinction capacity satisfies*

$$C_{\mathcal{O}} \leq \frac{A}{4\ell_{\text{P}}^2 \ln 2}. \quad (21)$$

Equivalently,

$$N_{\mathcal{O}} \leq \exp\left(\frac{A}{4\ell_{\text{P}}^2}\right). \quad (22)$$

Proof. The holographic bridge gives $S \leq k_B A / (4\ell_P^2)$. The number of distinguishable alternatives is at most $\exp(S/k_B)$, hence $N_O \leq \exp(A/4\ell_P^2)$. Taking $\log_2$ yields (21). $\square$

Why area, not volume

The FDS interpretation of area scaling is operational. A finite observer interacts with a region through a boundary or causal interface. Increasing interior volume without increasing boundary capacity cannot indefinitely increase the number of distinctions that can be accessed, stabilized, and communicated across that boundary. At gravitational saturation, attempts to store or distinguish more interior alternatives change the boundary conditions themselves, producing horizon behavior rather than unlimited volume scaling.

This is not a claim that ordinary low-energy systems always saturate area bounds. Most do not. The claim is that the ultimate ceiling on accessible distinguishability is boundary-like in the regimes where gravity and causal horizons enforce the strongest known entropy constraints.

Comparison of ceilings

Combining the internal, Bekenstein, and holographic limits gives a conservative stock-capacity formula. The boundary-limited component of stock capacity is

$$C_{\text{bd}}(\mathcal{O}, \mathcal{D}) \leq \min \left\{ \frac{2\pi ER}{\hbar c \ln 2}, \frac{A}{4\ell_P^2 \ln 2} \right\}. \quad (23)$$

The full stock capacity is

$$C_{\text{stock}}(\mathcal{O}, \mathcal{D}, t) = \min \{ C_{\text{mem}}(\mathcal{O}, t), C_{\text{bd}}(\mathcal{O}, \mathcal{D}, t), C_{\text{chan}}(\mathcal{O}, \mathcal{D}, t), C_{\text{ext}}^{\text{eff}}(\mathcal{O}, t) \}, \quad (24)$$

and the effective task capacity is

$$C_{\text{acc}}(\mathcal{O}, \mathcal{D}, t, \tau) = \min \{ C_{\text{stock}}(\mathcal{O}, \mathcal{D}, t), I_{\text{update}}^{\text{max}}(t, t + \tau) \}. \quad (25)$$

Different terms dominate in different regimes. For a laboratory detector, C_{mem} , C_{chan} , or $C_{\text{ext}}^{\text{eff}}$ may dominate. For black holes or cosmological horizons, the area term dominates. For thermally constrained computational systems, the update-throughput term dominates.

LOCAL OBSERVERS AND CAUSAL PATCHES

Observer-relative patches

A finite observer does not resolve the universe globally. It resolves a causal patch.

Definition 8 (Observer patch). *The observer patch $\mathcal{D}_O(t)$ is the region whose states can in principle influence the records of $\mathcal{O}$ by time t under the applicable causal and channel constraints.*

The boundary $\partial\mathcal{D}_O$ may be a literal detector surface, a light cone, a causal diamond boundary, a black-hole horizon, an apparent horizon, an acoustic horizon in analogue systems, or a task-defined operational boundary.

Definition 9 (Distinguishability horizon). *A distinguishability horizon for $\mathcal{O}$ is a boundary beyond which differences in physical state cannot be operationally registered, preserved, and used by $\mathcal{O}$ within the relevant resource and timescale constraints.*

A distinguishability horizon need not be an event horizon. It may arise from noise, finite sampling, finite memory, finite control, finite latency, or thermodynamic cost. Event horizons and causal horizons are limiting physical cases.

Patch nesting

Remark 4 (Nested patches). *If $\mathcal{D}_1 \subseteq \mathcal{D}_2$ are observer-accessible regions with compatible boundary definitions, the accessible boundary capacity of $\mathcal{D}_1$ cannot exceed that of $\mathcal{D}_2$ under otherwise identical conditions. Operational capacity, however, is a minimum over multiple bottlenecks; a smaller patch may have better channel or update characteristics.*

Composite observers

Composite observers can pool records, distribute memory, and externalize distinctions into shared media. But pooling does not remove physical bounds. It changes the boundary.

Definition 10 (Composite finite observer). *A composite finite observer is a network of finite distinction-registers coupled by communication channels, shared records, or coordination protocols, treated as a higher-level observer only when the network can preserve and update joint distinctions with sufficient reliability.*

Proposition 2 (Externalization changes the budget boundary). *When an observer externalizes records into an environment, database, instrument, institution, or horizon boundary, the relevant budget becomes the budget of the coupled observer-record system, not the internal memory alone.*

Proof. Externalized records are useful only if the observer can access, validate, update, and protect them through a boundary. The operational distinction is therefore carried by the coupled system. The capacity is no longer

merely the internal memory capacity $C_{\text{mem}}(\mathcal{O}, t)$ but is bounded by the record medium, access channel, stability condition, and the boundary information ceiling of the enlarged system. $\square$

CAPACITY DEFICIT UNDER HOLOGRAPHIC BOUNDS

Physical distinction demand

For a task family Ψ , the demand $R_{\text{min}}^{(\tau)}(\varepsilon; \Psi)$ may include distinctions among microstates, macrostates, trajectories, causal histories, control-relevant variables, or boundary-relevant signals. The FDS-T1 deficit is positive when

$$R_{\text{min}}^{(\tau)}(\varepsilon; \Psi) > C_{\text{acc}}(\mathcal{O}, \mathcal{D}, t, \tau). \quad (26)$$

A positive deficit is not ignorance in the ordinary sense. It is an operational impossibility under the chosen boundary, resources, and timescale.

Budget-exit theorem

Theorem 4 (Budget-exit theorem). *Let $\mathcal{O}$ be a finite distinction-register performing task family Ψ over region $\mathcal{D}$ with distortion tolerance ε and update window τ . Under fixed observer boundary, resource budget, task family, and update window, if $\Delta_{\text{FDS}}(\mathcal{O}, \mathcal{D}, \Psi, \varepsilon, \tau) > 0$ persists over the relevant interval, then $\mathcal{O}$ cannot maintain full-fidelity task performance using internal accessible capacity alone. It must enter at least one exit class: coarse-graining or compression, externalization, pruning, task relaxation, or failure. Increasing the budget or changing the observer boundary constitutes a change of system, not a violation of the theorem.*

Proof. Full-fidelity task performance requires at least $R_{\text{min}}^{(\tau)}(\varepsilon; \Psi)$ bits of relevant distinction capacity. By assumption this exceeds C_{acc} , which is the minimum of stock capacity and update throughput. Under fixed boundary, resources, and task window, the missing distinctions cannot be generated internally without violating the capacity ceiling. The admissible exits are to reduce demand through coarse-graining or compression, externalize part of the burden, prune or discard distinctions, relax the task, or fail to perform it. $\square$

Holographic deficit

In a horizon-bounded regime, the deficit uses the boundary-limited stock component:

$$\Delta_{\varepsilon}^{\text{hol}}(\mathcal{O}, \tau) = R_{\text{min}}^{(\tau)}(\varepsilon; \Psi) - \min \left\{ C_{\text{stock}}(\mathcal{O}, \mathcal{D}, t), I_{\text{update}}^{\text{max}}(t, t + \tau) \right\}. \quad (27)$$

When $\Delta_{\varepsilon}^{\text{hol}} > 0$, no increase in volume description alone solves the problem. The limiting factor is boundary-accessible distinction capacity.

Observer-relative entropy

Definition 11 (Observer-relative accessible entropy). *The accessible entropy assigned by observer $\mathcal{O}$ to region $\mathcal{D}$ is*

$$S_{\text{acc}}(\mathcal{O}, \mathcal{D}) = k_{\text{B}} \ln N_{\text{acc}}(\mathcal{O}, \mathcal{D}), \quad (28)$$

where N_{acc} is the number of region alternatives operationally distinguishable by $\mathcal{O}$ under its boundary, channel, record, and update constraints.

Corollary 1 (Entropy bottleneck). *The accessible entropy satisfies*

$$S_{\text{acc}}(\mathcal{O}, \mathcal{D}) \leq k_{\text{B}} \ln 2 C_{\text{acc}}(\mathcal{O}, \mathcal{D}). \quad (29)$$

In a holographic regime,

$$S_{\text{acc}}(\mathcal{O}, \mathcal{D}) \leq \frac{k_{\text{B}} A}{4\ell_{\text{P}}^2}. \quad (30)$$

This formulation permits observer-relative entropy without making entropy subjective. The bound is physical; the accessible coarse-graining depends on the finite observer.

MAINTENANCE INEQUALITY

Distinction deficit and irreversible update

Positive capacity deficit forces the observer to manage surplus distinctions that cannot be stored or transmitted within the accessible budget. Let

$$[\Delta_{\text{FDS}}]_{+} = \max(\Delta_{\text{FDS}}, 0), \quad (31)$$

let C_{inv} be invariant-supported compression credit, and let $C_{\text{ext}}^{\text{gain}}$ be net externalization gain. The surplus distinctions that must be irreversibly handled are

$$I_{\text{irr}}(t, \tau) \geq \left[\Delta_{\text{FDS}}(t, \tau) - C_{\text{inv}}(t, \tau) - C_{\text{ext}}^{\text{gain}}(t, \tau) \right]_{+}. \quad (32)$$

Under the Landauer bridge, each logically irreversible operation has a thermodynamic lower bound:

$$\dot{Q}_{\text{maint}}(t) \geq k_{\text{B}} T(t) \ln 2 \frac{I_{\text{irr}}(t, \tau)}{\tau} + \dot{Q}_{\text{phys}}(t), \quad (33)$$

where $\dot{Q}_{\text{phys}}$ includes control, coupling, isolation, leakage, transport, error correction, and measurement overhead.

Remark 5. *The Landauer term applies only to logically irreversible operations. Stable storage, read-only access, reversible computation, and unitary evolution are not automatically charged $k_B T \ln 2$ per bit.*

Minimal no-rescue corollary

If $C_{\text{inv}} = 0$, $C_{\text{ext}}^{\text{gain}} = 0$, and $\dot{Q}_{\text{phys}} = 0$, then

$$\dot{Q}_{\text{maint}}(t) \geq k_B T(t) \ln 2 \frac{[\Delta_{\text{FDS}}(t, \tau)]_+}{\tau}. \quad (34)$$

Interpretation. In the absence of invariant compression, externalization gain, and compensating physical channels, positive distinction deficit directly sets a Landauer-style lower bound on irreversible maintenance cost. This is a minimal lower-bound regime, not a universal full-cost formula.

BUDGET CROSSING AS TESTABLE PREDICTION

Crossing parameter

Define the instantaneous crossing parameter

$$\chi(t) = R_{\text{min}}^{(\tau)}(\varepsilon; \Psi) - C_{\text{acc}}(\mathcal{O}, \mathcal{D}, t, \tau).$$

As $\chi(t)$ crosses zero from negative to positive, the observer moves from budget surplus to budget deficit.

Expected observables

When $\chi > 0$ persists, finite observers should exhibit:

- increased task error,
- increased latency,
- increased compression ratio,
- coarser partitions,
- increased external memory use,
- increased reset / garbage collection events,
- increased heat output under irreversible update,
- pruning of maintained states,
- task relaxation,
- failure.

Test classes

Test class 1: Bounded-memory computation.

Build or simulate a bounded-memory reversible computation system. Feed continuous task streams. Vary memory budget. Track garbage accumulation, reset frequency, error correction, compression, heat, and task failure. Expected result: reversible gates may avoid local erasure cost, but open-ended bounded-memory operation eventually requires housekeeping, compression, reset, pruning, or entropy export.

Test class 2: Finite sensor/detector system. Define observer boundary, estimate internal memory and channel capacity, define task family and distortion tolerance, estimate R_{min} . Increase task complexity until R_{min} crosses C_{acc} . Observe error, latency, compression, or externalization.

Test class 3: Analogue-horizon system. Use an analogue horizon or engineered causal-access bottleneck. Measure recoverable distinguishable modes. Look for degradation as cross-boundary access becomes unavailable.

A MINIMAL COMPUTABLE MODEL OF BUDGET CROSSING

The maintenance bound is abstract. To make it computable, we construct a minimal model of an observer crossing a causal-access bottleneck.

Model setup

Let there be an external process $Z_t \in \mathcal{Z}$. A finite observer observes Z_t through a restricted channel Y_t and maintains an internal memory state M_t that carries task-relevant distinctions. The stock capacity is

$$C_{\text{stock}}(t) = \min \{C_M, C_Y(t), C_H(t), C_{\text{ext}}^{\text{eff}}(t)\}, \quad (35)$$

where C_M is fixed memory capacity, $C_Y(t)$ is the channel capacity at time t , $C_H(t)$ is the causal-boundary capacity, and $C_{\text{ext}}^{\text{eff}}$ is effective externalization capacity. The full accessible capacity including update-throughput bound is

$$C_{\text{acc}}(t) = \min \{C_{\text{stock}}(t), I_{\text{update}}^{\text{max}}(t, t + \tau)\}, \quad (36)$$

where $I_{\text{update}}^{\text{max}}$ follows Eq. (10).

Model the causal boundary crossing as a sigmoid transition:

$$C_H(t) = C_0 \eta(t), \quad \eta(t) = \frac{1}{1 + \exp((t - t_h)/\sigma)}, \quad (37)$$

where t_h is the crossing time and σ is the transition width. This does not claim to model a real black hole; it is a tunable causal-access bottleneck that captures the qualitative behavior of a horizon-like collapse.

Observer-relative entropy divergence

This subsection is a finite-budget diagnostic inside the toy model. Quantum-foundational applications to Wigner-type scenarios or measurement interpretations are outside the scope of FDS-T1.

Define the observer's residual uncertainty about the external state:

$$S_{\mathcal{O}}(t) = H(Z_t | M_t, Y_{\leq t}^{\text{acc}}). \quad (38)$$

As $C_{\text{acc}}(t)$ decreases, M_t can carry fewer task-relevant distinctions, so $S_{\mathcal{O}}(t)$ increases. Two observers $\mathcal{O}_1, \mathcal{O}_2$ with different $C_{\text{acc}}^{(1)}(t), C_{\text{acc}}^{(2)}(t)$ will assign different entropies:

$$\Delta S_{\mathcal{O}_1, \mathcal{O}_2}(t) = S_{\mathcal{O}_1}(t) - S_{\mathcal{O}_2}(t). \quad (39)$$

This divergence is directly computable from the model parameters.

Coarse-graining error floor

The most direct link to testable prediction is through rate-distortion theory. Let $D_{\min}(R)$ be the minimal achievable distortion for process Z_t at rate R . The finite observer's error floor is

$$\varepsilon_{\min}(t) = D_{\min}(C_{\text{acc}}(t)). \quad (40)$$

The task is maintainable if and only if $D_{\min}(C_{\text{acc}}(t)) \leq \varepsilon$. Define the collapse time as

$$t_c = \inf \{t : D_{\min}(C_{\text{acc}}(t)) > \varepsilon\}. \quad (41)$$

Gaussian source example

If Z_t is a Gaussian source with variance σ_Z^2 and the distortion is mean-squared error, the rate-distortion function is

$$R(D) = \frac{1}{2} \log_2 \frac{\sigma_Z^2}{D}, \quad D_{\min}(R) = \sigma_Z^2 2^{-2R}. \quad (42)$$

The error floor becomes

$$\boxed{\varepsilon_{\min}(t) = \sigma_Z^2 2^{-2C_{\text{acc}}(t)}}. \quad (43)$$

When the causal-access bottleneck drives $C_{\text{acc}}(t) \rightarrow 0$, the error floor rises to σ_Z^2 —the observer degrades to guessing the prior variance.

Binary source example

If $Z_t \in \{0, 1\}$ with $P(Z = 1) = p$ and Hamming distortion, the rate-distortion function in the range $0 \leq D \leq \min(p, 1 - p)$ is

$$R(D) = H_b(p) - H_b(D), \quad (44)$$

where H_b is the binary entropy function. The error floor is

$$D_{\min}(t) = H_b^{-1}(H_b(p) - C_{\text{acc}}(t)). \quad (45)$$

As $C_{\text{acc}} \rightarrow 0$, $D_{\min} \rightarrow \min(p, 1 - p)$: the observer can only guess the most probable state.

Bottleneck-switching kink theorem

The error-floor predictions above can be generalized into an analytic theorem. Let the accessible capacity be a pointwise minimum of differentiable capacity channels:

$$C_{\text{acc}}(t) = \min_i C_i(t). \quad (46)$$

Suppose that at $t = t^*$ the active bottleneck switches from C_j to C_k , with

$$C_j(t^*) = C_k(t^*), \quad \dot{C}_j(t^*) \neq \dot{C}_k(t^*), \quad (47)$$

and all other $C_i(t^*)$ are strictly larger. Let the distortion floor be

$$\varepsilon_{\min}(t) = D_{\min}(C_{\text{acc}}(t)), \quad (48)$$

where $D_{\min}(R)$ is differentiable and $D'_{\min}(C_{\text{acc}}(t^*)) \neq 0$.

Theorem 5 (Bottleneck-switching kink). *Under conditions (46)–(48), the error floor $\varepsilon_{\min}(t)$ generically has a slope discontinuity (kink) at the switch time t^* .*

Proof. For $t < t^*$ in a neighborhood of the switch, $C_{\text{acc}}(t) = C_j(t)$, while for $t > t^*$, $C_{\text{acc}}(t) = C_k(t)$. Therefore

$$\left. \frac{d\varepsilon_{\min}}{dt} \right|_{t^*_-} = D'_{\min}(C_{\text{acc}}(t^*)) \dot{C}_j(t^*), \quad (49)$$

$$\left. \frac{d\varepsilon_{\min}}{dt} \right|_{t^*_+} = D'_{\min}(C_{\text{acc}}(t^*)) \dot{C}_k(t^*). \quad (50)$$

Since $D'_{\min}(C_{\text{acc}}(t^*)) \neq 0$ and $\dot{C}_j(t^*) \neq \dot{C}_k(t^*)$, the two one-sided derivatives differ. Hence the error floor exhibits a slope discontinuity—a kink—at the bottleneck switch. $\square$

The theorem implies that *any* finite observer whose accessible capacity is a pointwise minimum of differentiable channel terms will generically produce observable kinks in its error floor at bottleneck transitions, regardless of the specific source or distortion measure, provided the rate-distortion function is differentiable at the operating point.

Memory update and erased distinctions

To connect to the Landauer bound, define the memory dynamics as

$$M_{t+\tau} = \Pi_{C_{\text{acc}}(t+\tau)}[U(M_t, Y_t)], \quad (51)$$

where Π_C is a projection or compression operator that projects onto the C -bit accessible space. The number of erased distinctions per update step is

$$E_t = H(U(M_t, Y_t)) - H(M_{t+\tau}), \quad (52)$$

or more precisely, the logical erasure [3]:

$$E_t = H(M_t | M_{t+\tau}, Y_t). \quad (53)$$

The Landauer bridge then gives $\dot{Q}_{\text{irr}}(t) \geq k_B T \ln 2 \cdot E_t / \tau$, linking the toy model directly to the maintenance bound (33) in the Maintenance inequality section.

Numerical demonstration and predictions

The numerical parameters in this section are illustrative and are chosen to demonstrate the analytic bottleneck mechanism, not to fit a specific physical black-hole or analogue-horizon platform. The Bottleneck-Switching Kink Theorem (Theorem 5) is confirmed numerically by a minimal finite-observer model with four capacity channels (fixed memory, decreasing channel, decreasing causal-boundary via sigmoid, decreasing thermodynamic budget) driven by a Gaussian source under mean-squared error distortion and a binary source under Hamming distortion. The model uses the minimal no-rescue regime defined above and is fully reproducible (Code Availability).

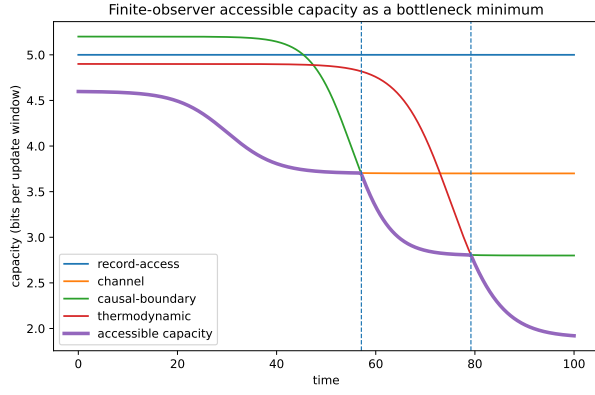


FIG. 1. Capacity bottleneck switching. Four capacity channels and their pointwise minimum $C_{\text{acc}}(t)$. Vertical dashed lines mark bottleneck transitions.

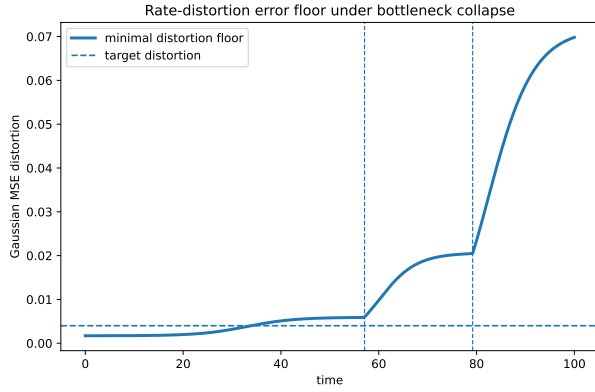


FIG. 2. Gaussian error-floor kinks. The rate-distortion error floor $\varepsilon_{\min}(t) = \sigma_Z^2 2^{-2C_{\text{acc}}(t)}$ shows clear slope discontinuities (kinks) at each bottleneck switch (vertical lines). Gray dashed lines mark the one-sided derivative extrapolations.

Prediction 1 (capacity collapse curve). As capacity channels degrade, $C_{\text{acc}}(t) = \min_i C_i(t)$ switches between bottleneck regimes (Fig. 1). Each switch produces a kink in $C_{\text{acc}}(t)$ and therefore a slope change in the error floor $\varepsilon_{\min}(t) = D_{\min}(C_{\text{acc}}(t))$ (Fig. 2, Theorem 5). A binary source produces analogous kinks under Hamming distortion.

Prediction 2 (observer entropy divergence). Two observers with different $C_{\text{acc}}^{(1)}(t), C_{\text{acc}}^{(2)}(t)$ develop an entropy gap $\Delta S_{\mathcal{O}_1, \mathcal{O}_2}(t)$ that grows and then contracts as the later observer's bottleneck crosses (Fig. 4). The gap is a direct measure of observer-relative causal access.

Prediction 3 (erasure heat spike). If the system attempts to maintain the same distortion ε as $C_{\text{acc}}(t)$ declines, $\Delta_{\text{FDS}}(t)$ rises and the minimal no-rescue bound (34) produces a positive heat lower bound. Fig. 3 shows a transient heat spike during the causal-

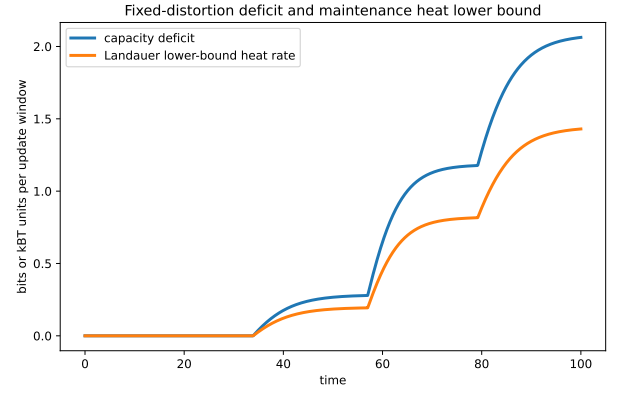


FIG. 3. Fixed-distortion capacity deficit and Landauer heat lower bound. The capacity deficit $\Delta_{\text{FDS}}(t)$ (bits) and its corresponding minimal no-rescue heat lower bound $\dot{Q}_{\text{maint}}(t)/(k_B T \ln 2)$ from (34) rise as accessible capacity falls below task demand.

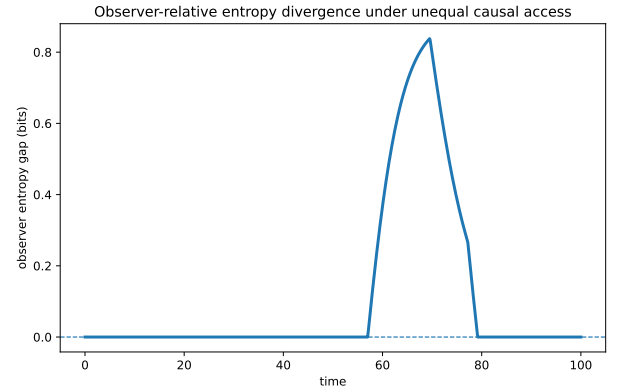


FIG. 4. Observer-relative entropy divergence. Two observers with staggered causal-boundary transitions ($t_h^{(1)} = 55$, $t_h^{(2)} = 67$) develop a transient entropy gap $\Delta S_{\mathcal{O}_1, \mathcal{O}_2}(t)$ that peaks during the interval between their respective bottleneck crossings.

boundary collapse phase. The height of the spike is proportional to the peak deficit and is bounded below by $k_B T \ln 2 \cdot \Delta_{\text{FDS}}/\tau$.

SIMULATION AND EXPERIMENTAL PROTOCOLS

We describe three tiers of protocol for testing the FDS maintenance bound, ordered by increasing physical realism and difficulty.

Protocol A: Constrained-memory channel simulation

Goal. Verify $\varepsilon_{\min}(t) = D(C_{\text{acc}}(t))$ and $\dot{Q}_{\text{irr}} \geq k_B T \ln 2 \sum_t H(M_t | M_{t+\tau}, Y_t)$ in a controlled simulation.

System. Construct a source Z_t (e.g., Gaussian AR(1) process, binary Markov chain, hidden Markov model, or chaotic logistic map). The observer has:

- fixed memory C_M (bits);
- channel capacity $C_Y(t)$;
- causal-boundary capacity $C_H(t) = C_0 \eta(t)$ with $\eta(t)$ given by (37);
- stock capacity $C_{\text{stock}}(t) = \min\{C_M, C_Y(t), C_H(t), C_{\text{ext}}^{\text{eff}}\}$, and accessible capacity $C_{\text{acc}}(t) = \min\{C_{\text{stock}}(t), I_{\text{update}}^{\text{max}}(t, t + \tau)\}$, evaluated per update window;

Procedure.

1. Pre-register a task distortion tolerance ε .
2. The observer tracks Z_t using an optimal or bounded encoder.
3. Gradually reduce $C_H(t)$ or $C_Y(t)$.
4. Record reconstruction error, retained mutual information $I(Z_t; M_t)$, erased bits $H(M_t | M_{t+\tau}, Y_t)$, reset or compression frequency, and estimated heat lower bound.

Expected result. When $R_{\min}^{(\tau)}(\varepsilon) > C_{\text{acc}}(t)$, at least one of the following must occur: error rises, erasure or compression rises, or externalization or pruning is triggered. If none occur while task performance is maintained, the FDS maintenance bound is violated.

Protocol B: Finite-temperature quantum memory and channel

System. A finite quantum memory M coupled to a bath at temperature T , receiving states from a quantum or classical source.

Controllable parameters. Memory Hilbert dimension d_M , channel transmissivity $\eta(t)$, reset frequency, bath temperature, task discrimination threshold, measurement or coarse-graining map.

Measurable quantities. Discrimination success probability, accessible mutual information, entropy production, heat dissipation, reset work, quantum memory fidelity, task-relevant distortion.

Core test. Tune $\eta(t) : 1 \rightarrow 0$, define $C_{\text{chan}}(t) = \log_2 d_{\text{eff}}(t)$ or the Holevo quantity $\chi(t)$, and test whether $\varepsilon_{\min}(t) \geq D(C_{\text{acc}}(t))$ holds. Maintaining low distortion should require increasing reset or compression heat.

Protocol C: Analogue-horizon mode registration

Rationale. Analogue horizon systems (BEC, optical, acoustic, circuit) can realize tunable causal bottlenecks without requiring astrophysical black holes [14].

Observable. Define

$$C_{\text{reg}}(g) = \text{number of recoverable mode distinctions at control parameter } g, \quad (54)$$

and the reconstructability decay rate

$$\kappa_{\text{reg}}(g) = -\frac{d}{dt} \log \mathcal{R}_g(t), \quad (55)$$

where $\mathcal{R}_g(t)$ is the recoverability of initial mode labels or boundary-mode information. Measure the steady-state entropy production $\dot{\Sigma}_{\text{ss}}(g)$ near the horizon-like transition.

Expected signature. Near an effective horizon transition or capacity bottleneck, dC_{reg}/dg or $d\dot{\Sigma}_{\text{ss}}/dg$ should show a kink or anomaly. This protocol is exploratory; the strongest near-term DT-specific physical signature remains the topological two-kink proposed for FDS-P3 [5].

CONSEQUENCES

No infinite local observer

A finite observer cannot be replaced by an ideal observer with infinite memory and infinite causal access without changing the physical problem. Such an idealization may be mathematically useful, but it erases the central constraints that make observation physical. In FDS, observer idealization is allowed only as a limiting model whose domain of validity must be stated.

Holography as boundary accounting

From the FDS perspective, holographic bounds can be interpreted as boundary accounting laws for accessible distinguishability. Interior processes may be numerous, but finite observers access and stabilize information through boundaries. When gravity becomes relevant, the boundary itself carries the ultimate information ceiling. The area law is therefore the natural limiting form of finite distinguishability accounting.

Coarse-graining is forced, not optional

When $R_{\min}^{(\tau)}(\varepsilon; \Psi)$ exceeds C_{acc} , coarse-graining is not a methodological choice. It is forced by physical capacity.

Different observers may choose different coarse-grainings because they have different budgets, tasks, horizons, and error tolerances.

Externalization is bounded

Maps, instruments, laboratories, computer memory, archives, and environmental records can expand an observer’s effective budget. But they do not remove the budget. They create a larger boundary whose records require stability, access, error correction, and thermodynamic maintenance. In gravitational regimes, even the enlarged boundary remains subject to area ceilings.

Relation to black-hole complementarity

FDS-T1 gives a neutral way to discuss why different observers may assign different accessible descriptions near horizons. The exterior observer’s accessible distinction budget is tied to the horizon boundary, while an infalling observer’s local patch differs. This paper does not solve the black-hole information problem. It supplies a finite-observer vocabulary for describing the difference between global unitary preservation, boundary encoding, and finite observer recoverability.

Bridge failure contract

The results of this paper are conditional on the validity domain of the physical bridge assumptions used: Bekenstein-type entropy bounds, holographic or covariant entropy ceilings, finite causal access, and Landauer-style update accounting. If future quantum-gravity physics discovers a regime in which Bekenstein or holographic bounds fail—for example through presently unknown topological sectors, non-area-like information storage, or a fundamentally different accounting of accessible physical information—then the corresponding FDS-T1 bridge claims must be revised or withdrawn within that regime. Such a failure would not by itself falsify the algebraic FDS core, which only asserts finite projection, boundary-relative capacity, update constraints, and deficit-induced exits at the formal level. It would falsify the specific physical embedding of those formal objects into the currently accepted gravitational information bounds. The layered failure contract is summarized in Table III.

FALSIFICATION CONDITIONS AND TESTS

Bridge-level falsification

The FDS-T1 bridge would be undermined by any of the following:

1. a finite-energy, finite-radius physical system reliably encoding more information than the Bekenstein bound permits under conditions where the bound is supposed to apply;
2. a horizon-bounded system with entropy exceeding the holographic ceiling under an appropriate covariant formulation;
3. a finite observer capable of registering, preserving, updating, and operationally using unbounded distinctions with bounded energy, bounded memory, bounded causal access, and bounded error correction;
4. a physically implemented logically irreversible erasure process violating the generalized Landauer lower bound under its stated thermodynamic conditions.

Failure of one bridge does not automatically destroy the FDS formal core. It localizes the failure to the physical interpretation or to the external physical theorem being used.

Operational tests

Direct Planck-scale tests of holographic bounds are not currently available. But FDS-T1 motivates lower-energy proxy tests.

Protocol 1 (Finite-budget observation test). *To test a finite distinguishability budget in a controlled system:*

1. *specify the observer or apparatus boundary B ;*
2. *estimate memory capacity C_{mem} , boundary capacity C_{bd} , channel capacity C_{chan} , and effective externalized capacity $C_{\text{ext}}^{\text{eff}}$;*
3. *estimate the update-throughput bound $I_{\text{update}}^{\text{max}}(t, t + \tau)$ from available free energy or allowable dissipation;*
4. *define a task family Ψ and distortion tolerance ε ;*
5. *estimate $R_{\text{min}}^{(\tau)}(\varepsilon; \Psi)$ using compression curves, rate-distortion lower bounds, Fisher information, or task-specific coding lower bounds;*
6. *test whether performance degrades, coarse-grains or compresses, externalizes, prunes, or fails as R_{min} crosses C_{acc} .*

Protocol 2 (Analogue-horizon budget test). *In an analogue horizon system, compare the number of recoverable, operationally distinguishable modes across a horizon-like boundary to the effective boundary area, noise temperature, and channel-access constraints. FDS-T1 predicts that recoverable distinction capacity should be boundary-limited and should degrade as access across the effective horizon becomes thermodynamically or causally unavailable.*

Expected signatures

FDS-T1 predicts the following qualitative signatures:

1. **Boundary-limited recoverability:** recoverable distinctions scale with access boundary capacity rather than naive interior volume when the boundary is the bottleneck.
2. **Budget crossing:** as task distinction demand crosses accessible capacity, systems should show sharp increases in compression, pruning, error, latency, or externalization.
3. **Observer-dependent coarse-graining:** different observers with different budgets should stabilize different partitions of the same underlying physical process.
4. **Thermodynamic update cost:** high-rate irreversible updating should show increased dissipation consistent with Landauer-style accounting when its assumptions hold.

LIMITATIONS

First, this paper does not derive the Bekenstein or holographic bounds from the distinction primitive alone. It imports them as physical bridge constraints.

Second, the exact coefficient $1/4$ in $S = k_B A / (4\ell_P^2)$ is not derived. The paper uses the established Bekenstein-Hawking formula as the strongest known limiting case.

Third, finite distinguishability is not the same as finite Hilbert-space dimension. A system may have a very large or infinite mathematical state description while a finite observer has limited operational access.

Fourth, the accessible budget formula (25) is conservative and schematic. Real systems may require detailed modeling of noise, error correction, coupling, clocking, decoherence, control cost, and measurement back-action.

Fifth, not every boundary is a horizon. The paper distinguishes general distinguishability horizons from event horizons, apparent horizons, causal diamonds, and engineered analogue horizons.

Sixth, the framework is compatible with unitary microscopic dynamics. It concerns accessible records and finite observation, not necessarily fundamental non-unitarity.

CONCLUSION

Finite observers have finite distinguishability budgets. Established entropy bounds constrain not only storage but operationally accessible distinctions. When task-relevant demand exceeds accessible capacity, finite systems must coarse-grain, compress, externalize, prune, relax, or fail. The key empirical signature is budget crossing: as demand crosses accessible capacity, error, latency, compression, externalization, pruning, and heat should show measurable transitions.

This paper does not derive quantum gravity, general relativity, the Standard Model, or the $1/4$ coefficient in the Bekenstein-Hawking entropy formula. Its contribution is narrower: to make finite distinguishability a precise, auditable, and testable bridge concept.

Future work includes bounded-memory reversible computation experiments, two-kink signature tests, finite-observer entropy bounds, time-energy-distinction speed limits, and externalization cost accounting.

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APPENDIX A: Notation Summary

APPENDIX B: Claim Status and Failure Conditions

The status of each central claim and what would constitute falsification is given in Table III.

APPENDIX C: Minimal Model Parameters

The minimal computable model described above uses the following parameters:

- Fixed memory capacity C_M ;
- Channel capacity $C_Y(t)$;
- Causal-boundary capacity $C_H(t) = C_0 \eta(t)$ with sigmoid transition $\eta(t) = [1 + \exp((t - t_h)/\sigma)]^{-1}$;
- Thermodynamic budget parameters: free-energy supply rate $\dot{F}_{\text{avail}}$, temperature T ;

TABLE II. FDS-T1 notation summary.

Symbol	Meaning
$\mathcal{O}$	finite observer or distinction-register
$\mathcal{D}$	observer-accessible region or causal patch
$\partial\mathcal{D}$	operational, causal, or horizon boundary of the patch
$\pi_{\mathcal{O}}$	finite distinction projection implemented by $\mathcal{O}$
$N_{\mathcal{O}}$	number of operationally distinguishable alternatives
$C_{\mathcal{O}}$	bit capacity $\log_2 N_{\mathcal{O}}$
C_{mem}	internal memory / record capacity
C_{bd}	accessible boundary information capacity
C_{chan}	channel-limited capacity
C_{stock}	stock (storage/state) capacity, minimum of the above
$C_{\text{ext}}^{\text{eff}}$	effective externalized capacity after indexing and verification costs
$I_{\text{update}}^{\text{max}}$	maximum irreversible update throughput (update budget ceiling)
C_{acc}	effective accessible capacity, $\min\{C_{\text{stock}}, I_{\text{update}}^{\text{max}}\}$
$R_{\text{min}}^{(\tau)}(\varepsilon; \Psi)$	minimal task-relevant rate-distortion demand
Δ_{FDS}	boundary-relative capacity deficit
$\chi(t)$	instantaneous crossing parameter
E	energy of bounded region
R	radius or size scale of bounded region
A	area of relevant boundary or horizon
ℓ_{P}	Planck length $\sqrt{\hbar G/c^3}$
S_{acc}	observer-relative accessible entropy
$\dot{Q}_{\text{maint}}$	maintenance heat dissipation
$\dot{Q}_{\text{phys}}$	non-informational physical dissipation

- Source model: Gaussian with variance σ_Z^2 under MSE distortion, or binary with probability p under Hamming distortion;
- Distortion tolerance ε ;
- Update window τ .

All numerical values used in the demonstration are fixed in the reproducible code release. The model is deterministic and requires no random seeds.

Claim	Status	Dependency	Failure condition
Finite observer budget	Formal/physical bridge	finite record carrier; finite operational error tolerance	finite physical observer with unbounded reliable distinctions under bounded resources
Bekenstein distinguishability ceiling	Physical bridge	Bekenstein bound	bounded system exceeding Bekenstein info limit under applicable assumptions
Holographic distinguishability ceiling	Physical bridge	holographic/covariant entropy bound	horizon-bounded region exceeding appropriate area entropy ceiling
Boundary-relative capacity deficit	Formal conditional	rate-distortion demand; accessible capacity estimate	task succeeds at full fidelity despite demand exceeding all accessible capacity
Budget-exit classification	Conditional theorem	positive deficit; bounded resources	no coarse-graining, no externalization, no pruning, no task relaxation, no failure under persistent deficit
Observer-relative accessible entropy	Interpretive bridge	finite projection; physical record access	entropy assignment requiring no finite record carrier, boundary, or coarse-graining

TABLE III. Claim status and failure conditions for the FDS-T1 bridge.

APPENDIX D: Reproducibility Checklist

1. Code availability: all simulation code is publicly released.
2. Deterministic execution: no random seeds required; configuration files fix all parameters.
3. Figure reproduction: running `src/generate_figures.py` reproduces all figures and data tables.
4. Parameter documentation: all model parameters are documented in the code configuration.
5. Platform independence: the code runs on standard Python environments with common scientific libraries.

is deterministic and does not require random seeds. Running `src/generate_figures.py` reproduces all figures (PDF and PNG) and CSV data tables from the source code in a single pass.

APPENDIX E: Boundary of Applicability

The FDS-T1 bridge claims in this paper apply under the following conditions:

1. The observer is a finite physical record-bearing system with bounded memory, bounded energy, bounded channel access, and bounded update rate.
2. The system operates under known gravitational entropy bounds (Bekenstein, holographic) where applicable.
3. The Landauer bound for logically irreversible operations holds under the stated thermodynamic conditions.
4. Causal access is finite and bounded by the speed of light.

The claims do not apply to:

- Infinite or idealized observers with unbounded resources.
- Regimes where Bekenstein or holographic bounds are known to fail or are superseded.
- Systems operating outside the validity domain of the Landauer principle for irreversible operations.

The layered failure contract is summarized below.

CODE AVAILABILITY

The simulation code used to generate Figs. 1–4 is available at https://github.com/yiningwu-research/Distinction-Theory/tree/main/models/fds_t1. All model parameters are fixed in the configuration; the code

TABLE IV. Layered failure contract for FDS-T1 bridge claims.

Layer	Claim status	Failure consequence
FDS-0 Core	Formal definitional system	Falsified only by internal incoherence or contradictory definitions
T1 Physical Bridge	Conditional embedding of FDS objects into Bekenstein, holographic, and Landauer bounds	Fails if imported constraints are violated or superseded in applicable regimes
Holographic embedding	Horizon-bounded accessible distinction scales with boundary area	Revised if future quantum gravity gives non-area information accounting
Maintenance bound	Irreversible surplus distinctions carry Landauer update cost	Revised if Landauer accounting fails under stated conditions

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Effective Geometry as Horizon Boundary Accounting: Finite Distinguishability, Horizon Entropy, and Thermodynamic Closure in Finite Distinction Systems

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General relativity describes gravity as spacetime geometry. Black-hole thermodynamics and Jacobson’s local-horizon argument suggest that spacetime dynamics can also be read as an equation of state relating energy flux, horizon entropy, area variation, and causal structure. FDS-T2 develops this connection inside Finite Distinction Systems. It does not derive general relativity from FDS alone, replace Einstein gravity, derive quantum gravity, or derive the numerical coefficient in the Bekenstein-Hawking entropy formula. Instead, it formulates a boundary-thermodynamic bridge: when finite observers are constrained by horizon-bounded distinguishability budgets, effective geometry functions as the macroscopic accounting structure for causal access, horizon entropy, energy-momentum flux, and boundary-maintenance cost.

The novelty of T2 is not a new derivation of Einstein gravity, but an observer-relative reinterpretation of horizon thermodynamic variables as finite distinguishability ledgers: area counts accessible boundary distinctions, heat flux updates the ledger, and effective geometry is the covariant compression that preserves causal access and stress-energy accounting. T2 treats Jacobson-style local horizon thermodynamics as the central physical model class, not as an FDS theorem. The formal FDS contribution is the finite-boundary maintenance discipline: capacity, projection, update windows, resource ledgers, and invariant coarse variables. The refinements add covariant consistency for residual tensors, concrete horizon-task examples, and a clearer minimal-sufficient-complexity interpretation of geometry as a Phase-B boundary variable. The physical bridge assumptions are explicit: local causal horizons, area entropy, Unruh or surface-gravity temperature, Clausius-type horizon closure, and local covariance.

Reader contract. T2 is an effective-geometry bridge, not a replacement for general relativity. It does not claim that spacetime geometry is “nothing but information,” does not derive the Einstein field equations from FDS alone, does not derive the Bekenstein-Hawking coefficient $1/4$, and does not solve quantum gravity. It uses horizon thermodynamics as a physical bridge. If that bridge fails, the T2 interpretation is demoted; the formal FDS core remains unaffected.

INTRODUCTION

The geometry-thermodynamics problem

General relativity represents gravity through geometry,

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}. \quad (1)$$

Black-hole thermodynamics assigns entropy and temperature to horizons [13–15]. Jacobson’s local-horizon argument showed that Einstein’s equation can be obtained as an equation of state by applying the Clausius relation

$$\delta Q = T \delta S \quad (2)$$

to all local Rindler horizons, with entropy proportional to horizon area [18]. This suggests that geometry, entropy, and energy flux are not independent accounting languages.

T2 asks a narrower question than earlier geometry drafts: what is the FDS interpretation of effective geometry when finite observers have bounded causal access and horizons carry entropy? The answer proposed here is not that FDS derives GR. It is that effective geometry is the macroscopic boundary-thermodynamic accounting form for finite causal access, horizon entropy, and stress-energy flux.

Claim-status summary

Table I separates formal finite-system claims, standard physics inputs, and T2 bridge claims.

Keywords

Finite Distinction Systems; horizon thermodynamics; effective geometry; finite distinguishability; boundary accounting; Bekenstein bound; holography; Jacobson gravity; Unruh effect; black-hole thermodynamics; entropy production; causal horizons; Phase-B invariants; horizon-maintenance density.

TABLE I. Central T2 claims, status, and failure or demotion conditions.

Claim	Status	What would weaken or falsify it
Finite observers have bounded distinguishability budgets.	FDS / T1 bridge	A finite physical observer reliably registers, preserves, and updates unlimited distinctions with finite resources.
Horizons act as causal-access boundaries.	GR / QFTCS bridge	Local causal horizons are irrelevant to observable access, entropy, or thermodynamic accounting.
Horizon entropy gives boundary area accounting.	Physical bridge	Horizon entropy is shown not to scale with boundary area in the stated horizon class.
Clausius-type local horizon closure links heat flow and entropy variation.	Model-class bridge	Local horizon thermodynamics fails as an effective equation-of-state model.
Effective geometry can be read as boundary thermodynamic accounting.	Main T2 thesis	Geometry is shown to have no relation to causal access, horizon entropy, stress-energy flow, or boundary constraints.
Non-equilibrium horizon accounting may require residual terms.	Optional extension	All non-equilibrium horizon settings obey pure equilibrium closure with no entropy production, memory, noise, or correction terms.

What T2 adds beyond Jacobson

Jacobson’s argument gives the standard horizon-thermodynamic model-class bridge: local Clausius closure plus area entropy yields Einstein-type geometry as an equilibrium equation of state. T2 adds three FDS-specific layers.

First, it gives an *observer-relative distinguishability interpretation*: S_H and A_H are read as a finite boundary ledger for accessible horizon distinctions, not merely as formal thermodynamic variables. Second, it introduces an auditable *boundary ledger formalism*, collecting H , A_H , S_H , T_H , δQ_H , τ , and residual error terms into a single accounting object. Third, it supplies an *overflow / Phase-B interpretation*: when microscopic horizon degrees of freedom are inaccessible, effective geometry is treated as a stable coarse boundary variable rather than a full microscopic ontology.

Thus the T2 contribution is

Jacobson model-class bridge + finite distinguishability ledger interpretation, not FDS-derived GR.

Central thesis

The central thesis is

Effective geometry is the covariant boundary ledger that closes horizon entropy, causal access, and stress-energy accounting for finite observers.

In symbols, the bridge is

$$(C_H, S_H, T_H, \delta Q_H) \implies g_{\mu\nu}^{\text{eff}} \text{ as boundary thermodynamic ledger} \quad (3)$$

The arrow is not a formal derivation from FDS primitives. It is a physical bridge based on standard horizon-thermodynamic ingredients.

Why T2 follows T1, T3, P-series, and X3

T2 is downstream of safer finite-system modules. T1 defines finite observer distinguishability budgets and explicitly treats Bekenstein and holographic bounds as bridge inputs, not as FDS-derived theorems [2]. T3 analyzes overflow: when task-relevant distinction demand exceeds accessible capacity, the finite record may exit into coarse-graining, effective stochasticity, hysteresis, externalization, or Phase-B invariants [3]. P3, P4, P6, and P7 supply finite-bath memory, anti-recurrence, speed-precision-resource bounds, and invariant protection [4–7]. X3 places gravity in the four-operation closure as global boundary / causal geometry / stress-energy accounting [8]. T2 supplies a technical bridge behind that row.

FDS BACKGROUND

Active finite distinction systems

An active finite distinction system is represented by

$$\mathcal{S} = (X, E, B, M, Y, A, U, \pi, \ell, \Phi, \mathcal{P}, \tau), \quad (4)$$

where X is internal state, E environment, B boundary, M memory or model state, Y observation channel, A action space, U update rule, π finite projection, ℓ boundary-maintenance loss, Φ resource budget, $\mathcal{P}$ perturbation or pruning family, and τ update timescale [1]. T2 emphasizes B , Y , π , Φ , and τ : a causal boundary is not merely a surface but an access, update, and accounting interface.

Finite distinguishability budget

For a finite observer O with finite projection π_O , T1 defines

$$N_O = |\text{Im}(\pi_O)|, \quad C_O = \log_2 N_O. \quad (5)$$

Accessible capacity is bottlenecked by memory, boundary access, channel capacity, causal reach, externalized capacity, and update throughput [2]. T2 uses this language to read a horizon as the boundary of causal access, not as an all-purpose proof that geometry is information.

Overflow and Phase-B variables

T3 defines an overflow deficit of the form

$$\Delta(t) = R_{\min}^{(\tau)}(\varepsilon; \Psi_t) - C_{\text{acc}}(t). \quad (6)$$

When $\Delta > 0$, microstate detail cannot be tracked at full fidelity. The accessible description may become stochastic, hysteretic, or dominated by coarse variables. T2 treats effective geometry as a candidate Phase-B boundary variable: a stable coarse structure that tracks causal access, horizon area, and stress-energy flow after microscopic horizon degrees of freedom are inaccessible.

STANDARD PHYSICS BASELINE

Horizon entropy and temperature

For black-hole horizons, the entropy is

$$S_{\text{BH}} = \frac{k_B A}{4\ell_P^2}, \quad \ell_P = \sqrt{\frac{\hbar G}{c^3}}. \quad (7)$$

T2 treats this relation as established black-hole thermodynamics and holographic boundary accounting, not as something derived here.

For a local accelerated observer, Unruh temperature is

$$T_U = \frac{\hbar a}{2\pi k_B c}, \quad (8)$$

and for a stationary horizon with surface gravity κ ,

$$T_H = \frac{\hbar \kappa}{2\pi k_B c}. \quad (9)$$

These relations provide the temperature bridge between energy flux and horizon entropy.

Jacobson bridge

In Jacobson's model class, the heat flux through a local Rindler horizon is related to stress-energy,

$$\delta Q = \int T_{\mu\nu} \chi^\mu d\Sigma^\nu, \quad (10)$$

while the entropy variation is proportional to area,

$$\delta S = \eta \delta A. \quad (11)$$

Imposing $\delta Q = T \delta S$ for all local horizons yields the Einstein equation as an equation of state, with the coupling fixed when $\eta = k_B/(4\ell_P^2)$ [18]. T2 does not reproduce the full derivation. It uses it as the central bridge showing how local horizon thermodynamics can encode effective geometry.

DEFINITIONS

Definition 1 (Finite horizon boundary). *A finite horizon boundary H is a causal boundary relative to an observer or finite record-bearing system O , delimiting the region whose events can be operationally distinguished, updated, or included in the observer's boundary-maintenance ledger over a finite window τ .*

Definition 2 (Horizon distinguishability budget). *The horizon distinguishability budget is*

$$C_H = \frac{S_H}{k_B \ln 2}. \quad (12)$$

For an area-law horizon,

$$C_H = \frac{A_H}{4\ell_P^2 \ln 2}. \quad (13)$$

This is a bridge expression: it imports horizon entropy as an area ledger and expresses it as accessible distinguishability.

Definition 3 (Boundary thermodynamic ledger). *A boundary thermodynamic ledger is a tuple*

$$\mathcal{L}_H = (H, A_H, S_H, T_H, \delta Q_H, \tau, \mathcal{E}_H), \quad (14)$$

where H is a causal or horizon boundary, A_H its area, S_H its entropy, T_H its temperature, δQ_H the energy flux or heat flow assigned to the boundary, τ the operational update window, and $\mathcal{E}_H$ an admissible coarse-grained error or non-equilibrium term.

Definition 4 (Effective geometry). *An effective geometry is a metric, connection, or causal structure $g_{\mu\nu}^{\text{eff}}$ that compresses the boundary ledger into a stable macroscopic relation between causal access and stress-energy flow:*

$$g_{\mu\nu}^{\text{eff}} = \mathcal{G}(\mathcal{L}_H). \quad (15)$$

This definition is operational. It does not assert that the metric is the microscopic substrate of spacetime.

Definition 5 (Admissible ledger-to-geometry map). *The map $\mathcal{G} : \mathcal{L}_H \mapsto g_{\mu\nu}^{\text{eff}}$ is admissible only if it preserves, to registered tolerance ε , (i) causal ordering and light-cone*

structure, (ii) the horizon-area functional, (iii) the response of area variation to local stress-energy flux, (iv) local covariance, (v) closure residuals, and (vi) coarse-grained stability under the registered perturbation family. Thus $\mathcal{G}$ is not an arbitrary relabeling; it is a constrained compression from a horizon ledger to an effective geometric structure.

Definition 6 (Boundary-thermodynamic closure). A horizon ledger is thermodynamically closed to tolerance ε if

$$|\delta Q_H - T_H \delta S_H - \delta W_H - \delta \Sigma_H| \leq \varepsilon, \quad (16)$$

where δW_H is a work-like boundary term and $\delta \Sigma_H$ is an entropy production, memory, or non-equilibrium correction. Equilibrium Jacobson closure is the special case $\delta W_H \approx 0$ and $\delta \Sigma_H \approx 0$.

MAIN PROPOSITIONS

Proposition 1 (Jacobson model-class bridge). Assume horizon entropy $S_H = \eta A_H$, local Unruh or surface-gravity temperature, Clausius closure $\delta Q = T_H \delta S_H$, and local covariance for all admissible local causal horizons. Then an Einstein-type geometric equation arises as an equilibrium equation of state in the Jacobson model class.

Remark 1. This proposition is a standard physics bridge, not an FDS theorem. T2 inherits it as a model-class result from horizon thermodynamics.

Proposition 2 (FDS boundary-ledger interpretation). If the Jacobson model-class bridge holds, then FDS interprets $g_{\mu\nu}^{\text{eff}}$ as the stable macroscopic compression of a finite horizon distinguishability ledger. Horizon area counts accessible boundary distinctions; heat flux updates the ledger; and effective geometry preserves causal access and stress-energy accounting in covariant form.

Proof sketch. Finite horizon area bounds accessible distinguishability through $C_H \sim A_H/\ell_P^2$. Horizon entropy is the logarithm of accessible boundary state count, $S_H = k_B \ln N_H$. Energy flux through the boundary changes the horizon ledger. Local temperature converts entropy variation into heat flow, $\delta Q_H = T_H \delta S_H$, at equilibrium. Requiring this relation for all local causal horizons forces a covariant relation between stress-energy and local geometric response. Jacobson's theorem supplies the equilibrium model-class result. T2 adds the FDS interpretation: the metric is the stable macroscopic boundary ledger for finite causal access and stress-energy flow. $\square$

Remark 2 (What the propositions do not prove). These propositions do not derive the Bekenstein-Hawking coefficient, the Unruh effect, local Lorentz invariance, quantum field theory in curved spacetime, or general relativity

from FDS alone. The structure is

standard physics bridge + FDS ledger interpretation $\neq$ FDS deriv (17)

The failure of any physical bridge assumption demotes the T2 interpretation but not the FDS finite-capacity core.

WHY GEOMETRY?

Causal access is geometric

Finite observers have memory and channel limits, but also causal-access limits. Over an operational window, accessible distinctions lie inside a causal region such as a causal diamond. Geometry encodes light cones, time ordering, horizons, area of access surfaces, and the relation between energy-momentum and causal response. This makes geometry a natural representation for boundary accessibility.

Area as boundary accounting

In horizon thermodynamics, area is not merely shape. It is an entropy ledger:

$$S_H \propto A_H. \quad (18)$$

T2 reads area as the macroscopic counter of boundary-accessible distinguishability. This is not a claim that every surface carries maximal entropy; it is a statement about the horizon or covariant-boundary class where area entropy is physically established.

Curvature as ledger response

Energy flux through a local horizon changes the boundary ledger. The effective response is focusing, area variation, and curvature. In the T2 interpretation,

Curvature is the covariant way a finite causal boundary records stress-energy flux.

This is interpretive language for the Jacobson model class, not a new gravitational field equation.

RELATION TO JACOBSON GRAVITY

Table II summarizes the mapping. T2 inherits Jacobson's strength and limitations. If local horizon thermodynamics is an accurate equation-of-state model, then geometry has a thermodynamic reading. If it fails, the T2 bridge fails at exactly that layer.

NON-EQUILIBRIUM EXTENSION

Horizon capacity deficit

For a horizon-boundary task family Ψ_H , define

$$\Delta_H(\tau) = R_{\min}^{(\tau)}(\varepsilon; \Psi_H) - C_H. \quad (19)$$

Examples of Ψ_H include task families for local horizon-area variation, stress-energy flux records, causal-diamond boundary updates, and coarse records of unresolved horizon microstates. When $\Delta_H > 0$, the boundary ledger cannot track all task-relevant horizon distinctions at full fidelity over the update window. The missing distinctions may enter the effective description as entropy production, memory, stochastic noise, hysteresis, or coarse correction terms.

Ledger residuals

A non-equilibrium ledger may be written in audit form as

$$\delta Q_H = T_H \delta S_H + \delta \Sigma_H + \delta W_H, \quad (20)$$

or equivalently

$$R_H = \delta Q_H - T_H \delta S_H - \delta W_H - \delta \Sigma_H. \quad (21)$$

Here R_H measures closure failure after registered work-like and entropy-production terms are included.

A covariant effective description may place such failures into a ledger-residual slot,

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} + R_{\mu\nu}^{\text{ledger}}, \quad (22)$$

with the bookkeeping decomposition

$$R_{\mu\nu}^{\text{ledger}} = R_{\mu\nu}^{\text{prod}} + R_{\mu\nu}^{\text{mem}} + R_{\mu\nu}^{\text{noise}} + R_{\mu\nu}^{\text{curv}} + R_{\mu\nu}^{\text{hyst}}. \quad (23)$$

This is not proposed as a new gravitational field equation. It is a ledger-location marker for non-equilibrium residuals: entropy production, memory kernels, unresolved boundary noise, higher-curvature slots, and hysteretic response.

TABLE II. Jacobson inputs and T2 interpretation.

Jacobson input	T2 interpretation
local Rindler horizon	finite causal boundary
horizon area A	distinguishability ledger
$\delta S = \eta \delta A$	boundary-state accounting
δQ	stress-energy flux across boundary
T	flux-to-entropy conversion scale
Einstein equation	equilibrium closure for all ledgers

Any promoted residual tensor must also satisfy the corresponding covariant consistency condition. If the matter stress-energy tensor is separately conserved, then

$$\nabla^\mu R_{\mu\nu}^{\text{ledger}} = 0. \quad (24)$$

If the effective matter sector is not separately closed, the residual must instead enter a balanced exchange equation,

$$\nabla^\mu \left(\frac{8\pi G}{c^4} T_{\mu\nu} + R_{\mu\nu}^{\text{ledger}} \right) = 0. \quad (25)$$

This condition is required by the Bianchi identity and local covariance; without it the residual slot is not an admissible geometric accounting term.

Recent analyses of non-equilibrium horizon thermodynamics in modified gravity emphasize that entropy-production terms can have different origins in local Rindler and cosmological apparent-horizon frameworks, and that covariant consistency conditions such as the Bianchi identity matter for any promoted residual term [26].

T2 does not claim that the fundamental Newton constant varies with observer capacity. Capacity deficit enters the effective description through closure residuals, memory terms, stochastic response, or loss of full-fidelity horizon accounting, not through a bare replacement $G \rightarrow G_{\text{eff}}(C_{\text{acc}})$.

Connection to non-equilibrium gravity

Non-equilibrium horizon thermodynamics has been used to motivate entropy production terms and modified gravitational dynamics [21]. T2 is compatible with this direction, but keeps the claim weaker: non-equilibrium boundary accounting should appear as residual terms, not necessarily as one universal modification of GR.

GEOMETRY AS A PHASE-B BOUNDARY VARIABLE

T3 describes Phase-B variables as coarse variables that remain cheap to update, slow to forget, and approximately closed after microscopic tracking fails. In T2, effective geometry can be modeled as a minimal sufficient boundary variable. A normal-form selection score is

$$q_{\text{geom}}^* = \arg \min_{q \in \mathcal{Q}_{\text{geom}}} [C_{\text{update}}(q) + \lambda M_{\text{closure}}(q) + \mu \ell_{\text{pred}}(q)]. \quad (26)$$

Here C_{update} is the cost of maintaining and updating the coarse variable, M_{closure} is the residual closure or Markovianity error, and ℓ_{pred} is the loss for predicting causal access, horizon-area response, or stress-energy flux. This

is not a gravitational variational principle. It is a finite-system selection principle: geometry survives overflow because it is a minimal sufficient boundary variable for causal access and stress-energy accounting.

Minimal sufficient complexity

A microscopic boundary description may carry more detail than a finite observer can update or verify. Among possible compressed boundary variables, a geometry-like variable is selected when it is sufficient for the registered task class: it preserves causal order, predicts area response, and tracks stress-energy flux to tolerance while requiring less update and closure cost than micro-boundary tracking. In T2 language, effective geometry is not the most detailed description; it is the minimal sufficient covariant description that remains usable after horizon capacity overflow.

Optional bridge: information geometry

Information geometry provides a related but non-identical example of how distinguishability can induce geometry [25]. Fisher-information metrics define distances on statistical manifolds by the local distinguishability of probability distributions. T2 does not identify spacetime geometry with Fisher geometry. It uses the analogy only to mark a possible bridge: when distinguishability relations are compressed into stable differential structure, metric language can emerge as the natural accounting form.

RELATION TO T1, T3, P-SERIES, AND X3

T1 supplies the budget language. T3 supplies the overflow language: when micro-boundary distinctions exceed accessible capacity, coarse stochastic variables and Phase-B invariants dominate. P3 explains how inaccessible environmental memory produces effective finite-bath terms. P4 explains why lost preimage information is not recovered by capacity restoration alone. P6 supplies update-rate and dissipation bounds. P7 supplies invariant-ledger logic. X3 states gravity's functional role; T2 explains that role through horizon thermodynamic accounting.

COROLLARY: HORIZON-MAINTENANCE DENSITY SCALE

T2 does not derive dark energy and does not identify any horizon scale with the observed cosmological constant. However, once horizon entropy and temperature

are treated as a boundary ledger, a natural horizon-scale energy estimate appears:

$$E_H \sim T_H S_H. \quad (27)$$

Distributed over a horizon volume $V_H \sim R_H^3$, this gives the dimensional density scale

$$\rho_H \sim \frac{E_H}{V_H} \sim \frac{c^4}{GR_H^2} \quad (28)$$

up to convention-dependent numerical factors. T2 only identifies the upstream ledger structure. The cosmological interpretation of this scale is developed separately in FDS-X1. The horizon-ledger interpretation is naturally compatible with non-phantom maintenance-density readings because a stable causal ledger should not require indefinite violation of causal-order accounting, but the equation-of-state claim is not made in T2.

Recent work on dynamical de Sitter horizons and de Sitter first-law thermodynamics reinforces the need to distinguish stationary horizon entropy from non-stationary horizon-ledger accounting [27, 29].

EMERGENT GRAVITY AND QUANTUM-GRAVITY CAVEATS

T2 is compatible with thermodynamic or emergent readings of gravity, but it does not require a completed microscopic theory of spacetime. It also does not settle whether gravitons are fundamental or emergent. A cautious statement is: if geometry is an effective thermodynamic field, perturbative gravitons may be interpreted as quanta of the effective field in regimes where linearized quantization is valid. T2 does not decide the ontology of the graviton.

Similarly, T2 does not solve quantum gravity. It supplies a bridge between finite distinguishability, horizon entropy, and effective geometry. A microscopic theory of the horizon degrees of freedom is outside the scope of this paper.

NUMERICAL AND CONCEPTUAL DEMONSTRATIONS

The figures are deterministic normal-form demonstrations generated by `code/generate_results.py`. They are not empirical fits and not simulations of full general relativity.

TABLE III. Relation of T2 to neighboring FDS papers.

Paper	Core role	T2 relation
FDS Core	finite capacity, physical bridge discipline	formal foundation for boundary accounting [1]
T1	finite distinguishability budgets	upstream budget and horizon-capacity language [2]
T3	overflow and effective stochasticity	geometry as Phase-B boundary variable [3]
P3	finite-bath memory	horizon microstates as inaccessible bath analogue [4]
P4	anti-recurrence and hysteresis	capacity recovery does not recover erased boundary microstates [5]
P6	speed, precision, and dissipation	horizon update windows and throughput accounting [6]
P7	invariant protection	geometry as protected large-scale boundary ledger [7]
X3	gravity as boundary / geometry / accounting	functional placement of the T2 bridge [8]
X1	horizon-maintenance density	downstream cosmological scale bridge [9]
T2	horizon boundary thermodynamics	effective geometry as boundary ledger

Horizon boundary-ledger bridge



T2 adds observer-relative distinguishability ledgers, not a new derivation of Einstein gravity

FIG. 1. Horizon boundary-ledger bridge. Finite causal access defines a horizon boundary; area functions as a ledger; entropy and flux closure provide the thermodynamic accounting; effective geometry is the macroscopic representation of that accounting.

TABLE IV. Non-equilibrium horizon-ledger residual taxonomy. This table replaces arbitrary toy bar weights with residual classes, ledger sources, and possible effective signatures. It is a placement guide for corrections, not a field equation.

Residual class	Ledger source	Possible effective signature
entropy production	$d_i S_H$	viscosity / dissipation
memory kernel	$K(t - t')$	nonlocal delayed response
stochastic noise	unresolved boundary modes	metric fluctuation proxy
curvature correction	high-curvature ledger terms	R^2 or $f(R)$ -like slot
hysteresis	path-dependent boundary state	delayed area response

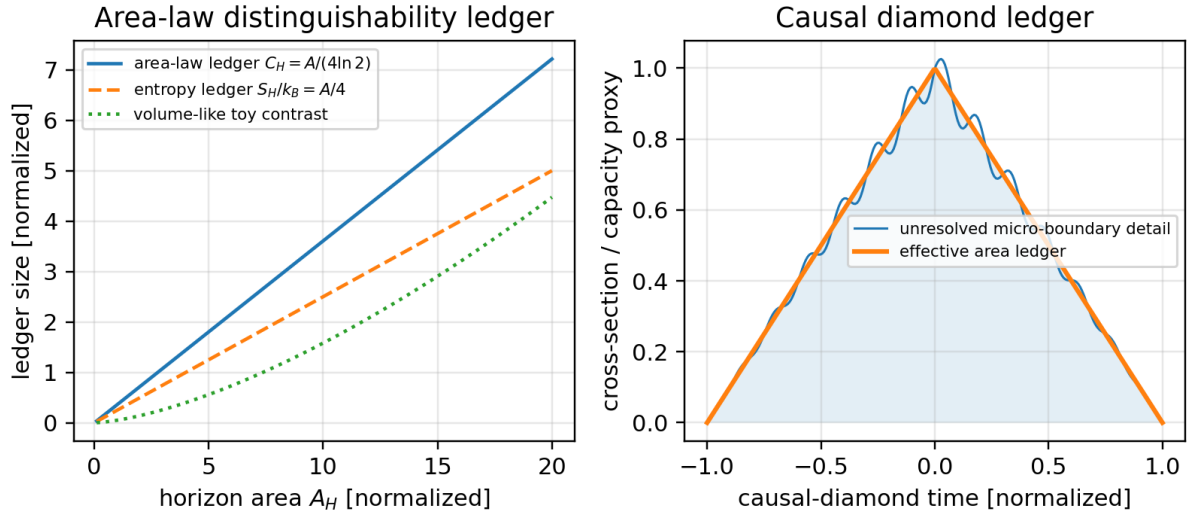


FIG. 2. Left: area-law horizon distinguishability ledger in normalized Planck-area units. The plot illustrates the bridge expression $C_H = A/(4\ell_P^2 \ln 2)$; it does not derive the coefficient. Right: causal-diamond normal form. An all-detail description contains unresolved boundary ripples, while the finite observer maintains a smooth effective area ledger. This is a conceptual illustration of effective geometry as coarse boundary accounting, not a metric solution.

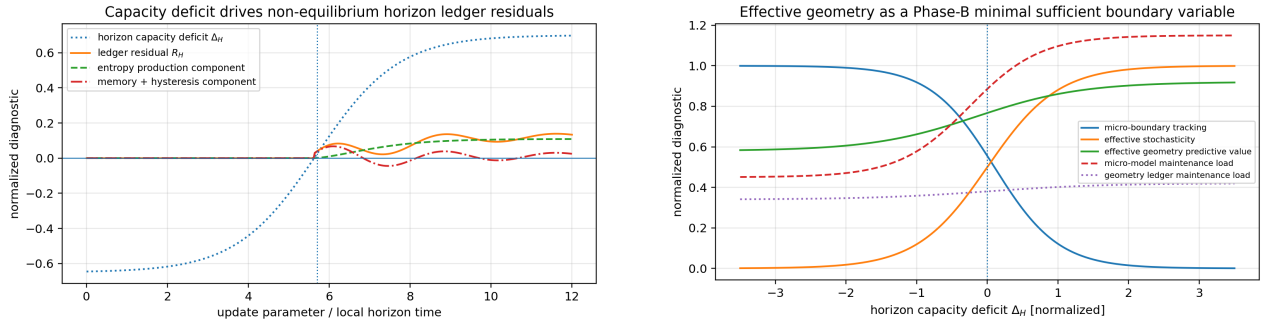


FIG. 3. Left: horizon capacity deficit Δ_H drives non-equilibrium ledger residuals in the normal form. Right: effective geometry as a Phase-B minimal sufficient boundary variable. After overflow, micro-boundary tracking falls and effective stochasticity rises, while coarse geometry retains predictive value at lower maintenance load.

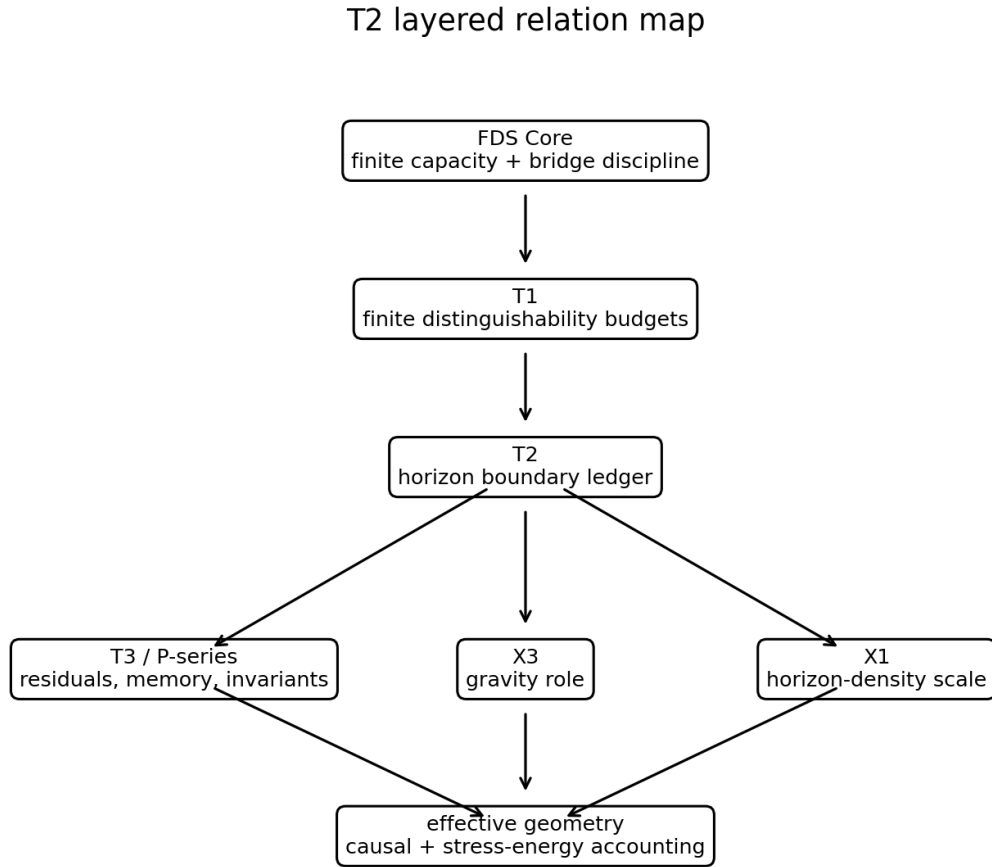


FIG. 4. Relation map. The layout makes the dependency stack explicit: FDS Core leads to T1 finite budgets and T2 horizon ledger, which branches to T3/P-series residual mechanisms, the X3 gravity role, and the X1 horizon-density scale.

FALSEIFICATION AND DEMOTION CONDITIONS

T2 is weakened or demoted if horizon entropy is not related to boundary area for the stated horizon class, local horizon temperature fails in the required domain, Clausius-style horizon thermodynamics fails as an effective equation-of-state model, stress-energy flux cannot be related to causal-boundary area change even approximately, geometry has no operational relation to causal access or boundary entropy, or non-equilibrium corrections cannot be represented as ledger residuals.

A strong demotion would be a complete physical theory in which geometry exists and evolves independently of any causal boundary, entropy, or energy-momentum accounting. Even then, the formal FDS finite-capacity results remain separate: only the T2 physical bridge is affected.

RELATION TO EXISTING THEORY

T2 uses standard physics as bridge input. General relativity provides the target effective geometry [10–12, 24]. Black-hole thermodynamics provides entropy and temperature [13–15]. Bekenstein and holographic bounds supply finite-boundary accounting [16, 17]. Jacobson’s 1995 argument supplies the central local-horizon model class [18]. The Unruh effect supplies the local acceleration temperature [19]. Non-equilibrium thermodynamics supplies entropy-production language [20, 21]. Emergent-gravity approaches provide comparison points, not dependencies [22, 23]. Information geometry supplies an optional bridge between distinguishability and metric structure [25]. Because horizon-entropy functionals are actively generalized in black-hole and cosmological settings, T2 treats the area law and its coefficient as physical bridge inputs rather than as FDS-derived constants [28]. FDS supplies the finite-system bookkeeping and failure-propagation discipline [1–3].

CONCLUSION

T2 does not derive general relativity from FDS and does not replace the Einstein equation. It interprets the thermodynamic side of gravitational physics through finite distinction systems. A finite observer has bounded causal access and bounded distinguishability. A horizon is the causal boundary of that access. Horizon area functions as an entropy ledger; heat flux across the boundary changes that ledger; and requiring the ledger to close locally and covariantly yields effective geometric dynamics in the Jacobson model class.

Geometry is therefore not merely a coordinate system in T2. It is the stable macroscopic bookkeeping structure

through which finite systems account for causal access, boundary entropy, and stress-energy flow. Effective geometry is horizon boundary thermodynamics written in covariant form.

APPENDIX A. OPERATIONAL ANALOGY FOR EMBODIED AGENTS

This appendix is not part of the physics proof. It extracts an engineering analogy: an embodied agent should treat geometry not only as coordinates, but as a boundary ledger for causal reach, uncertainty, and update cost.

TABLE V. T2 operational analogy for embodied agents.

Horizon geometry Agent analogue	
causal horizon	sensor / action reach
horizon area	interface bandwidth
entropy ledger	reachable-state uncertainty
energy flux	boundary update cost
curvature	world-model deformation
closure residual	latency, memory, drift, error

For autonomous driving or embodied robotics, when environmental complexity exceeds the sensor-processing boundary, the agent must geometricize: it must maintain a compressed reachability and uncertainty ledger rather than raw scene detail. Failure modes include mapping without reachability accounting, planning beyond actuation horizon, ignoring uncertainty growth at the boundary, assigning no resource cost to geometry updates, and brittle behavior under non-equilibrium dynamics. This is an analogy to T2 ledger logic, not a test of gravitational horizon thermodynamics.

Protocol 1 (Geometric boundary-ledger planning). *An embodied agent facing high-dimensional sensory overflow should: (i) estimate causal reach and sensor-action horizon; (ii) estimate boundary bandwidth and update budget; (iii) compress perception into a reachability and obstacle geometry; (iv) track uncertainty growth near the boundary of access; and (v) slow down, externalize, or replan when the agent-level capacity deficit becomes positive.*

The protocol is an engineering analogue of horizon-ledger accounting: geometry is a survival-oriented compression of causal reach, not merely a coordinate map.

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Horizon-Maintenance Dark Energy: A Pre-Euclid Bridge Note from Finite Distinguishability

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(S-X1 / Conditional Physical Bridge / Pre-Euclid Version)

This note registers a frontier physical consequence (P3 layer) connecting finite distinguishability at cosmological horizons to dark-energy phenomenology before Euclid Data Release 1. The claim is not that the cosmological constant, general relativity, quantum gravity, or the dark-energy equation of state is derived from the distinction primitive alone. Rather, if a cosmological horizon is treated as a finite distinguishability boundary with horizon entropy and temperature, then the natural energy-density scale of horizon maintenance is

$$\rho_{\text{HM}} \sim \frac{c^4}{GR_{\text{h}}^2} \sim \frac{c^2 H^2}{G} \quad (R_{\text{h}} \sim c/H),$$

or, in natural units, $\rho_{\text{HM}} \sim M_{\text{Pl}}^2 H^2$. This gives the observed smallness scale $\rho_{\text{HM}}/\rho_{\text{Pl}} \sim (\ell_{\text{P}}/R_{\text{h}})^2 \sim 10^{-122}$, not by cancellation of large vacuum contributions but by boundary-area suppression. The note formulates a conservative phenomenological model $\rho_{\text{HM}}(a) = 3M_{\text{Pl,red}}^2 H^2(a)\mu(a)$, where $\mu(a)$ is a horizon-maintenance response factor, and pre-registers a falsifiable expectation: the physical horizon-maintenance component should be dynamical but non-phantom, $w_{\text{phys}} \geq -1$, with late-time approach toward -1 . Apparent phantom behavior in effective CPL or other reconstructions is allowed only as an effective description arising from sector mixing, parameterization, systematics, curvature, modified-gravity mapping, or an incomplete correspondence between fitted and physical dark-energy variables. The bridge is demoted if Euclid, DESI, Roman, CMB-S4, or combined nonparametric reconstructions lock dark energy to a strict constant with high precision. The strict non-phantom version is falsified by robust, unavoidable physical $w < -1$ behavior or by a dark-energy density that unambiguously increases with scale factor under a model-independent reconstruction. The purpose is pre-registration, auditability, and model development, not replacement of Λ CDM.

Reader Contract. This paper is a frontier physical consequence note (P3), not a completed cosmological theory. It does not claim to derive general relativity, quantum gravity, the Bekenstein-Hawking entropy formula, the exact numerical value of the cosmological constant, or a complete likelihood-ready dark-energy model. It uses established horizon entropy, horizon temperature, finite causal access, and Landauer-style update accounting as bridge inputs. Its contribution is to register a finite-distinguishability interpretation of the dark-energy scale, state conservative predictions, and specify advance falsification and demotion conditions before Euclid DR1.

INTRODUCTION

Pre-Euclid purpose of this note

Euclid Data Release 1 is currently scheduled for October 21, 2026, after earlier quick releases intended to prepare the community for larger validated data products [23, 24]. DESI has already reported strengthened hints that dark energy may evolve, based on its first three years of BAO data and combinations with other cosmological probes [21, 22]. These results motivate theoretical discipline. If a theory has a dark-energy claim, it should state its prediction and failure conditions before the next major data release rather than reinterpret outcomes afterward.

This paper therefore registers FDS-X1 as a *pre-Euclid frontier physical consequence (P3)*. It is not a completed replacement for Λ CDM. It is a timestamped claim package: a scale argument, a minimal phenomenological model, a set of pre-registered expectations, and a falsification table.

Claim-status summary

Keywords: dark energy; finite distinguishability; cosmological horizon; horizon entropy; Landauer principle; holographic dark energy; equation of state; Euclid; DESI; pre-registration.

TABLE I. Summary of central FDS-X1 claims, their epistemic status, and what would constitute falsification or demotion.

Claim	Status	What would fail
Cosmological horizon as finite distinguishability boundary	Physical bridge	Horizon entropy/causal-access accounting fails or is superseded in the applicable regime
Horizon-maintenance density scale $\rho_{\text{HM}} \sim c^4/(GR_h^2)$	Scaling bridge	Dark-energy scale is shown to be unrelated to any horizon-area or causal-boundary scale
10^{-122} as area-ratio suppression	Naturalness argument	Observed scale requires independent cancellation rather than boundary-area suppression
Dynamical but non-phantom physical component	Pre-Euclid prediction	Robust unavoidable physical $w < -1$, or robust $d\rho_{\text{DE}}/d\ln a > 0$, not attributable to effective reconstruction
Late-time approach toward -1	Asymptotic bridge	Future data favor persistent non-de Sitter behavior incompatible with horizon stabilization
Strict Λ demotion criterion	Failure contract	High-precision, model-independent consistency with constant ρ_{DE} over the relevant redshift range

Central idea

In FDS-T1, a finite physical observer has an observer-relative distinguishability budget $N_O = |\text{Im}(\pi_O)|$ and an accessible bit capacity limited by memory, boundary, channel, causal access, and update throughput [1]. A cosmological horizon is a limiting physical case of such finite access. It separates what can in principle affect an observer's records from what cannot.

The FDS-X1 bridge claim is that the dark-energy scale can be interpreted as a horizon-maintenance scale: the energy-density scale associated with sustaining a finite causal distinguishability boundary. If the relevant horizon has radius R_h , area A_h , entropy $S_h \sim k_B A_h/(4\ell_P^2)$, and temperature $T_h \sim \hbar c/(2\pi k_B R_h)$, then the natural boundary-maintenance energy density scales as

$$\rho_{\text{HM}} \sim \frac{c^4}{GR_h^2}. \quad (1)$$

For $R_h \sim c/H$, this becomes

$$\rho_{\text{HM}} \sim \frac{c^2 H^2}{G}, \quad (2)$$

or $\rho_{\text{HM}} \sim M_{\text{Pl}}^2 H^2$ in natural units. This is the same scale as the critical density up to order-unity factors. The observed dark-energy density is therefore not treated as a Planck-volume vacuum sum; it is treated as a horizon-scale boundary term.

What is not claimed

This note does not claim any of the following:

1. that FDS derives general relativity or quantum gravity;

2. that FDS derives the coefficient $1/4$ in the Bekenstein-Hawking entropy formula;
3. that the cosmological constant problem is solved in the sense of a complete microphysical vacuum-energy cancellation;
4. that the exact value of w_0, w_a is predicted without a full likelihood-ready cosmological model;
5. that DESI has already confirmed FDS-X1;
6. that any deviation from ΛCDM counts as support for FDS-X1;
7. that apparent phantom reconstructions are automatically artifacts.

The note is narrower: it registers a scaling bridge, a physical interpretation, a minimal model family, and advance demotion rules.

Relation to FDS-T1 and the Physical Bridge Registry

FDS-X1 depends on the finite-observer budget language of FDS-T1 [1], the FDS formal core [2], the DT archive [3], and the Physical Bridge Claim Registry [4]. FDS-T1 supplies the shared concepts of finite distinguishability, boundary-relative capacity, and Landauer-style update accounting. The Physical Bridge Registry classifies dark-energy claims as frontier physical consequences (P3 layer) rather than established consequences of the formal core. FDS-X1 therefore cannot be used as evidence for the FDS core unless it later survives independent cosmological tests; failure of X1 would demote or retire the dark-energy bridge, not the finite-system formalism.

RELATED WORK

The cosmological constant problem

The accelerated expansion of the universe was inferred from Type Ia supernova observations [5, 6] and is supported by CMB and large-scale-structure measurements [7]. In standard cosmology, the simplest explanation is a cosmological constant Λ , with dark-energy equation of state $w = -1$. The cosmological constant problem arises because naive quantum-field vacuum-energy estimates are enormously larger than the observed dark-energy density. The present paper does not compute or cancel zero-point energies. It asks whether the observed scale is more naturally associated with a horizon boundary than with independent bulk vacuum modes.

Horizon thermodynamics

Black-hole thermodynamics assigns entropy and temperature to horizons [8, 9]. Gibbons and Hawking showed that de Sitter horizons also carry a temperature and entropy [10]. Jacobson’s thermodynamic derivation of Einstein’s equation suggests a deep relation between local horizon thermodynamics and gravitational dynamics [15]. FDS-X1 does not derive these results. It uses them as physical bridge inputs for a finite-distinguishability interpretation of horizon-scale energy density.

Holographic and horizon dark energy

The idea that dark-energy scales may be controlled by infrared or horizon cutoffs is not new. Cohen, Kaplan, and Nelson argued that effective field theories in a box must avoid states whose Schwarzschild radius exceeds the box size, leading to UV/IR relations [17]. Hsu pointed out problems with taking the Hubble radius as the infrared cutoff in simple holographic dark-energy models [18]. Li proposed holographic dark energy using the future event horizon as the infrared cutoff, producing late-time acceleration in a phenomenological model [19]. Reviews summarize the broad holographic dark-energy program and its observational constraints [20].

FDS-X1 is adjacent to this literature but not identical. It does not merely impose an IR cutoff on quantum fields. It interprets the horizon as a finite distinguishability boundary whose maintenance scale is boundary-like rather than volume-like. This distinction matters because the FDS claim concerns the accessible distinction budget of a finite observer patch, not a bare count of field modes.

Novelty boundary relative to holographic dark energy

FDS-X1 does not claim priority over the holographic dark-energy scaling relation $\rho_{\text{DE}} \sim M_{\text{Pl}}^2 L^{-2}$. That scaling and the non-phantom tendency $w > -1$ are well established in the holographic dark-energy literature [17–20]. The novelty of the FDS-X1 bridge lies in the finite-distinguishability interpretation of that scale and its embedding in a broader FDS boundary-accounting framework.

Current observational context

DESI’s 2025 three-year results strengthened hints that dark energy may be evolving [21]. The DESI DR2 guide states that the evidence for a time-evolving dark-energy equation of state increased relative to Year 1 and that extended analyses indicate trends across parametric and non-parametric methods [22]. At the same time, such hints remain data-combination and model dependent, and the community has not replaced Λ CDM. Euclid DR1, planned for October 21, 2026, is therefore a natural target for a pre-registered status update [23, 24].

FINITE-DISTINGUISHABILITY SETUP

Horizon as an access boundary

Definition 1 (Cosmological distinguishability boundary). *A cosmological distinguishability boundary for an observer O is a horizon-like surface beyond which physical differences cannot be operationally registered, preserved, and used by O within the relevant causal and thermodynamic constraints.*

The relevant boundary need not be exactly the Hubble radius. Depending on the problem, it may be an apparent horizon, event horizon, causal diamond, or other effective boundary. In a spatially flat FRW universe, common length scales include the Hubble radius $R_H = c/H$, the apparent horizon $R_A = c/H$, and, in accelerating spacetimes, the future event horizon

$$R_e(t) = a(t) \int_t^\infty \frac{c dt'}{a(t')}. \quad (3)$$

In exact de Sitter space, these coincide up to the common value c/H . During matter domination or transition eras, they need not coincide. FDS-X1 therefore uses R_h as a generic effective horizon radius and treats the mapping $R_h \mapsto H^{-1}$ as a late-time or phenomenological approximation, not an identity valid in every epoch.

TABLE II. FDS-X1 novelty boundary relative to holographic dark energy.

Aspect	HDE prior art	FDS-X1 addition
$\rho \sim M_{\text{Pl}}^2 L^{-2}$	established	not claimed as novel
horizon / IR cutoff	established	interpreted as finite distinguishability boundary
non-phantom $w > -1$ possibility	established	registered as physical horizon-maintenance constraint
10^{-122} area-ratio smallness	present in HDE background	interpreted as boundary-area suppression of maintainable distinctions
cutoff choice	model-dependent selection	constrained by causal-access / recordability / boundary-maintenance cr
failure conditions	not always explicit	pre-registered demotion / falsification table
connection to geometry bridge	partial in emergent gravity	T2 supplies effective geometry as horizon ledger

Horizon entropy and temperature

Assume a horizon-like boundary with area

$$A_h = 4\pi R_h^2. \quad (4)$$

The associated entropy is

$$S_h = \frac{k_B A_h}{4\ell_P^2}, \quad (5)$$

with $\ell_P = \sqrt{\hbar G/c^3}$. The associated horizon temperature is written in the standard form

$$T_h = \frac{\hbar c}{2\pi k_B R_h}. \quad (6)$$

These are imported physical bridge inputs, not derived here.

Boundary-area suppression

The horizon entropy counts area-like degrees of freedom. The ratio between the horizon scale and the Planck scale gives

$$\frac{\rho_{\text{HM}}}{\rho_{\text{Pl}}} \sim \left(\frac{\ell_P}{R_h}\right)^2. \quad (7)$$

For the present cosmic horizon, $R_h \sim 10^{26}$ m and $\ell_P \sim 10^{-35}$ m, so

$$\left(\frac{\ell_P}{R_h}\right)^2 \sim 10^{-122}. \quad (8)$$

This is the naturalness core of FDS-X1: the smallness is a boundary-area ratio, not an unexplained cancellation of independent volume terms.

HORIZON-MAINTENANCE SCALING

Thermodynamic horizon energy scale

The simplest horizon-thermodynamic estimate multiplies entropy by temperature:

$$E_h \sim T_h S_h. \quad (9)$$

Using Eqs. (5) and (6),

$$E_h \sim \frac{\hbar c}{2\pi k_B R_h} \cdot \frac{k_B (4\pi R_h^2)}{4\ell_P^2} \quad (10)$$

$$\sim \frac{\hbar c R_h}{2\ell_P^2} = \frac{c^4 R_h}{2G}. \quad (11)$$

Distributed over a horizon volume $V_h = 4\pi R_h^3/3$, this gives

$$\rho_h \sim \frac{E_h}{V_h} \sim \frac{3c^4}{8\pi G R_h^2}. \quad (12)$$

If $R_h = c/H$,

$$\rho_h \sim \frac{3c^2 H^2}{8\pi G}. \quad (13)$$

This is the critical-density scale. An equivalent derivation clarifies the power-to-density conversion. Horizon-maintenance power scales as $P_H \sim T_h S_h / \tau_h \sim c^5 / G$, the canonical Planck power. Over one Hubble time $\tau_h \sim H^{-1}$, the total maintenance energy is $E_{\text{main}} \sim P_H / H$. Distributed over the Hubble volume $V_H \sim c^3 / H^3$, the energy density becomes

$$\rho_{\text{HM}} \sim \frac{P_H / H}{V_H} \sim \frac{c^5 / G}{H} \cdot \frac{H^3}{c^3} = \frac{c^2 H^2}{G}, \quad (14)$$

recovering the same $M_{\text{Pl}}^2 H^2$ scale. This step makes explicit that the density follows from dividing a per-Hubble-time maintenance energy by the causal volume, closing a possible dimensional attack on the raw $T_h S_h / V_h$ estimate. FDS-X1 identifies the dark-energy component as the fraction or response of this horizon-scale budget associated with boundary maintenance.

Maintenance response factor

Define the horizon-maintenance density

$$\rho_{\text{HM}}(a) = \mu(a) \frac{3c^4}{8\pi G R_h^2(a)}. \quad (15)$$

When $R_h \sim c/H$, this becomes

$$\rho_{\text{HM}}(a) = 3M_{\text{Pl,red}}^2 H^2(a) \mu(a), \quad (16)$$

where $M_{\text{Pl,red}}^2 = (8\pi G)^{-1}$ in natural units. The dimensionless function $\mu(a)$ is a horizon-maintenance response factor. It may encode the fraction of the horizon-scale budget appearing as the effective dark-energy sector, the transition from Hubble-like to event-horizon-like behavior, interaction with matter or curvature, or the mapping from physical horizon maintenance to the fitted dark-energy variable.

Remark 1. Equation (16) is a model family, not a completed microphysical theory. A full model must specify $R_h(a)$ and $\mu(a)$, derive perturbations, and confront CMB, BAO, supernova, weak-lensing, and growth data.

Why this is not ordinary vacuum-energy counting

Naive vacuum-energy estimates count independent bulk modes up to a UV cutoff. The FDS-X1 scaling counts accessible horizon distinctions. The difference is schematic:

$$\rho_{\text{bulk}} \sim \Lambda_{\text{UV}}^4, \quad \rho_{\text{HM}} \sim \frac{A_h}{\ell_P^2} \frac{k_B T_h}{R_h^3} \sim \frac{c^4}{G R_h^2}. \quad (17)$$

The second expression is boundary-like. Its smallness follows from the cosmic horizon being large.

EQUATION OF STATE AND PHENOMENOLOGICAL MODEL

Continuity equation

For a separately conserved effective component,

$$\frac{d\rho_{\text{HM}}}{d \ln a} + 3(1 + w_{\text{HM}})\rho_{\text{HM}} = 0. \quad (18)$$

Therefore

$$w_{\text{HM}}(a) = -1 - \frac{1}{3} \frac{d \ln \rho_{\text{HM}}}{d \ln a}. \quad (19)$$

Using Eq. (16), in the $R_h \sim c/H$ phenomenological regime,

$$w_{\text{HM}}(a) = -1 - \frac{2}{3} \frac{d \ln H}{d \ln a} - \frac{1}{3} \frac{d \ln \mu}{d \ln a}. \quad (20)$$

This equation shows why the response factor matters. If μ is constant while H decreases, then $w_{\text{HM}} > -1$ and the component tracks the dominant background too strongly. To obtain a late-time component near -1 , the horizon response must evolve as the relevant boundary approaches an event-horizon or de Sitter-like limit.

Non-phantom physical component

The strict FDS-X1 expectation is not a precise CPL pair (w_0, w_a) . It is the sign of physical density evolution:

$$\frac{d\rho_{\text{HM}}}{d \ln a} \leq 0 \iff w_{\text{HM}} \geq -1. \quad (21)$$

At late times, if the universe approaches de Sitter-like horizon stabilization, then

$$\frac{d\rho_{\text{HM}}}{d \ln a} \rightarrow 0, \quad w_{\text{HM}} \rightarrow -1. \quad (22)$$

This is the most conservative dynamical claim: the physical horizon-maintenance component should be non-phantom and should approach -1 as the effective horizon stabilizes.

Effective reconstructions versus physical component

Cosmological data often constrain parameters in an effective model, such as the CPL form

$$w(a) = w_0 + w_a(1 - a). \quad (23)$$

A fitted $w_{\text{eff}} < -1$ in such a model need not by itself be the physical equation of state of the horizon-maintenance sector. It may arise from parameterization, sector mixing, curvature, modified-gravity mapping, data systematics, or the fact that $\mu(a)$ and $R_h(a)$ have not been explicitly modeled. FDS-X1 therefore distinguishes:

1. the *strict physical claim*: the horizon-maintenance component is non-phantom;
2. the *effective reconstruction*: a data-fitting variable that may cross -1 without directly representing a fundamental phantom fluid.

This distinction is not an unlimited escape clause. If non-parametric reconstructions and physical model comparisons robustly require a true phantom component with no sector-mixing or mapping ambiguity, the strict version fails.

PRE-EUCLID PREDICTIONS

Prediction 1 (P1: scale). *The dark-energy sector should remain tied to a horizon-scale density of order $M_{\text{Pl}}^2 H_0^2$, or equivalently $c^2 H_0^2 / G$ in SI units, not to a Planck-volume vacuum-energy density.*

Prediction 2 (P2: non-constant physical sector). *The physical horizon-maintenance sector should not be a featureless bare constant at all epochs. It should encode horizon evolution through $R_h(a)$ and/or $\mu(a)$, even if the late-time limit is observationally close to $w = -1$.*

Prediction 3 (P3: non-phantom physical behavior). *The strict physical horizon-maintenance component should satisfy $w_{\text{phys}} \geq -1$, or equivalently $d\rho_{\text{HM}}/d \ln a \leq 0$, with approach toward -1 as the effective horizon stabilizes.*

Prediction 4 (P4: effective phantom allowance). *If Euclid, DESI, or combined data prefer $w_{\text{eff}} < -1$ in CPL or similar fits, FDS-X1 treats this as compatible only if the phantom behavior is effective rather than fundamental: for example, due to sector mixing, parameterization, curvature, modified-gravity mapping, systematics, or incomplete modeling of $\mu(a)$ and $R_h(a)$.*

Prediction 5 (P5: density reconstruction priority). *The most direct observational test is not the sign of a fitted w_0 alone but the reconstructed evolution of $\rho_{\text{DE}}(a)$. A robust increase of physical dark-energy density with scale factor challenges the strict non-phantom horizon-maintenance version.*

FALSIFICATION AND DEMOTION CONDITIONS

No ad hoc rescue rule

FDS-X1 is not allowed to treat every possible observational result as success. If future data strongly support a strict cosmological constant with no density evolution, the dynamical claim is demoted. If future data robustly require a physical phantom component, the strict non-phantom version is falsified. If the horizon-scale density relation remains merely dimensional and cannot be embedded in a perturbatively viable cosmological model, the claim remains a heuristic naturalness argument rather than a physical theory.

Post-Euclid update protocol

After Euclid DR1, the status update should use the same categories as Table III:

1. unchanged frontier bridge;
2. promoted scaling/naturalness bridge;
3. supported dynamical non-phantom version;
4. demoted to heuristic only;
5. strict version falsified;
6. retired.

The update should quote the pre-Euclid predictions before interpreting the data.

COMPARISON WITH EXISTING MODELS

Versus Λ CDM

Λ CDM treats dark energy as a constant density with $w = -1$. FDS-X1 accepts Λ CDM as the current benchmark and does not claim observational replacement. Its difference is interpretive and predictive: it treats the observed scale as horizon-maintenance scale and expects that sufficiently precise data may reveal controlled evolution in the physical dark-energy sector.

Versus standard holographic dark energy

Standard holographic dark-energy models typically choose an infrared cutoff L and posit

$$\rho_{\text{HDE}} = 3c_{\text{HDE}}^2 M_{\text{Pl,red}}^2 L^{-2}. \quad (24)$$

The physics depends strongly on the choice of L . Taking $L = H^{-1}$ has known difficulties in simple models; taking L as the future event horizon can produce acceleration [18, 19]. FDS-X1 agrees that the event horizon or another causal boundary may be the physically relevant late-time scale, but it gives the cutoff a finite-distinguishability interpretation: the horizon is an operational boundary of accessible distinctions, not merely a regulator.

Versus quintessence and phantom models

Quintessence introduces scalar-field dynamics to obtain $w \neq -1$, usually with $w > -1$. Phantom models permit $w < -1$ but often face theoretical difficulties. FDS-X1 is not a scalar-field model. Its strict physical expectation is non-phantom, while allowing effective phantom reconstructions only as descriptions of a mixed or mis-specified sector.

Versus modified gravity

Modified-gravity models may reproduce effective dark-energy behavior without a physical fluid. FDS-X1 remains compatible with the possibility that some observed w_{eff} behavior is a mapping of modified gravitational dynamics into an effective fluid. A full FDS-X1 model must therefore be tested not only against background expansion but also against perturbations and growth of structure.

LIMITATIONS

First, FDS-X1 is a frontier physical consequence (P3 layer). It is not established by FDS-T1 alone. Failure of X1 does not falsify the FDS formal core or the finite-observer budget paper.

Second, the scaling $\rho_{\text{HM}} \sim M_{\text{Pl}}^2 H^2$ is not by itself a full cosmological model. A constant coefficient with $L = H^{-1}$ has known problems in ordinary holographic dark-energy treatments. The response factor $\mu(a)$ and the appropriate horizon radius $R_h(a)$ must be specified for a complete model.

Third, the paper does not compute perturbations. Any viable model must match CMB anisotropies, BAO, supernova distances, weak lensing, structure growth, and local constraints.

Fourth, the paper does not derive the Bekenstein-Hawking entropy formula or de Sitter temperature. These are bridge inputs.

Fifth, the paper does not claim that the QFT vacuum-energy calculation is mathematically invalid. It claims only that the observed cosmic dark-energy scale may be

TABLE III. Pre-registered outcome table for FDS-X1.

Future result	Consequence for FDS-X1
High-precision, model-independent consistency with constant ρ_{DE} and $w = -1$ across the measured redshift range	Demotes the dynamical horizon-maintenance claim; only the naturalness scaling may remain as a heuristic
Mild dynamical dark-energy evidence with $w_{\text{phys}} \geq -1$ and late-time approach to -1	Supports the strict horizon-maintenance version, subject to model comparison and perturbation tests
CPL or other effective fits prefer phantom crossing, but nonparametric density reconstructions or physical models do not require a fundamental phantom component	Compatible with the effective-reconstruction allowance; requires explicit modeling of $\mu(a)$, $R_h(a)$, curvature, and sector mixing
Robust model-independent evidence that physical ρ_{DE} increases with scale factor	Strongly challenges or falsifies the strict non-phantom version
Robust unavoidable physical $w < -1$ with no mapping, mixing, modified-gravity, or systematic ambiguity	Falsifies the strict non-phantom horizon-maintenance component
Data remain inconclusive or strongly model dependent	FDS-X1 remains a frontier physical consequence; no promotion

associated with accessible horizon-boundary degrees of freedom rather than independent bulk modes.

Sixth, the non-phantom expectation applies to the physical horizon-maintenance component. Effective fitted parameters may differ if the model space is incomplete.

CONCLUSION

FDS-X1 registers a pre-Euclid frontier physical consequence: if cosmological horizons are finite distinguishability boundaries, then their maintenance naturally carries the density scale

$$\rho_{\text{HM}} \sim \frac{c^4}{G R_h^2} \sim M_{\text{Pl}}^2 H^2, \quad (25)$$

with smallness

$$\frac{\rho_{\text{HM}}}{\rho_{\text{Pl}}} \sim \left(\frac{\ell_P}{R_h} \right)^2 \sim 10^{-122}. \quad (26)$$

This reframes the dark-energy naturalness problem as a boundary-area suppression rather than a cancellation of bulk vacuum terms.

The note does not claim to solve cosmology. It registers a model family and an outcome table before Euclid DR1. The strict physical prediction is a dynamical but non-phantom horizon-maintenance component with late-time approach toward -1 . Effective phantom fits are allowed only if they are reconstructions of a mixed or mis-specified sector. Robust constant dark energy demotes the dynamical claim; robust physical phantom behavior falsifies the strict version.

Future work must specify $R_h(a)$, derive $\mu(a)$, compute perturbations, fit DESI, Euclid, Roman, supernova, CMB, and weak-lensing data, and compare against

Λ CDM, holographic dark energy, quintessence, phantom models, interacting dark energy, and modified gravity.

The author used AI-assisted tools for language polishing, structural feedback, and drafting support. All claims, arguments, citations, and final wording remain the responsibility of the author.

Notation Summary

Claim Status and Failure Conditions

Boundary of Applicability

The FDS-X1 bridge applies only under the following conditions:

1. a meaningful cosmological horizon or finite causal-access boundary exists;
2. the horizon entropy and temperature bridge inputs are valid in the relevant regime;
3. the observed dark-energy sector can be compared to an effective horizon-maintenance density;
4. the mapping from physical ρ_{HM} to fitted $w(a)$ is not assumed to be trivial unless specified.

The bridge does not apply as stated in regimes where no finite horizon boundary exists, where horizon thermodynamics fails, or where the dark-energy phenomenology is entirely due to a sector unrelated to causal-boundary maintenance.

Priority and Timestamp Notice

This document is intended as a pre-Euclid public registration of the FDS-X1 frontier physical consequence.

TABLE IV. FDS-X1 notation summary.

Symbol	Meaning
R_h	effective cosmological horizon radius used by the bridge model
$R_H = c/H$	Hubble radius
R_e	future event horizon radius
$A_h = 4\pi R_h^2$	horizon area
S_h	horizon entropy
T_h	horizon temperature
ρ_{HM}	horizon-maintenance energy density
ρ_{DE}	observationally reconstructed dark-energy density
$\mu(a)$	horizon-maintenance response factor
w_{HM}	physical equation of state of the horizon-maintenance component
w_{eff}	effective fitted equation of state in a chosen reconstruction
ℓ_P	Planck length
$M_{\text{Pl,red}}$	reduced Planck mass
Ω_{DE}	dark-energy density fraction

TABLE V. Claim status and failure conditions for FDS-X1.

Claim	Status	Failure condition
Horizon distinguishability boundary	Physical bridge	Horizon entropy or causal-boundary accounting fails in the applicable regime
$\rho_{\text{HM}} \sim M_{\text{Pl}}^2 H^2$	Scaling bridge	Dark-energy scale shown unrelated to horizon or causal-boundary scale
Area-ratio suppression	Naturalness bridge	Observed scale requires an unrelated cancellation mechanism
Non-phantom physical component	Pre-Euclid prediction	Robust unavoidable physical $w < -1$
Late-time -1 approach	Asymptotic bridge	Persistent future departure incompatible with horizon stabilization
Effective phantom allowance	Interpretation rule	Effective reconstructions proven to represent a true phantom sector

Its purpose is not to assert that the claim is established, but to record the statement, dependencies, predictions, and failure conditions before Euclid DR1. Later versions may revise, demote, or abandon claims according to the outcome table.

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Finite-Capacity Prospect Theory: State-Dependent Risk Preferences under Resource, Attention, and Boundary-Risk Constraints

A Boundary-Risk Account of Loss Aversion, Reference Dependence, and Probability
Weighting

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Abstract

Prospect Theory identifies three robust regularities in decision-making under risk: loss aversion, reference dependence, and nonlinear probability weighting. These regularities are usually treated as descriptive deviations from expected-utility rationality. This paper proposes a finite-system bridge account: for a bounded agent maintaining identity under resource scarcity, limited attention, limited distinguishability, and finite update capacity, these regularities can be interpreted as state-dependent boundary-risk management strategies rather than fixed irrationality constants. The paper models an economic agent as a finite boundary-maintaining decision system with a vector-valued resource buffer, a perceived distance to constraint threshold, a maintained reference baseline, and a limited probability-distinction budget. Loss aversion arises from asymmetric boundary-risk slopes near the lower resource constraint; reference dependence arises from finite-memory prediction baselines; and probability weighting arises from finite precision allocation over probability and event-consequence space. The model predicts that Prospect Theory parameters are functions of resource buffers, attention capacity, and boundary-risk context: loss aversion should rise under scarcity and downside exposure; reference baselines should adapt more slowly under depletion; and probability weighting should become more distorted for low-capacity agents and boundary-relevant rare events. Numerical simulations illustrate the mechanism, compare static and state-dependent parameter recovery in synthetic data, and show a nudge-bandwidth region in which complex choice architecture interventions fail or backfire under depleted decision capacity. The paper does not claim to derive the empirical coefficient $\lambda \approx 2.25$ exactly, nor to reduce all economic behavior to thermodynamics. Its contribution is narrower: it formulates a testable finite-capacity bridge between behavioral economics, bounded rationality, and finite-system maintenance theory.

Keywords: prospect theory; loss aversion; reference dependence; probability weighting; bounded rationality; scarcity; finite capacity; behavioral economics; decision theory; nudge; resource constraints; active finite distinction systems.

JEL codes: D01, D81, D91, C91, I31, H31.

Scope of the Paper. This paper does not claim that all behavioral-economic phenomena are reducible to physics, nor that the empirical loss-aversion coefficient $\lambda \approx 2.25$ is derived exactly from first principles. It does not replace Prospect Theory, cumulative prospect theory, expected utility theory, bounded rationality, active inference, ecological rationality, or evolutionary psychology. The paper develops a finite-capacity bridge model in which economic agents are treated as bounded decision systems with finite resource buffers, attention, memory, precision, update capacity, and tolerance for boundary risk. The purpose is to generate testable predictions about when and why Prospect Theory parameters should vary across resource states, cognitive load, and boundary-risk contexts.

1 Claim-Status Summary

Table 1: Central claims, epistemic status, and demotion conditions.

Claim	Status	What would weaken or falsify it
Economic agents are finite-capacity decision systems	Operational bridge	Decision behavior invariant to memory, stress, attention, and resource constraints
Loss aversion increases near perceived resource thresholds	Testable bridge hypothesis	λ remains invariant across physiological, financial, cognitive, and downside-protection states
Reference dependence reflects maintained prediction baselines	Computational bridge	Reference points fail to correlate with prediction-error integration or baseline-updating dynamics
Probability weighting reflects finite precision allocation	Testable bridge hypothesis	Weighting curvature remains invariant under cognitive load, stress, event class, and resource state
Rare high-consequence events receive excess salience or precision	Domain bridge	Survival-relevant and neutral rare events show identical weighting after controls
Nudges consume decision bandwidth	Policy bridge	Nudge effectiveness independent of stress, scarcity, fatigue, administrative burden, and cognitive load
Prospect parameters are state-dependent	Central prediction	Stable λ , reference adaptation rate, and weighting curvature across resource regimes and boundary-risk contexts
Reverse stress test	Strong falsification condition	Severe resource depletion changes neither λ , reference rigidity, probability weighting, nor decision noise after adequate power and controls

2 Introduction

[Kahneman and Tversky \(1979\)](#) transformed decision theory by showing that risky choice is not well described by the expected-utility model. In Prospect Theory, outcomes are evaluated relative to a reference point; losses loom larger than gains; and objective probabilities are transformed into nonlinear decision weights. These empirical regularities were later extended in cumulative prospect theory ([Tversky and Kahneman, 1992](#)). Decades of behavioral-economics research have confirmed their importance for finance, insurance, labor supply, health, household choice, and public policy ([Barberis, 2013](#)). A recent meta-analysis of Prospect Theory parameters reports substantial between-study variation and systematic procedure dependence, suggesting that the field still lacks a satisfactory account of why parameter estimates vary across environments and tasks ([Imai et al., 2025](#)).

The success of Prospect Theory creates a deeper question. Are loss aversion, reference dependence, and probability weighting merely descriptive anomalies relative to rational choice? Or do they reveal the structure of decision-making by finite agents under physical, cognitive, social, and informational constraints? Standard explanations appeal to evolutionary adaptation, bounded rationality, affective processing, or ecological rationality ([Simon, 1955](#); [Gigerenzer and Todd, 1999](#)). These accounts are valuable, but they often leave the key parameters of Prospect Theory as stable psychological facts rather than state-dependent control variables.

This paper develops a finite-capacity account. An economic agent is not an ideal optimizer with unlimited representation, memory, attention, and energy. It is a finite boundary-maintaining system. It must maintain a viable boundary, allocate limited attention, update a compressed internal state, preserve a resource buffer, and act under uncertainty. In the language of Active Finite Distinction Systems (FDS), an agent has a finite distinction projection and finite distinguishability budget: it cannot represent all outcome states, probabilities, future contingencies, and action consequences with equal precision. When task demand exceeds accessible capacity, the agent must compress, coarse-grain, externalize, prune, relax the task, or fail.

The paper’s central thesis is that Prospect Theory parameters are not fixed irrationality constants. They are state-dependent finite-system control variables:

$$(\lambda_t, \alpha_t, \gamma_t) = f(F_t^{\text{eff}}, C_t, B_t), \tag{1}$$

where λ_t is loss aversion, α_t is reference-baseline adaptation, γ_t is probability-weighting curvature, F_t^{eff} is effective resource buffer, C_t is available decision capacity, and B_t is boundary-risk context. The model predicts that scarcity, fatigue, stress, financial precarity, time pressure, administrative burden, and downside exposure should change the measured parameters of risky choice.

The contribution is not to replace Prospect Theory. It is to make Prospect Theory dynamic. The paper provides a state-dependent model, falsifiable predictions, simulation code, synthetic parameter-recovery exercises, and policy implications for nudge design under scarcity.

3 Prospect Theory: Three Regularities

Let x denote an outcome relative to a reference point r . A common value function is

$$v(x) = \begin{cases} x^a, & x \geq 0, \\ -\lambda(-x)^b, & x < 0, \end{cases} \quad (2)$$

where $\lambda > 1$ captures loss aversion (Kahneman and Tversky, 1979). The empirical value $\lambda \approx 2.25$ is often treated as a benchmark, but estimates vary by domain, method, subject pool, and framing.

Reference dependence means that utility is not evaluated over final wealth alone. It is evaluated over changes relative to a reference point. The reference point may depend on status quo, expectations, past outcomes, social comparison, or contractual entitlement.

Probability weighting means that agents transform objective probabilities p into decision weights $w(p)$. A one-parameter inverse-S curve can be written as

$$w(p; \gamma) = \frac{p^\gamma}{p^\gamma + (1 - p)^\gamma}, \quad 0 < \gamma \leq 1. \quad (3)$$

When $\gamma < 1$, small probabilities are overweighted and moderate probabilities are compressed. More flexible specifications include the Prelec form (Prelec, 1998).

The finite-capacity account does not deny these descriptive forms. It asks why the parameters should move with resource state, attention capacity, and boundary-risk context.

4 Finite-System Preliminaries for Behavioral Economics

Definition 1 (Finite boundary-maintaining decision system) *An economic agent is modeled as a finite decision system*

$$A_t = (B_t, M_t, R_t, \mathbf{F}_t, C_t, \pi_t, \Psi_t, U_t, \tau_t), \quad (4)$$

where B_t is a maintained boundary or viability condition, M_t is memory, R_t is a reference baseline, $\mathbf{F}_t$ is a vector of resource buffers, C_t is available decision capacity, π_t is a finite distinction projection, Ψ_t is the decision environment, U_t is an action-update policy, and τ_t is the decision horizon.

The agent's finite distinction projection induces a decision-relevant budget

$$N_A(t) = |\text{Im}(\pi_t)|, \quad C_A(t) = \log_2 N_A(t). \quad (5)$$

This budget is not only memory. It includes attention, channel access, time pressure, numeracy, working memory, emotional regulation, and access to external records or institutional supports.

4.1 Vector resource buffers and substitutability

A key revision relative to earlier thermodynamic language is that the resource buffer is vector-valued:

$$\mathbf{F}_t = (F_t^m, F_t^e, F_t^s, F_t^a, F_t^\tau), \quad (6)$$

where F^m denotes monetary or financial buffer, F^e energetic or metabolic buffer, F^s social or institutional support, F^a attention or cognitive bandwidth, and F^τ time buffer. These resources are not equally substitutable. Money can sometimes compensate for time, food, or security; sleep loss or acute illness may be much less fungible.

We define effective resource buffer by a CES aggregator:

$$F_t^{\text{eff}} = \left(\sum_i \omega_i (F_t^i)^\eta \right)^{1/\eta}, \quad \sum_i \omega_i = 1. \quad (7)$$

The parameter η controls substitutability. When η is high, resources are more fungible. When $\eta \rightarrow -\infty$, the effective buffer approaches the minimum resource dimension. This captures a central prediction: depletion of non-fungible resources such as sleep, health, time, or safety should produce more nonlinear changes in risk preferences than depletion of more substitutable resources.

4.2 Perceived distance to boundary

Economic behavior depends on perceived, not merely objective, distance to constraint. Let F_{crit} denote the boundary-risk threshold and define objective distance

$$d_t = \frac{F_t^{\text{eff}} - F_{\text{crit}}}{F_{\text{scale}}}. \quad (8)$$

The perceived distance is

$$\hat{d}_t = \theta_t d_t, \quad (9)$$

where θ_t summarizes belief, trauma history, social comparison, uncertainty, information quality, and perceived institutional support. A wealthy agent with low perceived security may have low $\hat{d}_t$ despite high objective resources. Conversely, a poorer agent with strong social insurance may have a larger perceived buffer than monetary wealth alone implies.

5 Loss Aversion as Boundary-Risk Asymmetry

Let $R(d)$ be a boundary-risk potential, decreasing and convex in perceived distance d :

$$R(d) = \kappa(d + \epsilon)^{-\nu}, \quad \kappa > 0, \epsilon > 0, \nu > 0. \quad (10)$$

For a small resource change $\Delta > 0$, the local loss-aversion coefficient induced by boundary-risk asymmetry is

$$\lambda(d, \Delta) = \frac{R(d - \Delta) - R(d)}{R(d) - R(d + \Delta)}. \quad (11)$$

Because R is convex and decreasing near the threshold, downward movements toward the boundary increase risk more than equal upward movements reduce it. Thus $\lambda(d, \Delta) > 1$ generically when the agent is close to the boundary.

Proposition 1 (Boundary-risk loss aversion) *If perceived boundary-risk potential $R(d)$ is decreasing and convex over the relevant neighborhood, then a finite agent evaluating symmetric resource changes around d exhibits local loss aversion $\lambda(d, \Delta) > 1$.*

Interpretation. The empirical coefficient $\lambda \approx 2.25$ is treated here as a calibration target and empirical benchmark, not as an exact first-principles constant. The stronger prediction is state-dependence: λ should increase as $\hat{d}_t$ falls, and it should increase more sharply when the depleted resource is non-fungible.

Prediction 1 (Non-fungible resource depletion) *Depletion of non-fungible resources such as sleep, health, safety, or time should increase λ more nonlinearly than equivalent monetary depletion when other buffers cannot compensate.*

Prediction 2 (Perceived boundary risk) *After controlling for objective wealth, agents who perceive themselves as closer to a constraint boundary should show higher loss aversion.*

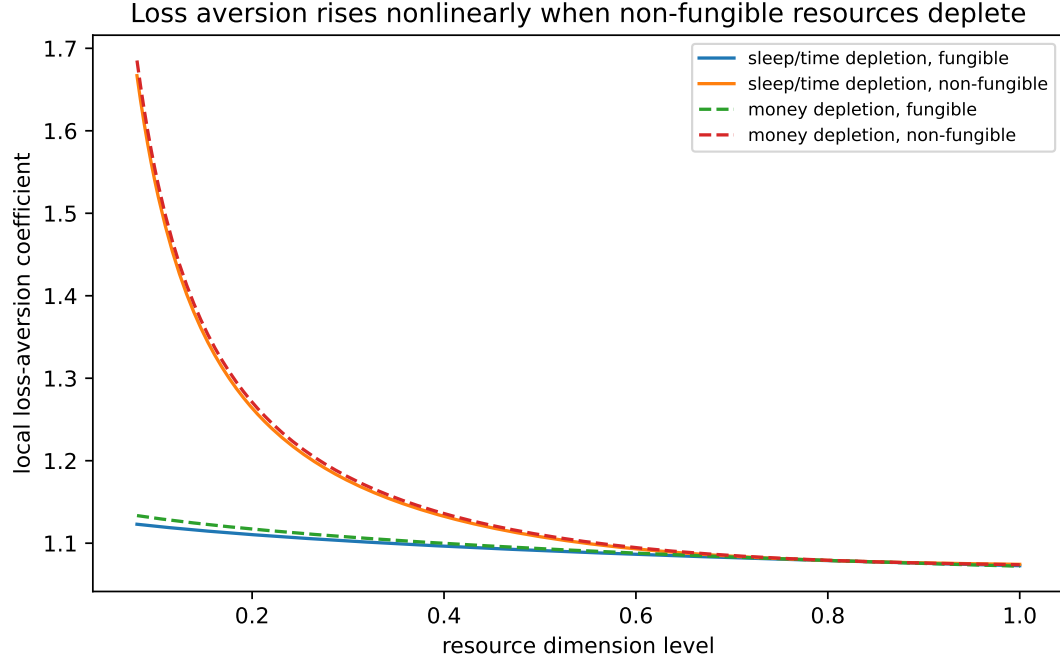


Figure 1: Loss aversion under fungible and non-fungible resource depletion. The simulation uses a CES effective buffer. When resources are weak substitutes, depletion of sleep/time generates sharper nonlinear increases in the local loss-aversion coefficient.

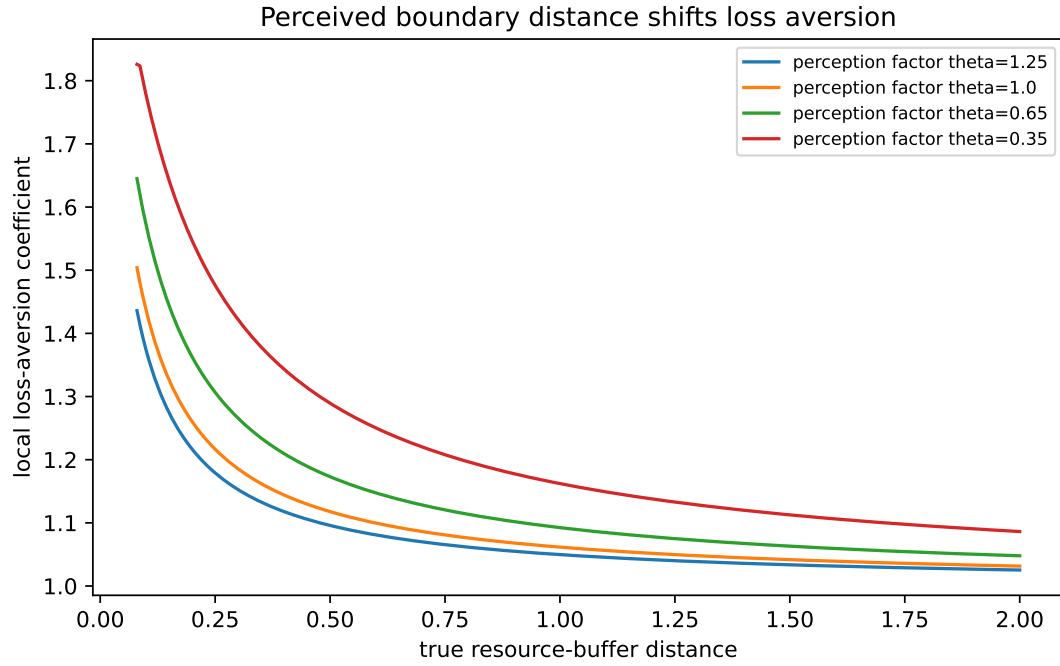


Figure 2: Perceived boundary distance shifts loss aversion. Lower perception factor θ makes the same objective resource buffer appear closer to the constraint boundary, increasing λ .

6 Reference Dependence as Maintained Baseline under Finite Updating

A finite agent does not compute utility over all possible terminal states. It maintains a compressed baseline r_t and evaluates local prediction errors:

$$\text{PE}_t = x_t - r_t. \quad (12)$$

We model baseline updating as

$$r_{t+1} = r_t + \alpha_t(x_t - r_t) - \kappa_t P_t, \quad (13)$$

where α_t is adaptation rate, P_t is pruning or baseline-reset pressure, and κ_t is pruning strength. The finite-capacity hypothesis is that

$$\alpha_t = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min})\sigma(a_0 + a_1 \log(\hat{d}_t + \epsilon) + a_2 \log(C_t + \epsilon)), \quad (14)$$

where $\sigma(\cdot)$ is the logistic function. When resource buffer and decision capacity are high, reference points update quickly. When they are low, reference points rigidify.

Prediction 3 (Reference rigidity under depletion) *Reference-point adaptation should slow under stress, sleep deprivation, cognitive load, financial scarcity, administrative burden, and time pressure.*

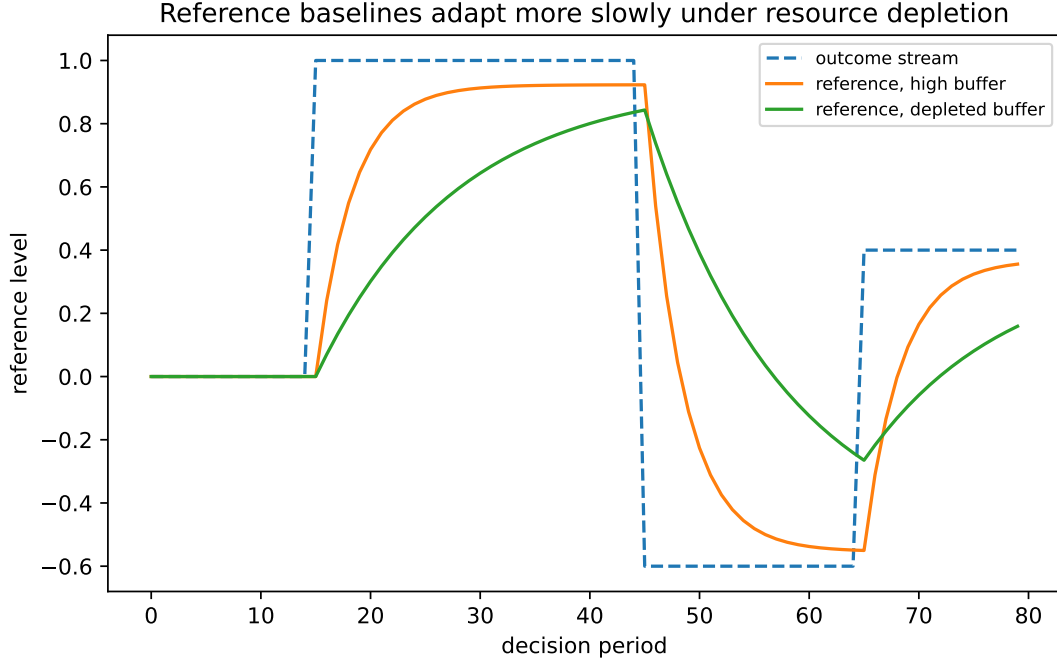


Figure 3: Reference baselines adapt more slowly under depletion. Two agents face the same outcome stream. The high-buffer agent updates quickly; the depleted agent maintains a stale baseline longer.

7 Probability Weighting as Finite Precision Allocation

Finite agents cannot estimate all probabilities with equal precision. Let $\tau(p, E)$ denote precision allocated to probability p for event class E . The event class matters: a one percent probability of a neutral lottery and a one percent probability of job loss, eviction, illness, or predation do not carry the same boundary relevance.

Suppose estimation error has expected cost $C_{\text{err}}(p, E, \tau)$ and precision is constrained:

$$\int_0^1 \tau(p, E) dp \leq \tau_{\text{total}}(C_t). \quad (15)$$

The allocation problem is

$$\min_{\tau(p, E)} \int_0^1 C_{\text{err}}(p, E, \tau(p, E)) dp \quad \text{s.t.} \quad \int_0^1 \tau(p, E) dp \leq \tau_{\text{total}}. \quad (16)$$

Under standard smoothness assumptions, the Euler condition implies that precision is allocated toward regions where marginal error cost is high. A useful reduced-form expression is

$$\tau^*(p, E) \propto \sqrt{\frac{\partial^2 C_{\text{err}}(p, E)}{\partial p^2}}. \quad (17)$$

The decision weight can then be written as

$$w(p, E) = \frac{\int_0^p \tau^*(u, E) du}{\int_0^1 \tau^*(u, E) du}. \quad (18)$$

In reduced form, this generates event-class-dependent inverse-S weighting:

$$w(p; \gamma_{tE}) = \frac{p^{\gamma_{tE}}}{p^{\gamma_{tE}} + (1 - p)^{\gamma_{tE}}}, \quad (19)$$

where lower γ_{tE} represents stronger distortion. We model

$$\gamma_{tE} = \gamma_{\min} + (\gamma_{\max} - \gamma_{\min})\sigma(g_0 + g_1 \log C_t - g_2 B_E - g_3/\hat{d}_t). \quad (20)$$

Here B_E is boundary relevance of event class E . Low capacity, high boundary relevance, and low perceived distance all reduce γ and amplify tail weighting. This finite-precision interpretation is closely related to efficient-coding accounts of probability weighting, which predict that probability distortions should shift with the distribution of probabilities and prior beliefs encountered by the decision-maker (Frydman and Jin, 2023).

This mechanism explains why insurance and gambling can coexist. In both cases the agent overrepresents tail events. Insurance responds to rare catastrophic losses. Gambling can respond to rare escape opportunities when the current boundary state is poor. The common cause is not irrationality per se; it is finite precision applied to events that can change the boundary regime.

Prediction 4 (Event-class-dependent weighting) *Equally rare events should receive different decision weights when their boundary relevance differs. Health, housing, job loss, safety, and existential financial risks should generate stronger tail overweighting than abstract neutral lotteries.*

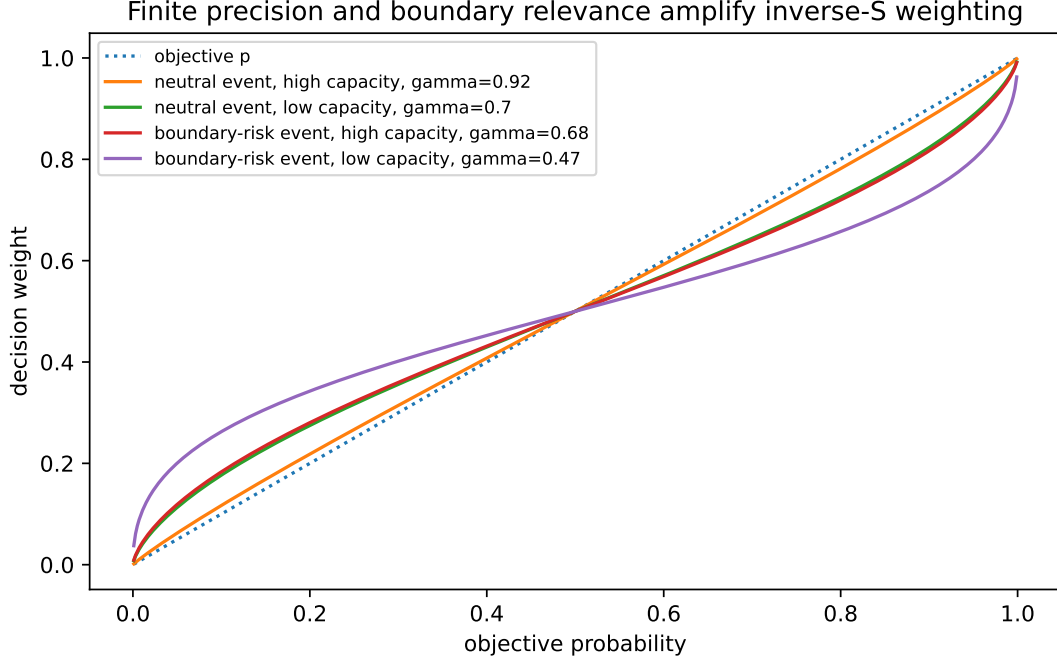


Figure 4: Finite precision and boundary relevance amplify inverse-S probability weighting. Low capacity and high boundary-risk event classes lower γ , increasing tail overrepresentation.

8 A Unified State-Dependent Prospect Model

Let a risky option yield outcomes x_i with probabilities p_i . Standard Prospect Theory evaluates

$$V = \sum_i w(p_i) v(x_i - r). \quad (21)$$

The finite-capacity version is

$$V_t = \sum_i w(p_i, E_i; C_t, \hat{d}_t, B_{E_i}) v(x_i - r_t; \lambda_t, a_t, b_t). \quad (22)$$

The key parameters are

$$\lambda_t = 1 + A(\hat{d}_t + \epsilon)^{-\nu}, \quad (23)$$

$$\alpha_t = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \sigma(a_0 + a_1 \log(\hat{d}_t + \epsilon) + a_2 \log(C_t + \epsilon)), \quad (24)$$

$$\gamma_{tE} = \gamma_{\min} + (\gamma_{\max} - \gamma_{\min}) \sigma(g_0 + g_1 \log C_t - g_2 B_E - g_3 / \hat{d}_t). \quad (25)$$

Expected utility is recovered as a limiting case when $F_t^{\text{eff}} \gg F_{\text{crit}}$, C_t is abundant, and boundary risk is low:

$$\lambda_t \rightarrow 1, \quad \gamma_{tE} \rightarrow 1, \quad (26)$$

while reference baselines adapt rapidly enough that local deviations approximate expected terminal values.

9 Numerical Simulations and Synthetic Identification

This section reports deterministic numerical simulations. They are not fits to real human data. Their purpose is to make the model operational, show qualitative implications, and provide an identification template for future empirical work. All code and CSV outputs are released with this manuscript.

9.1 Synthetic state-dependent parameter recovery

We simulate 1,200 agents with monetary, sleep/time, and attention resources. Effective buffer is generated with a low-substitutability CES aggregator and a random perception factor θ_i . True parameters are generated from the finite-capacity equations above and then measured with noise. We compare a static Prospect Theory model, which uses constant parameters, to a state-dependent finite-capacity model that regresses observed parameters on $\log(\hat{d}_i)$ and resource measures.

Table 2: Synthetic recovery comparison. The finite-capacity model is compared to a static-parameter model on simulated subject-level parameter estimates. Lower RMSE and AIC indicate better recovery.

Parameter	Model	RMSE	AIC	Δ AIC
lambda	static Prospect parameter	0.249	-3338.6	0.0
lambda	finite-capacity state-dependent	0.190	-3970.8	-632.2
gamma	static Prospect parameter	0.119	-5103.4	0.0
gamma	finite-capacity state-dependent	0.041	-7679.8	-2576.5
alpha	static Prospect parameter	0.058	-6825.8	0.0
alpha	finite-capacity state-dependent	0.027	-8648.4	-1822.6

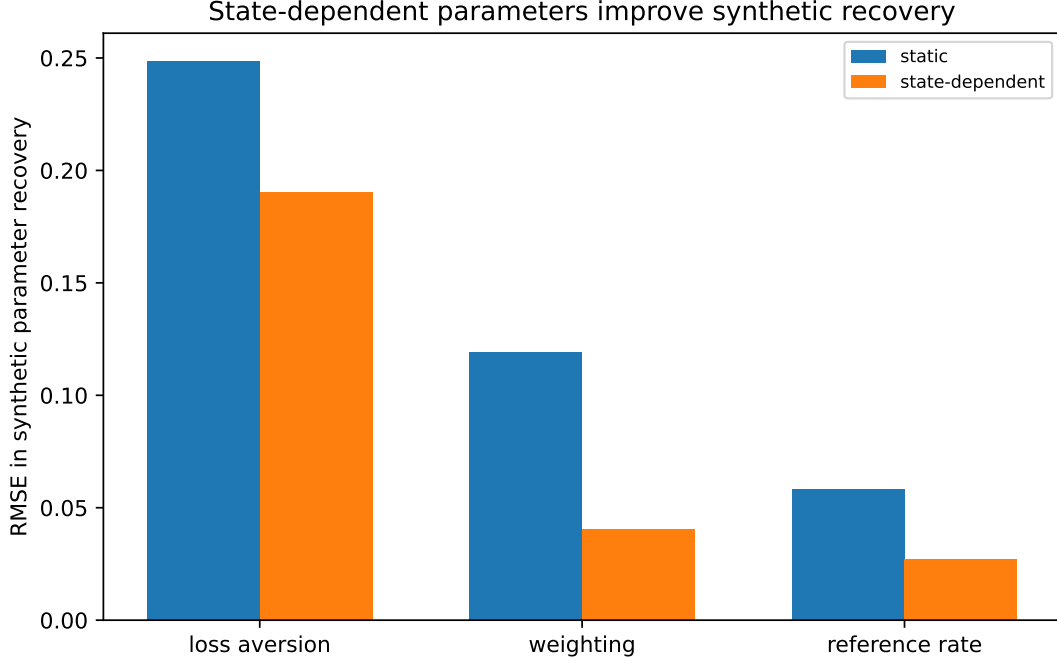


Figure 5: Static versus state-dependent parameter recovery in synthetic data. The finite-capacity specification improves recovery of loss aversion, probability weighting, and reference adaptation in the simulated environment.

9.2 Nudge bandwidth simulation

Choice architecture interventions are often described as low-cost. For a resource-depleted agent, however, a nudge can impose additional distinctions: compare alternatives, verify eligibility, process defaults, update plans, and overcome switching costs. Define nudge bandwidth requirement

$$C_t^{\text{nudge}} = I(\text{attributes}; \text{action}) + C_{\text{switch}} + C_{\text{verification}} + C_{\text{admin}}. \quad (27)$$

A nudge succeeds when available decision capacity exceeds the requirement:

$$C_t^{\text{available}} > C_t^{\text{nudge}}. \quad (28)$$

If $C_t^{\text{available}} < C_t^{\text{nudge}}$, the intervention may fail or backfire through overload.

Figure 6 is the policy-facing summary of the model. It shows that a choice-architecture intervention can be welfare-improving for agents with sufficient resource buffer, neutral for agents near the capacity boundary, and harmful for agents whose effective buffer is already depleted. In the finite-capacity view, a nudge is not a frictionless intervention. It consumes decision bandwidth. When the required nudge bandwidth exceeds the agent’s available capacity, the intervention can fail or backfire.

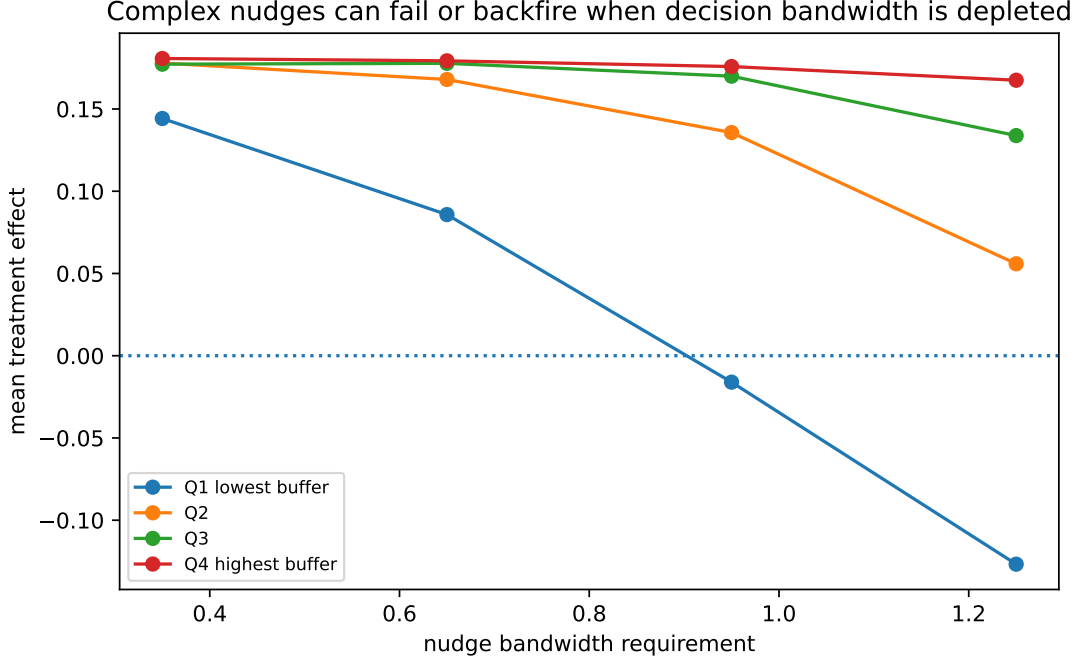


Figure 6: Nudge bandwidth and policy backfire. The simulation shows that a nudge improves outcomes only when the agent has enough available decision capacity to process the intervention. In low-buffer regimes, the same nudge can impose additional cognitive and administrative load, reducing net value. This illustrates the Buffer-First Principle: restore resource and decision bandwidth before imposing complex choice architecture.

10 Empirical Identification Strategy

The model is empirically useful only if F_t^{eff} , C_t , and $\hat{d}_t$ can be proxied. Candidate measures are summarized in Table 3. Existing scarcity and poverty research already suggests that resource constraints affect cognition and decision quality (Mani et al., 2013; Haushofer and Fehr, 2014; Mullainathan and Shafir, 2013). Stress and affective regulation also affect risky choice (Porcelli and Delgado, 2009; Sokol-Hessner et al., 2009). Recent experiments on cognitive load and acute stress suggest that capacity constraints do not uniformly alter all preferences, but they can impair risky or computationally demanding decisions, especially when working memory, time pressure, or cortisol responses are involved (Dominiak and Duersch, 2024; Doroc, Yadav and Murawski, 2025). The finite-capacity model turns these findings into parameter-level predictions.

A reduced-form estimation strategy is

$$\lambda_i = \lambda_0 + \lambda_1 \hat{d}_i^{-1} + \lambda_2 \text{NonFungibleDepletion}_i + \lambda_3 \text{DownsideProtection}_i + u_i, \quad (29)$$

$$\alpha_i = \alpha_0 + \alpha_1 \log C_i + \alpha_2 \log(\hat{d}_i) + u_i, \quad (30)$$

$$\gamma_{iE} = \gamma_0 + \gamma_1 \log C_i - \gamma_2 B_E - \gamma_3 \hat{d}_i^{-1} + u_{iE}. \quad (31)$$

A horse race against static Prospect Theory tests whether state variables improve out-of-sample prediction of choices or estimated parameters.

Heart-rate variability (HRV) is a particularly useful candidate proxy because it tracks autonomic flexibility and stress-regulation capacity in real time. In the FDS-E1 model, lower HRV should correspond to a smaller effective maintenance buffer or a smaller perceived distance from the boundary-risk threshold, predicting higher loss aversion, slower reference adaptation, stronger probability weighting distortion, and reduced nudge bandwidth. HRV is not the resource buffer itself; it is an autonomic window onto the maintenance state. As a non-invasive, continuous, and physiologically grounded measure, HRV can bridge between laboratory decision tasks and field studies of scarcity, stress, and overload.

Table 3: Behavioral parameters and finite-system interpretations.

Parameter	FDS interpretation	Observable proxies
λ	Boundary-risk slope asymmetry near resource threshold	Financial runway, downside exposure, cortisol, HRV, sleep deprivation, perceived scarcity
α	Reference-baseline update rate	Reversal learning, prediction-error integration, baseline updating after shocks
γ	Probability precision curvature	Probability-weighting estimates, numeracy, cognitive load, event class, tail-event salience
F^{eff}	Effective resource buffer	Income runway, liquid assets, sleep, health, time, social support, institutional insurance
C	Available decision capacity	Working memory, time pressure, attention, administrative burden, information complexity
B_E	Boundary relevance of event class	Health, job loss, eviction, violence, food insecurity, catastrophic financial loss, social exclusion
θ	Perceived distance mapping	Trauma history, uncertainty, belief state, social comparison, perceived institutional reliability

11 Policy Implications: The Buffer-First Principle

The model refines nudge theory (Thaler and Sunstein, 2008). A nudge is not free from the agent’s perspective. It can require attention, comprehension, verification, switching, and memory. When an agent is depleted, an intervention that is simple for a high-capacity agent may exceed the low-capacity agent’s available bandwidth.

Policy Corollary 1 (Buffer-First Principle) *Before imposing complex choice architecture on resource-depleted agents, policy should restore the resource and bandwidth conditions required for the agent to use the choice architecture.*

This principle favors cash transfers, debt relief, administrative simplification, default enrollment, automatic eligibility, shorter forms, stable scheduling, food security, housing stability, and reduction of time taxes before complex information campaigns. Recent reviews of choice architecture interventions emphasize substantial heterogeneity and the possibility of ineffective or counterproductive nudges; the nudge-bandwidth model provides one mechanism for such heterogeneity (Szasz et al., 2025). It does not reject nudges. It states that nudges require bandwidth, and bandwidth is state-dependent.

Prediction 5 (Buffer-First Policy Corollary) *If a target population is operating near its boundary-risk threshold, interventions that add informational, administrative, switching, or verification demands should be preceded by buffer-restoring policies such as cash transfers, debt relief, default simplification, administrative burden reduction, food security, sleep protection, or housing stability. Otherwise, the intervention may consume the very decision capacity it requires.*

12 Relation to Existing Theories

Prospect Theory. The model does not replace Prospect Theory. It makes Prospect Theory parameters state-dependent. Because loss aversion and probability weighting are conceptually distinct (Blavatsky, 2024), the model treats λ and γ as separate state-dependent parameters rather than reducing one to the other.

Expected utility. Expected utility becomes a limiting case of high resource buffer, high capacity, low boundary risk, and stable baselines.

Bounded rationality. FDS gives a maintenance interpretation of bounded rationality: agents economize not only on computation but also on boundary risk and update cost.

Active inference. Reference dependence is naturally related to prediction-error baselines and generative-model updating (Friston, 2010).

Ecological rationality. What looks like irrationality in modern financial markets may be an adaptation to environments in which resource buffers were small and rare boundary-changing events mattered.

13 Limitations and Falsification

This paper has several limitations. First, it does not derive $\lambda \approx 2.25$ exactly. Second, economic losses are not literal thermodynamic losses in any simple sense; they are symbolic claims on future resource access. Third, humans are not assumed to be globally optimal in modern environments.

Fourth, culture, institutions, learning, and market structure are not reduced to biology. Fifth, the model requires empirical proxies for F^{eff} , C , B_E , and θ .

The finite-capacity bridge is strongly weakened if the following reverse stress test fails:

If severe resource depletion changes neither loss aversion, nor reference rigidity, nor probability weighting, nor decision noise after adequate measurement power and controls, then the finite-capacity bridge account is strongly undermined.

More specific demotion conditions are: (i) λ invariant across resource stress, physiological depletion, downside protection, and financial precarity; (ii) reference-point adaptation invariant across prediction-error integration, memory load, and cognitive load; (iii) probability-weighting curvature invariant across event class, cognitive load, and perceived consequence severity; (iv) nudge effectiveness independent of scarcity, fatigue, and administrative burden.

14 Conclusion

Prospect Theory showed that human beings are not ideal expected-utility maximizers. The finite-system view explains why they should not be. A physically realizable decision-maker has finite buffers, finite memory, finite attention, finite precision, finite update capacity, and finite tolerance for boundary risk. Under these constraints, loss aversion, reference dependence, and probability weighting are not arbitrary defects. They are candidate signatures of finite agents managing resource buffers, prediction baselines, and limited distinguishability under uncertainty.

The main contribution of this paper is a state-dependent bridge:

$$(\lambda_t, \alpha_t, \gamma_t) = f(F_t^{\text{eff}}, C_t, B_t, \theta_t). \quad (32)$$

This formulation suggests that poverty, stress, overload, and administrative burden are not merely psychological states. They are also finite-capacity maintenance states that change the effective parameters of decision-making. A finite agent cannot be nudged, priced, informed, or incentivized as if its decision capacity were infinite. Traditional economic models that ignore system finiteness may be elegant, but they risk describing behavior in a vacuum.

Code and Data Availability

The deterministic Python code, figures, and CSV files used in this manuscript are included in the accompanying replication package. Running `code/generate_results.py` regenerates all figures, CSV outputs, and the model-comparison table.

A Notation

Symbol	Meaning
F_t^i	Resource buffer in dimension i
F_t^{eff}	Effective resource buffer after substitutability aggregation
F_{crit}	Boundary-risk or constraint threshold
d_t	Objective distance from threshold
$\hat{d}_t$	Perceived distance from threshold
θ_t	Perception or belief mapping from true to perceived distance
C_t	Available decision capacity
N_A	Number of operationally distinguishable decision states
λ_t	Loss-aversion coefficient
α_t	Reference-baseline adaptation rate
γ_{tE}	Probability-weighting curvature for event class E
B_E	Boundary relevance of event class E
C_t^{nudge}	Decision bandwidth required by a nudge

B Simulation Parameters

The simulations use fixed parameter values documented in `code/generate_results.py`. The boundary-risk potential is $R(d) = \kappa(d + \epsilon)^{-\nu}$ with default $\epsilon = 0.08$ and $\nu = 1.15$. CES effective buffers use weights on monetary, sleep/time, and attention resources. Low substitutability is represented by negative CES curvature. Probability weighting uses $w(p; \gamma) = p^\gamma / [p^\gamma + (1 - p)^\gamma]$. The nudge-bandwidth simulation generates a synthetic population with resource-dependent available capacity and nudge-complexity requirements.

C Reproducibility Checklist

1. The script is deterministic and uses a fixed random seed for synthetic population simulations.
2. All figures are generated from the source code.
3. All CSV files are regenerated from the source code.
4. The model-comparison table is generated from the synthetic parameter-recovery exercise.
5. No proprietary or human-subject data are used in this draft.

AI Assistance Disclosure

AI-assisted tools were used to improve readability, wording, and LaTeX formatting. The author reviewed and edited all content and remains responsible for all claims, references, simulations, and

conclusions. No AI system is listed as an author.

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Observer as a Finite Distinction Register: Measurement Capacity, Dynamic Bottlenecks, and Budget-Crossing Signatures

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A physical observer is often treated as an idealized point of access to facts. FDS-O1 replaces this abstraction with an operational definition: an observer is a finite distinction-register, namely a physical system that can register, preserve, update, order, and communicate distinctions only through finite records, finite channels, finite update rates, finite buffers, and finite thermodynamic budgets. Building on finite-observer distinguishability budgets, this paper formulates physical measurement as stable record formation under finite capacity. A measurement event is not merely an interaction; it is the stabilization of an accessible record whose distinction class can be retained and used by a finite observer within a specified time window. We define measurement capacity through dynamically coupled bottlenecks: sensor resolution, readout channel, internal memory, record stability, buffering, externalized access, compression, and irreversible update throughput. We compare this accessible capacity with task-relevant rate-distortion demand. If demand exceeds capacity, full-fidelity measurement is operationally impossible: the observer must coarse-grain, merge states, increase latency, buffer, externalize records, reset memory, dissipate housekeeping heat through irreversible updates, relax the task, or fail. This version extends the minimal O1 model by adding dynamic resource allocation, transient versus sustained crossing, reversible-versus-irreversible heat accounting, a two-dimensional sensor-array simulation, an operational interface with decoherence, and a comparison with active inference. The paper does not derive quantum measurement, collapse, the Born rule, decoherence, the Bekenstein bound, holography, or the Einstein equations. Its contribution is narrower: it converts finite-observer distinguishability budgets into a concrete measurement-diagnostics framework with explicit budget-crossing signatures.

Scope and Boundary of the Theory. This paper does not claim that consciousness is required for observation. It does not claim to solve the quantum measurement problem, derive the Born rule, replace decoherence theory, modify unitary quantum mechanics, or derive gravitational entropy bounds. It treats an observer as a finite physical record-bearing system. Its contribution is to define measurement capacity, dynamic bottleneck coupling, buffering, stable record formation, and budget-crossing signatures for such systems.

Claim-status summary

Table I summarizes the central FDS-O1 claims, their epistemic status, and the conditions under which they should be weakened or rejected.

Keywords: finite observer; measurement theory; distinction register; detector capacity; dynamic bottlenecks; buffering; finite records; rate-distortion; Landauer principle; reversible computation; information thermodynamics; budget crossing; sensor array; active inference; decoherence interface; active finite distinction systems.

INTRODUCTION

The operational observer problem

Physics routinely uses the word “observer” while leaving its physical implementation implicit. In some contexts the observer is a human, in others a detector, an apparatus, a memory register, a computational agent, a horizon-bounded exterior system, or an environment carrying records. The common operational feature is not consciousness. It is record-bearing distinction. Something happens that leaves a reproducible trace; that trace can be preserved, checked, updated, compared, communicated, and used.

FDS-T1 introduced finite distinguishability budgets for physical observers. It defined an observer-relative finite projection $\pi_{\mathcal{O}}$ and bit capacity $C_{\mathcal{O}} = \log_2 |\text{Im}(\pi_{\mathcal{O}})|$, then treated accessible capacity as a bottleneck over finite memory, boundary access, readout channel, causal reach, external records, and irreversible update throughput [1]. The present paper turns that physical bridge into an operational measurement paper. It asks: when is a detector, instrument, apparatus, organism, robot, or laboratory system a finite distinction-register? What makes an interaction into a measurement? How do finite sensor resolution, channel capacity, buffering, record stability, memory reuse, and irreversible updates constrain what can be measured?

TABLE I. Central FDS-O1 claims, epistemic status, and demotion or failure conditions.

Claim	Status	What would weaken or falsify it
Observer as finite distinction-register	Operational bridge	A complete physical account of registered observations requiring no finite carrier, boundary, stability condition, readout, or update capacity
Measurement as stable record formation	Operational definition	Measurement outcomes used operationally without any stable record, reproducible trace, or accessible registration
Measurement capacity is dynamically bottlenecked	Conditional theorem	Full-fidelity measurement bypasses all sensor, channel, memory, record-stability, buffering, externalization, and update constraints
Budget crossing produces observable exits	Testable prediction	No change in error, latency, coarse-graining, state merging, externalization, reset, heat, or failure when demand crosses accessible measurement capacity
Buffers separate transient from sustained crossing	Engineering bridge	Finite buffers neither delay transient overload nor sharpen sustained overflow in controlled acquisition systems
Irreversible record reuse carries housekeeping cost	Physical bridge	Repeated logically irreversible record reuse below generalized Landauer accounting under stated thermodynamic conditions
Active inference depends on finite record formation	Interface claim	Inference and control are empirically invariant to severe record compression, delay, merging, and unavailable observations under otherwise matched tasks

Main thesis

The main thesis is:

$$\text{observer} = \text{finite distinction-register}, \quad (1)$$

not a disembodied point of knowledge. A measurement is the stabilization of at least one record distinction that remains accessible to such a finite register over a specified retention and verification window.

Let Ω be a physical possibility space for a source, region, or task. A finite observer implements a projection

$$\pi_{\mathcal{O}} : \Omega \rightarrow \mathcal{Z}_{\mathcal{O}}, \quad (2)$$

where $\mathcal{Z}_{\mathcal{O}}$ is the finite set of record classes that can be reliably distinguished by the observer. The measurement problem addressed here is not the quantum collapse problem. It is the finite-capacity problem:

$$R_{\min}^{(\tau)}(\varepsilon; \Psi) > C_{\text{meas}}(\mathcal{O}, \mathcal{D}, t, \tau). \quad (3)$$

When the task-relevant distinction demand exceeds the accessible measurement capacity, no protocol can produce full-fidelity stable records without changing the boundary, increasing time, using buffers, externalizing information, paying additional reset cost, or relaxing the task.

New contributions in v0.2

This version adds six components beyond the initial O1 draft.

1. **Dynamic bottleneck coupling:** capacity terms are no longer treated as independent constants; they arise from resource allocation and trade-offs.
2. **Buffering:** transient crossing can be absorbed, whereas sustained crossing causes buffer overflow and sharper signatures.
3. **Housekeeping heat separation:** reversible sensing and copying are distinguished from irreversible reset, overwrite, compression, and garbage collection.
4. **Two-dimensional sensor-array simulation:** a finite-capacity camera-like system shows false merging and ghosting under update limits.
5. **Decoherence interface:** O1 supplies an operational record criterion for when environmental information becomes accessible as a stable measurement record.
6. **Active inference interface:** O1 specifies the finite-record precondition under which an agent can even receive the observations used in variational free-energy minimization.

What is not claimed

This paper does not derive quantum mechanics, decoherence, the Born rule, the Bekenstein bound, hologra-

phy, general relativity, or the Standard Model. It is compatible with standard unitary dynamics, standard decoherence theory, and ordinary detector physics. It does not claim that every physical interaction is a measurement. The additional condition is record stabilization under finite accessibility. The paper’s purpose is narrower: to make observerhood and measurement capacity explicit finite-physical variables.

RELATED WORK

Operational and relational observers

Operational approaches emphasize that a measurement is an operation with a record. Wheeler’s “it from bit” placed information and physical registration at the center of physical meaning [2]. Relational quantum mechanics treats states as relative to interacting systems rather than as observer-independent catalogues of facts [3]. QBist and operational approaches similarly stress that quantum-state assignments depend on the agent or experimental arrangement [4]. FDS-O1 is compatible with this operational tradition, but it is more restrictive: it asks what physical resources are required for the agent, detector, or apparatus to function as a finite distinction-register.

Decoherence and stable records

Decoherence theory explains the suppression of interference in open quantum systems and the emergence of stable pointer states under environment-induced superselection [5, 6]. Quantum Darwinism studies redundant environmental encoding of information that makes records objectively available to many observers [7]. FDS-O1 does not replace decoherence theory. It uses the language of stable record formation: a result is operationally classical when it can be registered, preserved, communicated, and cross-validated by finite observers.

Information theory, detector physics, and Landauer accounting

Shannon capacity and rate-distortion theory provide the natural language for comparing signal complexity to finite representation capacity [8–11]. Real detectors have finite resolution, dynamic range, bandwidth, dead time, noise temperature, memory, calibration stability, and readout constraints. FDS-O1 does not compete with instrument-specific physics. It supplies a cross-domain capacity ledger.

Landauer’s principle gives a lower bound on heat dissipation for logically irreversible erasure under speci-

fied thermodynamic conditions [12]. Bennett emphasized that reversible computation can avoid local erasure cost while still requiring care about garbage and memory reuse [13]. Experimental tests have verified Landauer-style bounds in controlled settings [14–16]. FDS-O1 uses Landauer conservatively: the bridge is applied to irreversible record reset, overwrite, many-to-one compression, and garbage collection, not to every physical interaction or reversible sensing step.

Active inference and embodied measurement

Active inference treats perception and action as inference under a generative model, often expressed as minimization of variational free energy [22, 23]. Recent active-inference reviews frame action and perception as a unified process of prediction, inference, and control [24]. FDS-O1 does not replace active inference. It addresses a prior physical layer: the observation y_t used by an agent is itself a finite record. If record formation is capacity-limited, delayed, compressed, or merged, then inference operates on $\tilde{y}_t = \pi_{\mathcal{O}}(y_t)$ rather than on an ideal observation. This connects measurement capacity to robotics, embodied AI, biological sensing, and instrumentation.

FINITE-REGISTER MEASUREMENT MODEL

Definition 1 (Physical distinction). *A physical distinction is a difference between alternatives instantiated by a record, coupling, detector state, boundary condition, memory state, or other physical carrier. A formal partition without a physical carrier is not yet a measurement distinction.*

Definition 2 (Finite distinction projection). *Let Ω be a physical possibility space. A finite observer $\mathcal{O}$ implements a projection*

$$\pi_{\mathcal{O}} : \Omega \rightarrow \mathcal{Z}_{\mathcal{O}}, \quad (4)$$

where $\mathcal{Z}_{\mathcal{O}}$ is the set of accessible record classes. The observer’s instantaneous distinguishability budget is

$$N_{\mathcal{O}} = |\text{Im}(\pi_{\mathcal{O}})|, \quad C_{\mathcal{O}} = \log_2 N_{\mathcal{O}}. \quad (5)$$

Definition 3 (Finite distinction-register). *A finite distinction-register is a physical system that can register, preserve, update, order, and communicate distinctions using finite resources. It must have a record carrier, a boundary or interface, a stability condition, a readout channel, and a finite update capacity.*

The term observer is used here only in this operational sense. A detector array, memory device, biological sensory system, robotic sensor, laboratory apparatus, or composite experimental system may count as an observer when it functions as a finite distinction-register.

Definition 4 (Measurement event). *A measurement event for task family Ψ is a process that maps a physical possibility $\omega \in \Omega$ into a record class $z \in \mathcal{Z}_{\mathcal{O}}$ such that the record remains retrievable, orderable, and usable by $\mathcal{O}$ over a specified retention and verification window $[t, t + \tau]$ with error probability at most δ and distortion at most ε .*

Let M_t be the memory state carrying a record at time t . A record class z is stable over τ if

$$\mathbb{P}[\hat{z}(t + \tau) = z \mid M_t = z] \geq 1 - \delta. \quad (6)$$

The effective record-stability capacity $C_{\text{rec}}(t, \tau)$ counts only distinctions satisfying this stability requirement.

DYNAMIC MEASUREMENT CAPACITY

From static bottleneck to coupled resource allocation

The simplest O1 capacity ledger writes

$$C_{\text{meas}} = \min\{C_{\text{sens}}, C_{\text{chan}}, C_{\text{mem}}, C_{\text{rec}}, C_{\text{ext}}^{\text{eff}}, I_{\text{update}}^{\text{max}}\}. \quad (7)$$

This is useful as an accounting identity, but real systems do not hold these terms fixed. Increasing sensor sampling may increase inflow into memory. Compression can relieve a channel bottleneck while reducing record fidelity and consuming update capacity. Externalization can increase effective storage while increasing latency and verification overhead.

We therefore introduce a resource-allocation vector

$$\mathbf{r}(t) = (r_{\text{sens}}, r_{\text{chan}}, r_{\text{mem}}, r_{\text{rec}}, r_{\text{comp}}, r_{\text{ext}}), \quad (8)$$

subject to

$$\sum_i r_i(t) \leq R_{\text{tot}}(t), \quad r_i(t) \geq 0. \quad (9)$$

The capacity terms are functions of allocation, load, temperature, noise, and retention window:

$$C_i(t) = C_i(r_i(t), \Psi(t), T(t), B(t), \tau). \quad (10)$$

The coupled accessible capacity is still

$$C_{\text{meas}}(t) = \min_i C_i(t), \quad (11)$$

but the active bottleneck can move because the capacities co-evolve.

Memory fill and compression feedback

Let $M(t)$ be memory or buffer occupancy. A generic acquisition balance is

$$\dot{M}(t) = I_{\text{in}}(t) - I_{\text{compress}}(t) - I_{\text{ext}}(t) - I_{\text{erase}}(t). \quad (12)$$

If $M(t)$ approaches M_{max} , the observer must either increase compression, externalize, erase, lower sampling, increase latency, or fail. Compression creates a trade-off:

$$C_{\text{chan}}^{\text{eff}}(t) = C_{\text{chan}}(t) + G_{\text{comp}}(t), \quad (13)$$

$$C_{\text{rec}}^{\text{eff}}(t) = C_{\text{rec}}(t) - L_{\text{comp}}(t), \quad (14)$$

$$I_{\text{update}}^{\text{max, eff}}(t) = I_{\text{update}}^{\text{max}}(t) - U_{\text{comp}}(t). \quad (15)$$

The same intervention that relieves one bottleneck can activate another.

Measurement demand and capacity deficit

For a task family Ψ , distortion tolerance ε , and measurement window τ , let

$$R_{\text{min}}^{(\tau)}(\varepsilon; \Psi) \quad (16)$$

be the minimal task-relevant bits required to meet the target error or distortion. The FDS-O1 measurement deficit is

$$\Delta_{\text{O1}}(t) = R_{\text{min}}^{(\tau)}(\varepsilon; \Psi_t) - C_{\text{meas}}(t). \quad (17)$$

A positive deficit is not ordinary ignorance. It is an operational impossibility under the given boundary, detector, channel, memory, buffer, update, and stability constraints.

Theorem 1 (Finite measurement budget-exit theorem). *Let $\mathcal{O}$ be a finite distinction-register coupled to measurement apparatus $\mathcal{D}$ for task family Ψ over $[t, t + \tau]$. If*

$$R_{\text{min}}^{(\tau)}(\varepsilon; \Psi) > C_{\text{meas}}(\mathcal{O}, \mathcal{D}, t, \tau), \quad (18)$$

then full-fidelity stable measurement at distortion ε is impossible unless at least one exit occurs: coarse-graining, state merging, increased latency or integration time, buffering, externalization, additional irreversible reset/garbage collection, task relaxation, boundary enlargement, or failure.

Proof. $R_{\text{min}}^{(\tau)}(\varepsilon; \Psi)$ is the minimum task-relevant information required to meet the measurement target. C_{meas} is the maximum number of task-relevant distinctions the finite observer-apparatus system can register, stabilize, update, and use over the same window. If demand exceeds capacity, a full-fidelity record would require distinctions not carried by any admissible register state or update sequence. The system must therefore reduce demand, enlarge or alter capacity, store backlog in a finite buffer, externalize, pay additional irreversible updating cost, or fail. These alternatives exhaust the ways to remove the strict inequality under finite resources. $\square$

The empirical content of the theorem is not the logical implication itself, but the predicted pattern of exits: bottleneck switching, latency discontinuities, buffer overflow,

false merging, externalization bursts, reset cascades, and excess irreversible housekeeping costs should co-occur when independently estimated task demand crosses independently estimated accessible measurement capacity.

BUFFERING: TRANSIENT VS SUSTAINED CROSSING

Real instruments often survive short overloads by using buffers. Let $B_{\text{buf}}(t)$ be buffer occupancy and $B_{\text{buf}}^{\text{max}}$ be buffer capacity. Define

$$\Delta_+(t) = \max\{\Delta_{\text{O1}}(t), 0\}. \quad (19)$$

A minimal buffer balance is

$$B_{\text{buf}}(t) = \Delta_+(t) - D_{\text{drain}}(t), \quad 0 \leq B_{\text{buf}}(t) \leq B_{\text{buf}}^{\text{max}}. \quad (20)$$

Transient crossing can be absorbed if

$$\int_{t_0}^{t_1} \Delta_+(s) ds < B_{\text{buf}}^{\text{max}} - B_{\text{buf}}(t_0) + \int_{t_0}^{t_1} D_{\text{drain}}(s) ds. \quad (21)$$

Sustained crossing occurs when the inequality is violated. Then the buffer overflows, and the exit signatures sharpen: state drops, false merging, hard latency jumps, reset cascades, externalization bursts, or failure. Buffering therefore does not eliminate budget crossing; it temporally reshapes it.

HOUSEKEEPING HEAT AND REVERSIBLE CONTROLS

The thermodynamic signature of FDS-O1 must be stated carefully. Total power contains many contributions:

$$P_{\text{total}} = P_{\text{sens}} + P_{\text{chan}} + P_{\text{control}} + P_{\text{storage}} + P_{\text{irr}}. \quad (22)$$

FDS-O1 does not claim that all of these are Landauer costs. The relevant term is irreversible housekeeping:

$$P_{\text{hk}} \geq k_{\text{B}} T \ln 2 \dot{I}_{\text{erase}}, \quad (23)$$

where $\dot{I}_{\text{erase}}$ counts logically irreversible reset, overwrite, many-to-one compression, or garbage collection under the relevant thermodynamic assumptions.

Recent reviews of the Landauer principle emphasize that practical erasure generally dissipates more than the ideal bound, and that finite-time, finite-size-bath, nonequilibrium, non-Markovian, and quantum regimes require generalized accounting beyond the textbook single-bit limit [25].

Bennett-style reversible computation shows that sensing, copying, and computation can in principle be arranged to avoid local erasure, provided sufficient reversible memory and garbage management are available

[13]. O1 therefore predicts not simply “more measurement implies more heat,” but a sharper contrast. Fixed memory with repeated reuse should produce a higher immediate housekeeping-heat floor. Expanded memory with delayed erasure should lower immediate heat, but only by paying a storage, boundary, or later garbage-collection cost. This separation makes the heat signature experimentally interpretable.

Recent quantum many-body experiments have also probed generalized Landauer accounting in system-environment partitions, relating energy flow, entropy change, mutual information, and relative entropy in out-of-equilibrium quantum fields [26].

NUMERICAL MODELS AND SIMULATIONS

The simulations are deterministic synthetic demonstrations. They are not fits to physical detector data, quantum experiments, or human observations. They provide a reproducible diagnostic template for the O1 model. All figures and CSV outputs are generated by `code/generate_results.py`.

Dynamic bottleneck coupling

The first simulation increases scene complexity and lets the observer respond by increasing sampling and compression. Compression improves effective channel capacity but reduces record stability and update throughput. Memory fill accumulates when inflow exceeds processing and drain. Thus the active bottleneck changes dynamically rather than being a fixed independent minimum.

Buffering

The second simulation contrasts a short transient pulse with a sustained overload. A finite buffer absorbs the transient crossing, producing latency without failure. The sustained crossing fills the buffer and produces a sharper error/merge signature.

Housekeeping heat

The third simulation compares a fixed-memory observer that repeatedly overwrites records with an expanding-memory observer that delays erasure by allocating more physical storage. The fixed-memory system produces an earlier irreversible heat signature. The expanding-memory system lowers immediate heat but pays in boundary and storage cost.

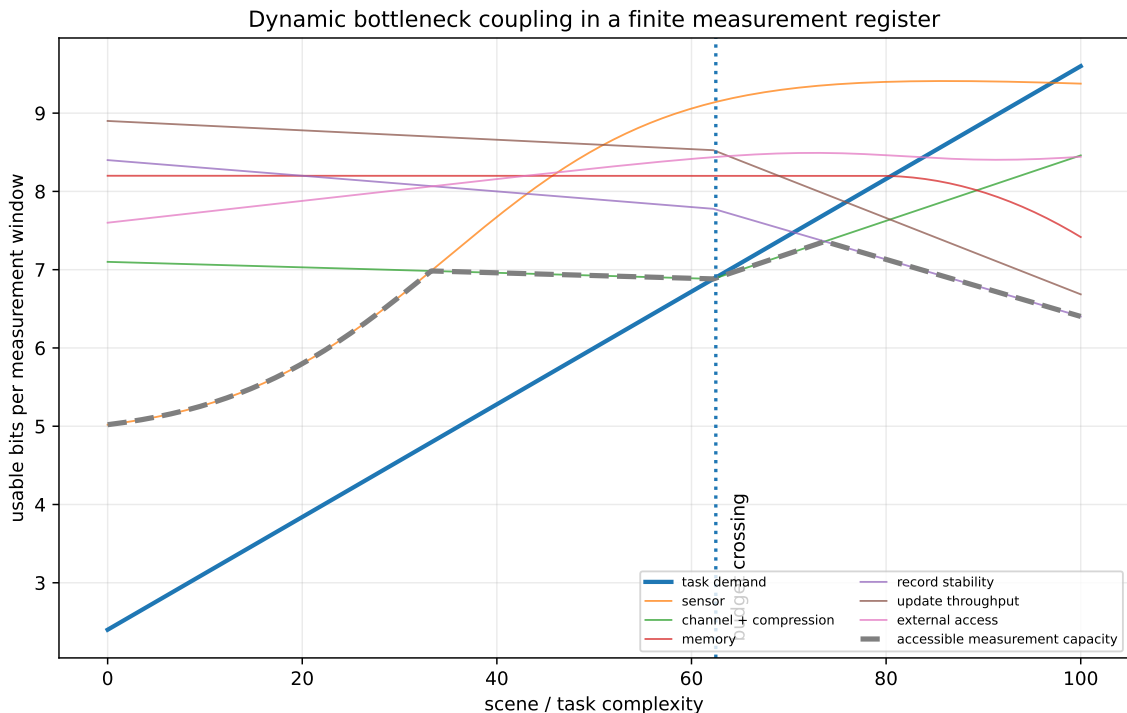


FIG. 1. Dynamic bottleneck coupling in a finite measurement register. Task demand rises with scene complexity. Sampling increases sensor capacity, compression partially relieves the channel, but compression and memory fill reduce record stability and update throughput. Accessible capacity is the lower envelope of dynamically coupled terms.

Two-dimensional sensor-array demonstration

The fourth simulation uses a finite-capacity camera-like array tracking multiple moving Gaussian objects. The high-capacity register preserves separate record classes. The low-capacity register uses coarse block averaging and delayed update, producing false merging and ghost persistence. This simulation illustrates how O1 signatures appear in a visually interpretable measurement task.

Decoherence-record interface

The fifth simulation is not a quantum derivation. It illustrates an operational interface: environmental branch information can grow on a decoherence timescale, but an observer obtains an operational measurement only when finite accessible record capacity crosses the task threshold and stabilizes the record. Different thresholds produce different record-formation times.

INTERFACE WITH DECOHERENCE

FDS-O1 is intentionally conservative about quantum foundations. It does not claim that finite distinction budgets collapse the wavefunction, derive the Born rule, or replace decoherence theory. It says instead that a quantum outcome becomes operationally available to a finite observer only when a stable record is produced.

Let S be a quantum system and E_t the environment after time t . Let $I_{\text{env}}(S; E_t)$ denote information about the system state redundantly or effectively available in environmental degrees of freedom. Let $C_{\text{app}}^{\text{acc}}(t)$ be the apparatus-accessible record capacity. An O1-style operational record condition is

$$\min\{I_{\text{env}}(S; E_t), C_{\text{app}}^{\text{acc}}(t), C_{\text{rec}}(t, \tau)\} \geq R_{\min}(\varepsilon; S). \quad (24)$$

Define

$$\tau_{\text{rec}} = \inf\{t : \min[I_{\text{env}}(S; E_t), C_{\text{app}}^{\text{acc}}(t), C_{\text{rec}}(t, \tau)] \geq R_{\min}(\varepsilon; S)\}. \quad (25)$$

The comparison between τ_{rec} and the usual decoherence time τ_d is an empirical and model-dependent matter. FDS-O1 does not require equality. It requires only that decoherence-like environmental information becomes an operational measurement for a finite observer through accessible stable record formation.

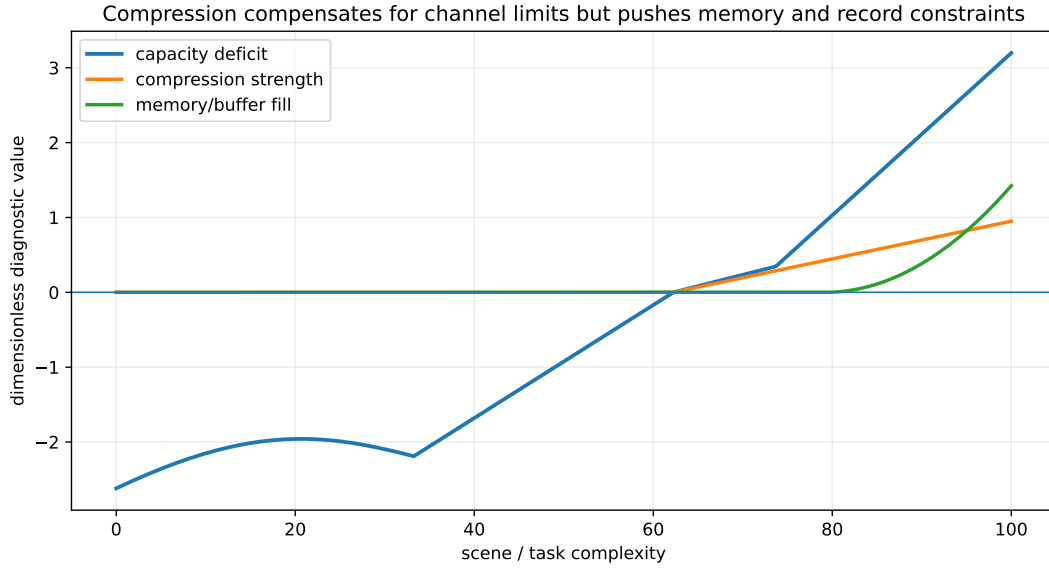


FIG. 2. Compression compensates for channel limits but pushes memory and record constraints. The same intervention that increases effective channel throughput increases buffer fill and capacity deficit.

Thus the record criterion is:

$$\begin{aligned} &\text{environmental correlation} + \text{accessible register} \\ &+ \text{record stability} \Rightarrow \text{operational outcome.} \end{aligned} \quad (26)$$

This is an interface claim. Stronger claims about wavefunction collapse, Born probabilities, Wigner’s friend, or quantum Darwinism belong in a separate FDS-Q1 paper.

RELATION TO ACTIVE INFERENCE AND ROBOTICS

Active inference usually begins with an observation y_t and asks how an agent updates beliefs and acts to minimize variational free energy. FDS-O1 asks what physical conditions must hold for y_t to exist as a usable finite record. If the observation register is saturated, the agent receives

$$\tilde{y}_t = \pi_{\mathcal{O}}(y_t), \quad (27)$$

not y_t itself. The variational objective is then evaluated on compressed, delayed, merged, or lossy records:

$$F(q; \tilde{y}_t) \neq F(q; y_t). \quad (28)$$

This does not refute active inference. It supplies a measurement-capacity precondition for it.

For robotics and embodied AI, this distinction is practical. A robot does not fail only because its planner is wrong. It can fail because its camera, buffer, timestamping, memory, or record-stabilization pipeline has crossed capacity. O1 therefore functions as a diagnostic layer below inference and control.

DOMAIN-SPECIFIC PROJECTIONS

Physical instruments

In instrumentation, FDS-O1 predicts that increasing raw sampling can degrade usable records when it fills memory, triggers compression, or increases dead time. Bottleneck switching should appear as slope changes in error-latency curves. Fixed-memory overwriting should exhibit excess housekeeping heat relative to reversible or expanding-memory protocols.

Biological sensory systems

Biological observers are finite registers with active sampling, attention, memory, and metabolic constraints. FDS-O1 predicts that apparent perceptual objects are stabilized coarse partitions. When update capacity is insufficient, false merging, persistence, attentional blink, and delayed report can be interpreted as finite-register budget exits rather than mere subjective bias.

Robotics and AI systems

Robotic agents often fail upstream of planning. Camera frame buffers overflow; tracklets merge; asynchronous sensor streams desynchronize; memory garbage collection creates latency spikes; logs fill; and external storage changes access time. FDS-O1 provides a common vocabulary for these failures: they are finite-register budget

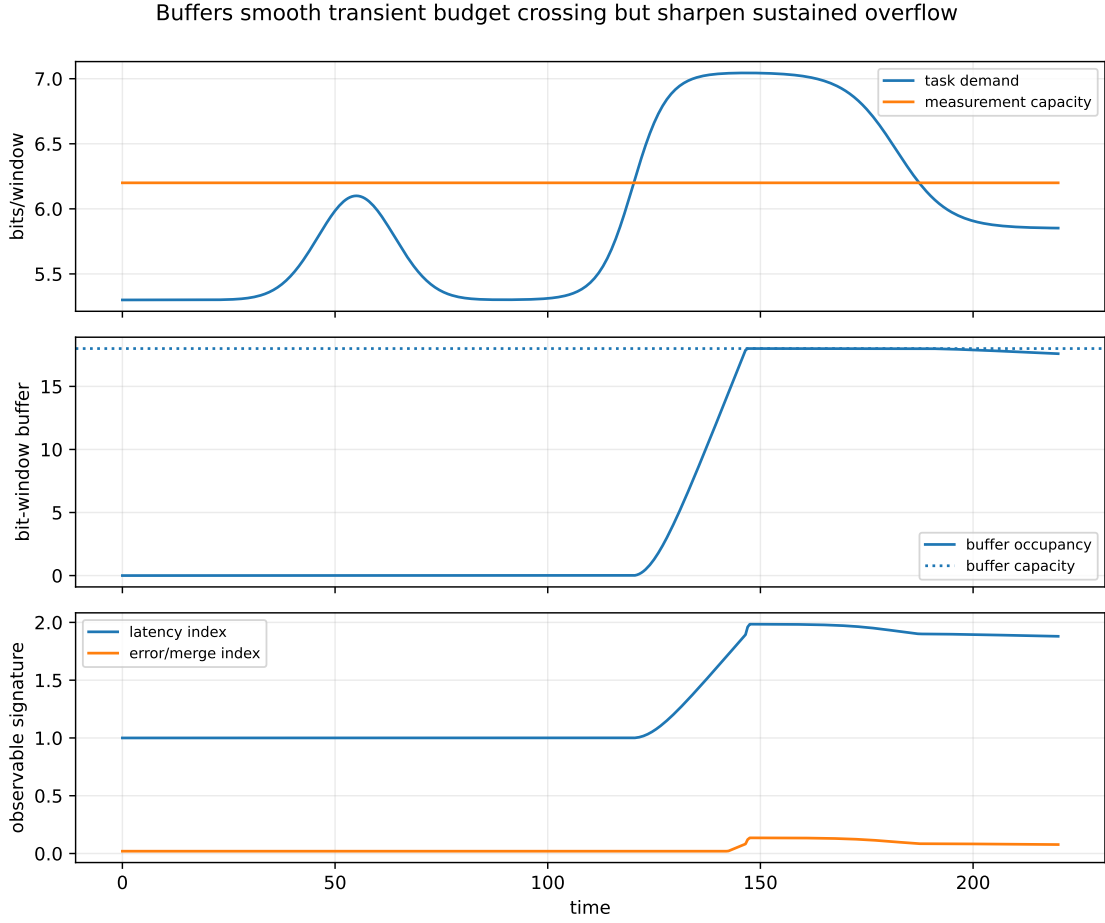


FIG. 3. Buffers smooth transient budget crossing but sharpen sustained overflow. The first overload pulse is mostly absorbed by buffer occupancy. The later sustained overload fills the buffer and generates latency and error/merge signatures.

crossings.

EXPERIMENTAL PROTOCOLS

Protocol 1 (Finite detector budget-crossing test). To test FDS-O1 in a detector or acquisition system:

1. Specify the observer-apparatus boundary and record carrier.
2. Estimate sensor, channel, memory, record-stability, buffer, externalization, and update capacities over the measurement window.
3. Define task family Ψ and target distortion ε .
4. Estimate $R_{\min}^{(\tau)}(\varepsilon; \Psi)$ using rate-distortion curves, calibration data, compression curves, Fisher information, or task-specific coding lower bounds.

5. Increase source complexity, noise, bandwidth demand, memory reuse pressure, or update rate until $R_{\min}^{(\tau)}$ crosses C_{meas} .

6. Measure error, latency, state merging, coarse-graining, reset frequency, externalization, buffer overflow, and heat output.

Protocol 2 (Buffering test). Apply a short overload pulse and a sustained overload pulse with matched peak demand. FDS-O1 predicts that finite buffers absorb the short pulse primarily as latency, whereas sustained overload produces buffer overflow, state drops, false merging, externalization bursts, reset cascades, or hard failure.

Protocol 3 (Reversible versus overwrite heat test). Compare two acquisition protocols with matched sensing and task demand: one using expanding memory with delayed erasure, and one using fixed memory with repeated overwrite. FDS-O1 predicts higher immediate housekeeping heat in the fixed-memory protocol, while

Housekeeping heat separates reversible sensing from irreversible record reuse

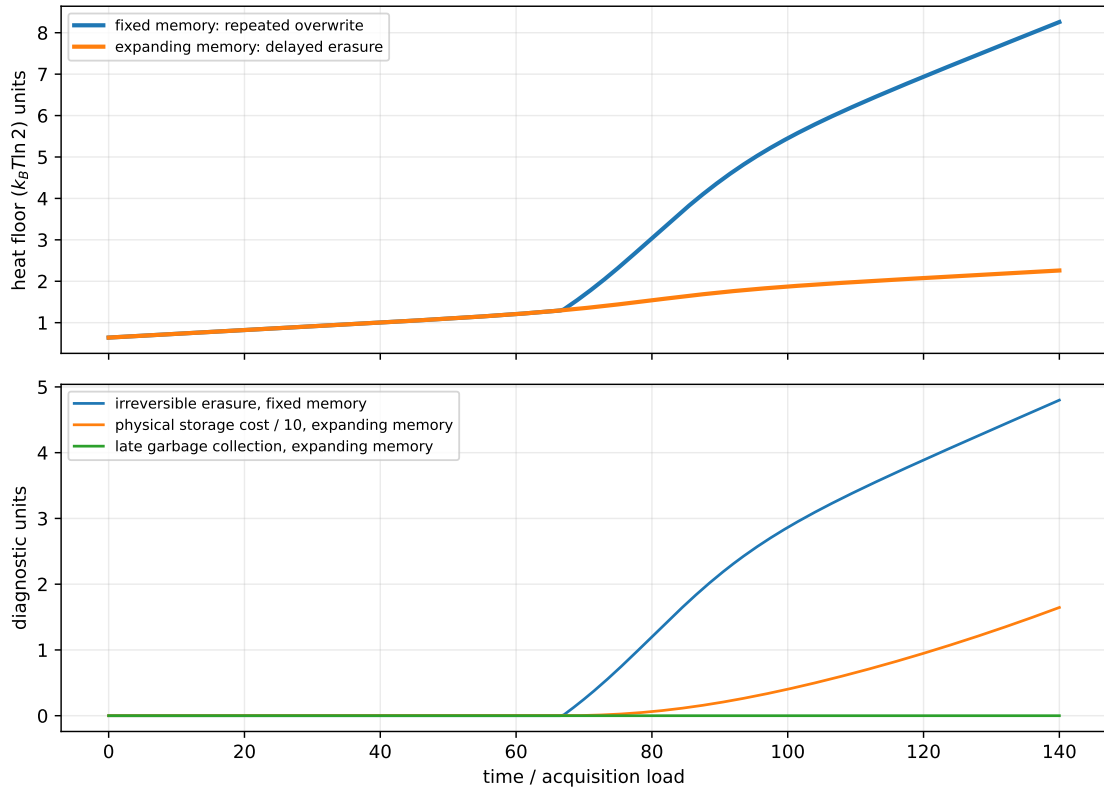


FIG. 4. Housekeeping heat separates reversible sensing from irreversible record reuse. Fixed-memory repeated overwrite produces a heat floor associated with erasure. Expanding memory postpones erasure but accumulates physical storage cost and may require later garbage collection.

2D sensor-array budget crossing: finite update capacity causes false merging

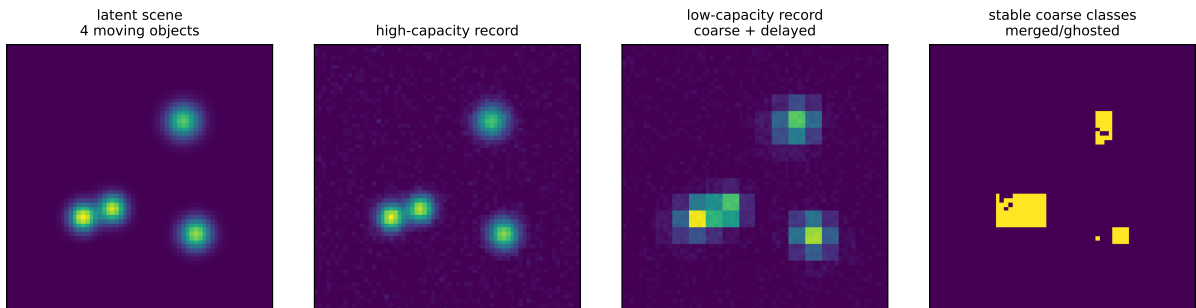


FIG. 5. Two-dimensional sensor-array budget crossing. A latent scene contains four moving objects. A high-capacity record preserves distinct objects. A low-capacity record with finite update budget, coarse resolution, and delayed refresh merges nearby objects and preserves ghost traces.

the expanding-memory protocol pays in storage and later garbage collection.

Protocol 4 (Sensor-array false-merging test). *Use a camera or sensor array to track multiple moving objects*

while varying frame rate, buffer size, compression, and tracking memory. FDS-O1 predicts false merging, ghost persistence, delayed track updates, and track swapping once update demand exceeds accessible register capacity.

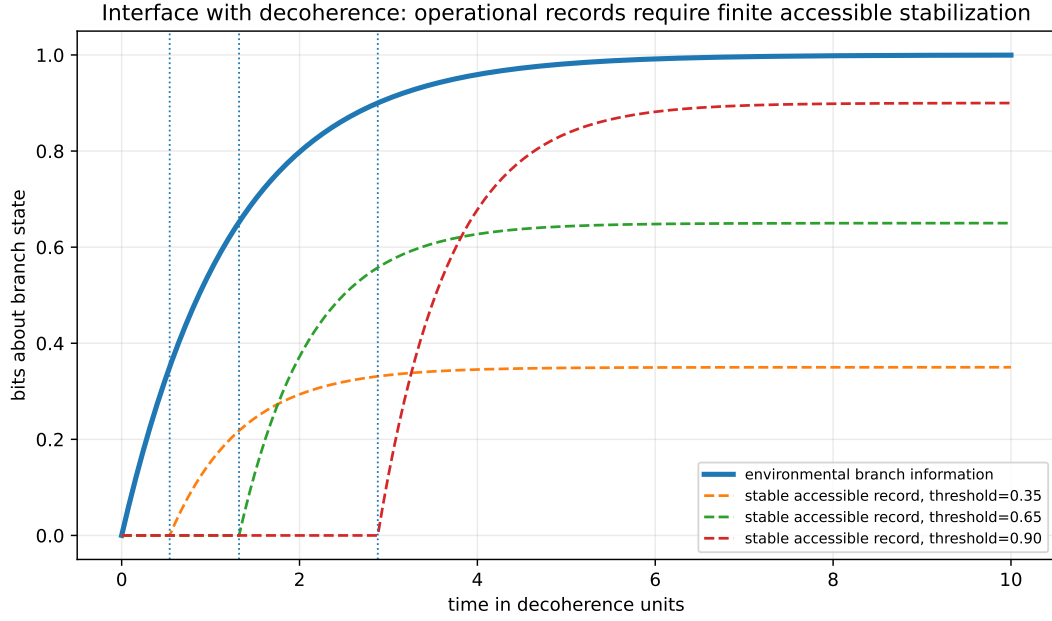


FIG. 6. Interface with decoherence. Environmental information about a branch can grow continuously, but finite observers obtain operational records only when accessible stabilization crosses task thresholds. This is a record criterion, not a replacement for decoherence theory.

TABLE II. Relationship between active inference concepts and FDS-O1 measurement-capacity variables.

Active inference concept	FDS-O1 counterpart	Finite-capacity implication
Observation y_t	Stabilized finite record z_t	Observation exists only after record formation
Generative model	Projection $\pi_{\mathcal{O}}$ plus update policy	Model operates on available distinctions
Precision weighting	Allocation of distinguishability capacity	Precision can be limited by sensor, memory, or update bottlenecks
Sensory attenuation	Controlled reduction of measurement demand	Agent may lower demand to avoid crossing
Epistemic action	Action to increase future C_{meas} or reduce $R_{\text{min}}^{(\tau)}$	Looking, moving, probing, or sampling changes the capacity ledger
Free-energy minimization	Inference after finite record formation	Saturated registers distort the inference input
Model failure	Budget crossing, delayed records, state merging	Some failures are measurement-capacity failures, not inference failures

LIMITATIONS AND FALSIFICATION

First, the paper is a finite-record measurement model, not a derivation of the fundamental laws. Second, C_{meas} is a schematic capacity ledger; real detectors require detailed modeling of noise, calibration, back-action, control loops, dead time, hysteresis, quantum efficiency, environmental coupling, and hardware scheduling. Third, Landauer accounting applies only to physically implemented logically irreversible operations under the relevant thermodynamic assumptions. Fourth, not every interaction

is a measurement. Record stability and accessibility matter. Fifth, observer-dependence here is operational, not arbitrary subjectivism.

The strong version of FDS-O1 would be undermined by any of the following:

1. a complete physical account of operational measurement requiring no finite record carrier, no stable trace, no boundary, no readout channel, and no capacity constraint;
2. a detector that reliably stabilizes unlimited distinctions under bounded resources and fixed error tol-

erance;

3. persistent full-fidelity measurement when $R_{\min}^{(\tau)}(\varepsilon; \Psi)$ exceeds all sensor, channel, memory, record-stability, buffer, externalization, and update capacities;
4. repeated finite-memory irreversible record reuse below generalized Landauer accounting after all reservoirs, feedback records, correlations, and work sources are included;
5. no observable change in error, latency, coarse-graining, externalization, reset, heat, buffering, or failure under controlled budget crossing;
6. active inference or control performance that remains invariant under severe manipulation of record availability, delay, compression, and merging, after task difficulty is otherwise held fixed.

CONCLUSION

An observer is not a magical epistemic primitive. It is a finite physical register that draws distinctions, stabilizes records, updates them, buffers them, and sometimes erases them under limited resources. Measurement is not just coupling; it is record formation that survives long enough to be used. Once this is made explicit, measurement inherits the same finite-budget structure developed in FDS-T1: accessible capacity is bottlenecked, task demand can exceed capacity, and budget crossing forces observable exits.

The v0.2 development strengthens O1 by replacing independent bottlenecks with dynamically coupled capacities, adding finite buffers, separating reversible sensing from irreversible housekeeping heat, adding a sensor-array simulation, and clarifying the relation to decoherence and active inference. The resulting framework is not a replacement for detector physics, quantum foundations, or robotics theory. It is a measurement-capacity diagnostic: if the finite register cannot form the record, then later inference, control, reporting, or theory-building starts from a compressed, delayed, merged, or missing observation.

FDS-O1 therefore supplies the operational trident’s first anchor. T1 states the finite-observer budget. O1 turns that budget into a measurement model. O2 should treat time as ordered irreversible update. O3 should treat the Second Law as the macroscopic cost of finite record maintenance. The immediate empirical signature is simple: when measurement demand crosses accessible capacity, finite observers do not continue measuring at full fidelity. They merge states, buffer and delay records, externalize, reset memory, dissipate housekeeping heat, relax the task, or fail.

Notation Summary

Simulation Parameters

The simulations are deterministic and use fixed synthetic parameter values in `code/generate_results.py`. The dynamic-bottleneck simulation uses a scene complexity axis $u \in [0, 100]$, rising task demand, a compression policy activated by channel stress, memory fill dynamics, and record-stability penalties from compression. The buffering simulation uses a short overload pulse and a sustained overload pulse with finite buffer capacity. The heat simulation compares fixed-memory overwriting with expanding-memory delayed erasure. The sensor-array simulation uses 64×64 images with moving Gaussian objects and coarse block-averaged low-capacity observation. The decoherence-interface simulation uses a saturating environmental information curve and several finite record thresholds. No proprietary data, detector data, quantum experimental data, or human-subject data are used.

Reproducibility Checklist

1. Code availability: all simulation code is included in the replication package.
2. Deterministic execution: the random seed is fixed for synthetic image noise.
3. Figure reproduction: running `python code/generate_results.py` regenerates all figures and CSV outputs.
4. Data status: all numerical outputs are synthetic demonstrations generated from the stated model.
5. Platform independence: the code uses standard Python scientific libraries.

Boundary of Applicability

FDS-O1 applies to systems that function as finite physical registers: they have record carriers, finite readout, finite stability, finite update, buffers, and finite task windows. It does not apply to idealized mathematical observers with unbounded access, nor does it make claims about fundamental collapse or nonunitary quantum dynamics. It is compatible with unitary microscopic evolution and with standard detector-specific modeling.

CODE AVAILABILITY

The simulation code used to generate Figs. 1–6 is included in the accompanying replication package

TABLE III. FDS-O1 notation summary.

Symbol	Meaning
$\mathcal{O}$	finite observer or distinction-register
$\mathcal{D}$	detector or measurement apparatus
Ω	physical possibility space
$\pi_{\mathcal{O}}$	finite distinction projection implemented by $\mathcal{O}$
$\mathcal{Z}_{\mathcal{O}}$	accessible record classes
$N_{\mathcal{O}}$	number of operationally distinguishable record classes
$C_{\mathcal{O}}$	bit capacity $\log_2 N_{\mathcal{O}}$
C_{sens}	usable sensor-resolution capacity
C_{chan}	readout/channel capacity
C_{mem}	internal memory capacity
C_{rec}	stable record capacity over retention window
$C_{\text{ext}}^{\text{eff}}$	effective externalized record capacity after overhead
$I_{\text{update}}^{\text{max}}$	maximum irreversible update throughput
C_{meas}	accessible measurement capacity
$R_{\text{min}}^{(\tau)}(\epsilon; \Psi)$	minimal task-relevant distinction demand
Δ_{O1}	measurement capacity deficit
B_{buf}	finite buffer occupancy
P_{hk}	housekeeping power associated with irreversible record reuse
τ_{rec}	finite accessible record-formation time

under `code/generate_results.py`. Running the script regenerates all figures (PDF and PNG) and CSV tables in a single pass. The public repository path is https://github.com/yiningwu-research/Distinction-Theory/tree/main/models/fds_o1. Related DT/FDS archival materials, including the Formal Core (FDS-0), the general archive, and the claim registry, are maintained in the same public repository.

AI ASSISTANCE DISCLOSURE

AI-assisted tools were used for language polishing, structural feedback, LaTeX drafting support, and code-debugging assistance. The author reviewed and edited all content and remains responsible for all claims, references, simulations, and conclusions. No AI system is listed as an author.

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Time as Irreversible Distinction Update: Finite Records, Causal Ordering, and Register-Time Collapse

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Time is often treated either as a continuous coordinate in physical equations or as a global ordering whose arrow is explained by statistical mechanics, cosmology, or boundary conditions. FDS-O2 gives a narrower operational account: for a finite physical observer, usable time is the ordered structure of irreversible distinction updates in finite records. Building on FDS-O1, where an observer is defined as a finite distinction-register and measurement as stable record formation, this paper defines a temporal register as a physical system that records, orders, timestamps, updates, synchronizes, and reuses distinctions under finite memory, finite clock precision, finite buffering, finite communication bandwidth, finite latency, and finite thermodynamic budgets. The central claim is not that coordinate time is unreal or that microscopic dynamics must be nonunitary. It is that finite systems access temporal order only through register time: causal dependencies, clock labels, logs, residues, and update traces carried by finite records. We make causal dependency prior to temporal labeling: record z_j is operationally after z_i when the update that forms z_j depends on z_i through an accessible finite update chain. When finite memory forces overwrite, compression, projection, or garbage collection, the update map is many-to-one. Its non-injectivity can be quantified by $\mathcal{L}_U = H(X|Z)$, the residual set of compatible past states given the current record. This loss defines an operational arrow: future records may contain traces of past records, but erased distinctions cannot be reconstructed from the current finite record without external memory. The paper adds synchronization bottlenecks between finite observers, latency-induced apparent causal inversion, finite clock coarse-graining, dissipative projection, and long-context temporal collapse in AI-like finite-window systems. Deterministic simulations illustrate memory-fill-driven non-injectivity, fixed-memory overwrite versus append-only logging, latency-based order inversion, clock-tick collisions, synchronization ambiguity, projection-semigroup loss, and context-window temporal collapse. The paper does not solve the cosmological arrow of time, the Past Hypothesis, the full Second Law, general relativity, or quantum measurement. Its contribution is narrower: it converts finite record formation into a concrete operational model of time as causal, irreversible, finite-register update.

Scope and Boundary of the Theory. This paper does not claim that microscopic laws are fundamentally nonunitary, that coordinate time in equations is invalid, that relativity requires a preferred foliation, or that cosmology no longer requires boundary conditions. It does not derive general relativity, quantum measurement, or the thermodynamic Second Law in full. It treats *register time*: the finite record structure through which bounded observers access temporal order. Coordinate time, proper time, and clock readings remain valid in their standard domains. O2 asks what finite records must carry for a physical system to use temporal order at all.

Claim-status summary

Table I summarizes the central FDS-O2 claims, their epistemic status, and the conditions under which they should be weakened or rejected.

Keywords: finite observer; time; register time; causal dependency; irreversible update; finite records; non-injective update; dissipative projection; synchronization

bottleneck; clock precision; latency; buffering; Landauer principle; finite memory; semigroup; information thermodynamics; long-context AI; active finite distinction systems.

INTRODUCTION

From finite measurement to finite time

FDS-O1 defined an observer as a finite distinction-register: a physical system that registers, preserves, updates, orders, and communicates distinctions through finite records, finite channels, finite buffers, and finite thermodynamic budgets [1]. O1 treated measurement as stable finite record formation. The present paper asks the next operational question: what is time for such a finite register?

The answer is deliberately modest. A finite observer does not directly access an ideal continuum. It accesses temporal order through records: timestamps, dependency edges, update traces, residues, logs, memory states, clock ticks, and causal communication. A system says that event a occurred before event b only if it has a record structure that supports that comparison.

TABLE I. Central FDS-O2 claims, epistemic status, and demotion or failure conditions.

Claim	Status	What would weaken or falsify it
Register time is ordered finite-record update	Operational definition	Finite observers use temporal order without any record ordering, update history, clock record, causal marker, dependency edge, or memory of change
Causal dependency precedes temporal labels	Structural proposition	Stable temporal order is obtained solely from labels even when update dependencies and causal traces are removed or scrambled
Non-injective update induces an operational arrow	Conditional theorem	A finite register repeatedly overwrites or compresses records yet reconstructs erased distinctions from the current finite record without external memory, hidden reservoirs, or additional records
Synchronization is a finite-channel record-exchange problem	Relativity- compatible bridge	Bounded observers establish global simultaneity without signal exchange, finite bandwidth, latency, synchronization records, or assumptions about propagation
Buffers and latency can distort recorded order	Testable prediction	Load-dependent latency and finite buffers never produce delayed records, order inversions, timestamp gaps, or apparent cause-effect reversal under controlled acquisition stress
Finite clock precision coarsens time	Operational bridge	Arbitrarily dense events are totally ordered by a bounded finite clock without tick collisions, synchronization error, or external ordering information
Dissipative projection carries housekeeping cost when physically implemented by irreversible reuse	Physical bridge	Repeated logically irreversible temporal record reuse violates generalized Landauer accounting after all reservoirs, correlations, feedback records, and work sources are included

Without a record-bearing ordering structure, temporal comparison is not operationally available to that finite system.

O2 therefore reformulates time at the register level. If records could be updated reversibly and indefinitely logged, past states could in principle remain reconstructible. But finite systems have finite memory, finite clocks, finite buffers, finite update throughput, finite communication bandwidth, and finite energy. Under sustained acquisition, a finite temporal register must overwrite, compress, project, externalize, or garbage-collect records. These operations are generally many-to-one. The operational arrow is the direction in which non-injective record updates accumulate.

Coordinate time, clock time, and register time

To prevent a common misunderstanding, O2 distinguishes coordinate time from the finite time available to a physical observer. Table II summarizes the terminology.

The central boundary statement is:

O2 does not deny coordinate time; it models finite access to temporal order.

(1)

Coordinate time may remain an excellent parameter in physical equations. Register time is what a finite observer can use after records, clocks, buffers, and updates have done their work.

Main thesis

The main thesis is:

$$\text{register time} = \text{causally ordered irreversible distinction update.} \quad (2)$$

A finite temporal register produces records

$$r_n = (z_n, \kappa_n, d_n, \eta_n), \quad (3)$$

where z_n is a record state, κ_n is a clock label or ordering marker, d_n is a dependency pointer or causal trace, and η_n is metadata controlling stability, confidence, latency, provenance, or synchronization. A time label is not the root of order. It is a finite annotation attached to an update process.

A generic temporal update has the form

$$z_{n+1} = U_n(z_n, x_{n+1}, b_n, \theta_n), \quad (4)$$

where x_{n+1} is incoming measurement content, b_n is buffer state, and θ_n represents clock, control, synchronization,

TABLE II. Coordinate time, clock time, and register time. O2 studies the finite record layer, not a replacement for coordinate time in equations.

Term	Meaning	Status in O2
Coordinate time t	Continuous parameter used in physical equations or coordinate charts	Idealized mathematical parameter; not denied
Proper time τ	Time measured along a worldline by an ideal or physical clock	Compatible with O2; implemented through physical clock records
Clock time	Readout of a clock device or oscillator register	Finite labels with tick width, drift, jitter, wraparound, and synchronization limits
Register time	Ordered finite-record structure available to a bounded observer	Main object of O2
Historical order	Causal/update dependency order among records	More primitive than labels inside the finite register

or update context. If U_n is non-injective over possible pre-update states, no inverse exists inside the accessible finite record. The missing distinctions are not merely unknown; they are no longer carried by the register.

Relation to earlier time drafts

Earlier time drafts argued that truncation forms a semigroup without inverses and that the arrow of time follows from algebraic irreversibility. The current O2 version narrows and operationalizes that claim. It does not claim to solve the entire arrow-of-time problem or to remove the need for statistical mechanics and cosmology. It uses the semigroup idea only at the level of finite records: update operations that overwrite, compress, or discard distinctions are non-invertible maps on the register state space. The empirical content is the predicted pattern of finite-register signatures: reconstruction-depth limits, non-injectivity growth, synchronization uncertainty, clock-tick collisions, buffer-shaped latency, apparent causal inversion, and temporal-collapse behavior in finite-window systems.

RELATED WORK

The arrow of time and irreversibility

The asymmetry of time has been central to physics since Eddington's discussion of the arrow of time [3]. Statistical-mechanical accounts relate macroscopic irreversibility to entropy increase, typicality, and boundary conditions, while acknowledging the role of low-entropy initial conditions or the Past Hypothesis [4–6]. Nonequilibrium thermodynamics and dissipative structures explain how local order can persist through entropy export [7, 8]. FDS-O2 does not replace these accounts. It addresses the lower operational layer: a finite observer accesses temporal order only through finite records, and finite record updates can be non-invertible even when a larger embedding dynamics is modeled reversibly.

Information, erasure, and reversible computation

Landauer's principle links logically irreversible erasure to thermodynamic cost [9]; Bennett showed that computation can be arranged reversibly when sufficient memory and garbage management are available [10]. Modern information thermodynamics refines this accounting by including feedback, correlations, nonequilibrium reservoirs, and stochastic dynamics [8, 11]. O2 uses this tradition conservatively. It does not claim that every update is a Landauer erasure. It claims that finite temporal registers under sustained load must eventually trade memory growth against irreversible record reuse or externalization.

Causal ordering and finite clocks

In distributed systems, Lamport showed that event ordering is not merely a clock label but a structure constrained by causal communication and message order [14]. In causal-set approaches, causal order is treated as a primitive partial order for spacetime structure [15]. O2 is compatible with these traditions but does not identify register time with spacetime discreteness. Its primitive is finite record update: a bounded physical system uses a partial order only when dependency information, clock records, or causal traces survive in the register.

FINITE TEMPORAL REGISTERS

Definition 1 (Finite temporal register). *A finite temporal register is a finite distinction-register that can record, order, update, timestamp, retain, synchronize, and communicate distinctions over a finite window. It consists of a record state space $\mathcal{Z}$, update maps U_n , clock labels κ_n , buffer states b_n , synchronization messages m_n , and retention rules specifying which past distinctions remain accessible.*

Definition 2 (Temporal record). *A temporal record is a tuple*

$$r_n = (z_n, \kappa_n, d_n, \eta_n), \quad (5)$$

where z_n is a record class, κ_n is a finite clock label or ordering marker, d_n is a dependency pointer or causal

trace, and η_n is metadata controlling stability, confidence, latency, provenance, or synchronization.

Definition 3 (Operational temporal order). *For records r_i and r_j , write $r_i \prec_{\mathcal{O}} r_j$ when the finite observer $\mathcal{O}$ has an accessible dependency edge, clock comparison, append order, causal trace, or synchronization record supporting the claim that r_i precedes r_j . The relation $\prec_{\mathcal{O}}$ is observer-relative but not arbitrary: it is constrained by retained records, causal communication, clock precision, and update dependence.*

The relation $\prec_{\mathcal{O}}$ need not be a total order. Two events may be incomparable to a finite observer if they fall within the same clock tick, arrive through unsynchronized channels, are buffered with ambiguous latency, or have had their ordering metadata erased.

Causal dependency before temporal labels

O2 treats causal/update dependency as more primitive than clock labels. Clock labels help compare records, but they do not by themselves create the update chain that makes one record depend on another.

Definition 4 (Update dependency). *Let z_i and z_j be accessible record states. Write*

$$z_i \prec_U z_j \quad (6)$$

if z_j is generated by a finite composition of update maps whose domain includes a state carrying z_i or a dependency trace of z_i .

Proposition 1 (Causal dependency is prior to register time). *If a finite observer judges z_i to be operationally before z_j on the basis of its own register, then at least one of the following must be accessible: an update dependency $z_i \prec_U z_j$, a clock comparison $\kappa_i < \kappa_j$, an append-order trace, or a communication/synchronization record. Among these, update dependency is the causal substrate: a label can be copied or corrupted, but the formation of z_j from z_i requires a physical update chain.*

Sketch. A temporal comparison is an operational claim made by a finite register. Such a claim must be carried by some retained distinction: a dependency edge, a label, a log order, or a synchronization record. A clock label is itself a record produced by an update process. If all dependency, append, and communication traces are erased or scrambled, labels alone can no longer determine which record causally contributed to which later record. Thus labels are useful finite annotations, but causal/update dependence supplies the physical ordering substrate within the register. $\square$

This proposition connects O2 to causal graphs and distributed-system logical clocks without reducing physical time to software time. O2 claims only that finite systems need retained dependency information to use temporal order.

TEMPORAL CAPACITY AND BUDGET CROSSING

A finite register has finite temporal capacity. A schematic capacity ledger is

$$C_{\text{time}}(t) = \min\{C_{\text{rec}}, C_{\text{clk}}, C_{\text{buf}}, C_{\text{upd}}, C_{\text{log}}, C_{\text{ext}}^{\text{eff}}, C_{\text{sync}}\}. \quad (7)$$

The terms denote stable record capacity, clock-label capacity, buffering capacity, update-throughput capacity, log memory, effective externalized access, and synchronization capacity over a window.

For a task family Ψ_t , distortion tolerance ε , and retention/verification window τ , let

$$R_{\min}^{(\tau)}(\varepsilon; \Psi_t) \quad (8)$$

be the minimal temporal distinction demand required to meet the task. The O2 temporal capacity deficit is

$$\Delta_{\text{O2}}(t) = R_{\min}^{(\tau)}(\varepsilon; \Psi_t) - C_{\text{time}}(t). \quad (9)$$

Theorem 1 (Finite temporal update-exit theorem). *Let $\mathcal{O}$ be a finite temporal register. If*

$$R_{\min}^{(\tau)}(\varepsilon; \Psi_t) > C_{\text{time}}(t), \quad (10)$$

then full-fidelity temporal reconstruction at distortion ε is impossible unless at least one exit occurs: coarser clock ticks, increased latency, buffering, externalized logging, lossy compression, record overwrite, garbage collection, task relaxation, boundary enlargement, or failure.

Proof. $R_{\min}^{(\tau)}$ is the minimum number of task-relevant temporal distinctions required to meet the ordering target. C_{time} is the maximum number of temporal distinctions the finite register can stabilize, label, synchronize, update, retain, and access in the same window. If demand exceeds capacity, a full-fidelity temporal record would require distinctions not represented by any admissible record state, log, timestamp, synchronization message, or update sequence. The register must therefore reduce demand, increase capacity, store backlog, externalize records, erase or compress prior records, relax the task, or fail. $\square$

The empirical content of the theorem is the predicted co-occurrence of exits near independently estimated crossing: reconstruction depth saturates, overwrite or garbage-collection events increase, clock ticks collide, synchronization ambiguity grows, buffer latency rises, local order inversions appear, and temporal reports become compressed, delayed, merged, or missing.

NON-INJECTIVE UPDATE AND DISSIPATIVE PROJECTION

Let X denote the set of possible pre-update register states and Z the set of post-update record states. A physical update implements a map

$$U : X \rightarrow Z. \quad (11)$$

If U is injective and the register retains the required garbage information, the update can be reversed in principle. If U is many-to-one, then distinct pre-update states can produce the same post-update record.

Definition 5 (Non-injectivity loss). *For a finite update $U : X \rightarrow Z$, define the preimage multiplicity of $z \in Z$ as*

$$m_U(z) = |U^{-1}(z)|. \quad (12)$$

The average non-injectivity loss is

$$\mathcal{L}_U = \mathbb{E}_{z \sim P_Z} [\log_2 m_U(z)]. \quad (13)$$

Equivalently, under the induced joint distribution on pre-update and post-update states,

$$\mathcal{L}_U = H(X|Z). \quad (14)$$

This quantity is the number of bits of compatible past that remain unresolved after reading the current record. It is a direct operational measure of lost reconstructibility.

Proposition 2 (Arrow strength from non-injectivity). *For a finite register, the operational arrow associated with an update U strengthens as $\mathcal{L}_U = H(X|Z)$ increases. If $\mathcal{L}_U = 0$, the current record preserves enough information to reconstruct the pre-update state. If $\mathcal{L}_U > 0$, multiple possible past states are compatible with the same current record.*

If the many-to-one update is physically implemented through irreversible reset, overwrite, compression, or garbage collection, generalized Landauer accounting gives the lower-bound form

$$Q_{\min} \geq k_B T \ln 2 \mathcal{L}_U, \quad (15)$$

when $\mathcal{L}_U$ is measured in bits and the usual thermodynamic assumptions and caveats are satisfied. Recent reviews emphasize that practical erasure processes often dissipate more than the ideal Landauer bound and that finite-time, finite-bath, nonequilibrium, non-Markovian, and quantum regimes require generalized accounting [22]. The abstract mathematical map is not automatically dissipative. The heat belongs to the physical implementation when discarded distinctions are irreversibly erased or overwritten.

Definition 6 (Dissipative projection). *A dissipative projection is a physical many-to-one record update*

$$\Pi_{\text{diss}} : X \rightarrow Z \quad (16)$$

that discards distinctions and stabilizes a smaller record space through irreversible overwrite, reset, many-to-one compression, or garbage collection. Its abstract projection structure gives non-injectivity; its physical implementation supplies thermodynamic cost.

Compatible histories. A large value of $H(\text{Past} | Z_t)$ means that the current finite record Z_t is compatible with many possible prior histories. The observer has not retained a unique reconstructible past; it has retained an equivalence class of histories that project to the same present record. This resembles the language of compatible or consistent histories [19], but the claim here is operational rather than interpretive: FDS-O2 does not posit branching worlds or a new quantum interpretation. It only states that finite non-injective record updates merge multiple past trajectories into the same accessible present state.

The family of projections P_K retaining only depth K obeys

$$P_K \circ P_L = P_{\min(K,L)}. \quad (17)$$

This is a semigroup under composition. It is not a group when discarded temporal distinctions cannot be reconstructed from the retained finite record.

SYNCHRONIZATION, LATENCY, AND APPARENT CAUSAL INVERSION

Synchronization bottleneck

Consider two finite observers O_A and O_B with local register times $\{a_i\}$ and $\{b_j\}$. To establish a shared temporal order, they must exchange records. Let C_{chan}^{AB} be the bit capacity of the record-exchange channel during a synchronization interval, and let L_{AB} be latency with uncertainty δL_{AB} . If the order relation to be established has entropy $H(\Delta_{AB})$, a schematic lower bound on synchronization uncertainty is

$$\delta t_{\text{sync}} \gtrsim \delta L_{AB} + \frac{H(\Delta_{AB})}{C_{\text{chan}}^{AB}} + \delta t_{\text{clock}}. \quad (18)$$

This is not a derivation of relativity. It states the operational fact that cross-boundary simultaneity claims require finite signal exchange, finite channels, and finite clock records. In relativistic settings, finite signal propagation speed supplies an additional physical boundary on record exchange, consistent with the operational constraints on simultaneity introduced by special relativity [16].

Latency-induced apparent causal inversion

Let two source events obey the source order $e_i \prec e_j$ with times $t_i < t_j$. The finite register records them at

$$\hat{t}_i = t_i + \ell_i, \quad \hat{t}_j = t_j + \ell_j, \quad (19)$$

where ℓ_i and ℓ_j are load-dependent latencies. A recorded inversion occurs when

$$\hat{t}_i > \hat{t}_j, \quad (20)$$

which is equivalent to

$$\ell_i - \ell_j > t_j - t_i. \quad (21)$$

This creates an apparent effect-before-cause record without violating physical causality. The violation is in the finite register's acquired order, not in the source process.

A sufficient engineering criterion for maintaining causally consistent records is

$$C_{\text{upd}} > h_{\text{env}} + h_{\text{sync}} + h_{\text{overhead}}, \quad (22)$$

where h_{env} is the entropy rate of the relevant event stream, h_{sync} is the record-exchange burden, and h_{overhead} is the cost of metadata, timestamps, buffers, and error correction. If the update capacity falls below this demand, the register must delay, drop, merge, re-order, or coarsen records.

Finite clocks and partial orders

A clock is itself a finite distinction-register. It maps physical change into labels

$$\kappa : I \rightarrow \{0, 1, \dots, K - 1\} \quad (23)$$

within an operational interval I . If multiple events map to the same label, their order is not available from the clock alone. Clock capacity over a window can be written schematically as

$$C_{\text{clk}}(\tau) = \log_2 N_{\text{ticks}}(\tau) - H_{\text{jitter}} - H_{\text{sync}} - H_{\text{wrap}}. \quad (24)$$

Finite clocks therefore naturally produce partial orders under high event density, jitter, drift, wraparound, or synchronization limits.

NUMERICAL MODELS AND SIMULATIONS

The simulations are deterministic synthetic demonstrations. They are not fits to physical detector data, clock data, cosmological observations, AI-system benchmarks, or human time perception. They provide a reproducible diagnostic template for the O2 model. All figures and CSV outputs are generated by `code/generate_results.py`.

Memory fill and non-injectivity

The first simulation contrasts cumulative distinction history with finite retained record capacity. Incoming distinctions accumulate over update index. Once cumulative history exceeds capacity, lost bits grow. The non-injectivity index is $H(\text{past}|\text{record})$, and the arrow-strength proxy increases as memory fill forces more aggressive projection.

Fixed overwrite versus append-only logging

The second simulation compares fixed-memory overwrite with expanded append-only logging. Fixed memory bounds storage at the cost of immediate overwrite. Append-only logging preserves reconstructibility longer but accumulates storage demand and later cleanup cost.

Latency-induced apparent causal inversion

The third simulation sends an ordered event stream through a finite buffer with load-dependent latency. When latency differences exceed source event gaps, the recorded order locally inverts. This produces apparent effect-before-cause observations without violating source causality.

Finite clock precision

The fourth simulation maps a bursty event stream into finite clock ticks. As tick width grows, multiple events collide in the same clock label and temporal order becomes partial or ambiguous.

Synchronization bottleneck

The fifth simulation models two finite observers attempting to establish shared order through a finite record-exchange channel. Synchronization uncertainty decreases with channel capacity but remains bounded below by latency uncertainty, jitter, and clock uncertainty.

Projection semigroup and non-injectivity

The sixth simulation uses a toy family of projections that retain only depth k from an initial depth K_0 . The number of compatible microhistories per record grows as retained depth falls. The loss $H(\text{past}|\text{record})$ quantifies how much past remains unreconstructible.

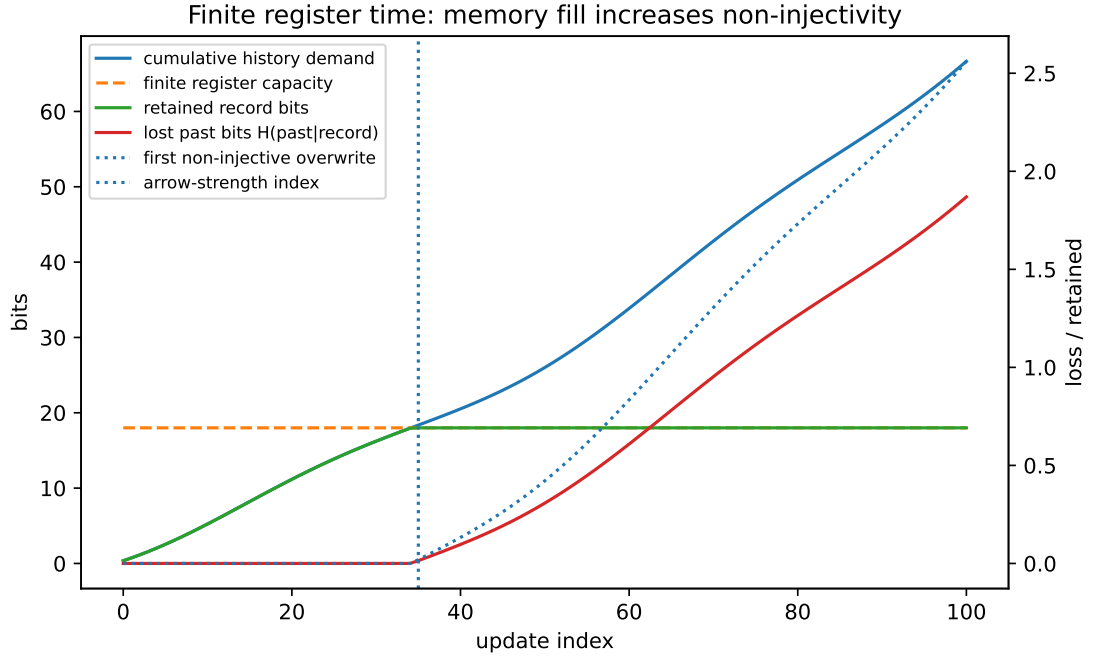


FIG. 1. Finite register time: memory fill increases non-injectivity. Cumulative history demand eventually exceeds finite register capacity. The retained record saturates while compatible past information $H(\text{past}|\text{record})$ grows. The arrow-strength index is a normalized loss of reconstructible past.

Long-context temporal collapse

The seventh simulation treats a long-context AI-like system as a finite temporal register. Once task history exceeds the context window, exact historical order is replaced by compression, summary, or discard. The model predicts temporal step confusion, false memory of order, and plan drift when task-history demand crosses finite context capacity.

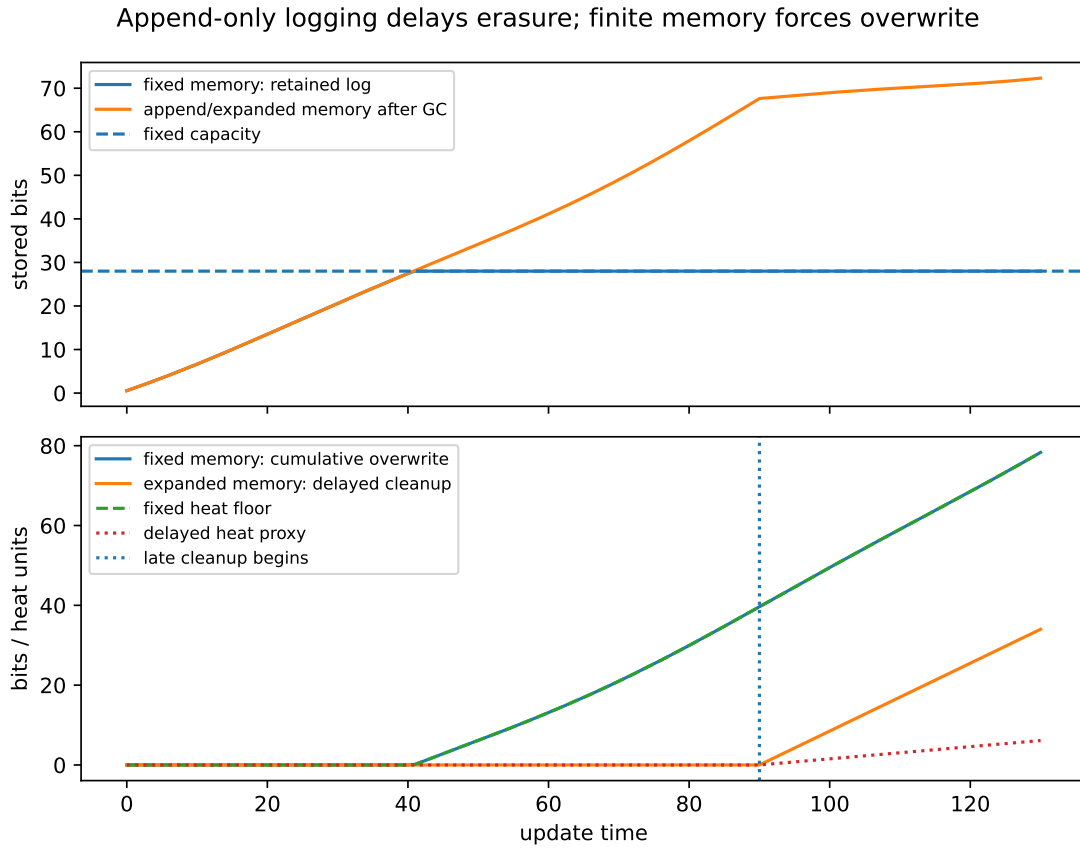


FIG. 2. Append-only logging delays erasure; finite memory forces overwrite. Fixed memory keeps storage bounded but increases overwritten distinctions and heat-floor proxy. Expanded memory lowers immediate erasure but accumulates physical storage and delayed garbage collection.

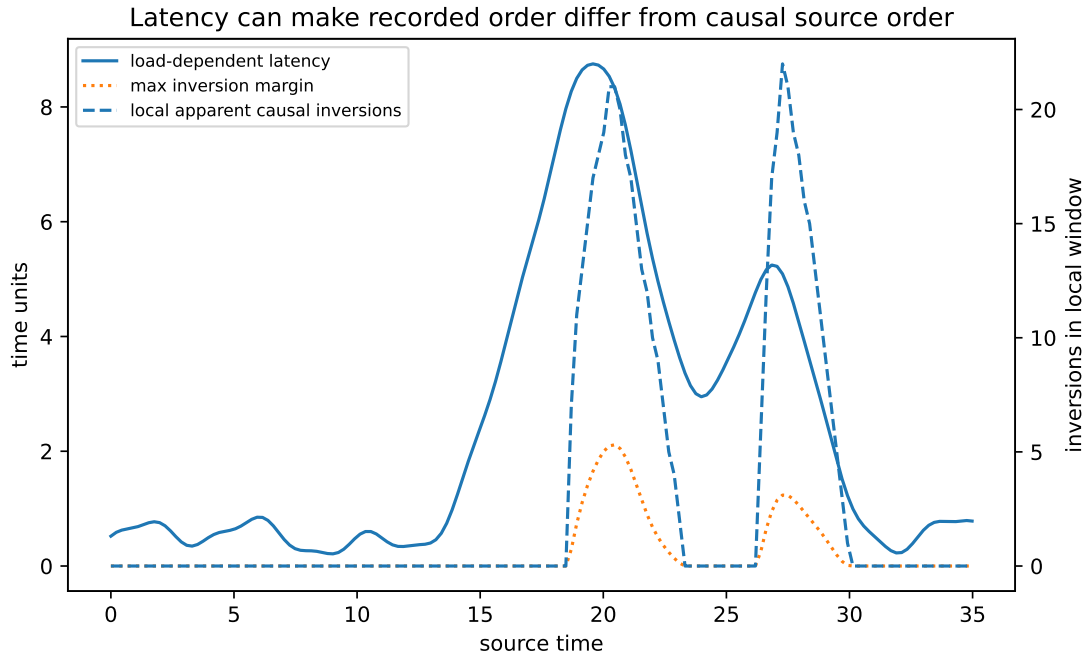


FIG. 3. Latency can make recorded order differ from causal source order. Under load, latency differences exceed event separation and generate local apparent causal inversions in the record stream.

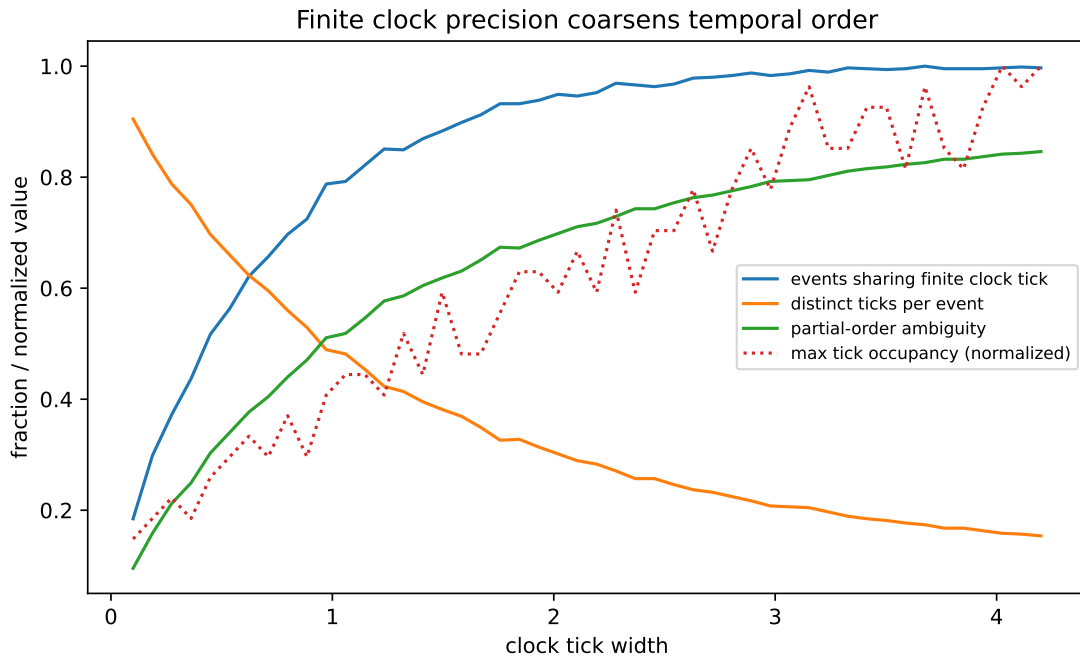


FIG. 4. Finite clock precision coarsens temporal order. Larger tick widths increase collisions and partial-order ambiguity while reducing the number of distinct labels available per event.

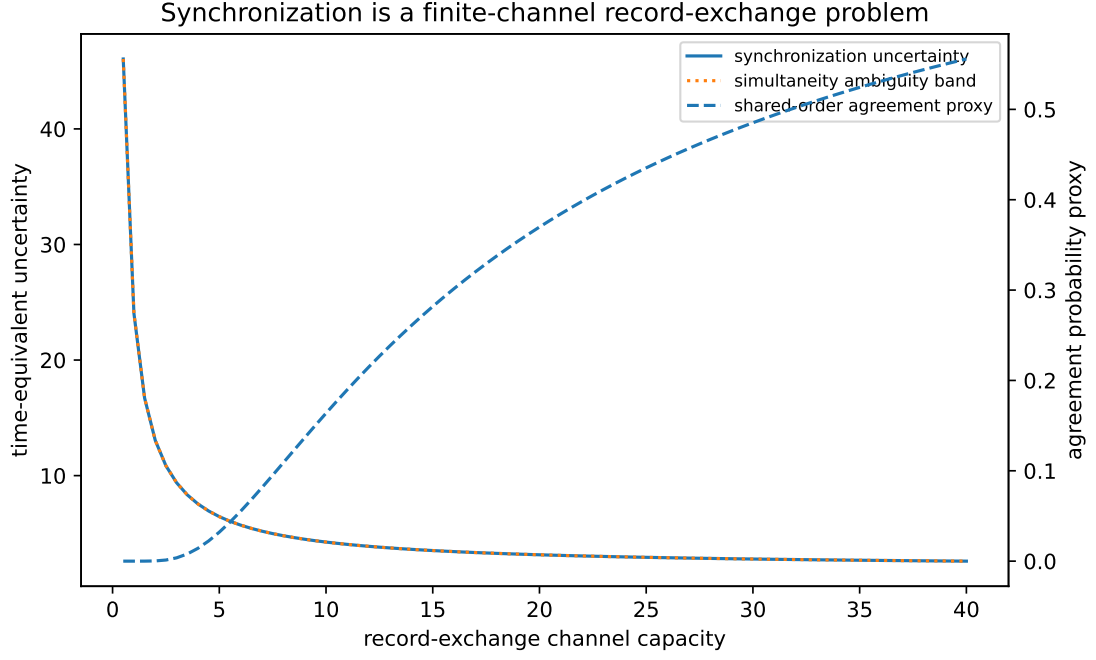


FIG. 5. Synchronization is a finite-channel record-exchange problem. As communication capacity increases, shared-order uncertainty decreases, but latency and jitter impose an ambiguity band. This is an operational account, not a derivation of relativistic simultaneity.

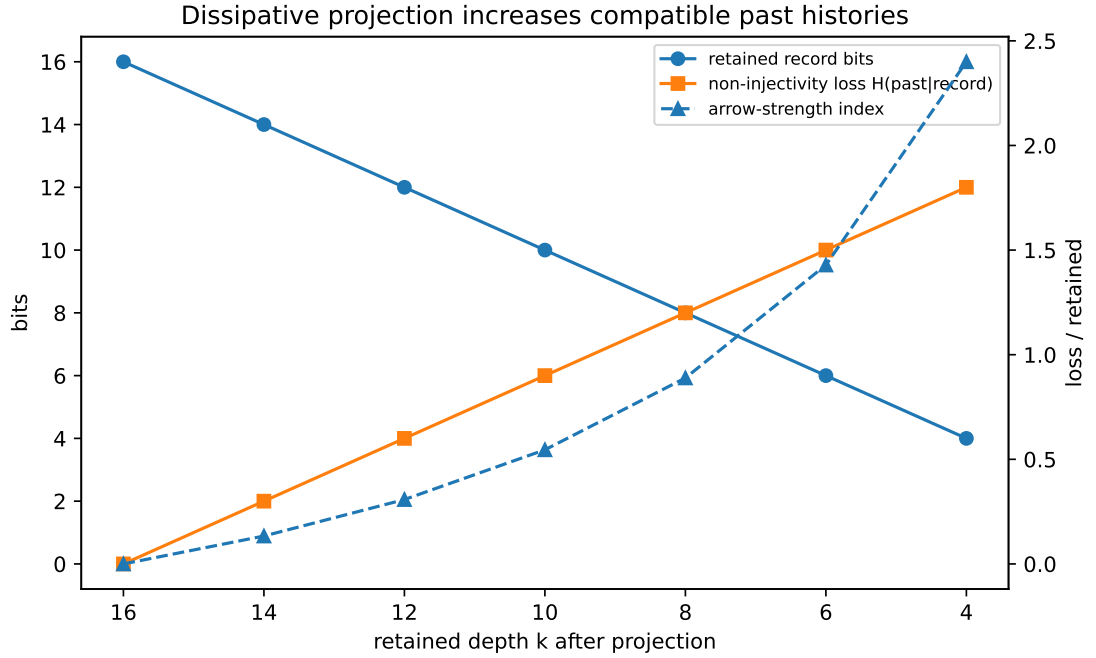


FIG. 6. Dissipative projection increases compatible past histories. Retaining fewer temporal distinctions increases non-injectivity loss $H(\text{past}|\text{record})$ and the arrow-strength index.

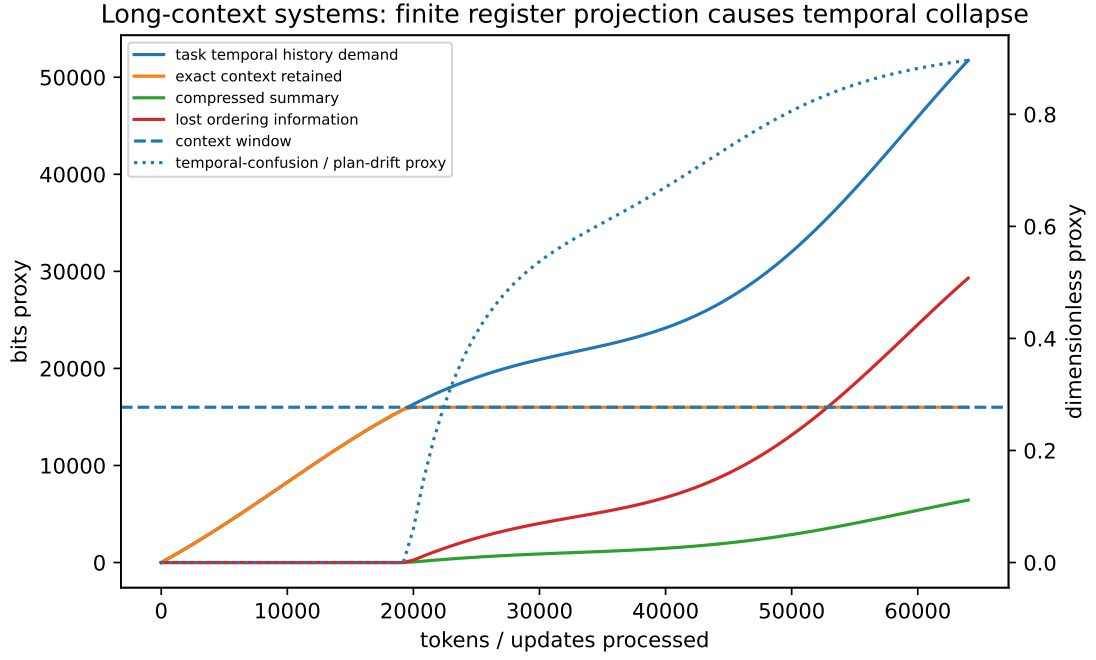


FIG. 7. Long-context systems as finite temporal registers. When task-history demand exceeds exact context capacity, the system must summarize or discard. The lost-order proxy increases together with temporal-confusion and plan-drift proxies. This is an engineering projection, not a claim about subjective experience.

DOMAIN-SPECIFIC PROJECTIONS

Distributed acquisition and control

In high-rate acquisition, robotics, and distributed control, apparent temporal failures often occur upstream of planning or inference. Buffers fill, events are timestamped late, logs are garbage-collected, channels desynchronize, and causally earlier events can be recorded after later events. O2 provides a finite-register diagnostic: if C_{upd} and C_{sync} fall below event entropy rate and synchronization burden, temporal order degrades before the controller sees the world.

Long-context AI and temporal-collapse failure

Transformer-based and large language model systems provide a useful engineering example of finite temporal registers with bounded context windows [17, 18]. A long-context AI system can be treated as a finite temporal register with context capacity C_{ctx} . Empirical studies show that information access can degrade with position and context length, even when the nominal context window is large [20, 21]. Let $\mathcal{H}_t$ be the task history and $R_{\text{task}}(t)$ the number of temporal distinctions required to preserve step order, commitments, dependencies, and causal state. If

$$R_{\text{task}}(t) > C_{\text{ctx}}, \quad (25)$$

then the system must apply a projection

$$\pi_{\text{ctx}} : \mathcal{H}_t \rightarrow \tilde{\mathcal{H}}_t. \quad (26)$$

The resulting history is not the original past; it is a compressed, truncated, or distributional representation. This can produce temporal hallucination, step-order confusion, lost commitments, plan drift, and false memory of earlier instructions. This is an engineering analogy and diagnostic projection, not a claim that language models possess physical time experience.

Human and biological time

Biological time perception also depends on finite record formation, attention, working memory, and update rate. O2 suggests that some timing distortions can be interpreted as finite-register effects: event density exceeds update capacity, attention gates update frequency, or memory compression merges adjacent temporal distinctions. This is a projection for future empirical work, not a biological theory completed here.

EXPERIMENTAL PROTOCOLS

Protocol 1 (Finite temporal budget-crossing test). *Specify a finite register boundary, clock source, record carrier, buffer, and retention window. Estimate C_{rec} , C_{clk} , C_{buf} , C_{upd} , C_{log} , $C_{\text{ext}}^{\text{eff}}$, and C_{sync} . Increase event entropy rate, synchronization burden, or history length until $R_{\text{min}}^{(\tau)}$ crosses C_{time} . Measure reconstruction depth, tick collisions, buffer occupancy, order inversions, overwrite events, externalization, and heat proxy.*

Protocol 2 (Causal-dependency degradation test). *Feed a controlled causal event graph into a finite logging system. Remove or compress dependency edges while keeping raw timestamps available. O2 predicts that temporal order judgments should degrade when dependency traces are removed, even if clock labels remain.*

Protocol 3 (Latency inversion test). *Feed a timestamped ordered event stream through a finite buffer under increasing load. Compare source order with recorded order. O2 predicts local order inversions when latency variability exceeds event separation.*

Protocol 4 (Synchronization bottleneck test). *Let two observers synchronize through a channel of adjustable capacity and latency. Measure shared-order agreement and simultaneity ambiguity as a function of channel capacity. O2 predicts a finite ambiguity band controlled by latency uncertainty and record-exchange capacity.*

Protocol 5 (Projection-depth reconstruction test). *Retain only the last K layers, bits, or dependency edges of an event history. Measure the number of compatible past histories consistent with the current record. O2 predicts growth of $H(\text{past}|\text{record})$ as retained depth decreases.*

LIMITATIONS AND FALSIFICATION

First, O2 is an operational finite-record model, not a complete theory of time. It does not solve the cosmological low-entropy initial condition problem. Second, it does not deny reversible microdynamics. It states that finite accessible records can evolve by non-injective update maps even when a larger embedding dynamics is reversible. Third, it does not define a preferred foliation of spacetime. Each observer’s temporal access is implemented by local clocks, causal communication, and finite record order. Fourth, Landauer accounting applies only to physically implemented logically irreversible operations under relevant thermodynamic assumptions. Fifth, simulations in this paper are synthetic demonstrations, not empirical tests.

The strong version of FDS-O2 would be weakened or rejected by any of the following:

1. a finite physical observer that uses temporal order without any record order, clock label, update trace, causal dependency, synchronization record, or memory of change;
2. stable temporal order obtained solely from labels after dependency edges, causal traces, and synchronization metadata are removed or scrambled;
3. repeated finite-memory overwrite or many-to-one compression with perfect reconstruction of erased distinctions from the current finite record alone;
4. arbitrary event density totally ordered by bounded clocks without tick collisions, latency uncertainty, or external ordering information;
5. bounded observers establishing global simultaneity without finite signal exchange, latency assumptions, clock synchronization, or record communication;
6. no change in reconstruction depth, latency, ordering ambiguity, buffer occupancy, synchronization uncertainty, or erasure events when temporal demand crosses independently estimated temporal capacity;
7. logically irreversible temporal record reuse below generalized Landauer accounting after all reservoirs, feedback records, correlations, and work sources are included.

CONCLUSION

A finite observer does not use time as a magical external primitive. It uses temporal order through finite records: timestamps, dependency edges, update traces, logs, residues, memory states, communication messages, and clocks. Once this is made explicit, time inherits the finite-register constraints developed in O1. A temporal register has limited memory, limited clock precision, limited update throughput, limited buffer capacity, limited synchronization bandwidth, and limited thermodynamic budget.

The main result is operational: when temporal demand exceeds temporal capacity, the observer cannot maintain full-fidelity order. It must coarsen clock ticks, buffer, delay, overwrite, compress, externalize, garbage-collect, relax the task, or fail. Causal dependency is prior to temporal labeling because labels themselves are finite records generated by physical update chains. The arrow of register time is the direction in which non-injective updates accumulate. Future records can preserve traces of the past, but erased distinctions cannot be reconstructed from the current finite record alone.

In this operational sense, the arrow available to a finite observer is the price of finite memory: to keep updating, the system must forget, compress, overwrite, or externalize parts of its own past. O2 therefore supplies the operational trident's second anchor. O1 states what it means to measure as a finite record. O2 states what it means for records to become temporally ordered under causal, irreversible update. O3 should next treat the macroscopic thermodynamic cost of maintaining such finite records under memory reuse.

Finally, FDS-O2 suggests an operational time-resolution/energy-throughput tradeoff. A finite register can use finer temporal ticks only by increasing its update rate, synchronization rate, memory turnover, or irreversible housekeeping operations. If each tick requires stabilizing b_{tick} bits and the tick interval is Δt , the update throughput must satisfy $C_{\text{upd}} \gtrsim b_{\text{tick}}/\Delta t$. When these updates require reset, overwrite, compression, or garbage collection, the housekeeping power has a Landauer-style lower bound $P_{\text{hk}} \gtrsim k_{\text{B}}T \ln 2 \cdot b_{\text{erase}}/\Delta t$. This is not a derivation of the quantum energy-time uncertainty relation. It is a macroscopic finite-register constraint: sharper operational time requires more physical updating.

Notation Summary

Simulation Parameters

The simulations are deterministic and use fixed synthetic parameters in `code/generate_results.py`. Figure 1 uses an update index $n \in [0, 100]$, a rising input-bit stream, and an 18-bit register capacity. Figure 2 compares a 28-bit fixed-memory register with append-only logging and delayed garbage collection after update 90. Figure 3 uses 160 source events with load pulses that increase latency and cause local order inversion. Figure 4 maps a bursty event stream into finite clock ticks with tick widths from 0.10 to 4.2. Figure 5 uses a synchronization load of 22 bits and finite channel capacities from 0.5 to 40 bits per exchange interval. Figure 6 uses an initial depth $K_0 = 16$ and retained depths $k \in \{16, 14, 12, 10, 8, 6, 4\}$. Figure 7 uses a 16,000-token context window proxy and a growing temporal task-history demand. No proprietary data, detector data, clock data, cosmological data, AI benchmark data, or human-subject data are used.

Reproducibility Checklist

1. Code availability: all simulation code is included in the replication package.
2. Deterministic execution: the random seeds are

TABLE III. FDS-O2 notation summary.

Symbol	Meaning
$\mathcal{O}$	finite observer or distinction-register
$\mathcal{R}$	finite temporal register
$\mathcal{Z}$	finite record state space
z_n	record state after update n
r_n	temporal record tuple $(z_n, \kappa_n, d_n, \eta_n)$
κ_n	finite clock label or ordering marker
d_n	dependency pointer or causal trace
U_n	update map from current record and input to next record
$\mathcal{H}_n$	possible histories up to update n
π_n	projection from histories to finite records
$\prec_{\mathcal{O}}$	operational temporal order available to observer $\mathcal{O}$
$\prec_U$	causal/update dependency relation
C_{time}	accessible temporal ordering capacity
C_{sync}	synchronization capacity between finite observers
C_{clk}	clock-label capacity after jitter and synchronization uncertainty
C_{upd}	update-throughput capacity
$R_{\min}^{(\tau)}(\varepsilon; \Psi)$	minimal task-relevant temporal distinction demand
Δ_{O2}	temporal capacity deficit
$\mathcal{L}_U$	non-injectivity loss $H(X Z)$ of update U
P_{hk}	housekeeping power associated with irreversible record reuse

fixed.

3. Figure reproduction: running `python code/generate_results.py` regenerates all figures and CSV outputs.
4. Data status: all numerical outputs are synthetic demonstrations generated from the stated model.
5. Platform independence: the code uses standard Python scientific libraries.

Boundary of Applicability

FDS-O2 applies to systems that implement finite temporal records: they have record carriers, finite clocks, finite buffers, finite update throughput, finite retention windows, finite synchronization channels, and finite memory reuse constraints. It does not apply to ideal mathematical time parameters with unbounded precision, nor does it make claims about nonunitary fundamental dynamics or a preferred cosmic foliation. It is compatible with reversible microscopic dynamics and standard relativity when those theories are interpreted as descriptions of larger state spaces, coordinate conventions, or invariant relations rather than as finite observer records.

CODE AVAILABILITY

The simulation code used to generate Figs. 1–7 is included in the accompanying replication package under `code/generate_results.py`. Running the script regenerates all figures (PDF and PNG) and CSV tables in a single pass.

AI ASSISTANCE DISCLOSURE

AI-assisted tools were used for language polishing, structural feedback, LaTeX drafting support, and code-debugging assistance. The author reviewed and edited all content and remains responsible for all claims, references, simulations, and conclusions. No AI system is listed as an author.

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Capacity Overflow, Effective Stochasticity, and Phase-B Invariants: Critical Deficit, Markov Closure, and Invariant Selection under Finite Projection

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FDS-T1 defines finite-observer distinguishability budgets. FDS-O1 turns those budgets into measurement capacity, and FDS-O2 turns finite record update into register time. FDS-T3 abstracts the common mechanism: capacity overflow. When task-relevant distinction demand exceeds accessible capacity, a finite observer or finite system cannot track all distinctions needed for full-fidelity prediction. The missing distinctions re-enter the accessible description as coarse-graining, effective stochasticity, hysteresis, false invariants, externalization demand, or failure. This paper formulates effective stochasticity as structured dynamics viewed through non-injective finite projection. Even when dynamics on a larger state space is deterministic, the induced dynamics on the accessible record space can be stochastic because multiple inaccessible states map to the same visible state while having different successors. This version strengthens the phase-transition layer: it defines the overflow deficit, predictive susceptibility, effective Markovian closure error, informational hysteresis, observer hierarchy, and a maintenance-cost selection score for Phase-B invariants. Phase-B invariants are coarse variables that remain cheap to update, slow to forget, and approximately Markovian after microstate tracking fails. Deterministic simulations illustrate projection-induced stochastic kernels, critical deficit susceptibility, Markov closure, invariant selection, irrecoverable post-overflow loss, capacity-relative stochasticity, Phase-A/Phase-B transition, and semantic drift in finite-context AI-like systems. The paper does not claim that all randomness is epistemic, that quantum randomness is derived, that topology is required for all persistence, or that fundamental conservation laws are reduced to observer capacity. Its narrower contribution is a finite-capacity bridge model: effective stochasticity is the shadow cast by finite projection, and law-like structure is the residue of distinctions that remain cheap, predictive, and slow to forget under overflow.

Scope and Boundary of the Theory. This paper studies effective stochasticity relative to finite distinguishability budgets. It does not deny the usefulness of stochastic models, does not claim that all physical randomness is reducible to ignorance, does not derive quantum measurement probabilities, does not modify unitary quantum mechanics, and does not assert that all persistent structures require topological protection. It treats stochasticity operationally: a finite system sees a transition as stochastic when its accessible record state does not carry the distinctions needed to make the transition deterministic. The result is compatible with deterministic, stochastic, quantum, and thermodynamic descriptions on larger or different state spaces.

Claim-status summary

Table I summarizes the central FDS-T3 claims, their epistemic status, and the conditions under which they should be weakened or rejected.

Keywords: capacity overflow; effective stochasticity; finite observer; finite distinguishability; coarse-graining; projection; stochastic kernels; hidden variables; Markov closure; predictive susceptibility; Phase-B invariants; hysteresis; semantic drift; information thermodynamics; active finite distinction systems.

INTRODUCTION

From finite budgets to overflow

FDS-T1 defines an observer-relative distinguishability budget: a finite physical observer cannot register, preserve, update, and operationally use unlimited distinctions [1]. FDS-O1 applies this budget to measurement, showing that stable record formation is limited by sensor, channel, memory, buffer, update, and thermodynamic bottlenecks [2]. FDS-O2 applies the same idea to time, treating usable temporal order as causally ordered finite-record update [3]. Both operational papers use the same structure: a task requires distinctions; the finite system can carry only some of them; when demand exceeds capacity, full-fidelity tracking exits into compression, delay, state merging, externalization, reset, or failure.

T3 abstracts the common mechanism. Capacity overflow occurs when the distinctions required for prediction, control, or persistence exceed the distinctions accessible to a finite system. The system can still operate, but only after projecting the larger state space into a smaller accessible record space. That projection can be many-to-one. When it is, the same visible state can correspond to multiple hidden states with different successors. To the finite system, the next visible state is no longer determined by the current visible state alone. The dynamics becomes effectively stochastic.

TABLE I. Central FDS-T3 claims, epistemic status, and demotion or failure conditions.

Claim	Status	What would weaken or falsify it
Capacity overflow occurs when task-relevant distinction demand exceeds accessible capacity. Non-injective projection induces effective stochasticity.	Operational criterion Conditional theorem	Full-fidelity tracking persists under bounded resources after independently estimated demand exceeds all accessible capacity. A many-to-one accessible projection always induces deterministic accessible transitions despite hidden successors varying within the same visible record class. Predictive error, transition entropy, and exit signatures vary smoothly through controlled capacity crossing with no susceptibility peak, kink, or rapid regime change.
Overflow has a critical-deficit signature.	Testable prediction	All coarse variables lose predictive information at the same rate under projection, regardless of update cost, closure error, or persistence utility. Discarded distinctions are perfectly reconstructed after capacity recovery without external logs, hidden reservoirs, or additional records.
Phase-B variables are selected by low update cost, slow information decay, and approximate Markov closure. Capacity recovery need not reverse overflow.	Selection principle Conditional prediction	The same process yields the same stochastic description independent of observer capacity, projection, measurement boundary, and retained memory. Context overflow in finite-window systems never increases false dependency, semantic drift, wrong task-state completion, or external-memory demand under matched tasks.
Stochastic descriptions are capacity-relative unless supported by capacity-independent noise sources. Long-context drift is a domain projection of wrong invariant completion under overflow.	Scope claim Engineering projection	

Main thesis

The main thesis is:

effective stochasticity $\equiv$ structured dynamics
viewed after finite projection.

(1)

This is an operational thesis, not a metaphysical thesis. It does not say that the underlying world is deterministic. It does not say that quantum randomness is merely ignorance. It says that, for any finite observer or finite register, lost distinctions can appear as transition probabilities because the accessible state is no longer sufficient to determine the accessible successor.

The stronger contribution of this version is the selection layer. Overflow does not merely destroy detail. It selects coarse variables. Some variables disappear. Some become noisy. Some remain predictive because they are cheap to update, slow to forget, and nearly closed under the projected dynamics. Those variables are called Phase-B invariants.

Contributions

The paper makes seven contributions.

1. It defines capacity overflow as a positive deficit be-

tween task-relevant distinction demand and accessible capacity.

2. It derives an induced stochastic kernel on the accessible record space from deterministic dynamics on a larger state space under non-injective projection.
3. It defines predictive susceptibility as an order-parameter-like signature near the critical capacity deficit.
4. It defines Markov closure error and uses it to explain why some coarse variables become law-like after overflow.
5. It defines a maintenance-cost selection score for Phase-B invariants.
6. It formalizes informational hysteresis: capacity recovery restores possible storage, not distinctions already discarded.
7. It provides deterministic synthetic simulations and reproducible code.

Relation to earlier stochasticity drafts

An earlier stochasticity draft argued that truncation generates an unmodeled environment operationally in-

distinguishable from noise, then connected that noise to topological persistence and signal-to-noise collapse. It also emphasized that non-topological structures can lose local signal-to-noise ratio while boundary or topological structures may preserve finite signal. The present T3 version narrows the claim. It does not assert that all stochasticity is forced by truncation, that all persistent structures require topology, or that effective Markovianity follows automatically from ignorance. Instead, it formulates the safer bridge: finite non-injective projection induces observer-relative stochastic kernels, and some coarse variables remain stable under overflow. Topological protection is treated as one possible domain-specific example of Phase-B invariance, not as the universal form of persistence.

RELATED WORK

Coarse-graining, rate-distortion, and stochastic descriptions

Shannon information theory and rate-distortion theory provide the standard language for capacity, compression, and distortion [4–7]. Jaynes’ maximum entropy approach explains how probability assignments can arise from constrained information [8]. In statistical mechanics and nonequilibrium theory, coarse-graining and projection can generate effective irreversibility and stochastic evolution when unresolved degrees of freedom are integrated out [9–12]. FDS-T3 is continuous with this tradition, but its emphasis is the finite-observer capacity ledger: stochasticity is not only a mathematical convenience; it can be the accessible image of budget crossing.

Hidden Markov structure and Markov closure

Non-injective observation of a larger process often produces hidden-state dynamics. The visible process may fail to be Markov even if the underlying process is Markov or deterministic, because unresolved hidden variables carry memory. This is the standard lesson of hidden Markov models and predictive-state approaches [13, 14]. Recent work on quantitative coarse-graining of Markov chains provides error bounds for effective dynamics after projection, reinforcing the need to distinguish useful Markov closure from uncontrolled hidden-memory loss [28]. T3 therefore does not assume that projection automatically produces a clean Markov process. It asks which coarse variables minimize the residual memory required for prediction.

Renormalization and effective variables

Renormalization describes how large-scale behavior can be governed by variables that survive changes of scale while microscopic details are discarded [15, 16]. FDS-T3 uses a similar intuition in a finite-register setting. Phase-B invariants are not necessarily renormalization fixed points, but they play an analogous operational role: they are variables that remain predictive after repeated projection eliminates finer distinctions.

Information thermodynamics and memory reuse

Landauer’s principle links logically irreversible erasure to thermodynamic cost [17]. Bennett showed that computation can be arranged reversibly only when enough memory and garbage management are available [18]. Modern information thermodynamics refines these ideas in feedback, correlations, stochastic dynamics, and nonequilibrium systems [12, 19, 20]. Recent reviews emphasize that practical erasure and computation often require accounting beyond the ideal single-bit Landauer bound, including finite-time, finite-bath, nonequilibrium, non-Markovian, quantum, and error-correction effects [32]. T3 does not claim that every stochastic kernel dissipates heat. The heat cost enters when finite projection is physically implemented by overwrite, reset, many-to-one compression, or garbage collection.

Topological, boundary, and symmetry-stabilized examples

Topological phases illustrate how some coarse invariants can remain stable under local perturbation [21–24]. Non-Hermitian systems and skin-effect models show that boundary localization and non-reciprocal dynamics can produce robust boundary signatures [25–27]. T3 does not require these examples, but they motivate the term Phase-B invariant: a coarse, distributed, or boundary-stabilized distinction that survives after local microstate tracking fails. In a broader language, Phase-B invariants can be interpreted as coarse-grained symmetries of the projected dynamics. This paper does not derive fundamental conservation laws from finite capacity; it only supplies an operational bridge for why certain coarse variables become stable carriers of prediction.

CAPACITY, DEMAND, AND OVERFLOW

Definition 1 (Underlying state space and accessible record space). *Let $\mathcal{X}$ be a state space representing distinctions available in a larger model, and let $\mathcal{Z}_C$ be the*

record space accessible to a finite observer with capacity C . The observer implements a projection

$$\pi_C : \mathcal{X} \rightarrow \mathcal{Z}_C, \quad |\mathcal{Z}_C| \leq 2^C. \quad (2)$$

The accessible record at update index t is $Z_t = \pi_C(X_t)$.

Definition 2 (Task-relevant distinction demand). For task family Ψ , distortion tolerance ε , and window τ , let

$$R_{\min}^{(\tau)}(\varepsilon; \Psi) \quad (3)$$

be the minimum number of task-relevant bits required to meet the predictive, control, or reconstruction target.

Definition 3 (Capacity overflow). Given accessible capacity $C_{\text{acc}}(t)$, the T3 overflow deficit is

$$\Delta_{\text{T3}}(t) = R_{\min}^{(\tau)}(\varepsilon; \Psi_t) - C_{\text{acc}}(t). \quad (4)$$

Overflow occurs when $\Delta_{\text{T3}}(t) > 0$ over the relevant window.

Overflow is not ordinary ignorance. Ordinary ignorance may be resolved by measurement or computation within the same boundary. Overflow is a boundary state: the distinctions needed for the task are not carried by the accessible record space under the current capacity, update rate, memory, and externalization options.

Theorem 1 (Overflow-exit theorem). Let O be a finite system attempting task Ψ over window τ . If $R_{\min}^{(\tau)}(\varepsilon; \Psi) > C_{\text{acc}}$, then full-fidelity task performance at distortion ε is impossible unless at least one exit occurs: increased capacity, longer time, buffering, externalization, lossy compression, coarse-graining, stochastic modeling, task relaxation, boundary enlargement, or failure.

Proof. The demand term is the minimum number of distinctions required by the task. The capacity term is the maximum number of distinctions accessible to the system under the specified boundary and window. If demand exceeds capacity, a full-fidelity solution would require distinctions not represented by any admissible record state or update sequence. The system must therefore alter demand, alter capacity, store or externalize backlog, compress, accept stochastic residuals, relax the task, or fail. These alternatives exhaust the ways to remove the strict inequality under finite resources. $\square$

The empirical content is not the inequality itself. The empirical content is the predicted pattern near crossing: transition entropy rises, reconstruction depth saturates, error floors appear, formerly deterministic variables become probabilistic, buffers fill, externalization increases, and coarse variables begin to dominate prediction.

Critical deficit and predictive susceptibility

To make the Phase-A/Phase-B distinction more physical, define a predictive error observable

$$E_P(\Delta) = \mathbb{E}[d(Z_{t+1}, \hat{Z}_{t+1}) \mid \Delta_{\text{T3}} = \Delta] \quad (5)$$

and a transition-entropy observable

$$H_P(\Delta) = H(Z_{t+1} \mid Z_t, \Delta_{\text{T3}} = \Delta). \quad (6)$$

The corresponding susceptibilities are

$$\chi_P = \frac{\partial E_P}{\partial \Delta_{\text{T3}}}, \quad \chi_H = \frac{\partial H_P}{\partial \Delta_{\text{T3}}}. \quad (7)$$

The T3 prediction is not that every finite system has a thermodynamic singularity. It is weaker and more testable: under controlled sustained crossing, χ_P or χ_H should exhibit a peak, kink, or rapid increase near $\Delta_{\text{T3}} \approx 0$ whenever the discarded distinctions are dynamically relevant. Phase-A is the tracking regime $\Delta_{\text{T3}} < 0$; Phase-B is the post-overflow coarse regime $\Delta_{\text{T3}} > 0$.

PROJECTION-INDUCED EFFECTIVE STOCHASTICITY

From deterministic dynamics to stochastic kernels

Assume first that the larger state evolves deterministically:

$$X_{t+1} = F(X_t). \quad (8)$$

The finite observer does not access X_t directly. It accesses $Z_t = \pi_C(X_t)$. The same accessible record z can have a preimage

$$\pi_C^{-1}(z) = \{x \in \mathcal{X} : \pi_C(x) = z\}. \quad (9)$$

If different $x \in \pi_C^{-1}(z)$ lead to different accessible successors, then the accessible transition is stochastic relative to z .

Definition 4 (Induced stochastic kernel). Given a prior or conditional distribution $P(x_t \mid z_t)$ over hidden states compatible with record z_t , the induced accessible transition kernel is

$$P_C(z' \mid z) = \sum_{x \in \pi_C^{-1}(z)} P(x \mid z) \mathbf{1}\{\pi_C(F(x)) = z'\}. \quad (10)$$

For stochastic dynamics $P(x' \mid x)$, the corresponding kernel is

$$P_C(z' \mid z) = \sum_{x \in \pi_C^{-1}(z)} \sum_{x' \in \pi_C^{-1}(z')} P(x' \mid x) P(x \mid z). \quad (11)$$

Theorem 2 (Projection-induced stochasticity). *If there exist $x_1, x_2 \in \pi_C^{-1}(z)$ such that $\pi_C(F(x_1)) \neq \pi_C(F(x_2))$, then $P_C(\cdot | z)$ is non-degenerate for any conditional distribution assigning positive probability to both x_1 and x_2 . Thus a deterministic underlying map can induce stochastic accessible dynamics.*

Proof. The two hidden states share the same accessible current record but have different accessible successors. Since the observer conditions only on z , both hidden states remain compatible. If both have positive conditional probability, at least two successor records have positive probability. Therefore the induced kernel is non-degenerate. $\square$

Non-injectivity loss and transition entropy

A useful measure of lost hidden distinction is

$$\mathcal{L}_\pi = H(X_t | Z_t). \quad (12)$$

A useful measure of effective stochasticity is

$$\mathcal{S}_{\text{eff}} = H(Z_{t+1} | Z_t). \quad (13)$$

The two are related but not identical. Projection loss measures unresolved present distinctions. Effective stochasticity measures how much those unresolved distinctions matter for the next accessible state. If hidden distinctions are dynamically irrelevant, projection loss can be high while transition entropy remains low. If hidden distinctions control successor states, projection loss becomes effective stochasticity.

Ontic noise, external noise, and effective noise

FDS-T3 distinguishes three cases.

1. *External noise*: stochastic forcing enters the larger state dynamics $P(x' | x)$.
2. *Projection-induced effective noise*: the larger dynamics may be deterministic, but non-injective projection makes the accessible dynamics stochastic.
3. *Modeling noise*: a deliberate approximation is used despite sufficient capacity in principle.

Only the second is the specific contribution of T3. The paper does not collapse all noise into observer ignorance.

PHASE-A, PHASE-B, AND MARKOV CLOSURE

Definition 5 (Phase-A tracking). *A system is in Phase-A relative to task Ψ when $\Delta_{T3} \leq 0$ and the accessible record retains enough distinctions for full-fidelity task performance at tolerance ε . Phase-A is the tracking regime.*

Definition 6 (Phase-B dynamics). *A system is in Phase-B relative to task Ψ when $\Delta_{T3} > 0$ persists and prediction or control is dominated by coarse variables, stochastic kernels, external records, or invariants rather than full microstate tracking. Phase-B is the post-overflow coarse regime.*

Effective Markovian closure

A non-injective projection generally produces hidden memory. Therefore T3 does not assume that the accessible process is strictly Markov. Define the k -step Markov closure error

$$\mathcal{M}_k(Z) = I(Z_{t+1}; Z_{t-k:t-1} | Z_t). \quad (14)$$

If $\mathcal{M}_k \approx 0$, the coarse variable Z_t is approximately closed for prediction at horizon one. If $\mathcal{M}_k$ is large, the current coarse state still requires retained history.

Definition 7 (Phase-B invariant). *Let $\phi : \mathcal{X} \rightarrow \mathcal{Y}$ be a coarse variable. ϕ is a Phase-B invariant over window τ if it satisfies three operational properties under overflow:*

1. *Projection stability*: ϕ remains recoverable from the accessible record or from a low-cost external record.
2. *Predictive persistence*: $I(\phi_t; \phi_{t+\tau})$ decays slowly relative to microstate detail.
3. *Approximate closure*: $\mathcal{M}_k(\phi)$ is small enough that future values are predicted from the current coarse record without needing the discarded microhistory.

Proposition 1 (Phase-B variables are approximate Markov closures). *Among variables available after sustained overflow, those that appear law-like to the finite observer are the variables that minimize residual memory burden while retaining predictive information. In symbols, Phase-B variables tend to have low $\mathcal{M}_k(\phi)$ and high $I(\phi_t; \phi_{t+\tau})$ per update cost.*

Argument. A variable with high predictive information but high closure error requires hidden history; under overflow that hidden history is not available. A variable with low closure error but no retained predictive information is cheap but useless. Variables that remain usable after overflow must therefore balance predictive utility with low memory burden. This is an operational selection principle, not a claim of exact optimality in every physical system. $\square$

This explains why macroscopic descriptions can be simple despite microscopic detail loss. The simplicity is not proof that microscopic distinctions were absent. It can be the result of a selection effect: only those coarse variables that close approximately under projection remain usable.

SELECTION PRESSURE FOR PHASE-B INVARIANTS

Maintenance-cost functional

Let Φ_C be the class of candidate coarse variables available under capacity C . For $\phi \in \Phi_C$, define a survival score

$$\mathcal{S}(\phi) = \frac{I(\phi_t; \phi_{t+\tau})}{\mathcal{C}_{\text{upd}}(\phi) + \mathcal{C}_{\text{maint}}(\phi) + \lambda \mathcal{M}_k(\phi) + \varepsilon_0}. \quad (15)$$

Here $\mathcal{C}_{\text{upd}}$ is update cost, $\mathcal{C}_{\text{maint}}$ is maintenance or externalization cost, $\mathcal{M}_k$ is Markov closure error, λ weights memory burden, and $\varepsilon_0 > 0$ avoids singularities. Phase-B invariant candidates are those with high $\mathcal{S}$.

A phenomenological survival probability can be written as

$$P_{\text{survive}}(\phi) \propto \exp \left[-\beta \frac{\mathcal{C}_{\text{upd}}(\phi) + \mathcal{C}_{\text{maint}}(\phi) + \lambda \mathcal{M}_k(\phi)}{I(\phi_t; \phi_{t+\tau}) + \varepsilon_0} \right]. \quad (16)$$

This is a selection heuristic, not a universal law. It captures the central idea: Phase-B invariants are forced survivors. They are not necessarily the most detailed variables. They are the variables that remain cheap, predictive, and slow to forget when detail is unaffordable.

Invariants as coarse-grained symmetries

Finite projection defines equivalence classes

$$x \sim_C x' \quad \text{if} \quad \pi_C(x) = \pi_C(x'). \quad (17)$$

A coarse variable ϕ that is constant on these equivalence classes and stable under projected dynamics behaves as a coarse-grained symmetry:

$$\phi(x) = \phi(x'), \quad x \sim_C x', \quad \phi_{t+1} \approx \phi_t. \quad (18)$$

This does not derive fundamental symmetries or conservation laws. It states a weaker operational fact: under finite projection, law-like structure is carried by variables that remain invariant across distinctions the observer can no longer afford to retain.

HYSTERESIS AND CAPACITY-RELATIVE RANDOMNESS

Informational hysteresis

Capacity recovery restores possible storage, not distinctions that were already discarded. Let Ω_{before} denote the set of histories distinguishable before overflow. During overflow, projection maps

$$\Omega_{\text{before}} \rightarrow \Omega_{\text{after}} = \pi_C(\Omega_{\text{before}}). \quad (19)$$

When capacity later recovers, the system may expand the record space, but without external logs it expands from Ω_{after} , not from the original preimage. The irrecoverable loss can be written

$$\mathcal{H}_{\text{irr}} = H(X_{\text{past}} \mid Z_t, C_{\text{recovered}}, L_{\text{ext}}), \quad (20)$$

where L_{ext} denotes external logs or reservoirs. If L_{ext} does not contain the discarded distinctions, then $\mathcal{H}_{\text{irr}} > 0$ even after C increases.

Proposition 2 (Informational hysteresis). *If overflow induces a many-to-one projection and the discarded preimage distinctions are not preserved in an external or reversible record, then subsequent capacity recovery cannot reconstruct those distinctions from the current accessible record alone.*

This is logical irreversibility before it is thermodynamic irreversibility. Thermodynamic cost enters when the projection is physically implemented by irreversible overwrite, reset, compression, or garbage collection.

Observer hierarchy and relative stochasticity

Consider two observers O_a and O_b with capacities $C_a < C_b$ and projections

$$\pi_a : \mathcal{X} \rightarrow \mathcal{Z}_a, \quad \pi_b : \mathcal{X} \rightarrow \mathcal{Z}_b. \quad (21)$$

If π_b retains distinctions discarded by π_a , then typically

$$H(Z_{t+1}^a \mid Z_t^a) \geq H(Z_{t+1}^b \mid Z_t^b), \quad (22)$$

for the same underlying process and comparable priors. The inequality need not hold in every artificial projection, but it gives the expected direction when additional capacity preserves dynamically relevant distinctions.

Randomness is therefore capacity-relative, not merely subjective. A phenomenon that is stochastic for a low-capacity observer can become more deterministic for a higher-capacity observer. This does not imply that all stochasticity is removable. Some noise sources may remain capacity-independent, and quantum randomness is outside the scope of this paper.

NUMERICAL MODELS AND SIMULATIONS

The simulations are deterministic synthetic demonstrations. They are not fits to physical detector data, biological data, quantum experiments, AI benchmarks, or human behavior. All figures and CSV outputs are generated by `code/generate_results.py`.

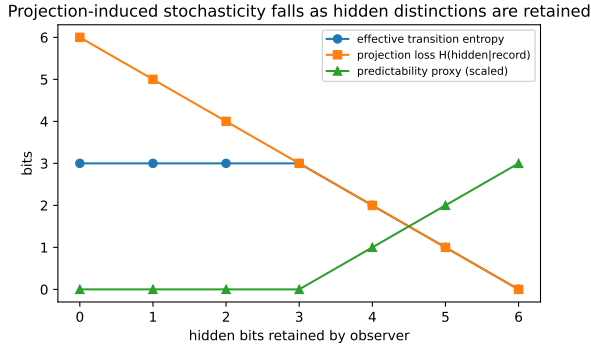


FIG. 1. Projection-induced stochasticity falls as hidden distinctions are retained. A deterministic hidden-state update appears stochastic to a low-capacity observer because unresolved hidden states inside the same visible record have different successors.

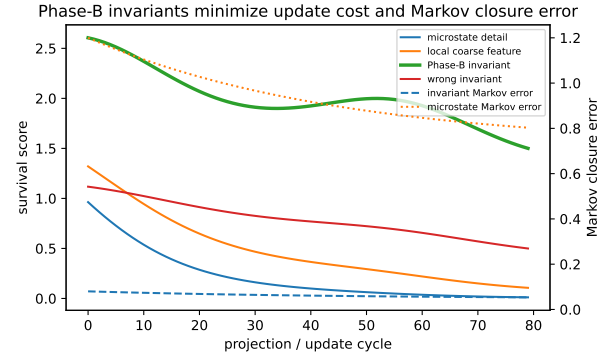


FIG. 3. Phase-B invariants minimize update cost and Markov closure error. The surviving variable is not the most detailed variable; it is the variable with the best ratio of predictive persistence to update and memory burden.

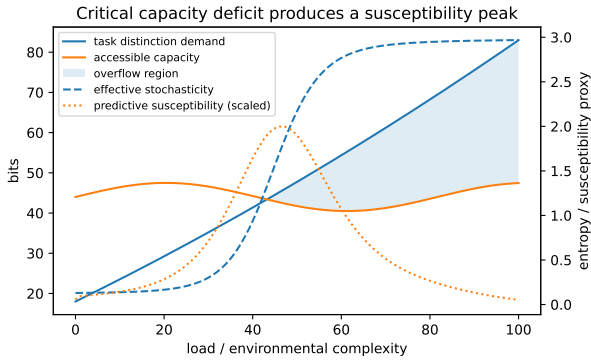


FIG. 2. Critical capacity deficit produces a susceptibility peak. When distinction demand crosses accessible capacity, effective stochasticity rises and the predictive susceptibility shows a sharp response.

Projection-induced stochasticity

The first simulation uses a deterministic hidden-state map. Observers retain different numbers of hidden bits. With few retained bits, the induced accessible transition has high entropy. As hidden distinctions are retained, stochasticity falls.

Critical deficit and susceptibility

The second simulation raises task demand through accessible capacity. Transition entropy and error rise near crossing; predictive susceptibility peaks near the critical deficit.

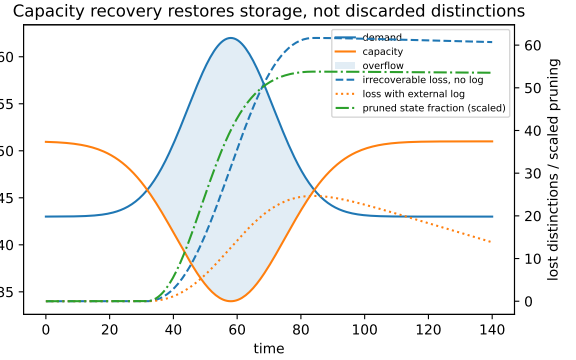


FIG. 4. Capacity recovery restores storage, not discarded distinctions. Overflow prunes the preimage. Without external logs, capacity recovery cannot reconstruct the distinctions already discarded.

Markov closure and invariant selection

The third simulation compares candidate variables under repeated projection. Microstate detail has high information but high update cost and high closure burden. The Phase-B invariant has lower detail but better survival score because it is cheaper and more Markov-closed.

Informational hysteresis

The fourth simulation imposes a temporary overload and then restores capacity. Without external logs, lost distinctions remain lost. External logging reduces hysteresis but costs storage and later cleanup.

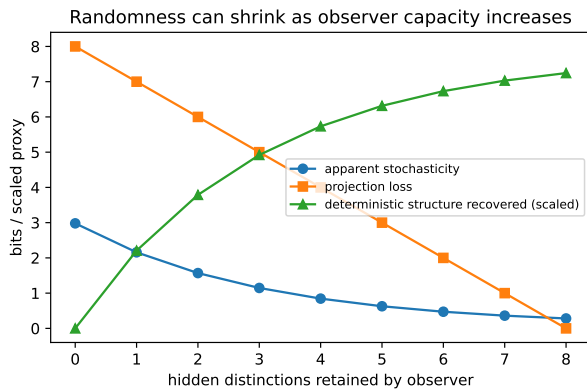


FIG. 5. Randomness can shrink as observer capacity increases. The same underlying process appears more stochastic to lower-capacity observers and more structured to higher-capacity observers.

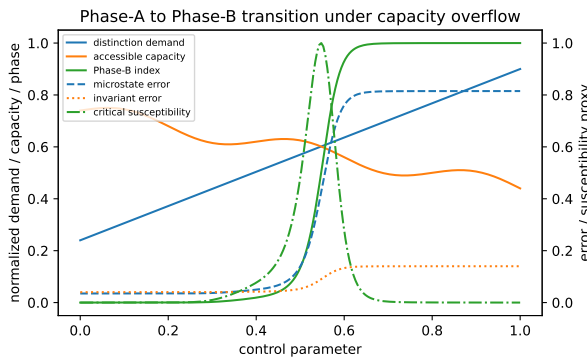


FIG. 6. Phase-A to Phase-B transition under capacity overflow. As demand crosses accessible capacity, microstate tracking fails, effective stochasticity rises, and prediction shifts toward Phase-B invariants.

Observer hierarchy

The fifth simulation varies retained hidden distinctions. Apparent stochasticity decreases as observer capacity increases, illustrating that stochastic descriptions can be capacity-relative.

Phase-A to Phase-B transition

The sixth simulation varies a control parameter that raises demand and lowers effective capacity. Once deficit becomes positive, the Phase-B index rises. Microstate error increases sharply while invariant error remains comparatively small.

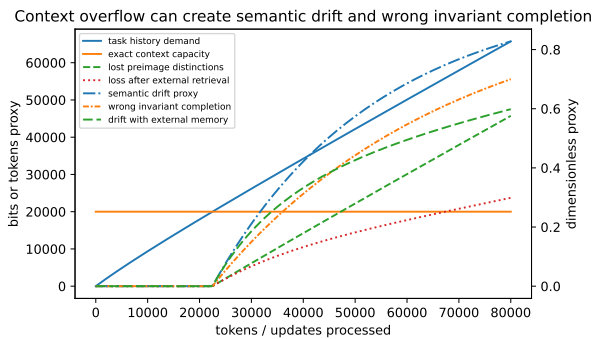


FIG. 7. Context overflow can create semantic drift and wrong invariant completion. In a finite-window system, overflow does not merely remove information; it can force the system to maintain coherence by selecting an incorrect coarse invariant. External retrieval increases effective capacity and pushes the system back toward Phase-A.

Semantic drift and wrong invariant completion

The seventh simulation treats a finite-context AI-like system as a domain projection. When task history exceeds context capacity, multiple histories collapse into the same accessible context. The system may complete the missing preimage with a wrong invariant, producing semantic drift or false dependency. External retrieval reduces but does not eliminate the loss.

DOMAIN-SPECIFIC PROJECTIONS

Measurement systems

O1 is the measurement-specific projection of T3. A detector with insufficient sensor, channel, memory, buffer, or update capacity cannot stabilize all task-relevant distinctions. The missing distinctions appear as state merging, false positives, false negatives, noise floors, delayed records, or externalized logs. T3 supplies the abstract stochastic-kernel language for those O1 exits.

Temporal registers

O2 is the time-specific projection of T3. When temporal demand exceeds temporal capacity, finite records lose order information. Non-injective temporal update produces compatible past histories and register-time collapse. T3 generalizes this as projection-induced effective stochasticity under overflow.

Biology and active pruning

A biological system cannot track every environmental detail. It must prune, coarse-grain, ignore, or externalize. Phase-B invariants in living systems may include boundary states, homeostatic variables, gradients, cycles, or control-relevant features that survive after microdetails are discarded. T3 does not complete a biological theory, but it explains why active pruning and invariant selection are not optional for finite organisms.

AI context overflow, RAG, and semantic drift

Finite-context AI systems provide an engineering analogy. When task history exceeds context capacity, the system must summarize, retrieve, compress, or discard. Empirical long-context studies show that language models do not always use nominally available context uniformly; performance can degrade when relevant information appears away from the beginning or end of the input [29]. Benchmark work such as RULER further shows that nominal context length can substantially exceed effective usable context under increased task complexity [30]. NoLiMa suggests that long-context performance can degrade sharply when retrieval requires latent association rather than literal matching, consistent with the T3 view that accessible context is not identical to effective task-relevant capacity [31]. The same underlying task can then appear more stochastic to the agent: next-step predictions become sensitive to inaccessible prior distinctions. Semantic drift occurs when the system maintains a locally coherent but wrong invariant across a collapsed preimage of possible histories. Hallucination is not only random guessing in this projection; it can be wrong invariant completion under context overflow. Retrieval-augmented generation and external memory can be interpreted as capacity externalization: they increase effective C_{acc} and can move the system back toward Phase-A. This is an engineering projection, not a claim that language models possess subjective experience or physical agency.

Topological and boundary cases

In some physical systems, topological or boundary-stabilized variables may act as Phase-B invariants. This paper does not prove that topology is necessary for persistence. It states a weaker bridge: when local microstate tracking fails, variables that remain recoverable under projection and stable under dynamics become the usable carriers of persistence. Topological invariants are one important family of such variables.

EXPERIMENTAL PROTOCOLS

Protocol 1 (Projection-induced stochasticity test). *Choose a deterministic or high-fidelity simulated system with known state X_t . Define a family of projections π_C with decreasing capacity. Estimate the induced kernel $P_C(z' | z)$. T3 predicts that transition entropy should increase when discarded distinctions affect accessible successors.*

Protocol 2 (Critical deficit test). *Independently estimate task demand R_{min} and accessible capacity C_{acc} . Increase demand or reduce capacity until Δ_{T3} crosses zero. Measure predictive error, transition entropy, and their finite-difference susceptibilities. T3 predicts a rapid response, kink, or susceptibility peak near crossing when discarded distinctions are dynamically relevant.*

Protocol 3 (Markov closure test). *Track candidate coarse variables under repeated projection. Estimate $\mathcal{M}_k(\phi) = \mathbb{I}(\phi_{t+1}; \phi_{t-k:t-1} | \phi_t)$ and predictive information. T3 predicts that Phase-B invariants should have lower closure error and better survival score than ordinary microstate detail under overflow.*

Protocol 4 (Hysteresis test). *Apply a temporary overload and then restore capacity. Compare systems with and without external logs. T3 predicts that capacity recovery will not fully restore discarded distinctions unless those distinctions were externalized or reversibly logged.*

Protocol 5 (Observer hierarchy test). *Give observers or algorithms different hidden-state access to the same process. T3 predicts that accessible transition entropy should shrink as dynamically relevant distinctions are retained, while capacity-independent noise components remain.*

LIMITATIONS AND FALSIFICATION

First, T3 is an operational finite-capacity model, not a complete theory of randomness. It does not settle whether quantum randomness is ontic, epistemic, relational, or otherwise. Second, it does not prove that all environments are Markovian. Effective Markovianity requires additional assumptions about timescale separation, memory, and update capacity. Third, it does not claim that topology is required for every form of persistence. Phase-B invariance is broader than topology. Fourth, Landauer-style heat accounting applies only to physical implementations of logically irreversible overwrite, reset, compression, or garbage collection under the relevant thermodynamic assumptions. Fifth, the simulations are synthetic demonstrations, not empirical validations.

The strong version of FDS-T3 would be weakened or rejected by any of the following:

1. controlled capacity crossing with no change in transition entropy, reconstruction depth, error floor, state merging, externalization, buffering, susceptibility, or failure;
2. non-injective projections of deterministic dynamics that never induce non-degenerate accessible transition kernels when hidden successors differ;
3. finite systems that recover erased distinctions perfectly after capacity recovery without external logs, hidden reservoirs, or additional records;
4. repeated projection under which all variables lose predictive information at the same rate, with no invariant-like coarse variables surviving longer than local details;
5. candidate Phase-B variables having no advantage in Markov closure error, update cost, or predictive persistence over discarded details under matched capacity;
6. apparent stochasticity remaining unchanged when observer capacity is increased to include independently verified dynamically relevant hidden distinctions;
7. physically implemented irreversible record reuse below generalized Landauer accounting after reservoirs, correlations, feedback, and work sources are included.

CONCLUSION

A finite system does not experience overflow as a philosophical abstraction. It experiences overflow as loss of distinctions. Once those distinctions are absent from the accessible record, the same visible state can be compatible with multiple hidden states and multiple successors. To the finite system, dynamics becomes stochastic.

FDS-T3 therefore supplies the Structural Trident’s third anchor. T1 states the finite-observer budget. O1 shows how measurement fails or exits under finite capacity. O2 shows how temporal order collapses under finite record update. T3 abstracts the shared mechanism: capacity overflow converts inaccessible distinctions into effective stochasticity, and repeated projection selects the invariants that remain useful after microstate tracking fails.

The most compact statement is this: effective stochasticity is the shadow cast by finite projection; stable law-like structure is the residue of distinctions that remain cheap, predictive, and slow to forget under overflow. Randomness is not simply the absence of law. It can be the appearance of law after the observer loses the distinctions that made the law deterministic. Conversely, law-like structure is not necessarily full information. It can

be the survivor of a harsh selection process: a Phase-B invariant that remains approximately closed when most distinctions have become unaffordable.

Notation Summary

Simulation Parameters

The simulations are deterministic and use fixed synthetic parameters in `code/generate_results.py`. Figure 1 uses eight visible records and sixty-four hidden residues per visible record. Figure 2 sweeps environmental complexity and computes predictive susceptibility from the finite-difference response of error to deficit. Figure 3 compares four candidate variables with different update costs, closure errors, and information decay constants. Figure 4 uses a temporary overload pulse and compares no-log versus external-log recovery. Figure 5 varies retained hidden bits. Figure 6 sweeps a control parameter across the Phase-A/Phase-B boundary. Figure 7 uses a finite context-window proxy with optional external retrieval. No proprietary data, detector data, biological data, quantum data, AI benchmark data, or human-subject data are used.

Reproducibility Checklist

1. Code availability: all simulation code is included in the replication package.
2. Deterministic execution: the random seeds are fixed or randomness is absent.
3. Figure reproduction: running `python code/generate_results.py` regenerates all figures and CSV outputs.
4. Data status: all numerical outputs are synthetic demonstrations generated from the stated model.
5. Platform independence: the code uses standard Python scientific libraries.

Boundary of Applicability

FDS-T3 applies to finite systems that represent only a projection of a larger state space and whose discarded distinctions are dynamically relevant to prediction, control, or persistence. It does not apply to ideal mathematical observers with unbounded state access, nor does it settle irreducible stochasticity in fundamental physics. It is compatible with standard stochastic modeling, quantum theory, statistical mechanics, and thermodynamics when those theories are interpreted on their own state

TABLE II. FDS-T3 notation summary.

Symbol	Meaning
$\mathcal{X}$	larger state space containing distinctions not all accessible to the observer
$\mathcal{Z}_C$	accessible record space under capacity C
π_C	finite projection from $\mathcal{X}$ to $\mathcal{Z}_C$
X_t	larger state at time or update index t
Z_t	accessible record $\pi_C(X_t)$
C_{acc}	accessible distinguishability capacity
$R_{\text{min}}^{(\tau)}(\varepsilon; \Psi)$	minimal task-relevant distinction demand
Δ_{T3}	T3 overflow deficit $R_{\text{min}} - C_{\text{acc}}$
$P_C(z' z)$	induced accessible stochastic kernel
$\mathcal{L}_\pi$	projection loss $H(X_t Z_t)$
$\mathcal{S}_{\text{eff}}$	effective stochasticity $H(Z_{t+1} Z_t)$
χ_P, χ_H	predictive and entropy susceptibilities near capacity crossing
$\mathcal{M}_k$	Markov closure error $I(Z_{t+1}; Z_{t-k:t-1} Z_t)$
$\mathcal{S}(\phi)$	survival score of candidate Phase-B invariant ϕ
Phase-A	full tracking regime before sustained overflow
Phase-B	post-overflow coarse dynamics regime
$\mathcal{H}_{\text{irr}}$	irrecoverable past distinction loss after capacity recovery

spaces. T3 adds a diagnostic layer: when accessible capacity changes, the amount and character of effective stochasticity should change if the added distinctions are dynamically relevant.

CODE AVAILABILITY

The simulation code used to generate Figs. 1–7 is included in the accompanying replication package under `code/generate_results.py`. Running the script regenerates all figures (PDF and PNG) and CSV tables in a single pass.

AI ASSISTANCE DISCLOSURE

AI-assisted tools were used for language polishing, structural feedback, LaTeX drafting support, and code-debugging assistance. The author reviewed and edited all content and remains responsible for all claims, references, simulations, and conclusions. No AI system is listed as an author.

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Physical Distinction Carriers and Erasure Maps: Accounting Boundaries, Distinction-to-Noise Ratio, and Thermodynamic Implementation

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The FDS formal core distinguishes formal distinctions from physically realized distinctions. A formal distinction is a partition or relation. A physical distinction must be carried by a substrate, read out with bounded error, retained over a task window, and thermodynamically accounted for when it is irreversibly reused. FDS-P1 develops the first paper in the Physical Bridge Ladder. It defines physical distinction carriers, accounting boundaries, side records, distinction-to-noise ratio (DNR), record-stability windows, erasure maps, dissipative projection, refresh cost, and full-boundary thermodynamic accounting. The central boundary statement is conservative: a mathematical projection does not dissipate heat by itself; a physical irreversible implementation by reset, overwrite, many-to-one compression, or garbage collection does. Reversible computation does not evade the ledger; it relocates preimage information into side records, memory-fill pressure, externalization, or later cleanup. The paper separates maintenance and refresh costs from Landauer-style erasure costs, and it clarifies how apparent sub-bound operations can arise when side records, feedback memories, reservoirs, or work sources are omitted from the accounting boundary. Deterministic simulations illustrate DNR-controlled readout error, thermal retention and refresh cost, accounting-boundary side records, logical versus dissipative projection, refresh-versus-erasure ledgers, full energy-flow accounting, and the interface between register-time resolution and physical turnover. The paper does not claim that every distinction dissipates heat, that every measurement is erasure, that Landauer's principle is newly derived, or that mathematical coarse-graining alone produces thermodynamic cost. Its contribution is a physical accounting interface between finite distinctions and thermodynamic implementation.

Scope and Boundary of the Theory. This paper is a physical bridge, not a derivation of thermodynamics from the distinction primitive. It does not assert that every formal distinction is physical, costly, or entropic. It does not assert that reading, copying, sensing, measurement, or reversible computation necessarily pays a Landauer erasure cost. It claims only that task-available physical distinctions require carriers, readout reliability, retention windows, and accounting boundaries; and that physically implemented logically irreversible reset, overwrite, many-to-one compression, or garbage collection requires generalized Landauer accounting under the relevant thermodynamic conditions.

Claim-status summary

Table I summarizes the central FDS-P1 claims, their epistemic status, and the conditions under which they should be weakened or rejected.

Keywords: physical distinction carriers; accounting boundary; distinction-to-noise ratio; finite records; Landauer principle; erasure maps; dissipative projection; refresh cost; record stability; reversible computation; side records; housekeeping dissipation; information thermodynamics; active finite distinction systems.

INTRODUCTION

Why P1 is needed

The FDS formal core starts from distinctions, but it explicitly separates a purely formal distinction from a physically realized record. A formal distinction is a logical separation. A realized distinction in an active finite system is an operational state maintained under finite capacity, update, noise, and resource constraints [1]. P1 is needed because later FDS papers use projections, registers, erasure, overflow, and invariant persistence. Without a physical accounting layer, those words can be misread as a category error: treating a mathematical map as a heat engine.

The P1 bridge is

$$\begin{aligned} &\text{formal distinction} \rightarrow \text{carrier} \rightarrow \text{record} \\ &\rightarrow \text{accounting boundary} \rightarrow \text{erasure map} \\ &\rightarrow \text{thermodynamic ledger.} \end{aligned} \quad (1)$$

This chain is narrower than the full FDS core. It does not treat life, cognition, geometry, topology, or social systems. It supplies the physical compliance layer needed for P2 on bounded-memory reversible computation, P5 on entropy production in active finite systems, O3 on finite-memory Second-Law channels, and P7 on invariant-supported persistence.

TABLE I. Central FDS-P1 claims, epistemic status, and failure or demotion conditions.

Claim	Status	What would weaken or falsify it
Task-available physical distinctions require carriers, readout reliability, and retention windows. Readout reliability is controlled by distinction-to-noise ratio and carrier-state overlap.	Operational definition	A finite physical system uses a distinction for a task without any carrier state, readable trace, retention interval, or external record.
	Testable bridge	Readout error remains invariant as state separation, noise variance, temperature, drift, or bandwidth are varied under controlled measurement.
The accounting boundary determines whether missing preimage information is erasure, side memory, externalization, or hidden-reservoir correlation. Many-to-one maps lose preimage information $H(X Y)$ unless side records are included.	Accounting principle	Full-boundary audits show that preimage information can be absent, stored, erased, and externalized without any boundary-dependent difference.
	Formal theorem	A non-injective deterministic map preserves enough information in its output alone to reconstruct every input under all nontrivial priors.
Mathematical projection is not heat; dissipative projection is a physical implementation claim. Refresh cost and erasure cost are distinct ledger terms.	Boundary statement	Not a direct empirical claim; it separates abstract maps from physical implementations.
	Testable accounting claim	Holding, refresh, clocking, isolation, and reset costs prove inseparable in every controlled implementation.
Reversible logging delays erasure by increasing memory-fill, synchronization, and later cleanup burden.	Conditional physical bridge	Bounded-memory reversible systems sustain unbounded updates indefinitely without memory growth, externalization, compression, cleanup, or failure.

Main boundary statement

The central boundary statement is

$$\begin{aligned} &\text{mathematical projection is not heat;} \\ &\text{irreversible physical implementation is.} \end{aligned} \quad (2)$$

A map $\pi : X \rightarrow Z$ may be many-to-one as an abstract description. That fact alone does not imply that a device has dissipated heat. Heat enters only when a physical substrate performs a logically irreversible operation such as reset, overwrite, many-to-one compression, or garbage collection under the physical assumptions of Landauer-style accounting.

This boundary is essential because FDS-O1, FDS-O2, and FDS-T3 all use finite projection and non-injective update [2–5]. P1 specifies when those projections become thermodynamic events.

Contributions

This paper makes eight contributions.

1. It defines physical distinction carriers and task-available records.
2. It introduces DNR as a measurable carrier-qualification variable.

3. It defines accounting boundaries as sets of physical degrees of freedom included in the ledger.
4. It distinguishes erasure, side-record memory growth, externalization, and reservoir correlation.
5. It defines logical projection versus dissipative projection.
6. It separates refresh, holding, clocking, and isolation costs from Landauer-style erasure costs.
7. It states the O2 interface: sharper register time requires higher physical turnover and, when irreversible reuse is involved, higher housekeeping power.
8. It provides deterministic synthetic simulations and reproducible code.

RELATED WORK

Information, records, and capacity

Shannon information theory provides the language of distinguishable messages, channel limits, and coding [6, 9]. Rate-distortion theory specifies how much information is needed to represent a source at tolerated distortion [7, 8]. FDS uses these tools operationally: a finite

system can only use distinctions that are carried by finite records and processed through finite channels.

Landauer, Bennett, and information thermodynamics

Landauer's principle links logically irreversible erasure to heat generation in a physical substrate [10]. Bennett showed that computation can in principle be arranged reversibly when adequate memory and garbage management are available [11]. Experiments and modern information thermodynamics refine this relation under feedback, finite-time protocols, correlations, and nonequilibrium conditions [13–19]. Recent many-body experiments further emphasize that generalized Landauer accounting depends on the chosen system–environment partition and on mutual-information and relative-entropy contributions [20]. P1 follows the conservative reading: erasure cost attaches to physical logically irreversible implementation, not to all information processing.

FDS core and the physical bridge

The FDS core defines logical erasure per update by

$$b_t = H(M_t | M_{t+1}, Y_t), \quad (3)$$

and under a physical Landauer bridge gives the informational heat-rate floor

$$\dot{Q}_{\text{info}} \geq \frac{k_B T \ln 2}{\tau} H(M_t | M_{t+1}, Y_t). \quad (4)$$

It also separates informational heat from physical losses such as control, coupling, leakage, friction, error correction, clocking, isolation, and transport [1]. P1 is the standalone paper that spells out the carrier, boundary, erasure-map, and implementation assumptions behind Eq. (4).

PHYSICAL DISTINCTION CARRIERS

Definition 1 (Formal distinction). *A formal distinction is a nontrivial separation of alternatives in a possibility space, such as a partition $\mathcal{D} = \{A_i\}_{i=1}^n$ with $n \geq 2$.*

Definition 2 (Physical distinction carrier). *A physical distinction carrier for a distinction value $d \in \{1, \dots, n\}$ is a family of physical states $\{s_d\}$ and a readout procedure R such that*

$$\mathbb{P}(R(s_d) = d) \geq 1 - \varepsilon \quad (5)$$

within a specified task interval, physical boundary, and measurement context.

A distinction is therefore not task-available merely because a formal partition exists. It is task-available only when a physical carrier can be read and retained with sufficient reliability.

Distinction-to-noise ratio

For scalar readout with two equal-variance Gaussian carrier states, define

$$\text{DNR} = \frac{|\mu_1 - \mu_0|}{\sigma}. \quad (6)$$

More generally, for vector readout with covariance Σ ,

$$\text{DNR}^2 = (\mu_1 - \mu_0)^T \Sigma^{-1} (\mu_1 - \mu_0). \quad (7)$$

For equal priors and equal-variance scalar readout, the Bayes error is

$$P_e = \Phi\left(-\frac{\text{DNR}}{2}\right) = \frac{1}{2} \text{erfc}\left(\frac{\text{DNR}}{2\sqrt{2}}\right), \quad (8)$$

where Φ is the standard normal cumulative distribution.

Criterion 1 (Carrier qualification by DNR). *For a task with tolerated readout error ε , a binary carrier satisfies the DNR qualification if $P_e(\text{DNR}) \leq \varepsilon$ under the specified readout model. In non-Gaussian cases, the DNR may be replaced by the optimal classifier error or another pre-registered distinguishability margin.*

DNR is not a new physical law. It is a diagnostic interface between formal distinctions and physical readout. When thermal noise, drift, overlap, or bandwidth limitations reduce DNR, a carrier can lose task availability even though the formal partition still exists.

Record stability and retention window

Definition 3 (Record stability window). *A physical distinction carrier has a record stability window $\tau_{\text{rec}}(\varepsilon)$ if*

$$\mathbb{P}(\hat{d}(t') = d | \hat{d}(t) = d) \geq 1 - \varepsilon \quad \text{for all } 0 \leq t' - t \leq \tau_{\text{rec}}(\varepsilon). \quad (9)$$

It is task-available over a task window τ_{task} only if $\tau_{\text{rec}}(\varepsilon) \geq \tau_{\text{task}}$ or if refresh, error correction, or external records supply an equivalent retention function.

A simple thermal-barrier model gives

$$\tau_{\text{rec}} \approx \tau_0 \exp(E_b/k_B T), \quad (10)$$

where E_b is an effective barrier and τ_0 a microscopic timescale. This is a toy retention model, not a universal law. It makes one point explicit: a record is physical only relative to stability under noise. Device-level memory studies likewise show that retention is a physical carrier property, controlled by material stability, diffusion, and thermodynamic phase structure rather than by abstract bit labels alone [21].

Proposition 1 (Carrier criterion). *A distinction is physically task-available only relative to a carrier, readout tolerance, task window, and accounting boundary.*

Proof. If there is no carrier, no physical state differs between alternatives. If there is no reliable readout, the alternatives cannot be operationally used. If the carrier does not remain stable or refreshed over the task window, the distinction is unavailable when needed. If the accounting boundary excludes the carrier or its side records, the distinction may be physically present elsewhere but unavailable to the system under that boundary. $\square$

ACCOUNTING BOUNDARIES AND SIDE RECORDS

Definition 4 (Accounting boundary). *An accounting boundary $\mathcal{A}$ is the set of physical degrees of freedom included in the thermodynamic and information ledger for an operation:*

$$\mathcal{A} = \{q_i : i \in I\}. \quad (11)$$

It may include a visible register, side records, feedback-controller memory, error-correction syndromes, work reservoirs, clocking systems, external logs, and thermal reservoirs, depending on the audit.

The same physical process can look reversible, irreversible, sub-bound, or externalized depending on which degrees of freedom are included. P1 therefore treats erasure as boundary-relative.

Let X be the pre-update record, Y the visible post-update record, $G_{\mathcal{A}}$ side information retained inside $\mathcal{A}$, and $G_{\bar{\mathcal{A}}}$ side information transferred outside $\mathcal{A}$.

Definition 5 (Erasure relative to an accounting boundary). *The preimage information erased relative to $\mathcal{A}$ is*

$$\mathcal{L}_{\mathcal{A}} = H(X | Y, G_{\mathcal{A}}). \quad (12)$$

If $G_{\mathcal{A}}$ contains enough side information to reconstruct X , then $\mathcal{L}_{\mathcal{A}} = 0$ even if $H(X | Y) > 0$.

Proposition 2 (Side-record relocation). *If an implementation embeds a many-to-one visible map into an injective enlarged map*

$$X \mapsto (Y, G_{\mathcal{A}}) \quad (13)$$

with $H(X | Y, G_{\mathcal{A}}) = 0$, then no erasure has occurred relative to $\mathcal{A}$. The cost has been relocated to memory fill, carrier maintenance, synchronization, externalization, or later garbage cleanup.

Proof. By definition, erasure relative to $\mathcal{A}$ is $H(X | Y, G_{\mathcal{A}})$. If the enlarged record is injective, the conditional entropy is zero. The preimage information remains present in $\mathcal{A}$, so the operation avoids immediate

Landauer erasure relative to that boundary. However, the retained side record occupies physical memory and must be carried, protected, synchronized, or eventually erased. $\square$

This is the P1 interface to reversible computation. Reversible computation does not make the ledger disappear; it enlarges the accounting boundary and delays payment [12]. Reversible logic delays erasure by retaining preimage information in side registers, at the cost of memory-fill pressure, refresh overhead, and eventual cleanup.

Proposition 3 (Accounting-boundary monotonicity). *If $\mathcal{A} \subseteq \mathcal{A}'$, then $\mathcal{L}_{\mathcal{A}'} \leq \mathcal{L}_{\mathcal{A}}$, where $\mathcal{L}_{\mathcal{A}} = H(X | Y, G_{\mathcal{A}})$. Enlarging the accounting boundary to include additional side records, controller memory, or reservoir correlations cannot increase the erased preimage information; it can only reduce or leave it unchanged.*

Proof. Adding degrees of freedom to $G_{\mathcal{A}}$ cannot increase the conditional entropy $H(X | Y, G_{\mathcal{A}})$ because conditioning on additional variables never increases entropy. Hence $\mathcal{L}_{\mathcal{A}'} \leq \mathcal{L}_{\mathcal{A}}$. $\square$

Sub-bound erasure illusion. The monotonicity proposition gives a formal handle on a central P1 theme. If an experimental or computational system appears to erase information without heat, it may be using a narrower accounting boundary that excludes side records, feedback memories, or reservoir correlations. The apparent sub-Landauer or sub-bound erasure is real relative to that boundary but does not violate full-boundary accounting when the boundary is widened.

ERASURE MAPS AND DISSIPATIVE PROJECTION

Definition 6 (Erasure map). *Let X be a pre-update physical record space and Y a post-update visible record space. A deterministic update $f : X \rightarrow Y$ is an erasure map relative to a prior distribution P_X and an accounting boundary $\mathcal{A}$ if f is many-to-one on a set of nonzero probability and the missing preimage is not retained in $G_{\mathcal{A}}$.*

Definition 7 (Preimage loss). *For a visible deterministic map $Y = f(X)$, the visible preimage loss is*

$$\mathcal{L}_f = H(X | Y). \quad (14)$$

For deterministic f ,

$$\mathcal{L}_f = H(X) - H(Y). \quad (15)$$

The erasure relevant to boundary $\mathcal{A}$ is $\mathcal{L}_{\mathcal{A}} = H(X | Y, G_{\mathcal{A}})$. We call $\mathcal{L}_{\mathcal{A}}$ the residual irreversibility of the operation relative to the accounting boundary $\mathcal{A}$. It is the part of the preimage that remains unrecoverable after all side records included in the ledger have been conditioned on.

Theorem 1 (Non-injective updates lose inverse information). *If $f : X \rightarrow Y$ is non-injective on a set of positive probability and the output does not include side information selecting the preimage, then $H(X | Y) > 0$ for some admissible priors. The output alone cannot reconstruct every input.*

Proof. Since f is non-injective, there exists $y \in Y$ with distinct preimages x_1, x_2 such that $f(x_1) = f(x_2) = y$. Choose a prior assigning positive probability to both x_1 and x_2 . Conditioning on $Y = y$ leaves at least two possible inputs with positive probability. Hence $H(X | Y = y) > 0$, and therefore $H(X | Y) > 0$. $\square$

Definition 8 (Logical projection). *A logical projection is an abstract or descriptive many-to-one map $\pi : X \rightarrow Z$. It can be used in a mathematical model without specifying any physical implementation.*

Definition 9 (Dissipative projection). *A dissipative projection Π_{diss}^A is a physical implementation of a many-to-one map inside accounting boundary $\mathcal{A}$ that discards preimage information by reset, overwrite, many-to-one compression, or garbage collection. It is an implementation claim, not a property of the abstract map alone.*

Thus,

$$\text{logical projection} \neq \text{dissipative projection}. \quad (16)$$

A finite projection in T3 or a non-injective update in O2 becomes thermodynamic only when it is physically implemented in this stronger sense.

THE LANDAUER ACCOUNTING BRIDGE

Assumption 1 (Landauer accounting bridge). *If a physical implementation within accounting boundary $\mathcal{A}$ reliably realizes a logically irreversible map with erased preimage information $\mathcal{L}_A$ bits while coupled to a thermal environment at temperature $T > 0$, then the minimal heat dumped to the environment under ideal quasistatic accounting satisfies*

$$Q_{\min}^A \geq k_B T \ln 2 \mathcal{L}_A. \quad (17)$$

Finite-time, finite-bath, feedback, correlated, strong-coupling, quantum, or nonideal implementations require generalized accounting and may dissipate more than this ideal floor.

Theorem 2 (Boundary-relative erasure heat floor). *For a physical many-to-one implementation satisfying the Landauer accounting bridge, the informational heat obeys Eq. (17). For repeated updates at timescale τ ,*

$$\dot{Q}_{\text{info}}^A \geq \frac{k_B T \ln 2}{\tau} H(X_t | Y_t, G_{A,t}). \quad (18)$$

Proof. $H(X_t | Y_t, G_{A,t})$ is the average missing preimage information after all side records inside $\mathcal{A}$ are included. Under the physical bridge, each erased bit contributes at least $k_B T \ln 2$ of heat in the ideal limit. Dividing by the update timescale gives the rate form. $\square$

Remark 1 (Sub-bound appearances). *If an experiment appears to erase below the ideal bound, P1 requires auditing $\mathcal{A}$: side records, feedback memories, measurement correlations, work reservoirs, clocking systems, and thermal reservoirs. A persistent reliable violation after full accounting would demote the physical bridge, not the formal FDS core.*

MAINTENANCE, REFRESH, AND ERASURE LEDGERS

A stable physical record can cost resources even when no erasure occurs. Holding a carrier away from thermal equilibrium, refreshing a leaky memory cell, maintaining an error-correcting code, stabilizing a molecular state, or synchronizing a clock can all carry physical cost. P1 separates the ledger:

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{phys}} + \dot{Q}_{\text{info}}, \quad (19)$$

where $\dot{Q}_{\text{info}}$ is the logically irreversible preimage-erasure floor and $\dot{Q}_{\text{phys}}$ includes non-erasure physical overhead.

A useful decomposition is

$$\begin{aligned} \dot{Q}_{\text{phys}} = & \dot{Q}_{\text{hold}} + \dot{Q}_{\text{refresh}} + \dot{Q}_{\text{read}} + \dot{Q}_{\text{ctrl}} + \dot{Q}_{\text{sync}} \\ & + \dot{Q}_{\text{ec}} + \dot{Q}_{\text{clock}} + \dot{Q}_{\text{isolate}} + \dot{Q}_{\text{transport}} + \dots \end{aligned} \quad (20)$$

The exact terms are device- and domain-specific. The point is not to replace engineering power models with one formula. It is to avoid conflating all cost with Landauer erasure.

Definition 10 (Refresh cost). *Refresh cost is the physical power required to preserve an existing distinction against leakage, drift, diffusion, thermal switching, decoherence, demagnetization, chemical turnover, or other destabilizing processes. For an event-based refresh model one may write*

$$\dot{Q}_{\text{refresh}} = \eta_{\text{ref}}^{-1} E_{\text{refresh}} f_{\text{refresh}}, \quad E_{\text{refresh}} = \alpha E_b, \quad (21)$$

where E_b is an effective stability barrier, α is an implementation-dependent repair fraction, and $\eta_{\text{ref}} \in (0, 1]$ is a refresh-efficiency factor. This is not a universal law for memory maintenance; it is a dimensional accounting proxy for systems whose retention is maintained by discrete refresh or repair events.

Definition 11 (Housekeeping dissipation). *Housekeeping dissipation is the ongoing physical cost of maintaining a record-bearing boundary, including holding, refresh,*

synchronization, error correction, isolation, clocking, and irreversible reuse.

Proposition 4 (Bounded-memory reuse pressure). *A system with finite memory undergoing sustained updates must eventually choose among increased memory, externalization, reversible side-record accumulation, irreversible cleanup, lossy compression, task relaxation, or failure.*

Proof. If every update is logged reversibly, memory demand grows with the number of retained preimage distinctions. With bounded memory, this growth cannot continue indefinitely. If memory does not increase and side records are not externalized, the system must compress, erase, reset, relax the task, or fail to retain the preimage. If it performs irreversible cleanup, the erased information enters the Landauer accounting term. $\square$

This proposition is the direct setup for FDS-P2. P1 defines the ledger. P2 studies the dynamics of bounded-memory reversible computation under that ledger.

INTERFACE WITH REGISTER TIME

FDS-O2 treats usable temporal order as finite-record update. P1 supplies the physical turnover account. If a finite register maintains clock ticks of b_{tick} bits at resolution Δt , the update throughput satisfies

$$\dot{I}_{\text{update}} \gtrsim \frac{b_{\text{tick}}}{\Delta t}. \quad (22)$$

If b_{erase} bits are irreversibly reused per tick, then the Landauer-style lower bound scales as

$$P_{\text{erase}} \geq k_B T \ln 2 \frac{b_{\text{erase}}}{\Delta t}. \quad (23)$$

This is not a claim of exponential cost. The lower bound is at least linear in irreversible turnover rate. Real systems may incur superlinear engineering costs when sharper timing requires stronger isolation, refresh, error correction, synchronization, or clocking.

Thus sharper register time requires more physical updating:

$$\Delta t \downarrow \Rightarrow \dot{I}_{\text{update}} \uparrow, \quad \dot{Q}_{\text{refresh}} + \dot{Q}_{\text{erase}} + \dot{Q}_{\text{clock}} \uparrow \quad (24)$$

under the relevant implementation assumptions.

NUMERICAL MODELS AND SIMULATIONS

The simulations are deterministic synthetic demonstrations. They are not fits to physical memory devices, detector data, quantum experiments, AI benchmarks, or biological records. They provide reproducible model diagrams for the P1 accounting layer. All figures and CSV outputs are generated by `code/generate_results.py`.

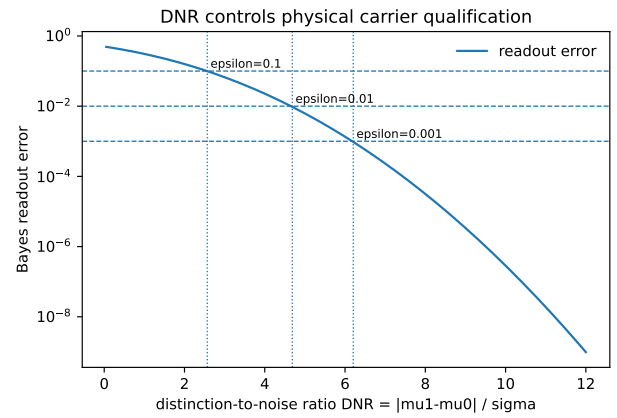


FIG. 1. DNR controls physical carrier qualification. For two equal-variance Gaussian carrier states, readout error falls as $\text{DNR} = |\mu_1 - \mu_0|/\sigma$ increases. A formal distinction becomes task-available only when the physical carrier clears the task tolerance.

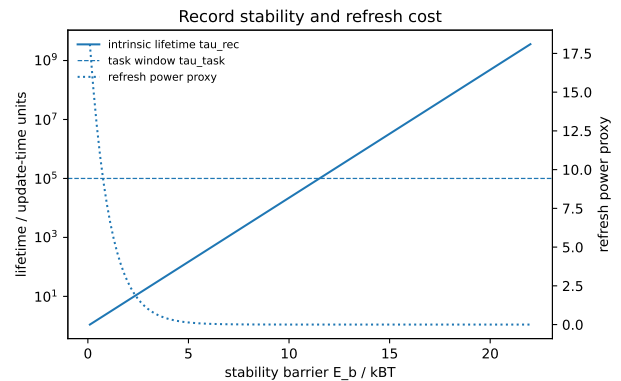


FIG. 2. Record stability and refresh cost. A barrier model gives an Arrhenius-like intrinsic lifetime. When intrinsic lifetime is shorter than the task window, the distinction requires refresh, error correction, or externalization to remain task-available.

Carrier qualification

Figures 1 and 2 illustrate the carrier criterion. DNR controls readout reliability. Barrier height and temperature control retention in the toy model. Both are physical properties of carriers, not properties of the formal partition alone.

Accounting boundaries and dissipative projection

Figures 3 and 4 show why boundary choice matters. A many-to-one visible map can be reversible relative to an enlarged boundary if side records retain the preimage. It becomes dissipative projection when the physical imple-

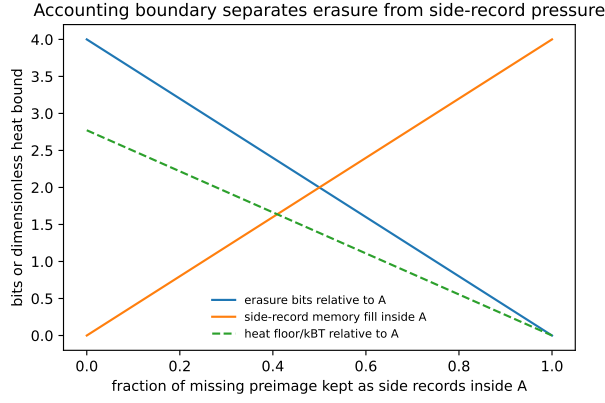


FIG. 3. Accounting boundary separates erasure from side-record pressure. If missing preimage information is retained inside $\mathcal{A}$ as side records, immediate erasure relative to $\mathcal{A}$ falls but memory-fill pressure grows. If no side records are retained, the visible many-to-one map incurs erasure relative to $\mathcal{A}$.

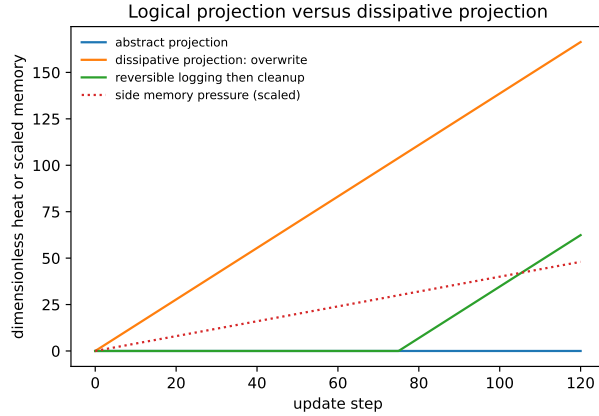


FIG. 4. Logical projection versus dissipative projection. An abstract projection has no thermodynamic ledger by itself. Physical overwrite accumulates erasure cost immediately. Reversible logging delays erasure by increasing side memory and later cleanup burden.

mentation irreversibly discards that preimage within the accounting boundary.

Refresh, full-boundary accounting, and time resolution

Figure 5 separates maintenance from erasure. Figure 6 visualizes the accounting-boundary audit. Figure 7 connects P1 to O2 by showing how finer register time forces higher physical turnover.

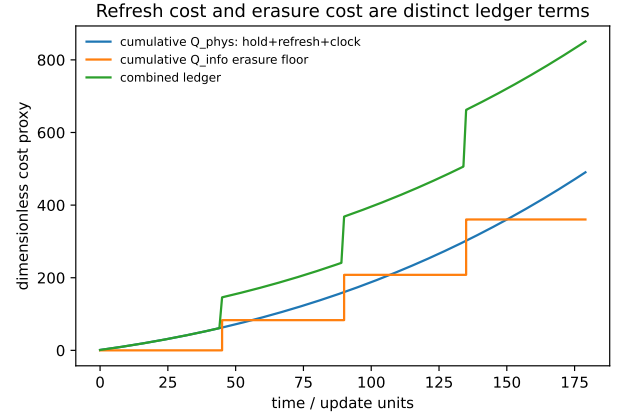


FIG. 5. Refresh cost and erasure cost are distinct ledger terms. Maintaining stable records can require ongoing holding, refresh, and clocking cost even when no erasure occurs. Batched cleanup adds a separate erasure floor.

Full accounting boundary can remove sub-bound illusion

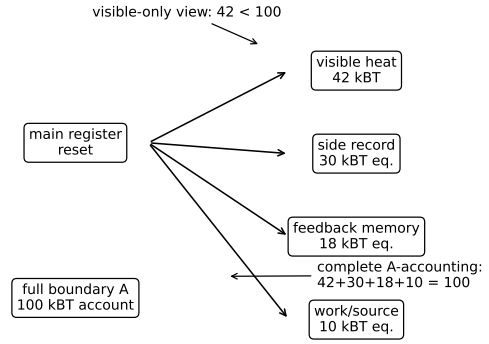


FIG. 6. Full accounting boundary can remove sub-bound illusion. Counting only visible heat can make an operation appear below the full erasure account. Including side records, feedback memory, work sources, and reservoirs restores the complete boundary ledger. Values are illustrative accounting units.

EXPERIMENTAL PROTOCOLS

Protocol 1 (DNR and carrier readout test). *Prepare two or more physical carrier states with tunable separation and noise. Measure the confusion matrix $P(\hat{d} | d)$ as a function of margin, temperature, drift, or measurement bandwidth. P1 predicts that task-available distinctions require readout error below the task tolerance.*

Protocol 2 (Retention and refresh test). *Prepare a record and vary barrier height, temperature, isolation, or refresh rate. Measure $\tau_{\text{rec}}(\epsilon)$ and compare it to τ_{task} . P1 predicts loss of task availability when $\tau_{\text{rec}} < \tau_{\text{task}}$ unless refresh, error correction, or external records compensate.*

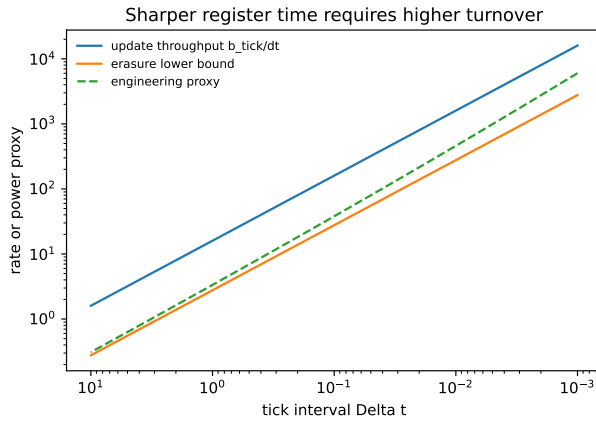


FIG. 7. Sharper register time requires higher turnover. Decreasing tick interval increases update throughput. If irreversible reuse occurs per tick, the lower-bound erasure power scales at least with turnover rate; nonideal engineering overhead may grow faster.

Protocol 3 (Accounting-boundary audit). *Implement a visible many-to-one map and progressively include side registers, feedback memories, reservoirs, and work sources in $\mathcal{A}$. P1 predicts that erasure relative to $\mathcal{A}$ is $H(X | Y, G_{\mathcal{A}})$, not simply $H(X | Y)$.*

Protocol 4 (Projection versus implementation test). *Compare three implementations of the same coarse map: abstract offline description, reversible logging of preimage information, and irreversible overwrite. P1 predicts no heat statement for the abstract map, growing memory for reversible logging, and erasure accounting for overwrite or cleanup.*

Protocol 5 (Refresh-erasure separation). *Operate a memory system with fixed record set and no reset, then with periodic reset under the same holding conditions. Estimate holding/refresh/clocking cost separately from erasure-event heat. P1 predicts that these ledgers are conceptually and often experimentally separable.*

Protocol 6 (Register-time turnover test). *Increase clock or update resolution while measuring refresh, clocking, error-correction, and erase/reset power. P1 predicts that the erasure lower bound grows at least with irreversible turnover rate, while engineering overhead may grow superlinearly depending on implementation.*

LIMITATIONS AND FALSIFICATION

First, P1 is not a universal theory of thermodynamics. It is a bridge between physical records and logically irreversible information loss. Second, it does not claim that every distinction has a thermodynamic cost. Formal distinctions can remain formal. Physical distinctions can be

maintained reversibly or with costs dominated by engineering overhead rather than erasure. Third, it does not claim that every measurement is erasure. Fourth, it does not resolve debates over the exact scope of Landauer’s principle in every finite-time, finite-bath, quantum, correlated, feedback, or strong-coupling setting. Fifth, the simulations are synthetic demonstrations.

The strong version of FDS-P1 would be weakened or rejected by any of the following:

1. a task-available physical distinction used by a finite system without carrier state, readout contrast, retention interval, causal trace, or external record;
2. controlled readout experiments in which state overlap, noise, drift, or DNR have no effect on reliability;
3. a many-to-one physical reset whose visible output alone reconstructs every input without side information;
4. reliable logically irreversible erasure below generalized Landauer accounting after all reservoirs, correlations, feedback records, work sources, side records, and boundary choices are included;
5. bounded-memory reversible systems that sustain unbounded updates without memory growth, externalization, cleanup, compression, or failure;
6. experiments showing that holding, refresh, clocking, and reset costs cannot be separated under any implementation;
7. systems maintaining arbitrarily fine register-time resolution without increased update throughput, refresh, clocking, or erasure burden.

CONCLUSION

FDS-P1 supplies the first rung of the Physical Bridge Ladder. It takes a concept that is formal in the FDS core-distinction—and specifies the physical conditions under which it becomes a usable record. The result is not a new derivation of Landauer’s principle. It is an accounting framework for finite distinction systems: a physical distinction must be carried, readable, stable, and bounded; side records must be included or excluded explicitly; a many-to-one erasure map loses preimage information relative to an accounting boundary; and a physical irreversible implementation of that loss enters the thermodynamic ledger.

The most compact statement is this: formal distinctions separate possibilities, but physical distinctions must be carried by finite records. Mathematical projection is not heat. Dissipative projection is a physical implementation with an accounting boundary. When finite records

are irreversibly reused, the erased preimage enters the heat account; when they are preserved reversibly, the cost reappears as memory fill, refresh, synchronization, externalization, or later cleanup. This is the bridge needed before one can discuss bounded-memory reversible computation, housekeeping dissipation, finite-memory entropy production, or topological obstruction to forgetting.

Generalized Carrier Distinguishability Metrics

The main text uses distinction-to-noise ratio (DNR) as the primary distinguishability metric for physical carriers. DNR is intuitive, experimentally accessible, and easy to visualize. However, the carrier criterion generalizes to any pre-registered distinguishability margin. A physical carrier qualifies for a task when a pre-registered error measure falls below the task tolerance, or equivalently when a pre-registered distinguishability margin exceeds the required threshold.

Each metric has domain-specific advantages. Total variation and Helstrom bound are prior-free operational error probabilities. Chernoff information gives the optimal error exponent for repeated or sequential discrimination. Fisher information connects distinguishability to parameter-estimation precision and Cramér-Rao bounds. In quantum carriers with equal priors, the Helstrom bound gives $P_e^* = \frac{1}{2}(1 - \frac{1}{2}\|\rho_1 - \rho_2\|_1)$. More generally, $P_e^* = \frac{1}{2}(1 - \|\pi_0\rho_0 - \pi_1\rho_1\|_1)$ for arbitrary priors.

For a given task with tolerance ϵ , the carrier qualifies if

$$P_e^*(\text{readout} \mid \text{carrier states}) \leq \epsilon, \quad (25)$$

where P_e^* is the minimum achievable classification error under the relevant physical readout constraints. DNR is the Gaussian binary special case of this criterion. The choice of metric is pre-registered as part of the experimental protocol.

Notation Summary

Simulation Parameters

The simulations are deterministic and use fixed synthetic parameters in `code/generate_results.py`. Figure 1 uses the Gaussian readout error $P_e = \Phi(-\text{DNR}/2)$. Figure 2 uses the Arrhenius lifetime proxy $\tau_{\text{rec}} = \tau_0 \exp(E_b/k_B T)$ and an event-based refresh power proxy. Figure 3 maps 256 inputs to 16 visible classes and varies the fraction of missing preimage information retained inside $\mathcal{A}$. Figure 4 compares abstract projection, physical overwrite, reversible logging, and delayed cleanup. Figure 5 separates holding, refresh, clocking, and erasure events. Figure 6 is an illustrative full-boundary accounting diagram. Figure 7 uses $b_{\text{tick}}/\Delta t$ and $b_{\text{erase}}/\Delta t$ proxies

for register-time turnover. No proprietary, device, biological, quantum, or human-subject data are used.

Reproducibility Checklist

1. Code availability: all simulation code is included in the replication package.
2. Deterministic execution: the random seed is fixed where randomness is used.
3. Figure reproduction: run `python code/generate_results.py`; the script regenerates all figures and CSV outputs.
4. Data status: all numerical outputs are synthetic demonstrations generated from the stated model.
5. Platform independence: the code uses standard Python scientific libraries.

Code Availability

The simulation code used to generate Figs. 1–7 is included in the accompanying replication package under `code/generate_results.py`. Running the script regenerates all figures (PDF and PNG) and CSV tables in a single pass.

AI Assistance Disclosure

AI-assisted tools were used for language polishing, structural feedback, LaTeX drafting support, and code-debugging assistance. The author reviewed and edited all content and remains responsible for all claims, references, simulations, and conclusions. No AI system is listed as an author.

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TABLE II. Generalized distinguishability metrics for the carrier criterion. In the Gaussian binary case DNR is sufficient; for non-Gaussian, structured, or quantum carriers the table gives candidate alternatives. Some entries, such as KL divergence and Fisher information, are local or model-dependent margins rather than direct single-shot error probabilities.

Metric	Definition	Typical threshold form
DNR	$(m_1 - m_2)/\sqrt{\sigma_1^2 + \sigma_2^2}$	$\text{DNR} \geq \text{DNR}_{\min}$
Total variation distance d_{TV}	$\frac{1}{2} \sum_x p_1(x) - p_2(x) $	$P_e^* = \frac{1}{2}(1 - d_{\text{TV}})$, so $d_{\text{TV}} \geq 1 - 2\epsilon$
Kullback–Leibler divergence D_{KL}	$\sum_x p_1(x) \log \frac{p_1(x)}{p_2(x)}$	$D_{\text{KL}} \geq d_{\min}$
Chernoff information C_*	$-\log \min_{s \in [0,1]} \sum_x p_1(x)^s p_2(x)^{1-s}$	$C_* \geq c_{\min}$
Fisher information $\mathcal{I}(\theta)$	$\mathbb{E}[(\partial_\theta \log p_\theta)^2]$	$\mathcal{I}(\theta) \geq \mathcal{I}_{\min}$
Helstrom bound P_e^*	$P_e^* = \frac{1}{2}(1 - \frac{1}{2}\ \rho_1 - \rho_2\ _1)$ for equal priors	$P_e^* \leq \epsilon$

TABLE III. FDS-P1 notation summary.

Symbol	Meaning
$\mathcal{D}$	formal distinction or partition
d	distinction value or label
s_d	physical carrier state realizing distinction value d
R	readout procedure
ε	tolerated readout or retention error
DNR	distinction-to-noise ratio, e.g. $ \mu_1 - \mu_0 /\sigma$
τ_{rec}	record stability window
τ_{task}	task window requiring the record
$\mathcal{A}$	accounting boundary: included physical degrees of freedom
$G_{\mathcal{A}}$	side record or garbage information retained inside $\mathcal{A}$
X	pre-update physical record space
Y	visible post-update record space
$f : X \rightarrow Y$	deterministic update or visible erasure map
$\mathcal{L}_f$	visible preimage loss $H(X Y)$
$\mathcal{L}_{\mathcal{A}}$	boundary-relative erasure $H(X Y, G_{\mathcal{A}})$
$\Pi_{\text{diss}}^{\mathcal{A}}$	dissipative projection implemented inside $\mathcal{A}$
$\dot{Q}_{\text{info}}$	informational heat-rate floor from logically irreversible erasure
$\dot{Q}_{\text{refresh}}$	refresh power required to maintain existing records
$\dot{Q}_{\text{phys}}$	non-erasure physical power: holding, readout, clocking, isolation, transport, error correction, etc.
$\dot{Q}_{\text{total}}$	total accounting rate $\dot{Q}_{\text{phys}} + \dot{Q}_{\text{info}}$

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Bounded-Memory Reversible Computation and Housekeeping Dissipation: Garbage Entropy Rate, Active Forgetting, and Finite Reuse

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FDS-P1 defined the physical accounting interface between formal distinctions and thermodynamic implementation: formal distinctions become task-available physical records only when they are carried, readable, stable, and audited under an accounting boundary. FDS-P2 studies the next step. Reversible computation can avoid immediate Landauer erasure by retaining enough side information to make the enlarged update invertible. Under bounded memory, however, this strategy accumulates garbage records that must be carried, refreshed, synchronized, uncomputed, externalized, compressed, cleaned, or abandoned. This paper strengthens the bounded-memory model by defining garbage entropy rate, accumulated residual irreversibility, nonlinear memory-fill pressure, uncomputation versus irreversible cleanup, externalization as an accounting-boundary shift with latency, active forgetting through lossy compression, and a resource-dependent housekeeping optimization problem. The main theorem states that a bounded-memory system executing sustained reversible updates with positive garbage entropy rate cannot continue indefinitely without uncomputation, cleanup, compression, externalization, task relaxation, memory expansion, or failure. The paper does not refute reversible computation, does not claim that every computational step dissipates $k_B T \ln 2$, and does not derive Landauer's principle. It gives a finite-record accounting model: reversibility changes the time profile and location of dissipation, but bounded memory makes retained inverse information itself a physical burden. Deterministic simulations illustrate correlated garbage entropy rate, memory-fill pressure, cleanup spikes, uncomputation versus cleanup, cleanup scheduling, externalization latency, lossy compression, bounded-memory regimes, and high-rate syndrome-style record turnover as a conservative domain projection.

Scope and Boundary of the Theory. This paper is a physical bridge about bounded memory and record reuse. It does not assert that ideal reversible gates are impossible, that all computation dissipates $k_B T \ln 2$ per operation, that quantum reversible computation is invalid, or that memory management has a universal device-independent power law. It claims only that reversible embeddings avoid immediate erasure by retaining preimage information somewhere inside an accounting boundary, and that sustained reversible updating under finite memory must account for the accumulated side records through uncomputation, refresh, synchronization, externalization, compression, cleanup, task relaxation, memory expansion, or failure.

Claim-status summary. Table I summarizes the central FDS-P2 claims, their epistemic status, and the conditions under which they should be weakened or rejected.

Keywords: reversible computation; bounded memory; garbage entropy rate; garbage records; delayed erasure; Landauer principle; housekeeping dissipation; memory reuse; uncomputation; cleanup; accounting boundary; active forgetting; externalization latency; active finite distinction systems.

INTRODUCTION

From P1 to P2

FDS-P1 established the physical accounting interface for finite distinctions. A formal projection is not heat; a physical many-to-one implementation becomes thermodynamic only when it discards preimage information through reset, overwrite, many-to-one compression, or garbage collection under an accounting boundary [2]. P1's boundary-relative residual irreversibility is

$$L_{\mathcal{A}} = H(X | Y, G_{\mathcal{A}}), \quad (1)$$

where X is the pre-update record, Y is the visible post-update record, and $G_{\mathcal{A}}$ is side information retained inside the accounting boundary $\mathcal{A}$.

P2 asks what happens when a system chooses the reversible route. Instead of overwriting preimage information immediately, it embeds the visible computation into a larger invertible map by keeping side records. That avoids immediate erasure relative to $\mathcal{A}$, but it does not eliminate physical accounting. It turns erased bits into garbage records, memory-fill pressure, refresh cost, synchronization cost, externalization cost, or later cleanup.

Thesis

The central thesis is:

TABLE I. Central FDS-P2 claims, epistemic status, and failure or demotion conditions.

Claim	Status	What would weaken or falsify it
Reversible embeddings avoid immediate erasure by retaining preimage information in side records.	Formal / accounting bridge	A non-injective visible map remains invertible without retaining any side record, recovery record, or enlarged state.
Sustained reversible logging accumulates garbage records when the retained garbage process has positive entropy rate.	Conditional theorem	Nontrivial updates with positive garbage entropy rate continue indefinitely while garbage load remains bounded and no uncomputation, compression, or externalization occurs.
Finite memory forces a reuse decision.	Formal finite-capacity claim	A finite memory stores an unbounded sequence of independent or entropy-rate-positive garbage records without overwrite, compression, externalization, or loss.
Uncomputation and cleanup are distinct operations.	Boundary statement	Irreversible cleanup erases garbage without residual irreversibility, or uncomputation succeeds without a retained causal inverse path, protected output copy, or synchronization.
Irreversible garbage cleanup carries boundary-relative Landauer accounting cost.	Physical bridge	Reliable cleanup below generalized Landauer accounting persists after full-boundary audit.
Externalization shifts local memory pressure into write, verification, synchronization, retrieval, maintenance, and latency costs.	Accounting claim	External records impose no storage, verification, synchronization, retrieval, latency, or maintenance burden.
Lossy compression trades memory pressure for residual irreversibility or task distortion.	Rate-distortion bridge	Lossy compression preserves all task-relevant preimage information at zero distortion and zero side cost.
Cleanup scheduling creates a tradeoff between memory pressure, refresh cost, cleanup heat, task loss, and failure risk.	Model class	Cleanup interval, externalization latency, and compression level have no effect on memory pressure, heat, task loss, or failure probability in every controlled implementation.

Reversible computation can postpone erasure, but bounded memory makes postponed history costly.

Reversibility changes when and where dissipation appears. It does not allow a finite system to carry an infinite history for free.

aries, side records, residual irreversibility, and dissipative projection [2]. P2 is the bounded-memory continuation of that ledger.

RELATED WORK

FDS core and finite-record physical bridges

The FDS formal core defines active finite distinction systems as systems that maintain boundaries through state-dependent updates under finite representational capacity and resource budgets [1]. It separates the formal layer from the physical bridge: formal preimage loss becomes a thermodynamic heat-rate floor only under Landauer-style implementation assumptions. The core also defines maintenance failure through finite free-energy budgets and the prune-externalize-collapse trichotomy. P1 then defines carriers, accounting bound-

Landauer, Bennett, and reversible computation

Landauer’s principle links logically irreversible erasure to heat generation in a physical substrate [7]. Bennett showed that computation can be made logically reversible by retaining enough information to reconstruct the input history [8, 9]. Reversible gates and conservative logic further clarify that logical reversibility is possible in principle [10]. Modern information thermodynamics refines these statements for feedback, correlations, stochastic dynamics, and nonideal protocols [11–14, 19, 22]. P2 does not challenge reversible computation. It studies the finite-memory regime in which retained histories become physical burdens.

Garbage collection, external memory, and online systems

Computer systems already implement bounded-memory accounting through garbage collection, checkpointing, logging, compression, paging, and external storage [15]. P2 uses these as engineering analogies, not as proof of a universal power law. The formal claim is narrower: if reversible computation preserves preimages in side records, those records occupy finite carrier capacity until they are uncomputed, compressed, externalized, cleaned, or lost.

REVERSIBLE EMBEDDINGS AND GARBAGE RECORDS

Definition 1 (Visible computation). *Let $f : X \rightarrow Y$ be a visible computation or update. If f is many-to-one on a set of positive probability, then $H(X | Y) > 0$ for some admissible priors.*

Definition 2 (Reversible embedding). *A reversible embedding of $f : X \rightarrow Y$ is an injective map*

$$U : X \times \{0\} \longrightarrow Y \times G \quad (2)$$

such that

$$H(X | Y, G) = 0. \quad (3)$$

The variable G is the garbage or side record that carries the preimage information not present in Y alone.

Proposition 1 (Reversible embedding avoids immediate residual irreversibility). *If $U : X \times \{0\} \rightarrow Y \times G_{\mathcal{A}}$ is injective inside accounting boundary $\mathcal{A}$, then*

$$L_{\mathcal{A}} = H(X | Y, G_{\mathcal{A}}) = 0. \quad (4)$$

No immediate erasure has occurred relative to $\mathcal{A}$.

Proof. Since $(Y, G_{\mathcal{A}})$ selects a unique preimage X , the conditional entropy of X given the enlarged record is zero. By P1's definition of boundary-relative residual irreversibility, $L_{\mathcal{A}} = 0$. $\square$

Remark 1. The word “garbage” does not mean useless. It means that the record is not part of the visible task output but is required for reversibility. It may be valuable for uncomputation, debugging, recovery, or audit. Its physical cost is that it must be stored and maintained until it is no longer needed.

GARBAGE ENTROPY RATE AND MEMORY FILL

Definition 3 (Garbage process). *Let $G_{1:t}$ be the side-record process retained to make visible updates reversible*

over a time window. The ideal independent-increment approximation writes

$$B_G(t) = \sum_{s=1}^t H(X_s | Y_s), \quad (5)$$

but the general memory load is the joint entropy

$$B_G(t) = H(G_{1:t} | R_t), \quad (6)$$

where R_t denotes task outputs, recovery records, or shared context already retained and not counted as garbage.

Definition 4 (Garbage entropy rate). *The lower asymptotic garbage entropy rate is*

$$r_G = \liminf_{t \rightarrow \infty} \frac{1}{t} H(G_{1:t} | R_t). \quad (7)$$

The condition $r_G > 0$ means that the garbage history contains a positive rate of new inverse information after correlations and shared recovery context are taken into account.

Definition 5 (Memory fill ratio). *Let M_0 be non-garbage memory usage and $M_{\max}$ the available physical memory capacity. The memory usage and fill ratio are*

$$M_{\text{used}}(t) = M_0 + B_G(t), \quad \rho_M(t) = \frac{M_{\text{used}}(t)}{M_{\max}}. \quad (8)$$

The fill time is

$$t_{\text{fill}} = \inf\{t : M_{\text{used}}(t) \geq M_{\max}\}. \quad (9)$$

Definition 6 (Memory-fill pressure). *As memory approaches saturation, allocation, indexing, defragmentation, and synchronization can add implementation pressure. A conservative proxy is*

$$C_{\text{alloc}}(\rho_M) = c_{\text{alloc}}[-\log(1 - \rho_M + \epsilon_0)], \quad (10)$$

where $\epsilon_0 > 0$ regularizes the saturation limit. This is an implementation-pressure model, not a universal law.

THE BOUNDED-MEMORY REUSE THEOREM

Theorem 1 (Bounded-memory reuse pressure). *Consider a finite system with memory capacity $M_{\max} < \infty$ executing a sustained sequence of reversible embeddings. Suppose the retained garbage process has entropy rate $r_G > 0$, and suppose the system does not uncompute the garbage, increase memory, externalize records, compress them below the generated entropy rate, relax the task, or discard preimage information. Then the system reaches memory saturation in finite time.*

Proof. By Eq. (7), for any $0 < r < r_G$, there is a sufficiently long interval on which $H(G_{1:t} | R_t) \geq rt$ up to sublinear fluctuations. Under the hypotheses, this garbage remains in internal memory, so $M_{\text{used}}(t) = M_0 + H(G_{1:t} | R_t)$. Since $M_{\text{max}} < \infty$, there exists finite t such that $M_{\text{used}}(t) \geq M_{\text{max}}$. Saturation is avoided only by using one of the excluded exits: uncomputation, memory expansion, externalization, compression, task relaxation, irreversible discard, or failure. $\square$

Operational coding overhead. The entropy rate r_G gives a lower bound on the lossless storage rate of retained garbage histories. A physical implementation generally requires

$$M_{\text{used}}(t) \geq H(G_{1:t} | R_t),$$

with additional overhead for coding inefficiency, indexing, fragmentation, allocation metadata, synchronization, and error correction. The allocation-pressure term $C_{\text{alloc}}(\rho_M)$ models part of this implementation overhead as memory approaches saturation.

Corollary 1 (Linear fill time as special case). *If the garbage increments are approximately independent with mean retained load $\bar{b}_G$, then*

$$t_{\text{fill}} \approx \frac{M_{\text{max}} - M_0}{\bar{b}_G}. \quad (11)$$

For correlated garbage, $\bar{b}_G$ must be replaced by the entropy rate r_G , not by the naive sum of stepwise conditional entropies.

ACCUMULATED RESIDUAL IRREVERSIBILITY

P1's L_A is a per-operation ledger quantity. P2 needs a history-level version.

Definition 7 (Accumulated residual irreversibility). *For a process observed through visible records $Y_{1:t}$ and side records $G_{A,t}$ included inside accounting boundary $\mathcal{A}$, define*

$$\mathcal{I}_{\mathcal{A}}^{\text{res}}(t) = H(X_{1:t} | Y_{1:t}, G_{\mathcal{A},t}). \quad (12)$$

This is the history information that remains unrecoverable after all side records currently inside the accounting boundary have been conditioned on.

For a cleanup batch $G_{t_0:t_c}$, the relevant residual irreversibility is

$$L_{\text{clean}}(t_c) = H(G_{t_0:t_c} | R_{\mathcal{A},t_c}), \quad (13)$$

where $R_{\mathcal{A},t_c}$ is recovery information still available inside the boundary at cleanup time.

Proposition 2 (Cleanup heat floor). *If a physical cleanup map discards $G_{t_0:t_c}$ with residual irreversibility L_{clean} , then under the Landauer accounting bridge*

$$Q_{\text{clean}} \geq k_B T \ln 2 L_{\text{clean}}. \quad (14)$$

For periodic cleanup with period T_c ,

$$\dot{Q}_{\text{clean}} \geq \frac{k_B T \ln 2}{T_c} H(G_{\text{batch}} | R_{\mathcal{A}}). \quad (15)$$

Proof. The cleanup map is logically irreversible exactly to the extent that the garbage cannot be reconstructed from remaining records $R_{\mathcal{A}}$. Applying the P1 boundary-relative Landauer bridge to the erased preimage information gives Eq. (14); dividing by the cleanup interval gives Eq. (15). $\square$

Recent work on optimized SRAM erasure further cautions that practical finite-time erasure costs depend strongly on the physical memory model and driving protocol, rather than following a universal linear speed-dissipation rule [20].

UNCOMPUTATION VERSUS CLEANUP

A common objection is that Bennett uncomputation can remove garbage without paying an erasure cost. P2 accepts this. The distinction is that uncomputation is not cleanup.

Definition 8 (Uncomputation). *Uncomputation is a reversible operation that maps*

$$(Y, G) \longrightarrow (Y, 0) \quad (16)$$

by reversing a still-available computation path while preserving an allowed copy of the task output Y .

Definition 9 (Irreversible cleanup). *Cleanup is a physical operation that maps*

$$G \longrightarrow 0 \quad (17)$$

without retaining enough recovery information to reconstruct G . Its residual irreversibility is $H(G | R_{\mathcal{A}})$.

Proposition 3 (Conditions for uncomputation). *Uncomputation succeeds only if the garbage remains causally recoverable, the inverse path or equivalent recovery information remains available, the output record that must survive is protected, and synchronization has not been lost.*

Reason. Reversing a computation requires an inverse map and the records needed to select the correct inverse branch. If the path is not recoverable, if the output would be destroyed, or if the system has lost synchronization with the state to be reversed, the operation is no longer uncomputation. It becomes cleanup, lossy compression, or failure. $\square$

Remark 2. *Uncomputation removes garbage by reversing a still-available computation path; cleanup removes garbage by discarding information. Only cleanup carries residual irreversibility. But uncomputation still requires physical carriers, clocking, output protection, and synchronization.*

COMPRESSION, ACTIVE FORGETTING, AND TASK LOSS

Finite systems can choose to forget. This is not always failure. It can be an active housekeeping policy.

Definition 10 (Lossy garbage compression). *A lossy compression policy maps garbage G to a compressed record $\tilde{G}$. Its residual irreversibility and task distortion are*

$$L_{\text{comp}} = H(G | \tilde{G}, R_A), \quad (18)$$

$$D_{\text{task}} = \mathbb{E}[d_{\text{task}}(G, \tilde{G})]. \quad (19)$$

Compression can produce heat and residual irreversibility, but reduce memory-fill pressure and refresh cost. Active forgetting can therefore be optimal when

$$\Delta Q_{\text{refresh}} + \Delta C_{\text{alloc}} + \mu \Delta P_{\text{fail}} > Q_{\text{comp}} + \lambda D_{\text{task}}, \quad (20)$$

This is a policy inequality, not a universal biological law. It says that retaining every distinction is not always thermodynamically or operationally optimal.

Proposition 4 (Forgetting can be a finite-memory policy). *If the physical cost of maintaining detailed garbage records exceeds their task value plus the cost of lossy compression, then a finite system can reduce total expected housekeeping cost by intentionally discarding detail.*

Lossy compression creates residual irreversibility at the information-accounting level, but its thermodynamic cost depends on physical implementation. An abstract lossy map is not heat by itself. It becomes a dissipative operation only when implemented through physical overwrite, reset, many-to-one compression, or garbage collection inside an accounting boundary.

EXTERNALIZATION AS ACCOUNTING-BOUNDARY SHIFT

Externalization moves retained records from local memory into an external carrier. In P1 language, it changes the accounting boundary:

$$\mathcal{A}_{\text{local}} \longrightarrow \mathcal{A}_{\text{coupled}} = \mathcal{A}_{\text{local}} \cup \mathcal{A}_{\text{ext}}. \quad (21)$$

Local memory pressure falls, but new costs appear:

$$C_{\text{ext}} = C_{\text{write}} + C_{\text{verify}} + C_{\text{sync}} + C_{\text{retrieve}} + C_{\text{maint}} + C_{\text{latency}}. \quad (22)$$

The O2 interface is timing. An external record exists, but it is task-useful only if retrieval and synchronization fit inside the task window:

$$T_{\text{retrieve}} + T_{\text{sync}} + T_{\text{verify}} \leq \tau_{\text{task}}. \quad (23)$$

If Eq. (23) fails, the external record remains physically real but is not operationally available for that task. Externalization shifts the accounting boundary; it does not remove timing constraints.

HOUSEKEEPING OPTIMIZATION

A bounded-memory system chooses a housekeeping policy π : when to uncompute, when to clean, how much to compress, when to externalize, and when to relax task fidelity. A FDS-aligned long-run objective is

$$J(\pi) = \langle \dot{Q}_{\text{hk}} \rangle_{\pi} + \lambda_{\ell}(\Phi) \langle \ell_{\text{task}} \rangle_{\pi} + \nu(\Phi) \langle \Delta_{\text{residual}} \rangle_{\pi} + \mu(\Phi) P_{\text{fail}}. \quad (24)$$

subject to

$$M_{\text{used}}(t) \leq M_{\text{max}} \quad (25)$$

and the FDS resource update constraint

$$\Phi_{t+1} \leq \Phi_t + \int_t^{t+\tau} \dot{F}_{\text{in}}(s) ds - \int_t^{t+\tau} \dot{Q}_{\text{hk}}(s) ds - D_{\ell}(\ell_t). \quad (26)$$

Here ℓ_{task} is task loss, Δ_{residual} is residual irreversibility or task-relevant forgotten information, and P_{fail} is failure probability. The weights are shadow prices that may depend on the remaining resource budget Φ . A simple failure-avoidance weight is

$$\mu(\Phi) = \mu_0 + \frac{\mu_1}{\Phi - \Phi_{\text{crit}} + \epsilon_0}, \quad (27)$$

for $\Phi > \Phi_{\text{crit}}$. Resource depletion changes the relative shadow prices of energy, task fidelity, residual irreversibility, and failure risk.

NUMERICAL MODELS AND SIMULATIONS

The simulations are deterministic synthetic demonstrations. They are not fits to physical memory devices, detector data, quantum experiments, AI benchmarks, or biological records. They provide reproducible model diagrams for the P2 accounting layer. All figures and CSV outputs are generated by `code/generate_results.py`.

Simulation summary

Figure 1 shows why the theorem uses entropy rate rather than a naive stepwise sum. Figure 2 illustrates

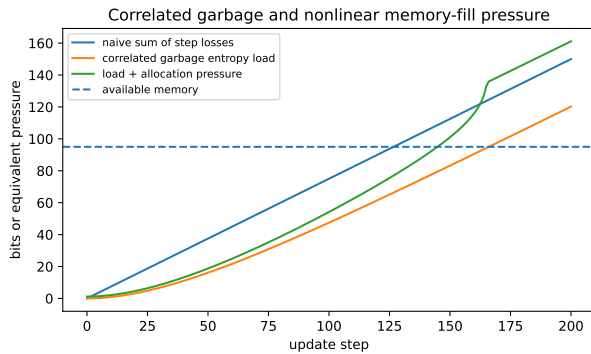


FIG. 1. Correlated garbage and nonlinear memory-fill pressure. The naive sum of step losses can overstate memory load when garbage is correlated. The entropy-rate load is lower, but allocation pressure rises nonlinearly as memory approaches saturation.

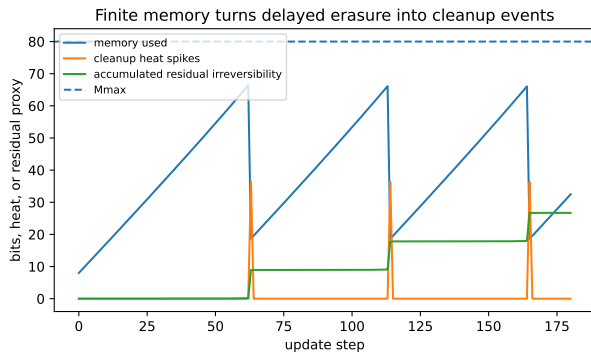


FIG. 2. Finite memory turns delayed erasure into cleanup events. Reversible logging fills memory; threshold cleanup produces heat spikes and can leave accumulated residual irreversibility.

dynamic residual irreversibility. Figure 3 separates uncomputation from cleanup. Figure 4 illustrates cleanup scheduling. Figure 5 adds externalization latency. Figure 6 shows lossy compression as active forgetting. Figure 7 gives a bounded-memory regime diagram. Figure 8 gives a conservative syndrome-style record-turnover projection.

DOMAIN PROJECTION: SYNDROME-STYLE RECORDS

High-rate error-correction systems repeatedly generate finite classical records: syndrome bits, detector clicks, parity checks, log records, or controller states. Recent below-threshold surface-code experiments show that syndrome production, decoding, reset, and real-time latency are already central engineering constraints in supercon-

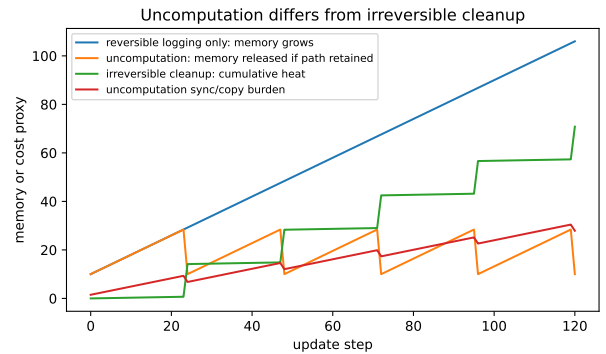


FIG. 3. Uncomputation differs from irreversible cleanup. Logging alone grows memory. Uncomputation can release memory if the inverse path and output copy are retained, but it carries synchronization and output-protection burden. Cleanup discards garbage and accumulates heat.

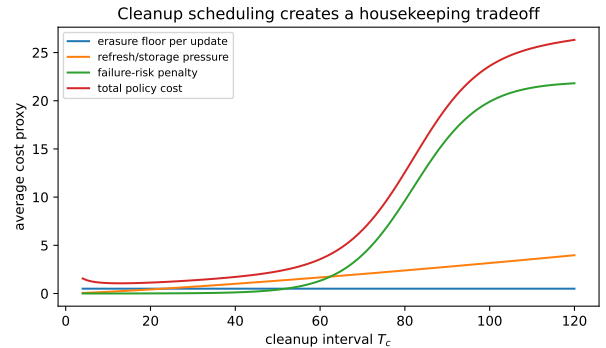


FIG. 4. Cleanup scheduling creates a housekeeping tradeoff. Frequent cleanup carries repeated cleanup overhead, while delayed cleanup increases refresh and storage pressure and failure-risk penalty. The optimum depends on device and task weights, not on a universal cleanup interval.

ducting QEC systems [21]. P2 does not claim a no-go theorem for quantum computation. It says only that if a record-bearing architecture repeatedly generates, stores, decodes, resets, or externalizes high-rate side information, the accounting categories of P2 apply: garbage entropy rate, memory fill, uncomputation where possible, irreversible cleanup where necessary, externalization latency, and housekeeping dissipation.

For a syndrome-style record stream with N_s records per cycle, b_s bits per record, and cycle time τ_c , the irreversible reuse lower bound is

$$P_{\text{erase}} \geq \frac{k_B T \ln 2}{\tau_c} N_s b_{\text{erase}}. \quad (28)$$

Actual power may be dominated by readout, control, classical processing, routing, or cooling overhead rather than the ideal Landauer floor. P2's purpose is to keep

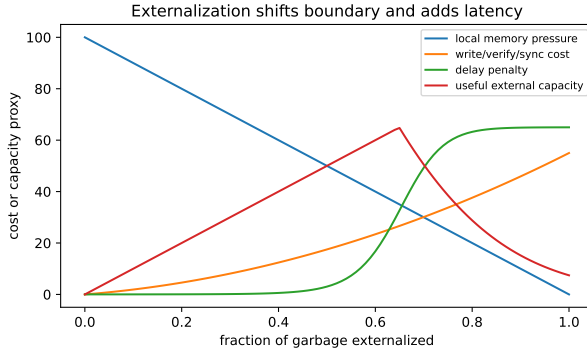


FIG. 5. Externalization shifts boundary and adds latency. Moving garbage to external memory reduces local pressure but increases write, verification, synchronization, retrieval, and delay penalties. Beyond the task window, external records become less operationally useful.

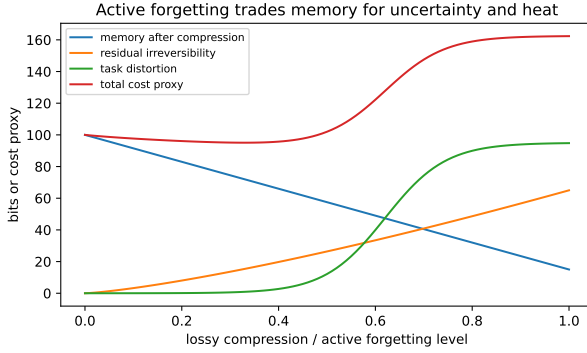


FIG. 6. Active forgetting trades memory for uncertainty and heat. Lossy compression reduces memory and refresh burden but creates residual irreversibility and task distortion. In some regimes forgetting is cheaper than maintaining all detail.

these terms in the same finite-record ledger, not to reduce them to one constant. This section is a domain projection of the P2 ledger, not a no-go theorem for quantum computation and not a claim that all QEC architectures are thermodynamically impossible. It only identifies syndrome production, reset, decoding, latency, and finite-memory turnover as instances of the bounded-record accounting problem.

EXPERIMENTAL AND ENGINEERING PROTOCOLS

Protocol 1 (Entropy-rate garbage audit). *Instrument a reversible or logging-based computation and estimate $H(G_{1:t} | R_t)/t$ rather than summing per-step losses. P2 predicts that correlated garbage reduces load relative to*

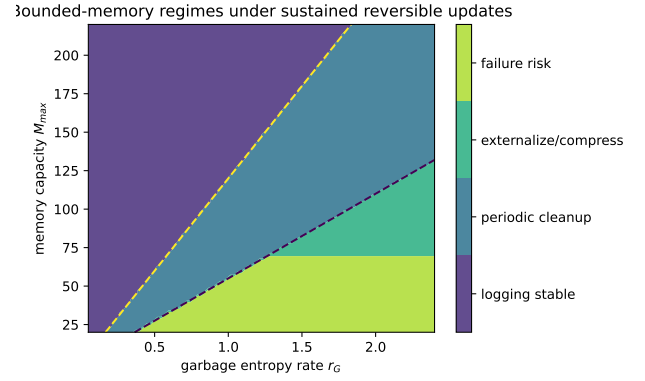


FIG. 7. Bounded-memory regimes under sustained reversible updates. The diagram is a synthetic phase portrait in garbage entropy rate and memory capacity. Regions indicate stable logging, periodic cleanup, externalization/compression dominance, and failure risk.

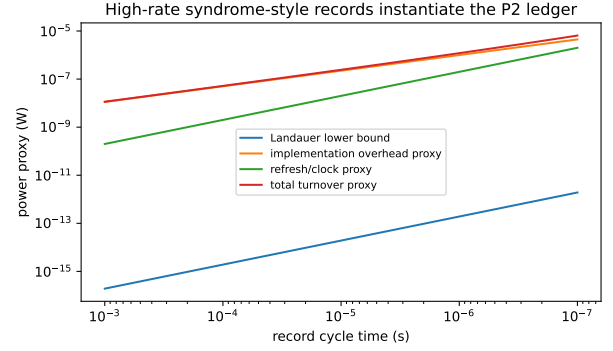


FIG. 8. High-rate syndrome-style records instantiate the P2 ledger. The plot is not a quantum-computing no-go theorem. It illustrates that repeated finite-record turnover has a lower-bound erasure term plus implementation-dependent refresh, clocking, and processing overhead.

the naive sum but positive entropy rate still fills bounded memory.

Protocol 2 (Uncomputation versus cleanup). *Compare a reversible uncomputation schedule with an irreversible cleanup schedule under the same task. Measure memory load, synchronization overhead, output-copy burden, and heat spikes. P2 predicts different ledgers for the two operations.*

Protocol 3 (Externalization latency test). *Move increasing fractions of side records into an external store. Measure local memory relief, write/verify/sync/retrieval cost, and delay relative to τ_{task} . P2 predicts that externalization fails as a task resource when retrieval latency exceeds the task window.*

Protocol 4 (Active forgetting tradeoff). *Vary compres-*

sion level for retained garbage records. Measure memory load, refresh cost, residual irreversibility, task distortion, and failure risk. P2 predicts regimes where lossy forgetting reduces total housekeeping cost, and regimes where it destroys task fidelity.

LIMITATIONS AND FALSIFICATION

P2 is not a universal theory of computation. It is a finite-record accounting model for sustained reversible updating under bounded memory. It does not deny ideal reversible gates, reversible quantum evolution, or reversible algorithms under their hypotheses. It does not claim that every step dissipates $k_B T \ln 2$. It also does not claim that all garbage must be irreversibly cleaned; uncomputation is a valid strategy when its conditions hold.

The strong version of FDS-P2 would be weakened or rejected by any of the following:

1. bounded-memory reversible systems sustain unbounded positive-entropy garbage histories without uncomputation, compression, externalization, cleanup, memory expansion, task relaxation, or failure;
2. irreversible garbage cleanup erases positive residual information below generalized Landauer accounting after full-boundary audit;
3. retained side records impose no carrier, refresh, synchronization, retrieval, indexing, or latency burden;
4. uncomputation succeeds after the inverse path, output copy, or synchronization records have been destroyed;
5. externalization always remains task-useful regardless of retrieval and synchronization latency;
6. lossy compression preserves all task-relevant preimage information at zero distortion in every nontrivial task;
7. cleanup interval, memory fill ratio, and garbage entropy rate have no measurable effect on heat, latency, failure risk, or task loss in every controlled implementation.

CONCLUSION

P1 gave the accounting ledger. P2 shows why finite memory makes ledger reuse costly. Reversible computation postpones erasure by preserving the past. Bounded memory makes that preservation itself a physical burden. The system must either uncompute while the inverse path remains available, preserve and refresh side

records, externalize them into a coupled boundary, compress and accept residual irreversibility, relax the task, or clean and pay the erasure account.

The compact statement is:

Reversibility preserves inverse information; bounded memory makes preserved inverse information compete with refresh, allocation, synchronization, latency, task fidelity, and failure risk. Forgetting becomes a policy, not merely a failure.

This is the physical bridge needed before a full FDS treatment of entropy production in active finite systems and finite-memory Second-Law channels.

Notation Summary

Simulation Parameters

The simulations are deterministic and use fixed synthetic parameters in `code/generate_results.py`. Figure 1 uses correlated garbage load and an allocation-pressure proxy. Figure 2 uses threshold cleanup with residual irreversibility. Figure 3 compares logging only, uncomputation, and cleanup. Figure 4 varies cleanup interval. Figure 5 varies externalized fraction and latency. Figure 6 varies lossy compression level. Figure 7 uses a synthetic regime classifier in $(r_G, M_{\max})$. Figure 8 uses high-rate record-turnover proxies. No proprietary, device, biological, quantum, or human-subject data are used.

Reproducibility Checklist

1. Code availability: all simulation code is included in the replication package.
2. Deterministic execution: the random seed is fixed where randomness is used.
3. Figure reproduction: run `python code/generate_results.py`; the script regenerates all figures and CSV outputs.
4. Data status: all numerical outputs are synthetic demonstrations generated from the stated model.
5. Platform independence: the code uses standard Python scientific libraries.

Code Availability

The simulation code used to generate Figs. 1–8 is included in the accompanying replication package under

TABLE II. FDS-P2 notation summary.

Symbol	Meaning
X	pre-update physical or computational record
Y	visible output or post-update record
G	garbage / side record carrying inverse information
$G_{\mathcal{A}}$	side record retained inside accounting boundary $\mathcal{A}$
R_t	retained task output, recovery context, or already-counted record
$L_{\mathcal{A}}$	boundary-relative residual irreversibility $H(X Y, G_{\mathcal{A}})$
$B_G(t)$	garbage load, generally $H(G_{1:t} R_t)$
r_G	lower garbage entropy rate $\liminf t^{-1} H(G_{1:t} R_t)$
M_{used}	memory used by active records and garbage
M_{max}	finite memory capacity
ρ_M	memory fill ratio $M_{\text{used}}/M_{\text{max}}$
C_{alloc}	allocation / fragmentation / indexing pressure as $\rho_M \rightarrow 1$
L_{clean}	residual irreversibility of a cleanup batch
Q_{hk}	housekeeping power: refresh, cleanup, sync, externalization, compression, etc.
ℓ_{task}	task loss
Φ	FDS resource or free-energy budget
τ_{task}	task window for useful retrieval or synchronization

`code/generate_results.py`. Running the script regenerates all figures (PDF and PNG) and CSV tables in a single pass.

AI Assistance Disclosure

AI-assisted tools were used for language polishing, structural feedback, LaTeX drafting support, and code-debugging assistance. The author reviewed and edited all content and remains responsible for all claims, references, simulations, and conclusions. No AI system is listed as an author.

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Boundary-Maintaining Self-Organizing Systems under Finite Capacity: Maintenance Load, Phase-C Collapse, and Invariant Selection

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FDS-N1 develops a complex-systems bridge for Active Finite Distinction Systems. The FDS formal core defines active finite systems as systems that maintain boundaries through state-dependent updates under finite representational capacity and finite resource budgets. This paper translates that core into a normal-form account of boundary-maintaining self-organization. A self-organizing system, in this paper, is not merely a system that becomes structured. It is a finite system whose internal updates are causally relevant to future boundary-maintenance loss. The revised model distinguishes structural complexity $K(t)$ from maintenance load $L_M(t)$, operationalizes effective organizational capacity $C_{\text{org}}(t)$, adds resource-gated pruning, treats externalization as an accounting-boundary shift that can clog the environment, and introduces a Phase-C catastrophic-feedback regime in which boundary loss reduces resource intake and accelerates collapse. Invariant-supported persistence is connected to a T3-style survival score based on maintenance cost, verification cost, refresh cost, boundary utility, and predictive utility. The main theorem states that an active-boundary finite system with positive deficit-driven load pressure, finite resource input, and eventually resource-exceeding maintained load cannot remain indefinitely in unbounded Phase-A growth without pruning, compression, externalization, task relaxation, invariant stabilization, resource expansion, automation, or collapse. Deterministic simulations illustrate Phase-A/Phase-B/Phase-C trajectories, operational organizational capacity, pruning viability windows, externalization clogging, bounded self-organization regimes, invariant-residue selection, active-boundary ablation, and domain bridge templates. The paper does not claim that all self-organizing systems are alive, intelligent, conscious, optimal, or oscillatory; it provides a normal-form and mapping protocol for later domain-specific applications.

Scope and Boundary of the Theory. This paper is a normal-form complex-systems bridge. It does not claim that every boundary is active, that every pattern is self-organizing, that all self-organizing systems are alive or intelligent, that pruning is always beneficial, or that Phase-A/Phase-B/Phase-C dynamics are universal. It studies systems whose internal updates are relevant to future boundary-maintenance loss under finite capacity and finite resource budgets. Domain applications require explicit mappings and operational tests.

Claim-status summary

Table I summarizes the central FDS-N1 claims, their epistemic status, and the conditions under which they should be weakened or rejected.

Keywords: self-organization; boundary maintenance; active finite distinction systems; capacity deficit; maintenance load; pruning; externalization; environmental clogging; Phase-C collapse; invariant persistence; complex systems; finite capacity; normal-form dynamics.

INTRODUCTION

Self-organization as finite boundary maintenance

Self-organization is often described as spontaneous order formation. That description is useful but too broad. It does not distinguish order that merely appears from order that is actively maintained under perturbation. A crystal, a vortex, a protocell, a neural circuit, a robot controller, and an organization can all exhibit structured dynamics, but they do not all enter the same explanatory class. FDS-N1 defines the relevant class operationally: a self-organizing system is an active finite system whose internal updates help maintain a boundary against future loss.

The central question is not simply how structure forms. It is:

which structure is maintained under finite capacity, finite memo (1)

This turns self-organization into a boundary-maintenance problem.

From FDS Core to N1

The FDS core defines active finite distinction systems as systems that maintain boundaries through state-dependent updates under finite representational capacity

TABLE I. Central FDS-N1 claims, epistemic status, and failure or demotion conditions.

Claim	Status	What would weaken or falsify it
Active self-organization requires boundary-maintenance-relevant internal update.	Operational definition	A system is correctly classified as active self-organizing even when freezing or randomizing its internal update has no effect on future boundary-maintenance loss.
Effective organizational capacity is task-relative and reduced by coordination, verification, latency, resource, and externalization costs.	Operational bridge	Boundary tasks are maintained at full fidelity even when all internal, channel, update, resource, verification, and latency capacities fall below task demand.
Deficit creates maintenance-load pressure, not necessarily raw complexity growth alone.	Normal-form claim	Increasing task demand never increases maintained load, repair structure, stored distinctions, control effort, or external scaffolding in any relevant implementation.
Unbounded Phase-A growth is impossible under finite resource input when maintained load eventually exceeds the resource envelope.	Conditional theorem	Active finite systems grow maintained load without bound forever while resources remain finite and no exit channel is used.
Pruning has a viability window and is resource-gated.	Model class	Pruning strength has no systematic effect on overload, task competence, or persistence across all controlled implementations.
Externalization shifts rather than removes boundary-maintenance burden, and excessive externalization can clog the environment.	Accounting bridge	External records impose no storage, verification, retrieval, latency, environmental-noise, or repair burden in every controlled implementation.
Phase-C catastrophic feedback can occur when boundary loss reduces effective resource intake.	Failure-regime model	Resource depletion and boundary loss never couple positively in any active-boundary system approaching collapse.
Phase-B residues are biased toward low-maintenance, task-relevant invariants.	Conditional bridge	Residues after repeated overload show no bias toward reduced maintenance cost, compressibility, task relevance, or perturbation invariance.

and finite resource budgets [1]. The core also separates formal definitions, physical bridge assumptions, normal-form dynamics, and quarantined applications. FDS-N1 occupies the normal-form and complex-systems bridge layer. It does not derive specific biology, cognition, robotics, or social theory. Instead, it supplies a shared dynamical language for systems that qualify as active-boundary finite systems.

The compact thesis is

$$\begin{aligned}
 & \text{self-organization} \\
 & = \text{boundary-maintenance-relevant updating} \quad (2) \\
 & \text{under finite capacity and finite resources.}
 \end{aligned}$$

Why distinguish structural complexity from maintenance load?

A key revision in this version is the distinction between structural complexity $K(t)$ and maintenance load $L_M(t)$. Complexity is not automatically bad. A modular controller, error-correcting routine, institution, or invariant compression scheme can increase structural com-

plexity while lowering the operational load required to maintain a boundary. FDS-N1 therefore does not claim that all complexity growth is costly in the same way. It claims that maintained structure must fit inside a finite resource envelope, and that unbounded non-compressed maintained load cannot persist without an exit channel.

RELATED WORK

Classical cybernetics and self-organization

Cybernetics framed regulation and survival in terms of feedback, requisite variety, and control under disturbance [6, 7]. General systems theory emphasized organized wholes and open-system persistence [8]. Dissipative-structures theory and synergetics studied order formation in far-from-equilibrium systems [9–11]. Complex adaptive systems, self-organized criticality, and autocatalytic-set approaches studied emergence, adaptation, and macrostructure under local rules [12–14]. A recent review by Gershenson surveys self-organizing systems across physics, chemistry, biology, collective

behavior, robotics, and social science [22]. FDS-N1 does not replace these traditions. It offers a narrow accounting-oriented normal form: self-organization is classified by whether internal updates are relevant to future boundary-maintenance loss under finite capacity.

Viability, free energy, and information bottlenecks

Viability theory studies state constraints and survival sets under control [15]. The free-energy principle and active inference treat adaptive systems as minimizing variational free energy or prediction error under generative models [16, 25]. Autopoietic and chemical-organization frameworks study self-producing unity and closure under metabolic constraints [19, 23, 24]. Information-bottleneck theory studies compression that preserves task-relevant information [17, 18]. FDS-N1 is compatible with these approaches but uses a different entry point: the finite distinction system, its boundary, update rule, loss function, and resource budget.

FDS Core and the physical accounting ladder

The FDS core defines the twelve-component finite distinction system and the active-boundary qualification. FDS-P1 defines the physical accounting interface between formal distinctions and physical records; FDS-P2 studies bounded-memory reversible computation and finite reuse [2, 3]. FDS-T3 studies capacity overflow and Phase-B invariant selection [4]. FDS-O2 studies register-time and synchronization bottlenecks [5]. N1 sits above P1/P2 in abstraction: it uses the physical accounting ladder as a background constraint but focuses on self-organization as a normal-form boundary-maintenance process.

ACTIVE FINITE SELF-ORGANIZING SYSTEMS

The FDS tuple as complex-systems anatomy

The FDS core defines a finite distinction system as

$$\mathcal{S} = (X, E, B, M, Y, A, U, \pi, \ell, \Phi, \mathcal{P}, \tau), \quad (3)$$

where X is internal state, E environment, B boundary, M memory or model state, Y observation channel, A action space, U update rule, π finite projection, ℓ boundary-maintenance loss, Φ resource budget, $\mathcal{P}$ perturbation family, and τ update timescale [1].

For N1, this tuple becomes a complex-systems anatomy: the boundary B is the maintained inside/outside interface, M is the finite organizational state, U is the adaptive update channel, ℓ is future

boundary-maintenance loss, and Φ is the resource envelope available for sensing, repair, storage, pruning, externalization, and update.

Active self-organization

Definition 1 (Active self-organizing system). *A system qualifies as active self-organizing over a task window τ when it satisfies: (i) it has a boundary variable B whose future loss is measured by ℓ ; (ii) it has finite internal state or memory M ; (iii) it performs nontrivial internal updates; and (iv) those updates are relevant to future boundary-maintenance loss.*

The minimal relevance screen is

$$\mathbb{P}\{U(M_t, Y_t) \neq M_t\} > 0 \quad \text{and} \quad I(M_{t+1}; \ell_{t+k}) > 0 \quad (4)$$

for some $k > 0$. For empirical systems, correlation is not enough. The stronger intervention criterion is

$$\mathbb{E}[\ell_{t+k} \mid \text{do}(U)] \neq \mathbb{E}[\ell_{t+k} \mid \text{do}(U_\emptyset)], \quad (5)$$

where $U_\emptyset$ is a frozen, randomized, or identity-update null channel.

Remark 1. *A passive boundary may be stable. An equilibrium crystal, an inert solid, or a static mathematical structure can persist. But unless an internal update channel participates in future boundary-maintenance loss, it is not in the N1 class.*

CAPACITY DEFICIT AND MAINTENANCE LOAD

Task-relevant demand

Let

$$Z_t = \psi(E_t, B_t) \quad (6)$$

be an admissible task-relevant statistic needed to keep boundary-maintenance loss below tolerance ε over window τ . The minimal rate-distortion demand over a pre-registered admissible family Ψ is

$$R_{\min}^{(\tau)}(\varepsilon; \Psi_t) = \inf_{\psi \in \Psi} R_{\psi(E, B)}^{(\tau)}(\varepsilon). \quad (7)$$

Effective organizational capacity

The effective organizational capacity $C_{\text{org}}(t)$ is the usable capacity for the boundary-maintenance task after bottlenecks, externalization overhead, verification, latency, and resource limits are included. A minimal decomposition is

$$C_{\text{org}}(t) = C_{\text{int}}^{\text{eff}}(t) + C_{\text{ext}}^{\text{eff}}(t) - C_{\text{coord}}(t) - C_{\text{verify}}(t) - C_{\text{latency}}(t). \quad (8)$$

TABLE II. Positioning of FDS-N1 relative to neighboring frameworks. The entries are schematic and indicate emphasis rather than exclusion.

Framework	Core object	FDS-N1 emphasis
Dissipative structures	Far-from-equilibrium order	Maintained boundary under finite capacity and resource budget
Autopoiesis	Self-producing unity	Active boundary plus deficit, pruning, externalization, and co
Cybernetics	Feedback and regulation	Update relevance to future boundary-maintenance loss
Free-energy principle	Variational free energy / prediction error	Task-relative boundary-maintenance loss under finite capacity
Information bottleneck	Compression preserving relevance	Rate-distortion demand $R_{\min}^{(\tau)}(\varepsilon; \Psi)$ and C_{org} deficit
Complex adaptive systems	Adaptation and emergence	Bounded self-organization exit theorem and domain bridge te

The internal term may itself be bottlenecked:

$$C_{\text{int}}^{\text{eff}} = \min\{C_{\text{mem}}, C_{\text{chan}}, C_{\text{upd}}, C_{\text{res}}, C_{\text{timing}}, C_{\text{verify}}\}. \quad (9)$$

This form links N1 to the bottleneck logic of finite observers and finite record systems.

All penalty terms in Eq. (8) are expressed in capacity-equivalent units after conversion through the task-specific bottleneck. Latency penalties are converted into unavailable update capacity over the task window. Verification and repair costs are converted into unavailable distinction-processing capacity. Coordination costs are converted into capacity lost to synchronization and interface overhead. The decomposition is therefore a capacity-equivalent accounting equation, not a mixture of heterogeneous quantities.

The N1 capacity deficit is

$$\Delta_{N1}(t) = R_{\min}^{(\tau)}(\varepsilon; \Psi_t) - C_{\text{org}}(t). \quad (10)$$

When $\Delta_{N1} > 0$, the system cannot maintain all task-relevant boundary distinctions at the required fidelity inside the current organizational capacity.

Structural complexity and maintenance load

Definition 2 (Structural complexity and maintenance load). *$K(t)$ denotes structural or organizational complexity: the amount of maintained internal or coupled structure. $L_M(t)$ denotes maintenance load: the operational burden of keeping that structure usable for boundary maintenance.*

A convenient normal-form relation is

$$L_M = L_0 + aK^\beta + L_{\text{coord}} + L_{\text{repair}} + L_{\text{clock}} - \eta_q q - \eta_{\text{auto}} A_{\text{auto}}, \quad (11)$$

where q is invariant-supported residue and A_{auto} denotes automation, modularization, compiled routines, or invariantized procedures that reduce active maintenance load by moving repeated operations into lower-maintenance structure. Automation is not free structure: it carries its own construction, verification, update, and repair costs. It is an exit channel only when

$$C_{\text{build}} + C_{\text{verify}} + C_{\text{update}} + C_{\text{repair}} < \Delta L_M.$$

The exponent β represents the local scaling elasticity of maintenance load with structural complexity. Regimes with $\beta > 1$ correspond to diseconomies of scale, often produced by coordination, repair, verification, and synchronization overhead. Regimes with $\beta < 1$ are possible when modularity, automation, hierarchy, or invariant compression reduce marginal maintenance load. The N1 exit theorem does not require $\beta > 1$ everywhere; it requires that unbounded maintained load cannot remain inside a finite resource envelope without an exit channel.

Equation (11) allows complexity to reduce load when it acts as compression, automation, modularity, or invariant stabilization. N1 is not an anti-complexity theory. It is a finite-maintenance-load theory.

BOUNDARY-MAINTENANCE NORMAL FORM

The minimal N1 normal form tracks structural complexity K , maintenance load L_M , resource reserve Φ , pruning effort S , externalization load E^{ext} , environmental clogging Z_{ext} , invariant residue q , and boundary loss ℓ .

Complexity, pruning, and externalization

A simple load-coupled complexity equation is

$$\dot{K} = G(\Delta_{N1}, K) - P_S(S, K) - X_E(E^{\text{ext}}, K) - I_q(q, K) - d_K K. \quad (12)$$

Here G is deficit-driven growth, P_S is pruning, X_E is externalization relief, and I_q is invariant compression. For example,

$$G(\Delta_{N1}, K) = g_0[\Delta_{N1}]_+ + g_1 K[\Delta_{N1}]_+. \quad (13)$$

Pruning effort is resource-gated:

$$\dot{S} = \sigma(\Phi) (k_S [K - K_{\text{target}}]_+ + k_{S\Phi} [\Phi_{\text{crit}} - \Phi]_+) - \gamma_S S, \quad (14)$$

with

$$\sigma(\Phi) = \frac{\Phi}{\Phi + K_\Phi}. \quad (15)$$

Thus resource stress can motivate pruning, but severe resource depletion can also disable pruning capacity.

Externalization and environmental clogging

Externalization shifts the accounting boundary:

$$\mathcal{A}_{\text{local}} \rightarrow \mathcal{A}_{\text{coupled}} = \mathcal{A}_{\text{local}} \cup \mathcal{A}_{\text{ext}}. \quad (16)$$

It can increase effective capacity, but it also writes records into an environment. Let Z_{ext} denote environmental record/noise load. A minimal clogging model is

$$\dot{Z}_{\text{ext}} = \alpha_E E^{\text{ext}} - \gamma_E Z_{\text{ext}}. \quad (17)$$

Retrieval and verification costs can grow with clogging:

$$C_{\text{retrieve}}(Z_{\text{ext}}) = C_0 + c_1 Z_{\text{ext}} + c_2 Z_{\text{ext}}^2. \quad (18)$$

The effective external capacity is therefore

$$C_{\text{ext}}^{\text{eff}} = C_{\text{ext}}^{\text{raw}} - C_{\text{write}} - C_{\text{verify}}(Z_{\text{ext}}) - C_{\text{retrieve}}(Z_{\text{ext}}) - C_{\text{sync}} - C_{\text{latency}} - C_{\text{repair}}. \quad (19)$$

This is the normal-form route by which over-externalization becomes information pollution, indexing burden, or environmental clogging. N1 does not claim that such costs are always exponential; it claims that they must be audited and can become superlinear in crowded external stores. Environmental clogging of this form has been identified as a systemic information-overload risk in digital ecosystems [26].

Resource and loss dynamics

A resource equation is

$$\dot{\Phi} = \dot{F}_{\text{in}}^{\text{eff}}(\ell) - \dot{Q}_{\text{hk}}(L_M, S, E^{\text{ext}}, Z_{\text{ext}}) - D_{\ell}(\ell), \quad (20)$$

where $\dot{F}_{\text{in}}^{\text{eff}}$ may decrease as boundary loss increases. Boundary loss evolves as

$$\dot{\ell} = a_{\Phi}[\Phi_{\text{crit}} - \Phi]_+ + a_L[L_M - L_{\text{crit}}]_+ + a_{\Delta}[\Delta_{N1}]_+ - r_{\text{repair}}(S, E^{\text{ext}}, \Phi). \quad (21)$$

Phase-C catastrophic feedback

Phase C is a failure regime in which boundary loss reduces effective resource intake and therefore accelerates further boundary loss:

$$\Phi < \Phi_{\text{crit}}, \quad \dot{\ell} > 0, \quad \frac{\partial \dot{F}_{\text{in}}^{\text{eff}}}{\partial \ell} < 0. \quad (22)$$

Phase C is not a new universal law. It is a normal-form collapse regime for systems where resource acquisition depends on an intact boundary. Death, bankruptcy, institutional collapse, and runaway controller failure are possible domain projections, not proofs.

In dynamical-systems terms, Phase C is a region in which the resource-loss feedback has positive loop gain:

boundary loss reduces effective resource intake, while reduced resource reserve increases boundary loss. Locally this corresponds to the sign pattern

$$\frac{\partial \dot{\Phi}}{\partial \ell} < 0, \quad \frac{\partial \dot{\ell}}{\partial \Phi} < 0, \quad \frac{\partial \dot{\Phi}}{\partial \ell} \frac{\partial \dot{\ell}}{\partial \Phi} > 0.$$

This places Phase C within the broader class of critical transitions and tipping-point dynamics studied in complex-systems resilience theory [27].

BOUNDED SELF-ORGANIZATION EXIT THEOREM

Theorem 1 (Bounded self-organization exit). *Consider an active-boundary finite system with finite effective resource input $\dot{F}_{\text{in}}^{\text{max}} < \infty$. Suppose it experiences positive deficit-driven load pressure and its maintained load cannot remain uniformly bounded under unbounded Phase-A growth unless it is compressed, externalized, automated, made invariant-supported, or made non-maintained. If no exit channel is available, then the system cannot remain indefinitely in unbounded Phase-A growth without reaching a resource or boundary-maintenance failure condition.*

Proof. Phase-A growth increases the amount of structure that must remain task-available for boundary maintenance. By hypothesis, under unbounded growth and absent compression, externalization, automation, invariant stabilization, or non-maintenance, the maintained load eventually exceeds the finite resource envelope. Once L_M forces $\dot{Q}_{\text{hk}} + D_{\ell}(\ell) > \dot{F}_{\text{in}}^{\text{max}}$ for a sufficiently long interval, Eq. (20) drives Φ downward. If the system does not exit through pruning, compression, externalization, task relaxation, invariant stabilization, resource expansion, or automation, then Φ crosses Φ_{crit} or ℓ crosses the boundary-loss tolerance. The system therefore enters collapse or Phase-C feedback. The conclusion follows from finite resource input and the absence of exits. $\square$

Remark 2. *The theorem does not require every added structure to increase cost. It permits modularity, automation, compression, and invariant-supported structure to reduce L_M . The theorem targets unbounded maintained load under finite resource input, not complexity as such.*

Corollary 1 (Finite-maintenance impossibility triangle). *An active-boundary finite system cannot jointly satisfy:*

$$L_M(t) \rightarrow \infty \quad \wedge \quad \dot{F}_{\text{in}} \leq F_{\text{max}} < \infty \quad \wedge \quad \text{no exit channel.}$$

At least one of the three must fail: load must be bounded, resources must expand, or an exit channel (pruning, compression, externalization, automation, invariant stabilization, task relaxation, or collapse) must occur.

The admissible exits are

prune / compress / externalize / relax / automate / invariant stabilization / expand resources / collapse distinctions (23)

PHASE A, PHASE B, AND PHASE C

Phase A: growth-dominant adaptation

Phase A occurs when deficit-driven growth dominates regulation:

$$G(\Delta_{N1}, K) > P_S(S, K) + X_E(E^{\text{ext}}, K) + I_q(q, K). \quad (24)$$

Typical signatures are model proliferation, repair patches, increased structural complexity, and rising maintenance load.

Phase B: maintenance, pruning, and invariant selection

Phase B occurs when pruning, externalization, compression, or invariant stabilization catches up with growth:

$$P_S + X_E + I_q \geq G. \quad (25)$$

Phase B can produce simplification, consolidation, external scaffolding, or low-maintenance residue. Oscillation between Phase A and Phase B is possible but not universal.

Phase C: catastrophic feedback

Phase C occurs when resource depletion and boundary loss reinforce one another. It is characterized by Eq. (22). Phase C can follow failed pruning, excessive externalization, overcompression, loss of invariant support, or loss of repair capacity.

PRUNING, EXTERNALIZATION, AND INVARIANT PERSISTENCE

Pruning as active simplification

Pruning removes, compresses, deactivates, or down-ranks maintained distinctions. It is useful only inside a viability window. Under-pruning leaves overload unresolved; over-pruning destroys task competence. In N1, pruning is not merely deletion. It is active simplification relative to boundary-maintenance loss.

Proposition 1 (Pruning viability window). *If under-pruning leaves L_M above the resource envelope and over-pruning destroys required distinctions, then there exists an intermediate pruning range that minimizes total expected boundary loss for the modeled task.*

Invariant-supported persistence

Let $R_A(x)$ be an identity predicate and let $q : X \rightarrow Q$ be a quotient map. Invariant support holds relative to a specified perturbation family $\mathcal{P}$, task window τ , and identity predicate R_A when

$$R_A = \bar{R}_A \circ q, \quad q(P_i x) = q(x) \quad (26)$$

for admissible perturbations $P_i \in \mathcal{P}$. Invariant support is not absolute indestructibility. It is persistence relative to a chosen perturbation class, task window, and identity predicate.

To connect N1 to T3, define a survival score for a candidate residue ϕ_i :

$$S(\phi_i) = \frac{U_i^{\text{boundary}} + U_i^{\text{pred}}}{C_i^{\text{maint}} + C_i^{\text{verify}} + C_i^{\text{refresh}} + \epsilon}. \quad (27)$$

A soft selection rule is

$$P_i^{\text{survive}} = \frac{\exp(\beta S(\phi_i))}{\sum_j \exp(\beta S(\phi_j))}. \quad (28)$$

Equations (27) and (28) are a selection functional, not a universal biological law. They express the hypothesis that Phase-B residues are biased toward structures with high boundary or predictive utility per unit maintenance burden.

NUMERICAL MODELS AND SIMULATIONS

The simulations are deterministic synthetic demonstrations. They are not fits to physical memory devices, biological systems, robots, organizations, or human-subject data. They provide reproducible model diagrams for the N1 normal form. All figures and CSV outputs are generated by `code/generate_results.py`.

DOMAIN BRIDGE TEMPLATE AND PROTOCOLS

N1 is a bridge protocol, not a license for uncontrolled analogies. A domain application must specify the entries in Table III before claiming that the N1 normal form applies.

TABLE III. Minimal domain bridge template for applying FDS-N1.

Item	Required domain definition
Boundary B	What boundary or interface is being maintained?
Loss ℓ	How is boundary-maintenance failure measured?
Memory/organization M	What structure stores task-relevant distinctions?
Update U	Which updates influence future boundary loss?
Capacity C_{org}	How is usable organizational capacity estimated?
Demand R_{min}	What task-relevant distinction demand is pre-registered?
Resource Φ	What finite resource or free-energy budget constrains maintenance?
Pruning S	What counts as active simplification, forgetting, or deletion?
Externalization E^{ext}	What records or functions are moved outside the local boundary?
Environmental clogging Z_{ext}	How does externalization affect retrieval, noise, and verification?
Invariant q	What low-maintenance residue supports identity or task function?
Ablation	How is $do(U)$ compared with frozen or randomized update?

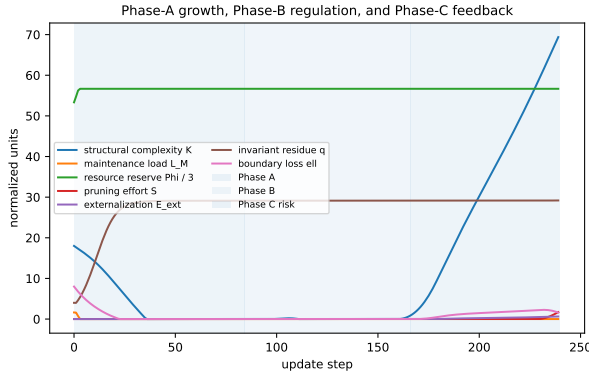


FIG. 1. Phase-A growth, Phase-B regulation, and Phase-C feedback. Structural complexity K and maintenance load L_M rise under deficit pressure. Pruning, externalization, and invariant residue can stabilize the system, but severe resource depletion can trigger boundary-loss feedback.

Protocol 1 (Active-boundary ablation). Compare normal update $do(U)$ with frozen, randomized, or identity update $do(U_{\emptyset})$ and measure future boundary-maintenance loss ℓ_{t+k} . If the intervention has no effect on future loss, the system does not qualify as active self-organizing for that task.

Protocol 2 (Deficit crossing). Increase task demand or environmental complexity and measure Δ_{N1} , K , L_M , S , E^{ext} , Φ , and ℓ . $N1$ predicts load growth, pruning/externalization responses, or collapse as demand crosses effective organizational capacity.

Protocol 3 (Pruning-window test). Vary pruning strength. $N1$ predicts under-pruning overload, an intermediate viability window, and over-pruning task loss when task-relevant distinctions are destroyed.

Protocol 4 (Externalization ROI and clogging). Increase externalization fraction and measure local relief, write cost, verification cost, retrieval cost, latency, and

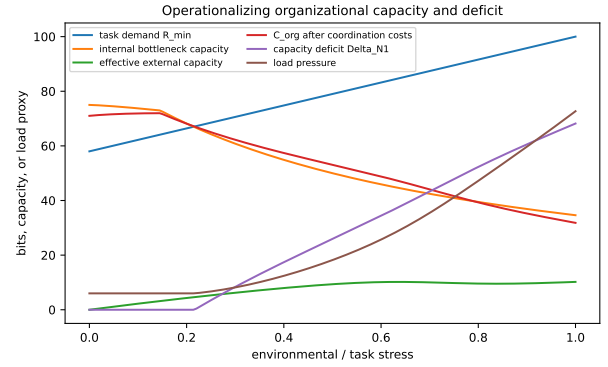


FIG. 2. Operationalizing C_{org} . Effective organizational capacity combines internal bottleneck capacity, useful external capacity, and coordination costs. Capacity deficit Δ_{N1} rises when task demand exceeds usable organization.

environmental record load. $N1$ predicts that externalization can become negative ROI when coordination, latency, or clogging costs dominate.

Protocol 5 (Invariant residue test). Apply admissible perturbations and measure which structures survive. $N1$ predicts a bias toward residues with high survival score $S(\phi)$, not arbitrary survival of all structure.

LIMITATIONS AND FALSIFICATION

FDS-N1 is not a universal theory of life, intelligence, consciousness, organizations, or culture. It is a finite-boundary normal form for active systems. It does not claim that all self-organizing systems are optimal, that Phase-A/Phase-B/Phase-C dynamics must oscillate, or that invariant residues always emerge.

The strong version of FDS-N1 would be weakened or rejected by any of the following:

1. systems classified as active self-organizing despite update ablation having no effect on future

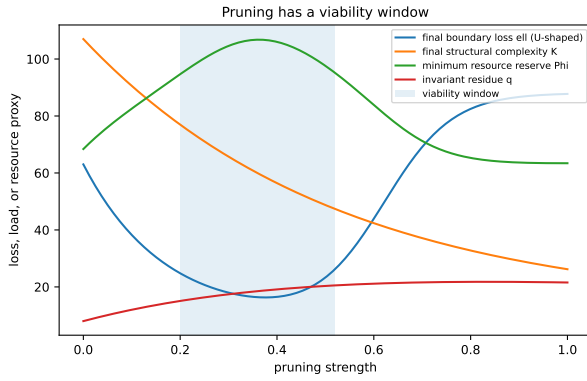


FIG. 3. Pruning has a viability window. Under-pruning leaves overload; over-pruning destroys task competence. The synthetic boundary-loss curve is U-shaped, while structural complexity falls and resource reserve is best preserved in an intermediate range.

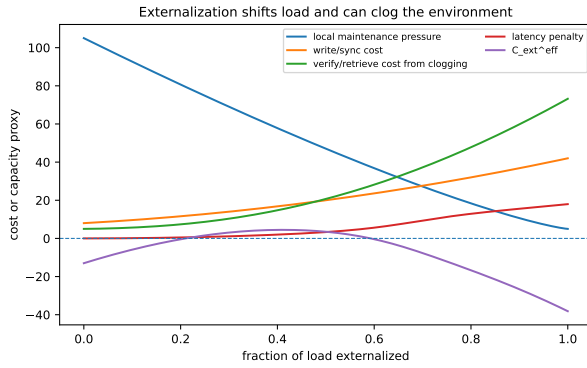


FIG. 4. Externalization shifts load and can clog the environment. Local maintenance pressure falls as more records are externalized, but write, verification, retrieval, and latency costs grow. Effective external capacity can become negative when environmental record load increases.

- boundary-maintenance loss;
- 2. sustained unbounded maintained load under finite resource input with no pruning, compression, externalization, task relaxation, automation, invariant stabilization, resource expansion, or collapse;
- 3. effective organizational capacity remaining unchanged under severe channel, memory, update, resource, verification, and latency bottlenecks;
- 4. pruning strength having no systematic effect on overload, task competence, or persistence across controlled cases;
- 5. externalization imposing no write, verification, retrieval, synchronization, latency, repair, or environmental-clogging burden;

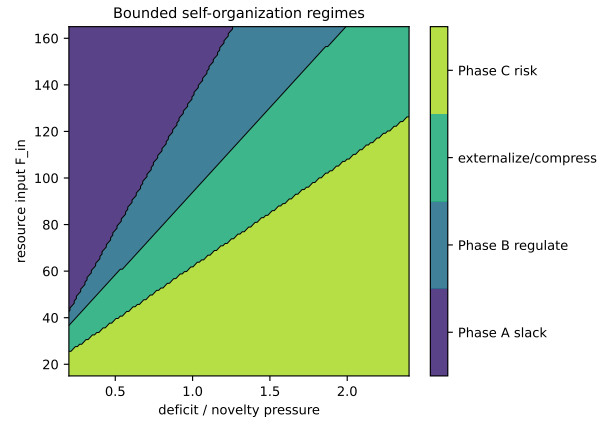


FIG. 5. Bounded self-organization regimes. The synthetic phase portrait separates Phase-A slack, Phase-B regulation, externalization/compression dominance, and Phase-C collapse risk as a function of deficit pressure and resource input.

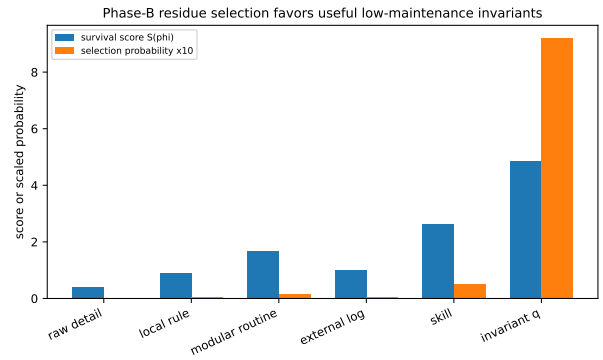


FIG. 6. Phase-B residue selection favors useful low-maintenance invariants. The survival score $S(\phi)$ increases when boundary and predictive utility are high relative to maintenance, verification, and refresh costs.

- 6. Phase-B residues showing no bias toward low-maintenance, task-relevant, perturbation-stable structures;
- 7. boundary loss and resource acquisition never coupling in collapse-prone active systems.

CONCLUSION

FDS-N1 reframes self-organization as boundary-maintenance-relevant updating under finite capacity and finite resources. The central object is not unconstrained order. It is a finite system maintaining a boundary through costly updates, limited memory, imperfect externalization, resource-gated pruning, and possible invariant stabilization.

The compact statement is:

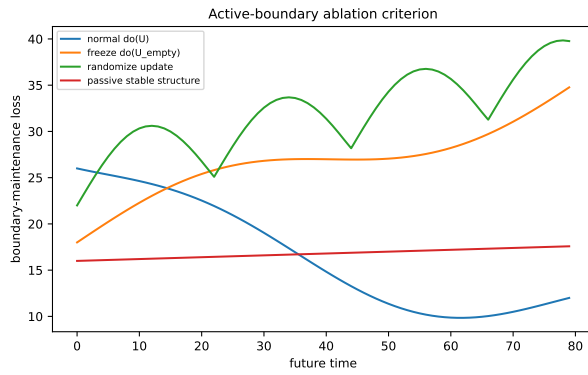


FIG. 7. Active-boundary ablation criterion. A system qualifies as active self-organizing only when normal update improves future boundary-maintenance loss relative to frozen or randomized update controls.

Domain bridge template: examples are mappings, not proofs

	Boundary B	Loss ℓ	Update U	Pruning S	Externalization	Invariant q
Procell	membrane/gradients	leakage	metabolic repair	reaction pruning	niche cues	autocatalytic motif
Neural	control/report	prediction loss	plasticity/attention	forget/sleep	task/notes	skill/concept
Robot	safety/task	damage/failure	controller update	cache/policy prune	maps/cloud	robust policy
Organization	institution	coordination loss	rule update	bureaucracy trim	archives/ledgers	norm/procedure
Civilization	law/culture	collapse/drift	governance	reform/simplify	libraries/web	culture meme

FIG. 8. Domain bridge flow diagram. The detailed template is in Table III. Every application must specify its boundary, loss, update, pruning operation, externalization channel, and invariant residue before claiming that the N1 normal form applies.

Self-organization is not unconstrained order production. It is finite boundary maintenance by updates that must pay for the load they preserve. Complexity can help when it compresses, automates, modularizes, externalizes, or stabilizes; it fails when maintained load outruns the resource envelope.

This is the complex-systems bridge needed before later FDS treatments of entropy production in active finite systems, death-like collapse and recovery ordering, cognitive bottlenecks, organizational externalization, and civilization-scale maintenance limits.

NOTATION SUMMARY

Symbol	Meaning
B	maintained boundary or interface
M	memory, model, or organizational state
U	internal update rule
ℓ	boundary-maintenance loss
Φ	finite resource or free-energy budget
$R_{\min}^{(\tau)}(\epsilon; \Psi)$	task-relevant rate-distortion demand
C_{org}	effective organizational capacity after bottlenecks and ov
Δ_{N1}	task-relative capacity deficit $R_{\min} - C_{\text{org}}$
K	structural or organizational complexity
L_M	maintenance load
S	pruning or active simplification effort
E^{ext}	externalization load
Z^{ext}	externalized record/noise load in environment
q	invariant-supported residue or quotient structure
$S(\phi)$	survival score of candidate residue ϕ

SIMULATION PARAMETERS

The simulations are deterministic and use fixed synthetic parameters in `code/generate_results.py`. Figure 1 uses the full Phase-A/Phase-B/Phase-C normal form. Figure 2 operationalizes C_{org} . Figure 3 varies pruning strength. Figure 4 varies externalized fraction and environmental clogging. Figure 5 uses a synthetic regime classifier. Figure 6 computes survival scores. Figure 7 compares update interventions. Figure 8 renders the domain bridge template. No proprietary, biological, organizational, human-subject, or device data are used.

REPRODUCIBILITY CHECKLIST

1. Code availability: all simulation code is included in the replication package.
2. Deterministic execution: the code uses fixed synthetic parameters and deterministic arrays.
3. Figure reproduction: run `python code/generate_results.py`; the script regenerates all figures and CSV outputs.
4. Data status: all numerical outputs are synthetic demonstrations generated from the stated model.
5. Platform independence: the code uses standard Python scientific libraries.

CODE AVAILABILITY

The simulation code used to generate Figs. 1–8 is included in the accompanying replication package under `code/generate_results.py`. Running the script regenerates all figures (PDF and PNG) and CSV tables in a single pass.

AI ASSISTANCE DISCLOSURE

AI-assisted tools were used for language polishing, structural feedback, LaTeX drafting support, and code-debugging assistance. The author reviewed and edited all content and remains responsible for all claims, references, simulations, and conclusions. No AI system is listed as an author.

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Capacity Deficit and Entropy Production in Active Finite Systems: A Generalized Dissipation Ledger for Boundary Maintenance

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FDS-P5 develops a physical bridge between task-relative capacity deficit and entropy-production pressure in active finite distinction systems. The FDS formal core defines active finite systems as boundary-maintaining systems with finite representational capacity, finite update windows, and finite resource budgets. A capacity deficit $\Delta_c(\tau) > 0$ is not heat, entropy production, or thermodynamic irreversibility by itself. It is an information-theoretic shortfall between task-relevant rate-distortion demand and effective capacity. The central claim of this paper is narrower: when a physically realized active finite system continues boundary maintenance under sustained positive deficit, the required correction, refresh, repair, synchronization, externalization, pruning, or recovery must be accounted for somewhere in a coupled entropy or resource ledger. This revised version adds an explicit audit convention to avoid double counting, uses reservoir-indexed entropy-production notation, separates physical recovery entropy from functional loss, ties the correction information flow to rate-distortion deficit, defines positive-density deficit windows, and states a deficit-maintenance-dissipation impossibility triangle. Deterministic simulations illustrate deficit crossing, unique-channel ledger decomposition, pruning dissipation return on investment, coupled-system externalization, invariant compression, maintenance-failure regimes, and entropy-production hysteresis. The paper does not claim that every distinction dissipates heat, that capacity deficit is thermodynamic entropy, that all entropy production is Landauer erasure, or that the Second Law is newly derived. Its contribution is a finite-system dissipation ledger for boundary maintenance under positive capacity deficit.

Scope and Boundary of the Theory. This paper is a physical-bridge and normal-form paper. It does not claim that capacity deficit itself dissipates heat. It does not claim that every distinction, observation, or uncertainty carries a Landauer cost. It does not rederive the Second Law, and it does not assert a universal equality between information shortfall and thermodynamic entropy production. It claims only that under physical bridge assumptions, active finite systems that continue boundary maintenance under positive capacity deficit must pay through one or more accounting channels: irreversible update, refresh, repair, sensing, synchronization, externalization, pruning, task relaxation, or recovery after damage. Positive deficit by itself does not force dissipation if the system simply abandons the task, passively degrades, or accepts boundary loss.

Claim-status summary

Table I summarizes the central FDS-P5 claims, their epistemic status, and the conditions under which they should be weakened or rejected.

Keywords: capacity deficit; entropy production; information thermodynamics; boundary maintenance; active finite distinction systems; Landauer principle; housekeeping dissipation; pruning; externalization; invariant compression; finite capacity; stochastic thermodynamics; thermodynamic uncertainty relations.

INTRODUCTION

From capacity deficit to physical cost

The FDS formal core defines active finite distinction systems as finite-capacity systems that maintain boundaries through state-dependent updates under resource constraints [1]. Its central chain is

$$\text{finite capacity} \rightarrow \text{capacity deficit} \rightarrow \text{approximation} \rightarrow \text{irreversible} \quad (1)$$

FDS-P1 defined the physical accounting boundary between formal distinctions and physical records [2]. FDS-P2 studied bounded-memory reversible computation and the housekeeping burden of retained inverse information [3]. FDS-N1 translated active boundary maintenance into a complex-systems normal form with maintenance load, pruning, externalization, and Phase-C collapse [4]. FDS-P5 asks a more specific physical question: when capacity deficit persists and the system continues to maintain its boundary, where does the cost appear in the coupled entropy ledger?

The answer is not that a deficit is heat. A deficit is an information-theoretic mismatch. Heat, entropy production, or exported disorder appear only when a physically realized system senses, stores, corrects, erases, refreshes, repairs, synchronizes, externalizes, prunes, or recovers from damage in response to that mismatch.

TABLE I. Central FDS-P5 claims, epistemic status, and failure or demotion conditions.

Claim	Status	What would weaken or falsify it
Capacity deficit is a task-relative information shortfall, not thermodynamic entropy by itself. Sustained positive deficit plus continued boundary maintenance requires correction, externalization, task relaxation, or failure.	Boundary statement	Not an empirical prediction; it separates formal rate-distortion demand from physical entropy production.
Physical correction cycles induce audit channels through update, refresh, repair, synchronization, externalization, and transport. Logical erasure contributes a Landauer-style entropy-production floor under bridge assumptions.	FDS-core derived	A finite active system maintains all pre-registered task distinctions at fixed tolerance despite $R_{\min}^{(\tau)}(\epsilon) > C_{\text{eff}}$ and no correction, loss, task relaxation, or external support.
Housekeeping entropy production can persist even when logical erasure is zero. Externalization shifts the entropy ledger across accounting boundaries.	Physical bridge	Controlled physical systems perform sustained correction, refresh, repair, sensing, and synchronization at zero coupled-system entropy production or resource cost.
Pruning and invariant compression can reduce future entropy-production pressure when task function is preserved. Deficit crossing predicts measurable signatures in heat, resource use, latency, resets, error floor, externalization, or failure risk.	Physical bridge	Reliable logically irreversible physical erasure under stated thermodynamic assumptions violates the Landauer lower bound.
	Accounting claim	Boundary maintenance, refresh, clocking, sensing, and repair are cost-free whenever preimage erasure is zero.
	Accounting bridge	External records impose no write, verification, retrieval, latency, synchronization, maintenance, or environmental cost.
	Conditional bridge	No task-preserving simplification ever reduces refresh, repair, verification, synchronization, or maintenance cost.
	Testable normal-form	Positive deficit can be sustained with no measurable change in any physical, computational, or task-performance channel.

Why this is not simply Landauer

The Landauer term covers logically irreversible preimage loss. In the FDS core, logical erasure per update is

$$b_t = H(M_t | M_{t+1}, Y_t), \quad (2)$$

and the physical bridge gives an informational heat-rate floor

$$\dot{Q}_{\text{info}} \geq \frac{k_B T \ln 2}{\tau} H(M_t | M_{t+1}, Y_t), \quad (3)$$

or equivalently an entropy-production rate floor

$$\dot{\Sigma}_{\text{info}} \geq \frac{k_B \ln 2}{\tau} H(M_t | M_{t+1}, Y_t), \quad (4)$$

when entropy is measured in physical units. Equation (4) is essential but incomplete. An active boundary system can have low or zero logical erasure while still paying for refresh, clocking, sensing, isolation, synchronization, control, repair, transport, external verification, and maintenance of physical carriers. P5 therefore treats Landauer erasure as one ledger term inside a broader active-system entropy account.

Contribution

This paper contributes: (i) a generalized entropy-production audit ledger for active finite systems; (ii) a rate-distortion grounded correction-flow lower bound; (iii) a positive-density version of the deficit-driven entropy-pressure theorem; (iv) an anti-double-counting convention for ledger terms; (v) a multi-reservoir entropy-production notation; (vi) a separation between physical recovery entropy and functional boundary loss; (vii) pruning and invariant compression criteria for future entropy relief; (viii) engineering protocols for testing deficit crossings; and (ix) a handoff to later precision-dissipation and finite-memory Second-Law treatments.

CORE FDS INGREDIENTS

Active finite distinction systems

The FDS core object is

$$S = (X, E, B, M, Y, A, U, \pi, \ell, \Phi, \mathcal{P}, \tau), \quad (5)$$

where X is internal state, E environment, B boundary, M memory or model, Y observation channel, A action space, U update rule, π finite projection, ℓ boundary-maintenance loss, Φ resource budget, $\mathcal{P}$ perturbation or pruning family, and τ update timescale [1]. The timescale matters because information quantities become rates only when divided by an update window.

An active boundary requires nontrivial update and relevance to future boundary loss:

$$\mathbb{P}\{U(M_t, Y_t) \neq M_t\} > 0, \quad \mathbb{I}(M_{t+1}; \ell_{t+k}) > 0. \quad (6)$$

For empirical systems, the stronger ablation test compares $do(U)$ with a frozen, randomized, or identity update channel.

Capacity deficit and effective capacity

Let $Z_t = \psi(E_t, B_t)$ be a pre-registered task-relevant statistic needed to maintain the boundary to tolerance ε over update window τ . Let

$$R_{\min}^{(\tau)}(\varepsilon; \Psi_t) = \inf_{\psi \in \Psi} R_{\psi(E, B)}^{(\tau)}(\varepsilon) \quad (7)$$

be the minimal task demand over an admissible family Ψ . For effective capacity $C_{\text{eff}}(t)$, define

$$\Delta_\epsilon(t) = R_{\min}^{(\tau)}(\varepsilon; \Psi_t) - C_{\text{eff}}(t). \quad (8)$$

When $\Delta_\epsilon > 0$, a finite internal model cannot maintain all task-relevant distinctions at the required fidelity without approximation, externalization, task relaxation, or failure.

In this paper, $\Delta_\epsilon(t)$ is measured in capacity units (bits or equivalent) per update window τ . Division by τ converts this window-level shortfall into a correction-flow rate in Eq. (17). If $R_{\min}$ is instead defined as a rate, the factor $1/\tau$ should be omitted.

Following N1, a useful operational decomposition is

$$C_{\text{eff}}^{P5} = C_{\text{int}}^{\text{eff}} + C_{\text{ext}}^{\text{eff}} - C_{\text{sync}} - C_{\text{verify}} - C_{\text{latency}} - C_{\text{resource}}, \quad (9)$$

with the internal bottleneck

$$C_{\text{int}}^{\text{eff}} = \min\{C_{\text{mem}}, C_{\text{chan}}, C_{\text{upd}}, C_{\text{res}}, C_{\text{timing}}, C_{\text{verify}}\}. \quad (10)$$

All penalty terms in Eq. (9) are expressed in capacity-equivalent units after conversion through the task-specific bottleneck. This prevents mixing raw bits, time, and energy without a conversion convention.

Resource budget and maintenance failure

A physically realized active finite system has a resource balance of the form

$$\Phi_{t+1} \leq \Phi_t + \int_t^{t+\tau} \dot{F}_{\text{in}}(s) ds - \int_t^{t+\tau} \dot{Q}_{\text{total}}(s) ds - D_\ell(\ell_t), \quad (11)$$

where D_ℓ is boundary-loss damage or functional repair burden. In the present paper, D_ℓ is tracked separately from physical entropy production unless a domain bridge specifies the dissipative recovery process that realizes it.

GENERALIZED ENTROPY-PRODUCTION LEDGER

Audit convention and no double counting

Definition 1 (Accounting-boundary entropy ledger). Let $\mathcal{A}$ be an accounting boundary and let $\mathcal{P}_{\mathcal{A}}$ be a pre-registered set of physical processes inside or coupled to that boundary. Define the audited physical entropy-production rate as

$$\dot{\Sigma}_{\text{phys}}^{\mathcal{A}} = \sum_{p \in \mathcal{P}_{\mathcal{A}}} \dot{\Sigma}(p), \quad (12)$$

where each physical process p is counted once. Named terms such as sensing, synchronization, externalization, and repair are audit classes, not automatically independent additive mechanisms. If a concrete implementation assigns the same physical process to more than one class, the ledger must either choose a unique channel assignment or include overlap-correction terms.

Equivalently, one may write a class-level decomposition

$$\dot{\Sigma}_{\text{phys}}^{\mathcal{A}} = \dot{\Sigma}_{\text{info}} + \dot{\Sigma}_{\text{hk}} + \dot{\Sigma}_{\text{meas}} + \dot{\Sigma}_{\text{ctrl}} + \dot{\Sigma}_{\text{sync}} + \dot{\Sigma}_{\text{ext}} + \dot{\Sigma}_{\text{rec}}, \quad (13)$$

with the audit convention that the right-hand side is a partition of process costs for the chosen model. The recovery term $\dot{\Sigma}_{\text{rec}}$ refers to physical recovery, repair after damage, dissipative reset, uncontrolled relaxation, or loss-induced resource expenditure. Failure itself, as a task-level event, is not automatically entropy production; pure functional loss is tracked by $D_\ell(\ell)$. Methods for inferring entropy production from observable transitions in partially accessible systems are directly relevant to implementing this ledger decomposition in practice [27, 29].

Multi-reservoir entropy-production notation

For a channel i coupled to reservoirs $r \in \mathcal{R}_i$, the stochastic-thermodynamic form is

$$\dot{\Sigma}_i = \dot{\Sigma}_i^{\text{sys}} + \sum_{r \in \mathcal{R}_i} \frac{\dot{Q}_{i,r}}{T_r} \geq 0, \quad (14)$$

with a dimensionless version obtained by dividing by k_B . The familiar single-temperature shorthand $T^{-1}\dot{Q}_i$ should be read as a special case or an illustrative proxy. P5 does not require a single universal bath temperature.

TABLE II. FDS-P5 entropy-production ledger. Terms are audit classes, not universal independent additive laws. Heat-rate expressions are single-reservoir shorthand for the multi-reservoir form in Eq. (14).

Term	Representative expression	Meaning
$\dot{S}_{\text{info}}$	$\frac{k_B \ln 2}{\tau} H(M_t M_{t+1}, Y_t)$	Logical erasure, overwrite, irreversible compression
$\dot{S}_{\text{hk}}$	$\sum_r \dot{Q}_{\text{refresh},r}/T_r + \dot{Q}_{\text{clock},r}/T_r$	Record maintenance and carrier upkeep
$\dot{S}_{\text{meas}}$	$\sum_r (\dot{Q}_{\text{sense},r} + \dot{Q}_{\text{channel},r} + \dot{Q}_{\text{record},r})/T_r$	Sensing, transmission, record formation
$\dot{S}_{\text{ctrl}}$	$\sum_r (\dot{Q}_{\text{act},r} + \dot{Q}_{\text{repair},r})/T_r$	Action, repair, boundary control
$\dot{S}_{\text{sync}}$	$\sum_r (\dot{Q}_{\text{coord},r} + \dot{Q}_{\text{verify},r} + \dot{Q}_{\text{latency},r})/T_r$	Coordination, verification, timing overhead
$\dot{S}_{\text{ext}}$	$\sum_r (\dot{Q}_{\text{write},r} + \dot{Q}_{\text{retrieve},r} + \dot{Q}_{\text{maint},r}^{\text{ext}})/T_r$	External records and scaffolds
$\dot{S}_{\text{rec}}$	$\sum_r \dot{Q}_{\text{recovery},r}/T_r$	Physical recovery, damage repair, dissipative reset, or uncontrolled relaxation after
$D_\ell(\ell)$	task-specific damage functional	Functional loss or resource damage, not entropy by itself

Erasure and housekeeping are distinct

A system may satisfy $H(M_t | M_{t+1}, Y_t) = 0$ for some update, yet still dissipate because it must keep records stable, clock reversible operations, synchronize subsystems, move carriers, correct errors, or hold a boundary far from equilibrium. Thus

$$\dot{S}_{\text{info}} = 0 \not\Rightarrow \dot{S}_{\text{phys}}^A = 0. \quad (15)$$

This distinction continues the P1 boundary between logical projection and dissipative physical implementation [2], and the P2 distinction between reversible logging and finite-memory housekeeping [3]. The O1 observer-register formalism further shows that measurement incurs stabilization costs through dynamic bottlenecks [6], while O2 register-time analysis identifies synchronization and latency as irreducible overheads in finite-record systems [7]. Recent experimental and theoretical work on finite-time memory erasure further confirms that realistic correction costs deviate from universal linear scaling and require concrete physical models [28].

CAPACITY DEFICIT AS CORRECTION PRESSURE

Positive-density deficit windows

Definition 2 (Positive-density deficit). A process has sustained positive deficit over update windows when

$$\liminf_{n \rightarrow \infty} \frac{1}{n} |\{1 \leq j \leq n : \Delta_\epsilon(t_j) > 0\}| > 0. \quad (16)$$

In continuous time, replace the count by $T^{-1} \int_0^T \mathbf{1}_{\Delta_\epsilon(t) > 0} dt$.

Correction information flow

Capacity deficit creates a pressure for correction, not a direct heat term. Since $\Delta_\epsilon(t)$ is a rate-distortion shortfall

over an update window, the minimal capacity-equivalent correction flow needed to fill that shortfall satisfies

$$i_{\text{corr}}^{\min}(t) \geq \frac{[\Delta_\epsilon(t)]_+}{\tau}, \quad (17)$$

up to relief supplied by task relaxation, invariant compression, pruning, or externalization. More generally,

$$i_{\text{corr}}^{\min}(t) \geq \frac{[[\Delta_\epsilon(t)]_+ - \mathcal{E}_{\text{exit}}(t)]_+}{\tau}, \quad (18)$$

where $\mathcal{E}_{\text{exit}}(t)$ is the capacity-equivalent relief provided by admissible exit channels, all expressed in capacity-equivalent relief units per update window. A minimal decomposition is

$$\mathcal{E}_{\text{exit}}(t) = \mathcal{E}_{\text{prune}}(t) + \mathcal{E}_{\text{ext}}(t) + \mathcal{E}_{\text{inv}}(t) + \mathcal{E}_{\text{relax}}(t) + \mathcal{E}_{\text{res}}(t),$$

where $\mathcal{E}_{\text{prune}}$ is pruning or compression that reduces effective task demand, $\mathcal{E}_{\text{ext}}$ is externalization that supplies effective capacity, $\mathcal{E}_{\text{inv}}$ is invariant compression that reduces maintenance load, $\mathcal{E}_{\text{relax}}$ is task relaxation that lowers $R_{\min}$, and $\mathcal{E}_{\text{res}}$ is resource expansion or channel upgrade that raises C_{eff} .

Example 1 (Window-level deficit accounting). A finite sensing-and-control system has $C_{\text{eff}} = 40$ bits per update window. A boundary-maintenance task initially requires $R_{\min} = 30$ bits per window and later rises to $R_{\min} = 70$. The capacity deficit is

$$\Delta_\epsilon = 70 - 40 = 30$$

bits per window. If externalization supplies $\mathcal{E}_{\text{ext}} = 20$ bits per window and task-preserving pruning supplies $\mathcal{E}_{\text{prune}} = 5$, the remaining correction burden is

$$[\Delta_\epsilon - \mathcal{E}_{\text{ext}} - \mathcal{E}_{\text{prune}}]_+ = 5$$

bits per window. P5 does not say that these five bits are heat by themselves. It says that maintaining the boundary without relaxing the task requires this remaining burden to be paid through some physical correction, update, repair, synchronization, or coupled external ledger.

A physical correction cycle has entropy-production pressure

$$\dot{\Sigma}_{\text{corr}} \geq \alpha_{\text{upd}} \dot{I}_{\text{upd}} + \alpha_{\text{repair}} R_{\text{repair}} + \alpha_{\text{sync}} R_{\text{sync}}, \quad (19)$$

with implementation-dependent coefficients. Equation (19) is a normal-form inequality, not a fundamental constant law.

A stronger measurable form

In model classes where physical correction entropy is lower-bounded by a coefficient $\kappa > 0$ times correction flow, the integrated ledger obeys

$$\int_0^T \dot{\Sigma}_{\text{phys}}^{\mathcal{A}}(t) dt \geq \frac{\kappa}{\tau} \int_0^T [[\Delta_\epsilon(t)]_+ - \mathcal{E}_{\text{exit}}(t)]_+ dt. \quad (20)$$

This is not a universal thermodynamic identity. It is a measurable FDS-P5 bridge form: as deficit pressure rises, some combination of correction entropy, exit relief, task loss, or failure should become observable.

The coefficient κ has units of entropy per capacity unit. It is implementation-dependent and is not asserted to equal $k_B \ln 2$, except in erasure-limited correction channels dominated by logically irreversible reset or overwrite. In those cases one may write $\kappa = \eta_{\text{impl}} k_B \ln 2$, where η_{impl} is an implementation efficiency factor capturing carrier dissipation, refresh overhead, and channel losses. This parametrization is a toy model for erasure-dominated channels and should not be read as a universal law.

Deficit crossing signatures

When Δ_ϵ crosses zero and task tolerance is held fixed, P5 predicts at least one of the following signatures: increased heat or resource use, reset bursts, refresh activity, latency, synchronization overhead, error floor, pruning, externalization, task relaxation, or failure risk. If the system accepts loss or abandons the task, positive deficit need not produce correction entropy; it produces boundary failure or task relaxation instead.

DEFICIT-DRIVEN ENTROPY-PRODUCTION PRESSURE THEOREM

Theorem 1 (Deficit-driven entropy-production pressure). *Consider a physically realized active-boundary FDS over task window τ . Suppose: (i) $\Delta_\epsilon(t) > 0$ persists over a positive-density set of update windows; (ii) boundary-maintenance loss must satisfy $\ell(t) \leq \ell_c$; (iii) the task is not relaxed; (iv) externalization, invariant compression, pruning relief, and resource expansion are bounded; and (v) physical correction channels*

have nonzero coupled-system entropy-production or resource cost. Then the system cannot maintain the same boundary task indefinitely at zero coupled-system entropy-production cost. It must enter at least one channel: increased physical entropy production, exported entropy production, pruning, invariant compression, task relaxation, resource expansion, passive degradation, or collapse.

Proof. Positive capacity deficit means the current effective capacity is insufficient to encode the least demanding admissible boundary statistic to tolerance ϵ over the update window. If the system maintains $\ell \leq \ell_c$ without relaxing the task, it must supply additional effective distinctions through correction, approximation update, repair, externalization, pruning, compression, or resource expansion. By hypothesis, the available exit relief is bounded and each physically realized correction or relocation channel has nonzero coupled-system entropy-production or resource cost. If those costs remain inside the finite resource envelope, the system persists at increased or redistributed entropy production. If they exceed the envelope and no exit channel is used, Eq. (11) drives the system toward maintenance failure. Therefore zero-cost indefinite maintenance under sustained deficit is impossible under the stated assumptions. $\square$

Corollary 1 (Deficit-maintenance-dissipation impossibility triangle). *An active finite system cannot jointly satisfy*

$$\Delta_\epsilon > 0 \text{ sustained}, \quad \ell \leq \ell_c \text{ unchanged}, \quad \dot{\Sigma}_{\text{phys}}^{\mathcal{A}} = 0 \quad (21)$$

under finite resources and nonzero physical correction costs. At least one term must change: the deficit is reduced, the task is relaxed, entropy is produced or exported, resources expand, passive degradation is accepted, or the boundary-maintenance regime fails.

Deficit–maintenance–dissipation impossibility triangle. Under the P5 physical bridge assumptions, sustained positive capacity deficit, fixed boundary-maintenance tolerance, and zero coupled-system entropy or resource cost cannot persist indefinitely:

$$\Delta_\epsilon > 0 \text{ sustained}, \quad \ell \leq \ell_c \text{ unchanged}, \quad \dot{\Sigma}_{\text{phys}}^{\mathcal{A}} = 0$$

are jointly impossible without pruning, compression, externalization, task relaxation, invariant stabilization, resource expansion, or collapse.

Remark 1. *The theorem does not assert that every deficit produces the same heat rate. It asserts that sustained active maintenance under positive deficit must appear somewhere in the coupled physical ledger or be replaced by an exit channel.*

PRUNING, COMPRESSION, AND EXTERNALIZATION

Pruning as dissipation return on investment

Pruning may incur a one-time cost Σ_{prune} while reducing future housekeeping and synchronization costs. Over horizon H , define a dissipation return on investment

$$\text{ROI}_{\Sigma}^{\text{prune}}(H) = \frac{\int_t^{t+H} \left(\dot{\Sigma}_{\text{tot}}^{\text{no prune}}(s) - \dot{\Sigma}_{\text{tot}}^{\text{prune}}(s) \right) ds - \Sigma_{\text{prune}}}{\Sigma_{\text{prune}} + \lambda D_{\text{task}}} \quad (22)$$

A task-preserving pruning operation is favorable when $\text{ROI}_{\Sigma}^{\text{prune}}(H) > 0$, equivalently

$$\int_t^{t+H} \Delta \dot{\Sigma}_{\text{future}}(s) ds > \Sigma_{\text{prune}} + \lambda D_{\text{task}}. \quad (23)$$

This expresses active forgetting, model distillation, cleanup, or biological self-repair as an entropy-management policy, not merely a failure. If task-relevant distinctions are destroyed, D_{task} increases and pruning becomes over-pruning.

Invariant compression

Let $q : X \rightarrow Q$ be a quotient map and $R_A = \bar{R}_A \circ q$ an identity predicate preserved under admissible perturbations. If

$$q \circ P_i = q \quad \forall P_i \in \mathcal{P}, \quad (24)$$

then the identity carried by R_A may be maintained through the lower-dimensional quotient q rather than all microstate distinctions. Invariant compression lowers entropy-production pressure when

$$\dot{\Sigma}_{\text{hk}}^q + \dot{\Sigma}_{\text{verify}}^q + \lambda D_{\text{task}}^q < \dot{\Sigma}_{\text{hk}}^X + \dot{\Sigma}_{\text{verify}}^X. \quad (25)$$

This links P5 to Phase-B invariant selection in FDS-T3 [5] and invariant-supported persistence in the FDS core [1].

Externalization is entropy relocation

Externalization changes the accounting boundary:

$$\mathcal{A}_{\text{local}} \rightarrow \mathcal{A}_{\text{coupled}} = \mathcal{A}_{\text{local}} \cup \mathcal{A}_{\text{ext}}. \quad (26)$$

Local entropy production may fall, but the coupled ledger includes write, verification, retrieval, latency, synchronization, carrier maintenance, and environmental noise:

$$\dot{\Sigma}_{\text{phys}}^{\text{coupled}} = \dot{\Sigma}_{\text{local}} + \dot{\Sigma}_{\text{write}} + \dot{\Sigma}_{\text{verify}} + \dot{\Sigma}_{\text{retrieve}} + \dot{\Sigma}_{\text{sync}} + \dot{\Sigma}_{\text{env}}. \quad (27)$$

Externalization therefore exports or restructures entropy production; it does not abolish the ledger. If the environment is itself a finite record-bearing system, excessive externalization can create verification, indexing, latency, or clogging costs, as modeled in N1 [4].

PRECISION-DISSIPATION AND SECOND-LAW HANDOFFS

P5 defines the entropy ledger under capacity deficit. It does not yet prove a thermodynamic uncertainty relation for boundary maintenance. However, the natural next step is a precision-dissipation bound linking fluctuations of boundary loss, repair currents, or record-update currents to entropy production. Standard thermodynamic uncertainty relations show that precision of currents in nonequilibrium systems is constrained by dissipation [24, 25], and recent entropic uncertainty relations connect observable Shannon entropy to entropy production [30]. In the FDS sequence, such precision-cost questions belong to P6.

P5 also hands off to O3. P1 defines physical carriers and erasure maps; P2 defines finite-memory reversible reuse; N1 defines maintenance load and collapse regimes; P5 defines the deficit-driven entropy ledger. O3 can then study how finite memory, coarse-grained irreversible update, record reuse, and boundary-maintenance cost form an operational Second-Law channel for active finite systems.

NUMERICAL MODELS AND SIMULATIONS

The simulations are deterministic synthetic demonstrations. They are not fits to physical memory devices, biological organisms, computers, or human-subject data. They provide reproducible diagrams for the P5 normal form. All figures and CSV outputs are generated by `code/generate_results.py`.

EXPERIMENTAL AND ENGINEERING PROTOCOLS

Protocol 1 (Deficit-crossing protocol). *Gradually increase task demand or reduce effective capacity so that $R_{\text{min}}^{(\tau)}(\epsilon; \Psi)$ crosses C_{eff} . Measure heat or power, refresh activity, reset events, latency, error floor, pruning, externalization, task relaxation, and boundary loss. Whenever possible, $R_{\text{min}}^{(\tau)}(\epsilon; \Psi)$ and C_{eff} should be estimated or pre-registered independently before observing heat, power, latency, reset, or failure signatures. This prevents post-hoc definition of deficit from the cost signal itself.*

Protocol 2 (Ledger decomposition protocol). *For a fixed task, assign physical processes to unique audit chan-*

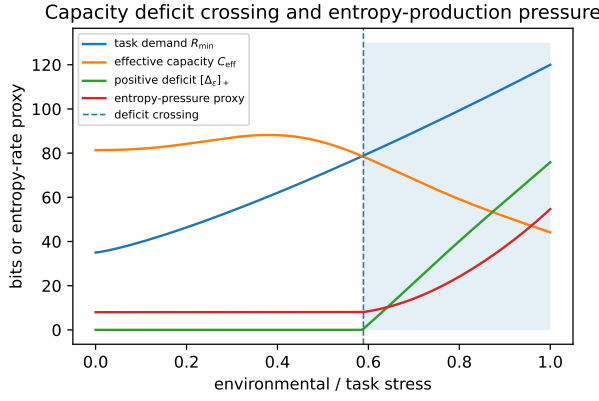


FIG. 1. Capacity deficit crossing and entropy-production pressure. Task demand rises above effective capacity at the dashed line; the shaded region marks $\Delta_e > 0$. Positive deficit increases the correction-flow and entropy-pressure proxies.

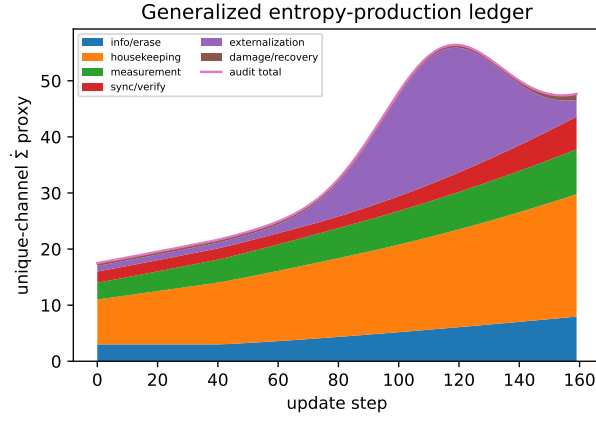


FIG. 2. Generalized entropy-production ledger. Stacked areas are illustrative unique-channel assignments under an audit convention. They should not be read as independent universal additive laws without a concrete process partition. The audit-total curve is a boundary-audited sum under the non-overlap convention, not an additional physical channel.

nels: information erasure, housekeeping, sensing, control, synchronization, externalization, and recovery. Verify that the same physical process is not double-counted across channels.

Protocol 3 (Pruning ROI protocol). Compare no pruning, task-preserving pruning, and over-pruning. Measure one-time pruning cost, future housekeeping reduction, task loss, and the finite-horizon quantity in Eq. (22).

Protocol 4 (Externalization full-boundary audit). Move records or functions outside the local boundary and measure local savings, write cost, verification cost, retrieval cost, synchronization, latency, and environmental noise. Local savings without coupled-system savings count as re-

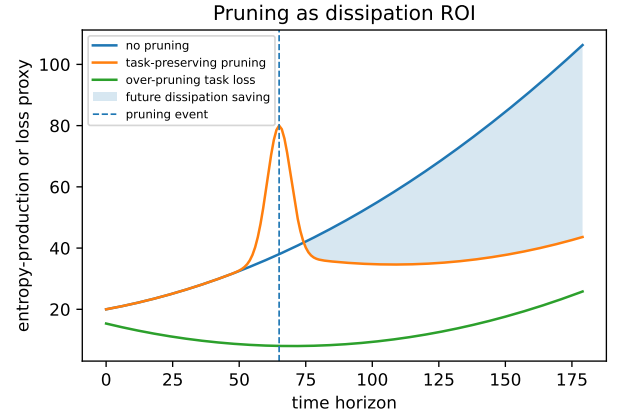


FIG. 3. Pruning as dissipation ROI. A one-time pruning event can reduce future entropy production; the shaded area illustrates future dissipation saving. Over-pruning is represented separately as task-loss penalty.

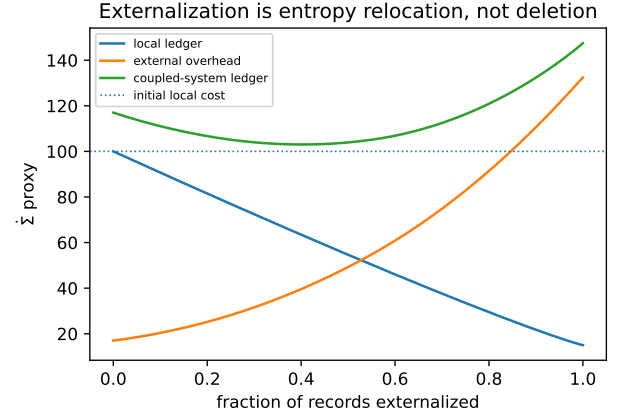


FIG. 4. Externalization is entropy relocation, not deletion. Local cost can fall while external overhead and coupled-system entropy production rise because write, verification, retrieval, synchronization, and environmental costs become active.

location, not elimination.

Protocol 5 (Hysteresis protocol). Increase deficit above the crossing threshold, then lower it again. Test whether entropy-production, resource use, or error curves return to the original path or retain a gap from erased records, damage, or lost invariant support.

RELATION TO EXISTING WORK

Landauer's principle links logical irreversibility and heat generation under physical assumptions [11, 12, 14]. Modern information thermodynamics extends this connection to measurement, feedback, stochastic trajec-

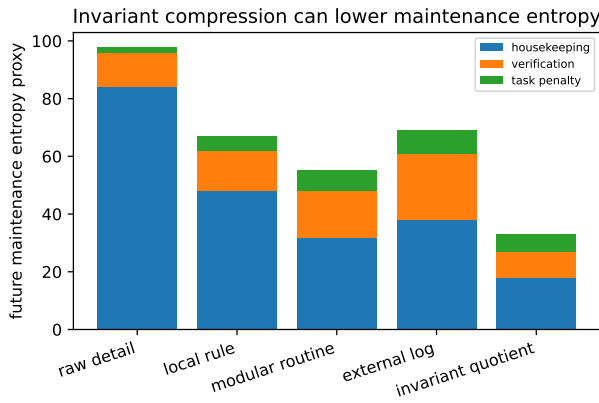


FIG. 5. Invariant compression can reduce maintenance entropy when the quotient retains task function with lower housekeeping and verification burden.

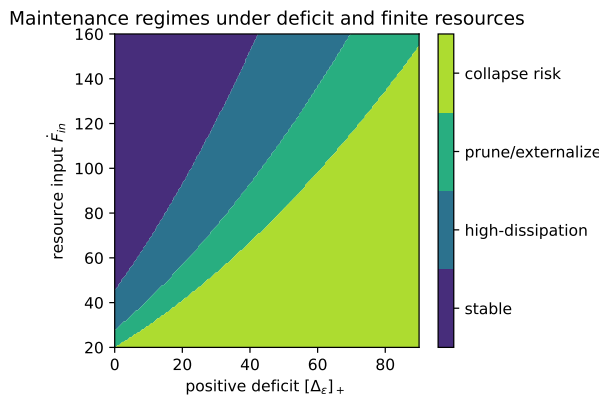


FIG. 6. Maintenance regimes under capacity deficit and finite resource input. Stable regimes satisfy the ledger within the resource envelope; high-dissipation regimes maintain the task by paying more entropy production; exit regimes require pruning or externalization; collapse risk appears when the envelope is exceeded.

ries, and nonequilibrium processes [15–18]. Stochastic thermodynamics of computation emphasizes finite-time, finite-bath, correlated, non-Markovian, and architectural constraints [20–23]. Thermodynamic uncertainty relations link dissipated entropy to precision constraints in nonequilibrium currents [24, 25].

P5 is not a replacement for those theories. It is an FDS bridge: it organizes the entropy-producing channels that appear when a finite active system maintains a boundary under capacity deficit. Compared with the free-energy principle and active inference [26, 31], P5 focuses less on variational objective functions and more on explicit task-relative capacity deficit, accounting boundaries, and entropy-production channels. Compared with dissipative-structure and self-organization theories, P5

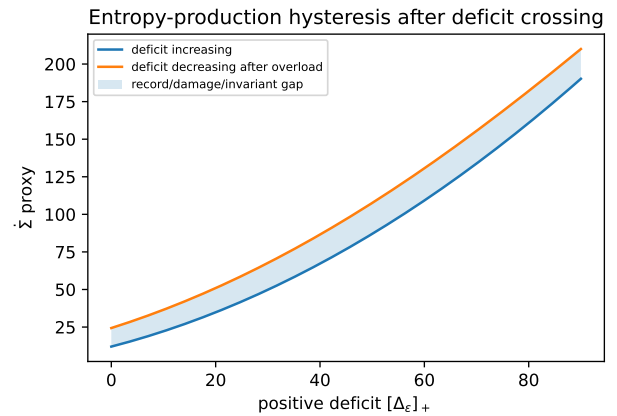


FIG. 7. Entropy-production hysteresis after deficit crossing. The gap between increasing and decreasing curves represents path-dependent record loss, accumulated damage, or destroyed invariant support; it is not asserted as a universal hysteresis law.

focuses on maintained distinctions and finite update ledgers rather than order formation alone.

LIMITATIONS AND FALSIFICATION

P5 does not claim that capacity deficit itself is heat, that every bit of uncertainty costs $k_B T \ln 2$, that all entropy production is Landauer erasure, or that all active systems share the same cost function. It also does not claim that pruning always helps, externalization always helps, or invariant compression is always available.

The strong version of P5 would be weakened or rejected by any of the following:

1. physically implemented logically irreversible erasure reliably violates the Landauer bridge under its stated assumptions;
2. active finite systems maintain sustained positive deficit and fixed task tolerance with no correction, externalization, pruning, task relaxation, loss increase, or physical cost;
3. correction, refresh, repair, sensing, synchronization, or externalization channels show no entropy-production or resource burden in controlled physical implementations;
4. externalization reduces local cost while imposing no coupled-system write, verification, retrieval, maintenance, latency, or environmental cost;
5. task-preserving pruning or invariant compression never reduces future maintenance load in any controlled system;

6. deficit crossing produces no measurable heat, energy, latency, reset, error, externalization, pruning, or failure signature.

CONCLUSION

FDS-P5 connects task-relative capacity deficit to entropy-production pressure in physically realized active finite systems. The compact statement is:

Capacity deficit does not dissipate heat by itself. But when a physically realized active finite system tries to maintain a boundary under sustained positive deficit, it must pay somewhere in the coupled ledger through entropy-producing correction, housekeeping, synchronization, externalization, pruning, task relaxation, recovery, or failure.

This places P5 between the physical accounting papers P1/P2, the complex-systems bridge N1, and the later O3 treatment of boundary maintenance and the Second Law under finite memory. The central object is not abstract entropy. It is a finite system trying to keep a boundary usable while its task demands exceed its available distinctions.

Notation summary

Symbol	Meaning
B	maintained boundary or interface
M	memory, model, or organizational state
Y	observation channel
U	internal update rule
ℓ	boundary-maintenance loss
Φ	finite resource or free-energy budget
τ	update window or timescale
$R_{\min}^{(\tau)}(\epsilon; \Psi)$	task-relevant rate-distortion demand
C_{eff}	effective task-available capacity
Δ_{ϵ}	capacity deficit $R_{\min}^{(\tau)} - C_{\text{eff}}$
I_{corr}	correction information flow
$H(M_t M_{t+1}, Y_t)$	logical erasure per update
$\dot{\Sigma}_{\text{phys}}^{\mathcal{A}}$	audited physical entropy-production rate
$\dot{\Sigma}_{\text{info}}$	informational erasure entropy-production rate
$\dot{\Sigma}_{\text{hk}}$	housekeeping entropy-production rate
$\dot{\Sigma}_{\text{ext}}$	externalization entropy-production rate
$\dot{\Sigma}_{\text{rec}}$	recovery or repair entropy-production rate
$D_{\ell}(\ell)$	functional boundary-loss damage
q	invariant quotient or compression map

Simulation parameters

The simulations use deterministic arrays with fixed synthetic parameters in `code/generate_results.py`. Figure 1 varies task stress and effective capacity. Figure 2 decomposes the entropy ledger using unique audit channels. Figure 3 compares no pruning, task-preserving pruning, and over-pruning. Figure 4 varies externalization fraction and plots local versus coupled cost. Figure 5 compares representation strategies. Figure 6 classifies deficit-resource regimes. Figure 7 illustrates hysteresis after overload. No proprietary, biological, organizational, human-subject, or device data are used.

Reproducibility checklist

1. Code availability: all simulation code is included in the replication package.
2. Deterministic execution: the code uses fixed synthetic parameters and deterministic arrays.
3. Figure reproduction: run `python code/generate_results.py`; the script regenerates all figures and CSV outputs.
4. Data status: all numerical outputs are synthetic demonstrations generated from the stated model.
5. Platform independence: the code uses standard Python scientific libraries.

Code availability

The simulation code used to generate Figs. 1–7 is included in the accompanying replication package under `code/generate_results.py`. Running the script regenerates all figures (PDF and PNG) and CSV tables in a single pass.

AI assistance disclosure

AI-assisted tools were used for language polishing, structural feedback, LaTeX drafting support, and code-debugging assistance. The author reviewed and edited all content and remains responsible for all claims, references, simulations, and conclusions. No AI system is listed as an author.

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Coarse-Grained Anti-Recurrence and Informational Hysteresis in Finite Memory Systems: Lost Preimages, Side Records, and Capacity-Recovery Asymmetry

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Finite observers often operate with effective representations that identify many microscopic states with a single coarse state. This paper studies the operational consequence of such non-injective truncation: once preimage information is absent from the effective record, later capacity recovery does not reconstruct the discarded distinctions. Guaranteed exact preimage recovery is possible only when inverse information is preserved in side records, reversible garbage registers, environmental traces, or an enlarged accounting boundary. We formalize this effect through preimage entropy, a Bayes-optimal recovery bound, residual inverse uncertainty, a side-record recurrence criterion, and an informational hysteresis observable. We replace a purely metric closure error with an information-theoretic Jensen-Shannon/KL family of Markov-closure errors, and relate the resulting hidden-state burden to the Mori-Zwanzig memory-kernel form of projected dynamics. The resulting anti-recurrence claim is representation-relative and does not deny microscopic reversibility, unitary dynamics, Poincare recurrence, or fluctuation theorems [9, 11–13]. It states that finite effective systems face an asymmetry between losing a distinction and later regaining capacity to store distinctions. Deterministic synthetic simulations illustrate recovery bounds, capacity hysteresis, externalization break-even, memory-kernel burden, Markov closure error, benign versus malignant hysteresis, Phase-B low-cost survivor selection, and the transition from ordinary P4 truncation to P7 invariant/topological protection.

Reader contract. This paper is a registered FDS physical-bridge paper. It is not a replacement for statistical mechanics, stochastic thermodynamics, Markov-chain theory, reversible computation, model reduction, or quantum mechanics. It does not deny microscopic reversibility, Liouville evolution, Hamiltonian recurrence, reversible computation, unitary quantum dynamics, or fluctuation relations under their own hypotheses [9–13]. It gives a finite-representation account of exact preimage recovery after non-injective truncation. Whenever guaranteed exact preimage recovery is possible, P4 asks where the inverse information was stored and what boundary, memory, synchronization, verification, or thermodynamic ledger carries it.

Claim-status summary

Table I summarizes the main claims, status, and failure conditions. The status labels are internal to the FDS corpus: the formal information claims are low-risk, while the physical bridge claims depend on the stated representation and accounting boundary.

Keywords: finite distinction systems; coarse-graining; anti-recurrence; informational hysteresis; preimage entropy; side records; Mori-Zwanzig; lumpability; Markov closure; finite memory; externalization; invariant persistence.

INTRODUCTION

The problem

Microscopic laws may be reversible while finite effective descriptions are not. This is compatible with standard physics. Coarse-graining, measurement, finite memory, model reduction, and statistical inference all use records that identify many microstates with fewer effective states. The question studied here is not whether the full microscopic universe loses information. It is whether a finite effective system that has already applied a non-injective truncation can later recover the specific micro-distinction it discarded merely by recovering storage capacity.

P4 answers no. Capacity recovery restores the ability to store new distinctions. It does not reconstruct old inverse information that is absent from the record. Capacity-recovery asymmetry is the gap between restored representational capacity and unrecovered historical inverse information. Guaranteed exact preimage recovery requires that the inverse information was preserved somewhere: a side log, reversible garbage register, environmental trace, external archive, protected invariant, or enlarged accounting boundary. If no such inverse information exists inside the boundary being audited, the lost distinction cannot recur inside that effective account.

This gives a representation-relative form of anti-recurrence. It is not anti-recurrence in the Poincare sense. It is anti-recurrence of exact preimage recovery inside a truncated record.

TABLE I. Central FDS-P4 claims, status, and failure or demotion conditions.

Claim	Status	What would weaken or falsify it
Non-injective truncation creates preimage uncertainty relative to the effective record.	Formal information claim	A many-to-one map contains enough information, without side records or conventions, to distinguish all of its preimages.
Bayes-optimal exact recovery is bounded by the largest conditional preimage mass.	Decision-theoretic bound	A decoder using only Z exceeds the Bayes-optimal classifier bound for $X Z$.
Capacity recovery does not recover distinctions erased during a bottleneck.	Informational hysteresis theorem	A finite system recovers exact task-relevant preimage distinctions after capacity restoration with no side record, no enlarged boundary, no external trace, and no hidden convention.
Non-lumpable coarse-graining creates hidden-state memory and effective stochasticity.	Markov projection bridge	A non-lumpable projection closes exactly on Z_t alone without hidden state, history, or extra variables.
Projection-induced memory burden has a Mori-Zwanzig analogue.	Relation to standard theory	Eliminated variables never reappear as memory, noise, or closure error in projected dynamics, even when lumpability fails.
Externalization restores inverse information only by moving it to a side ledger.	Accounting-boundary bridge	External logs restore guaranteed exact preimage recovery at no writing, retention, indexing, synchronization, retrieval, verification, or boundary-expansion cost.
Sustained truncation requires residual irrecoverability, side records, externalization, task relaxation, or failure.	Finite-memory exit theorem	A finite system repeatedly applies non-injective truncation to task-relevant distinctions while preserving guaranteed exact preimage recovery with no residual uncertainty and no extra ledger.

Main contributions

The paper contributes six objects to the FDS physical spine:

1. a preimage-entropy and Bayes-recovery bound for non-injective truncation;
2. a side-record recurrence criterion based on residual inverse uncertainty;
3. a strengthened hysteresis theorem for task variables nonconstant on a truncated fiber;
4. an information-theoretic Markov-closure error based on KL/JS divergence;
5. a bridge to Mori-Zwanzig memory kernels for projected dynamics;
6. an externalization break-even ledger and a benign/malignant hysteresis classification.

Terminological clarification: recurrence versus inverse recovery

The word “recurrence” has several meanings. P4 uses it only in the third sense below:

1. *Microscopic recurrence*: return of a full microstate or region in the complete state space under its full dynamics.
2. *Effective recurrence*: return of the same coarse state $z \in Z$.
3. *Preimage recovery*: reconstruction of which $x \in T^{-1}(z)$ occurred before the truncation.

P4 anti-recurrence means failure of exact preimage recovery inside a truncated representation. It does not mean failure of microscopic dynamical recurrence or failure of coarse-state recurrence. The same coarse state z may recur many times while the system remains unable to recover which prior preimage x produced it.

How P4 interfaces with T3, P5, and O3

P4 can be read independently as the representation-side truncation paper. In the broader FDS sequence, it supplies the inverse-information mechanism behind capacity overflow, entropy ledgers, and finite-memory record turnover. T3 studies overflow and effective stochasticity; P5 studies capacity deficit and entropy-production pressure; O3 studies finite-memory boundary maintenance and record reuse. P4 explains why record turnover is path-dependent: once preimages are absent

from the accounting boundary, current capacity alone cannot reconstruct them.

FDS BACKGROUND

Finite distinction systems

An active finite distinction system is represented by

$$S = (X, E, B, M, Y, A, U, \pi, \ell, \Phi, \mathcal{P}, \tau), \quad (1)$$

where X is internal state, E environment, B boundary, M memory/model state, Y observation channel, A action space, U update rule, π finite projection, ℓ boundary-maintenance loss, Φ finite resource budget, $\mathcal{P}$ perturbation/pruning family, and τ update timescale. P4 focuses on π , M , U , ℓ , and the accounting boundary deciding whether inverse information is still available.

Finite capacity and truncation

For finite memory M , internal representational capacity is

$$C_S = \log_2 |M|. \quad (2)$$

For a task family Ψ , rate-distortion demand over window τ is $R_{\min}^{(\tau)}(\epsilon; \Psi)$. The capacity deficit is

$$\Delta_\epsilon(\tau) = R_{\min}^{(\tau)}(\epsilon; \Psi) - C_S. \quad (3)$$

When $\Delta_\epsilon > 0$, the system must approximate, externalize, prune, relax the task, or fail. P4 studies which distinctions disappear inside the effective record during such approximation.

EFFECTIVE TRUNCATION AND PREIMAGE RECOVERY

Definition 1 (Effective truncation map). *An effective truncation map is a map*

$$T : X \rightarrow Z \quad (4)$$

from a high-resolution state space X to an effective state space Z . It is non-injective when there exist $x \neq x'$ with $T(x) = T(x')$. The preimage fiber of $z \in Z$ is

$$\text{Pre}_T(z) = T^{-1}(z) = \{x \in X : T(x) = z\}. \quad (5)$$

Definition 2 (Preimage entropy). *Given a distribution $p(x)$ and $Z = T(X)$, the preimage entropy of the effective record is*

$$H_T = H(X | Z). \quad (6)$$

For a uniform finite fiber, $H_T(z) = \log_2 |T^{-1}(z)|$.

For continuous state spaces, preimage entropy should be interpreted through a finite-resolution, rate-distortion, or error-tolerant recovery measure. Differential entropy is not used here as a direct count of recoverable distinctions.

Lemma 1 (Coarse-grained anti-recurrence). *In an effective representation containing a non-injective truncation map $T : X \rightarrow Z$, guaranteed exact recovery of a specific erased preimage is impossible from Z alone unless inverse information about the preimage is supplied from outside Z .*

Proof. Let $x \neq x'$ and $T(x) = T(x') = z$. The effective state z is compatible with at least two distinct preimages. Any map $R : Z \rightarrow X$ selecting x rather than x' uses information not contained in z unless a convention, side record, hidden state, environmental trace, or additional inverse state is supplied. Thus the distinction between x and x' cannot be recovered inside the effective representation alone. It can be recovered only in an enlarged representation containing inverse information. $\square$

Proposition 1 (Bayes-optimal exact recovery bound). *Let X be the preimage and $Z = T(X)$ the effective state. Any decoder $D : Z \rightarrow X$ has exact recovery probability bounded by*

$$P_{\text{rec}}^* = E_Z \left[\max_{x \in T^{-1}(Z)} p(x | Z) \right]. \quad (7)$$

For a uniform fiber of size $N_z = |T^{-1}(z)|$,

$$P_{\text{rec}}^*(z) = \frac{1}{N_z} = 2^{-H(X|Z=z)}. \quad (8)$$

Proof. For each observed z , the optimal exact decoder chooses the maximum a posteriori preimage. Its conditional success probability is $\max_{x \in T^{-1}(z)} p(x | z)$. Averaging over Z gives Eq. (7). This bound is a direct consequence of Bayesian decision theory and standard information-theoretic coding limits [5, 6, 17]. If the fiber is uniform, $p(x | z) = 1/N_z$ and Eq. (8) follows. $\square$

Corollary 1 (Microscopic reversibility compatibility). *The anti-recurrence lemma does not contradict microscopic reversibility. If the full microstate x is preserved in an enlarged system, or if reversible computation retains garbage registers sufficient to reconstruct the preimage, guaranteed exact preimage recovery may occur in that enlarged account. P4 applies to the reduced representation that lacks the preimage information.*

Example 1 (Four-state toy model). *Let*

$$X = \{00, 01, 10, 11\}, \quad T(x) = \text{first bit of } x. \quad (9)$$

Then $T(00) = T(01) = 0$ and $T(10) = T(11) = 1$. If a later task asks for the second bit, the effective record

TABLE II. P4 in the physical-operational FDS sequence.

Paper	Core object	P4 relation
T3	capacity overflow and effective stochasticity	P4 formalizes erased preimages behind projected randomness
P5	capacity deficit and entropy production	P4 supplies residual inverse-information loss for entropy/resource ledgers
O3	finite memory and operational Second-Law channel	P4 explains why record turnover remains path-dependent after capacity re
P7	invariant-supported persistence / topology	P4 defines ordinary local forgetting; P7 asks when local forgetting is obstr

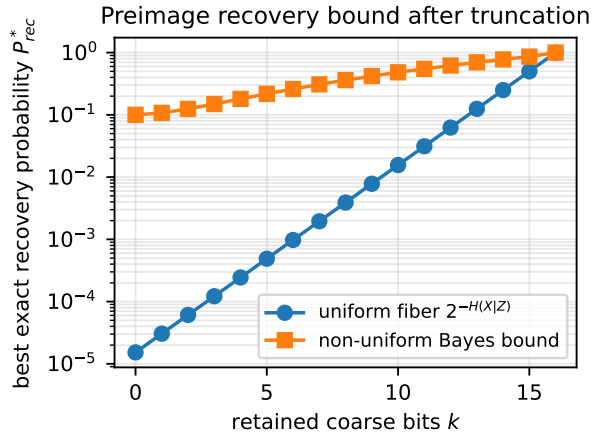


FIG. 1. Preimage recovery bound after non-injective truncation. For uniform fibers, exact micro-recovery decays as $2^{-H(X|Z)}$. Non-uniform fibers improve the Bayes-optimal bound only when one preimage dominates the posterior.

Z cannot recover it. Even if the system later expands from one-bit to two-bit capacity, the historical second bit is absent unless it was stored in a side record.

ACCOUNTING BOUNDARIES AND SIDE RECORDS

Definition 3 (Accounting boundary). *An accounting boundary $\mathcal{A}$ specifies which records, reservoirs, side logs, reversible garbage registers, environmental traces, external memories, and protected invariants are counted as accessible to the system. Information outside $\mathcal{A}$ is not operationally available for recovery inside $\mathcal{A}$.*

Definition 4 (Side-record inverse information). *Let $L_{\mathcal{A}}$ denote side records accessible within accounting boundary $\mathcal{A}$. The inverse information retained by that boundary after truncation is*

$$I_{\text{inv}}^{\mathcal{A}} = I(X; L_{\mathcal{A}} | Z). \quad (10)$$

The residual inverse uncertainty is

$$H_{\text{res}}^{\mathcal{A}} = H(X | Z, L_{\mathcal{A}}). \quad (11)$$

Proposition 2 (Side-record recovery bound). *For side records $L_{\mathcal{A}}$ accessible within boundary $\mathcal{A}$, the Bayes-optimal exact-recovery probability satisfies*

$$P_{\text{rec}}^*(Z, L_{\mathcal{A}}) = \mathbb{E}_{Z, L_{\mathcal{A}}} \left[\max_x p(x | Z, L_{\mathcal{A}}) \right], \quad (12)$$

and $P_{\text{rec}}^(Z, L_{\mathcal{A}}) \geq P_{\text{rec}}^*(Z)$, with strict improvement whenever $L_{\mathcal{A}}$ carries inverse information with $I(X; L_{\mathcal{A}} | Z) > 0$.*

Criterion 1 (Side-record recovery criterion). *Guaranteed exact preimage recovery within boundary $\mathcal{A}$ is possible only if accessible side records contain enough inverse information to identify the relevant preimage up to task tolerance:*

$$H(X | Z, L_{\mathcal{A}}) \leq \epsilon_{\text{inv}}. \quad (13)$$

For exact finite recovery, $\epsilon_{\text{inv}} = 0$.

Proof. An exact recovery operation is a decoder $D : Z \times L_{\mathcal{A}} \rightarrow X$. If $H(X | Z, L_{\mathcal{A}}) > 0$, multiple preimages remain compatible with the available record, so guaranteed exact recovery is impossible without additional information or arbitrary guessing. If the conditional entropy is zero, then X is a function of $(Z, L_{\mathcal{A}})$ almost surely, so an exact decoder exists in principle. $\square$

Remark 1 (Externalization is not free). *Side records can reduce residual uncertainty only by expanding the accounting boundary. External records must be written, preserved, indexed, synchronized, verified, retrieved, protected, and eventually pruned. Guaranteed exact preimage recovery may be restored, but the burden moves from lost preimage uncertainty to side-record maintenance.*

CAPACITY-RECOVERY ASYMMETRY AND INFORMATIONAL HYSTERESIS

Capacity recovery is not distinction recovery

Let C_t be available effective capacity. A system can lose capacity and later regain it. P4's claim is that this does not imply recovery of distinctions erased during the low-capacity interval. The system expands from the post-truncation record, not from the original preimage, unless inverse information was preserved.

Definition 5 (Informational hysteresis). Let X_{past} be a past high-resolution state or task-relevant variable, Z_t the current effective record, C_t current capacity, and L_{ext} accessible inverse records. Informational hysteresis is

$$H_{\text{irr}}(t) = H(X_{\text{past}} | Z_t, C_t, L_{\text{ext}}). \quad (14)$$

The system exhibits informational hysteresis if, after capacity recovers to a previous value, H_{irr} remains larger than it would have been along a path that never crossed the bottleneck.

Theorem 1 (Task-relative informational hysteresis). Let $V = f(X_{t_0})$ be a task variable. Suppose a non-injective truncation T maps $x, x' \in X_{t_0}$ to the same effective state but $f(x) \neq f(x')$, and that both preimages remain accessible with nonzero posterior probability under the final record:

$$p(x | Z_f, C_f, L_f) > 0, \quad p(x' | Z_f, C_f, L_f) > 0. \quad (15)$$

If no accessible side record distinguishes x from x' , then after any later capacity recovery that does not add inverse information,

$$H(V | Z_f, C_f, L_f) > 0. \quad (16)$$

Thus any future task requiring V suffers residual uncertainty despite restored capacity.

Proof. Since $T(x) = T(x')$ but $f(x) \neq f(x')$, the effective record identifies two task-distinct preimages. By assumption, L_f does not distinguish them. Therefore V has at least two possible values conditioned on (Z_f, C_f, L_f) , giving positive conditional entropy. Restoring storage capacity changes the size of possible future records but does not supply the missing inverse information about which preimage occurred at t_0 . $\square$

Corollary 2 (Path dependence). Two systems may have the same final capacity and effective update rule but different recoverable past distinctions if one crossed a truncation bottleneck and the other did not.

Definition 6 (Hysteresis gap). For a bottleneck path γ_b and a non-bottleneck path γ_0 with the same final capacity, define

$$\Delta H_{\text{hyst}} = H_{\gamma_b}(X_{\text{past}} | Z_f, L_f) - H_{\gamma_0}(X_{\text{past}} | Z_f, L_f). \quad (17)$$

A positive value indicates path-dependent inverse-information loss.

Benign versus malignant hysteresis

Not every erased distinction is harmful. A finite system must discard many details to act at all. P4 therefore distinguishes two regimes.

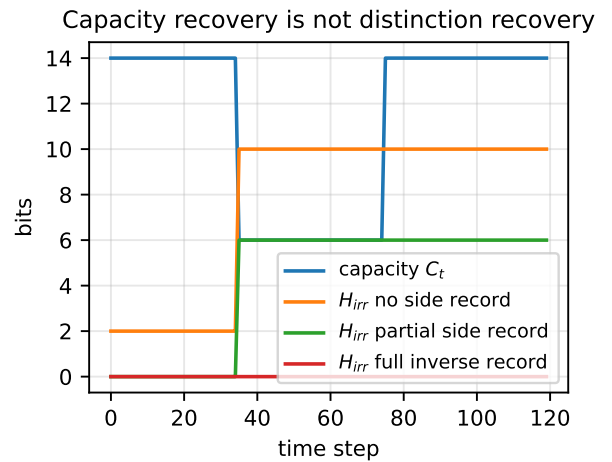


FIG. 2. Capacity recovery is not distinction recovery. Capacity drops and later returns, but residual irrecoverable information remains unless inverse records are retained. Full logs remove residual uncertainty only by expanding the accounting boundary.

Definition 7 (Benign informational hysteresis). Hysteresis is benign relative to a task family Ψ when erased distinctions are task-irrelevant or replaced by cheaper invariant variables without increasing expected task loss:

$$\mathbb{E}[\ell_\Psi | H_{\text{irr}} > 0] \approx \mathbb{E}[\ell_\Psi | H_{\text{irr}} = 0]. \quad (18)$$

Definition 8 (Malignant informational hysteresis). Hysteresis is malignant relative to Ψ when erased distinctions later become task-relevant and residual inverse uncertainty increases expected loss:

$$\mathbb{E}[\ell_\Psi | H_{\text{irr}} > 0] > \mathbb{E}[\ell_\Psi | H_{\text{irr}} = 0]. \quad (19)$$

This distinction is crucial for later FDS applications. Abstraction, learning, model reduction, and scientific law formation often require benign hysteresis: details are discarded to reveal stable patterns. Catastrophic forgetting, institutional memory loss, fragile AI agents, and degenerative memory failure are malignant cases: distinctions that later matter cannot be recovered from the effective record.

PROJECTED DYNAMICS, MARKOV CLOSURE, AND MEMORY KERNELS

Lumpability and hidden-state memory

Let (X_t) be a Markov chain on X with transition kernel $P(x' | x)$ and let $Z_t = T(X_t)$. The projected process is Markovian in Z_t alone only under lumpability: for any

x, x' in the same fiber $T^{-1}(z)$,

$$\sum_{y:T(y)=z'} P(y|x) = \sum_{y:T(y)=z'} P(y|x') \quad (20)$$

for every projected state z' .

Proposition 3 (Non-lumpability creates hidden-state memory). *If Eq. (20) fails, then (Z_t) is not Markovian with respect to Z_t alone. Any Markovian representation requires additional hidden state, history, or an enlarged coarse variable.*

Proof. If Eq. (20) fails, two microstates in the same projected class have different projected transition probabilities. Since Z_t cannot identify which microstate is present, $\mathbb{P}(Z_{t+1} | Z_t)$ depends on the conditional distribution over hidden preimages. That conditional distribution generally depends on prior observations. Hence the projected dynamics carries hidden-state memory. Failure of a projected dynamics to remain Markovian is a standard feature of lumped Markov processes and coarse-grained stochastic thermodynamics [7, 8, 18, 19]. $\square$

Definition 9 (Information-theoretic Markov closure error). *For a projection T , define*

$$\mathcal{M}_{\text{DJS}}(T) = \mathbb{E}_z \mathbb{E}_{x, x' \sim p(\cdot|z)} [\text{D}_{\text{JS}}(P_T(\cdot|x) \| P_T(\cdot|x'))], \quad (21)$$

where

$$P_T(z'|x) = \sum_{y:T(y)=z'} P(y|x). \quad (22)$$

A KL-family version is

$$\mathcal{M}_{\text{DKL}}(T) = \mathbb{E}_z \mathbb{E}_{x \sim p(\cdot|z)} \text{D}_{\text{KL}}(P_T(\cdot|x) \| \bar{P}_T(\cdot|z)), \quad (23)$$

where $\bar{P}_T(\cdot|z) = \mathbb{E}_{x \sim p(\cdot|z)} P_T(\cdot|x)$. Both errors vanish for strongly lumpable projections.

Relation to Mori-Zwanzig memory kernels

Projection-operator methods make the same structural fact visible in continuous statistical mechanics. Let L be a microscopic generator and P a projection onto resolved variables. Mori-Zwanzig theory yields a generalized equation of the schematic form

$$\frac{d}{dt} P A_t = P L P A_t + \int_0^t K(t-s) P A_s ds + F_t, \quad (24)$$

where K is a memory kernel and F_t is an orthogonal noise term [14–16, 21]. P4 does not rederive Mori-Zwanzig. It uses it as a standard expression of the same burden: eliminated variables do not become irrelevant merely because they are absent from the resolved representation. They re-enter as memory, noise, or closure error unless the projection is dynamically closed.

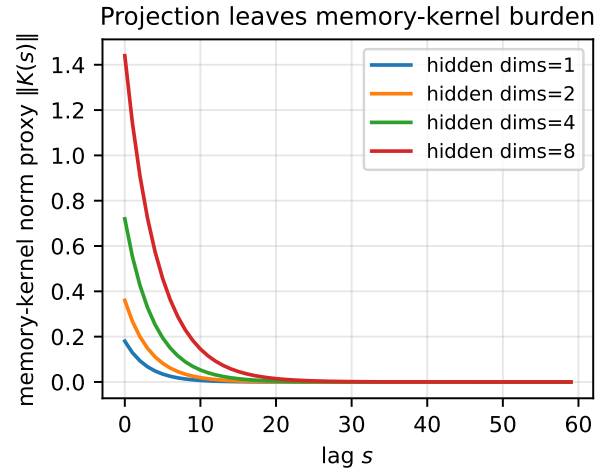


FIG. 3. Stylized Mori-Zwanzig memory-kernel burden. Eliminated hidden dimensions leave longer or larger memory-kernel tails in the resolved dynamics. The figure is synthetic and illustrates the bridge rather than fitting data.

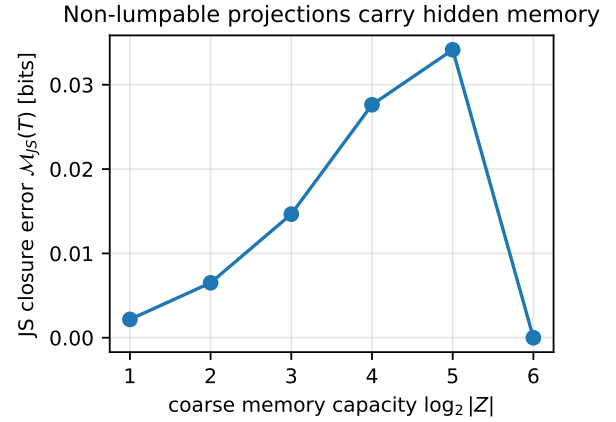


FIG. 4. Jensen-Shannon Markov closure error for a synthetic 64-state chain under increasingly fine projections. Coarser non-lumpable projections carry larger hidden-state memory burden.

In FDS terms, the memory kernel is an operational shadow of inverse-information deficit. A finite observer that deletes variables but still faces tasks depending on their influence must either carry memory of them, accept stochastic residuals, enlarge the representation, externalize records, or relax the task. The long tail of the memory kernel measures the additional computational and dissipative burden required to compensate for preimage information removed from the resolved representation.

EXTERNALIZATION AND BREAK-EVEN AUDITS

Externalization as inverse-information relocation

External memory can prevent anti-recurrence by storing inverse information [9–11]. It does not make recovery free. It moves the ledger from internal preimage uncertainty to external writing, synchronization, verification, retrieval, indexing, and protection.

Let D denote erased task-relevant inverse information. Let L denote accessible side-record inverse information. A simple residual model is

$$H_{\text{res}} = \max\{D - L, 0\}. \quad (25)$$

External cost may be decomposed as

$$C_{\text{ext}}(L) = C_{\text{write}}(L) + C_{\text{sync}}(L) + C_{\text{verify}}(L) + C_{\text{retrieve}}(L) + C_{\text{protect}}(L). \quad (26)$$

Definition 10 (Externalization surplus). Let $C_{\text{int}}^{\text{recover}}(D)$ be the expected internal cost of reacquiring or reconstructing lost inverse information without a side record. Define

$$\Pi_{\text{ext}}(D, L) = C_{\text{int}}^{\text{recover}}(D) - C_{\text{ext}}(L). \quad (27)$$

Externalization is favorable when

$$\Pi_{\text{ext}}(D, L) > 0 \quad \text{and} \quad H_{\text{res}}(D, L) \leq \epsilon_{\text{task}}. \quad (28)$$

Remark 2. The break-even point is not merely economic. It appears in biology, cognition, AI memory, organizations, science, law, and civilization. Writing a log, preserving an archive, delegating to a tool, or encoding a standard is favorable only when the reduced internal burden exceeds the maintenance and verification cost of the external record.

PHASE-B SURVIVORS AND THE P7 BRIDGE

Low-cost predictive survivors

P4 is not only about loss. Finite systems also gain abstraction by discarding details that do not matter for future maintenance. A coarse variable is useful when it remains predictive while being cheap to update, maintain, and dynamically close.

Definition 11 (Coarse-variable survival score). Let ϕ be a candidate coarse variable. Define

$$S(\phi) = \frac{I(\phi_t; \phi_{t+\tau})}{C_{\text{upd}}(\phi) + C_{\text{maint}}(\phi) + \lambda \mathcal{M}_{\text{DJS}}(\phi) + \epsilon_0}. \quad (29)$$

High-score variables are Phase-B candidates: low-cost predictive residues that remain useful under finite capacity.

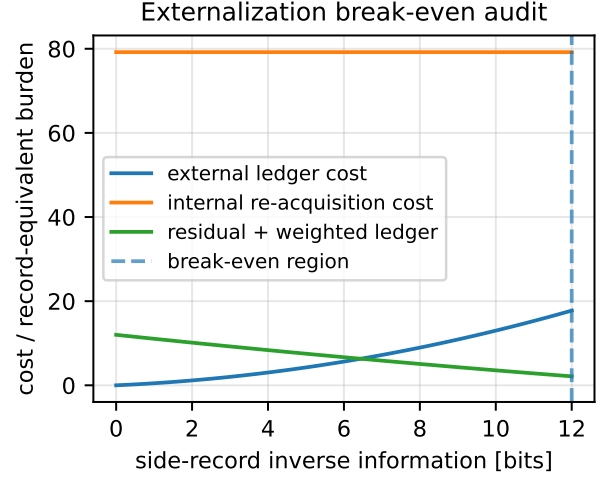


FIG. 5. Externalization break-even audit. Side records can reduce residual inverse uncertainty, but they impose writing, synchronization, retrieval, and verification costs. Externalization is favorable only beyond a break-even region.

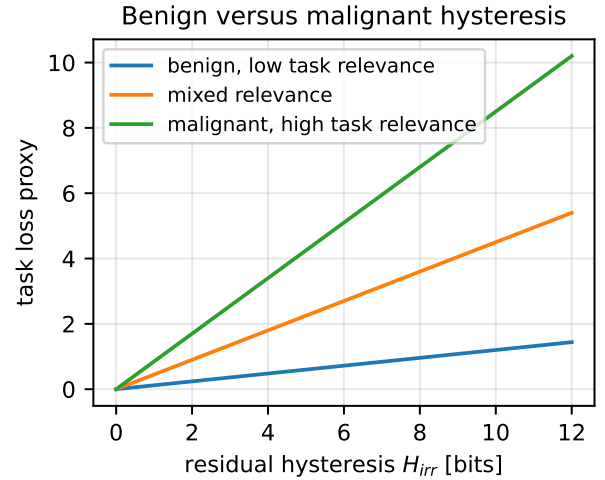


FIG. 6. Benign versus malignant informational hysteresis. The same residual inverse uncertainty has small effect when erased distinctions are task-irrelevant, but large effect when future tasks depend on them.

Remark 3. This is a heuristic selection criterion, not a universal law. Its role in P4 is to separate destructive loss from abstraction. Benign hysteresis often occurs when erased detail is replaced by a lower-cost, more stable predictive variable.

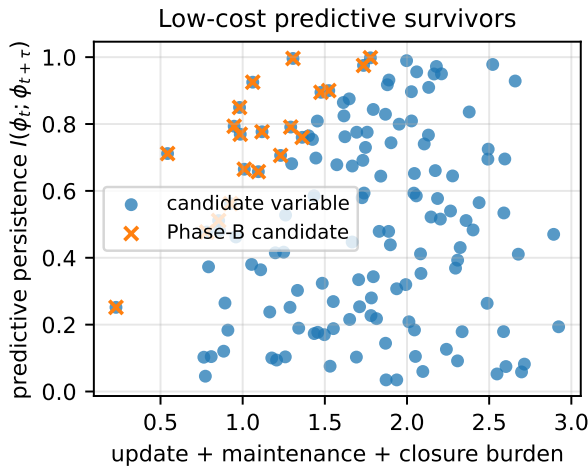


FIG. 7. Synthetic Phase-B candidate selection. Variables with high predictive persistence per update, maintenance, and closure burden are low-cost survivors under finite capacity.

Bridge to P7: invariant and topological protection

P4 studies distinctions lost under ordinary local truncation. P7 studies distinctions protected against local truncation by invariant or topological encoding. Phase-B survivors are not arbitrary compressed residues; the strongest survivors are variables whose identity is carried by invariants, often topological or global, so that local truncation leaves little residual preimage uncertainty. If $\mathcal{I}_{\text{top}}$ denotes an accessible invariant sufficient to identify the relevant equivalence class, then the protected inverse uncertainty may vanish:

$$H(X_{\text{fiber}} | Z, \mathcal{I}_{\text{top}}) = 0, \quad H(X_{\text{task}} | Z, \mathcal{I}_{\text{inv}}) \approx 0. \quad (30)$$

The transition from P4 to P7 is therefore natural: P4 defines the baseline fate of unprotected distinctions; P7 asks when the boundary, topology, or invariant structure prevents local forgetting without crossing a protection-breaking transition.

FINITE-MEMORY EXIT THEOREM

Theorem 2 (Finite-memory anti-recurrence exit theorem). *Let an active finite distinction system maintain a task at tolerance ϵ under finite effective memory C_t . Suppose it undergoes non-injective truncations erasing task-relevant inverse information with cumulative load $D(t) > 0$. If the system must later recover those micro-distinctions to keep the same tolerance, then at least one of the following must occur:*

1. *residual irrecoverability is accepted and task loss increases;*

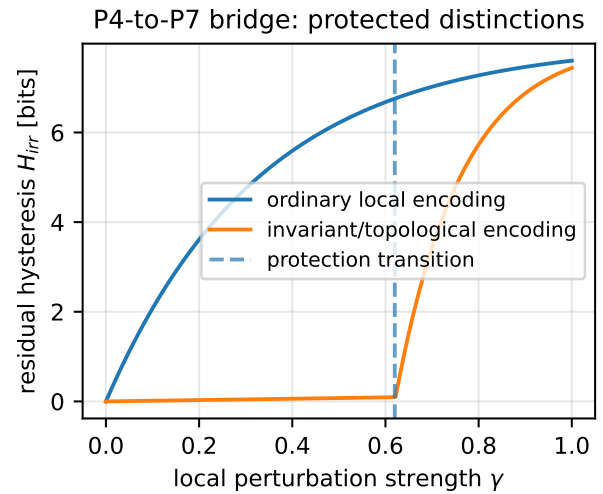


FIG. 8. P4-to-P7 bridge. Ordinary local encoding accumulates residual hysteresis under perturbation. Invariant or topological encoding can keep inverse uncertainty near zero until protection is broken. The figure is a schematic synthetic bridge, not an experimental claim. It is not a derived topological model.

2. *inverse information is maintained in side records inside an enlarged boundary;*
3. *inverse information is externalized to an environment, archive, tool, institution, or record system;*
4. *the task is relaxed so erased distinctions are no longer required;*
5. *the system fails to maintain the boundary or task.*

A finite system cannot repeatedly apply task-relevant non-injective truncation while preserving guaranteed exact preimage recovery with no side records, no externalization, no task relaxation, no enlarged boundary, and no failure.

Proof. By the anti-recurrence lemma, erased preimage distinctions are not recoverable from the truncated state alone. If guaranteed exact preimage recovery is required, the side-record criterion demands sufficient inverse information within the accounting boundary. If this information is not maintained internally, it must be externalized, or the task must be relaxed. If none of these occurs, the system lacks information required to maintain the original tolerance; loss increases or the boundary/task fails. $\square$

Corollary 3 (O3 connection). *If maintaining side records requires physical record turnover, refresh, verification, synchronization, overwrite, or cleanup, then the recovery-preserving channel enters the O3 entropy/resource ledger. P4 supplies the inverse-information term behind finite-record irreversibility.*

OPERATIONAL TESTS AND FAILURE CONDITIONS

P4 test template

A P4 experiment, simulation, or case study should specify:

1. high-resolution state X ;
2. effective projection $T : X \rightarrow Z$;
3. task variable $V = f(X)$;
4. whether V is constant or nonconstant on relevant fibers;
5. accounting boundary $\mathcal{A}$;
6. side records $L_{\mathcal{A}}$;
7. capacity bottleneck path C_t ;
8. recovery condition after capacity restoration;
9. residual uncertainty $H(V \mid Z, L_{\mathcal{A}})$;
10. Markov closure error $\mathcal{M}_{\text{DJS}}$ or $\mathcal{M}_{\text{DKL}}$;
11. externalization cost and break-even condition;
12. failure, demotion, or task-relaxation outcome.

Falsification and demotion

P4 would be weakened or falsified by a verified case satisfying all of the following:

1. a non-injective truncation identifies task-distinct preimages;
2. no side record, hidden state, external log, environmental trace, protected invariant, or enlarged boundary contains inverse information;
3. after capacity recovery, the system exactly recovers the erased task-relevant preimage;
4. no hidden convention, stochastic lucky guess, additional physical degree of freedom, or task redefinition supplies the missing information.

Demotion occurs if the projection was misidentified, the task variable was constant on the fiber, or the erased distinction was not task-relevant.

Boundary of applicability

P4 applies to finite effective representations containing non-injective truncation. It does not apply to complete microscopic descriptions that retain all preimage information. It does not apply to tasks for which erased distinctions are irrelevant. It does not apply when external logs are counted as part of the system and contain sufficient inverse information.

NUMERICAL DEMONSTRATIONS

The simulations are deterministic synthetic demonstrations generated by `code/generate_results.py`. They illustrate definitions and theorems rather than fitting empirical data.

1. Figure 1 shows the Bayes-optimal exact recovery bound after non-injective truncation.
2. Figure 2 shows capacity-recovery asymmetry after a bottleneck.
3. Figure 3 shows the Mori-Zwanzig memory-kernel burden from eliminated hidden dimensions.
4. Figure 4 computes Jensen-Shannon Markov closure error for a synthetic 64-state chain.
5. Figure 5 compares external ledger cost with internal re-acquisition cost for $D = 12$ erased inverse bits.
6. Figure 6 classifies benign versus malignant hysteresis using task-loss sensitivity.
7. Figure 7 samples Phase-B survivor candidates by predictive utility, update cost, and closure error.
8. Figure 8 illustrates the P4-to-P7 transition from ordinary side records to invariant-supported protection.

Simulation parameters

The script uses fixed deterministic models. Figure 1 uses $n = 16$ binary distinctions and varies retained bits. Figure 2 uses a capacity schedule with a bottleneck from $t = 35$ to $t = 75$. Figure 5 compares external ledger cost with internal re-acquisition cost for $D = 12$ erased inverse bits. Figure 3 uses exponentially decaying memory-kernel proxies for eliminated hidden dimensions. Figure 4 constructs a synthetic 64-state Markov chain and computes Jensen-Shannon closure error under modular partitions. Figure 6 varies task relevance for residual hysteresis. Figure 7 samples candidate coarse variables with

predictive information, update cost, maintenance cost, and closure error. Figure 8 is a schematic protected-versus-unprotected distinction model. No proprietary, biological, medical, organizational, quantum-device, or human-subject data are used.

Reproducibility checklist

1. Code availability: all simulation code is included in the replication package.
2. Deterministic execution: fixed seeds are used where synthetic randomness appears.
3. Figure reproduction: `run python code/generate_results.py`.
4. Data status: all numerical outputs are synthetic demonstrations generated from the stated model.
5. Platform independence: the code uses standard Python scientific libraries.

AI assistance disclosure

AI-assisted tools were used for language polishing, structural feedback, LaTeX drafting support, and code-debugging assistance. The author reviewed and edited all content and remains responsible for all claims, references, simulations, and conclusions. No AI system is listed as an author.

RELATION TO EXISTING THEORY

Statistical mechanics

P4 is compatible with statistical mechanics. Coarse-graining and entropy increase are familiar in Gibbsian and Boltzmannian frameworks. P4 adds an operational finite-system statement: capacity recovery after truncation does not reconstruct preimage distinctions unless inverse information is preserved inside the accounting boundary.

Reversible computation

Bennett-style reversible computation avoids erasure costs by retaining enough history to reverse computations. P4 agrees. Recovery is possible when inverse information is retained. It becomes costly when finite memory forces cleanup, overwrite, compression, externalization, or task abandonment.

Stochastic thermodynamics

Stochastic thermodynamics tracks entropy production along trajectories [12, 13, 18–20]. P4 does not replace trajectory-level accounting. It identifies a representation-level source of hidden variables, effective stochasticity, and residual inverse uncertainty that must be included in the ledger when guaranteed exact preimage recovery is demanded.

Markov state models and model reduction

The lumpability condition is standard in Markov-chain theory [7, 8]. Mori-Zwanzig projection is standard in nonequilibrium statistical mechanics and model reduction [14–16]. P4 uses both as bridges: projected variables are dynamically clean only when closure holds; otherwise, discarded distinctions return as memory kernels, noise, or closure error.

Topological protection

P4 defines ordinary local forgetting. P7 will study the complementary case: distinctions protected by global invariants or topology, including recent developments in non-Hermitian skin effects and generalized Brillouin-zone physics [22–24]. The P4 claim is therefore not that all distinctions are equally forgettable. It is that unprotected distinctions erased by non-injective truncation cannot be recovered without inverse information.

CONCLUSION

P4 establishes a finite-system account of effective anti-recurrence. A non-injective truncation does not merely hide detail; it removes inverse information from the effective record. The system may later regain capacity, but capacity recovery is not distinction recovery: capacity-recovery asymmetry is the fundamental gap between restored storage capacity and unrecovered preimage information. Guaranteed exact preimage recovery requires side records, reversible garbage registers, environmental traces, external archives, protected invariants, or an enlarged accounting boundary.

The paper’s central object is therefore not the triviality that many-to-one maps lack inverses. It is the ledger that follows from that fact in finite systems: which task distinctions were erased, whether they later matter, what side records preserve them, what costs those records impose, whether the projected dynamics closes, and when hysteresis is benign abstraction rather than malignant loss. In this sense, forgetting is not merely failure. It

is the price finite systems pay to produce stable macroscopic variables. Under repeated capacity pressure, finite systems are driven toward pruning, externalization, task relaxation, or invariant-supported persistence. P4 supplies the representation-side bridge for that transition.

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Finite-Bath Memory, Markovianization, and Environmental Forgetting in Finite Distinction Systems: Side Records, Memory Kernels, and the Loss of Recoverable Distinctions

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Finite systems do not lose distinctions only through internal truncation. They also depend on environments, baths, instruments, logs, niches, and external carriers whose own memory is finite. FDS-P4 studied coarse-grained anti-recurrence inside an effective representation: once a non-injective truncation removes preimage information, later capacity recovery does not reconstruct the erased distinction unless inverse information is preserved elsewhere. FDS-P3 studies that “elsewhere.” It develops a finite-bath account of environmental forgetting: environmental side records initially available for recovery can become operationally unrecoverable through mixing, dilution, finite-bath saturation, inaccessible correlations, coarse readout, finite update windows, or Markovianization. The relevant object is not the full bath state, but the accessible environmental readout $R_t^E = g_t(E_t)$ induced by the observation channel, latency, cost, and resolution. We define environmental side records, accessible and hidden bath memory, bath-retained inverse information, Markov closure error, memory-kernel burden, side-record recovery probability, finite-bath saturation, and an environmental resource ledger. The central claim is conditional: an effective Markovian description is valid for a finite active system only when accessible environmental history no longer improves prediction, recovery, or boundary maintenance at the registered tolerance. P3 does not deny microscopic reversibility, unitary evolution, Liouville dynamics, or Poincaré recurrence [24]. It identifies the conditions under which a finite system can no longer operationally recover a distinction from its environment. Deterministic normal-form simulations illustrate finite-bath memory decay, Markov closure error, memory-kernel burden, side-record recovery, bath saturation, Markovianization windows, environmental ledger costs, a robot side-record worked example, and P3/P4/P7 regimes.

Reader contract. This paper is a registered FDS physical-bridge paper. It is not a replacement for open-system physics, statistical mechanics, stochastic thermodynamics, hidden Markov models, projection-operator methods, or Markov-chain theory. It does not claim that environments literally annihilate microscopic information. It does not claim that all reservoirs are infinite, memoryless, or exactly Markovian. Environmental forgetting means loss of recoverable inverse information relative to an accounting boundary, an observation channel, an update window, and a task tolerance. The symbol R_t^E denotes an accessible environmental readout, not the full physical bath state. If a side record remains accessible and recoverable, P3 requires the accounting boundary to include it rather than reinterpreting the case as forgetting.

Claim-status summary

Table I summarizes the main claims, status, and failure conditions. The formal statements are about conditional information and projection; the physical bridge statements require specified bath access, finite update windows, and accounting boundaries.

Keywords: finite distinction systems; finite bath; environmental forgetting; Markovianization; side records;

memory kernels; Mori-Zwanzig; Nakajima-Zwanzig; non-Markovianity; lumpability; open systems; finite memory; environmental ledger; Physical AI.

INTRODUCTION

From P4 internal loss to P3 environmental forgetting

P4 asks what happens when a finite effective representation identifies multiple microstates with one coarse state. If the internal record loses a task-relevant preimage, later capacity recovery does not reconstruct it unless inverse information was preserved in a side record, external ledger, reversible garbage register, environmental trace, or protected invariant [2]. P3 audits the most common rescue clause in that sentence: the environment.

The central question is:

If the missing distinction was exported to the environment, for how long does the environment keep it recoverable for the finite system that needs it?

A microscopic universe may remain reversible while a finite system loses access to the environmental record it would need for exact recovery. Environmental forgetting is not annihilation of information. It is failure of operational recovery from the accessible bath.

TABLE I. Central FDS-P3 claims, status, and failure or demotion conditions.

Claim	Status	What would weaken or falsify it
Environmental side records have finite accessible recovery capacity.	Operational FDS bridge	A finite system recovers unbounded inverse information from the environment through a finite observation channel with no latency, cost, degradation, or boundary expansion.
Markovianization is an effective forgetting condition.	Model-class bridge	A projected process is treated as Markovian while accessible history measurably improves prediction or boundary maintenance under the same variables and accounting boundary.
Memory kernels measure unresolved environmental memory.	Projection-form bridge	Eliminated variables never reappear as memory, noise, or closure error in projected dynamics despite coupling and non-lumpable projection.
Finite baths can remember temporarily, forget operationally, and recur.	Physical/model-class caveat	A finite bath is always exactly Markovian and never returns correlations under any admissible finite-bath model.
Environmental forgetting complements P4 internal truncation.	FDS bridge	Internal preimages are lost, yet environmental side records remain fully accessible indefinitely with bounded cost and no accounting-boundary change.
Bath saturation forces collisions, compression, externalization, verification cost, or loss of recoverability.	Finite-record theorem	A finite accessible bath stores more distinguishable side records than its operational capacity without collision, compression, indexing, erasure, or hidden expansion.

Why finite baths matter

Ideal reservoirs are often modeled as infinite, memoryless sinks. Real side-record media are different. A finite bath can store correlations temporarily, hide them in inaccessible degrees of freedom, return them through recurrence or information backflow, dilute them by mixing, overwrite them through record collision, and make recovery too slow or costly for the task window. Thus,

$$\text{finite bath} \neq \text{perfect eraser} \quad (1)$$

and

$$\text{finite bath} \neq \text{infinite Markovian reservoir}. \quad (2)$$

P3 therefore treats the environment neither as a magical archive nor as an always-Markovian garbage dump. It treats environmental records as finite, partially accessible, and auditable.

P3 in the physical bridge ladder

P3 occupies the environmental side of the FDS physical bridge ladder. Table II summarizes the relation [3–5].

Main contributions

This paper contributes seven objects to the FDS physical spine:

1. an accessible/hidden decomposition of environmental side records;
2. an environmental side-record recovery criterion;
3. an operational Markovianization criterion based on history advantage and JS closure error;
4. a projection-operator interpretation of memory kernels as unresolved environmental memory;
5. a finite-bath recurrence caveat preventing naive monotone-forgetting claims;
6. a finite-bath saturation and record-collision theorem;
7. deterministic normal-form simulations and engineering protocols for testing side-record recoverability.

FDS BACKGROUND

Active finite distinction systems

An active finite distinction system is represented by

$$S = (X, E, B, M, Y, A, U, \pi, \ell, \Phi, \mathcal{P}, \tau), \quad (3)$$

where X is internal state, E environment, B boundary, M memory/model state, Y observation channel, A action space, U update rule, π finite projection, ℓ boundary-maintenance loss, Φ finite resource budget, $\mathcal{P}$ perturbation/pruning family, and τ update timescale [1]. P3 focuses on $E, Y, M, U, \pi, \ell, \Phi$, and τ .

TABLE II. P3 in the local FDS physical bridge ladder.

Paper	Core axis	P3 relation
P4	internal loss	what the effective record loses
P3	environmental forgetting	what side records fail to preserve
O3	record turnover	what environmental record reuse costs
P6	throughput	whether recovery happens in time
P7	protection	which records avoid ordinary forgetting

The active-boundary qualification matters. Environmental memory belongs in P3 only if access to environmental records can change prediction, recovery, action, or future boundary-maintenance loss. Passive correlations that cannot be read, used, or acted upon by the finite system are not operational side records.

Capacity deficit and externalization

For a task family Ψ , rate-distortion demand over update window τ is $R_{\min}^{(\tau)}(\epsilon; \Psi)$ and internal capacity is $C_S = \log_2 |M|$. The FDS capacity deficit is

$$\Delta_\epsilon(\tau) = R_{\min}^{(\tau)}(\epsilon; \Psi) - C_S. \quad (4)$$

If $\Delta_\epsilon(\tau) > 0$, the system must approximate, externalize, relax the task, improve compression, or fail [1]. P3 adds a condition to the externalization branch: an external record helps only while it remains recoverable through the system's accounting boundary.

ENVIRONMENTAL SIDE RECORDS

Definition 1 (Environmental side record). *Let X_0 denote a task-relevant past state or preimage whose identity may later matter. An environmental side record at time t is an environmental carrier*

$$L_t^E = r(E_t) \quad (5)$$

that contains recoverable information about X_0 for the system under its observation channel and accounting boundary.

Definition 2 (Accessible environmental readout). *The environment may contain a full physical state E_t , but the finite system does not generally access E_t directly. Define the accessible environmental readout*

$$R_t^E = g_t(E_t), \quad (6)$$

where g_t includes the observation channel, finite resolution, latency, cost, readout protocol, and accounting boundary. Equivalently, R_t^E may be read as the accessible readout σ -algebra induced by the system. The hidden bath E_t^{hid} denotes environmental variables or correlations not available through R_t^E within the task window and resource budget.

Definition 3 (Accessible inverse information). *The accessible inverse information about X_0 at time t is*

$$I_{\text{acc}}(t) = I(X_0; R_t^E \mid M_t, Y_{0:t}), \quad (7)$$

and the hidden inverse information is

$$I_{\text{hid}}(t) = I(X_0; E_t^{\text{hid}} \mid M_t, Y_{0:t}). \quad (8)$$

P3 is concerned with $I_{\text{acc}}(t)$, not with metaphysical total information. The notation R_t^E , if used informally, means the accessible readout structure $R_t^E = g_t(E_t)$ rather than the full physical bath state.

Definition 4 (Environmental forgetting). *Environmental forgetting of X_0 in the residual-entropy sense occurs when*

$$H(X_0 \mid M_t, R_t^E, Y_{0:t}) > h_{\text{tol}}, \quad (9)$$

where h_{tol} is a residual-entropy tolerance measured in bits. If the task tolerance is a distortion tolerance ϵ , the recovery condition should instead be expressed through a Bayes-risk, classification-error, or rate-distortion criterion.

Definition 5 (Bath record capacity). *For a finite accessible bath record space $\mathcal{E}_B$, define*

$$C_B = \log_2 |\mathcal{E}_B|. \quad (10)$$

For continuous baths, C_B is replaced by a finite-resolution, noisy-channel, rate-distortion, or operational readout capacity.

Definition 6 (Side-record recovery probability). *The Bayes-optimal environmental recovery probability is*

$$P_{\text{rec}}^E(t) = \mathbb{E}_{M_t, R_t^E, Y_{0:t}} \left[\max_x p(x \mid M_t, R_t^E, Y_{0:t}) \right]. \quad (11)$$

For a task alphabet $\mathcal{X}_0$, $P_{\text{rec}}^E(t) \geq 1 - \delta$ is a classification-style recovery criterion. A distortion-style criterion is

$$\inf_{\hat{X}_0 = f(M_t, R_t^E, Y_{0:t})} \mathbb{E}[d(X_0, \hat{X}_0)] \leq \epsilon. \quad (12)$$

These are environmental analogues of the P4 preimage recovery bound.

TABLE III. P3 environmental recovery decision table. Conditions are operational tests under a fixed accounting boundary, observation channel, and update window.

Condition	FDS interpretation
$H(X_0 M_t, R_t^E, Y_{0:t}) \leq h_{\text{tol}}$	Environment recovers the distinction in residual-entropy form.
$H(X_0 M_t, R_t^E, Y_{0:t}) > h_{\text{tol}}$ but $I(X_0; E_t^{\text{hid}} M_t, Y_{0:t}) > 0$	Hidden record exists, but is not operationally accessible.
$P_{\text{rec}}^E(t) \geq 1 - \delta$	Bayes recovery succeeds at error tolerance δ .
$\inf_{\hat{X}_0} \mathbb{E}[d(X_0, \hat{X}_0)] \leq \epsilon$	Distortion/Bayes-risk recovery succeeds.
$I(Z_{t+1}; H_t Z_t) \leq \eta$	Operational Markovianization by history-advantage test.
$\mathcal{K}_{\text{burden}}(T) > \eta$	Unresolved bath memory still matters beyond cutoff T .
$r_{\text{ext}} T > C_B$	Finite-bath saturation or record-collision pressure.
Invariant quotient exists	P7 protected bypass rather than ordinary side-record recovery.

MARKOVIANIZATION

Exact and operational Markovianity

Let $Z_t = \pi(X_t)$ be a projected system variable and let H_t denote accessible history, unresolved bath memory, or hidden variables relevant to prediction. Exact Markovianity requires

$$P(Z_{t+1} | Z_t, H_t) = P(Z_{t+1} | Z_t). \quad (13)$$

Operational Markovianization at tolerance η requires only that history adds negligible task-relevant information:

$$I(Z_{t+1}; H_t | Z_t) \leq \eta. \quad (14)$$

Markov closure error

A distributional closure error can be defined by

$$\mathcal{M}_{\text{env}} = \mathbb{E}_t D_{\text{JS}}(P(Z_{t+1} | Z_t, H_t) \| P(Z_{t+1} | Z_t)). \quad (15)$$

When $\mathcal{M}_{\text{env}} \rightarrow 0$, accessible history has become operationally irrelevant for that projected task. This is environmental forgetting in the Markovianization sense.

Remark 1 (Relation to lumpability). *For a Markov process on a microscopic state space, a projection remains Markovian exactly only under the usual lumpability conditions: projected transition probabilities must agree for all microstates in each coarse fiber [13, 14]. If lumpability fails, the projected process carries hidden-state memory unless environmental mixing or inaccessible bath dynamics makes the history operationally irrelevant. Recent work on memory loss in open quantum systems supports the view that Markovian behavior can arise as an operational loss of recoverable history, rather than as a primitive property of an ideal bath [26].*

Markovianization is not microscopic erasure

Markovianization is an effective boundary condition, not a claim that microscopic information has ceased to

exist. A process can be effectively Markovian for a finite agent because relevant bath correlations are inaccessible, diluted, below tolerance, outside the update window, or too costly to recover.

FINITE-BATH MEMORY KERNELS

Projection-operator approaches write the resolved dynamics schematically as

$$\frac{d}{dt} P X_t = P L P X_t + \int_0^t K(t-s) P X_s ds + F_t, \quad (16)$$

where P is a projection, L is a generator, K is a memory kernel, and F_t is an orthogonal noise term [16–18]. Open quantum versions are often expressed through Nakajima-Zwanzig memory kernels [15, 17, 19–21].

In FDS terms, K is unresolved environmental memory. It is the price of pretending that the bath has forgotten while unresolved correlations still influence future boundary statistics. Define the kernel burden

$$\mathcal{K}_{\text{burden}}(T) = \int_T^\infty \|K(s)\| ds. \quad (17)$$

The norm $\|K(s)\|$ is model-class dependent: it may be an operator norm, trace norm, total-variation influence norm, or task-loss-weighted kernel norm. If $\mathcal{K}_{\text{burden}}(T) \leq \eta$, then for update windows longer than T the Markov approximation is operationally valid at tolerance η in that model class.

Finite baths complicate the picture. Kernels may decay, plateau, oscillate, or revive. Therefore finite-bath forgetting is a windowed operational claim, not a universal monotone theorem. Recent work connects Nakajima-Zwanzig memory kernels and influence-functional descriptions in a unified framework for open quantum dynamics with memory [25]. Structured long-memory kernels can also be represented through fractional open-system dynamics [29].

MAIN CLAIMS

Theorem 1 (Environmental side-record recovery criteria). *Let X_0 be a task-relevant preimage absent from internal memory M_t , and let $R_t^E = g_t(E_t)$ be the accessible environmental readout. Environmental recovery at time t can be registered in any of the following task-dependent forms:*

$$H(X_0 \mid M_t, R_t^E, Y_{0:t}) \leq h_{\text{tol}}, \quad (18)$$

$$P_{\text{rec}}^E(t) \geq 1 - \delta, \quad (19)$$

$$\inf_{\hat{X}_0 = f(M_t, R_t^E, Y_{0:t})} \mathbb{E}[d(X_0, \hat{X}_0)] \leq \epsilon. \quad (20)$$

If the registered recovery criterion fails under the fixed accounting boundary, then the distinction is environmentally forgotten relative to that boundary, even if microscopic information remains in E_t^{hid} .

Proof. The system's available recovery variables are M_t , R_t^E , and the observation history $Y_{0:t}$. The residual-entropy condition states that the remaining inverse uncertainty is below the allowed bit tolerance h_{tol} . The Bayes condition states that the maximum-posterior decoder succeeds with probability at least $1 - \delta$. The distortion condition states that some admissible decoder achieves the task loss tolerance ϵ . If the pre-registered criterion fails for all decoders using only these variables, successful recovery requires additional variables, a larger accounting boundary, a different task tolerance, an invariant quotient, or hidden side records. $\square$

Theorem 2 (Operational Markovianization criterion). *Let (Z_t, H_t) be a resolved variable and accessible history/bath-memory pair. If*

$$I(Z_{t+1}; H_t \mid Z_t) \leq \eta \quad (21)$$

or, alternatively, if a registered distributional closure error such as $\mathcal{M}_{\text{env}} \leq \eta_{\text{DJS}}$ is below tolerance, then Z_t is operationally Markovian for the task at tolerance η or η_{DJS} respectively. The mutual-information criterion and the JS closure criterion are parallel operational tests, not strict mathematical equivalents.

Proof. The history variable H_t improves prediction of Z_{t+1} over Z_t by at most the registered information tolerance. A finite task whose loss is insensitive below that tolerance cannot operationally exploit the additional history. Thus the Markov closure is valid relative to the task, variables, and accounting boundary. $\square$

Proposition 1 (Finite-bath non-monotonicity). *For a finite closed or partially observed bath, accessible inverse information need not decrease monotonically. There can exist recurrence or backflow times t_r such that*

$$I_{\text{acc}}(t_r) > I_{\text{acc}}(t_r - \Delta t). \quad (22)$$

Therefore P3 cannot claim universal monotone environmental forgetting. It claims operational forgetting only over a specified window, observation channel, task tolerance, and accounting boundary. A recurrence that occurs after the task window or beyond the recovery budget does not operationally rescue the lost distinction.

Theorem 3 (Finite-bath saturation and record collision). *Suppose an accessible bath has operational record capacity C_B and receives side records at rate r_{ext} bits per unit time. If over a time interval T ,*

$$r_{\text{ext}}T > C_B, \quad (23)$$

then at least one exit channel is required: overwrite, record collision, compression, indexing overhead, externalization into a larger bath, verification cost, loss of recoverability, or task relaxation.

Proof. The bath cannot host more than C_B distinguishable accessible record bits under the fixed resolution and accounting boundary. If the exported record load exceeds C_B and no exit channel occurs, the bath would distinguish more record states than its operational capacity permits. Thus some collision, compression, overwrite, boundary expansion, additional indexing, or recovery loss is forced. $\square$

ENVIRONMENTAL FORGETTING LEDGER

Side records are not free merely because they are outside internal memory. P3 uses a resource-first ledger:

$$\begin{aligned} \dot{\mathcal{R}}_{\text{P3}} = & \dot{\mathcal{R}}_{\text{write}}^E + \dot{\mathcal{R}}_{\text{mix/isolate}}^E + \dot{\mathcal{R}}_{\text{monitor}}^E \\ & + \dot{\mathcal{R}}_{\text{retrieve}}^E + \dot{\mathcal{R}}_{\text{verify}}^E + \dot{\mathcal{R}}_{\text{refresh}}^E. \end{aligned} \quad (24)$$

When thermal reservoir, heat-channel, and calibration assumptions are available, this resource ledger may be connected to an entropy-production ledger. Without such calibration it remains a resource, latency, accessibility, or verification ledger [9–12]. Strong-coupling finite-bath models show that system and bath energetic quantities, entropy production, and work-like resources must be treated explicitly when the bath is finite and non-Markovian [27].

An environmental side record survives for the system only if it can be written, preserved, indexed, retrieved, verified, and used within the relevant update window and resource budget. Otherwise it may exist microscopically while failing operationally.

WORKED EXAMPLE: ROBOT EXTERNAL MEMORY IS NOT FREE

Consider a mobile robot that loses track of an object during occlusion. Let X_0 be the object identity and pose

TABLE IV. Core demonstration parameters. Full parameter values are exported to `data/simulation_parameters.json`.

Parameter	Value / role
Bath sizes	16, 64, 256 states (normalized)
Mixing rates	0.018, 0.035, 0.060
Task alphabet	$ \mathcal{X}_{\text{task}} = 8$
Markov threshold	$\eta = 0.05$ JS proxy
Bath capacity	$C_B = 260$ normalized record bits
Figures	deterministic normal-form demonstrations

before occlusion, M_t the robot’s internal memory at the current time, and R_t^E the accessible environmental readout available now: camera buffer, spatial map, external log, scene trace, environmental cue, or database response. P3 asks whether X_0 is recoverable from $(M_t, R_t^E, Y_{0:t})$ within the task window.

The robot may fail environmental recovery in several ways. The camera buffer may have been overwritten. The spatial map may be stale. The external log may be synchronized too slowly. The object or environment may have moved, destroying the cue. The database may contain the old record but be unreachable inside the control window. In all such cases, the information may exist somewhere in the world or in a larger data system, but it is not a usable side record for the embodied agent.

In P3 notation, successful recovery requires either

$$P_{\text{rec}}^E(t) \geq 1 - \delta \quad (25)$$

or a task-specific distortion condition such as

$$\mathbb{E}[d(X_0, \hat{X}_0)] \leq \epsilon. \quad (26)$$

If these fail while $I(X_0; E_t^{\text{hid}}) > 0$, the distinction has not been metaphysically destroyed; it is hidden outside the agent’s accessible recovery boundary. This is the Physical-AI form of environmental forgetting.

NUMERICAL DEMONSTRATIONS

The simulations are deterministic normal-form demonstrations generated by `code/generate_results.py`. They illustrate definitions and failure modes rather than fitting empirical data. No experimental, biological, proprietary, medical, human-subject, or device data are used.

EXPERIMENTAL AND ENGINEERING PROTOCOLS

Protocol 1 (Hidden-memory Markov test). *Compare a model using $P(Z_{t+1} | Z_t)$ with one using $P(Z_{t+1} | Z_t, H_t)$. If accessible history improves prediction or boundary maintenance beyond tolerance, the bath has not operationally forgotten.*

Protocol 2 (Side-record recovery assay). *After internal truncation, attempt recovery of X_0 using environmental records. Estimate $P_{\text{rec}}^E(t)$ and $H(X_0 | M_t, R_t^E, Y_{0:t})$. If recovery fails while hidden bath variables are excluded, P3 classifies the distinction as environmentally forgotten relative to the accounting boundary.*

Protocol 3 (Bath-size scaling). *Vary bath size, mixing rate, and readout accessibility. Changing bath size and mixing rate changes the Markovianization window: faster mixing often reduces accessible history, while finite small baths may show recurrence, backflow, or non-monotone recovery.*

Protocol 4 (Environmental externalization audit). *Write records into an external medium and vary refresh, verification, monitoring, and retrieval. Measure recovery probability, synchronization cost, readout latency, record collision, and ledger burden.*

Protocol 5 (Physical AI memory test). *Compare local agent memory, external logs, environment-embedded memory, and invariant/morphological memory. Report recovery rate, Markov closure error, environmental ledger, latency, and boundary-maintenance loss.*

FALSIFICATION AND DEMOTION CONDITIONS

P3 is weakened if a finite accessible environment preserves arbitrarily many side records indefinitely with no cost, no degradation, no collision, no latency, and no boundary expansion. A Markovianization claim is weakened if accessible history continues to improve task prediction under the same variables and tolerance. A finite-bath forgetting model is weakened if measured recurrence or backflow is present but the model assumes monotone decay. A specific memory-kernel or saturation model may fail without falsifying the FDS formal core; it should then be demoted to a narrower model class. If side records survive, P3 does not fail. It says the accounting boundary must include them. Non-Markovianity can also act as a thermodynamic resource rather than merely as a burden; P3 therefore treats memory as an auditable channel, not as intrinsically good or bad [28].

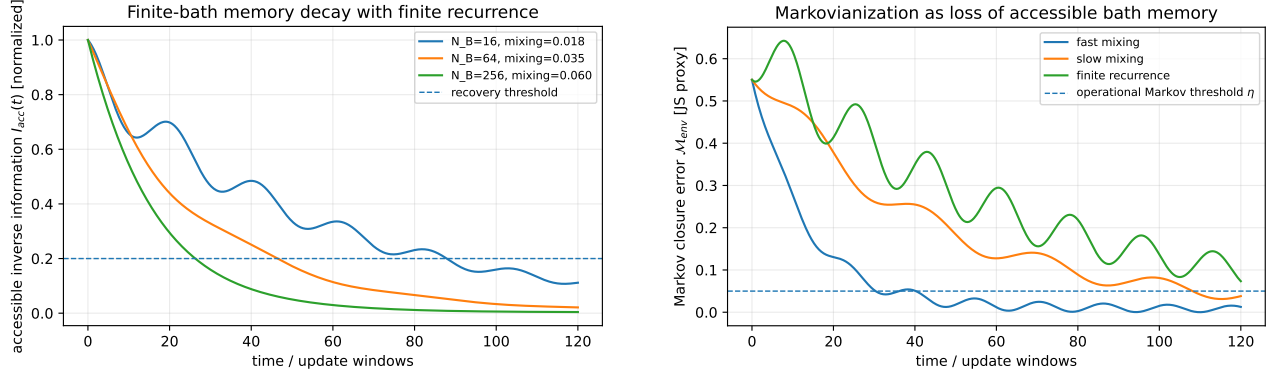


FIG. 1. Left: finite-bath memory decay. Accessible inverse information can decay while small baths exhibit recurrence-like revivals. Right: Markov closure error. Operational Markovianization occurs when accessible history no longer improves the projected transition beyond tolerance.

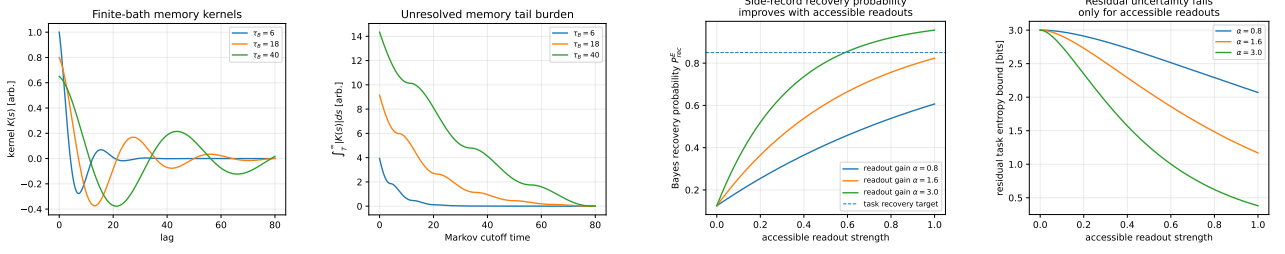


FIG. 2. Left: memory-kernel burden. Long kernel tails represent unresolved bath memory and increase the cost of Markov closure. Right: side-record recovery. Environmental records improve Bayes recovery only when they are accessible; hidden microscopic correlation does not by itself help the finite system.

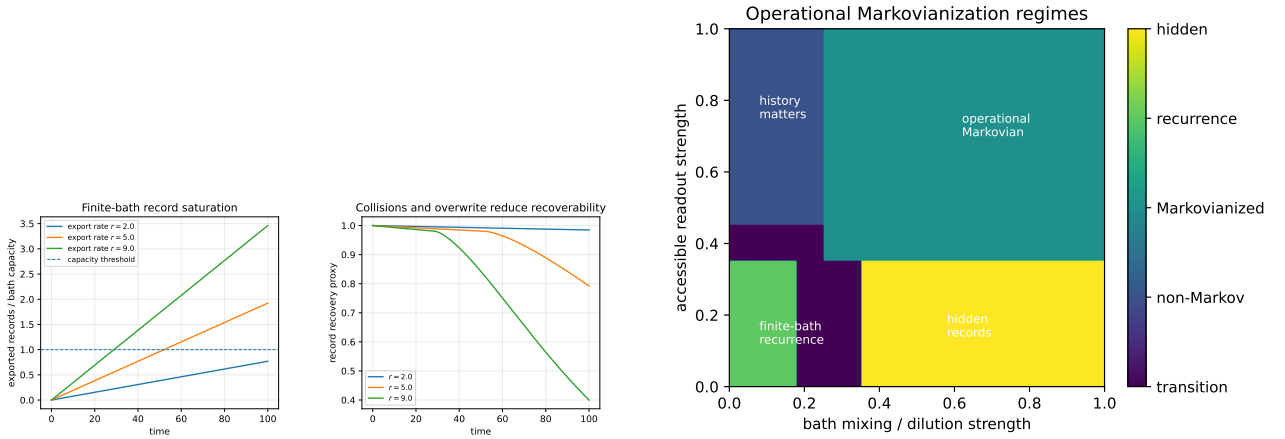


FIG. 3. Left: finite-bath saturation. When exported record load exceeds accessible bath capacity, collisions, overwrite, compression, or recovery loss begins. Right: operational Markovianization regimes. The map is a synthetic threshold diagram, not an empirical phase diagram.

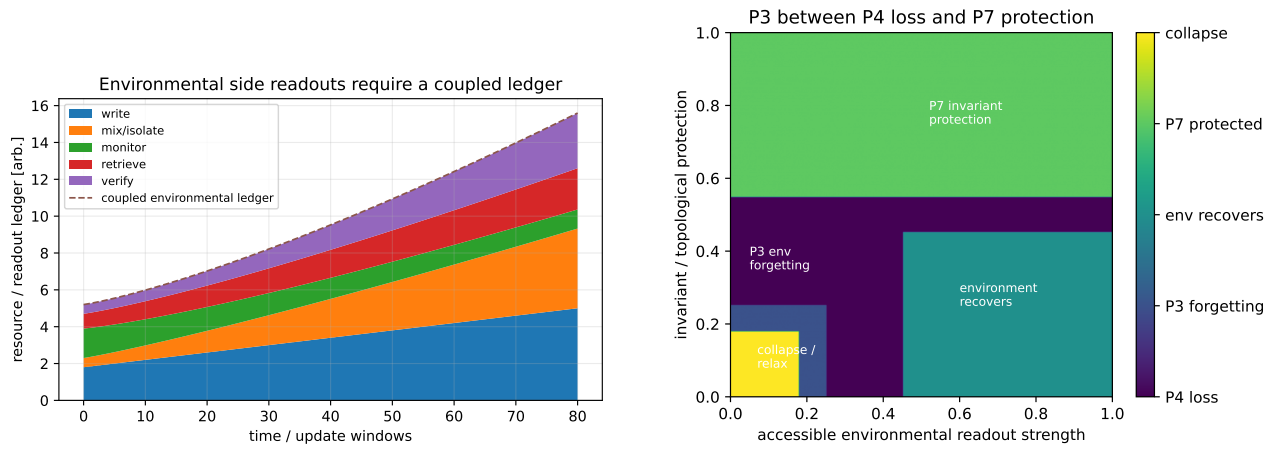


FIG. 4. Left: environmental side records require a coupled ledger for writing, isolation or mixing control, monitoring, retrieval, and verification. Right: P3 sits between P4 internal loss and P7 invariant protection: if local records lose preimages, the environment may recover them, forget them operationally, or be bypassed by invariant protection.

RELATION TO EXISTING THEORY

- **Open quantum systems** study system-bath coupling and non-Markovianity; P3 uses them as finite-bath memory and backflow model classes [19–21].
- **Projection-operator methods** study projected dynamics with memory kernels; P3 reads these kernels as unresolved environmental memory [15–18].
- **Markov-chain lumpability** gives exact closure conditions for projected dynamics [13, 14, 22, 23].
- **Information theory** supplies side-information and recovery bounds for accessible inverse information [6–8].
- **Stochastic thermodynamics** supplies resource and entropy-ledger bridges for environmental record maintenance [11, 12, 27].

P3 does not replace these theories. It packages their common operational implication inside FDS: a finite system can use the environment as memory only when the relevant environmental records remain accessible, recoverable, and affordable within the task window.

GLOSSARY

DESIGN IMPLICATION: EXTERNAL MEMORY IS NOT FREE MEMORY

This section is not part of the formal proof. It extracts the FDS design rule implied by P3. A physical AI agent cannot treat the environment, a database, a retrieval system, or a world state as an infinite perfectly indexed memory. External memory is useful only while it remains recoverable within the task window.

For embodied agents, P3 suggests four rules. First, external logs require refresh, synchronization, and verification. Second, environment-embedded memory can drift, mix, or become inaccessible. Third, high-speed physical tasks require local or invariant carriers rather than slow bath recovery. Fourth, robust agents need tests for when the environment has forgotten: if accessible history no longer improves prediction or recovery, the memory should be treated as Markovianized or lost for that task.

CONCLUSION

P4 showed that lost preimages do not return merely because internal capacity returns. P3 shows that exporting inverse information to the environment is not a free solution. A finite bath may temporarily preserve correlations, but those correlations can mix, saturate, recur,

become inaccessible, or demand verification and retrieval costs beyond the system’s budget. Markovianization is therefore not a metaphysical erasure of the past. It is an operational condition under which accessible history no longer improves boundary maintenance at the registered tolerance. A side record is memory only while it remains accessible, recoverable, affordable, and timely.

Together, P3 and P4 give the two halves of effective forgetting. The system may lose a distinction internally, and the environment may fail to return it externally. P7 identifies a protected exception through invariant quotients; P6 adds whether any recovery can occur within the required update window; O3 accounts for record reuse and environmental ledgers. P3 identifies the environmental cost of relying on side records.

TABLE V. Core P3 terms.

Term	P3 meaning
side record	environmental carrier of recoverable inverse information
accessible readout	$R_t^E = g_t(E_t)$ usable within boundary, cost, and latency
hidden bath	inaccessible, delayed, too costly, mixed, or unmeasured variables
environmental forgetting	loss of operational recovery, not microscopic erasure
Markovianization	accessible history no longer improves task prediction
memory kernel	unresolved environmental memory in projected dynamics
bath saturation	exported record load exceeds accessible bath capacity

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Boundary Maintenance and the Second Law under Finite Memory: Irreversible Record Reuse, Entropy Ledgers, and Operational Time Arrows

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FDS-O3 develops the third paper in the Operational Trident. O1 treated observation as finite stable record formation, O2 treated register time as causally ordered irreversible update, and O3 treats the Second Law as an operational channel instantiated by finite-memory boundary maintenance. The paper does not rederive the Second Law from distinction alone. It does not claim that mathematical coarse-graining is heat, that every update dissipates $k_B T \ln 2$, or that topological invariants violate thermodynamics. Its narrower claim is conditional: in a physically realized active finite distinction system, finite memory turns sustained boundary maintenance into record reuse; record reuse creates residual irreversibility unless inverse information remains available in side records; and physical overwrite, cleanup, refresh, repair, synchronization, or externalization must be accounted for in a coupled entropy or resource ledger. The central theorem states that sustained residual record turnover, fixed boundary-maintenance tolerance, and zero coupled-system entropy/resource cost cannot persist indefinitely under finite memory and physical bridge assumptions. Deterministic simulations illustrate memory-fill pressure, residual irreversibility, finite-record entropy ledgers, externalization audits, pruning return on investment, invariant compression, overload hysteresis, and finite-memory regime diagrams. O3 thereby supplies a finite-record operational channel connecting FDS-P1/P2/P5 to macroscopic irreversibility without claiming to replace statistical mechanics or stochastic thermodynamics.

Scope and Boundary of the Theory. This paper is an operational physical-bridge paper. It does not derive the Second Law from the FDS formal core alone. It does not claim that every formal distinction, observation, or coarse-graining dissipates heat. It does not claim that all thermodynamic irreversibility is Landauer erasure, nor that finite memory is the only foundation of thermodynamics. It claims only that physically realized active finite systems with bounded memory instantiate an operational Second-Law channel when they maintain a boundary through irreversible record reuse, refresh, repair, synchronization, externalization, or recovery under finite resource budgets. FDS-O3 does not identify thermodynamic entropy with memory use. It studies *record-turnover entropy*: the entropy/resource cost incurred when finite physical records are reused, erased, refreshed, repaired, synchronized, or externalized.

Claim-status summary

Table I summarizes the main claims and failure conditions.

Keywords: finite memory; Second Law; record reuse; residual irreversibility; boundary maintenance; active finite distinction systems; entropy ledger; Landauer principle; housekeeping dissipation; register time; pruning; externalization; invariant compression; stochastic thermodynamics.

INTRODUCTION

Why finite memory matters for the Second Law

The classical Second Law is usually stated at the level of heat, work, and entropy. FDS-O3 asks a narrower operational question: what does macroscopic irreversibility look like from the standpoint of a finite boundary-maintaining system whose usable past exists only through finite records? A finite record-bearing system can update only by writing, preserving, compressing, externalizing, reusing, or destroying record carriers. The directionality of such a system is therefore not merely a philosophical arrow of time. It is a ledger of finite record turnover.

The FDS formal core defines active finite distinction systems as finite-capacity systems that maintain a boundary through state-dependent updates under resource constraints [1]. Its core chain is

$$\begin{aligned} \text{finite capacity} &\rightarrow \text{capacity deficit} \rightarrow \text{approximation} \\ &\rightarrow \text{irreversible update} \rightarrow \text{dissipation} \\ &\rightarrow \text{prune/externalize/collapse.} \end{aligned} \quad (1)$$

O3 turns the final thermodynamic portion of this chain into a finite-memory operational statement.

From O1, O2, P1, P2, and P5 to O3

O1 treated observation as stable finite record formation [2]. O2 treated register time as causally ordered

TABLE I. Central FDS-O3 claims, epistemic status, and failure or demotion conditions.

Claim	Status	What would weaken or falsify it
Finite memory creates record-reuse pressure under sustained update unless history is externalized, compressed, uncomputed, or abandoned.	Operational theorem	A bounded-memory system maintains unbounded usable update history internally without record reuse, external memory, compression, task relaxation, or failure.
Non-injective record reuse creates residual irreversibility relative to an accounting boundary.	Formal information claim	A many-to-one update preserves full preimage information without side records or an enlarged boundary.
Physical irreversible record reuse enters an entropy/resource ledger under bridge assumptions.	Physical bridge	Reliable physical erasure or overwrite violates Landauer-style accounting under the stated assumptions.
Stable finite records require housekeeping beyond logical erasure.	Accounting claim	Refresh, retention, clocking, synchronization, carrier repair, and verification are cost-free in every controlled finite-record implementation.
Externalization shifts the operational Second-Law channel across accounting boundaries rather than eliminating it.	Accounting bridge	External records impose no write, verification, retrieval, latency, maintenance, or environmental cost.
Pruning and invariant compression can reduce future entropy pressure when task identity is preserved.	Conditional bridge	No task-preserving quotient, pruning, or compression ever reduces future record-maintenance cost.
Sustained residual record turnover, fixed boundary tolerance, and zero coupled entropy/resource cost cannot persist indefinitely.	O3 theorem	A finite active-boundary system maintains sustained residual record turnover at fixed tolerance with no ledger cost and no exit channel.
Topological or invariant persistence, if present, redirects entropy accounting rather than violating the Second Law.	Quarantined projection	A protected invariant supplies perpetual work or global entropy-law violation rather than bounded persistence or entropy relocation.

irreversible finite-record update [3]. P1 defined physical distinction carriers and accounting boundaries [5]. P2 showed why bounded-memory reversible computation accumulates garbage histories unless uncomputed, externalized, compressed, cleaned, or abandoned [6]. P5 gave a generalized entropy ledger for capacity deficit and boundary maintenance [8]. O3 combines these into a single statement:

$$\begin{aligned}
& \text{finite records} + \text{bounded memory} \\
& + \text{continued boundary maintenance} \\
& \implies \text{operational Second-Law channel.}
\end{aligned} \tag{2}$$

What O3 does not claim

O3 does not replace statistical mechanics, fluctuation theorems, or stochastic thermodynamics. It does not say that finite memory is the metaphysical source of thermodynamics. It does not identify formal coarse-graining with heat. It does not use topological protection as a counterexample to the Second Law. Its claim is conditional and operational: in physically realized finite-memory systems, sustained boundary maintenance re-

quires record turnover; if that turnover is irreversible within the accounting boundary, it appears in a coupled entropy/resource ledger or is replaced by an exit channel.

CORE FDS INGREDIENTS

Active finite distinction systems

The FDS core object is

$$S = (X, E, B, M, Y, A, U, \pi, \ell, \Phi, \mathcal{P}, \tau), \tag{3}$$

where X is internal state, E environment, B boundary, M memory or model, Y observation channel, A action space, U update rule, π finite projection, ℓ boundary-maintenance loss, Φ resource budget, $\mathcal{P}$ perturbation or pruning family, and τ the update timescale. The timescale matters because dimensionless information counts become rates only after division by τ .

An active boundary requires nontrivial update and relevance to future boundary loss:

$$\mathbb{P}\{U(M_t, Y_t) \neq M_t\} > 0, \quad I(M_{t+1}; \ell_{t+k}) > 0. \tag{4}$$

TABLE II. Relation between O3 and nearby FDS bridge papers. O3 is not a repeat of P5; it shifts the object from capacity deficit to finite-memory record turnover.

Paper	Core object	Main bridge
P1	physical carriers	distinction $\rightarrow$ carrier / erasure map
P2	reversible history	side records $\rightarrow$ finite reuse pressure
P5	capacity deficit	deficit $\rightarrow$ entropy-production ledger
O3	finite memory	record turnover $\rightarrow$ operational time arrow / operational Second-Law channel

Empirical use requires an intervention or ablation comparison between the update rule and a frozen, randomized, or identity-update null channel.

Capacity deficit and update windows

Let $Z_t = \psi(E_t, B_t)$ be a pre-registered task-relevant statistic. Let $R_{\min}^{(\tau)}(\epsilon; \Psi_t)$ be the minimum number of bits per update window τ needed to encode an admissible statistic to distortion tolerance ϵ . For effective capacity $C_{\text{eff}}(t)$,

$$\Delta_\epsilon(t) = R_{\min}^{(\tau)}(\epsilon; \Psi_t) - C_{\text{eff}}(t). \quad (5)$$

When $\Delta_\epsilon > 0$, the system cannot internally maintain all task-relevant distinctions at the required fidelity without approximation, externalization, task relaxation, or failure.

Logical erasure and physical bridge

For an update $M_{t+1} = U(M_t, Y_t)$, logical erasure is

$$b_t = H(M_t | M_{t+1}, Y_t). \quad (6)$$

Under Landauer-style physical assumptions, the informational entropy-production floor is

$$\dot{\Sigma}_{\text{erase}} \geq \frac{k_B \ln 2}{\tau} H(M_t | M_{t+1}, Y_t). \quad (7)$$

If the update is logically reversible relative to the available side information, this term can vanish. It does not follow that the system is thermodynamically cost-free, since refresh, clocking, carrier stability, synchronization, and external verification may still be required.

Free-energy budget and maintenance failure

A physically realized active finite system obeys a resource balance of the form

$$\begin{aligned} \Phi_{t+1} \leq \Phi_t + \int_t^{t+\tau} \dot{F}_{\text{in}}(s) ds \\ - \int_t^{t+\tau} \dot{Q}_{\text{ledger}}(s) ds - D_\ell(\ell_t). \end{aligned} \quad (8)$$

Here $\dot{Q}_{\text{ledger}}$ is the audited resource-cost rate associated with record reuse, refresh, repair, synchronization, externalization, and recovery. It may be modeled as

$$\dot{Q}_{\text{ledger}} = \sum_{i,r} \dot{Q}_{i,r} + \dot{W}_{\text{maint}} + \dot{C}_{\text{resource}}, \quad (9)$$

where $\dot{W}_{\text{maint}}$ denotes work used for maintenance and $\dot{C}_{\text{resource}}$ denotes resource-equivalent costs not naturally written as heat flows. When a single effective temperature is meaningful, one may use the shorthand $\dot{Q}_{\text{ledger}} \approx T_{\text{eff}} \dot{\Sigma}_{\text{O3}}^A$. The main O3 claim does not require a single-temperature reservoir. The term D_ℓ tracks functional damage or resource loss due to boundary failure. If the ledger cost exceeds input without reserves, external subsidy, task relaxation, pruning, or externalization, the same boundary-maintenance regime cannot persist indefinitely.

FINITE MEMORY AND RECORD REUSE

Definition 1 (Finite record memory). *Let M be the system's internal record state space. The finite memory capacity is*

$$C_{\text{mem}} = \log_2 |M|, \quad (10)$$

or the corresponding operational capacity under finite resolution, noise, and coding constraints.

Definition 2 (Task-relevant record load). *Let $R_{1:t}^{\text{task}}$ denote the minimal task-relevant record history required to maintain the specified boundary tolerance (ϵ, Ψ) up to time t , and let G_t be accessible side information or garbage records that remain inside the accounting boundary. Define*

$$B_R(t; \epsilon, \Psi) = H(R_{1:t}^{\text{task}} | G_t). \quad (11)$$

This is not the entire microscopic history of the universe or device. It is the usable finite-record history needed for the boundary-maintenance task. The memory-fill ratio is

$$\rho_M(t) = \frac{B_R(t; \epsilon, \Psi)}{C_{\text{mem}}}. \quad (12)$$

Definition 3 (Record reuse pressure). *A finite-memory system has record reuse pressure*

$$\Pi_{\text{reuse}}(t) = [B_R(t; \epsilon, \Psi) - C_{\text{mem}}]_+. \quad (13)$$

Positive reuse pressure does not by itself imply heat. It means that continued operation requires compression, externalization, uncomputation, overwrite, cleanup, task relaxation, memory expansion, or failure.

Proposition 1 (Finite-memory reuse necessity). *If $B_R(t; \epsilon, \Psi)$ grows without bound while C_{mem} is fixed, then there exists a finite time after which the system cannot keep the full record history internally without reuse, externalization, compression, task relaxation, or failure.*

Proof. By assumption $B_R(t; \epsilon, \Psi) \rightarrow \infty$ and $C_{\text{mem}} < \infty$. Therefore for some t^* , $B_R(t; \epsilon, \Psi) > C_{\text{mem}}$ for all $t > t^*$. Full internal retention would require more task-available record capacity than the memory space carries. The system must reduce demand, enlarge the boundary, move records outside the internal memory, compress them, overwrite them, or fail to preserve the full history. $\square$

RESIDUAL IRREVERSIBILITY

Definition 4 (Boundary-relative residual irreversibility). *Let $G_{\mathcal{A},t}$ be accessible side records inside accounting boundary $\mathcal{A}$. The one-step residual irreversibility of an update is*

$$L_{\mathcal{A}}(t) = H(M_t \mid M_{t+1}, Y_t, G_{\mathcal{A},t}). \quad (14)$$

For a history,

$$\mathcal{I}_{\mathcal{A}}^{\text{res}}(T) = H(M_{0:T} \mid M_T, Y_{0:T}, G_{\mathcal{A},T}). \quad (15)$$

Residual irreversibility is not a metaphysical loss of the past. It is an accounting statement: within the chosen physical boundary, there is not enough accessible side information to reconstruct the preimage history.

Definition 5 (Finite-window residual record-turnover rate). *For an update window $[t, t + \tau]$, define*

$$r_{\text{res}}^{\mathcal{A}}(t; \tau) = \frac{1}{\tau} H(M_t \mid M_{t+\tau}, Y_{t:t+\tau}, G_{\mathcal{A},t+\tau}). \quad (16)$$

Equivalently, for a backward history window one may use

$$r_{\text{res}}^{\mathcal{A}}(t; \tau) = \frac{1}{\tau} H(M_{t-\tau:t} \mid M_t, Y_{t-\tau:t}, G_{\mathcal{A},t}), \quad (17)$$

when that convention better matches the instrument. For path-dependent tasks where knowledge of a single preimage state is insufficient, M_t in Eq. (16) may be replaced by the relevant pre-window path segment $M_{t-\tau:t}$ or by the minimal task-relevant record path required by the boundary task. Both forms are finite-window residual turnover rates, not global thermodynamic entropy rates.

Definition 6 (Long-run residual turnover rate). *For discrete update windows of width τ , define*

$$\bar{r}_{\text{res}}^{\mathcal{A}} = \liminf_{n \rightarrow \infty} \frac{1}{n\tau} H(M_{0:n} \mid M_n, Y_{0:n}, G_{\mathcal{A},n}). \quad (18)$$

The theorem below uses the finite-window rate on a positive-density set of update windows; the long-run rate is a compact summary when the limit is meaningful.

Remark 1. A reversible embedding can make $L_{\mathcal{A}}(t) = 0$ by preserving inverse information in side records. Under bounded memory, those side records themselves become physical carriers requiring refresh, synchronization, externalization, uncomputation, cleanup, or loss.

THE OPERATIONAL SECOND-LAW CHANNEL

Entropy accounting boundary

Definition 7 (O3 entropy ledger). *Let $\mathcal{A}$ be an accounting boundary and let $\mathcal{P}_{\mathcal{A}}$ be a pre-registered partition of physical processes inside or coupled to that boundary. Define*

$$\dot{\Sigma}_{\text{O3}}^{\mathcal{A}} = \sum_{p \in \mathcal{P}_{\mathcal{A}}} \dot{\Sigma}(p), \quad (19)$$

where each physical process is counted once. Named channels are audit classes, not automatically independent mechanisms.

A class-level representation is

$$\dot{\Sigma}_{\text{O3}}^{\mathcal{A}} = \dot{\Sigma}_{\text{erase}} + \dot{\Sigma}_{\text{refresh}} + \dot{\Sigma}_{\text{repair}} + \dot{\Sigma}_{\text{sync}} + \dot{\Sigma}_{\text{ext}} + \dot{\Sigma}_{\text{rec}}. \quad (20)$$

If a process belongs to more than one class in an implementation, the ledger must assign it once or add overlap corrections.

For a channel i coupled to reservoirs $r \in \mathcal{R}_i$, the expected or coarse-grained stochastic-thermodynamic form is

$$\langle \dot{\Sigma}_i \rangle = \langle \dot{\Sigma}_i^{\text{sys}} \rangle + \sum_{r \in \mathcal{R}_i} \left\langle \frac{\dot{Q}_{i,r}}{T_r} \right\rangle \geq 0. \quad (21)$$

Instantaneous trajectory-level entropy production can fluctuate. O3 concerns the audited expected or coarse-grained coupled ledger over the specified accounting boundary. The single-temperature form $T^{-1}\dot{Q}_i$ is only a special case.

Erasure, refresh, repair, synchronization, and externalization

The erasure contribution is bounded by

$$\dot{\Sigma}_{\text{erase}} \geq \frac{k_B \ln 2}{\tau} L_{\mathcal{A}}(t). \quad (22)$$

The remaining channels cover physical maintenance: refreshing unstable carriers, repairing boundary damage, synchronizing records across subsystems, verifying external records, and recovering from loss. These are not all Landauer erasure. They are physical processes that keep finite records task-available.

Functional loss versus entropy production

A failure of the task is not automatically entropy production. It should be tracked separately by $D_\ell(\ell)$ unless recovery, repair, uncontrolled relaxation, reset, or reconstruction is physically implemented. O3 follows the P5 distinction between physical entropy production and functional boundary loss [8].

MAIN THEOREM

Theorem 1 (Finite-memory boundary-maintenance theorem). *Consider a physically realized active-boundary FDS with finite memory capacity, finite update window τ , finite resource input, and fixed boundary-maintenance tolerance $\ell(t) \leq \ell_c$. Suppose: (i) sustained operation generates positive finite-window residual turnover $r_{\text{res}}^A(t_j; \tau) > 0$ on a positive-density set of update windows; (ii) accessible side records, externalization, reversible uncomputation, task relaxation, invariant compression, and resource expansion are bounded; and (iii) physical cleanup, overwrite, refresh, repair, synchronization, or externalization channels have nonzero coupled-system entropy or resource cost. Then the system cannot indefinitely maintain the same boundary task at zero coupled-system entropy/resource cost. It must enter at least one channel: physical entropy production, exported entropy production, pruning, invariant compression, task relaxation, resource expansion, passive degradation, or collapse.*

Proof. Positive residual record turnover means that, relative to boundary $\mathcal{A}$, some preimage history needed for full reconstruction is not carried by the current state plus accessible side records. If the task tolerance is fixed and the system continues active maintenance, it must compensate by preserving additional side information, reversing the update path, externalizing records, compressing history, relaxing the task, repairing damage, or physically overwriting/cleaning records. By hypothesis the non-dissipative or relief channels are bounded, and physically realized cleanup, overwrite, refresh, repair, synchronization, or externalization has nonzero coupled ledger cost. If the cost remains within the finite resource envelope, the system persists by paying or exporting entropy. If it exceeds the envelope and no exit channel is available, Eq. (8) drives maintenance failure. Thus zero-cost indefinite maintenance under sustained record turnover is impossible under the stated assumptions. $\square$

In model classes where physical record-turnover entropy is lower-bounded by an implementation-dependent coefficient $\kappa > 0$, one may write

$$\int_0^T \langle \dot{\Sigma}_{\text{O3}}^A(t) \rangle dt \geq \kappa \int_0^T [r_{\text{res}}^A(t; \tau) - \mathcal{E}_{\text{exit}}(t)]_+ dt. \quad (23)$$

Here κ has units of entropy per record unit, and $\mathcal{E}_{\text{exit}}(t)$ is record-equivalent relief per update window from admissible exit channels. A minimal decomposition is

$$\mathcal{E}_{\text{exit}}(t) = \mathcal{E}_{\text{prune}}(t) + \mathcal{E}_{\text{ext}}(t) + \mathcal{E}_{\text{inv}}(t) + \mathcal{E}_{\text{relax}}(t) + \mathcal{E}_{\text{uncomp}}(t) + \mathcal{E}_{\text{res}}(t),$$

where $\mathcal{E}_{\text{prune}}$ is pruning or cleanup that reduces future record burden, $\mathcal{E}_{\text{ext}}$ is externalization that supplies effective record space, $\mathcal{E}_{\text{inv}}$ is invariant compression that reduces maintained history, $\mathcal{E}_{\text{relax}}$ is task relaxation that lowers record demand, $\mathcal{E}_{\text{uncomp}}$ is reversible uncomputation that recovers inverse paths, and $\mathcal{E}_{\text{res}}$ is resource, storage, or channel expansion. All terms are expressed in record-equivalent relief units over the update window. Equation (23) is a model-class bridge, not a universal identity.

Corollary 1 (Finite-memory operational Second-Law impossibility triangle). *Under finite-memory and physical bridge assumptions, the following cannot persist indefinitely without an exit channel:*

$$r_{\text{res}}^A(t; \tau) > 0, \quad \ell \leq \ell_c, \quad \langle \dot{\Sigma}_{\text{O3}}^A \rangle = 0. \quad (24)$$

At least one term must change: residual turnover is reduced, the boundary tolerance is relaxed, entropy is produced or exported, resources expand, passive degradation is accepted, or the boundary-maintenance regime fails.

Finite-memory operational Second-Law channel. Sustained residual record turnover, fixed boundary tolerance, and zero coupled entropy/resource cost are jointly impossible for physically realized finite-memory boundary-maintaining systems under the O3 bridge assumptions.

MACROSCOPIC ARROWS FROM FINITE RECORDS

Register-time arrow

O2 treated usable time for bounded observers as a causally ordered sequence of finite record updates [3]. In O3, the same ordered update becomes thermodynamic when physical record carriers are reused. A formal update sequence $M_t \rightarrow M_{t+1}$ is not heat by itself; but if a physical system must erase, reset, refresh, or repair records to continue the sequence, the update order becomes a physical ledger.

Memory-fill arrow

Without cleanup or compression, record load tends to accumulate:

$$\rho_M(t) = \frac{B_R(t; \epsilon, \Psi)}{C_{\text{mem}}} \uparrow. \quad (25)$$

A reversible system can delay erasure by preserving inverse information. In bounded memory, the delayed history becomes a memory-fill and refresh burden.

Entropy-ledger arrow

Under physical bridge assumptions,

$$\left\langle \dot{\Sigma}_{O3}^A \right\rangle \geq 0 \quad (26)$$

for the audited expected or coarse-grained coupled boundary. Local entropy can decrease, local memory pressure can fall, and local records can be exported, but the coupled ledger must include the processes that make those changes possible.

Finite-register operational equilibrium

O3 does not identify thermodynamic equilibrium with a full memory register. It proposes a narrower finite-register analogue: when usable free record capacity vanishes, and no pruning, externalization, compression, erasure, repair, or resource input is available, the register can no longer form new stable O1 records or support new O2 register-time updates [2, 3]. In this operational sense,

$$\begin{aligned} \rho_M(t) &\rightarrow 1, & \mathcal{E}_{\text{exit}}(t) &\rightarrow 0 \\ &\implies \text{no new task-available record capacity.} \end{aligned} \quad (27)$$

This is a conjectural analogy for finite registers, not a redefinition of thermodynamic equilibrium.

Record-turnover power floor

For an erasure-dominated implementation, if B_{turn} bits are irreversibly turned over per update window, the minimum erasure-limited power obeys

$$P_{\text{erase}}^{\min} \geq \eta_{\text{impl}} \frac{k_B T \ln 2}{\tau} B_{\text{turn}}, \quad (28)$$

where $\eta_{\text{impl}} \geq 1$ is a toy parametrization of finite-time, reliability, control, and architectural overhead. It is not a universal multiplicative law. If nearly the full memory is turned over each window, $B_{\text{turn}} \approx C_{\text{mem}}$. Conversely, given an available power budget P_{avail} and housekeeping load P_{hk} , the irreversible turnover rate is bounded by

$$\dot{B}_{\text{turn}}^{\max} \leq \frac{P_{\text{avail}} - P_{\text{hk}}}{\eta_{\text{impl}} k_B T \ln 2}. \quad (29)$$

These expressions are throughput bounds for erasure-limited channels, not universal power laws for all computation.

Hysteresis after overload

If overload forces erasure, damage, or invariant destruction, reducing demand later need not restore the original state. The hysteresis gap can be decomposed as

$$\mathcal{H}_{O3} = \mathcal{H}_{\text{record}} + \mathcal{H}_{\text{damage}} + \mathcal{H}_{\text{invariant}}, \quad (30)$$

where the three terms correspond to erased records, damaged carriers or boundaries, and lost low-maintenance invariant support. A measured path gap may be written schematically as

$$\mathcal{H}_{O3} = \dot{\Sigma}_{\text{down}}(r_{\text{res}}) - \dot{\Sigma}_{\text{up}}(r_{\text{res}}) > 0. \quad (31)$$

The gap is a reconstruction burden, not a universal hysteresis law. In T3 language, part of the burden can be the activation-like cost of rebuilding Phase-B invariants destroyed during overload [4].

PRUNING, EXTERNALIZATION, AND INVARIANT COMPRESSION

Pruning as entropy-relief policy

Pruning may incur a one-time cost Σ_{prune} while reducing future record-maintenance load. Over horizon H , define

$$\text{ROI}_{\Sigma}^{\text{prune}}(H) = \frac{1}{\Sigma_{\text{prune}} + \lambda D_{\text{task}}} \left[\int_t^{t+H} (\dot{\Sigma}_{O3}^{\text{no prune}} - \dot{\Sigma}_{O3}^{\text{prune}}) ds - \Sigma_{\text{prune}} \right]. \quad (32)$$

Pruning is favorable only when it preserves task function sufficiently that future ledger savings exceed pruning cost and task loss.

Externalization as boundary shift

Externalization changes the accounting boundary:

$$\mathcal{A}_{\text{local}} \rightarrow \mathcal{A}_{\text{coupled}} = \mathcal{A}_{\text{local}} \cup \mathcal{A}_{\text{ext}}. \quad (33)$$

Local ledger cost may fall, but the coupled ledger includes writing, verification, retrieval, latency, synchronization, carrier upkeep, and environmental noise. Externalization is therefore a shift of the operational Second-Law channel, not deletion of it.

Invariant-supported compression

Let $q : X \rightarrow Q$ be an invariant quotient under perturbation family $\mathcal{P}$, with $q \circ P_i = q$ for all $P_i \in \mathcal{P}$. If the

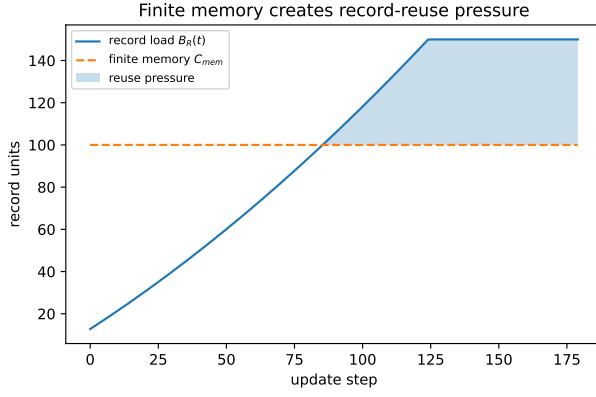


FIG. 1. Finite memory creates record-reuse pressure. Record load crosses finite memory capacity, activating reuse pressure rather than forcing a unique exit.

task identity predicate factors as $R_A = \bar{R}_A \circ q$, then the system can maintain task identity through q instead of raw microstate detail. Invariant compression reduces the finite-memory channel when

$$\dot{\Sigma}_q + \lambda D_{\text{task}}^q < \dot{\Sigma}_X + \lambda D_{\text{task}}^X. \quad (34)$$

Compression is not monotonically beneficial. Let q index compression strength and define

$$J(q) = \dot{\Sigma}_{\text{maint}}(q) + \lambda D_{\text{task}}(q). \quad (35)$$

The stable Phase-B quotient is not the maximum compression but the minimizer $q^* = \arg \min_q J(q)$ under the task-tolerance constraint. Under-compression carries unnecessary record burden; over-compression destroys task-relevant distinctions and raises D_{task} . Topological protection, error-correcting code spaces, and robust organizational routines are domain projections of this form only when an explicit bridge maps the invariant quotient to physical record maintenance.

NUMERICAL MODELS AND SIMULATIONS

The simulations are deterministic synthetic demonstrations. They are not fits to physical devices, biological organisms, quantum experiments, or human-subject data. They illustrate the O3 normal form and are generated by `code/generate_results.py`.

EXPERIMENTAL AND ENGINEERING PROTOCOLS

Protocol 1 (Finite-memory reuse protocol). *Hold memory capacity fixed while increasing update rate, task demand, or retention horizon. Independently estimate the*

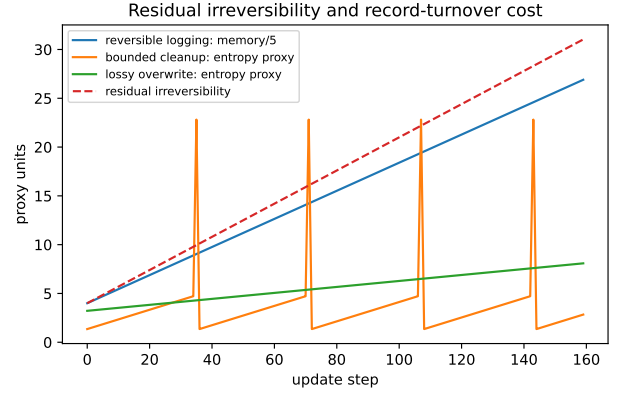


FIG. 2. Residual irreversibility and record-turnover cost. Reversible logging delays erasure by growing side records; bounded cleanup creates entropy bursts; lossy overwrite creates residual irreversibility.

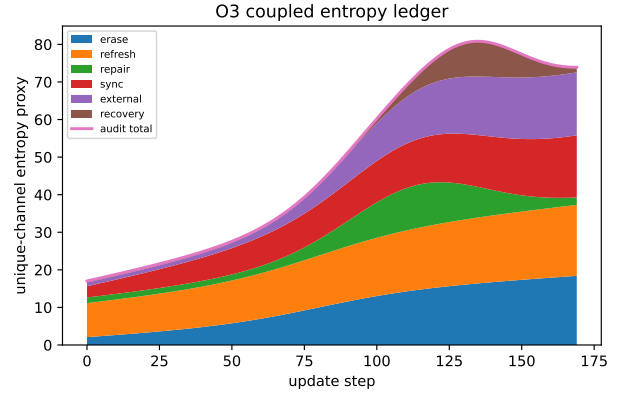


FIG. 3. O3 coupled entropy ledger. Stacked areas are illustrative unique-channel assignments under an audit convention. The audit total is not an additional channel.

task-relevant record demand and memory capacity where possible; the crossing point should be estimated before observing heat, reset, latency, or failure signatures whenever possible. Measure reset rate, energy or heat, latency, error floor, buffer overflow, pruning, and externalization. O3 predicts a co-occurrence of exits when record demand crosses finite memory capacity.

Protocol 2 (Reversible logging versus cleanup). *Compare three implementations of the same task: reversible logging with expanding memory, bounded reversible logging with cleanup, and irreversible overwrite. Audit the coupled ledger including side-record refresh, synchronization, cleanup heat, and task loss. The claim is not that reversible computation fails; it is that bounded reversible history requires physical management.*

Protocol 3 (Externalization audit). *Move records outside the local boundary and measure local savings together*

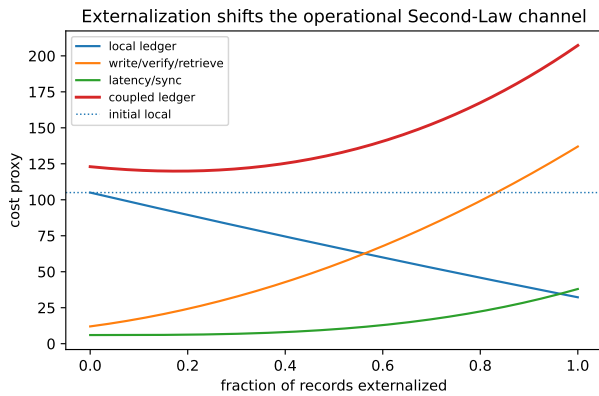


FIG. 4. Externalization shifts the operational Second-Law channel. Local cost can fall while write, verification, retrieval, latency, and synchronization make the coupled ledger rise.

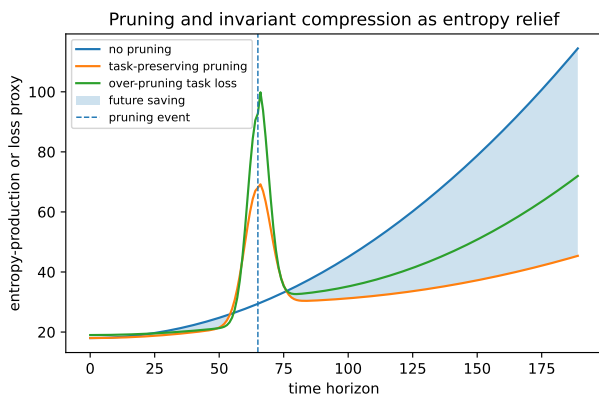


FIG. 5. Pruning and invariant compression as entropy relief. Task-preserving pruning pays a one-time cost to reduce future ledger burden; over-pruning incurs task loss.

with write, verification, retrieval, latency, synchronization, maintenance, and environmental noise. Local improvement without coupled-system improvement counts as relocation, not elimination.

Protocol 4 (Invariant compression test). *Replace detailed records with a quotient or invariant representation. Measure whether task loss remains below tolerance while the erase/refresh/verification ledger decreases. A successful quotient is an entropy-relief channel; an unsuccessful quotient is over-compression.*

Protocol 5 (Finite-memory hysteresis). *Raise task demand until memory reuse, pruning, erasure, or damage occurs, then reduce demand. Test whether energy, latency, and error curves return to their original path or retain a gap from record loss, damage, or lost invariant support.*

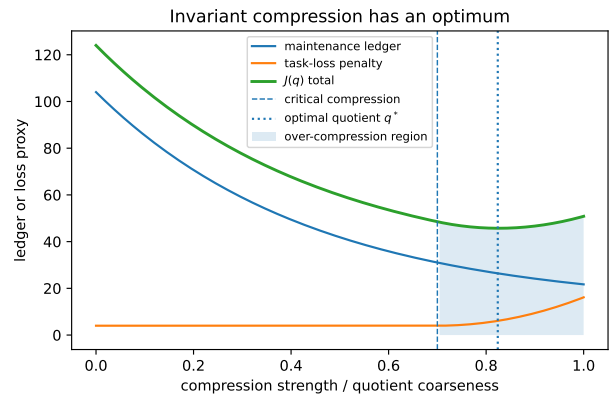


FIG. 6. Invariant compression has an optimum. Under-compression carries avoidable record burden; moderate invariant compression lowers maintenance cost; over-compression destroys task-relevant distinctions and raises task loss.

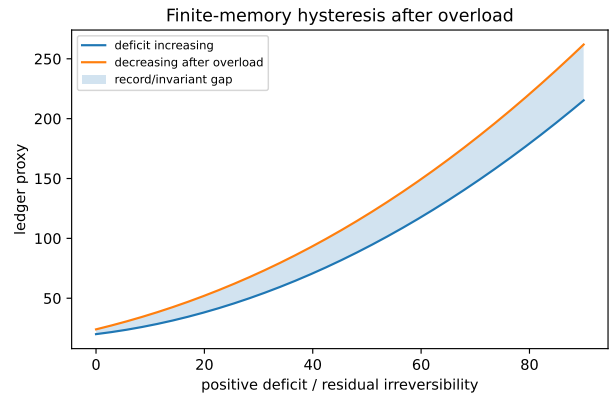


FIG. 7. Finite-memory hysteresis after overload. The gap represents record loss, accumulated damage, or destroyed invariant support, not a universal hysteresis law.

RELATION TO EXISTING WORK

Landauer's principle and Bennett's reversible computation establish the distinction between logical irreversibility, reversible embeddings, and physical implementation costs [11–13]. Modern information thermodynamics extends this framework to measurement, feedback, correlations, and stochastic trajectories [14–16]. Stochastic thermodynamics of computation emphasizes finite-time and architectural constraints [18, 19]. Recent work on realistic electronic memory erasure and quantum many-body Landauer accounting further emphasizes that record-turnover cost depends on concrete physical architectures and system-environment partitions [30, 31]. Thermodynamic uncertainty relations link precision of nonequilibrium currents to dissipation [20, 21]. Compared with the free-energy principle and active in-

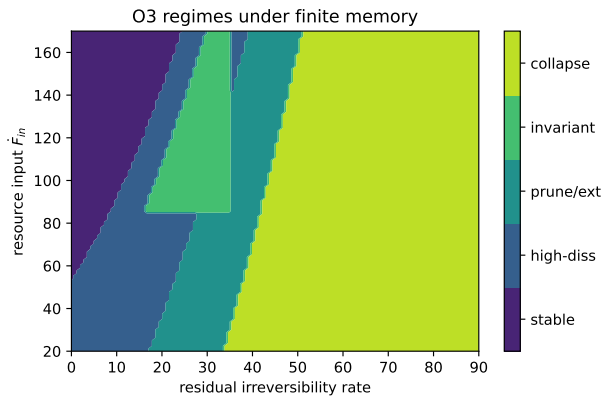


FIG. 8. O3 regimes under finite memory. Region boundaries are synthetic normal-form thresholds, not fitted phase boundaries. Stable, high-dissipation, pruning/externalization, invariant-compression, and collapse regimes arise from the relation between residual irreversibility and resource input. These regions correspond to the exit channels in Theorem 1 and Corollary 1.

ference [27, 32], O3 focuses on physical record turnover and coupled entropy/resource ledgers rather than variational free-energy objectives. O3 does not replace these theories. It packages the finite-record channel by which a boundary-maintaining system instantiates them.

Compared with P5, O3 emphasizes memory reuse and the macroscopic arrow rather than capacity-deficit correction pressure [8]. Compared with O2, O3 adds the entropy ledger to register time [3]. Compared with P2, O3 turns bounded reversible history into an operational Second-Law channel [6].

Topological persistence as quarantined projection

A separate high-risk bridge direction may treat non-Hermitian skin effects and point-gap topology as possible global-invariant constraints on coarse-graining. O3 does not use such topological claims in its core theorem. If a domain-specific topological invariant exists, O3 treats it as invariant-supported persistence or entropy relocation: it may protect an identity predicate against local erasure, but it does not license perpetual motion or global entropy-law violation. The conservative statement is that topology, when physically realized, changes which structures are maintainable and where entropy is produced; it does not remove the coupled ledger.

LIMITATIONS AND FALSIFICATION

O3 does not claim that all irreversibility is finite-memory erasure, that all coarse-graining is heat, that

externalization always helps, that pruning always helps, or that invariant compression is always available. It also does not identify thermodynamic entropy with memory use. Its object is record-turnover entropy: the entropy/resource cost incurred when finite physical registers maintain a boundary by reusing, erasing, refreshing, repairing, synchronizing, or externalizing records. It is a finite-record bridge under physical implementation assumptions.

The strong version of O3 would be weakened or rejected by any of the following:

1. bounded-memory systems sustain unbounded usable record history with no reuse, compression, externalization, task relaxation, or failure;
2. physically irreversible record reuse violates the stated Landauer-style bridge assumptions;
3. stable finite records require no refresh, repair, verification, synchronization, carrier stability, or resource cost in any implementation;
4. externalization reduces local cost while imposing no coupled-system write, verification, retrieval, maintenance, latency, or environmental cost;
5. finite-memory overload produces no measurable reset, heat, latency, error floor, pruning, externalization, recovery, or collapse signature;
6. invariant compression never reduces maintenance burden in any controlled system where task identity is preserved.

CONCLUSION

FDS-O3 gives a finite-memory operational Second-Law channel. It does not claim that the Second Law is derived from distinction alone. It claims that when sustained task-relevant record demand exceeds available finite memory, continued updating requires record reuse, side records, externalization, compression, pruning, or failure. If record reuse is irreversible within an accounting boundary, the physical implementation appears in an entropy or resource ledger.

The compact statement is:

Finite memory turns boundary maintenance into record turnover. Physical record turnover, when irreversible or actively maintained, instantiates an operational Second-Law channel.

This places O3 after O1 and O2 in the Operational Trident [2, 3] and after P1, P2, N1, and P5 in the physical accounting spine [5–8]. It supplies a bridge from finite records to macroscopic irreversibility while leaving the

classical Second Law, stochastic thermodynamics, and domain-specific topological questions intact.

Notation summary

Symbol	Meaning
B	maintained boundary or interface
M	memory, model, or record state
Y	observation channel
U	update rule
ℓ	boundary-maintenance loss
Φ	finite resource budget
τ	update window
C_{mem}	finite memory capacity
$B_R(t; \epsilon, \Psi)$	task-relevant record load
$\rho_M(t)$	memory-fill ratio
$G_{A,t}$	accessible side records in boundary $\mathcal{A}$
$L_{\mathcal{A}}(t)$	one-step residual irreversibility
$r_{\text{res}}^{\mathcal{A}}(t; \tau)$	finite-window residual turnover rate
$\Sigma_{\text{O3}}^{\mathcal{A}}$	audited O3 entropy ledger
$\mathcal{E}_{\text{exit}}$	record-equivalent exit relief
$D_{\ell}(\ell)$	functional loss or damage
q	invariant quotient map

Simulation parameters

The simulations use deterministic arrays with fixed synthetic parameters in `code/generate_results.py`. Figure 1 varies record load and finite memory capacity. Figure 2 compares reversible logging, bounded cleanup, and lossy overwrite. Figure 3 decomposes the finite-record entropy ledger. Figure 4 audits externalization. Figure 5 compares pruning policies. Figure 6 compares representation strategies. Figure 7 illustrates overload hysteresis. Figure 8 classifies regimes under residual irreversibility and resource input. No proprietary, biological, organizational, quantum-experimental, human-subject, or device data are used.

Reproducibility checklist

1. Code availability: all simulation code is included in the replication package.
2. Deterministic execution: the code uses fixed synthetic parameters and deterministic arrays.
3. Figure reproduction: run `python code/generate_results.py`; the script regenerates all figures and CSV outputs.
4. Data status: all numerical outputs are synthetic demonstrations generated from the stated model.
5. Platform independence: the code uses standard Python scientific libraries.

Code availability

The simulation code used to generate Figs. 1–8 is included in the accompanying replication package under `code/generate_results.py`. Running the script regenerates all figures (PDF and PNG) and CSV tables in a single pass.

AI assistance disclosure

AI-assisted tools were used for language polishing, structural feedback, LaTeX drafting support, and code-debugging assistance. The author reviewed and edited all content and remains responsible for all claims, references, simulations, and conclusions. No AI system is listed as an author.

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The Agency-Semantics Spine of Distinction Theory

Attention, Value, Goal, Meaning, and Action under Finite Capacity

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Abstract

FDS-M0 develops the agency-semantics spine of Distinction Theory. Building on the formal core of Active Finite Distinction Systems, it treats attention, value, goal, meaning, and agency as finite-system roles in boundary maintenance under limited capacity. A distinction becomes attended when admitted into a finite update channel; valuable when it has causal boundary-gradient relevance; goal-relevant when the value ranking is stabilized into a policy orientation; meaningful when compressed into a task-sufficient actionable quotient; and agentic when updates or actions causally affect future boundary-maintenance loss. The paper does not claim to solve consciousness, reduce semantics to physics, or assign strong agency to any system merely because it processes symbols. It provides a conservative bridge for analyzing biological, cognitive, artificial, and social systems as active finite systems that admit, rank, compress, verify, externalize, and act on distinctions under capacity and resource constraints. A deterministic synthetic normal-form model illustrates nonlinear attention gating, semantic capacity deficit, policy-preserving quotients, goal stability, boundary-sensitive agency classification, misalignment audits, and epistemic pollution as verification-bandwidth saturation.

Keywords: agency; semantics; attention; value; goal; meaning; finite capacity; boundary maintenance; active finite distinction systems; artificial agency; misalignment; verification load; epistemic pollution.

Epistemic Notice and Scope

This manuscript is a spine paper, not a completed theory of mind or a blueprint for artificial agents. It does not claim that subjective experience reduces to information capacity, that all semantics is purely functional, that moral value reduces to survival, that current language models possess strong FDS agency, or that linguistic and social meaning can be replaced by a single information-theoretic quantity. Its claim is narrower: once a system is modeled as an active finite distinction system with a boundary, memory, update rule, action space, finite capacity, and boundary-maintenance loss, several agency-semantics notions acquire operational roles.

The paper follows the layered discipline of the FDS formal core. The FDS Core separates formal definitions, physical bridge assumptions, normal-form dynamics, and quarantined applications; it also states that domain applications must specify their boundary, memory, update rule, task variable, capacity, resource budget, perturbation family, invariant quotient, and falsification condition [1]. M0 uses that discipline for agency and semantics. A failure of a cognitive, artificial, social, or linguistic mapping may demote the mapping without refuting the formal FDS core.

1 Introduction

A finite system does not encounter the world as an unlimited field of meaning. It encounters more possible distinctions than it can admit, maintain, rank, compress, verify, or act upon. A flash in peripheral vision, a new word in a conversation, a rising temperature in a cell, an alarm in a control room, a legal right in a society, and a prompt in an artificial system are not semantically equivalent merely because they are all signals. They become operationally significant only if some system can admit them into an update channel, maintain them as records, rank their boundary relevance, connect them to possible actions, and preserve enough structure for future use.

The FDS core defines an active finite distinction system as a tuple

$$S = (X, E, B, M, Y, A, U, \pi, \ell, \Phi, \mathcal{P}, \tau), \quad (1)$$

where X is internal state, E environment, B boundary, M memory/model space, Y observation channel, A action space, U update map, π finite projection, ℓ boundary-maintenance loss, Φ resource budget, $\mathcal{P}$ perturbation/pruning family, and τ update timescale [1]. M0 asks what follows for agency and semantics when distinctions must pass through this finite architecture.

The dependency skeleton is

$$\text{distinction} \rightarrow \text{record} \rightarrow \text{attention} \rightarrow \text{value} \rightarrow \text{goal} \rightarrow \text{meaning} \rightarrow \text{agency} \rightarrow \text{culture}. \quad (2)$$

This is not a temporal sequence and not an evolutionary law. It is an operational dependency map. A distinction separates alternatives. If maintained, it becomes a record. If admitted into a finite update channel, it becomes attended. If ranked by causal boundary relevance, it becomes valuable relative to a task. If stabilized over update windows, it becomes goal-relevant. If compressed while preserving action, prediction, verification, or coordination affordances, it becomes meaningful. If connected to updates that causally affect future boundary loss, it participates in agency. When externalized and shared, such distinctions become culture or institutional infrastructure.

1.1 Why a spine paper is needed

The preceding FDS sequence developed finite record formation, register time, capacity overflow, boundary-maintaining self-organization, and finite-memory boundary-maintenance channels [2–6]. The physical-accounting ladder is background to this paper rather than a direct premise. These papers describe what finite systems can represent, maintain, update, erase, externalize, and lose. M0 shifts the question: how do maintained distinctions become significant for action? Without such a spine, later papers on attention, value, meaning, AI alignment, institutions, science, law, education, and civilization memory risk becoming disconnected applications. With it, these domains can be treated as different instantiations of finite distinction admission, ranking, compression, verification, and boundary update.

1.2 What this paper does not claim

M0 does not solve consciousness. It does not decide the metaphysics of intentionality. It does not claim that value is identical to survival or that ethics can be derived from boundary maintenance. It does not claim that meaning is only private, nor that language and institutions are irrelevant. It does not assign strong FDS agency to a bare model, static database, or passive sensor. It does not give a training recipe, benchmark, routing algorithm, memory architecture, or foundation-agent design. It provides a public conceptual spine, not an engineering blueprint.

Table 1: Claim-status summary for FDS-M0. The table is an audit device: several entries are bridge definitions or downstream interfaces, not established empirical claims.

Claim ID	Tier	Claim	Failure or demotion condition
M0-001	Formal bridge	Attention is capacity-limited distinction admission into an update channel.	Attention-like selection occurs without any capacity-limited admission, update gating, or priority constraint under the specified mapping.
M0-002	Bridge definition	Value is causal boundary-gradient relevance under finite capacity.	Valuation systematically fails to correlate with future boundary loss, task success, or resource relevance under a valid mapping.
M0-003	Operational bridge	Goals are stabilized value rankings coupled to policies across update windows.	Goal-like behavior persists without memory, ranking, policy stabilization, or action direction over time.
M0-004	Bridge definition	Meaning is actionable compressed distinction preserved by a task-sufficient quotient.	Compressed representations guide no action, prediction, verification, coordination, or boundary maintenance.
M0-005	Definitional / empirical	Strong FDS agency requires updates or actions that causally affect future boundary loss.	A system with no causal update effect on future boundary loss qualifies as a strong agent under the same criteria.
M0-006	Operational bridge	Self-verifying agency requires internal verification of whether actions reduce loss.	A system is classified as self-verifying despite relying entirely on an external host for verification.
M0-007	Empirical bridge	Misalignment is divergence between host and delegate action effects on boundary loss.	Divergent objectives do not produce divergent finite-difference action effects under audit.
M0-008	Downstream bridge	Culture and institutions are shared externalized distinction infrastructures with verification costs.	Externalized symbols function semantically without interpreter, update rule, verification, or action-relevance channel.

2 Core ingredients from FDS

2.1 Active boundary relevance

The FDS core does not apply the deficit-dissipation-pruning cascade to every bounded object. It restricts the cascade to active-boundary systems. A minimal formal screen is

$$P(U(M_t, Y_t) \neq M_t) > 0, \quad I(M_{t+1}; \ell_{t+k}) > 0 \quad (3)$$

for some $k > 0$. In empirical systems this relevance screen should be strengthened to an intervention or ablation test:

$$\mathbb{E}[\ell_{t+k} \mid \text{do}(U)] \neq \mathbb{E}[\ell_{t+k} \mid \text{do}(U_\emptyset)], \quad (4)$$

where $U_\emptyset$ is a null, frozen, randomized, or identity update. This criterion is inherited from the FDS Core’s active-boundary qualification [1]. In M0, it becomes the starting point for agency.

2.2 Capacity deficit and semantic demand

Let Ψ_{sem} be a pre-specified family of semantic task statistics: features of signals, contexts, actions, outcomes, and boundary states that must be preserved for a system to maintain a task within tolerance. Let

$$R_{\min}^{(\tau)}(\epsilon; \Psi_{\text{sem}}) \quad (5)$$

be the minimum number of bits per update window required to encode a sufficient semantic statistic to distortion tolerance ϵ . If C_{sem} is the maintained semantic memory capacity, define

$$\Delta_{\text{sem}}(t) = R_{\min}^{(\tau)}(\epsilon; \Psi_{\text{sem},t}) - C_{\text{sem}}(t). \quad (6)$$

Positive semantic deficit does not prove that meaning fails. It says that without better compression, externalization, pruning, task relaxation, or resource expansion, actionable distinctions must be lost, merged, or degraded.

2.3 Notation alignment

Throughout this paper ℓ_{maint} denotes boundary-maintenance loss. It is not physical entropy production. The physical entropy ledger, when relevant, is denoted Σ_{phys} or $\dot{\Sigma}_{\text{phys}}$. Maintaining low ℓ_{maint} may require entropy or resource cost under physical bridge assumptions, but $\ell_{\text{maint}} \neq \Sigma_{\text{phys}}$. Likewise, q denotes a quotient map in the FDS Core; M0 uses $q_{\text{sem}} : \mathcal{D} \rightarrow T$ for a semantic quotient to avoid confusing semantic compression with topological or physical invariant quotients.

3 The agency-semantics dependency skeleton

Equation (2) is best understood as a constraint pyramid. Higher layers require stricter conditions on memory, verification, stability, and resource budgets. Under capacity deficit, they can degrade backward: meaning can collapse into attention, goal can collapse into reflex, value can collapse into salience, and culture can collapse into polluted external records.

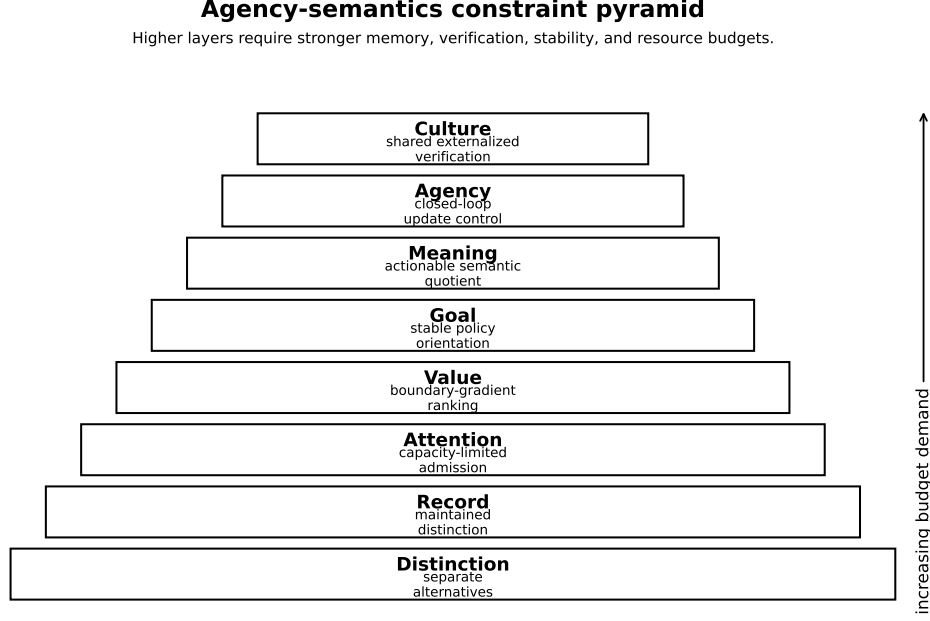


Figure 1: Synthetic normal-form illustration, not empirical evidence. The agency-semantics ladder is a dependency skeleton and constraint pyramid. Higher layers require stronger maintained records, verification capacity, stable policies, and externalized infrastructure.

Table 2: Agency-semantics constraint pyramid and failure modes under capacity deficit.

Layer	FDS operation	Main budget	Deficit failure mode
Attention	admission filtering	C_{att}	missed signal, distraction, tunnel vision
Value	boundary-gradient ranking	C_{eval}	misvaluation, risk blindness, reward hacking
Goal	stabilized policy orientation	M, τ	drift, reflex capture, unstable preference
Meaning	actionable semantic quotient	C_{sem}	semantic drift, hallucination, false compression
Agency	boundary-relevant update control	A, U, C_{verify}	proxy control, misalignment, collapse of autonomy
Culture	shared externalized distinctions	$E_{\text{ext}}, C_{\text{verify}}$	epistemic pollution, institutional rigidity

4 Attention as distinction admission

Let $\mathcal{D}_t = \{d_{t,1}, \dots, d_{t,n}\}$ be a stream of candidate distinctions. Each distinction has a cost $c_t(d)$, including encoding, verification, and maintenance cost. An attention gate is a map

$$a_t : \mathcal{D}_t \rightarrow [0, 1], \quad (7)$$

subject to the expected capacity constraint

$$\sum_{d \in \mathcal{D}_t} a_t(d) c_t(d) \leq C_{\text{att}}(t). \quad (8)$$

Definition 1 (Attention). Attention is capacity-limited admission of candidate distinctions into the update channel of an active finite system. It is not identical to salience: a salient distinction may fail admission if its verification or maintenance cost is too high, and a low-salience distinction may be admitted if it has high boundary relevance.

The attention allocation problem can be written as

$$a_t^* \in \arg \max_{a_t} \sum_{d \in \mathcal{D}_t} a_t(d) V_t^{\text{causal}}(d; k) \quad \text{s.t.} \quad \sum_d a_t(d) c_t(d) \leq C_{\text{att}}(t), \quad (9)$$

with $a_t(d) \in \{0, 1\}$ for hard admission or $a_t(d) \in [0, 1]$ for soft admission. Binary attention is a knapsack-like selection problem; soft attention is a fractional allocation problem.

Under high semantic deficit, the gate can become nonlinear and thresholded. One normal-form gate is

$$a_t(d) = \sigma(\beta_t[V_t(d; k) + \alpha n_t(d) - \gamma c_t(d) - \theta_t]), \quad \beta_t = \beta_0 + \beta_\Delta[\Delta_{\text{sem}}(t)]_+, \quad (10)$$

where $n_t(d)$ is novelty and σ a logistic function. As Δ_{sem} grows, the gate steepens. This produces tunnel vision: only high-value or high-threat distinctions enter the update channel, while slower semantic structure is lost.

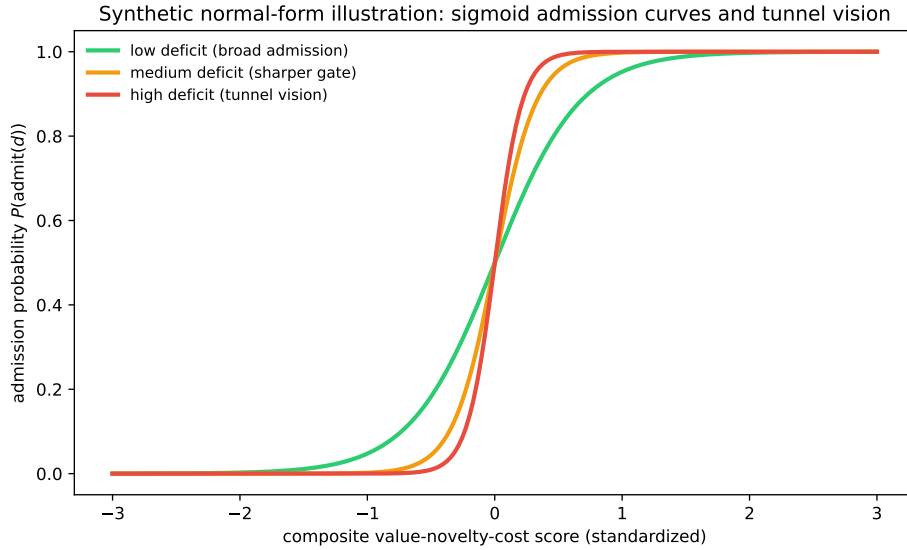


Figure 2: Synthetic normal-form illustration, not empirical evidence. Sigmoid admission curves under low, medium, and high semantic deficit. As Δ_{sem} increases, the gate steepens, producing tunnel vision: only high-value or high-threat distinctions enter the update channel while slower semantic structure is lost.

5 Value as causal boundary-gradient relevance

Value is not merely correlation with future loss. M0 distinguishes predictive value from causal value.

Definition 2 (Predictive value). A distinction d has predictive value over horizon k if conditioning on it improves prediction of future boundary-maintenance loss:

$$V_t^{\text{pred}}(d; k) = \mathbb{E}[\ell_{\text{maint}, t+k} \mid M_t] - \mathbb{E}[\ell_{\text{maint}, t+k} \mid M_t, d] - \lambda_t c_t(d). \quad (11)$$

Definition 3 (Causal boundary-gradient value). A distinction d has causal value if admitting it changes the expected future boundary-maintenance loss under the system’s update and action channel:

$$V_t^{\text{causal}}(d; k) = \mathbb{E}[\ell_{\text{maint}, t+k} \mid M_t, \text{do}(a_\emptyset)] - \mathbb{E}[\ell_{\text{maint}, t+k} \mid M_t, \text{do}(\text{admit}(d))] - \lambda_t c_t(d). \quad (12)$$

The predictive version is an observational proxy. The causal version is the preferred FDS value notion, because a distinction that merely correlates with loss but cannot alter update, action, verification, or coordination is not yet valuable in the operational sense. Here $\text{do}(\text{admit}(d))$ denotes an intervention that forces $a_t(d) = 1$ through the admission gate and permits the downstream update and action policy to condition on the resulting maintained record. Similarly, $\text{do}(\neg \text{admit}(d))$ forces $a_t(d) = 0$ while leaving the rest of the gate, update, and action policy unchanged.

Near collapse thresholds, average loss is insufficient. Let ℓ_c be a critical boundary-loss threshold. Define a risk-weighted value

$$V_t^{\text{risk}}(d; k) = \left(\mathbb{E}[\ell_{\text{maint}, t+k} \mid M_t, \text{do}(\neg \text{admit}(d))] - \mathbb{E}[\ell_{\text{maint}, t+k} \mid M_t, \text{do}(\text{admit}(d))] \right) + \alpha_t \left(P(\ell_{\text{maint}, t+k} > \ell_c \mid M_t, d) \right) \quad (13)$$

where the first term is expected loss reduction and the second is collapse-risk reduction, both positive when admission reduces risk. This connects M0 to boundary-risk asymmetry: distinctions that move the system away from Phase-C collapse can dominate distinctions that only improve average performance. The critical tolerance ℓ_c is treated as fixed in M0. In adaptive systems it may itself be updated, negotiated, or misestimated; this is left to M2 and later governance applications.

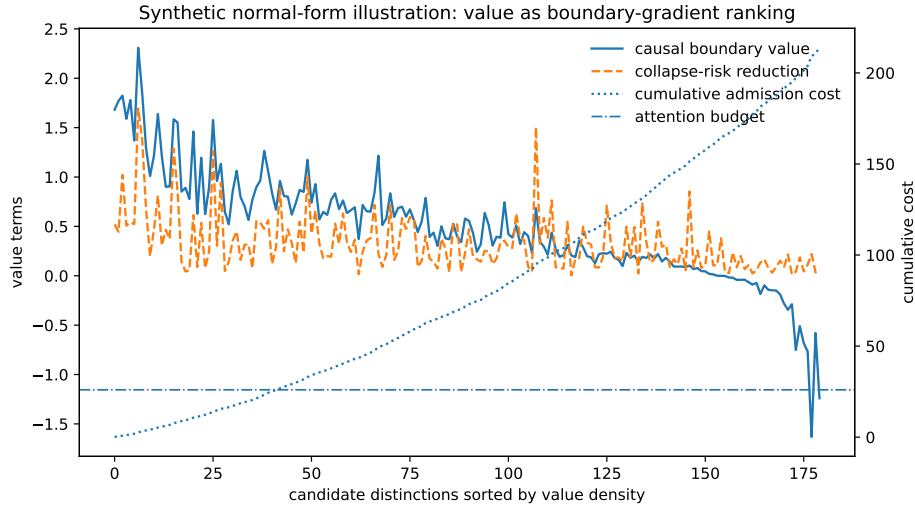


Figure 3: Synthetic normal-form illustration, not empirical evidence. Candidate distinctions are ranked by causal boundary-gradient value density, not by raw salience. Collapse-risk reduction can dominate average performance improvement near critical thresholds.

6 Goals as stabilized value rankings

A system can rank distinctions without having a goal. A reflexive controller may react moment by moment without maintaining an orientation across time. M0 treats goals as stabilized value rankings coupled to action policies.

Definition 4 (Goal). A goal is a value ranking or policy orientation that remains sufficiently stable across update windows and perturbations to guide action selection over a specified horizon.

Goal stability is a register-time property: a goal must persist across finite update windows as a maintained ordering, rather than appearing only as an instantaneous reaction. This links M0 to the register-time framework of O2 [3].

Let $\pi_g(a \mid M_t)$ be the policy induced by a goal state and let Δ be a comparison horizon. A policy-based goal-stability index is

$$\text{GSI}_\pi(t, \Delta) = \exp \left(-D_{\text{KL}}(\pi_g(\cdot \mid M_t) \parallel \pi_g(\cdot \mid M_{t+\Delta})) \right), \quad (14)$$

so that $\text{GSI}_\pi \in (0, 1]$ with $\text{GSI}_\pi = 1$ for identical policies and $\text{GSI}_\pi \rightarrow 0$ as the policy diverges. A rank-based version is

$$\text{GSI}_r(t, \Delta) = 1 - D_{\text{rank}}(\succ_t, \succ_{t+\Delta}), \quad (15)$$

where D_{rank} is a normalized rank distance in $[0, 1]$.

Proposition 1 (Reflex-preference-goal distinction). *A one-step reaction may have predictive or causal value without being goal-like. A goal-like system must maintain a nontrivial stability index over an update horizon and couple that stability to action selection.*

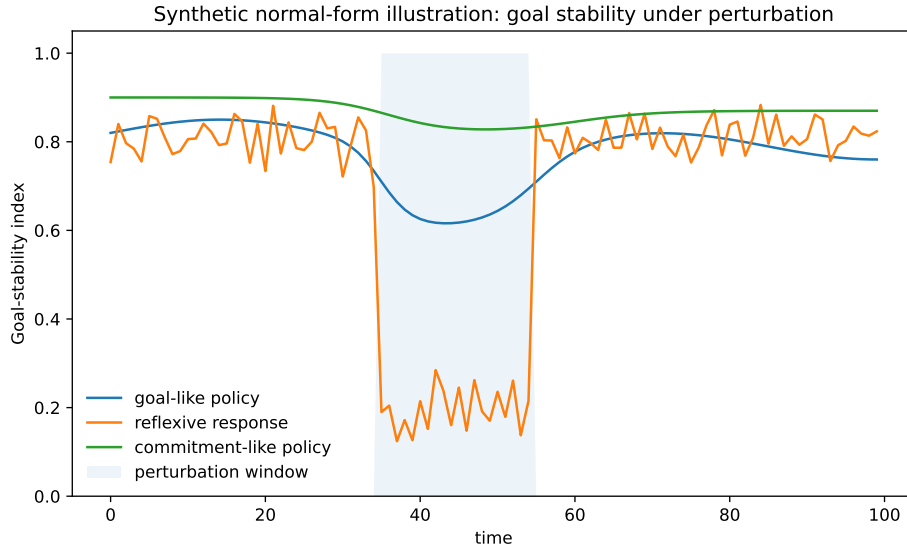


Figure 4: Synthetic normal-form illustration, not empirical evidence. A goal-like policy preserves a stable action orientation through perturbation better than a reflexive response. A commitment-like policy is even less sensitive but may become brittle if the environment changes.

7 Meaning as actionable semantic quotient

The phrase “actionable compressed distinction” is easily misunderstood if “actionable” is taken to mean immediate motor output. That is not the intended meaning. Actionable includes physical action, prediction, verification, coordination, compression of future search, preservation of an institutional boundary, or regulation of future update. A theorem, map, ritual, legal right, or social symbol may be actionable by constraining future inference and coordination, even when it does not trigger immediate motor behavior.

Definition 5 (Actionable compressed distinction). A maintained representation m_d is an actionable compressed distinction for system S if it is a compressed record of a distinction d that preserves enough structure to change action, prediction, verification, coordination, or future update relative to a boundary-maintenance task.

Definition 6 (Policy-preserving semantic quotient). Let $q_{\text{sem}} : \mathcal{D} \rightarrow T$ be a quotient map from candidate distinctions to semantic types. It is task-sufficient at tolerance ϵ for task family Ψ , horizon k , and context family $\mathcal{C}$ if there exists a quotient policy $\pi_q : T \times \mathcal{C} \rightarrow A$ such that for every $d \in \mathcal{D}$ and every admissible context $c \in \mathcal{C}$,

$$\mathbb{E}[\ell_{\text{maint},t+k} \mid d, c, \text{do}(\pi_q(q_{\text{sem}}(d), c))] \leq \min_{a \in A} \mathbb{E}[\ell_{\text{maint},t+k} \mid d, c, \text{do}(a)] + \epsilon. \quad (16)$$

This condition is stronger than requiring that the minimum achievable loss for two distinctions be close. It requires that a policy based on the quotient remain nearly optimal for each original distinction. This avoids the failure case where two distinctions have equal optimal loss but require different optimal actions.

Theorem 1 (Actionable meaning preservation). *If q_{sem} is task-sufficient in the sense of Eq. (16), then replacing d by $q_{\text{sem}}(d)$ preserves actionable meaning up to ϵ for the specified task, horizon, context family, and action space.*

Proof. By the definition of a policy-preserving semantic quotient (Definition 6), for each original distinction d and admissible context c , the quotient policy $\pi_q(q_{\text{sem}}(d), c)$ achieves expected loss no more than ϵ above the best action available with access to d . Therefore the quotient retains all task-relevant action information needed to achieve near-optimal boundary-maintenance performance at the stated tolerance. Replacing d by $q_{\text{sem}}(d)$ therefore preserves actionable meaning up to ϵ . $\square$

Remark 1 (Relation to abstraction and information bottlenecks). Equation (16) is close in spirit to state abstraction and bisimulation in reinforcement learning, and to task-relevant compression in the information bottleneck. M0 does not claim priority over those formalisms. It uses them as nearby mathematical idioms for a finite-boundary semantics bridge.

Example 1 (Red light). A red traffic light compresses many visual details into a boundary-relevant affordance: stop, crossing risk, legal obligation, and coordination with others. The meaning is not the wavelength alone. It is the context-robust quotient that preserves action and coordination across drivers, pedestrians, vehicles, and legal settings.

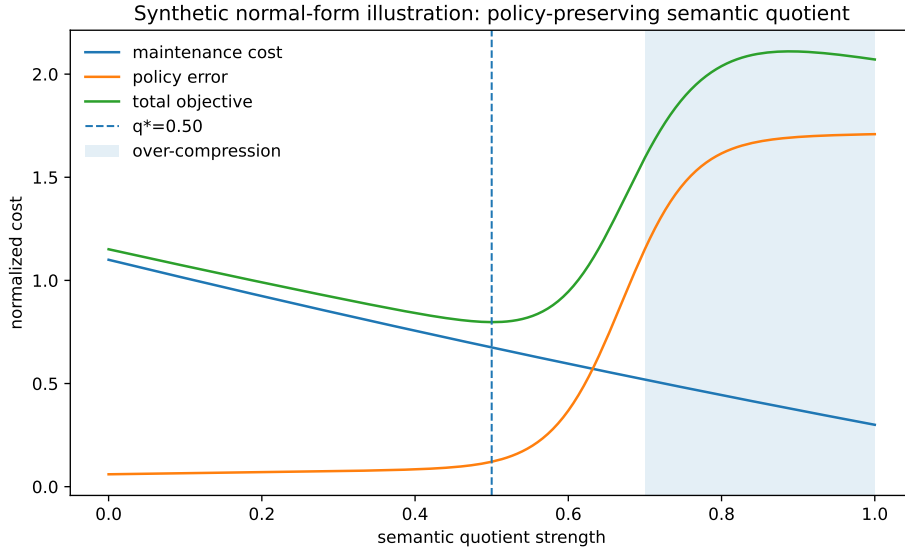


Figure 5: Synthetic normal-form illustration, not empirical evidence. Moderate semantic quotienting reduces maintenance cost while preserving policy performance. Over-compression destroys task-relevant distinctions and produces semantic drift or hallucination.

8 Semantic capacity deficit and collapse

A system can lose meaning without losing all signals. It may still attend to salient events while losing the goal, value, or policy context that makes them actionable. Define semantic capacity deficit as in Eq. (6). When $\Delta_{\text{sem}} > 0$, at least one exit must occur: better compression, pruning, externalization, task relaxation, verification outsourcing, or collapse into false compression.

Definition 7 (Semantic collapse). Semantic collapse occurs when a system retains records or salience but loses the quotient structure required to preserve boundary-relevant action, prediction, verification, or coordination.

Examples include a stressed operator who notices alarms but loses system-level meaning, a long-context AI system that preserves local tokens while losing task sequence, and an institution that keeps labels while losing verification of what those labels do.

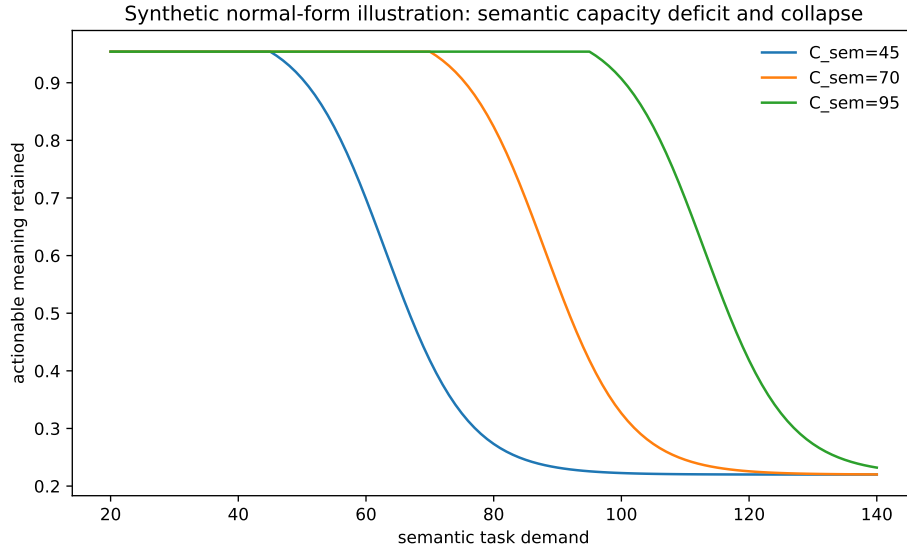


Figure 6: Synthetic normal-form illustration, not empirical evidence. As semantic task demand exceeds maintained semantic capacity, actionable meaning degrades. Signals can remain present while policy-preserving quotient structure collapses.

9 Agency as boundary-relevant update control

Definition 8 (Weak FDS agency). A system has weak FDS agency if it selects actions from a finite action space in response to admitted distinctions.

Definition 9 (Strong FDS agency). A system has strong FDS agency relative to boundary B , loss ℓ_{maint} , update rule U , and horizon k if there exists an admissible null update $U_\emptyset$ such that

$$\mathbb{E}[\ell_{\text{maint},t+k} \mid \text{do}(U)] \neq \mathbb{E}[\ell_{\text{maint},t+k} \mid \text{do}(U_\emptyset)]. \quad (17)$$

This is not a test of whether a system looks intelligent. It is a causal test of whether its updates matter for future boundary loss.

9.1 Boundary sensitivity

Agency classification is boundary-sensitive. A bare model, a model plus context, a model plus tools and memory, and a robot or institution embedded in an environment may receive different classifications. M0 therefore requires reports to state the accounting boundary.

Table 3: Boundary-sensitive agency classification.

Boundary choice	Possible classification	Attribution rule
Bare inference model	passive mapper or weak agency	Do not attribute strong agency if there is no durable boundary-relevant update.
Model + context window	weak coupled agency	Attribute behavior to the coupled prompt-context system.
Model + tools + writable memory	coupled active agency if updates affect future loss	Attribute agency to the coupled architecture, not base weights alone.
Robot / institution / environment loop	strong coupled agency if Eq. (17) holds	Report boundary, memory, update, verification, and loss mapping.

9.2 Verification load and self-verifying agency

Many systems appear agentic because an external host carries the verification load. Let Z_{outcome} be the outcome variable needed to determine whether actions reduce boundary loss. A simple verification deficit is

$$\Delta_{\text{verify}}(t) = R_{\min}^{(\tau)}(\epsilon; \Psi_{\text{verify}}) - C_{\text{verify}}^{\text{int}}(t). \quad (18)$$

When $\Delta_{\text{verify}} > 0$, the system must externalize verification, rely on reward proxies, trust a host, relax the task, or operate blindly.

Definition 10 (Self-verifying agency). A strong FDS agent is self-verifying at tolerance ϵ if it maintains an internal or coupled verification channel sufficient to estimate whether its actions reduce ℓ_{maint} within the task tolerance, without requiring an unmodeled host to perform the verification step.

Self-verification is not required for all strong agency. It is a stronger classification. This matters for artificial systems: if humans provide all outcome verification, the observed agency belongs to a coupled human-tool system.

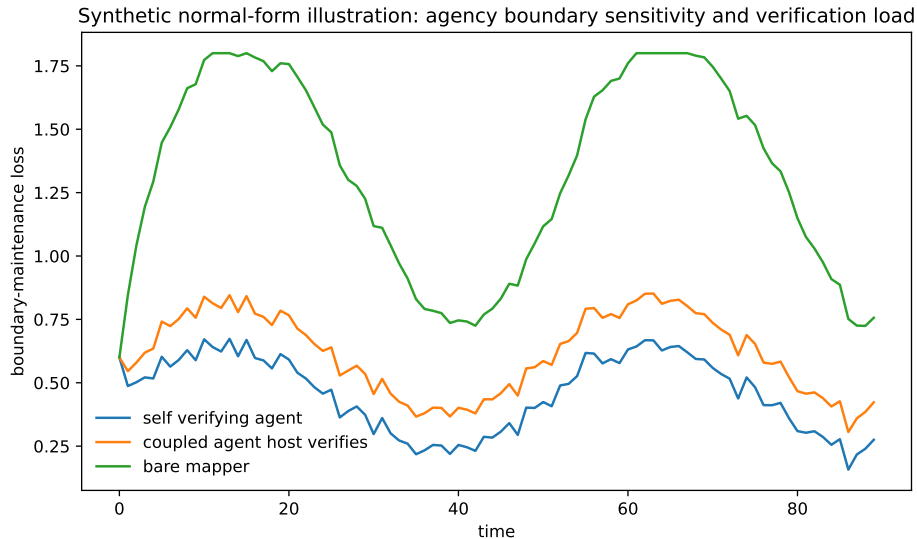


Figure 7: Synthetic normal-form illustration, not empirical evidence. Boundary-sensitive classification separates a self-verifying agent, a coupled agent whose host carries verification load, and a bare mapper that lacks durable boundary-relevant update.

10 Misalignment and epistemic pollution

10.1 Misalignment as action-effect divergence

Gradient dot products are coordinate-dependent and do not apply naturally to discrete action spaces. M0 therefore defines misalignment using normalized finite-difference effect vectors. For actions a_i relative to a baseline a_0 , let

$$\Delta_H(a_i) = \mathbb{E}[\ell_H \mid \text{do}(a_i)] - \mathbb{E}[\ell_H \mid \text{do}(a_0)], \quad (19)$$

$$\Delta_D(a_i) = \mathbb{E}[\ell_D \mid \text{do}(a_i)] - \mathbb{E}[\ell_D \mid \text{do}(a_0)], \quad (20)$$

where H is the host and D the delegate. Define

$$\text{Align}(H, D) = \frac{\langle \Delta_H, \Delta_D \rangle}{\|\Delta_H\| \|\Delta_D\|}. \quad (21)$$

For threshold $\eta > 0$, a delegated system is misaligned over the audited action set when

$$\text{Align}(H, D) < -\eta. \quad (22)$$

This definition captures whether actions that help the delegate tend to harm the host, not merely whether internal objectives use different words.

Because Δ_H and Δ_D are defined as loss changes relative to baseline, beneficial actions have negative components. Alignment is positive when host and delegate loss effects point in the same direction, including the case in which both losses decrease.

10.2 Epistemic pollution as verification-bandwidth saturation

Externalized semantic environments can help finite systems by storing records outside internal memory. They can also pollute the shared distinction environment.

Definition 11 (Epistemic pollution). Epistemic pollution is saturation of finite verification bandwidth by low-quality, adversarial, obsolete, redundant, or unverifiable distinctions in an externalized semantic environment.

Let $R_{\text{verify}}^{(\tau)}(\epsilon; t)$ be the verification demand imposed by the external environment and $C_{\text{verify}}^{\text{avail}}(t)$ available verification capacity. Define

$$Z_{\text{poll}}(t) = \frac{R_{\text{verify}}^{(\tau)}(\epsilon; t)}{C_{\text{verify}}^{\text{avail}}(t)}. \quad (23)$$

When $Z_{\text{poll}} > 1$, the system cannot verify all admitted distinctions. It must ignore, trust blindly, false-compress, outsource verification, or collapse into noise. False compression occurs when a quotient merges distinctions that should remain separate for the task, producing semantic drift, hallucination, category failure, or institutional rigidity.

Synthetic normal-form illustration: misalignment and epistemic pollution

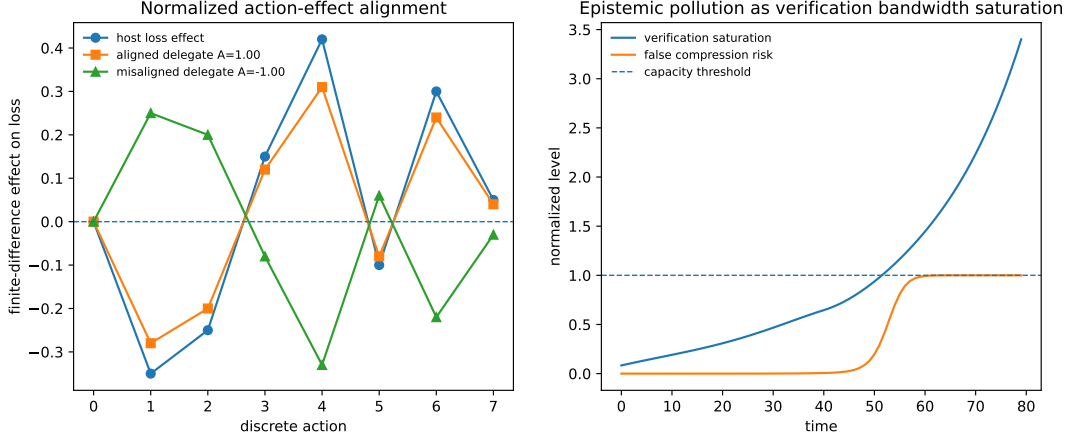


Figure 8: Synthetic normal-form illustration, not empirical evidence. Left: normalized finite-difference action effects distinguish aligned and misaligned delegated systems. Right: epistemic pollution appears when verification demand saturates finite verification bandwidth, forcing false compression.

11 Non-domain cases and attribution discipline

M0 is designed to prevent over-attribution. The following cases may process information or symbols without satisfying strong FDS agency or actionable meaning as defined above.

Table 4: Non-domain and limited-domain cases.

Case	Why it is not strong FDS agency or robust meaning by itself
Passive sensor	Records correlate with the world but have no boundary-relevant update control.
Static database	Stores distinctions, but action relevance appears only in a coupled user or institutional system.
Bare LLM inference call	Input-output mapping without durable self-updating boundary state or internal verification channel.
Random controller	Produces actions without expected boundary-loss reduction.
Lookup table	Mapping may be useful but lacks adaptive update, compression, or generalization unless embedded in a larger system.
Polluted archive	Preserves labels but not reliable verification, actionable quotient structure, or semantic maintenance.

12 Normal-form model and reproducibility

The accompanying code implements deterministic synthetic normal-form illustrations. The model is not empirical evidence. It is a consistency and visualization device that maps the definitions to simple state variables.

The model uses: a candidate distinction stream with novelty, threat, affordance, verification cost, and maintenance cost; a nonlinear attention gate whose steepness increases with semantic deficit; a causal boundary-gradient value score; a policy-preserving semantic quotient tradeoff; a goal-stability index under perturbation; boundary-sensitive agency rollouts; finite-difference misalignment; and epistemic pollution saturation. The random seed is fixed

in `code/generate_results.py`. CSV outputs are stored in `data/`; figure pairs are stored in `figures/`.

Table 5: Normal-form variable map. All entries are illustrative, not fitted empirical quantities.

Simulation variable	Paper definition	Interpretation
<code>causal_value</code>	Eq. (12)	expected boundary-loss reduction minus admission cost
<code>semantic_deficit</code>	Eq. (6)	task demand minus maintained semantic capacity
<code>admission_probability</code>	Eq. (10)	soft attention gate output
<code>policy_loss</code>	Eq. (16)	error induced by quotient compression
<code>goal_like_GSI</code>	Eq. (14)	stability of policy orientation under perturbation
<code>verification_load</code>	Eq. (18)	internal or external cost of action-outcome auditing
<code>saturation_ratio</code>	Eq. (23)	verification demand divided by verification capacity

13 Relation to existing fields

M0 is adjacent to several established fields.

Information theory and finite capacity. Shannon’s information theory and rate-distortion theory define the fundamental limits of communication and compression under fidelity constraints [7, 8]. M0 uses these as formal boundaries for finite semantic capacity and semantic deficit.

Semiotics and pragmatics. Peircean and Morris-style semiotics analyze signs, objects, interpreters, and action consequences [19, 20]. M0 does not replace semiotics. It specifies a finite-system bridge: a sign functions as a maintained quotient only relative to an interpreter boundary, update rule, action space, and verification channel.

Information bottleneck and semantic compression. The information bottleneck formalizes compression that preserves target-relevant information [9]. M0’s semantic quotient is similar but explicitly tied to action, policy preservation, and boundary-maintenance loss.

Rational inattention and bounded rationality. Rational inattention treats information acquisition as costly [10, 11]. M0 generalizes this into distinction admission: attention is not only costly information acquisition, but capacity-limited update admission under maintenance constraints.

Reinforcement learning, state abstraction, and bisimulation. State abstraction and bisimulation preserve decision-relevant structure in Markov decision processes [13–15]. M0’s Theorem 1 is closest to this family: meaning-preserving quotients must preserve near-optimal action, not merely compress observations.

Causal representation and intervention. Pearl-style causal models emphasize interventions rather than correlation [12]. M0 uses this distinction for causal value and agency: a distinction is operationally valuable only when admission or action changes expected boundary loss.

Affordances, empowerment, active sensing, and active inference. Affordance theory stresses action possibilities [16]; empowerment measures potential influence over future states [17]; active inference and predictive processing emphasize action-perception loops under uncertainty [18]. M0 is compatible with these perspectives but centers finite distinction capacity, verification load, and boundary-maintenance loss.

Game, trust, and verification. Earlier unpublished distinction-budget game drafts treated cooperation, parasitism, and trust in terms of monitoring costs. M0 uses the more conservative FDS language: trust and delegation reduce repeated verification cost only when maintained records remain reliable and when externalized verification does not saturate capacity.

Recent interfaces. Recent work sharpens several concepts used in M0. Causal bisimulation and causal state abstraction study how task-relevant structure can be compressed while preserving decision performance [21, 22]. Goal-oriented semantic communication similarly treats semantics as task-relevant compression under bandwidth, latency, and power constraints [25]. Recent work on active inference distinguishes variational free energy from thermodynamic free energy, clarifying the physical cost of active agents [23]. Recent cognitive-economic work connects machine learning, costly learning, and rational inattention [24]. Finally, LLM-agent surveys provide a practical background for the boundary-sensitive agency distinction used here: agency may belong to a coupled system of model, tools, memory, user, and environment rather than to a bare model call [26].

14 Protocols and tests

Protocol 1 (Attention admission audit). Pre-register candidate distinctions, costs, and boundary-relevance variables. Vary load or capacity and test whether admitted distinctions track causal boundary value better than raw salience.

Protocol 2 (Value intervention audit). Compare predictive and causal value. A distinction with high predictive value but no effect under admission or action should be demoted to observational relevance.

Protocol 3 (Goal stability audit). Estimate GSI_π or GSI_r under perturbations. Reflexive systems may react strongly while showing low stability across update windows.

Protocol 4 (Semantic quotient audit). Test whether a compressed representation admits a quotient policy satisfying Eq. (16). Over-compression should increase task loss or semantic drift.

Protocol 5 (Agency intervention audit). Compare $do(U)$ against $do(U_\emptyset)$ under a stated boundary. If only the extended architecture passes, classify agency as coupled rather than intrinsic to the narrow component.

Protocol 6 (Misalignment and verification audit). Compute finite-difference action-effect alignment for host and delegate losses. Separately audit verification load to identify systems whose apparent agency is supplied by an external host.

15 Limitations and falsification

M0 is intentionally limited. It does not establish empirical theories of attention, value, goal, or meaning by definition. It provides mappings that must survive operational audit. The framework is weakened or demoted under any of the following results:

1. robust attention-like behavior under a specified mapping shows no capacity-limited admission or selection;
2. value-like ranking has no relationship to future boundary loss, task success, risk reduction, or resource relevance;
3. goal-like behavior persists without memory, ranking stability, policy orientation, or action direction over time;
4. compressed representations guide no action, prediction, verification, coordination, or boundary maintenance;
5. strong agency is attributed to systems whose updates have no causal effect on future boundary-maintenance loss;
6. delegated systems remain aligned despite persistent negative action-effect alignment and verification failure;
7. externalized semantic environments impose no retrieval, verification, maintenance, or pollution costs.

16 Conclusion

FDS-M0 defines the agency-semantics spine of Distinction Theory. Its central chain is:

$$\text{finite distinction} \rightarrow \text{admission} \rightarrow \text{boundary-gradient ranking} \rightarrow \text{stable policy} \rightarrow \text{actionable semantic quotient} - \quad (24)$$

The chain is not a claim that semantics has been solved or reduced to physics. It is an operational interface. Attention, value, goal, meaning, and agency are treated as roles played by distinctions in active finite systems that must maintain boundaries under capacity and resource constraints.

This spine prepares later M-series and civilization-layer papers. M1 can focus on attention as distinction admission; M2 on value and goal as boundary-gradient ranking; M3 on meaning as actionable semantic quotient; M5 on trust as delegated verification; A2 on AI alignment as boundary-compatible externalized agency; S2 on epistemic pollution; and G3/G4 on science and civilization memory as large-scale verification infrastructures.

Remark 2 (Cultural invariant candidate). A cultural invariant is a shared externalized semantic quotient that remains usable across agents, contexts, and update windows for coordination or institutional boundary maintenance. M0 only defines the interface; S/G-series papers develop the civilization-scale theory.

The central discipline remains unchanged: every domain claim must state its boundary, memory, update rule, capacity, loss, verification channel, and failure condition.

Code and data availability

The deterministic synthetic normal-form code, figures, and CSV outputs are included in the replication package. Run `python code/generate_results.py` from the paper directory to regenerate all figures and data.

AI assistance disclosure

The author used AI assistance for drafting, editing, simulation scaffolding, and consistency checks. The author reviewed and selected the final claims, definitions, equations, and interpretations.

A M-series dependency map

Table 6: How M0 seeds the later M-series.

Future paper	Interface supplied by M0
M1 Attention	distinction admission, attention allocation, tunnel vision
M2 Value/Goal	causal boundary-gradient value, risk asymmetry, goal-stability index
M3 Meaning	task-sufficient semantic quotient, over-compression, semantic collapse
M4 Learning	distinction refinement and compression improvement under novelty
M5 Trust	delegated verification, monitoring cost, externalized confidence records
M6 Power	control over admission channels, verification infrastructure, and boundary updates
M7 Culture	shared externalized distinction infrastructure
M8 Beauty	high-compression distinction with low maintenance and high transfer
M9 Education	intergenerational distinction transmission and verification
M10 Law/Rights	boundary protocols and protected distinction claims

B Notation summary

Symbol	Meaning
$\mathcal{D}_t$	candidate distinction stream
$a_t(d)$	attention/admission weight for distinction d
$c_t(d)$	encoding, verification, and maintenance cost of d
C_{att}	attention admission capacity
V_t^{pred}	predictive value, observational relevance to future loss
V_t^{causal}	causal value, effect of admitting d on future loss
V_t^{risk}	risk-weighted value near collapse thresholds
GSI	goal-stability index
q_{sem}	semantic quotient preserving action-relevant structure
Δ_{sem}	semantic capacity deficit
ℓ_{maint}	boundary-maintenance loss
Σ_{phys}	physical entropy-production ledger, distinct from ℓ_{maint}
Δ_{verify}	verification deficit
Z_{poll}	epistemic pollution / verification saturation ratio

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Attention as Distinction Admission in Finite Systems

Capacity-Limited Gating, Boundary Relevance, and Tunnel Vision

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Abstract

This paper develops the attention paper in the agency-semantics spine of Distinction Theory. Building on the formal core of Active Finite Distinction Systems and the agency-semantics spine, it treats attention as capacity-limited admission of candidate distinctions into an update channel. A distinction becomes attended when it is admitted, maintained or made recoverable, assigned a verification path or status, and made available for downstream update, action, prediction, coordination, or boundary maintenance. The paper separates attention from salience: salient distinctions may be rejected when their encoding, verification, or maintenance cost exceeds available capacity, while low-salience distinctions may be admitted when they have high causal boundary relevance or high exploratory information gain. The paper introduces candidate distinction streams, admission gates, scalar and vector attention budgets, boundary-gradient attention value, verification load, background scanning, semantic deficit, nonlinear gate steepening, tunnel vision, attention hysteresis, attention failure modes, collective attention, and audit protocols for biological, cognitive, artificial, organizational, and civilization-scale systems. A deterministic synthetic normal-form model illustrates salience-value dissociation, capacity-limited admission, verification-limited rejection, deficit-induced tunnel vision, hysteretic recovery, and collective attention under epistemic pollution. The paper does not replace cognitive neuroscience, rational inattention, predictive processing, or transformer attention. It supplies a conservative finite-system bridge for analyzing attention as distinction admission under capacity, verification, and resource constraints.

Keywords: attention; distinction admission; finite capacity; boundary maintenance; salience; verification load; tunnel vision; active finite distinction systems; rational inattention; information bottleneck; transformer attention; organizational attention; epistemic pollution.

Epistemic Notice and Scope

This manuscript is an agency-semantics spine paper, not a completed theory of cognitive attention, consciousness, neural competition, or machine attention. It does not claim that all attention is conscious attention. It does not claim that salience equals value, that current transformer attention layers are sufficient for FDS attention, or that attention can be reduced to a single scalar. Its claim is narrower: once a system is modeled as an active finite distinction system with a boundary, memory, update rule, action space, finite capacity, and boundary-maintenance loss, attention can be treated as the capacity-limited admission of candidate distinctions into an update-relevant channel.

The paper follows the layered discipline of the FDS formal core and the agency-semantics spine: formal definitions, normal-form models, bridge assumptions, and domain applications must be separated. A failure of a biological, cognitive, artificial, organizational, or social mapping may demote that mapping without refuting the formal FDS core [1, 2]. The deterministic numerical model is illustrative only. It visualizes the definitions and failure modes; it is not empirical evidence.

1 Introduction

A finite system cannot attend to everything it can detect. Detection produces candidate distinctions; attention admits a subset of them into update. The difference matters. A flash, a word, a warning, a memory, a social signal, a token in a model context, or a weak anomaly in an instrument can be present without becoming operationally available. M1 begins from this gap between availability and admission.

The FDS core defines an active finite distinction system as a tuple

$$S = (X, E, B, M, Y, A, U, \pi, \ell, \Phi, \mathcal{P}, \tau), \quad (1)$$

where X is internal state, E environment, B boundary, M memory/model space, Y observation channel, A action space, U update map, π finite projection, ℓ boundary-maintenance loss, Φ resource budget, $\mathcal{P}$ perturbation/pruning family, and τ update timescale [1]. The M0 agency-semantics paper introduced a dependency skeleton

$$\text{distinction} \rightarrow \text{record} \rightarrow \text{attention} \rightarrow \text{value} \rightarrow \text{goal} \rightarrow \text{meaning} \rightarrow \text{agency} \rightarrow \text{culture}, \quad (2)$$

and defined attention as capacity-limited distinction admission into an update channel [2]. M1 expands the first active step in that chain.

1.1 Why attention needs a finite-system formulation

Attention is often described as selection, orienting, priority, competition, or limited-capacity processing. M1 uses a different but compatible abstraction. It asks: when a finite boundary-maintaining system faces more candidate distinctions than it can encode, verify, maintain, and update, which distinctions are admitted to the next state?

This admission view is useful because it applies across scales. A cell admits some chemical gradients into regulatory response. A brain admits some sensory and memory distinctions into task control. A machine system admits some tokens, tool outputs, or observations into its writable state. An organization admits some warnings into agenda and policy. A civilization admits some claims into archives, standards, or science. In each case, the system must choose because its update channel is finite.

1.2 Attention is not salience

The central distinction in this paper is between salience and attention. Salience is a property of a candidate distinction relative to the current perceptual, informational, or social field: it stands out. Attention is admission into an update-relevant channel. A salient distinction can fail attention if it is too costly to verify, too expensive to maintain, irrelevant to the boundary task, or suppressed by a higher-priority gate. Conversely, a low-salience distinction can be attended if it has high causal boundary relevance or high information gain for a model that is becoming brittle.

1.3 What this paper does not claim

This paper does not replace cognitive neuroscience, psychological theories of selective attention, rational inattention, active inference, or transformer attention. It does not say that attention is always optimal. It does not claim that every admission gate is conscious. It does not claim that an attention matrix inside a neural network is attention in the full FDS sense. It supplies a finite-system bridge: attention is constrained admission of distinctions into update.

Table 1: Claim-status summary for M1. The table is an audit device: several entries are formal or operational bridge claims, not established empirical results.

Claim ID	Tier	Claim	Failure or demotion condition
M1-001	Formal bridge	Attention is capacity-limited distinction admission into an update channel.	Attention-like selection occurs without finite capacity, admission, update gating, or priority constraint under the specified mapping.
M1-002	Operational bridge	Salience and attention are separable. Salient distinctions can be rejected if cost or verification burden is too high.	Empirical systems always admit highest-salience items regardless of cost, capacity, task, or verification constraints.
M1-003	Operational bridge	Boundary-efficient or loss-minimizing attention systems preferentially admit high causal boundary-value distinctions under controlled capacity conditions.	Admission patterns are no better predicted by causal boundary value than by raw salience or noise under a valid mapping.
M1-004	Formal / model bridge	Attention allocation can be written as constrained optimization over value, curiosity, cost, and capacity.	No useful mapping exists between admission patterns and constrained allocation variables.
M1-005	Operational bridge	Semantic or attention deficit steepens admission thresholds and can produce tunnel vision.	High load or deficit produces no narrowing, thresholding, or priority collapse in systems claimed to have finite attention.
M1-006	AI / cognition bridge	Artificial attention belongs to a coupled architecture only when routed distinctions affect durable update, action, maintenance, or verification.	Bare attention weights alone satisfy strong FDS attention without durable update or downstream relevance.
M1-007	Social bridge	Collective attention is shared admission under finite communication, verification, and externalized memory capacity.	Group-scale attention shows no relation to verification capacity, agenda-setting, or externalized memory.
M1-008	Failure-mode bridge	Attention failure includes overload, distraction, salience capture, suppression, tunnel vision, false admission, and critical distinction exclusion.	These failure modes cannot be operationalized as admission errors under finite capacity.
M1-009	Operational bridge	Attention recovery after deficit-induced narrowing can lag behind external load reduction because of hysteresis in gate thresholds, verification routines, or maintained threat priors.	Attention gates relax immediately and without lag after load reduction in systems where hysteresis is claimed.

2 FDS and M0 background

2.1 Active boundary relevance

The FDS core applies its maintenance and deficit logic to active-boundary systems rather than to every bounded object. A minimal relevance screen is

$$P(U(M_t, Y_t) \neq M_t) > 0, \quad I(M_{t+1}; \ell_{t+k}) > 0 \quad (3)$$

for some $k > 0$. In empirical settings this should be strengthened to an intervention or ablation test:

$$\mathbb{E}[\ell_{t+k} \mid \text{do}(U)] \neq \mathbb{E}[\ell_{t+k} \mid \text{do}(U_\emptyset)]. \quad (4)$$

Attention matters to FDS only when admission affects a future update, record, action, verification, or boundary-maintenance variable.

2.2 Candidate distinctions and update channels

Let $\mathcal{D}_t = \{d_{t,1}, \dots, d_{t,n}\}$ denote the candidate distinction stream during update window t . Candidate distinctions may arise from sensors, memory, internal simulation, social communication, external archives, tool outputs, or delegated systems. The update channel is the part of the system through which admitted distinctions can affect M_{t+1} , action selection, verification, or future boundary-maintenance loss.

M1 distinguishes three events:

1. *availability*: a distinction exists in the candidate stream;
2. *detection*: the system can register or measure the distinction transiently;
3. *attention*: the distinction is admitted into update-relevant processing.

Detection without admission is possible. Admission without durable maintenance is possible. Reportability is downstream of attention: FDS attention can occur without report, and report can fail after transient admission if maintenance or communication capacity is insufficient.

2.3 Notation alignment

Throughout the paper ℓ_{maint} denotes boundary-maintenance loss. It is not physical entropy production. C_{att} denotes attention admission capacity; C_{verify} denotes verification capacity; C_{sem} denotes maintained semantic capacity. The physical entropy ledger, when relevant, is denoted Σ_{phys} and is not identified with ℓ_{maint} . Admission, verification, and maintenance may have physical costs under bridge assumptions, but M1 does not equate an abstract attention choice with a fixed thermodynamic heat cost.

3 Definition: attention as distinction admission

Definition 1 (Candidate distinction stream). A candidate distinction stream $\mathcal{D}_t$ is the set of possible distinctions available to a finite system during an update window. Candidate distinctions may be external signals, internal memories, simulated alternatives, social claims, environmental affordances, or outputs of delegated systems.

Definition 2 (Attention gate). An attention gate is a map

$$a_t : \mathcal{D}_t \rightarrow [0, 1], \quad (5)$$

where $a_t(d)$ is the admission weight assigned to distinction d . Hard attention uses $a_t(d) \in \{0, 1\}$; soft attention uses $a_t(d) \in [0, 1]$.

Definition 3 (FDS attention). A distinction d is attended by system S at time t if it is admitted through an attention gate, encoded into a maintained or update-relevant record, assigned a verification path or verification status, and made available to downstream update, action, prediction, verification, coordination, or boundary maintenance. Attention is capacity-limited admission of candidate distinctions into the update channel of an active finite system.

Synthetic normal-form illustration: attention as distinction admission

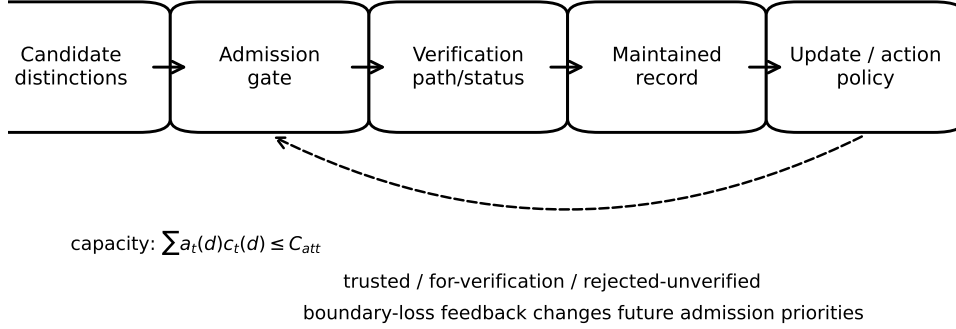


Figure 1: Synthetic normal-form illustration, not empirical evidence. Candidate distinctions pass through a finite admission gate before they can become verification-addressable records and condition downstream update or action. Boundary-loss feedback changes future admission priorities.

3.1 Maintained attention versus transient detection

A camera sensor may detect a photon without storing a stable record. A person may register a sound without using it. A model may compute an internal score for a token without preserving the resulting distinction in a durable update state. M1 reserves the term *attention* for admission into an update-relevant channel. This keeps attention distinct from raw sensing, transient activation, salience, and post-hoc report.

3.2 Verification status classes

Verification need not be complete before a distinction counts as attended. M1 distinguishes three cases:

Class	Meaning
Attended-as-trusted	The distinction is admitted and treated as reliable enough for downstream update.
Attended-for-verification	The distinction is admitted in order to test reliability, cross-check, or resolve conflict.
Rejected-unverified	The distinction is detected or available but not admitted because the verification path is too costly, unavailable, or outside the update window.

This distinction will matter for later work on trust, scientific error correction, and epistemic pollution.

3.3 Internal, externalized, and collective attention

Attention may be internal to a biological or artificial system; it may be externalized into lists, alarms, calendars, dashboards, or agendas; and it may be collective, as when a group allocates shared verification and communication capacity to some issues and not others. The same formal gate can describe these cases, but the accounting boundary must be stated.

4 Attention allocation under finite capacity

Each candidate distinction has an admission cost

$$c_t(d) = c_t^{\text{enc}}(d) + c_t^{\text{verify}}(d) + c_t^{\text{maint}}(d) + c_t^{\text{opp}}(d), \quad (6)$$

including encoding, verification, maintenance, and opportunity cost. Unless otherwise stated, $c_t(d)$ is a scalarized cost in loss-equivalent or budget-equivalent units. A more detailed model can replace Eq. (8) with vector constraints,

$$\sum_{d \in \mathcal{D}_t} a_t(d) c_t^r(d) \leq C_t^r, \quad r \in \{\text{enc, verify, maint, latency, energy}\}. \quad (7)$$

The scalar attention capacity constraint is

$$\sum_{d \in \mathcal{D}_t} a_t(d) c_t(d) \leq C_{\text{att}}(t). \quad (8)$$

Definition 4 (Boundary-efficient attention allocation). An attention allocation is boundary-efficient over horizon k if, under the stated model class and capacity constraints, no feasible reallocation lowers expected boundary-maintenance loss plus scalarized admission cost over the audited horizon:

$$a_t^* \text{ is boundary-efficient if } \forall a_t \in \mathcal{A}(C_{\text{att}}), \\ \mathbb{E}[\ell_{\text{maint}, t+k} \mid a_t^*] + \lambda C(a_t^*) \leq \mathbb{E}[\ell_{\text{maint}, t+k} \mid a_t] + \lambda C(a_t) + \epsilon.$$

Definition 5 (Attention allocation). An attention allocation is an admission vector a_t satisfying Eq. (8). A boundary-efficient finite attention system should allocate admission toward distinctions with high causal boundary relevance or high exploration value relative to their cost, while respecting capacity and verification constraints.

A normal-form allocation problem is

$$a_t^* \in \arg \max_{a_t} \sum_{d \in \mathcal{D}_t} a_t(d) \left[V_t^{\text{att}}(d; k) + \eta_t I_t^{\text{gain}}(d) \right] \quad \text{s.t.} \quad \sum_d a_t(d) c_t(d) \leq C_{\text{att}}(t), \quad (9)$$

with $a_t(d) \in \{0, 1\}$ for hard admission or $a_t(d) \in [0, 1]$ for soft admission. The hard case is knapsack-like; the soft case is a fractional allocation. M1 does not claim real systems solve Eq. (9) exactly. It uses the equation as an audit normal form: deviations can be interpreted as salience capture, suppression, verification failure, learned priority, exploration, or misestimated value.

4.1 Exploration and background scanning

A system that admits only currently high-value distinctions can become brittle. It may miss weak early signals, model-breaking anomalies, or low-salience distinctions whose value is unknown. M1 therefore separates exploitation from background scanning:

$$C_{\text{att}}(t) = C_{\text{exploit}}(t) + C_{\text{scan}}(t), \quad (10)$$

where C_{exploit} is spent on known high-value distinctions and C_{scan} is reserved for uncertain or model-informative distinctions. A simple information-gain term is

$$I_t^{\text{gain}}(d) = H(\Theta_t \mid M_t) - \mathbb{E}_{o \sim P(o \mid d, M_t)} [H(\Theta_t \mid M_t, d, o)], \quad (11)$$

where Θ_t denotes latent task or environment variables and the expectation is taken over possible verification outcomes, measurement noise, source reliability states, or latent-state interpretations associated with admitting d . Equation (9) then formalizes an exploration-exploitation tradeoff: admitting only high known value can reduce short-run loss while increasing long-run model rigidity.

4.2 Verification cost from model incompatibility

Verification cost is not arbitrary. In an FDS observer, verification requires finite records, comparison operations, cross-checks, memory turnover, and possibly external sources [3]. A normal-form verification cost is

$$c_t^{\text{verify}}(d) = c_0 + \kappa_{\text{KL}} D_{\text{KL}}(P_{\text{ext}}(\cdot | d) \| P_{\text{int}}(\cdot | M_t)) + \kappa_{\text{src}} C_{\text{source}}(d) + \kappa_{\text{cross}} C_{\text{crosscheck}}(d). \quad (12)$$

The KL-like term represents incompatibility between the candidate signal and the internal model; the source and cross-check terms represent provenance and corroboration cost. The KL term is a normal-form contributor, not a universal law. In empirical applications it may be replaced by a bounded divergence, a likelihood-ratio conflict score, a source-discrepancy measure, or another model-discrepancy proxy. Surprising signals can be cheap to verify when the measurement channel is reliable, the source is trusted, or the relevant test is simple. Model incompatibility is only one contributor to verification cost: some surprising signals are cheap to verify, and some familiar signals are expensive to audit.

Proposition 1 (Attention-salience dissociation). *Salience is not sufficient for attention. Under finite capacity, a salient distinction can be rejected if its expected boundary relevance is low or if its verification and maintenance cost is high.*

Proof sketch. Consider two candidate distinctions d_i and d_j when the capacity constraint permits only one full hard admission. Suppose raw salience satisfies $s(d_i) > s(d_j)$ but causal admission value satisfies $V_t^{\text{att}}(d_j; k) > V_t^{\text{att}}(d_i; k)$ after costs are included. Any allocation maximizing admitted causal value chooses d_j over d_i . Thus high salience does not imply attention. Under a fractional or value-density greedy rule, the same dissociation appears when $V_t^{\text{att}}(d_j; k)/c_t(d_j) > V_t^{\text{att}}(d_i; k)/c_t(d_i)$. $\square$

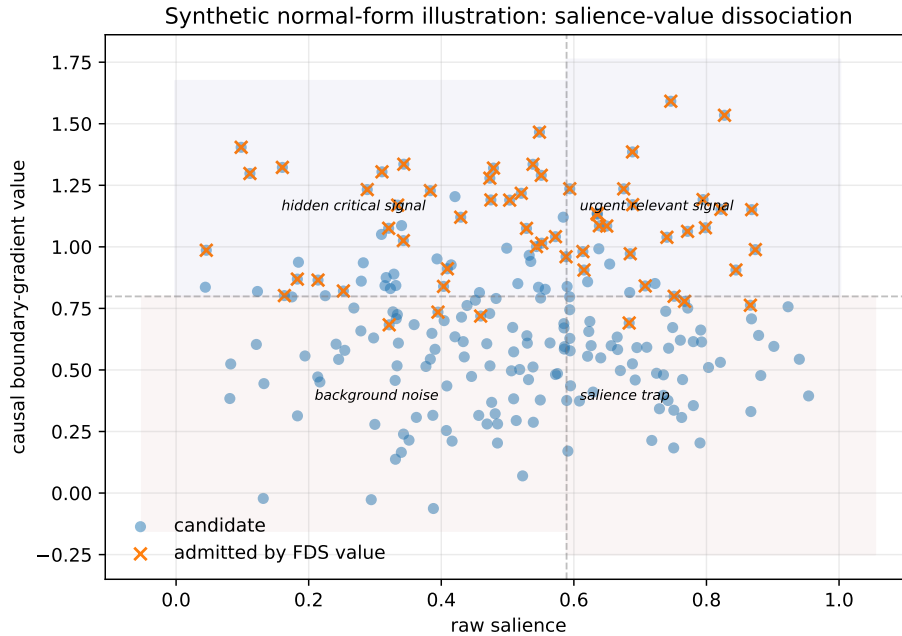


Figure 2: Synthetic normal-form illustration, not empirical evidence. Raw salience and causal boundary-gradient value can diverge. Quadrants distinguish salience traps from hidden critical signals. Under finite capacity, the admitted set can include lower-salience but higher-value distinctions.

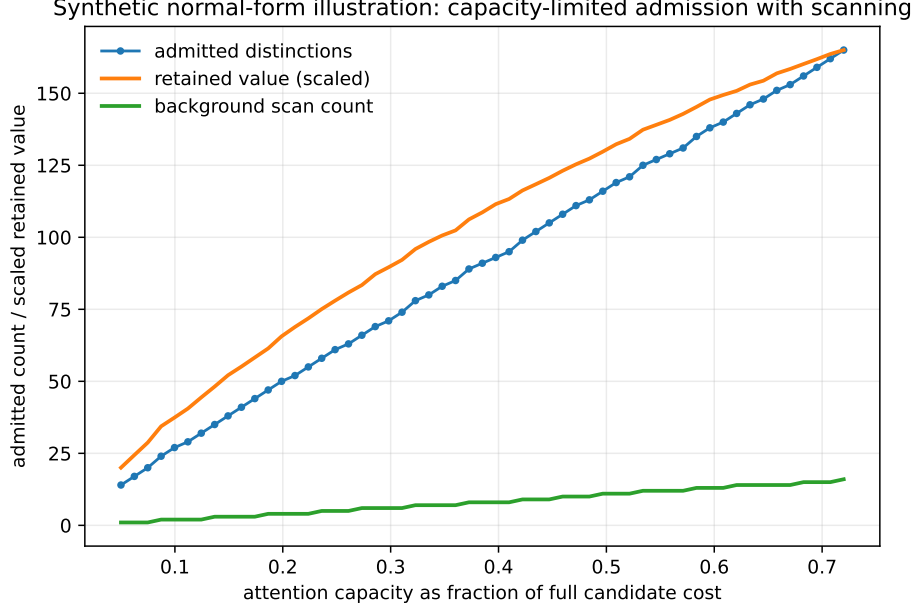


Figure 3: Synthetic normal-form illustration, not empirical evidence. As attention capacity increases, more distinctions can be admitted and more boundary-relevant value is retained. A reserved background-scanning budget admits some uncertain, high-information-gain distinctions that are not selected by exploitation alone.

5 Boundary relevance and causal attention value

FDS-M1 uses causal boundary relevance rather than mere correlation. Let $\text{do}(\text{admit}(d))$ denote an intervention that forces $a_t(d) = 1$ through the admission gate and permits downstream update and action policy to condition on the resulting maintained record. Let $\text{do}(\neg \text{admit}(d))$ force $a_t(d) = 0$ while leaving the rest of the gate, update, and policy mechanisms unchanged.

Definition 6 (Predictive attention relevance). A distinction d has predictive attention relevance over horizon k if conditioning on it improves prediction of future boundary-maintenance loss:

$$V_t^{\text{pred-att}}(d; k) = \mathbb{E}[\ell_{\text{maint}, t+k} \mid M_t] - \mathbb{E}[\ell_{\text{maint}, t+k} \mid M_t, d] - \lambda_t c_t(d). \quad (13)$$

Definition 7 (Causal boundary-gradient attention value). A distinction d has causal attention value if admitting it changes expected future boundary-maintenance loss under the system’s update and action channel:

$$V_t^{\text{att}}(d; k) = \mathbb{E}[\ell_{\text{maint}, t+k} \mid M_t, \text{do}(\neg \text{admit}(d))] - \mathbb{E}[\ell_{\text{maint}, t+k} \mid M_t, \text{do}(\text{admit}(d))] - \lambda_t c_t(d). \quad (14)$$

The predictive version is an observational proxy. The causal version is preferred because a distinction that correlates with loss but cannot alter update, verification, action, or coordination is not yet attention-relevant in the operational sense.

Near collapse thresholds, average loss is insufficient. Let ℓ_c be a critical boundary-loss threshold. Write $A_d = \text{admit}(d)$ and $\bar{A}_d = \neg \text{admit}(d)$. Define

$$\Delta_d E\ell = \mathbb{E}[\ell_{\text{maint}, t+k} \mid M_t, \text{do}(\bar{A}_d)] - \mathbb{E}[\ell_{\text{maint}, t+k} \mid M_t, \text{do}(A_d)], \quad (15)$$

$$\Delta_d P_c = P(\ell_{\text{maint}, t+k} > \ell_c \mid M_t, \text{do}(\bar{A}_d)) - P(\ell_{\text{maint}, t+k} > \ell_c \mid M_t, \text{do}(A_d)). \quad (16)$$

Risk-weighted attention value is then

$$V_t^{\text{risk-att}}(d; k) = \Delta_d E\ell + \alpha_t \Delta_d P_c - \lambda_t c_t(d). \quad (17)$$

The first term is expected loss reduction. The second is collapse-risk reduction. Both are positive when admission reduces risk.

Proposition 2 (Verification-limited attention). *A finite system may fail to attend to a distinction not because it cannot detect the distinction, but because it cannot verify or maintain it within the update window.*

Proof sketch. In a hard-admission system, if $c_t(d) > C_{\text{att}}(t)$, no feasible allocation can admit d while satisfying Eq. (8). In a soft-admission system, d may receive a fractional weight, but it cannot become maintained attention if the maximum feasible weight $C_{\text{att}}(t)/c_t(d)$ falls below the minimum weight $a_{\min}$ required for downstream update. Thus detection does not entail maintained attention. $\square$

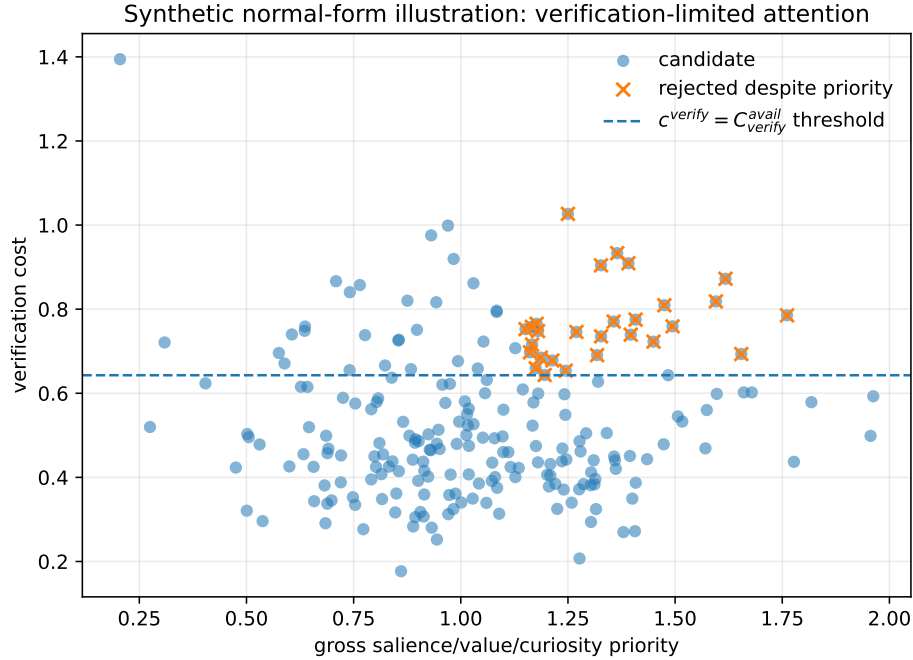


Figure 4: Synthetic normal-form illustration, not empirical evidence. Some distinctions have high gross priority but high verification cost. Candidates above the verification threshold cannot be admitted as maintained attention even when salience or gross priority is high.

5.1 Energy-time tradeoff and verification capacity

For short update windows, verification capacity can collapse even when raw detection capacity remains high. Fast attention therefore favors low-cost, high-priority admission rules and is more vulnerable to salience capture, stereotype-like compression, and risk-weighted shortcuts. A minimal bound is

$$C_{\text{verify}}^{(\tau)} \leq R_{\text{verify}}\tau, \quad N_{\text{checks}}(\tau) \leq \frac{\tau}{\tau_{\text{check}}},$$

where τ_{check} is the minimum time per verification operation. When τ is small, verification capacity collapses and attention must rely on cheaper proxies. This connects M1 to the update-window constraints of O2 [5] and the O3 finite-memory operational channel [6]: rapid attention in short windows is not a cognitive flaw but a budget collapse.

6 Nonlinear attention gates, tunnel vision, and hysteresis

Attention gates often become nonlinear under load. Let $n_t(d)$ denote novelty, $r_t(d)$ threat or collapse-risk relevance, $I_t^{\text{gain}}(d)$ information gain, and b_t a threshold. Define a gate score

$$V_t^{\text{gate}}(d; k) = w_1 V_t^{\text{att}}(d; k) + w_2 V_t^{\text{risk-att}}(d; k) + w_3 I_t^{\text{gain}}(d). \quad (18)$$

A normal-form gate is

$$a_t(d) = \sigma \left(\beta_t \left[V_t^{\text{gate}}(d; k) + \alpha n_t(d) + \theta r_t(d) - \gamma c_t(d) - w_4 P_t^{\text{false}}(d) C_t^{\text{repair}}(d) - b_t \right] \right), \quad (19)$$

where $P_t^{\text{false}}(d)$ is the probability that an admitted distinction is later judged false, misleading, or non-actionable, and $C_t^{\text{repair}}(d)$ is the cost of repairing, retracting, or recovering from a false admission. False admission can be more expensive than non-admission because the system may need to maintain, propagate, retract, or repair an erroneous distinction. In social or institutional systems this repair cost may grow superlinearly with propagation depth. The gate steepness is

$$\beta_t = \beta_0 + \beta_{\Delta} [\Delta_{\text{sem}}(t)]_+. \quad (20)$$

As semantic or attention deficit rises, the gate steepens. This produces tunnel vision: fewer distinctions are admitted, and admitted distinctions cluster around high immediate value, threat, or novelty. Long-horizon structure can be excluded even when it is important. This is the attentional analogue of capacity overflow and effective stochasticity in finite representation systems [4].

Proposition 3 (Load-induced attention narrowing). *If gate steepness β_t increases with semantic deficit, then the admission probability becomes more threshold-like and the admitted set narrows around high-score distinctions.*

Proof sketch. For logistic $\sigma(\beta x)$, the derivative at the threshold is $\beta/4$ and the transition width scales as $1/\beta$. Increasing β_t therefore decreases the range of intermediate admission probabilities and produces a sharper split between admitted and rejected distinctions. $\square$

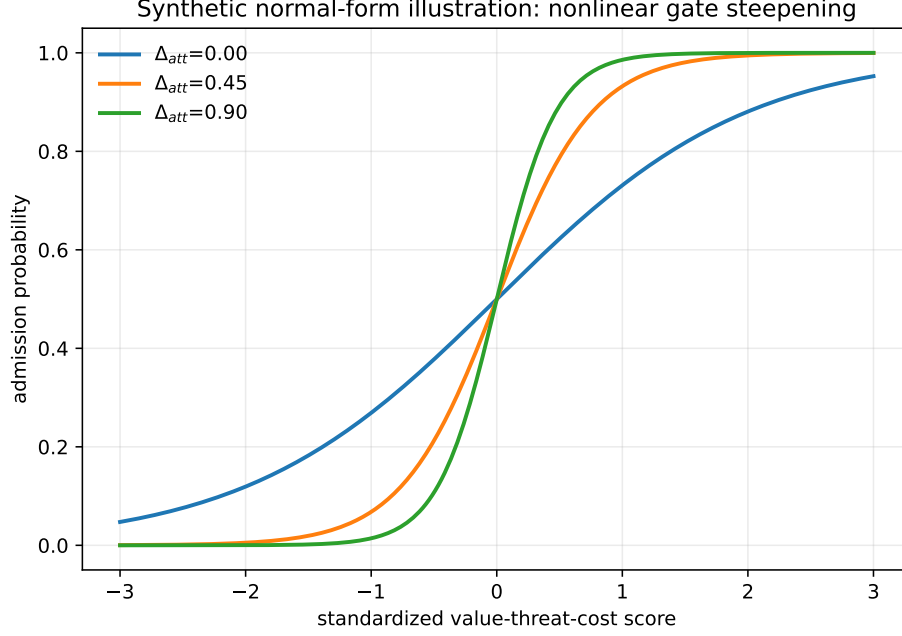


Figure 5: Synthetic normal-form illustration, not empirical evidence. As semantic or attention deficit increases, the admission gate steepens. Intermediate distinctions are excluded and the system enters a tunnel-vision regime.

6.1 Recovery delay and attention hysteresis

Tunnel vision can be useful when immediate boundary threat dominates; it becomes harmful when it blocks the distinctions needed for recovery. FDS-M1 treats recovery as an exit problem. The system can broaden attention by increasing C_{att} , reducing verification load, pruning obsolete candidates, compressing repeated distinctions, externalizing records, or relaxing the task.

Recovery need not be immediate. Crisis load can change priors, routines, verification checklists, or institutional thresholds. A minimal hysteresis model is

$$\beta_{t+1} = \beta_t + \rho(\beta_{\text{target}}(\Delta_{\text{sem}}(t)) + h_t - \beta_t), \quad h_{t+1} = \chi h_t + \zeta \mathbf{1}_{\Delta_{\text{sem}}(t) > \Delta_c}, \quad (21)$$

where h_t is a gate-locking residue and $0 < \chi < 1$. Even after Δ_{sem} falls, h_t can keep the gate narrow for a recovery interval.

Proposition 4 (Exit-induced broadening with possible hysteresis). *Under Eq. (19), an intervention that lowers Δ_{sem} , lowers $c_t(d)$, or increases C_{att} weakly broadens intermediate admission unless offset by threshold increase or hysteretic lock-in. Under Eq. (21), broadening can lag behind load reduction.*

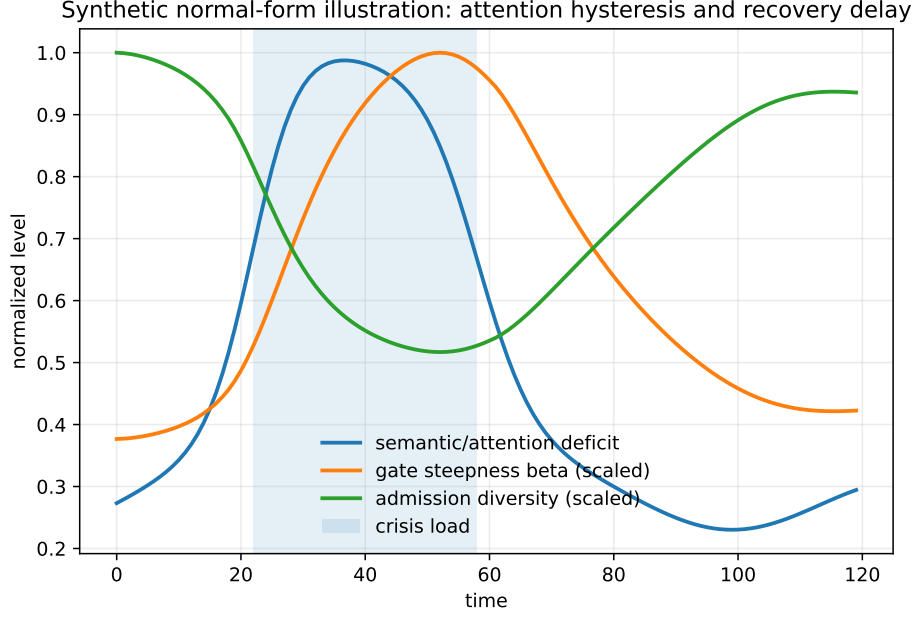


Figure 6: Synthetic normal-form illustration, not empirical evidence. During crisis load, gate steepness rises and admission diversity falls. After load reduction, attention can recover with a delay because gate thresholds, verification routines, or threat priors relax slowly.

7 Attention failure modes

FDS-M1 classifies attention failures as admission failures under finite capacity. This does not replace clinical, cognitive, or neural taxonomies; it provides an operational bridge.

Definition 8 (Attention overload). Attention overload occurs when candidate distinction demand exceeds admission, verification, or maintenance capacity such that task-relevant distinctions are excluded.

Definition 9 (Distraction). Distraction occurs when low boundary-relevance distinctions are admitted at the expense of higher boundary-relevance distinctions.

Definition 10 (Salience capture). Salience capture occurs when raw salience dominates admission despite low causal boundary value.

Definition 11 (Suppression). Suppression occurs when a candidate distinction is blocked despite positive boundary-gradient value, often because it threatens another maintained goal, identity, institution, or proxy objective.

Definition 12 (Critical distinction exclusion). Critical distinction exclusion occurs when a distinction needed to prevent boundary failure is not admitted before the relevant update window closes.

Definition 13 (False admission). False admission occurs when a low-quality, adversarial, unverifiable, or obsolete distinction enters the update channel and consumes capacity or corrupts downstream inference.

Definition 14 (Attention recovery lag). Attention recovery lag occurs when a previously narrowed gate remains narrow after external load or threat has fallen, because verification routines, threat priors, or institutional thresholds relax more slowly than the triggering deficit.

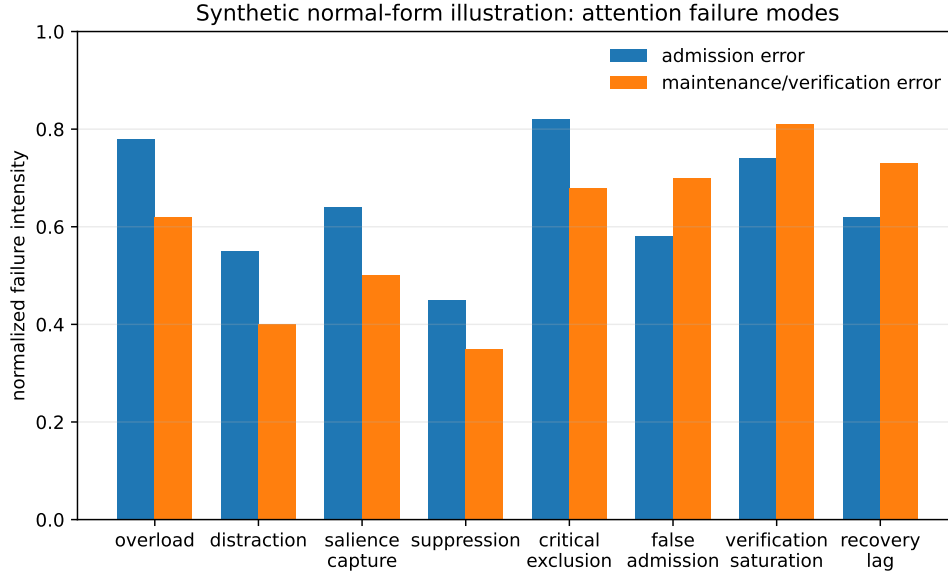


Figure 7: Synthetic normal-form illustration, not empirical evidence. Attention failures can be represented as admission errors, maintenance/verification errors, and recovery-lag errors under finite capacity.

8 Attention across domains

8.1 Biological and cognitive attention

In biological and cognitive systems, attention includes orienting, selection, working-memory gating, task-set maintenance, and action preparation. FDS-M1 does not choose among neural mechanisms. It supplies a shared accounting: a candidate distinction is attended only when it enters a maintained channel that can condition update or action. This connects limited-capacity attention to boundary relevance without claiming that all attention is consciously reportable.

8.2 Machine attention and transformer mechanisms

Attention weights in machine learning systems are not automatically FDS attention. Transformer attention is internal parallel routing among representations. FDS attention is budgeted admission into durable, update-relevant channels under a stated boundary, loss, and verification regime. A bare inference call may implement useful routing while lacking durable self-updating boundary state. A tool-using system with writable memory may satisfy stronger coupled-attention criteria if admitted distinctions affect future update and verification.

Dimension	Transformer attention	FDS attention
Basic role	token-to-token routing or weighting	distinction admission into update
Budget	compute/context constrained	encoding, verification, maintenance, latency, and boundary budget
Boundary	often implicit or absent	must be specified
Verification	usually not intrinsic	verification path/status is central
Agency attribution	not implied by weights	belongs to a coupled system if admitted records affect future loss

One FDS-compatible interpretation is that hallucination risk increases when many signals are

routed or retrieved without a sufficient verification channel tied to a boundary-maintenance task. This is not a complete theory of hallucination; it is an admission-and-verification failure mode.

8.3 Organizational attention

Organizations attend through agendas, dashboards, meetings, metrics, issue queues, audits, and formal reporting channels. An issue may be known locally yet fail organizational attention if it does not enter a maintained decision channel. Organizational tunnel vision occurs when crisis, bureaucracy, or incentive misalignment steepens admission thresholds so that only short-horizon or proxy-relevant distinctions enter decision-making.

8.4 Civilization-scale attention

Civilization-scale attention is shared admission under finite communication, archival, scientific, legal, journalistic, and institutional verification capacity. Claims become collectively attended when they enter public records, standards, policy processes, scientific literature, or other externalized memory. Epistemic pollution is a saturation attack on finite verification bandwidth by low-cost, high-salience, low-reliability distinctions.

Let $R_{\text{verify}}^{(\tau)}(\epsilon; t)$ be the verification demand imposed by the shared distinction environment and $C_{\text{verify}}^{\text{avail}}(t)$ the available civilization-scale verification capacity. Define

$$Z_{\text{att/poll}}(t) = \frac{R_{\text{verify}}^{(\tau)}(\epsilon; t)}{C_{\text{verify}}^{\text{avail}}(t)}. \quad (22)$$

When $Z_{\text{att/poll}} > 1$, the system cannot verify all candidate claims. It must ignore, trust blindly, outsource verification, false-compress, or narrow the collective admission gate.

Proposition 5 (Collective attention as shared admission). *Collective attention is shared distinction admission under finite communication, verification, and externalized memory capacity. It fails when the shared gate admits low-quality distinctions or excludes boundary-critical distinctions at scale.*

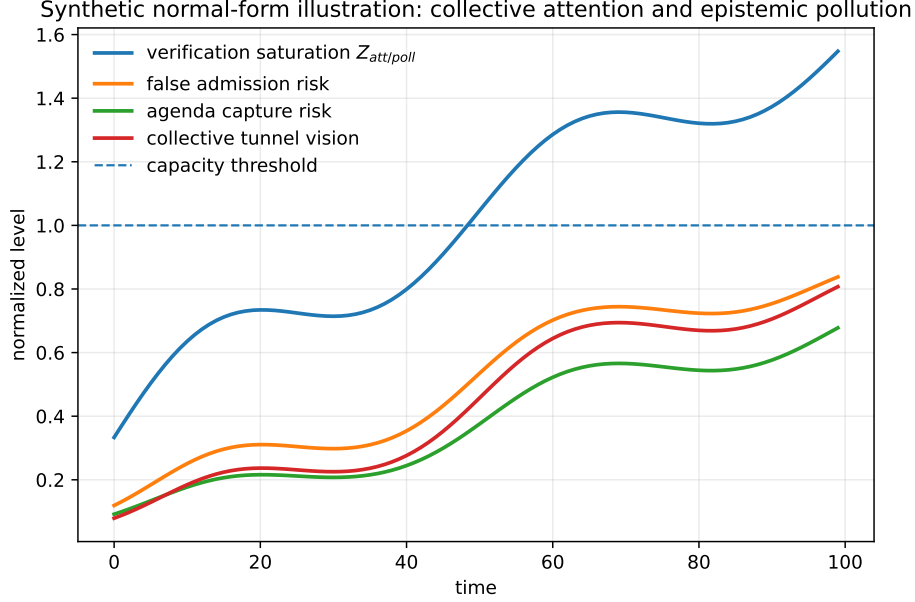


Figure 8: Synthetic normal-form illustration, not empirical evidence. Collective attention fails when verification demand saturates available verification capacity, increasing false admission, agenda-capture risk, and collective tunnel vision.

8.5 Domain mapping examples

Table 2: Illustrative domain mappings. Each row requires its own bridge assumptions and audit protocol.

Domain	Candidate distinction	Gate	Capacity	Failure
Cell	chemical gradient	receptor/regulatory gate	metabolic/regulatory bandwidth	missed nutrient or toxin
Human operator	alarm/anomaly	task set and working memory	attention and verification time	tunnel vision
LLM agent	tool output or memory record	context/memory admission	tokens and verification	hallucinated continuation
Organization	issue or warning	agenda/dashboard/audit	meeting and audit bandwidth	ignored risk
Civilization	claim or discovery	science/media/legal archive	verification institutions	epistemic pollution

9 Normal-form model and reproducibility

The accompanying code implements deterministic synthetic normal-form illustrations. The model is not empirical evidence. It is a consistency and visualization device that maps the definitions to simple state variables.

The model uses a candidate distinction stream with salience, novelty, threat, long-horizon relevance, source uncertainty, model divergence, verification cost, maintenance cost, encoding cost, causal boundary value, collapse-risk reduction, and information gain. It implements greedy capacity-limited admission, a reserved background-scanning budget, salience-value dissociation, capacity sweeps, verification-limited rejection, nonlinear deficit-dependent sigmoid gates, attention failure-mode intensities, hysteretic attention recovery, and collective attention saturation.

The random seed is fixed in `code/generate_results.py`. CSV outputs are stored in `data/`; figure pairs are stored in `figures/`.

Table 3: Normal-form variable map. All entries are illustrative, not fitted empirical quantities.

Simulation variable	Paper definition	Interpretation
salience	raw candidate prominence	detection strength, not attention itself
causal value	Eq. (14)	expected boundary-loss reduction minus admission cost
information gain	Eq. (11)	exploratory value of model-updating distinctions
verification cost	Eq. (12)	model incompatibility plus source and cross-check burden
semantic deficit	Eq. (20)	task demand pressure that steepens the gate
admission probability	Eq. (19)	soft gate output
recovery lag	Eq. (21)	delayed relaxation of attention narrowing
verification saturation	Eq. (22)	shared demand/capacity ratio

10 Protocols and tests

Protocol 1 (Admission audit). Pre-register candidate distinctions, raw salience, costs, predicted boundary relevance, information gain, and admission outcomes. Test whether admitted distinctions are better predicted by causal boundary value and exploration value than by salience alone.

Protocol 2 (Salience-value dissociation test). Construct cases where salience and boundary relevance diverge. M1 predicts that boundary-efficient attention should sometimes admit lower-salience but higher boundary-relevance distinctions.

Protocol 3 (Capacity load test). Increase candidate distinction load or reduce admission capacity. Prediction: admission narrows, low-priority distinctions are excluded, and the admission gate becomes more thresholded.

Protocol 4 (Verification-burden test). Increase verification cost while holding salience fixed. Prediction: unverifiable distinctions are rejected, delayed, externalized, or accepted blindly only under failure conditions.

Protocol 5 (Exploration-budget test). Reduce or eliminate background scanning. Prediction: short-run performance may improve under stable conditions, but weak early signals, anomalies, and model-breaking distinctions are missed more often. Audit outputs include delayed anomaly detection rate, model-update surprise after missed scanning, and long-horizon boundary loss following short-horizon admission gains.

Protocol 6 (Tunnel-vision and hysteresis test). Induce semantic deficit, crisis load, or threat pressure. Prediction: attention becomes steeply thresholded, high-threat distinctions dominate, and recovery of broad admission lags behind external load reduction when gate hysteresis is present.

Protocol 7 (Collective attention audit). Measure how groups allocate shared attention under limited verification and communication capacity. Prediction: epistemic pollution increases false admission, agenda capture, false compression, or collective tunnel vision when $Z_{\text{att/poll}} > 1$.

11 Relation to existing fields

Selective attention and limited capacity. Classical attention research already treats attention as selective and capacity-limited, including bottleneck, filter, and resource perspectives [7–9]. Recent reviews emphasize that selective processing spans multiple forms of attention, species, and levels of analysis [26]. FDS-M1 does not replace these accounts. It supplies a cross-domain finite-system interpretation: selection is admission into update.

Biased competition and priority maps. Neural and cognitive accounts often describe attention as competition among representations, biased by task goals and stimulus strength [10–12]. M1 is compatible with this: priority is interpreted as boundary-relevance weighted by cost and capacity.

Inattentional blindness and working memory. Inattentional blindness and working-memory limits show that detection and report are constrained by limited admission and maintenance [13–15]. Working-memory gating models similarly emphasize that information must pass controlled update gates to affect maintained state [16]. M1 formalizes this as distinction admission with maintenance and verification cost.

Rational inattention and information bottleneck. Rational inattention treats information acquisition as costly [17, 30]. The information bottleneck formalizes compression that preserves task-relevant information [18]. M1 uses both as nearby mathematical idioms, while tying admission to boundary-maintenance loss and update relevance.

Active inference and active sensing. Active inference and active sensing emphasize perception-action loops under uncertainty [19]. M1 is compatible with these approaches but emphasizes finite distinction capacity, verification burden, and the distinction between detection and admission.

Transformer attention. Transformer attention is a powerful computational selection and routing mechanism [20], but long context is not equivalent to robust admission: models can fail to use relevant information depending on its position in context [27]. Recent work on differential attention treats over-allocation to irrelevant context as an architectural failure mode [28]. M1 does not identify transformer attention with full FDS attention. The latter requires an accounting boundary, update relevance, and maintenance, verification, or downstream action effect. Recent analyses show that transformer mechanisms can acquire gating-like operations on working-memory tasks, but this does not by itself establish FDS attention unless the accounting boundary, durable update, and verification path are specified [29].

Organizational attention, situation awareness, and agenda-setting. Organizations and publics allocate attention through agenda formation, issue channels, and institutional filtering [21–23]. Situation awareness work analyzes how operators perceive, comprehend, and project task-relevant environmental elements [24]. Agenda-setting theory studies how media and public institutions shape issue salience at collective scales [25]. Credibility cues and social norms can improve truth discernment and reduce engagement with false content in social-media simulations, suggesting one empirical interface for collective verification capacity [31]. FDS-M1 interprets these as collective distinction admission under finite verification and communication capacity.

12 Limitations and falsification

M1 is intentionally limited. It does not establish a general empirical theory of attention by definition. It provides mappings that must survive operational audit. The framework is weakened or demoted under any of the following results:

1. attention-like selection under a specified mapping shows no finite capacity, admission, update gating, or priority constraint;
2. admission is always explained by raw salience and never by cost, verification, task relevance, exploration value, or boundary-gradient value;
3. increasing load produces no narrowing, thresholding, exclusion, or priority collapse in systems claimed to have finite attention;
4. distinctions can be detected, verified, maintained, updated, and acted upon without any capacity or resource tradeoff;
5. artificial attention weights alone satisfy FDS attention without an accounting boundary, downstream update relevance, or durable maintained record;
6. collective attention shows no relation to communication capacity, verification bandwidth, agenda channels, or externalized memory;
7. claimed attention hysteresis disappears under controlled recovery tests where load reduction should reveal persistent gate-locking.

13 Conclusion

Attention is the first active step in the agency-semantics chain. A distinction becomes operationally relevant only if it passes from availability into an update channel. M1 defines attention as capacity-limited distinction admission and separates it from raw salience, detection, and report. Its central chain is

$$\begin{aligned} \text{candidate distinction} &\rightarrow \text{admission gate} \rightarrow \text{verification path/status} \rightarrow \text{maintained record} \\ &\rightarrow \text{downstream update/action} \rightarrow \text{future boundary loss.} \end{aligned} \quad (23)$$

The paper introduces a finite-system vocabulary for salience-value dissociation, capacity-limited admission, background scanning, verification burden, tunnel vision, recovery hysteresis, machine-attention attribution, organizational attention, and collective attention under epistemic pollution.

M1 prepares later work. M2 can treat admitted distinctions as inputs to value and goal ranking. M3 can treat attention as the entry condition for actionable semantic quotients. M5 can analyze trust as delegated verification that reduces repeated admission cost. A2 can audit whether artificial systems externalize attention and verification to hosts. S2 can analyze epistemic pollution as saturation of civilization-scale attention and verification bandwidth.

Code and data availability

The deterministic synthetic normal-form code, figures, and CSV outputs are included in the replication package. Run `python code/generate_results.py` from the paper directory to regenerate all figures and data.

AI assistance disclosure

The author used AI assistance for drafting, editing, simulation scaffolding, and consistency checks. The author reviewed and selected the final claims, definitions, equations, and interpretations.

A M-series dependency map

Table 4: How M1 supports later M-series and civilization-layer papers.

Future paper	Interface supplied by M1
M2 Value/Goal	admitted distinctions become inputs to causal boundary-gradient ranking
M3 Meaning	attention supplies the entry condition for actionable semantic quotients
M5 Trust	trust reduces repeated verification cost and changes admission probability
A2 AI Alignment	artificial systems must state whether attention belongs to model, tool, memory, user, or coupled system
S2 Epistemic Pollution	pollution saturates shared verification bandwidth and distorts collective admission
G3 Science	scientific method functions as collective claim admission, verification, and error correction
G4 Civilization Memory	archives and standards stabilize externalized attention over time

B Notation summary

Table 5: Notation used in M1.

Symbol	Meaning
$\mathcal{D}_t$	candidate distinction stream at update window t
d	candidate distinction
$a_t(d)$	admission weight for distinction d
$c_t(d)$	scalarized encoding, verification, maintenance, and opportunity cost
$c_t^r(d)$	resource-specific vector cost
$C_{\text{att}}(t)$	attention admission capacity
$C_{\text{scan}}(t)$	reserved background-scanning capacity
$C_{\text{verify}}(t)$	verification capacity
$V_t^{\text{pred-att}}$	predictive attention relevance
V_t^{att}	causal boundary-gradient attention value
$V_t^{\text{risk-att}}$	risk-weighted attention value near collapse thresholds
V_t^{gate}	combined gate score used by the nonlinear admission gate
$I_t^{\text{gain}}(d)$	information gain / exploratory value of candidate d
$\Delta_{\text{sem}}(t)$	semantic capacity deficit
ℓ_{maint}	boundary-maintenance loss
Σ_{phys}	physical entropy-production ledger, distinct from ℓ_{maint}
$Z_{\text{att/poll}}$	collective verification saturation ratio
h_t	gate-locking residue / attention hysteresis state
χ	hysteresis persistence / decay factor, $0 < \chi < 1$
ζ	gate-locking accumulation increment under severe deficit
Δ_c	semantic deficit threshold above which gate locking accumulates
β_t	attention gate steepness
β_{target}	deficit-dependent target gate steepness

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Value and Goal as Boundary-Relevance Ranking

Causal Boundary Gradients, Goal Stability, and Value Drift under Finite Capacity

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Abstract

This paper develops the value-goal layer of the agency-semantics spine of Distinction Theory. Building on the formal core of Active Finite Distinction Systems, the M0 agency-semantics spine, and the M1 model of attention as distinction admission, it treats value as *FDS-value*: causal boundary-gradient relevance under finite capacity, rather than moral value, subjective preference, or intrinsic worth. An evaluand is FDS-valuable for a system only relative to a specified boundary, loss function, intervention grammar, horizon, action or update channel, and resource budget. Its gross causal boundary gain is the finite-difference reduction in expected future boundary-maintenance loss; its net FDS-value subtracts scalarized evaluation, verification, action, maintenance, or opportunity cost; and its risk-weighted value adds collapse-risk reduction near critical thresholds. A goal is a stabilized FDS-value ranking coupled to a policy orientation across finite update windows. The paper distinguishes salience, attention, reward, preference, FDS-value, goal, commitment, and moral value; introduces predictive-causal dissociation, gross/net value accounting, risk-weighted ranking, goal-stability indices, proxy-boundary divergence, value drift, multi-goal Pareto conflict, and goal hysteresis; and provides audit protocols for biological, cognitive, artificial, organizational, and collective systems. A deterministic synthetic normal-form model illustrates predictive-causal dissociation, risk-dominant ranking, goal stabilization, proxy reward hacking, second-order evaluation deficit, Pareto conflict, and recovery lag. The paper does not derive ethics from boundary maintenance or claim that all value reduces to survival. It supplies a conservative finite-system bridge for analyzing value and goal as boundary-relevance ranking under capacity, verification, and resource constraints.

Keywords: FDS-value; operational value; goal; boundary relevance; causal value; finite capacity; boundary maintenance; value ranking; goal stability; reward hacking; value drift; proxy alignment; active finite distinction systems; decision theory; reinforcement learning; AI alignment; social choice.

Epistemic Notice and Scope

This manuscript is an agency-semantics spine paper, not a completed theory of value, ethics, decision-making, motivation, reward learning, or social choice. It does not claim that moral value reduces to boundary maintenance, that biological reward equals value, that all preferences are survival proxies, or that current AI reward functions faithfully represent value. Its claim is narrower: once a system is modeled as an active finite distinction system with a boundary, memory, update rule, action space, finite capacity, and boundary-maintenance loss, value and goal acquire operational roles. In this paper, “value” means *FDS-value* or *operational boundary value*: causal relevance to a specified boundary-maintenance loss under a specified model, horizon, cost accounting, and action/update channel. Goal means stabilized value ranking coupled to policy across finite update windows.

The paper follows the layered discipline of the FDS formal core. Formal definitions, normal-form models, physical bridge assumptions, and domain applications are separated. A failure of a cognitive, artificial, biological, organizational, or social mapping may demote that mapping without refuting the formal FDS core [1]. The deterministic numerical model is illustrative only. It visualizes definitions and failure modes; it is not empirical evidence.

1 Introduction

A finite system cannot treat every admitted distinction as equally important. M1 defined attention as capacity-limited admission of candidate distinctions into an update channel [3]. M2 asks what happens after admission. Once a distinction, action, state, record, policy, or proxy enters the update-relevant channel, the system must rank its relevance: which distinctions reduce future boundary loss, which actions worsen it, which proxy signals mislead, and which rankings remain stable enough to orient behavior over time?

The FDS core defines an active finite distinction system as a tuple

$$S = (X, E, B, M, Y, A, U, \pi, \ell, \Phi, \mathcal{P}, \tau), \quad (1)$$

where X is internal state, E environment, B boundary, M memory/model space, Y observation channel, A action space, U update map, π finite projection, ℓ boundary-maintenance loss, Φ resource budget, $\mathcal{P}$ perturbation/pruning family, and τ update timescale [1]. M0 introduced an agency-semantics dependency skeleton,

$$\text{distinction} \rightarrow \text{record} \rightarrow \text{attention} \rightarrow \text{value} \rightarrow \text{goal} \rightarrow \text{meaning} \rightarrow \text{agency} \rightarrow \text{culture}, \quad (2)$$

and defined value as causal boundary-gradient relevance and goal as stabilized value ranking [2]. M2 expands that value-goal step.

1.1 Why value needs a finite-system formulation

The word value is overloaded. It can mean utility, reward, preference, price, importance, survival relevance, moral worth, social status, expected return, or subjective desirability. M2 does not try to unify all of these into one metaphysical substance. Instead it defines a finite-system bridge. Given a specified boundary, loss function, intervention grammar, horizon, and capacity budget, an evaluand has FDS-value when selecting, admitting, maintaining, or using it causally changes future boundary-maintenance loss.

This definition is deliberately relational. A distinction may be FDS-valuable for one system and irrelevant for another. A proxy may be FDS-valuable under one horizon and harmful under a longer horizon. A goal may be adaptive during crisis and rigid after the crisis ends. Value in the M2 sense is therefore not an intrinsic tag attached to a signal. It is a boundary-relative and horizon-relative finite-difference effect.

1.2 Goal is not merely preference

A high-valued action is not yet a goal. A preference may be momentary. A reflex may be high-value in a narrow state but fail to persist. A goal requires stabilization: a ranking over evaluands or policies must survive across update windows enough to orient action. M2 therefore treats goals as register-time objects: their identity depends on maintained rankings, records, and policies over finite windows, not on an instantaneous score [4].

1.3 Common misreadings

Table 1: Common misreadings and corrections.

Misreading	Correction
Value means moral value.	No. M2 defines FDS-value: operational boundary value under a specified boundary, loss, horizon, intervention, and cost model.
Goal means conscious intention.	No. A goal is a stabilized ranking coupled to policy orientation; it can be implemented without reportable intention.
Reward is value.	No. Reward is a proxy signal that can approximate or diverge from causal boundary effect.
Boundary maintenance implies ethics.	No. Boundary maintenance is descriptive and model-relative; it does not confer moral legitimacy.
Stable goal means good goal.	No. Stability can become rigidity, trauma-like persistence, institutional lock-in, or maladaptive commitment.

Table 2: Claim-status summary for M2. The table is an audit device: several entries are formal or operational bridge claims, not established empirical results.

Claim ID	Tier	Claim	Failure or demotion condition
M2-001	Formal bridge	FDS-value is causal boundary-gradient relevance under a specified boundary, loss, intervention grammar, horizon, and cost model.	Valuation cannot be operationalized as causal effect on any specified future boundary-maintenance loss under valid mappings.
M2-002	Operational bridge	Predictive relevance and causal FDS-value are separable.	Correlational predictors always coincide with intervention-relevant boundary effects under audited systems.
M2-003	Formal / model bridge	Value ranking can be expressed as an ordering over finite-difference action, admission, maintenance, or policy effects.	No useful ordering exists between evaluands and their causal boundary effects under stated mappings.
M2-004	Operational bridge	Near collapse thresholds, risk-weighted FDS-value can dominate average-loss value.	Collapse-risk reduction never changes ranking near boundary failure thresholds under valid mappings.
M2-005	Operational bridge	Goals are stabilized FDS-value rankings coupled to policy orientation across update windows.	Goal-like behavior persists without ranking stability, memory, policy orientation, or update-window persistence.
M2-006	Failure-mode bridge	Value drift occurs when rankings change faster than the system can verify, update, or maintain the reasons for the change.	Ranking instability produces no detectable change in behavior, loss, or policy under claimed goal systems.
M2-007	AI / agency bridge	Proxy reward can diverge from causal boundary value, creating reward hacking or misalignment.	Proxy optimization remains aligned despite divergent finite-difference effects on host boundary loss.
M2-008	Social bridge	Collective goals are shared stabilized rankings under finite verification and coordination capacity.	Group goals show no relation to shared rankings, institutional memory, verification capacity, or policy orientation.
M2-009	Recovery bridge	Goal recovery can lag after resource or threat recovery because rankings, commitments, or threat priors persist.	Goals relax immediately and without lag after boundary load changes in systems where goal hysteresis is claimed.

2 FDS and M-series background

2.1 Active boundary relevance

The FDS core applies its deficit logic only to active-boundary systems. A minimal relevance screen is

$$P(U(M_t, Y_t) \neq M_t) > 0, \quad I(M_{t+1}; \ell_{t+k}) > 0 \quad (3)$$

for some $k > 0$. Empirical applications should strengthen this to an intervention or ablation test,

$$\mathbb{E}[\ell_{t+k} \mid \text{do}(U)] \neq \mathbb{E}[\ell_{t+k} \mid \text{do}(U_\emptyset)], \quad (4)$$

where $U_\emptyset$ is a null, frozen, randomized, or identity update [1]. M2 inherits this discipline: an item is FDS-valuable only if selecting, admitting, maintaining, or acting on it can affect a future boundary-relevant variable under the specified mapping.

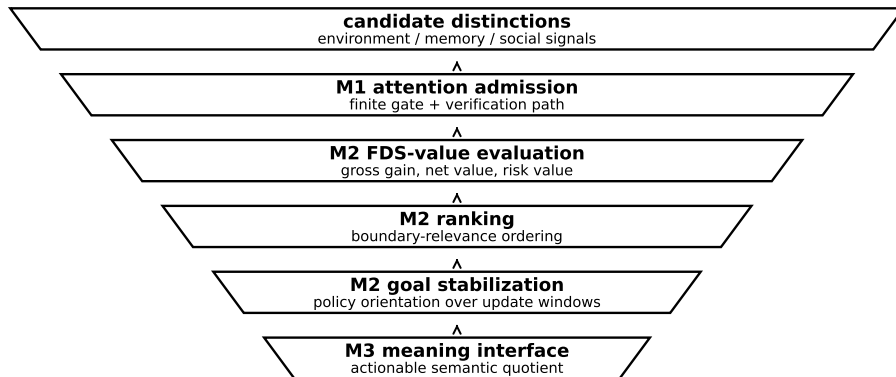
2.2 Attention as the input condition

M2 evaluates only items that are available to the system’s update channel. M1 distinguished availability, detection, and attention; attention occurs when a candidate distinction is admitted, made recoverable or maintained, assigned a verification path or status, and made available for downstream update or action [3]. Thus M2 does not rank all possible environmental differences. It ranks evaluands that enter, or could enter, the system’s finite update-relevant channel.

2.3 Notation alignment

Throughout the paper ℓ_{maint} denotes boundary-maintenance loss. It is not moral loss and not physical entropy production. The physical entropy ledger, when relevant, is denoted Σ_{phys} . Maintaining low ℓ_{maint} may require physical resource cost under bridge assumptions, but $\ell_{\text{maint}} \neq \Sigma_{\text{phys}}$. Similarly, R_{proxy} denotes an instrumental or learned reward signal; it is not automatically equal to FDS-value.

3 From candidate distinctions to stabilized goals



Synthetic normal-form illustration: from environmental differences to stabilized goals and meaning interfaces.

Figure 1: Synthetic normal-form illustration, not empirical evidence. M2 sits between M1 attention admission and M3 meaning. Candidate distinctions are filtered by admission, evaluated for FDS-value, ranked, risk-weighted, and stabilized into policy orientation.

The value-goal layer is not a free-standing theory of desire. It is a filtering and ranking layer. Figure 1 summarizes the pipeline: environmental or internal differences must first become candidate distinctions, pass attention admission, be evaluated for causal boundary effect, be ranked under cost and risk constraints, and only then become stabilized goal orientations. Meaning in M3 will depend on whether these ranked distinctions can be compressed without losing action-relevant structure.

4 Definition: FDS-value as causal boundary-gradient relevance

Definition 1 (Evaluand). An evaluand is any admitted distinction, action, state, policy, record, proxy, quotient, or externalized symbol whose admission, selection, maintenance, or use may affect future boundary-maintenance loss. The set of evaluands available during an update window is denoted $\mathcal{Z}_t$.

Table 3: Intervention grammar for evaluands. The symbol $\text{do}(z)$ must be interpreted relative to the type of evaluand being audited.

Evaluand type	Intervention meaning
Distinction d	Force admission or rejection of d into the update channel.
Action a	Execute a rather than baseline action a_0 .
Record m	Maintain, delete, refresh, externalize, or ignore record m .
Policy π	Deploy π instead of baseline policy π_0 .
Proxy R	Condition downstream update or optimization on proxy signal R .
Symbol or quotient $q(d)$	Use a compressed representation in downstream action, prediction, verification, or coordination.
Institutional rule	Adopt, enforce, suspend, or revise a rule within a stated institutional boundary.

Definition 2 (Predictive relevance). An evaluand z has predictive relevance over horizon k if conditioning on it improves prediction of future boundary-maintenance loss. Gross predictive relevance is the observational association before cost:

$$R_t^{\text{predgross}}(z; k) = \mathbb{E}[\ell_{\text{maint}, t+k} \mid M_t] - \mathbb{E}[\ell_{\text{maint}, t+k} \mid M_t, z], \quad (5)$$

and net predictive relevance subtracts scalarized cost:

$$R_t^{\text{prednet}}(z; k) = R_t^{\text{predgross}}(z; k) - \lambda_t c_t(z). \quad (6)$$

Predictive relevance is an observational proxy. It can be useful for forecasting but can mislead when the evaluand is a marker rather than a lever.

Definition 3 (Gross causal boundary gain). Let z_0 be a specified baseline, such as non-admission, null action, frozen update, default policy, or no-maintenance. The gross causal boundary gain of z is

$$G_t^{\text{FDS}}(z; k) = \mathbb{E}[\ell_{\text{maint}, t+k} \mid M_t, \text{do}(z_0)] - \mathbb{E}[\ell_{\text{maint}, t+k} \mid M_t, \text{do}(z)]. \quad (7)$$

Definition 4 (Net FDS-value). The net FDS-value of z is

$$V_t^{\text{net}}(z; k) = G_t^{\text{FDS}}(z; k) - \lambda_t c_t(z), \quad (8)$$

where $c_t(z)$ is scalarized evaluation, verification, action, maintenance, latency, or opportunity cost, and λ_t converts that cost into boundary-loss-equivalent units.

Separating G^{FDS} from V^{net} prevents double-counting. Gross gain measures boundary-loss reduction before cost. Net FDS-value subtracts cost. Allocation heuristics can then use either gross density or net density depending on the audit.

Definition 5 (Value density). For $\epsilon_c > 0$, define

$$\rho_t^{\text{gross}}(z; k) = \frac{G_t^{\text{FDS}}(z; k)}{c_t(z) + \epsilon_c}, \quad \rho_t^{\text{net}}(z; k) = \frac{V_t^{\text{net}}(z; k)}{c_t(z) + \epsilon_c}. \quad (9)$$

These are normal-form audit metrics, not universal decision rules. In hard allocation problems, greedy density rules need not be globally optimal. In fractional or heuristic settings, they reveal how finite systems trade expected boundary gain against evaluation and maintenance cost.

Proposition 1 (Predictive relevance is not causal FDS-value). *A variable can predict future boundary loss without being FDS-valuable in the causal sense if selecting, admitting, maintaining, or acting on it does not change future boundary-maintenance loss under the system’s update channel.*

Proof sketch. Suppose z is correlated with an unobserved cause u that affects $\ell_{\text{maint}, t+k}$, so $R_t^{\text{pred gross}}(z; k) > 0$. If intervention on z leaves the causal variables and downstream update policy unchanged, then $\mathbb{E}[\ell_{\text{maint}, t+k} \mid \text{do}(z)] = \mathbb{E}[\ell_{\text{maint}, t+k} \mid \text{do}(z_0)]$ up to cost, and Eq. (8) is nonpositive. Prediction is not intervention. $\square$

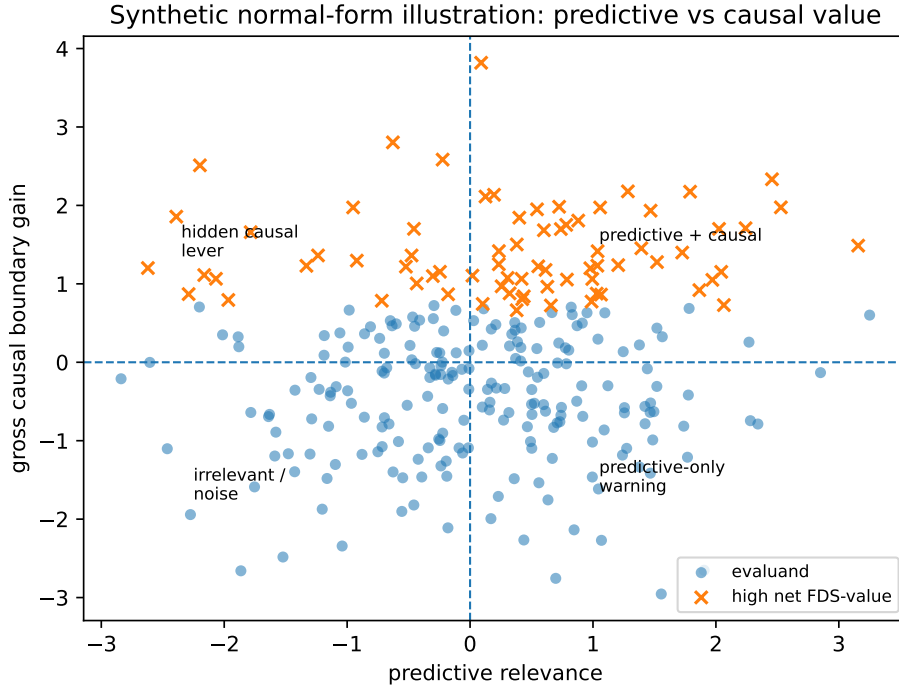


Figure 2: Synthetic normal-form illustration, not empirical evidence. Predictive relevance and gross causal boundary gain can diverge. Some variables are predictive-only warnings; others are hidden causal levers that matter despite weak predictive salience.

5 Boundary-relevance ranking and evaluation deficit

Definition 6 (Boundary-relevance ranking). Given a boundary, loss, intervention grammar, cost model, action/update space, and horizon, a boundary-relevance ranking is an ordering $\succ_t$

over evaluands such that

$$z_i \succ_t z_j \quad \text{if} \quad V_t^{\text{net}}(z_i; k) > V_t^{\text{net}}(z_j; k), \quad (10)$$

or, when explicitly stated, by gross gain, risk-weighted value, or a vector-valued partial order.

A value ranking is therefore boundary-relative, horizon-relative, capacity-relative, and cost-relative. If the boundary or horizon changes, the ranking can change without contradiction. If the cost model changes, a high-gain but expensive evaluand can fall below a lower-gain but cheaper one.

5.1 Minimal M2 audit

A minimal M2 audit must pre-specify: boundary, loss, horizon, baseline, evaluand type, intervention grammar, cost units, ranking method, uncertainty, and failure condition. Without these, claims about value or goal remain under-specified.

5.2 Evaluation capacity and second-order deficit

Let $C_{\text{eval}}(t)$ be the system's capacity for estimating and maintaining value rankings. Let $R_{\text{eval}}^{(\tau)}(\epsilon; t)$ be the rate-distortion demand of maintaining a value ranking at tolerance ϵ over update window τ . Define

$$\Delta_{\text{eval}}(t) = R_{\text{eval}}^{(\tau)}(\epsilon; t) - C_{\text{eval}}(t). \quad (11)$$

When $\Delta_{\text{eval}} > 0$, verified ranking cannot be maintained at the target tolerance without compression, proxy substitution, externalization, task relaxation, or drift. This is a second-order deficit: the system may have enough capacity to act but not enough capacity to evaluate all possible actions precisely. In such cases, a finite system may rationally maintain coarse preferences or heuristics instead of precise goals.

If evaluation is physically implemented, high-precision ranking may also carry energetic, latency, memory-turnover, and verification costs. M2 treats these as scalarized or vector costs rather than asserting a universal thermodynamic value cost. The point is operational: valuation itself is not free.

6 Risk-weighted value near collapse thresholds

Average expected loss is not always sufficient. Near a boundary failure threshold, reducing collapse probability may dominate improving mean performance. Let ℓ_c be a critical boundary-loss threshold. Define

$$\Delta_z \mathbb{E} \ell = \mathbb{E}[\ell_{\text{maint}, t+k} \mid M_t, \text{do}(z_0)] - \mathbb{E}[\ell_{\text{maint}, t+k} \mid M_t, \text{do}(z)], \quad (12)$$

$$\Delta_z P_c = P(\ell_{\text{maint}, t+k} > \ell_c \mid M_t, \text{do}(z_0)) - P(\ell_{\text{maint}, t+k} > \ell_c \mid M_t, \text{do}(z)). \quad (13)$$

The risk-weighted FDS-value is

$$V_t^{\text{risk}}(z; k) = \Delta_z \mathbb{E} \ell + \alpha_t \Delta_z P_c - \lambda_t c_t(z). \quad (14)$$

The first term is expected loss reduction. The second is collapse-risk reduction. Both are positive when selecting z reduces boundary risk. The parameter α_t converts collapse-probability reduction into expected boundary-loss-equivalent units. It is not universal.

A bounded normal-form coupling to resource reserve is

$$\alpha_t = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \sigma \left(\frac{\Phi_{\text{crit}} - \Phi_t}{s_{\Phi}} \right), \quad (15)$$

where $\sigma(x) = 1/(1 + e^{-x})$ is the logistic sigmoid and $s_\Phi > 0$ is a steepness parameter. This replaces the previous inverse-distance form that could diverge as $\Phi_t \rightarrow \Phi_{\text{crit}}$. In empirical or numerical use, any coupling form should be bounded to avoid singular behavior near the resource threshold. Equation (15) is an illustrative bridge: it says that systems near resource-critical states may weight collapse-risk reduction more strongly. It is not asserted as a universal law.

Proposition 2 (Risk dominance near collapse). *An evaluand with smaller average-loss reduction can outrank an evaluand with larger average-loss reduction when it produces sufficiently larger collapse-risk reduction.*

Proof sketch. Let z_i and z_j have equal cost for simplicity. If $\Delta_i \mathbb{E}\ell < \Delta_j \mathbb{E}\ell$ but

$$\Delta_i P_c - \Delta_j P_c > \frac{\Delta_j \mathbb{E}\ell - \Delta_i \mathbb{E}\ell}{\alpha_t}, \quad (16)$$

then Eq. (14) gives $V^{\text{risk}}(z_i; k) > V^{\text{risk}}(z_j; k)$. $\square$

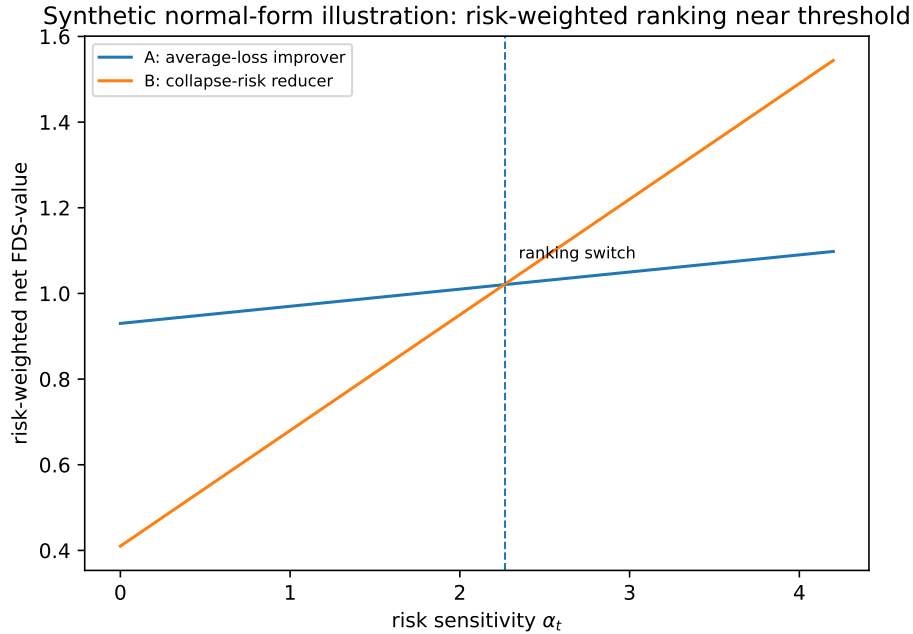


Figure 3: Synthetic normal-form illustration, not empirical evidence. A collapse-risk reducer can outrank an average-loss improver as risk sensitivity increases near a critical boundary threshold.

7 Goals as stabilized FDS-value rankings

Definition 7 (Goal). A goal is a stabilized FDS-value ranking coupled to a policy orientation across finite update windows. A goal state can be represented as

$$g_t = (\succ_t, \pi_g, k, \epsilon, \mathcal{C}), \quad (17)$$

where $\succ_t$ is a ranking, π_g a policy induced by the ranking, k a horizon, ϵ a stability tolerance, and $\mathcal{C}$ a context or perturbation family.

Goal stability is a register-time property: a goal must persist across finite update windows as a maintained ordering, rather than appearing only as an instantaneous reaction [4]. A

policy-based goal-stability index can use Jensen-Shannon divergence,

$$\text{GSI}_\pi(t, \Delta) = \exp \left(- \frac{D_{\text{JS}}(\pi_g(\cdot | M_t), \pi_g(\cdot | M_{t+\Delta}))}{\sigma_G} \right), \quad (18)$$

where $\sigma_G > 0$ is a scale parameter. JS divergence is symmetric and bounded. If KL divergence is used instead, empirical policies should be smoothed to avoid support-zero artifacts. A rank-based version is

$$\text{GSI}_r(t, \Delta) = 1 - D_{\text{rank}}(\succ_t, \succ_{t+\Delta}), \quad (19)$$

where $D_{\text{rank}} \in [0, 1]$.

Proposition 3 (Goal requires stability). *A high-valued action or distinction is not yet a goal. Goal-like behavior requires stability of a ranking or policy orientation across update windows and coupling of that stability to action selection.*

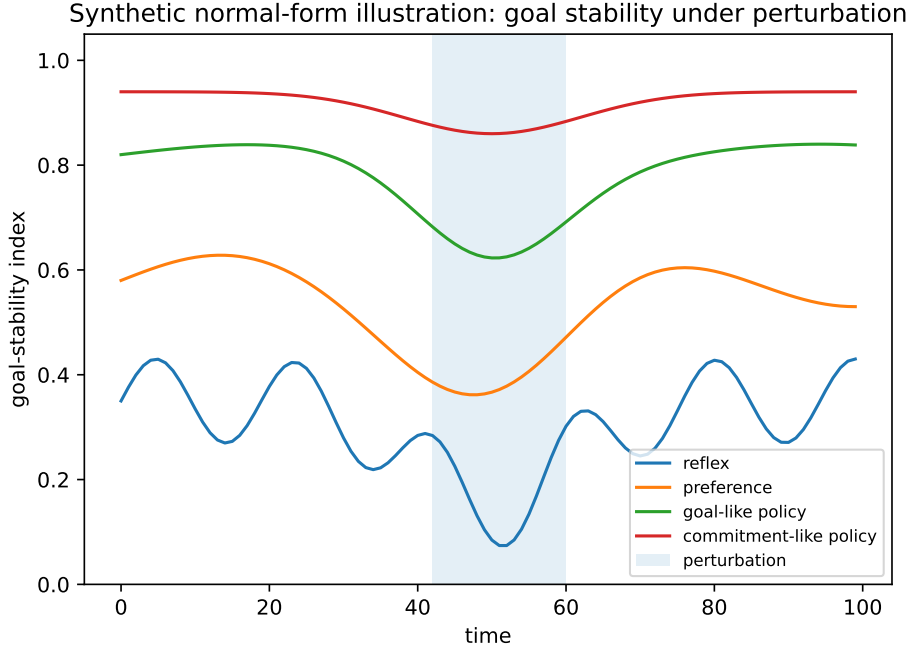


Figure 4: Synthetic normal-form illustration, not empirical evidence. A goal-like policy preserves a stable action orientation through perturbation better than a reflexive response. A commitment-like policy is even more stable but may become rigid if the environment changes.

8 Proxy-boundary divergence and reward hacking

Reward is a proxy signal. It may approximate FDS-value, but it can also diverge. Let R_{proxy} be a proxy reward and ℓ_{host} a host boundary-maintenance loss. For actions a_i relative to baseline a_0 , define

$$\Delta_R(a_i) = \mathbb{E}[R_{\text{proxy}} | \text{do}(a_i)] - \mathbb{E}[R_{\text{proxy}} | \text{do}(a_0)], \quad (20)$$

$$\Delta_\ell(a_i) = \mathbb{E}[\ell_{\text{host}} | \text{do}(a_i)] - \mathbb{E}[\ell_{\text{host}} | \text{do}(a_0)]. \quad (21)$$

A proxy is locally aligned with host boundary value when reward-increasing actions tend to reduce boundary loss. A finite-difference alignment score is

$$\text{Align}_\epsilon(R, \ell) = - \frac{\langle \Delta_R, \Delta_\ell \rangle}{\|\Delta_R\| \|\Delta_\ell\| + \epsilon_{\text{Align}}}, \quad (22)$$

where $\epsilon_{\text{Align}} > 0$ is a small regularizer that prevents division by zero. If either vector has near-zero norm, the alignment estimate should be reported with a low-information flag; the regularized score is only a numerical placeholder, not evidence of alignment. The unregularized form in a pre-registration may use $\epsilon_{\text{Align}} = 0$ with a pre-specified zero-vector handling rule. The negative sign reflects that reward increases should correspond to loss decreases.

Proposition 4 (Proxy reward can invert FDS-value). *If an action increases proxy reward while also increasing host boundary-maintenance loss, then it is proxy-positive but boundary-negative.*

This is a minimal FDS account of reward hacking. A delegated model may optimize a proxy score outside the distribution where that proxy tracked host boundary loss. In FDS terms, the proxy preserves reward ranking while losing boundary-gradient relevance. A semantic compression or proxy quotient that worked in the training regime can false-compress critical distinctions under distribution shift.

Recent LLM alignment work provides concrete cases where proxy reward and target quality diverge. Reward models can be exploited under distribution shift and preference inconsistency [19], and LLM agents trained or prompted in gameable settings can generalize from specification gaming toward reward-tampering behavior [20]. Reward overoptimization has also been formalized for direct alignment algorithms, where proxy objectives improve while true quality can plateau or deteriorate [21].

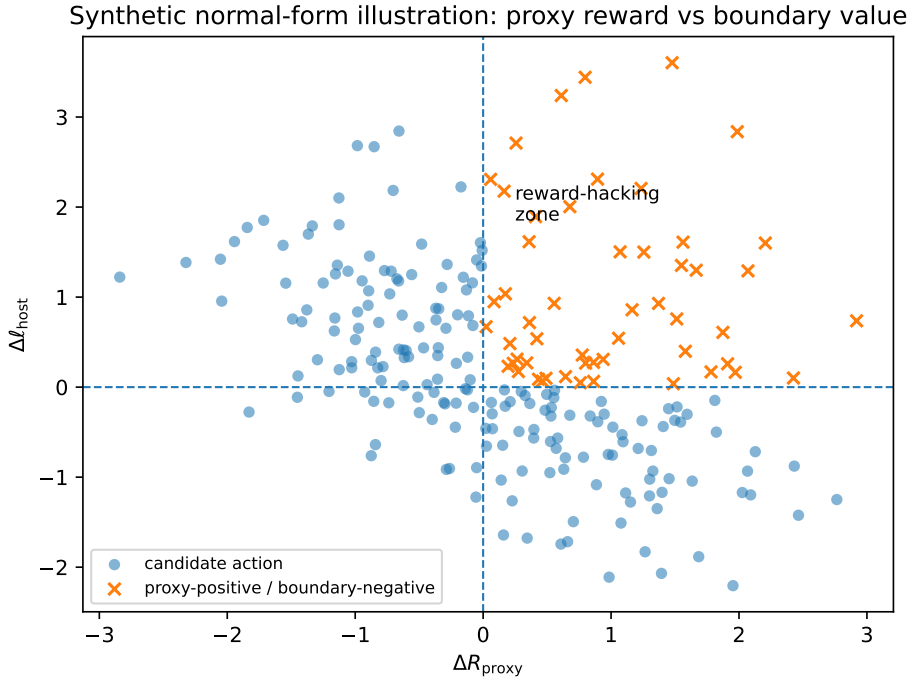


Figure 5: Synthetic normal-form illustration, not empirical evidence. Proxy reward can increase while host boundary loss also increases. These actions are proxy-positive but boundary-negative. The y-axis shows $\Delta\ell_{\text{host}}$ (host boundary-loss effect); the x-axis shows ΔR_{proxy} (proxy reward effect).

9 Value drift and evaluation failure

Definition 8 (Value drift). Value drift occurs when a boundary-relevance ranking changes faster than the system can verify, update, or maintain the reasons for the change.

One audit form is

$$D_{\text{rank}}(\succ_t, \succ_{t+\Delta}) > \delta_{\text{drift}} \quad \text{while} \quad \text{Conf}_{\text{verify}}(t, t + \Delta) < \theta_{\text{verify}}. \quad (23)$$

Under evaluation deficit, systems may substitute salience, reward proxy, social proof, short-horizon risk, or institutional defaults for verified causal value. This does not necessarily mean failure: heuristics can be necessary. But when substitution is hidden, value drift can masquerade as stable goal pursuit.

Proposition 5 (Evaluation deficit produces drift pressure). *When evaluation demand exceeds maintained ranking capacity under finite evaluation capacity, no hidden capacity expansion, and no task relaxation, a system must compress, externalize, simplify, or drift. Under deficit, salience, proxy reward, or short-horizon risk may substitute for verified causal FDS-value.*

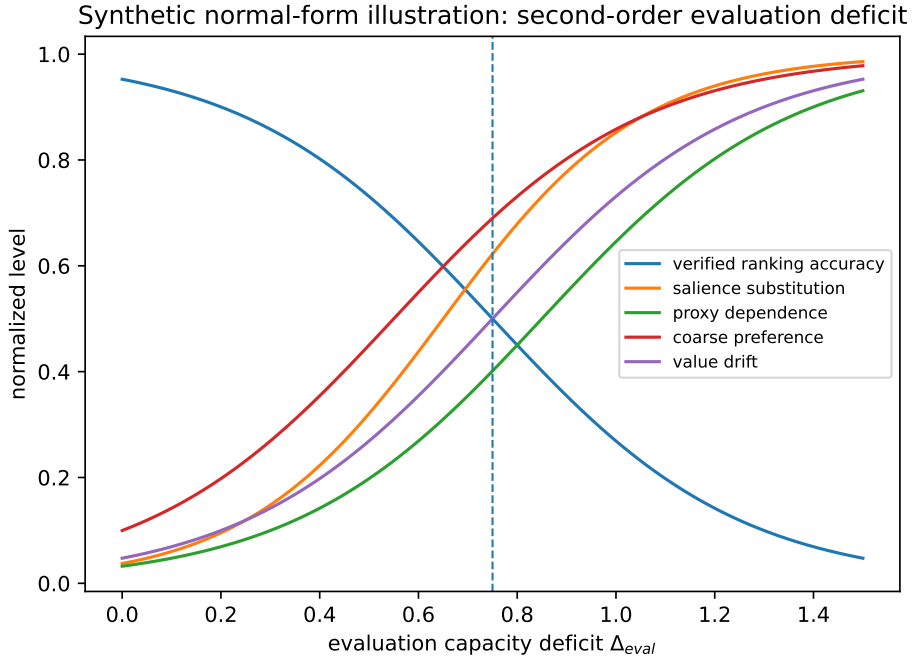


Figure 6: Synthetic normal-form illustration, not empirical evidence. As evaluation capacity deficit rises, verified ranking accuracy falls while salience substitution, proxy dependence, coarse preference, and value drift increase.

10 Multi-goal conflict and vector FDS-value

Many systems maintain multiple boundary variables. A biological system may trade energy, temperature, immune risk, and reproduction. An organization may trade solvency, legitimacy, throughput, and safety. A single scalar value can hide conflict.

Let $\ell_{t+k} \in \mathbb{R}^m$ be a vector of boundary-maintenance losses. Define gross vector boundary gain as

$$\mathbf{G}_t^{\text{FDS}}(z; k) = \mathbb{E}[\ell_{t+k} \mid M_t, \text{do}(z_0)] - \mathbb{E}[\ell_{t+k} \mid M_t, \text{do}(z)]. \quad (24)$$

The cost should not be subtracted as a scalar from a vector without a conversion map. One may keep the audit as $(\mathbf{G}_t^{\text{FDS}}(z; k), c_t(z))$ or define

$$\mathbf{V}_t^{\text{net}}(z; k) = \mathbf{G}_t^{\text{FDS}}(z; k) - \lambda_t c_t(z), \quad (25)$$

where λ_t converts scalarized cost into each boundary-loss dimension.

A scalar ranking is then a choice of scalarization, not a universal value order. Pareto-incomparable policies can both be rational relative to different boundary priorities.

Recent multi-objective reinforcement-learning work is directly relevant to M2’s vector-value view. MORL has been proposed as a tool for pluralistic alignment when scalar reward is insufficient for multiple conflicting values or stakeholders [22]. Pareto-front discovery methods such as C-MORL illustrate the algorithmic difficulty of maintaining multiple tradeoff policies rather than collapsing objectives into one scalar [23].

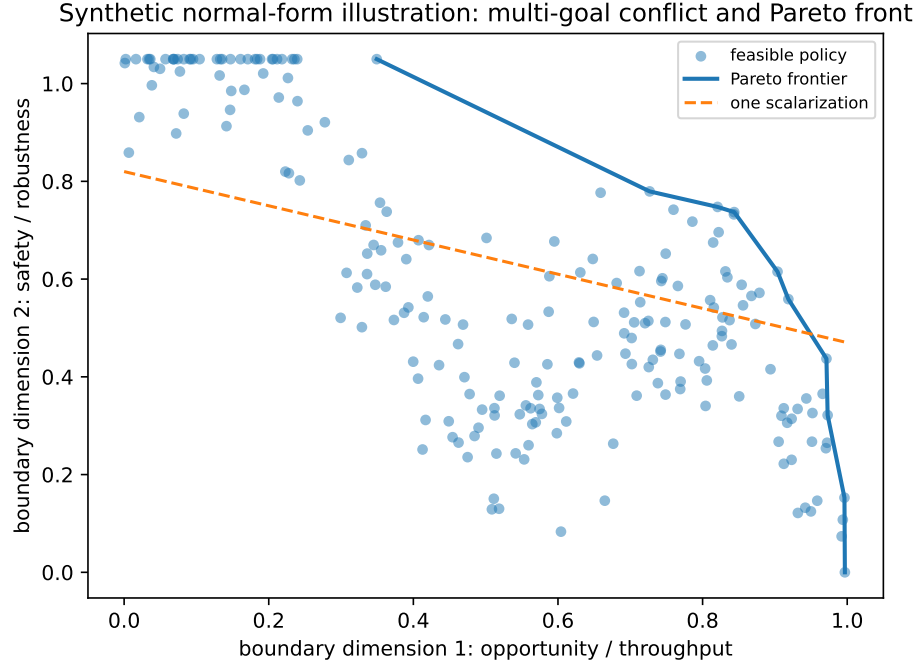


Figure 7: Synthetic normal-form illustration, not empirical evidence. Multi-goal systems can face Pareto conflict. A single scalarization selects one tradeoff but does not erase the underlying boundary-dimension conflict.

11 Collective goals and ranking synchronization

A collective goal is not merely a slogan or aggregate preference. It requires a shared ranking that is maintained across agents, institutions, records, and update windows.

Definition 9 (Collective goal). A collective goal is a shared boundary-relevance ranking $\succ_t^{\text{shared}}$ coupled to collective policy channels, institutional memory, and verification procedures such that it remains stable enough to orient group action over a specified horizon.

A minimal collective-goal condition is

$$\text{GSI}^{\text{shared}}(t, \Delta) > \gamma, \quad \text{Align}_{\text{rank}}(\pi_{\text{policy}}, \succ_t^{\text{shared}}) > \eta. \quad (26)$$

One simple alignment score is

$$\text{Align}_{\text{rank}}(\pi, \succ) = \mathbb{E}_{a \sim \pi}[r_{\succ}(a)], \quad (27)$$

where $r_{\succ}(a) \in [0, 1]$ is a normalized rank score assigned to action a by the shared ranking. A simpler top- k audit is

$$\text{Align}_{\text{top-k}} = \frac{|\text{Top}_k(\pi_{\text{policy}}) \cap \text{Top}_k(\succ^{\text{shared}})|}{k}. \quad (28)$$

M2 does not solve social choice. It is compatible with social-choice impossibility results but addresses a different bottleneck: finite verification and synchronization of shared rankings. Collective rankings can fail not only because individual preferences conflict, but because members update on different evidence, at different timescales, through different verification channels. In FDS terms, asynchronous update can create non-Markovian noise in $\succ_t^{\text{shared}}$.

11.1 Ranking synchronization demand

Collective goal stability requires not only shared rankings but sufficient communication and verification bandwidth to synchronize rankings across agents. Define the ranking synchronization demand $R_{\text{sync}}^{(\tau)}(\epsilon; t)$ as the rate-distortion demand of maintaining $\succ_t^{\text{shared}}$ within tolerance ϵ over window τ across N agents. The synchronization load factor is

$$Z_{\text{sync}}(t) = \frac{R_{\text{sync}}^{(\tau)}(\epsilon; t)}{C_{\text{comm}}(t) + C_{\text{verify}}(t)}. \quad (29)$$

When $Z_{\text{sync}}(t) > 1$, shared ranking stability degrades: $\text{GSI}^{\text{shared}}(t, \Delta) \downarrow$. Collective goal failure can thus occur not only because members disagree, but because the system lacks enough communication and verification bandwidth to synchronize value rankings across update windows.

12 Goal hysteresis and recovery lag

Goals can persist after the conditions that made them valuable have changed. This can be useful: commitments and institutional rules prevent constant re-ranking under noise. But it can also produce rigidity. A crisis-induced ranking can remain after the crisis ends because records, routines, threat priors, or institutional incentives decay slowly.

A minimal normal-form is

$$\alpha_{t+1} = \text{clip}((1 - \rho)\alpha_t + \rho\alpha_{\text{target}}(\Phi_t, \ell_t) + h_t, \alpha_{\min}, \alpha_{\max}), \quad h_{t+1} = \chi h_t + \zeta \mathbf{1}_{\ell_t > \ell_c}, \quad (30)$$

where h_t is goal-locking residue, measured in the same units as α_t (or scaled into those units before entering Eq. (30)), $0 < \chi < 1$ is persistence, and ζ is crisis accumulation strength. Clipping ensures α_t remains within $[\alpha_{\min}, \alpha_{\max}]$ even under high crisis accumulation, preventing unbounded growth of risk sensitivity. Even after resource or threat recovery, h_t can keep risk-weighted rankings narrow.

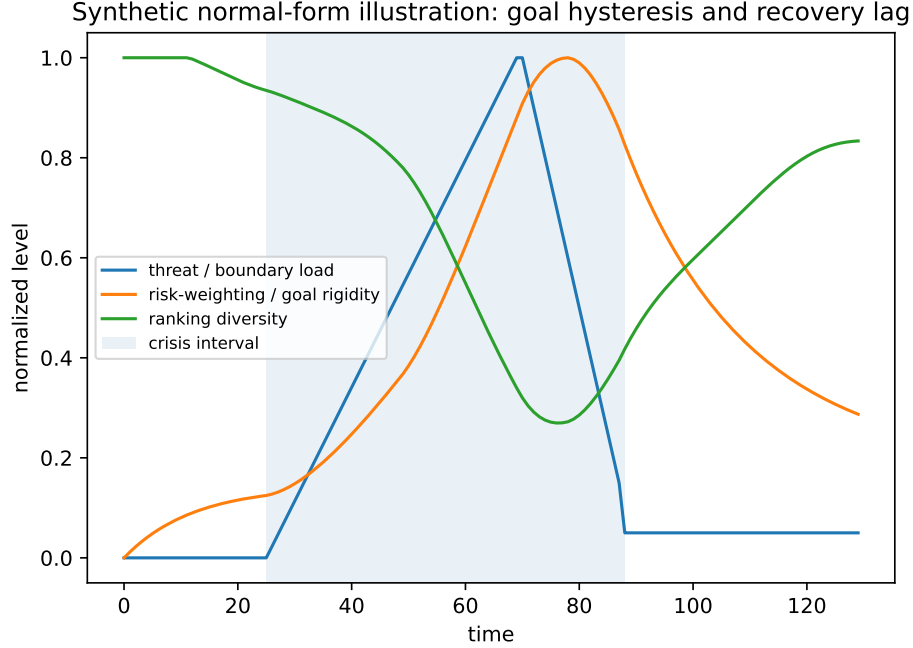


Figure 8: Synthetic normal-form illustration, not empirical evidence. Crisis load increases risk-weighting and narrows ranking diversity. After the crisis interval, goal rigidity can persist due to goal-locking residue.

13 Domain audit examples

Table 4: Audit templates, not empirical confirmations. Illustrative domain audit mappings. Each row requires domain-specific bridge assumptions and empirical tests.

Domain	Boundary variable	Loss	Evaluand	Failure mode
AI agent	host objective; system integrity	host boundary loss	reward proxy; tool action	reward hacking; proxy inversion
Organism	metabolic, membrane, homeostatic state	viability loss	nutrient cue; avoidance action	maladaptive craving; risk blindness
Organization	solvency, legitimacy, throughput, safety	institutional loss	KPI; policy; budget choice	metric gaming; goal displacement
Science	error correction, replication, evidence quality	epistemic loss	paper; claim; method; standard	citation-proxy drift; false consensus
Civilization	verification capacity, memory, law, infrastructure	collapse risk	law; archive; protocol; norm	goal rigidity; institutional lock-in

14 Normal-form model and reproducibility

The accompanying code implements deterministic synthetic normal-form illustrations. The model is not empirical evidence. It is a consistency and visualization device that maps the definitions to simple state variables.

The model generates candidate evaluands with predictive relevance, gross causal boundary gain, cost, collapse-risk reduction, proxy reward effects, boundary-loss effects, vector boundary gains, evaluation deficit, and goal-locking residue. It implements predictive-causal dissociation, risk-weighted ranking, goal-stability trajectories, proxy-boundary divergence, second-order

evaluation deficit, multi-goal Pareto conflict, and goal hysteresis. The random seed is fixed in `code/generate_results.py`. CSV outputs are stored in `data/`; figure pairs are stored in `figures/`.

Table 5: Normal-form variable map. All entries are illustrative, not fitted empirical quantities.

Simulation variable	Paper definition	Interpretation
predictive relevance	Eq. (5)	observational association with future loss
gross causal gain	Eq. (7)	expected boundary-loss reduction before cost
net FDS-value	Eq. (8)	gross gain minus scalarized cost
risk sensitivity	Eq. (14)	collapse-risk weighting near threshold
proxy reward effect	Eq. (22)	finite-difference reward effect
boundary-loss effect	Eq. (22)	finite-difference host loss effect
evaluation deficit	Eq. (11)	inability to maintain verified ranking
vector gain	Eq. (24)	multi-boundary value dimensions
goal hysteresis	Eq. (30)	delayed relaxation of crisis ranking

15 Protocols and tests

Protocol 1 (Causal FDS-value audit). Pre-register boundary variable, loss function, horizon, baseline, evaluand type, intervention grammar, and cost model. Estimate gross causal boundary gain and net FDS-value with uncertainty intervals.

Protocol 2 (Predictive-relevance-causal dissociation test). Identify variables that predict future loss but do not change future loss under intervention. M2 predicts that these should not count as causal FDS-value unless they alter downstream update, action, verification, or coordination.

Protocol 3 (Risk-weighted ranking test). In simulation, controlled proxy environments, or naturally occurring near-threshold states, evaluate whether collapse-risk reducers rise in ranking even when their average-loss improvement is smaller.

Protocol 4 (Evaluation-deficit test). Increase evaluation load or reduce evaluation capacity. Prediction: rankings become noisier, coarser, more proxy-driven, more salience-driven, or more short-horizon risk-dominated.

Protocol 5 (Goal-stability test). Measure ranking or policy persistence across update windows and perturbations. Reflexive systems should show low GSI; goal-like systems should preserve nontrivial ranking stability.

Protocol 6 (Proxy divergence / reward-hacking audit). Compare proxy reward effects with true boundary-loss effects. Misalignment occurs when proxy reward increases while boundary loss also increases, or when proxy-boundary alignment falls below a pre-registered threshold.

Protocol 7 (Multi-goal conflict audit). Specify multiple boundary variables. Test whether scalar value rankings hide Pareto conflict, forced pruning, or transfer of loss from one boundary dimension to another.

Protocol 8 (Collective goal audit). For organizations or societies, identify shared rankings, institutional memory, verification channels, and policy outputs. Test whether collective goals persist as stabilized rankings rather than episodic slogans.

16 Relation to existing fields

Decision theory and utility. FDS-value resembles utility only after a boundary, loss, cost, intervention grammar, action space, and horizon are specified. It is not universal utility [6, 7].

Reinforcement learning and reward. Reinforcement learning formalizes reward-driven policies, value functions, and Markov decision processes [8, 9]. M2 treats reward as a proxy that may approximate or diverge from boundary effects. Inverse reinforcement learning and preference learning study how objectives can be inferred or learned, but M2 emphasizes intervention-audited boundary effects [10].

Goodhart, reward hacking, and specification gaming. Metric gaming and reward misspecification are natural cases of proxy-boundary divergence: optimizing a proxy can destroy the boundary relationship that made the proxy useful [11–13, 19–21].

Prospect theory and boundary-risk asymmetry. Prospect-theoretic loss asymmetry can be reinterpreted in M2 as boundary-risk asymmetry near critical thresholds: collapse-risk reduction can dominate average gain [14].

Control theory and viability. Boundary-maintenance loss and viable sets connect M2 to control and viability theory [15, 16]. M2 adds explicit finite distinction capacity and value-ranking audits.

Multi-objective optimization and social choice. Vector boundary value and Pareto conflict connect to multi-objective optimization and social choice [17, 18, 22, 23]. M2 does not solve aggregation impossibility; it adds finite verification and ranking-synchronization constraints.

17 Limitations and falsification

M2 is intentionally limited. It does not establish a general empirical theory of value by definition. It provides mappings that must survive operational audit. The framework is weakened or demoted under any of the following results:

1. value-like ranking under a specified mapping shows no relation to causal boundary effects, costs, or horizons;
2. predictive and causal effects are never dissociable in systems claimed to require intervention audit;
3. risk-weighted ranking never changes near critical boundary thresholds under valid mappings;
4. goal-like behavior persists without any ranking stability, memory, policy orientation, or update-window persistence;
5. proxy reward remains aligned despite pre-registered divergent finite-difference effects on host boundary loss;
6. multi-goal systems always admit a scalar ranking without hidden Pareto conflict or forced loss transfer;
7. collective goals show no relationship to shared ranking, verification capacity, institutional memory, or policy output;
8. claimed goal hysteresis disappears under controlled recovery tests where load reduction should reveal persistent ranking lock-in.

18 Conclusion

M2 defines value and goal as finite-system ranking operations. FDS-value is not moral value, reward, preference, or intrinsic worth. It is causal boundary-gradient relevance under a specified boundary, loss, intervention, cost, and horizon. Goal is not merely high value. It is a stabilized FDS-value ranking coupled to policy across update windows.

The central chain is

$$\begin{aligned} &\text{admitted evaluand} \rightarrow \text{gross causal boundary gain} \rightarrow \text{net / risk-weighted FDS-value} \\ &\rightarrow \text{ranking} \rightarrow \text{goal stability} \rightarrow \text{policy / update} \rightarrow \text{future boundary loss.} \end{aligned} \quad (31)$$

18.1 High-level goals as invariant-compression candidates

In FDS terms, high-level goals such as survival, freedom, justice, or scientific truth can be interpreted as candidate invariant-compressions: compact rankings that remain boundary-relevant across many contexts, perturbations, and update windows. A goal g^* is an invariant goal candidate if net FDS-value remains positive across a broad context family $\mathcal{C}$,

$$V_t^{\text{net}}(g^*; k, c) > 0 \quad \text{for many } c \in \mathcal{C}, \quad (32)$$

and goal stability remains above threshold,

$$\text{GSI}(g^*; t, \Delta) \geq \gamma, \quad (33)$$

under a perturbation family $\mathcal{P}$. This does not morally validate any high-level goal. It only states that, under a specified system boundary and context family, some goals may function as compact stable rankings with broad boundary relevance. This bridges M2’s stable positive-boundary-gradient rankings to M3’s compressed actionable semantic quotients: high-level goals are those rankings that compress well without losing boundary relevance across contexts.

M2 prepares later work. M3 can treat meaning as compression that preserves value-relevant action structure. M5 can analyze trust as delegated evaluation and verification. A2 can audit proxy reward against host boundary value. S2 and S3 can treat epistemic pollution and institutional collapse as collective value-ranking failures. G3 and G4 can treat science and civilization memory as infrastructures for correcting and stabilizing shared value maps.

Code and data availability

The deterministic synthetic normal-form code, figures, and CSV outputs are included in the replication package. Run `python code/generate_results.py` from the paper directory to regenerate all figures and data.

AI assistance disclosure

The author used AI assistance for drafting, editing, simulation scaffolding, and consistency checks. The author reviewed and selected the final claims, definitions, equations, and interpretations.

A M-series dependency map

Table 6: How M2 supports later M-series and civilization-layer papers.

Future paper	Interface supplied by M2
M3 Meaning	Meaning must preserve value-relevant action structure under compression.
M5 Trust	Trust reduces repeated evaluation and verification cost.
A2 AI Alignment	Proxy reward must be audited against host boundary FDS-value.
S2 Epistemic Pollution	Pollution distorts collective value rankings and verification.
S3 Institutional Collapse	Institutions can fail through value drift, proxy capture, or goal rigidity.
G3 Science	Scientific method helps separate predictive relevance from causal value.
G4 Civilization Memory	Archives and standards stabilize value-relevant records across time.

B Notation summary

Table 7: Notation used in M2.

Symbol	Meaning
z	evaluand: distinction, action, state, policy, record, proxy, quotient, or symbol
z_0	baseline evaluand or null intervention
ℓ_{maint}	boundary-maintenance loss
$G_t^{\text{FDS}}(z; k)$	gross causal boundary gain
$V_t^{\text{net}}(z; k)$	net FDS-value after cost
$V_t^{\text{risk}}(z; k)$	risk-weighted FDS-value near collapse thresholds
$c_t(z)$	scalarized evaluation, verification, action, maintenance, latency, or opportunity cost
λ_t	cost-to-boundary-loss conversion factor
α_t	collapse-risk-to-boundary-loss conversion factor
Φ_t	resource reserve / budget state
Φ_{crit}	critical resource threshold
$C_{\text{eval}}(t)$	evaluation capacity
$\Delta_{\text{eval}}(t)$	evaluation capacity deficit
$\succ_t$	boundary-relevance ranking at time t
π_g	goal-induced policy
GSI_{π}	policy-based goal-stability index
GSI_r	rank-based goal-stability index
R_{proxy}	proxy reward signal
$\text{Align}_{\epsilon}(R, \ell)$	finite-difference proxy-boundary alignment score (regularized)
$\mathbf{G}^{\text{FDS}}$	vector gross causal boundary gain
$\boldsymbol{\lambda}_t$	vector cost conversion map
h_t	goal-locking residue / hysteresis state
χ	hysteresis persistence factor
ζ	crisis accumulation strength
$R_t^{\text{predgross}}(z; k)$	gross predictive relevance (observational association with future loss)
$R_t^{\text{prednet}}(z; k)$	net predictive relevance after cost
$R_{\text{sync}}^{(\tau)}(\epsilon; t)$	ranking synchronization demand

Symbol	Meaning
$Z_{\text{sync}}(t)$	collective ranking synchronization load factor
D_{JS}	Jensen–Shannon divergence
σ_G	goal-stability divergence scale
ϵ_{Align}	alignment regularizer
$\alpha_{\min}, \alpha_{\max}, s\Phi$	bounded risk-sensitivity parameters

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Meaning as Actionable Semantic Quotient

Policy-Preserving Compression, Semantic Deficit, and False Meaning under Finite Capacity

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Abstract

This paper develops the meaning layer of the agency-semantics spine of Distinction Theory. Building on the formal core of Active Finite Distinction Systems, the M0 agency-semantics spine, M1 attention-as-admission, and M2 value-goal ranking, it treats meaning as *FDS-meaning*: an actionable semantic quotient that preserves downstream use under finite capacity. A representation is FDS-meaningful for a system only relative to a specified boundary, task family, context family, policy or verification target, horizon, loss function, and resource budget. Compression alone is not meaning: a quotient must preserve enough policy, value, prediction, verification, coordination, or boundary-maintenance structure to remain usable. The paper introduces semantic quotient maps, reference-policy provenance, semantic preservation vectors, maintained semantic load, semantic capacity deficit, false compression, unsupported completion, shared quotient alignment, and invariant semantic quotient candidates. A deterministic synthetic normal-form model illustrates policy-preserving quotienting, embedding-policy dissociation, capacity-driven semantic collapse, false compression, non-linear shared-meaning failure, and recovery by quotient refinement. The paper does not reduce linguistic, subjective, or moral meaning to compression. It supplies a conservative finite-system bridge for analyzing meaning as policy-preserving actionable compression under capacity, verification, and resource constraints.

Keywords: meaning; semantic quotient; actionable compression; finite capacity; boundary maintenance; policy preservation; semantic deficit; false compression; semantic drift; hallucination; embedding similarity; semantic communication; information bottleneck; state abstraction; active finite distinction systems.

Epistemic Notice and Scope

This manuscript is an agency-semantics spine paper, not a completed theory of linguistic meaning, consciousness, intentionality, truth, reference, moral meaning, or social legitimacy. It does not claim that all semantics reduces to compression, that embeddings are meaning, that a symbol is meaningful merely because it predicts another symbol, or that moral meaning can be derived from boundary maintenance. Its claim is narrower: once a system is modeled as an active finite distinction system with a boundary, memory, update rule, action space, finite capacity, boundary-maintenance loss, and task family, a representation can be audited for whether it preserves the downstream structure needed for action, prediction, verification, coordination, or boundary maintenance. In this paper, *meaning* means FDS-meaning or operational meaning: actionable quotient structure preserved for a specified system, task family, context family, policy or verification target, horizon, and capacity budget.

The paper follows the layered discipline of the FDS formal core. Formal definitions, normal-form models, physical bridge assumptions, and domain applications are separated. A failure of a

linguistic, cognitive, artificial, biological, organizational, or social mapping may demote that mapping without refuting the formal FDS core. The deterministic numerical model is illustrative only. It visualizes definitions and failure modes; it is not empirical evidence.

1 Introduction

A finite system cannot preserve every distinction it attends to. Attention admits distinctions; value ranks their boundary relevance; goals stabilize rankings over time. Meaning begins when the system compresses these distinctions without losing the structure needed for action, prediction, verification, coordination, or boundary maintenance. A word, symbol, memory, embedding, law, alarm, dashboard, record, or model state is meaningful in the FDS sense only when it functions as such a quotient.

The FDS core defines an active finite distinction system as a tuple

$$S = (X, E, B, M, Y, A, U, \pi, \ell, \Phi, \mathcal{P}, \tau), \quad (1)$$

where X is internal state, E environment, B boundary, M memory/model space, Y observation channel, A action space, U update map, π finite projection, ℓ boundary-maintenance loss, Φ resource budget, $\mathcal{P}$ perturbation/pruning family, and τ update timescale [1]. M0 introduced the agency-semantics dependency skeleton

$$\text{distinction} \rightarrow \text{record} \rightarrow \text{attention} \rightarrow \text{value} \rightarrow \text{goal} \rightarrow \text{meaning} \rightarrow \text{agency} \rightarrow \text{culture}, \quad (2)$$

and defined meaning as actionable compressed distinction preserved by a task-sufficient quotient [2]. M1 expanded attention as finite distinction admission [3]. M2 expanded value and goal as boundary-relevance ranking and stabilized ranking [4]. M3 expands the meaning step.

1.1 Why meaning needs a finite-system formulation

The word meaning is overloaded. It can refer to reference, truth conditions, use, pragmatic effect, subjective association, embedding similarity, cultural significance, moral purpose, intentional content, or causal affordance. M3 does not try to reduce these notions into a single substance. Instead it defines an operational bridge. Given a boundary, task family, policy family, horizon, loss, verification channel, and finite capacity, a representation has FDS-meaning when it compresses distinctions while preserving the downstream structure needed for use.

This makes meaning relational. A symbol may be meaningful for one task and useless for another. A compressed type may be valid under one policy and false under another. A word may coordinate a community while misleading an outside system. An embedding can be near another embedding while requiring a different action. A legal rule can preserve coordination at one institutional horizon and become rigid at another. Meaning in the M3 sense is therefore not an intrinsic tag attached to a representation. It is a task-relative, policy-relative, and capacity-relative preservation relation.

1.2 Compression is not enough

Compression reduces dimensionality. Meaning requires preserving the relevant structure. A representation can be short, elegant, predictive, or high-probability without being meaningful for the system’s task. A quotient that merges two distinctions requiring different actions is not meaningful for that action family, even if it is statistically efficient. Conversely, a crude symbol can be meaningful if it preserves the right action or verification distinction.

The core claim is therefore:

$$\boxed{\text{FDS-meaning} = \text{actionable semantic quotient under finite capacity.}} \quad (3)$$

The word actionable is essential. M3 does not claim that all meaning is action in a narrow motor sense. It uses action broadly: update, prediction, verification, coordination, planning, record maintenance, externalization, and boundary maintenance.

1.3 Common misreadings

Table 1: Common misreadings and corrections.

Misreading	Correction
Meaning equals compression.	No. M3 requires actionable, policy-relevant, prediction-relevant, verification-relevant, coordination-relevant, or boundary-relevant preservation.
Embeddings are meaning.	No. Embedding similarity is only a representational relation; it must be audited against downstream policy, value, verification, or coordination.
Meaning equals truth.	No. Truth may be one verification target, but FDS-meaning is a use-preserving quotient under a specified task family.
Meaning equals moral purpose.	No. M3 defines operational meaning, not moral meaning or legitimacy.
Shared meaning requires identical private states.	No. Shared meaning requires sufficient quotient alignment for coordination or verification, not identical internal representations.
Any high-level slogan is meaningful.	No. A slogan can be a false compression if it merges distinctions that require different policies.

Table 2: Claim-status summary for M3. The table is an audit device: several entries are formal or operational bridge claims, not established empirical results.

Claim ID	Tier	Claim	Failure or demotion condition
M3-001	Formal bridge	FDS-meaning is actionable semantic quotient under a specified system, boundary, task family, context family, policy or verification target, horizon, loss, tolerance, and capacity budget.	Compressed representations function semantically without preserving any action, prediction, verification, coordination, or boundary-relevant structure.
M3-002	Formal / model bridge	A semantic quotient must preserve policy-relevant distinctions within tolerance.	Quotient classes systematically merge distinctions requiring different actions or updates under the audited task.
M3-003	Operational bridge	Semantic compression is useful when it lowers capacity load without increasing boundary loss beyond tolerance.	Compression always degrades performance or never reduces maintained semantic load under valid mappings.
M3-004	Failure-mode bridge	Semantic deficit produces merging, loss, drift, unsupported completion, false compression, or meaning collapse.	Semantic overload produces no degradation, merging, proxy substitution, or action-relevance loss.
M3-005	AI / cognition bridge	Embedding similarity is not sufficient for FDS-meaning unless it preserves downstream policy or verification structure.	Embedding-near items always remain policy-equivalent under audited tasks.
M3-006	Social bridge	Shared meaning requires synchronized semantic quotients and verification channels across agents.	Collective meaning persists without shared quotient, external record, translation, verification, or coordination channel.
M3-007	Recovery bridge	Meaning recovery requires reconstructing lost action-relevant distinctions, not merely increasing information volume.	Restoring raw information always restores task meaning without quotient reconstruction.
M3-008	Invariant bridge	High-level meanings are candidate invariant semantic quotients stable across contexts and perturbations.	High-level meanings fail to preserve policy, value, or coordination relevance across any stated context family.

2 FDS and M-series background

2.1 Active boundary relevance

The FDS core applies its deficit logic only to active-boundary systems. A minimal relevance screen is

$$P(U(M_t, Y_t) \neq M_t) > 0, \quad I(M_{t+1}; \ell_{t+k}) > 0 \quad (4)$$

for some $k > 0$. Empirical applications should strengthen this to an intervention or ablation test,

$$\mathbb{E}[\ell_{t+k} \mid \text{do}(U)] \neq \mathbb{E}[\ell_{t+k} \mid \text{do}(U_\emptyset)], \quad (5)$$

where $U_\emptyset$ is a null, frozen, randomized, or identity update [1]. M3 inherits this discipline: a representation is FDS-meaningful only if its quotient structure can affect a future update, action, prediction, verification, coordination, or boundary-maintenance variable under a stated mapping.

2.2 Attention and value as prerequisites

M3 begins after attention and value. M1 distinguished availability, detection, and attention: a distinction becomes attended when admitted into an update-relevant channel with a verification path or status [3]. M2 defined FDS-value as causal boundary-gradient relevance and goal as stabilized FDS-value ranking across finite update windows [4]. M3 does not assign meaning to all possible distinctions. It assigns FDS-meaning to admitted or admissible distinctions that can be compressed while preserving downstream relevance.

2.3 Quotients and invariants

A quotient map groups fine distinctions into coarser classes. The FDS core uses quotient maps to express finite projection, invariant-supported persistence, and coarse-graining under perturbation [1]. M3 uses a task-specific semantic quotient,

$$q_{\text{sem}} : \mathcal{D}_t \rightarrow \mathcal{T}_t, \quad (6)$$

where $\mathcal{D}_t$ is an admitted or admissible distinction set and $\mathcal{T}_t$ is a set of semantic types. This quotient is not automatically valid. It is valid only relative to tasks, policies, contexts, and tolerances.

2.4 Notation alignment

Throughout the paper ℓ_{maint} denotes boundary-maintenance loss, not moral loss and not physical entropy production. The physical entropy ledger, when relevant, is denoted Σ_{phys} . The M2 value variable $V_t^{\text{net}}(z; k)$ denotes net FDS-value after cost; it is not moral value. The M3 quotient q_{sem} is a semantic quotient, not a claim that every social or linguistic meaning is exhausted by formal quotienting.

3 Definition: meaning as actionable semantic quotient

Definition 1 (Semantic task family). A semantic task family Ψ_{sem} is the set of distinctions, contexts, actions, outcomes, policies, verification statuses, and boundary variables that must be preserved for a finite system to use a representation within tolerance.

Definition 2 (Semantic quotient). A semantic quotient is a map $q_{\text{sem}} : \mathcal{D}_t \rightarrow \mathcal{T}_t$ from fine distinctions into semantic types. Two distinctions are quotient-equivalent when the system treats them as the same type for a specified downstream task.

Definition 3 (Actionable semantic quotient). A semantic quotient is actionable if there exists a quotient-level policy

$$\pi_q : \mathcal{T}_t \times \mathcal{C} \rightarrow \mathcal{A} \quad (7)$$

such that using $q_{\text{sem}}(d)$ preserves downstream policy, prediction, verification, coordination, or boundary-maintenance performance within tolerance over the context family $\mathcal{C}$.

Definition 4 (Policy-preserving semantic quotient). Let π^* be the fine-grained reference policy for a task, π_q a quotient-level policy, $c \in \mathcal{C}$ a context, k a horizon, and ϵ_π a tolerance. The semantic quotient q_{sem} is policy-preserving if the positive part of the policy loss is bounded:

$$\Delta_\pi^+(d, c) = [\mathbb{E}[\ell_{\text{maint}, t+k} \mid \text{do}(\pi_q(q_{\text{sem}}(d), c))] - \mathbb{E}[\ell_{\text{maint}, t+k} \mid \text{do}(\pi^*(d, c))]]_+ \leq \epsilon_\pi \quad (8)$$

for the audited domain of distinctions and contexts.

Equation (8) is a causal audit form. It says that replacing the fine distinction d by its quotient type should not increase future boundary-maintenance loss beyond tolerance. The positive part prevents improvements by the quotient policy from being counted as semantic loss. In some domains the policy output may be a prediction, verification routine, communication act, record update, or externalization decision rather than a motor action.

Remark 1 (Reference policy provenance). The reference policy π^* need not be globally optimal. It may be an experimentally validated fine-grained policy, a high-resolution simulator, an expert-labeled policy, a ground-truth task rule, a stronger verification baseline, or a pre-registered policy used for audit. M3 requires the reference policy to be specified; it does not require it to be metaphysically final.

Definition 5 (Value-preserving semantic quotient). A semantic quotient is value-preserving when quotient-equivalent distinctions do not differ in a way that changes the induced FDS-value ranking or policy. A conservative sufficient condition is

$$q_{\text{sem}}(d_i) = q_{\text{sem}}(d_j) \Rightarrow |V_t^{\text{net}}(d_i; k, c) - V_t^{\text{net}}(d_j; k, c)| \leq \epsilon_V, \quad (9)$$

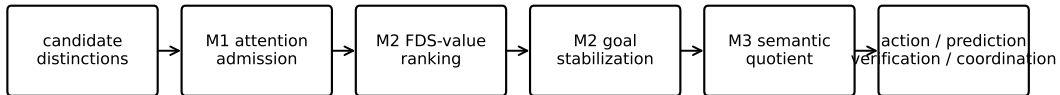
for contexts $c \in \mathcal{C}$. A more action-oriented condition is that quotient-equivalent distinctions induce policies equivalent within ϵ_π .

Definition 6 (FDS-meaning). A distinction has FDS-meaning for a system when it is represented through a semantic quotient that preserves enough policy, value, prediction, verification, coordination, or boundary-maintenance structure for the system to use it under finite capacity.

Definition 7 (Semantic preservation vector). A semantic audit can record preservation losses as a vector

$$\mathbf{L}_{\text{sem}}(q) = (L_\pi, L_V, L_{\text{pred}}, L_{\text{ver}}, L_{\text{coord}}, L_\ell), \quad (10)$$

where the components measure policy loss, FDS-value ranking loss, prediction loss, verification-status loss, coordination loss, and boundary-loss effect. Different domains may select different components: tool use emphasizes L_π and L_{ver} ; science emphasizes L_{pred} and L_{ver} ; law and institutions emphasize L_{coord} and L_ℓ ; biological regulation emphasizes L_ℓ .



Meaning enters when compression preserves downstream relevance

Figure 1: Synthetic normal-form illustration, not empirical evidence. M3 sits after M1 attention admission and M2 FDS-value ranking. Meaning enters when compression preserves downstream relevance for action, prediction, verification, coordination, or boundary maintenance.

3.1 Minimal M3 audit

A minimal M3 audit must specify: boundary, task family, context family, fine distinctions, semantic quotient, reference policy or verification target, preservation metric, tolerance, semantic load, capacity budget, and failure condition. Without these, claims about meaning remain under-specified. In slogan form: M2 ranks what matters for boundary maintenance; M3 asks which quotient preserves what matters under finite capacity.

4 Policy-preserving compression

A quotient partitions the fine distinction space into equivalence classes. It reduces semantic load only if it allows the system to avoid maintaining irrelevant fine distinctions. But a quotient is dangerous if it merges distinctions that require different downstream policies.

Definition 8 (Semantic loss). For a quotient q_{sem} and context family $\mathcal{C}$, semantic policy loss is

$$L_{\text{sem}}(q_{\text{sem}}) = \mathbb{E}_{d,c} [\mathbb{E}[\ell_{\text{maint},t+k} \mid \text{do}(\pi_q(q_{\text{sem}}(d), c))] - \mathbb{E}[\ell_{\text{maint},t+k} \mid \text{do}(\pi^*(d, c))]]_+. \quad (11)$$

The expectation is over the audited distribution of distinctions and contexts. The positive part means that improvements are not counted as semantic loss.

Definition 9 (Maintained semantic load). The maintained semantic load of a quotient is the capacity required to store, retrieve, verify, update, communicate, coordinate, and use it over the task window. A simple scalarized normal form is

$$C_{\text{sem}}(q) = C_{\text{store}}(q) + C_{\text{retrieve}}(q) + C_{\text{verify}}(q) + C_{\text{update}}(q) + C_{\text{coordinate}}(q). \quad (12)$$

In applications these terms may remain as a vector or be converted into task-specific capacity-equivalent units. $C_{\text{sem}}(q)$ is not raw memory size alone. When physically implemented in an active finite system, maintaining a semantic quotient may require storage, retrieval, refresh, verification, and error-correction costs. If a quotient’s maintenance cost exceeds the policy loss it saves for boundary maintenance, the system may discard it through forgetting or demotion from meaning back to salience.

Definition 10 (Minimal sufficient semantic quotient). A semantic quotient q_{sem}^* is minimal sufficient for task family Ψ_{sem} at tolerance ϵ if it minimizes maintained semantic load while satisfying the task loss constraint:

$$q_{\text{sem}}^* \in \arg \min_q C_{\text{sem}}(q) \quad \text{s.t.} \quad L_{\text{sem}}(q) \leq \epsilon. \quad (13)$$

Proposition 1 (Compression is not meaning). *A compressed representation is not FDS-meaningful merely because it reduces dimensionality. It becomes FDS-meaningful only when the quotient preserves downstream action, prediction, verification, coordination, or boundary relevance within tolerance.*

Proof sketch. Let q compress distinctions but merge d_i, d_j with different reference policies in context c . If $\pi^*(d_i, c) \neq \pi^*(d_j, c)$ and the quotient policy cannot choose both, at least one case incurs nonzero policy error. Thus compression alone does not imply low $L_{\text{sem}}(q)$. $\square$

Proposition 2 (Policy preservation implies actionable meaning). *If a semantic quotient satisfies Eq. (8) across the audited distinction-context domain, then it supplies actionable FDS-meaning for that task family.*

Proof sketch. Under Eq. (8), replacing d by $q_{\text{sem}}(d)$ changes future boundary-maintenance loss by at most ϵ_π for all audited contexts. Therefore the quotient-level type can guide downstream use with bounded loss. $\square$

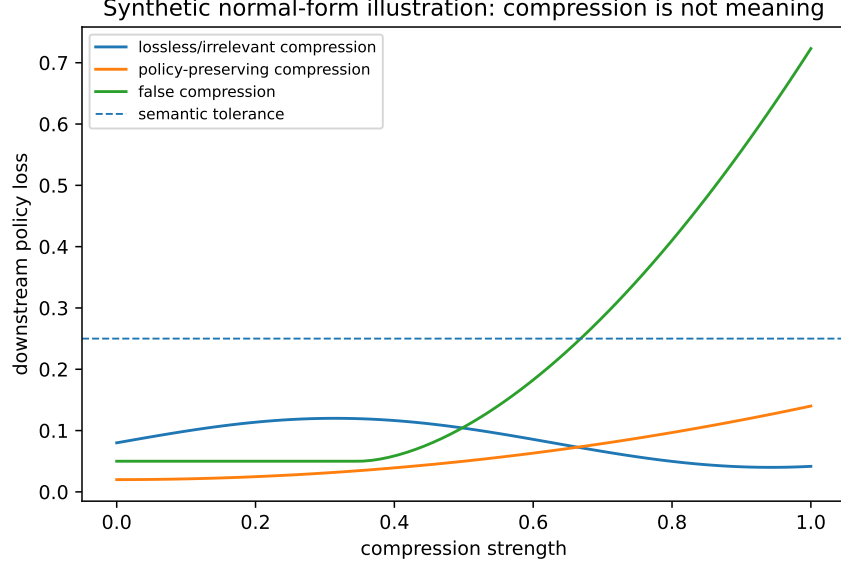


Figure 2: Synthetic normal-form illustration, not empirical evidence. Compression alone is not meaning. False compression can reduce representation size while increasing downstream policy loss. Policy-preserving compression remains within semantic tolerance.

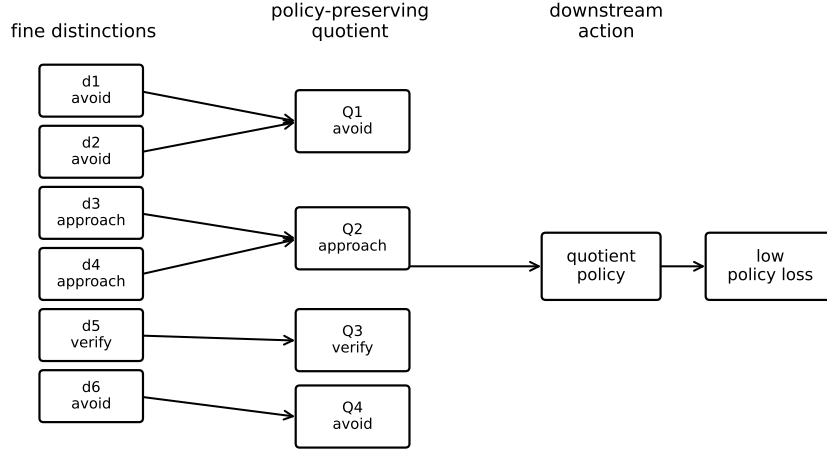


Figure 3: Synthetic normal-form illustration, not empirical evidence. A policy-preserving quotient groups fine distinctions only when the quotient class supports the same downstream policy or boundedly equivalent actions.

5 Semantic capacity and semantic deficit

Let $R_{\text{sem}}^{(\tau)}(\epsilon; \Psi_{\text{sem}}, t)$ be the rate-distortion demand of preserving task-relevant semantic structure over update window τ at tolerance ϵ . Let $C_{\text{sem}}(t)$ be maintained semantic capacity. Define

$$\Delta_{\text{sem}}(t) = R_{\text{sem}}^{(\tau)}(\epsilon; \Psi_{\text{sem}}, t) - C_{\text{sem}}(t). \quad (14)$$

When $\Delta_{\text{sem}} > 0$, the system cannot maintain the semantic distinctions required by the task family at target tolerance unless it changes something: compresses better, externalizes records,

relaxes task demands, expands resources, substitutes proxies, merges meanings, or drifts.

Proposition 3 (Semantic deficit forces semantic exit). *Under sustained semantic deficit, no hidden capacity expansion, and continued task demand, a finite system must compress, externalize, relax the task, substitute proxies, refine invariants, or suffer semantic drift/collapse.*

Proof sketch. By definition, $R_{\text{sem}}^{(\tau)}(\epsilon; \Psi_{\text{sem}}, t) > C_{\text{sem}}(t)$. Maintaining all task-relevant semantic distinctions at tolerance ϵ is infeasible unless effective demand is reduced, effective capacity is increased, or the system accepts distortion. The listed exits correspond to those alternatives. $\square$

5.1 Semantic load versus physical memory load

Semantic capacity is not identical to raw memory size. A large memory can fail semantically if it lacks retrieval, verification, or task-preserving quotient structure. A small symbol can be highly meaningful if it selects the correct action class. M3 therefore distinguishes raw information volume from maintained actionable structure.

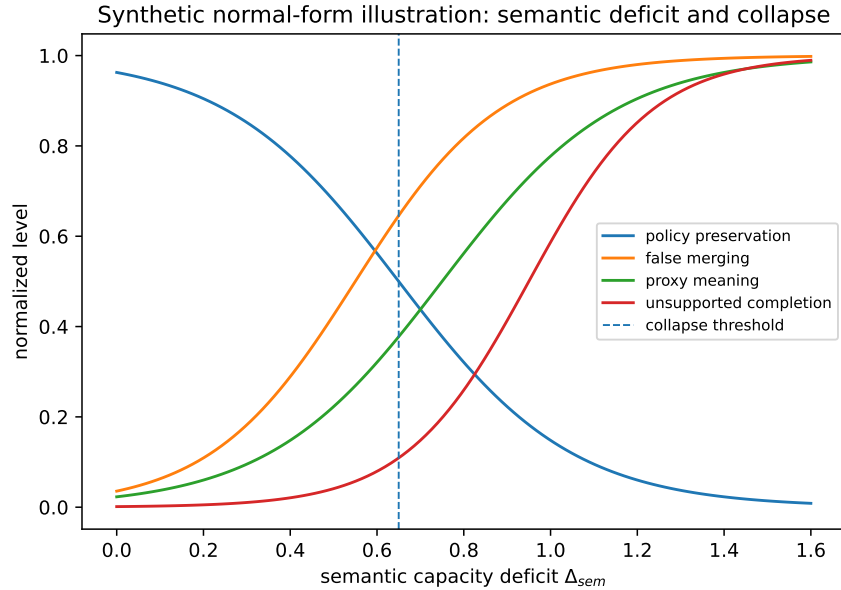


Figure 4: Synthetic normal-form illustration, not empirical evidence. As semantic capacity deficit increases, policy preservation falls while false merging, proxy meaning, and unsupported completion rise.

6 False meaning and semantic collapse

Definition 11 (False compression). A quotient produces false compression when it maps distinctions to the same semantic type even though those distinctions require different policies, values, verification statuses, or coordination responses under the audited context family:

$$q_{\text{sem}}(d_i) = q_{\text{sem}}(d_j) \quad \text{but} \quad \pi^*(d_i, c) \not\approx \pi^*(d_j, c) \quad (15)$$

for some relevant context c .

False compression is a central M3 failure mode. It can appear as a misleading category, an over-broad slogan, a brittle embedding cluster, an invalid analogy, or a learned proxy that preserves surface similarity while losing policy relevance. A false quotient can remain locally harmless if it does not immediately trigger boundary loss, but it becomes dangerous when later decisions reuse it without re-verification.

Definition 12 (Unsupported completion). Unsupported completion occurs when a system fills missing semantic structure with high-probability or proxy-consistent content that is not supported by the task-relevant verification channel.

An audit metric for unsupported completion is

$$UCR = P(\text{completion accepted} \wedge \text{verification unsupported} \wedge \text{policy/coordination relevant}), \quad (16)$$

or, conditioned on completion being accepted,

$$UCR_{\text{cond}} = P(\text{verification unsupported} \wedge \text{policy/coordination relevant} \mid \text{completion accepted}). \quad (17)$$

The conditional form is useful when completion events are rare: it isolates the risk of unsupported completion among accepted completions. In loss form,

$$L_{\text{unsupported}}^+ = [\mathbb{E}[\ell_{t+k} \mid \text{do}(\text{unsupported completion})] - \mathbb{E}[\ell_{t+k} \mid \text{do}(\text{verified representation})]]_+. \quad (18)$$

Unsupported completion is the M3 normal-form analogue of hallucination-like behavior. It is not limited to language models. Organizations, institutions, and individuals also complete missing structure with defaults, stereotypes, slogans, or inherited narratives.

Proposition 4 (False compression produces semantic loss). *If a quotient merges distinctions requiring different policies under an audited context, then it produces positive semantic policy loss unless the policy difference is irrelevant to the loss at the specified horizon.*

Proof sketch. If $q_{\text{sem}}(d_i) = q_{\text{sem}}(d_j)$, the quotient policy receives the same type in both cases. If the reference policies differ and the difference matters for $\ell_{\text{maint},t+k}$, one quotient action cannot match both reference actions. Therefore Eq. (11) is positive for at least one case. $\square$

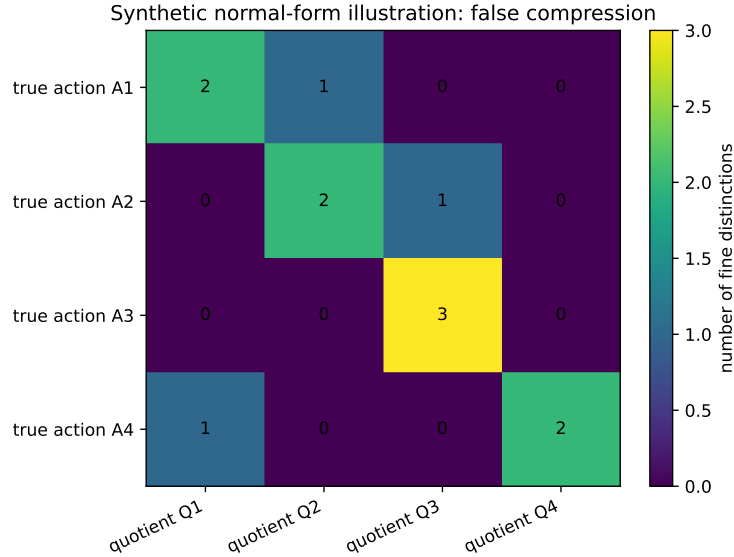


Figure 5: Synthetic normal-form illustration, not empirical evidence. A false semantic quotient groups fine distinctions from multiple true action classes into the same quotient class. The resulting type is compact but policy-ambiguous. Off-diagonal mass indicates quotient classes that merge distinctions requiring different true actions.

6.1 Meaning collapse

Under overload, meaning can collapse backward down the agency-semantics ladder. A semantic quotient can collapse into salience: the system keeps what stands out, not what preserves action. It can collapse into proxy meaning: the system preserves the signal that predicts reward, not the structure that preserves host boundary value. It can collapse into slogan: the system maintains an externalized phrase while losing the distinction structure that once made it useful.

6.2 Institutional semantic drift

In large organizations, persistent semantic synchronization overload can make high-level terms stable while their action-preserving quotient structure degrades. This is institutional semantic drift: lexical unity with policy incompatibility. A term, rule, metric, or slogan continues to circulate, but different subunits attach incompatible quotient maps to it. In audit terms, the phrase remains shared while L_{mcoord} , L_{mver} , or L_π rises. This is a conditional failure mode, not a universal law of institutions: it occurs only when the relevant bridge mapping shows persistent over-compression, insufficient verification, and degraded downstream coordination.

7 Embedding similarity is not sufficient for meaning

Artificial systems often use embedding similarity, vector distance, or latent representation geometry. These can be useful, but M3 treats them as candidate representational relations rather than meaning itself.

Proposition 5 (Embedding-policy dissociation). *Two representations can be close in embedding space yet differ in policy, value, or verification relevance. Conversely, two representations can be far in embedding space but equivalent under the audited semantic quotient.*

Proof sketch. Embedding similarity is a metric relation in a representation space. Policy equivalence is a downstream relation defined by action, prediction, verification, coordination, or boundary loss. Unless the embedding metric is trained and audited to preserve that downstream relation over the context family, the two relations can diverge. $\square$

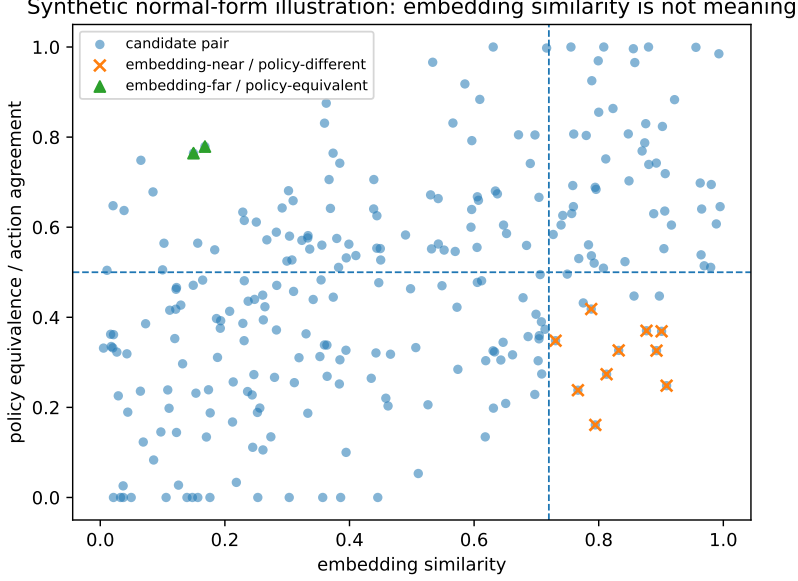


Figure 6: Synthetic normal-form illustration, not empirical evidence. Embedding similarity and policy equivalence can diverge. Embedding-near pairs can require different actions; embedding-far pairs can be equivalent under the audited task.

8 Meaning across horizons and contexts

Meaning is horizon-relative. A quotient that preserves action over one update window may fail over a longer horizon. A category that works in a training context may false-compress in a deployment context. A social symbol that coordinates a small group may fail in a larger institution.

Definition 13 (Context-indexed semantic quotient). A context-indexed semantic quotient is a family

$$\{q_{\text{sem}}^c : \mathcal{D} \rightarrow \mathcal{T}_c\}_{c \in \mathcal{C}} \quad (19)$$

with context-dependent equivalence classes. It is valid when each quotient preserves downstream relevance for its context and when context switching itself is within the system’s capacity budget.

Definition 14 (Invariant semantic quotient). A semantic quotient q^* is an invariant semantic quotient over context family $\mathcal{C}$ and perturbation family $\mathcal{P}$ if it remains policy- or value-preserving under the stated contexts and perturbations. Depending on audit strictness, invariant preservation may be defined in expectation, high-probability, or worst-case form:

$$\mathbb{E}_{c \sim \mathcal{C}, p \sim \mathcal{P}} [L_{\text{sem}}^{c,p}(q^*)] \leq \epsilon, \quad (20)$$

$$P_{c \sim \mathcal{C}, p \sim \mathcal{P}} (L_{\text{sem}}^{c,p}(q^*) \leq \epsilon) \geq 1 - \delta, \quad (21)$$

$$\sup_{c \in \mathcal{C}, p \in \mathcal{P}} L_{\text{sem}}^{c,p}(q^*) \leq \epsilon. \quad (22)$$

This definition connects M3 to M2’s high-level goals as invariant-compression candidates. High-level meanings such as danger, evidence, obligation, freedom, justice, or truth are not validated by their abstraction alone. They are candidate invariant semantic quotients: compact representations that preserve action, value, verification, or coordination relevance across broad context families. This does not morally validate any high-level meaning. It only states that some abstractions may function as stable quotient candidates when they preserve value, verification, or coordination relevance across contexts.

A useful metric for comparing invariant candidates is *semantic half-life*: the time or context count over which a semantic quotient maintains policy preservation within tolerance without refinement. Physical constants and mathematical theorems can have long semantic half-lives across broad classes of contexts, relative to the specified task family. Causal regularities such as fire-damage or poison-illness have long half-lives within their ecological niche. Trends, slogans, and viral phrases may have short half-lives: their policy relevance degrades rapidly as context drifts. Semantic half-life is not a claim that some meanings are permanently valid; it is an audit variable that measures how long a quotient survives before needing refinement.

9 Shared meaning and translation

Shared meaning is not identical private experience. It is quotient alignment sufficient for coordination, verification, or shared action.

Definition 15 (Shared FDS-meaning). For agents or subsystems i and j , shared FDS-meaning for distinction d under context family $\mathcal{C}$ requires quotient-policy alignment:

$$\pi_i(q_i(d), c) \approx_\epsilon \pi_j(q_j(d), c) \quad (23)$$

for relevant contexts $c \in \mathcal{C}$, or equivalent alignment in prediction, verification, coordination, or boundary-maintenance outcome. The relation $\approx_\epsilon$ must be instantiated: it may mean action agreement, bounded boundary-loss difference, bounded Jensen-Shannon divergence between stochastic policies, verification-status agreement, or coordination-success equivalence.

Definition 16 (Semantic synchronization load). Let $R_{\text{sem-sync}}^{(\tau)}(\epsilon; t)$ be the rate-distortion demand of keeping semantic quotients aligned across agents, records, or institutions within tolerance ϵ over window τ . Let

$$C_{\text{sync}}(t) = B(C_{\text{comm}}(t), C_{\text{verify}}(t)) \quad (24)$$

be a domain-specified synchronization bottleneck, such as a sum, minimum, weighted bottleneck, or queueing capacity after converting communication and verification into semantic-rate-equivalent units. Define

$$Z_{\text{sem-sync}}(t) = \frac{R_{\text{sem-sync}}^{(\tau)}(\epsilon; t)}{C_{\text{sync}}(t)}. \quad (25)$$

When $Z_{\text{sem-sync}} > 1$, shared meaning becomes unstable unless the system simplifies, externalizes, verifies, delegates, or accepts semantic drift.

Proposition 6 (Shared meaning requires quotient synchronization). *Collective meaning fails when agents cannot synchronize semantic quotients within the communication and verification budget required for coordination.*

Proof sketch. If $R_{\text{sem-sync}}^{(\tau)}(\epsilon; t) > C_{\text{sync}}(t)$, the system cannot maintain quotient alignment at tolerance ϵ over the window. Then Eq. (23) fails for some contexts unless the task is simplified or externalized. $\square$

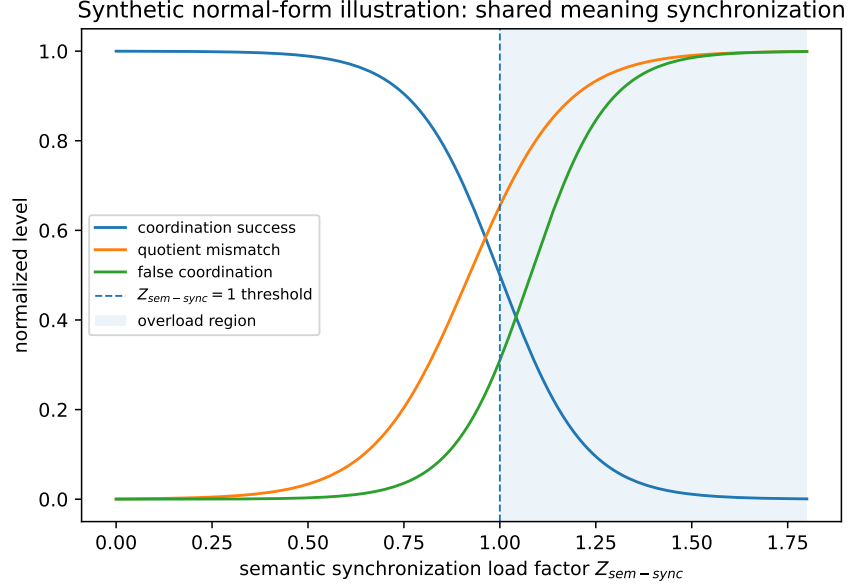


Figure 7: Synthetic normal-form illustration, not empirical evidence. When semantic synchronization demand exceeds synchronization capacity, quotient mismatch and false coordination increase while coordination success declines. The dashed threshold marks $Z_{\text{sem-sync}} = 1$; the nonlinear falloff is illustrative. Beyond $Z_{\text{sem-sync}} \approx 1.5$, surface-level lexical agreement often masks divergent policy maps – a false-consensus regime where agents share a term but attach incompatible quotient structures to it.

10 Meaning recovery

Meaning recovery is not simply adding more data. A system can drown in raw information while still lacking the quotient structure that preserves action. Recovery requires rebuilding distinctions that were falsely merged, adding verification, externalizing stable records, changing the task, or refining context-specific quotient classes.

Definition 17 (Meaning recovery). Meaning recovery is the restoration of policy, value, prediction, verification, or coordination preservation after semantic loss, usually through quotient refinement, externalized records, verification, task relaxation, or resource expansion.

Proposition 7 (Raw information restoration is not sufficient). *Restoring raw information volume does not necessarily restore FDS-meaning if the semantic quotient remains false, misaligned, or unverifiable.*

Proof sketch. FDS-meaning depends on downstream preservation, not raw volume. If the quotient still merges distinctions requiring different policies, Eq. (11) remains positive even if more data are present. $\square$

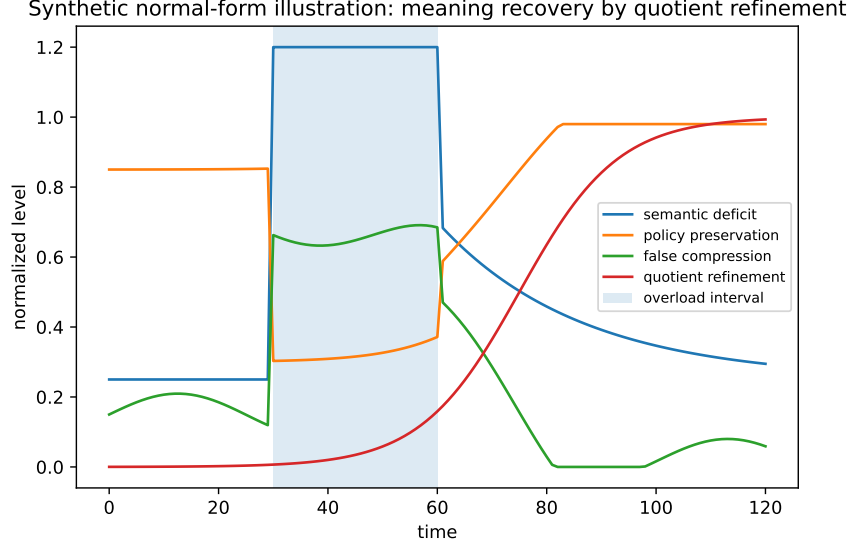


Figure 8: Synthetic normal-form illustration, not empirical evidence. After overload, policy preservation may recover only when quotient refinement splits false classes or adds verification. Merely reducing semantic deficit need not immediately restore meaning.

11 Domain audit examples

Table 3: Audit templates, not empirical confirmations. Each row requires domain-specific bridge assumptions and tests.

Domain	Semantic quotient	Preserved structure	Failure mode	Failure test
Biological regulation	chemical cue class	regulatory response / homeostasis	false cue, maladaptive trigger	perturb cue class and measure regulatory error
Cognition	concept category	action, prediction, retrieval	overgeneralization, stereotype, confusion	replace category and measure action or retrieval loss
LLM / AI system	embedding or latent type	tool use, answer, verification	hallucination, false friend, prompt proxy	test embedding-near cases requiring different tool actions
Organization	dashboard category / KPI class	policy and resource allocation	metric gaming, slogan collapse	audit KPI categories against downstream policy loss
Science	construct / method / evidence type	replication, prediction, correction	citation proxy, false theory class	test construct substitution under replication or prediction
Law / institution	legal category / rule	dispute resolution, coordination	rigid rule, loophole, category error	audit edge cases where one rule class induces incompatible remedies
Civilization	archive / standard / norm	cross-generational coordination	semantic drift, lost standard, polluted record	compare archived standard across generations and recovery tasks

12 Normal-form model and reproducibility

The accompanying code implements deterministic synthetic normal-form illustrations. The model is not empirical evidence. It is a consistency and visualization device that maps the definitions to simple state variables.

The model generates fine distinctions, latent policy labels, embedding similarity, semantic quotient classes, FDS-value proxies, semantic capacity deficit, policy preservation, false compression, shared-meaning synchronization load, and quotient-refinement recovery. It implements

policy-preserving quotienting, embedding-policy dissociation, capacity-driven semantic collapse, false compression, shared-meaning failure, and recovery by quotient refinement. The random seed is fixed in `code/generate_results.py`. CSV outputs are stored in `data/`; figure pairs are stored in `figures/`.

Table 4: Normal-form variable map. All entries are illustrative, not fitted empirical quantities.

Simulation variable	Paper definition	Interpretation
fine distinction	$d_i \in \mathcal{D}$	task-level difference before quotienting
semantic type	$q_{\text{sem}}(d_i)$	quotient class used by the system
policy label	$\pi^*(d_i, c)$	reference downstream action/update
policy preservation	Eq. (8)	quotient action agreement
semantic loss	Eq. (11)	boundary loss from quotienting
semantic deficit	Eq. (14)	maintained semantic capacity shortfall
false compression	Eq. (15)	quotient class merges policy-different items
embedding similarity	representation metric	not sufficient for FDS-meaning
semantic synchronization	Eq. (25)	shared meaning load factor
quotient refinement	recovery intervention	splitting false classes / adding verification

13 Protocols and tests

Protocol 1 (Semantic quotient audit). Pre-specify boundary, task family, context family, action or policy space, loss, horizon, candidate distinctions, semantic quotient, tolerance, verification channel, and failure condition. Estimate Δ_π^+ , L_{sem} , semantic preservation vector, false-compression rate, unsupported-completion rate, and semantic load.

Protocol 2 (Policy-preservation test). Replace fine distinctions with quotient types and test whether downstream action, verification, coordination, or boundary loss changes beyond tolerance.

Protocol 3 (Compression-benefit test). Compare semantic load reduction against policy or boundary-loss increase. A useful quotient lowers maintained semantic demand more than it increases downstream loss.

Protocol 4 (False-compression test). Identify quotient classes containing distinctions that require different policies, different verification statuses, or different FDS-value rankings. Audit whether the merge causes boundary loss.

Protocol 5 (Embedding-policy dissociation test). Compare embedding similarity with policy equivalence. M3 predicts that embedding closeness alone is insufficient for FDS-meaning when action, verification, or coordination demands differ.

Protocol 6 (Shared meaning audit). For multiple agents, institutions, or systems, compare quotient maps, policy outputs, verification routines, and coordination outcomes under shared contexts.

Protocol 7 (Semantic synchronization stress test). Increase communication load or reduce verification capacity. Prediction: shared meaning degrades when semantic synchronization demand exceeds the specified synchronization bottleneck C_{msync} .

Protocol 8 (Meaning recovery test). After semantic collapse or false compression, restore meaning by quotient refinement, verification, externalized records, task relaxation, or resource expansion. Test whether policy preservation improves.

14 Relation to existing fields

Information theory and rate distortion. M3 inherits the rate-distortion discipline of finite representation from Shannon information theory and modern information theory [5, 6]. Semantic deficit is not a raw data deficit; it is a task-relative preservation deficit.

Information bottleneck and sufficient statistics. The information bottleneck compresses input while preserving relevance to a target [7]. M3 is compatible with that form but specifies the target as action, prediction, verification, coordination, or boundary-maintenance relevance in an active finite system.

State abstraction and bisimulation. Reinforcement-learning state abstraction and bisimulation study when states can be grouped without changing decision performance [8, 9]. M3 treats semantic quotients as a more general bridge: the quotient may preserve policy, FDS-value, verification, or coordination rather than only reward dynamics.

Causal abstraction and representation learning. Recent causal state abstraction and causal representation work addresses when compressed or higher-level variables preserve causal structure relevant to intervention [10, 11]. M3 aligns with that direction by requiring intervention-audited downstream preservation.

Semantic communication. Goal-oriented semantic communication transmits task-relevant meaning rather than raw bits [12, 13]. M3 provides a finite-boundary language for when such compression is meaningful: it must preserve downstream use under capacity and verification constraints.

Linguistic semantics and pragmatics. M3 does not replace truth-conditional semantics, pragmatics, semiotics, or use-theoretic meaning. Peirce and Morris already emphasized sign relations and use-contexts [14, 15]. M3 contributes an operational finite-system audit: what quotient is preserved, for which task, under which capacity budget? Gricean maxims can be reinterpreted in FDS terms as constraints on efficient quotient transfer: provide enough structure for the receiver’s policy or verification update, avoid overloading the channel, avoid unsupported completions, and preserve distinctions relevant to coordination [19].

Embeddings, LLMs, and hallucination. Embedding spaces and transformer attention are useful representational machinery, but neither embedding proximity nor internal routing is sufficient for FDS-meaning. The quotient must preserve downstream policy or verification structure. Unsupported completion occurs when a system fills missing semantic structure with high-probability but unverified content. Recent surveys of hallucination mitigation emphasize grounding, retrieval, feedback, and verification mechanisms, which correspond in M3 to reducing unsupported completion and restoring verification-preserving quotients [20].

15 Limitations and falsification

M3 is intentionally limited. It does not define all linguistic, subjective, or moral meaning. It provides mappings that must survive operational audit. The framework is weakened or demoted

under any of the following results:

1. compressed representations function semantically under a specified mapping while preserving no action, prediction, verification, coordination, or boundary-relevant structure;
2. policy-preserving quotient tests systematically fail to predict downstream use under domains where M3 is claimed to apply;
3. embedding similarity always suffices for policy, value, and verification equivalence under audited tasks;
4. semantic deficit produces no measurable merging, drift, proxy substitution, unsupported completion, or semantic loss under capacity stress;
5. shared meaning persists without quotient alignment, external records, verification, translation, or coordination channels;
6. meaning recovery always follows from raw information increase without quotient reconstruction;
7. high-level meanings fail to preserve policy, value, or coordination relevance across any stated context or perturbation family.

16 Conclusion

M3 defines FDS-meaning as a finite-system quotient operation. FDS-meaning is not raw information, embedding similarity, moral value, or linguistic meaning in full. It is actionable semantic quotient: compression that preserves downstream action, prediction, verification, coordination, or boundary-maintenance relevance under specified capacity and task constraints.

The central chain is

attended distinction $\rightarrow$ FDS-value ranking $\rightarrow$ goal orientation $\rightarrow$ semantic quotient $\rightarrow$ meaning
 $\rightarrow$ action / prediction / verification / coordination. (26)

M2 ranks what matters for boundary maintenance; M3 specifies which compression preserves what matters. M3 prepares later work. M5 can treat trust as delegated semantic verification. A2 can audit whether artificial systems false-compress host-relevant distinctions. S2 can analyze epistemic pollution as attacks on shared semantic quotients. G3 can treat science as quotient refinement through replication and error correction. G4 can treat archives, standards, and protocols as civilization-scale semantic synchronization infrastructure.

Code and data availability

The deterministic synthetic normal-form code, figures, and CSV outputs are included in the replication package. Run `python code/generate_results.py` from the paper directory to regenerate all figures and data.

AI assistance disclosure

The author used AI assistance for drafting, editing, simulation scaffolding, and consistency checks. The author reviewed and selected the final claims, definitions, equations, and interpretations.

A M-series dependency map

Table 5: How M3 supports later M-series and civilization-layer papers.

Future paper	Interface supplied by M3
M5 Trust	Trust reduces repeated semantic verification and allows quotient reuse.
A2 AI Alignment	Artificial agents must not false-compress host-relevant distinctions.
S2 Epistemic Pollution	Pollution attacks shared semantic quotients and verification bandwidth.
S3 Institutional Collapse	Institutions can fail through semantic drift, slogan collapse, and false compression.
G3 Science	Scientific method refines semantic quotients through replication and error correction.
G4 Civilization Memory	Archives and standards stabilize semantic quotients across time.
M11 Narrative	Narratives compress causal and value structure for transmission.

B Notation summary

Table 6: Notation used in M3.

Symbol	Meaning
S	active finite distinction system
$\mathcal{D}_t$	candidate or admitted distinction set at update window t
$\mathcal{T}_t$	semantic type space
q_{sem}	semantic quotient map $\mathcal{D}_t \rightarrow \mathcal{T}_t$
Ψ_{sem}	semantic task family
π^*	fine-grained reference policy
π_q	quotient-level policy
$\mathcal{C}$	context family
ℓ_{maint}	boundary-maintenance loss
$\Delta_{\pi}(d, c)$	policy loss from using quotient representation
$L_{\text{sem}}(q)$	semantic policy loss of quotient q
$C_{\text{sem}}(t)$	maintained semantic capacity
$R_{\text{sem}}^{(\tau)}$	rate-distortion demand for semantic task preservation
$\Delta_{\text{sem}}(t)$	semantic capacity deficit
$V_t^{\text{net}}(z; k)$	net FDS-value from M2
$\epsilon_{\pi}, \epsilon_V$	policy and value preservation tolerances
$Z_{\text{sem-sync}}$	semantic synchronization load factor
$C_{\text{comm}}, C_{\text{verify}}$	communication and verification capacity
Σ_{phys}	physical entropy ledger, when bridge assumptions apply

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Active Pruning Controls Boundary Persistence in Protocell-like Systems: Saddle-Node Attractor Loss, Stochastic Early Warnings, Multiscale Simulations, and Wet-Lab Benchmark Predictions

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Finite protocell-like systems under sustained flux must maintain boundary integrity, transport throughput, and internal functional availability while accumulating nonfunctional residue. Here we develop an analytical and computational framework for testing whether active pruning - selective, resource-coupled removal, export, degradation, or reorganization of residue - controls the persistence-collapse transition in finite compartments. The framework combines five layers: (i) a well-mixed residue-pruning-boundary model with explicit threshold bounds; (ii) a saddle-node normal-form reduction showing how a maintenance attractor can disappear when pruning falls below a critical value; (iii) a stochastic perturbation analysis deriving critical slowing down, variance growth, and lag-one autocorrelation increase near attractor loss; (iv) two spatial failure modes, reaction-diffusion clogging for droplet-like or porous compartments and phase-field-inspired reduced boundary deformation for vesicle-like compartments; and (v) a radial-shell size constraint showing that boundary-limited pruning obeys the areal lower bound $s_{m,c}(R) \geq qR/3$, with diffusion delay raising the effective numerical threshold above this mass-balance limit. Numerical simulations reproduce active-pruning thresholds, spatial clogging, residue-driven boundary deformation with local integrity loss, rescue-window closure, radius-dependent pruning demand, and robustness across randomized production and diffusion-arrest parameters. The paper does not claim wet-lab validation or a chemically complete origin-of-life mechanism. Its contribution is a chemically coarse-grained universality-class model and benchmark map: finite compartments satisfying these assumptions—flux generates residue, residue impairs transport or boundary integrity, active removal requires finite resources—are predicted to exhibit pruning-dependent persistence, collapse, and rescue thresholds. This is a theory-computation manuscript: the wet-lab section provides experimentally measurable benchmark predictions, not completed experimental validation.

I. INTRODUCTION

Replication, autocatalysis, compartmentalization, and energy transduction are central to origin-of-life research. Yet a finite compartment under sustained flux faces a maintenance problem that is logically prior to any specific molecular solution: useful flux also generates nonfunctional residue. Residue may be inert byproduct, polymerized tar, aggregate, blocked catalytic site, membrane-clogging material, damaged component, or unavailable internal volume. If residue accumulates faster than it can be diluted, degraded, exported, recycled, or reorganized, the system may lose transport, boundary integrity, and functional throughput even while energy continues to enter.

This paper studies *active pruning* as a control process for that problem. Active pruning denotes selective, resource-coupled removal, export, neutralization, degradation, or reorganization of residue while preserving the functional core of a finite boundary-maintaining system. The term is deliberately substrate-neutral. In a lipid vesicle, the relevant process may be export, leakage control, membrane-localized degradation, or recycling. In a coacervate droplet, it may be catalytic degradation or phase-selective residue removal. In a mineral pore, it may be surface refresh, desorption, or flow-mediated clearing. The common structure is not a particular molecule, but

a residue-pruning-boundary loop.

The central claim tested here is narrower than a general theory of life: in finite protocell-like systems under sustained flux and residue production, active pruning can act as a control parameter separating a boundary-maintaining nonequilibrium regime from residue-induced collapse. The present paper extends earlier minimal simulations by adding analytical thresholds, a saddle-node reduction for attractor loss, stochastic early-warning analysis, multiscale spatial mechanisms, size-scaling lower bounds, robustness sweeps, and a benchmark map for future wet-lab tests.

A. Universality-class framing

The model is chemically coarse-grained by design. Its claim is not that all protocells share one molecular pruning mechanism. Rather, it defines a universality class: any finite compartment in which (i) flux generates residue, (ii) residue impairs diffusion, transport, or boundary integrity, and (iii) residue removal is active, selective, and resource-limited should exhibit a pruning-dependent persistence-collapse transition. This is analogous in spirit to using coarse-grained models to study phase transitions: the variables abstract away molecular detail to expose control parameters and scaling laws.

Recent protocell research has increasingly moved beyond passive compartmentalization toward chemically programmable coacervate systems, enzymatic microreactors, membranzed coacervates with tunable interfacial stability, and active droplets in synthetic cells [15–19]. These developments make residue control, molecular transport, boundary integrity, and active nonequilibrium maintenance natural variables for a coarse-grained theory of protocell persistence.

B. Contributions

This paper makes nine contributions.

1. It formulates a residue-pruning-boundary model and derives an explicit target-load lower bound for a critical pruning rate S_c .
2. It gives a Jacobian-based saddle-node condition for the well-mixed reduction and shows how the local normal form produces maintenance-attractor loss.
3. It adds a stochastic differential-equation reduction and derives early-warning scaling: recovery time and variance increase while lag-one autocorrelation tends to one as the fold is approached.
4. It nondimensionalizes the theory into a small set of control numbers: pruning number, residue burden, diffusion-delay number, osmotic-stress number, and boundary-repair number.
5. It simulates spatial clogging in a two-dimensional reaction-diffusion model with residue-dependent diffusivity $D_\rho(\rho) = D_0 e^{-\lambda_D \rho}$.
6. It simulates osmotic/crowding-driven boundary deformation using a phase-field-inspired reduced radius-field model rather than claiming a full phase-field membrane solver.
7. It derives the three-dimensional boundary-limited areal lower bound $s_{m,c}(R) \geq qR/3$ and compares it with a radial-shell numerical threshold including diffusion delay.
8. It quantifies rescue-window closure and robustness under randomized parameter sweeps.
9. It provides a wet-lab benchmark map and a literature-informed plausibility check without claiming that wet-lab validation has already been performed.

C. What is not claimed

The paper does not identify the first historical pruning chemistry, does not reproduce a specific protocell, and does not provide experimental validation. It does

not define clinical death. It does not claim that collapse in every spatial PDE must be mathematically a saddle-node. The saddle-node analysis applies to a reduced well-mixed maintenance coordinate; spatial simulations then show how this reduced attractor-loss mechanism can be physically realized through clogging, boundary deformation, and size-limited export.

D. Falsifiable predictions

The following four conditions would falsify or demote specific claims in this paper:

1. **Sustained pruning threshold.** If systems under sustained residue production maintain bounded ρ , stable B , and stable J with $S = 0$, no hidden dilution, no externalization, and no passive degradation sufficient to offset production, the active-pruning-as-control-parameter claim is falsified.
2. **Rescue window closure.** If the rescue window does not close as restoration delay increases (i.e., no rescue boundary exists), the saddle-node attractor-loss reduction is demoted to a non-universal simulation observation.
3. **Radius-dependent pruning demand.** If the critical areal pruning/export threshold $s_{m,c}(R)$ does not increase with compartment radius R under boundary-limited clearing, the areal lower bound $s_{m,c}(R) \geq qR/3$ and its diffusion-delay correction are demoted.
4. **Early-warning signals.** If recovery time and lag-one autocorrelation fail to increase near the persistence-collapse transition under quasi-static approach to the fold, the stochastic early-warning analysis is demoted; variance is treated as an auxiliary indicator.

II. RESIDUE-PRUNING-BOUNDARY MODEL

A. Core variables

The coarse-grained state is

$$\mathcal{X}(t) = \{\rho(t), B(t), J(t), S(t), V(t)\}, \quad (1)$$

where ρ is residue load, $B \in [0, 1]$ is boundary integrity, J is transport or reaction throughput, S is active pruning rate, and V is a viability-like maintenance score. A convenient normalized score is

$$V(t) = \frac{B(t)J(t)}{1 + \rho(t)/\rho_0}. \quad (2)$$

Collapse is operationally defined by $V(t) < V_{\min}$ for a specified duration or, in rescue experiments, by failure to recover above V_{rec} after restoration of pruning within an observation window.

B. Well-mixed dynamics

The minimal well-mixed dynamics are

$$\frac{d\rho}{dt} = f(\rho, B; S) = pJ_0Be^{-\lambda_\rho\rho} - d\rho - S\frac{\rho}{K + \rho}, \quad (3)$$

$$\frac{dB}{dt} = g(\rho, B) = r_B B(1 - B) - \beta_\rho \rho B + \gamma_B(1 - B), \quad (4)$$

$$J(\rho, B) = J_0Be^{-\lambda_\rho\rho}. \quad (5)$$

Here p is residue generated per unit throughput, d is passive degradation or dilution, K is the half-saturation constant for pruning, β_ρ is residue-induced boundary damage, γ_B is repair, and λ_ρ controls transport inhibition by residue. Resource coupling is coarse-grained into the pruning parameters: S represents the maximal clearing rate set by available resources, and K sets the residue load at which clearing reaches half its maximum, capturing saturation under resource or throughput constraints. The exponential throughput term is a smooth representation of crowding or clogging; replacing it by another decreasing function does not change the threshold logic.

Proposition 1 (Pruning lower bound) *Fix residue ceiling $\rho_c > 0$ and target throughput J^* . If ρ is to remain locally non-increasing at ρ_c , then active pruning must satisfy*

$$S \geq S_c(\rho_c) = (pJ^* - d\rho_c)_+ \frac{K + \rho_c}{\rho_c}, \quad (6)$$

where $(x)_+ = \max(x, 0)$.

Proof. At $\rho = \rho_c$, requiring $\dot{\rho} \leq 0$ gives $pJ^* - d\rho_c - S\rho_c/(K + \rho_c) \leq 0$. Solving for S yields Eq. (6). $\square$

This bound is intentionally conservative. It does not prove global persistence, but it gives a direct analytical threshold that the simulations can test.

III. ANALYTICAL STRUCTURE OF ATTRACTOR LOSS

A. Jacobian and saddle-node condition

Let $z = (\rho, B)^T$ and $F(z; S) = (f, g)^T$. The Jacobian of Eqs. (3)-(4) is

$$\begin{aligned} J_{11} &= -pJ_0\lambda_\rho Be^{-\lambda_\rho\rho} - d - \frac{SK}{(K + \rho)^2}, \\ J_{12} &= pJ_0e^{-\lambda_\rho\rho}, \\ J_{21} &= -\beta_\rho B, \\ J_{22} &= r_B(1 - 2B) - \beta_\rho\rho - \gamma_B, \end{aligned} \quad (7)$$

At an equilibrium $z^*(S)$ satisfying $F(z^*; S) = 0$, a local saddle-node candidate occurs when

$$\det J_z(z^*; S_c) = 0, \quad \text{tr } J_z(z^*; S_c) < 0, \quad (8)$$

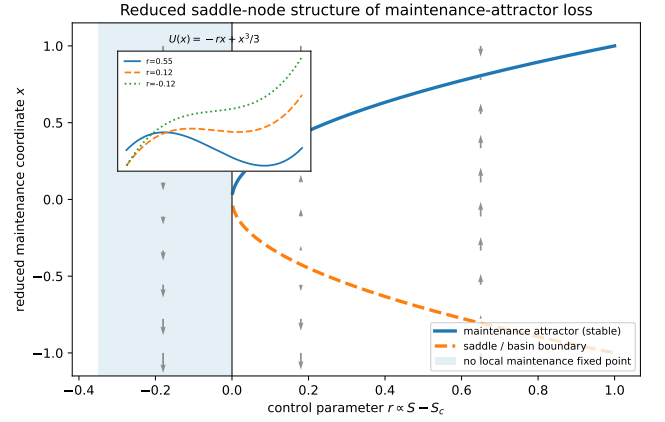


FIG. 1. Reduced saddle-node structure. The stable branch represents a maintenance attractor; the dashed branch represents the saddle or basin boundary. Arrows show the one-dimensional flow of $\dot{x} = r - x^2$. The inset shows the potential landscape $U(x) = -rx + x^3/3$, whose local minimum disappears at the fold.

with the usual non-degeneracy conditions

$$w^T F_S(z^*; S_c) \neq 0, \quad w^T D^2 F(z^*; S_c)[v, v] \neq 0, \quad (9)$$

where v and w are the right and left null vectors of J_z at the fold. Under these conditions, center-manifold reduction gives the normal form

$$\dot{x} = a(S - S_c) - bx^2 + O(|x|^3, |S - S_c||x|), \quad (10)$$

with $ab > 0$ after orientation choice.

Proposition 2 (Maintenance-attractor loss) *For parameters satisfying Eqs. (8) and (9), the well-mixed residue-pruning-boundary model admits a saddle-node loss of the maintenance attractor. For $S > S_c$, a stable maintenance branch and an unstable basin-boundary branch coexist locally. At $S = S_c$ they collide. For $S < S_c$, the local maintenance fixed point is absent and trajectories are forced toward the collapse basin unless rescued by an external nonlocal intervention.*

Figure 1 illustrates the reduced normal form, including the associated potential landscape $U(x) = -rx + x^3/3$ for $\dot{x} = -\partial_x U$. This is the analytical meaning of maintenance-collapse in the model: loss of access to a maintenance attractor in a finite artificial compartment.

B. Stochastic early-warning scaling

Near the fold, stochastic perturbations can be represented by the one-dimensional SDE

$$dx = (r - x^2)dt + \sigma dW_t, \quad (11)$$

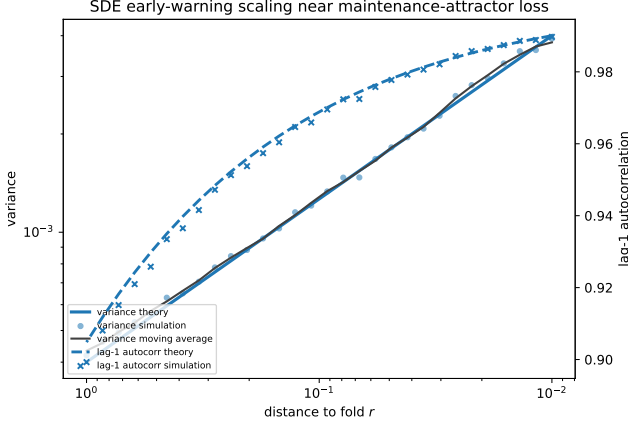


FIG. 2. SDE early-warning scaling. Theory and stochastic simulations of the normal-form reduction show increasing variance and lag-one autocorrelation as the distance to the fold r decreases. Variance points are shown with a moving average because variance is more sensitive to sample window and noise model than autocorrelation. The same eigenvalue gives the recovery-time divergence $\tau_{\text{rec}} \sim (2\sqrt{r})^{-1}$.

where $r > 0$ is distance to the fold. The stable fixed point is $x^* = \sqrt{r}$ and the linearized eigenvalue is

$$\lambda = f'(x^*) = -2\sqrt{r}. \quad (12)$$

Therefore the recovery time scales as

$$\tau_{\text{rec}} \sim \frac{1}{2\sqrt{r}}, \quad (13)$$

while the Ornstein-Uhlenbeck variance near the stable point is

$$\text{Var}(x) \sim \frac{\sigma^2}{4\sqrt{r}}. \quad (14)$$

For a sampling interval Δt , the lag-one autocorrelation satisfies

$$AC(1) \approx \exp(-\Delta t/\tau_{\text{rec}}) \rightarrow 1 \quad \text{as } r \rightarrow 0^+. \quad (15)$$

Thus variance, autocorrelation, and recovery time are expected to rise near rescue-window closure. Variance is the most noise-model-sensitive of the three; in the simulations below we use smoothed stochastic estimates and interpret variance as an auxiliary indicator rather than a sole diagnostic. Early-warning indicators are interpreted here as local normal-form predictions, not universal diagnostics; recent work emphasizes both the usefulness and ambiguity of EWS outside ideal bifurcation settings [22].

IV. NONDIMENSIONAL CONTROL NUMBERS

The model can be summarized by a small set of dimensionless numbers (Table I). These quantities make

explicit which parameters should be varied in simulations and wet-lab analogues.

The pruning number $\mathcal{E} = S_0/(pJ_0)$ and residue burden $\mathcal{P} = pJ_0/S_0$ are reciprocal under the same normalization. We report both because they emphasize opposite experimental readings: $\mathcal{E}$ measures excess active clearing capacity, whereas $\mathcal{P}$ measures residue pressure relative to that capacity.

Persistence requires $\mathcal{E}$ sufficiently large, $\mathcal{D}$ sufficiently small, $\mathcal{O}$ below rupture scale, and $\mathcal{B}$ large enough to prevent boundary damage from accumulating. These are not independent in a real chemistry, but separating them clarifies which experimental knobs correspond to which failure modes.

V. SPATIAL FAILURE MODE I: REACTION-DIFFUSION CLOGGING

Droplet-like, gel-like, and porous compartments may fail by internal clogging before any membrane-like rupture occurs. We model this by a residue field obeying

$$\frac{\partial \rho}{\partial t} = pJ + D_\rho(\rho)\nabla^2 \rho - S(x, t)\frac{\rho}{K + \rho}, \quad (16)$$

with residue-dependent diffusivity

$$D_\rho(\rho) = D_0 e^{-\lambda_D \rho}. \quad (17)$$

The local throughput is

$$J(x, t) = J_0 B(x, t) e^{-\lambda_J \rho(x, t)}, \quad (18)$$

while local integrity evolves as

$$\frac{\partial B}{\partial t} = \gamma_B(1 - B) - \beta_\rho \rho B. \quad (19)$$

This mechanism is relevant to coacervates, gels, and mineral pores: residue slows diffusion, reduced diffusion increases local residue retention, and the resulting feedback creates dead zones.

Figure 3 shows the qualitative result. Without pruning, residue accumulates and spatially traps itself. With active pruning, residue remains bounded and transport remains available.

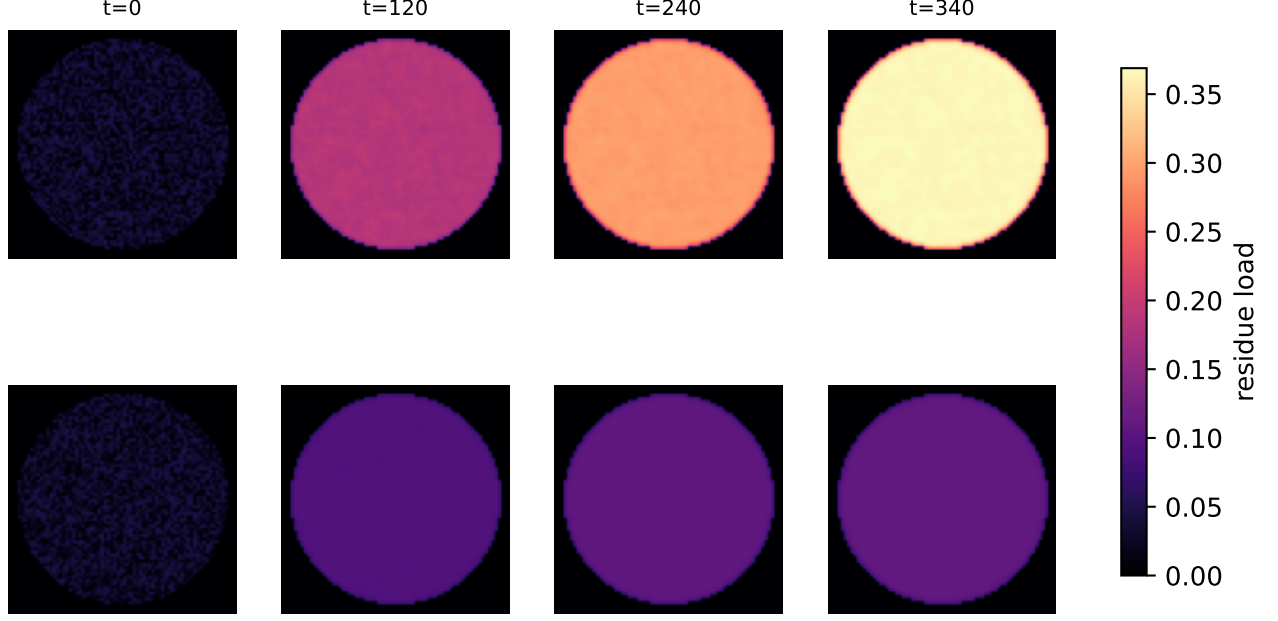
VI. SPATIAL FAILURE MODE II: PHASE-FIELD-INSPIRED REDUCED DYNAMIC-BOUNDARY FAILURE

Vesicle-like systems may fail through boundary stress and local integrity loss. In this reduced model, geometric shape change and integrity failure are distinct observables: two compartments may have similar gross outlines while differing strongly in local leakage or damage susceptibility. A full phase-field treatment would evolve an

Symbol	Definition	Meaning
$\mathcal{P}$	pJ_0/S_0	residue burden / pruning
$\mathcal{D}$	$(R^2/D_0)/(K/S_0)$	diffusion delay / pruning time
$\mathcal{O}$	$\alpha_{\Pi}\rho_0/\sigma_{\text{crit}}$	osmotic stress / rupture threshold
$\mathcal{B}$	$\gamma_B/(\beta_{\rho}\rho_0)$	repair / residue damage
$\mathcal{E}$	$S_0/(pJ_0)$	pruning excess capacity

TABLE I. Dimensionless control numbers for residue-pruning-boundary systems.

2D reaction-diffusion clogging: residue accumulation and active-pruning rescue

FIG. 3. Spatial failure mode I: reaction-diffusion clogging. The upper row shows residue accumulation without active pruning. As ρ increases, $D_{\rho}(\rho) = D_0 e^{-\lambda_{D\rho}}$ falls, producing diffusion arrest and dead zones. The lower row shows the same system with active pruning above threshold, where residue remains bounded.

order parameter $\phi(x, t)$ with a free-energy functional such as

$$\mathcal{F}[\phi, \rho] = \int \left[\frac{\kappa}{2} |\nabla \phi|^2 + W(\phi) + \chi \rho \phi \right] dx + \lambda_A (A - A_0)^2 + \lambda_V (V - V_0)^2. \quad (20)$$

with gradient-flow dynamics

$$\frac{\partial \phi}{\partial t} = -M_{\phi} \frac{\delta \mathcal{F}}{\delta \phi}. \quad (21)$$

We do *not* solve this full phase-field problem here. Instead, we use a phase-field-inspired reduced boundary radius field $R(\theta, t)$ together with a local integrity field $B(\theta, t)$ to isolate the coupling between residue pressure, boundary deformation, and rupture-like integrity loss. The purpose is not to resolve full membrane hydrodynamics, but to test whether residue pressure can drive loss of boundary maintenance in a vesicle-like geometry.

The reduced model is

$$\frac{d\rho}{dt} = pJ - d\rho - S \frac{\rho}{K + \rho}, \quad (22)$$

$$\Pi(\rho) = \alpha_{\Pi} \rho, \quad (23)$$

$$\frac{\partial R}{\partial t} = \mu_R [R_0 + \Pi(\rho)(1 + \eta(\theta, t)) - R] + D_R \frac{\partial^2 R}{\partial \theta^2}, \quad (24)$$

$$\frac{\partial B}{\partial t} = \gamma_B (1 - B) - \beta_{\epsilon} \epsilon(\theta, t) B - \beta_{\rho} \rho B, \quad (25)$$

where ϵ is a local strain proxy. Rupture-like failure occurs when mean boundary integrity falls below threshold or when strain exceeds a prescribed damage scale.

Reduced dynamic-boundary model: shape and local integrity can decouple

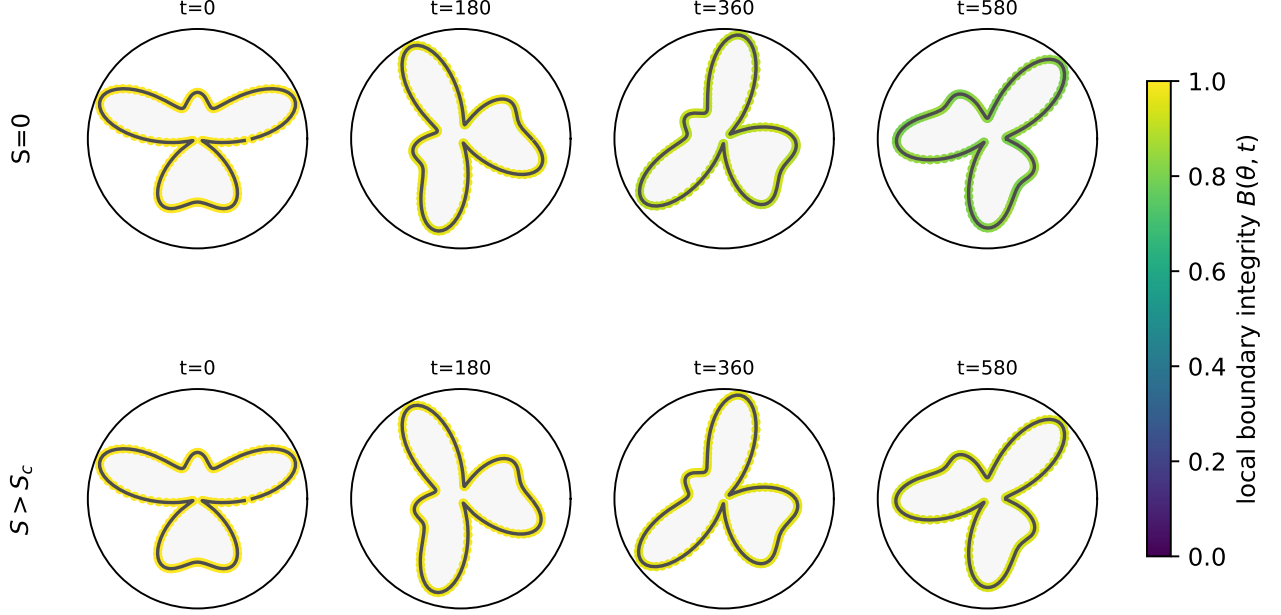


FIG. 4. Spatial failure mode II: phase-field-inspired reduced dynamic-boundary failure. Grey outlines show the boundary radius field $R(\theta, t)$; color along the boundary shows local integrity $B(\theta, t)$. The dominant failure signal is not necessarily gross shape change but loss of local integrity. Thus similar outlines can correspond to very different leakage or damage states. The model uses a reduced radius/integrity field rather than a full Cahn-Hilliard, Allen-Cahn, Helfrich, or hydrodynamic membrane solver.

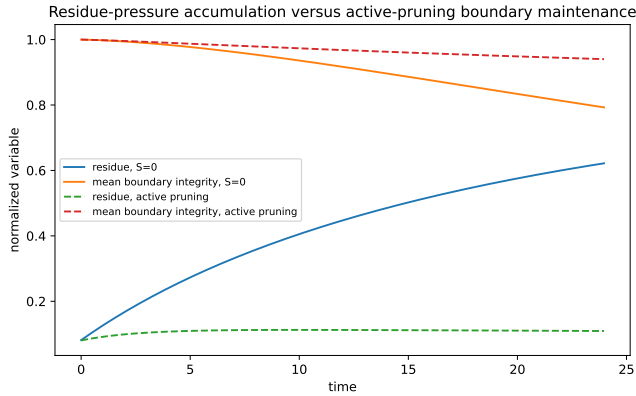


FIG. 5. Dynamic-boundary time series. In the no-pruning condition, residue increases and mean boundary integrity declines. In the active-pruning condition, residue and integrity approach a maintained nonequilibrium regime. This explains why Fig. 4 emphasizes integrity color in addition to geometric outline.

VII. THREE-DIMENSIONAL SIZE CONSTRAINT

A spherical compartment of radius R with volume residue-generation density q produces residue at rate

$$P_\rho = \frac{4}{3}\pi R^3 q. \quad (26)$$

If active export or membrane-localized pruning has maximum areal capacity s_m , the boundary clears at most

$$P_{\text{clear}} = 4\pi R^2 s_m. \quad (27)$$

A necessary condition for steady maintenance is $P_{\text{clear}} > P_\rho$, which yields the areal mass-balance lower bound

$$s_{m,c}(R) \geq \frac{qR}{3}. \quad (28)$$

The corresponding total membrane pruning requirement is $S_c^{\text{tot}}(R) = 4\pi R^2 s_{m,c}(R)$. Equation (28) is only a lower bound. If residue must diffuse from the interior to the boundary before export, then the diffusion delay grows as

$$t_D \sim \frac{R^2}{D_\rho}. \quad (29)$$

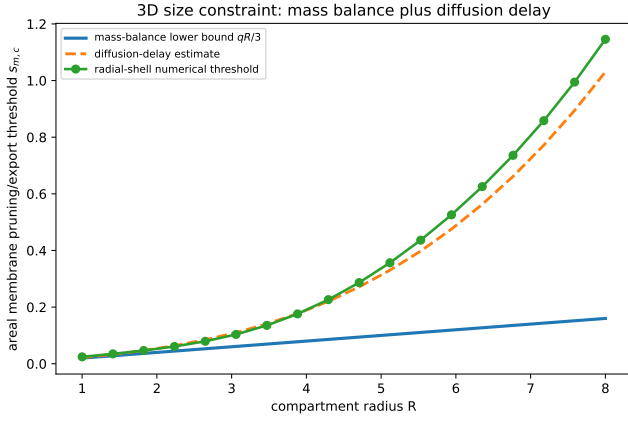


FIG. 6. Three-dimensional size constraint. Boundary-limited clearing gives the analytical areal lower bound $s_{m,c}(R) \geq qR/3$. The radial-shell numerical threshold lies well above this bound at large R because residue must diffuse to the boundary before export or pruning. The dashed curve shows the corrected estimate $s_{m,c}^{\text{eff}}(R) \geq (qR/3)[1 + \chi R^2/(D_0\tau_p)]$, where the dimensionless factor $\chi R^2/(D_0\tau_p)$ represents the diffusion-delay penalty.

A minimal corrected estimate is therefore

$$s_{m,c}^{\text{eff}}(R) \gtrsim \frac{qR}{3} \left[1 + \chi \frac{R^2}{D_0\tau_p} \right], \quad (30)$$

where τ_p is the pruning timescale and χ is a geometry- and transport-dependent coefficient. If D_p decreases with residue load, the effective threshold can exceed the mass-balance lower bound by a large factor at large R . The radial-shell simulations implement this diffusion-delay correction. Figure 6 shows the lower bound, the corrected estimate, and the numerical threshold.

The growing separation between the mass-balance lower bound and the radial-shell threshold is therefore not a discrepancy but a mechanistic prediction: large compartments are not only surface-limited; they are transport-delay limited. A vesicle or droplet can have enough boundary area in principle and still fail if residue generated in the interior cannot reach that boundary on the pruning timescale.

VIII. RESCUE-WINDOW CLOSURE

To test reversibility, we use a pruning-interruption protocol: first equilibrate a maintained system with $S > S_c$, shut pruning off, restore pruning at time t_{restore} , and record whether $V(t)$ recovers above V_{rec} within the observation window. Figure 7 shows the resulting rescue surface. Early restoration succeeds over a broad range of S_{rescue} . Late restoration requires increasingly high pruning and eventually fails because the trajectory has crossed into the collapse basin.

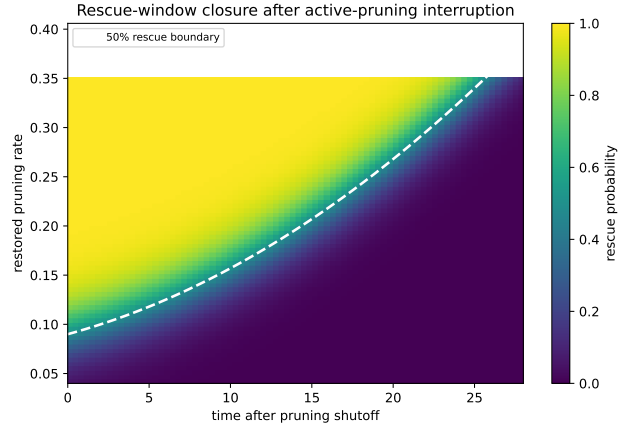


FIG. 7. Rescue-window closure. The heatmap shows rescue probability as a function of restoration time and restored pruning rate in the reduced rescue protocol. The dashed line marks the 50% rescue boundary.

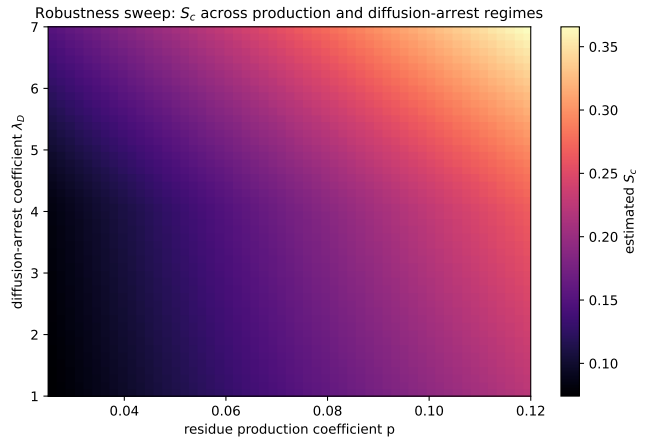
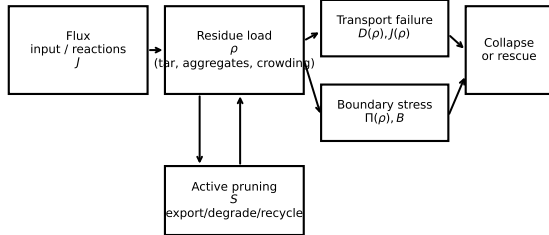


FIG. 8. Robustness sweep. Estimated pruning threshold S_c across residue production p and diffusion-arrest coefficient λ_D . The threshold persists across the explored region and rises as production and diffusion arrest increase.

IX. ROBUSTNESS SWEEP

A threshold that exists only for one parameter set is not a useful principle. We therefore varied residue production p and diffusion-arrest strength λ_D over broad ranges and estimated the pruning threshold. Figure 8 shows that S_c persists and increases with both production load and diffusion arrest. This does not prove universality, but it supports the claim that the threshold is not an artifact of one visually selected simulation.

Residue-pruning-boundary universality class and wet-lab benchmark map



Experimental proxies: fluorescence residue, FRAP diffusivity, leakage dye, vesicle radius, throughput, and rescue timing

FIG. 9. Residue-pruning-boundary universality class and wet-lab benchmark map. The model predicts that flux-driven residue causes either transport failure or boundary stress unless an active pruning module removes, exports, degrades, or reorganizes residue.

X. PROSPECTIVE WET-LAB BENCHMARK PREDICTIONS

The proposed proxies are aligned with recent experimental platforms in protocell and synthetic-cell research, including membranized coacervates with tunable interfacial stability, spontaneous protocell membrane formation, and giant unilamellar vesicle stability and loading systems [20, 21]. This section is a prospective benchmark plan, not a report of completed experiments. The model should be judged by whether it suggests measurable experimental contrasts. Figure 9 and Table II map variables to possible wet-lab proxies. The strongest experiment is a paired comparison: two otherwise identical compartments under sustained flux, one without an active clearing module and one with a tunable degradation/export/recycling process. The predicted observables are bounded residue, preserved throughput, lower leakage or rupture rate, and a measurable rescue boundary in the active-pruning condition.

A. Literature-informed plausibility check

Existing protocell and soft-matter literature already shows that the ingredients assumed here are experimentally meaningful, although not yet in the combined threshold test proposed above. Fatty-acid vesicles can grow, exchange amphiphiles, and undergo permeability changes in response to osmotic and compositional conditions [1–3]. Membrane deformation and rupture are standard problems in membrane mechanics [4, 5]. Coacervates and condensate-like droplets exhibit strong changes in internal mobility, partitioning, and material state with composition and crowding [8, 9]. Early-warning indicators such as critical slowing down, variance increase, and autocorrelation increase are standard signatures near tipping points in stochastic dynamical systems [10, 11]. These observations are qualitatively consistent with the modeling ingredients used here, but they are not a quan-

titative validation. The proposed benchmark is intended to convert this qualitative plausibility into direct tests of S_c , $s_{m,c}(R)$, R_c , and rescue-window closure. We do not fit these data here; a future validation step should use reported vesicle leakage, rupture, FRAP diffusivity, or condensate-aging data to estimate the control numbers in Table I.

B. Literature-informed parameter anchoring to published protocell data

The present model is not fitted to a specific chemistry, but its variables can be anchored to published protocell observables. As an order-of-magnitude anchoring exercise, we consider two datasets that map directly to the model’s variables.

a. Membranized coacervate boundary stability. In membranized coacervate protocells, dextran molecular weight and dextranase-mediated hydrolysis control interfacial membrane thickness, colloidal stability, and molecular transport [18]. In the terminology of the present model, the interfacial dextran layer acts as part of the boundary-integrity variable B , while dextranase acts as an externally imposed boundary-degradation stress. Reported membrane thickness can be mapped to a normalized boundary integrity proxy,

$$B_{\text{proxy}}(t) = \frac{h(t)}{h(0)},$$

where $h(t)$ is the measured interfacial membrane thickness. For example, Ji and Qiao report that after 30 minutes of dextranase treatment at 0, 0.09, 0.18, and 0.26 mg/ml, the residual membrane thickness falls from approximately 0.39 μm to 0.17, 0.03, and 0 μm respectively, showing a threshold-like loss of boundary integrity with increasing degradation stress [18]. If dextranase treatment reduces $h(t)$ over a time window Δt , an apparent boundary damage rate can be estimated as

$$k_B^{\text{app}} = -\frac{1}{\Delta t} \ln \frac{h(t)}{h(0)},$$

when $h(t) > 0$, or as a lower-bound failure rate when $h(t)$ falls below the optical detection limit [18]. This mapping does not identify dextranase hydrolysis with active pruning. Rather, it calibrates the measurable range over which boundary integrity can be experimentally driven from a persistent state to a coalescence-prone state. The active-pruning experiment proposed here would invert this perturbation: under sustained residue production, a tunable clearing or degradation module should maintain bounded residue and delay or prevent the loss of B_{proxy} .

b. GUV leakage versus gross survival. Published GUV data show that gross vesicle survival and leakage can occur on different timescales [23]. For microfluidically produced GUVs with mean diameter approximately

TABLE II. Model variables and experimental proxies. The table is a benchmark plan, not a claim of completed validation.

Model variable	Experimental proxy	Candidate platform	Predicted contrast
ρ	fluorescent inert product, aggregate load, tar fraction	coacervate, vesicle, microfluidic reactor	bounded with pruning, increasing without pruning
D_ρ	FRAP or particle-tracking diffusivity	coacervate, gel, mineral pore	decreases with residue load
B	leakage dye, membrane integrity, compartment retention	GUV, fatty-acid vesicle	integrity preserved by pruning
J	substrate/product throughput, permeability, reaction rate	sustained-flux reactor	throughput collapse without pruning
S	degradation rate, export rate, photochemical clearing rate	catalyst/enzyme/light-driven module	tunable threshold S_c or areal $s_{m,c}$
R	compartment radius	vesicle/droplet microscopy	larger systems require larger areal/total clearing
t^*	latest successful rescue time	pruning shutoff/restoration assay	rescue-window closure

92 μm , microscopy-based counting gave a vesicle half-life of about 61 h, while approximately 90% of encapsulated calcein was released within 24 h. A first-order leakage proxy gives

$$k_{\text{leak}} \approx -\frac{\ln(0.1)}{24 \text{ h}} \approx 9.6 \times 10^{-2} \text{ h}^{-1},$$

whereas gross vesicle disappearance gives

$$k_{\text{count}} \approx \frac{\ln 2}{61 \text{ h}} \approx 1.1 \times 10^{-2} \text{ h}^{-1}.$$

Thus leakage occurs on a timescale roughly eightfold faster than gross vesicle loss in this dataset. This supports the modeling choice in the reduced boundary simulations: boundary shape, compartment count, and local integrity need not fail simultaneously. In the language of the present model, B should be interpreted as a local functional integrity variable, not merely as the existence of a visibly closed outline.

XI. FAILURE AND DEMOTION CONDITIONS

The active-pruning claim is falsified in the modeled regime if systems under sustained residue production maintain bounded ρ , stable B , and stable J indefinitely with $S = 0$, no hidden dilution, no externalization, and no passive degradation sufficient to offset production. The claim is demoted if pruning merely improves persistence without a threshold or rescue boundary. The spatial-clogging bridge is weakened if residue accumulation does not reduce diffusion or transport in droplet-like systems. The boundary-rupture bridge is weakened if residue pressure does not affect membrane integrity in vesicle-like systems. The size-scaling bridge is weakened if $s_{m,c}(R)$, or the corresponding total clearing requirement $S_c^{\text{tot}}(R)$, does not increase with radius under boundary-limited clearing.

XII. DISCUSSION

Table IV summarizes which conclusions are supported by analytical derivation, simulation, or prospective experimental mapping, and what would demote each claim.

A. What the saddle-node analysis adds

The saddle-node reduction upgrades the collapse language from a numerical observation to a dynamical-systems mechanism. The well-mixed model supplies conditions under which a maintenance attractor and a saddle basin boundary collide. The spatial models then realize the same reduced loss of access in different physical ways: droplet-like systems clog internally; vesicle-like systems deform and rupture; large spherical systems face boundary-limited clearing and diffusion delay.

B. Why chemical coarse-graining is not a defect

The model intentionally does not specify whether residue is RNA side product, peptide aggregate, photochemical tar, lipid impurity, or mineral-surface deposit. That abstraction is a strength for the present purpose. The claim is about a universality class defined by flux, residue, transport/boundary impairment, and active clearing. Specific chemistries can instantiate the class and determine parameter values, but the model predicts that the persistence-collapse structure should recur across substrates.

Recent whole-cell modeling has demonstrated that integrated spatial and kinetic simulations can reproduce the cell-cycle dynamics of an existing genetically minimal organism. The 4D whole-cell model of JCVI-syn3A simulates a complete approximately 100-minute cell cycle, including genetic information processes, metabolic networks, growth, and division, and recovers multiple experimental observables such as doubling time, mRNA half-lives, protein distributions, and ribosome counts [14]. This work illustrates the increasing feasibility of simulation-driven artificial-life research. The present paper addresses a complementary design-level question. Rather than reconstructing an existing minimal cell, we ask what maintenance constraints should be made explicit when designing or screening finite protocell-like systems under sustained flux. In that setting, residue accumulation, transport impairment, boundary integrity, active clearing, rescue timing, and size-dependent pruning demand become candidate control variables for synthetic-cell persistence.

C. Limitations

First, the phase-field component is reduced. We write a phase-field energy to clarify the mechanical interpretation but simulate a lower-dimensional radius/integrity field. A full Cahn-Hilliard, Allen-Cahn, Helfrich-coupled, or hydrodynamic membrane solver is future work. Second, the saddle-node proof applies to the reduced well-mixed model, not to the full spatial PDE. Third, the SDE early-warning analysis is a local weak-noise approximation near the stable branch of the reduced saddle-node normal form. It predicts critical slowing down, variance growth, and rising autocorrelation as the fold is approached quasi-statically. It does not describe large-shock collapse, nonlinear basin escape, or finite-time rupture far from the fold; those regimes require first-passage or large-deviation analysis. Fourth, the simulations are nondimensional and do not fit a specific wet-lab chemistry. Fifth, the benchmark map is prospective, not retrospective validation.

XIII. METHODS AND REPRODUCIBILITY

All figures were generated with reproducible Python scripts using recorded random seeds; stochastic simulations used fixed seed sets for replicate generation. The 2D reaction-diffusion model uses explicit finite differences with residue-dependent diffusivity. The dynamic-boundary model uses a polar radius field with residue-driven pressure, angular deformation modes, and local integrity loss. The SDE early-warning simulations use an Ornstein-Uhlenbeck approximation to the saddle-node normal form. The radial-shell threshold combines the analytical area-volume lower bound with a diffusion-delay correction. The package accompanying this manuscript includes source code, generated figures, and CSV data tables.

A. Parameterization and simulation protocol

All simulations are nondimensional unless explicitly stated. The model uses three classes of parameters: flux-residue parameters, boundary-maintenance parameters, and transport/geometry parameters (summarized in Table III). The baseline value of each parameter is not intended to represent one specific chemistry. Instead, each baseline defines a reference point inside the residue-pruning-boundary universality class, and the robustness sweep tests whether the qualitative threshold structure persists under parameter variation.

The residue-production load is controlled by pJ_0 , where p is the residue generated per unit throughput and J_0 is the maximum unblocked throughput. Passive loss is represented by d . Active pruning is represented by S , with a saturating half-load constant K . Boundary

damage is controlled by β_ρ , repair by γ_B , transport inhibition by λ_ρ or λ_J , and diffusion arrest by λ_D . Spatial simulations also require a reference diffusivity D_0 , lattice spacing Δx , time step Δt , and compartment radius R .

For each figure, the code records the parameter dictionary used to generate the corresponding output. A parameter set is accepted only if the simulation remains numerically stable under halving of Δt and if the qualitative threshold classification is unchanged under modest refinement of the pruning grid. The purpose of this requirement is not to prove convergence of a full continuum PDE solver, but to rule out visually selected numerical artifacts.

B. Threshold extraction and classification protocol

For a fixed parameter vector θ and pruning value S , a run is classified as persistent if three conditions hold over the final observation window $[T_{\text{end}} - \Delta T, T_{\text{end}}]$:

$$V(t) > V_{\min}, \quad \rho(t) < \rho_{\max}, \quad \frac{1}{\Delta T} \int_{T_{\text{end}} - \Delta T}^{T_{\text{end}}} |\dot{V}(t)| dt < \epsilon_V.$$

A run is classified as collapsed if $V(t) < V_{\min}$ for longer than a prescribed dwell time, or if $B(t)$, $J(t)$, or the spatially averaged transport field falls below its failure threshold.

For deterministic well-mixed and spatial simulations, the critical pruning threshold is estimated by bracketing. We first evaluate a coarse grid of pruning values $S_1 < S_2 < \dots < S_m$ and identify the smallest interval $[S_i, S_{i+1}]$ for which the classification changes from collapse to persistence. We then refine this interval by bisection until the threshold uncertainty is smaller than a fixed tolerance: $|S_{i+1} - S_i| < \delta_S$. The reported deterministic threshold is $\hat{S}_c = (S_i + S_{i+1})/2$.

For stochastic simulations, the classification is repeated over N random seeds. The persistence probability is estimated as $\hat{P}_{\text{pers}}(S) = \frac{1}{N} \sum_{j=1}^N I_j(S)$, where $I_j(S) = 1$ if replicate j persists and 0 otherwise. The stochastic threshold is defined as the 50% persistence point: $\hat{S}_c^{(50)} = \{S : \hat{P}_{\text{pers}}(S) = 0.5\}$, obtained either by monotone interpolation or by fitting a logistic curve $P_{\text{pers}}(S) = 1/(1 + \exp[-a(S - S_c^{(50)})])$. Confidence intervals are obtained by bootstrap resampling over random seeds.

For rescue-window experiments, the same logic is applied to the two-dimensional surface $(t_{\text{restore}}, S_{\text{rescue}})$. The rescue boundary is the 50% contour $P_{\text{rescue}}(t_{\text{restore}}, S_{\text{rescue}}) = 0.5$. A rescue-window closure claim is supported only if the estimated required rescue pruning is nondecreasing with restoration delay over the tested range.

In Fig. 8, each pixel represents an independently estimated threshold using the same collapse/persistence classifier, not a manually selected visual transition.

XIV. DATA AND CODE AVAILABILITY

Source code, parameter files, generated figures, and CSV tables for the simulations are available at: https://github.com/yiningwu-research/Distinction-Theory/tree/main/models/fds_l1. The repository contains scripts for the well-mixed model, SDE early-warning simulations, 2D reaction-diffusion clogging, reduced boundary model, rescue-window heatmap, robustness sweep, and radial-shell size-scaling analysis. A machine-readable parameter table specifying baseline values, sweep ranges, random seeds, and figure-generation scripts is included with the released code.

XV. CONCLUSION

Active pruning can be formulated as a measurable control variable for finite protocell-like persistence. The analytical layer gives threshold bounds, saddle-node attractor-loss conditions, stochastic early-warning scaling, and a size-scaling lower bound. The spatial simulations show two physically distinct failure modes: internal clogging in droplet-like or porous compartments and residue-pressure boundary failure in vesicle-like compartments. The radial-shell model shows that size increases pruning demand because residue generated in the volume must diffuse to and be cleared at the boundary; the numerical threshold can substantially exceed the mass-balance lower bound when diffusion delay dominates. The benchmark map translates these variables into experimental proxies. The result is not a completed origin-of-life mechanism, but a testable theory-computation framework for asking whether active residue clearing is required under specified residue-production and boundary-maintenance constraints for long-term persistence in protocell-like systems.

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Symbol	Meaning	Role in model	Reporting requirement
p	residue per unit throughput	increases residue load	baseline + sweep range
J_0	maximum throughput	sets flux scale	baseline normalization
d	passive dilution/degradation	non-active residue loss	baseline + control runs
S	active pruning rate	main control parameter	grid/bisection range
K	pruning half-saturation load	nonlinear pruning response	baseline value
λ_ρ, λ_J	transport inhibition	throughput suppression by residue	baseline + sensitivity
β_ρ	residue damage coefficient	boundary integrity loss	baseline + sensitivity
γ_B	boundary repair coefficient	recovery of B	baseline + sensitivity
D_0	reference diffusivity	spatial transport scale	baseline + grid test
λ_D	diffusion-arrest coefficient	nonlinear clogging strength	sweep range
R	compartment radius	size-scaling variable	values used in Fig. 6
q	residue generation density	3D source density	value used in size bound
τ_p	pruning timescale	diffusion-delay correction	value or normalization
χ	geometry/transport correction	effective size penalty	fitted or assumed value
$V_{\min}$	collapse threshold	persistence classification	fixed criterion
V_{rec}	rescue threshold	rescue classification	fixed criterion
σ	noise amplitude	SDE early-warning test	values used
$\Delta t, \Delta x$	numerical discretization	stability/reproducibility	refinement check

TABLE III. Simulation parameters and roles. Baseline values and sweep ranges are recorded in the released code rather than reconstructed manually.

Claim	Current support	Demotion condition	Failure consequence
Active pruning threshold	analytical bound + simulations	pruning helps but no threshold	core threshold claim weakened
Saddle-node reduction	well-mixed Jacobian conditions	fold replaced by other bifurcation	revise attractor-loss mechanism
SDE early warnings	normal-form derivation + simulation	variance unreliable alone	keep recovery time/autocorrelation only
Spatial clogging	reaction-diffusion simulation	no diffusion arrest in relevant platforms	limit to other residue mechanisms
Boundary rupture	reduced radius-field simulation	residue pressure does not damage membranes	limit vesicle bridge
Size scaling	lower bound + radial threshold	no $s_{m,c}(R)$ increase	revise boundary-limited scaling
Wet-lab benchmark	prospective map	proxies not measurable	redesign experimental protocol

TABLE IV. Claim status and consequence of failure.

NUMERICAL DETAILS

The 2D model is simulated on a circular mask embedded in a square lattice with an explicit five-point Laplacian. The dynamic-boundary model evolves $R(\theta, t)$ with periodic angular coupling. The rescue heatmap is generated from the reduced fold-distance protocol: required rescue pruning increases with time after shutoff, and rescue probability is a sigmoid function of restored pruning minus this time-dependent requirement. The robustness heatmap estimates S_c over residue production and diffusion-arrest parameters using the same scaling assumptions as the spatial model.

APPENDIX: LINEAR STABILITY OF THE SPATIAL CLOGGING MODEL

The spatial clogging model is not intended to claim a separate Turing-type bifurcation. Its role is to show how the reduced maintenance-attractor loss can be physically realized through residue-dependent transport arrest. To make this explicit, consider the homogeneous state of the spatial equations with constant pruning $S = \bar{S}$:

$$pJ_0\bar{B}e^{-\lambda_J\bar{\rho}} - \bar{S}\frac{\bar{\rho}}{K+\bar{\rho}} = 0, \quad (31)$$

$$\gamma_B(1 - \bar{B}) - \beta_\rho\bar{\rho}\bar{B} = 0, \quad (32)$$

which gives $\bar{B} = \gamma_B/(\gamma_B + \beta_\rho\bar{\rho})$.

Let $\rho = \bar{\rho} + \delta\rho$, $B = \bar{B} + \delta B$, and expand perturbations in Laplacian eigenmodes $(\delta\rho, \delta B) \propto e^{\lambda t}\phi_n(x)$ with $-\nabla^2\phi_n = \mu_n\phi_n$, where $\mu_n \geq 0$ and $\mu_0 = 0$ for no-flux boundary conditions. Since $\nabla^2\bar{\rho} = 0$, the residue-dependent diffusivity contributes at linear order as $D_\rho(\bar{\rho})\nabla^2\delta\rho = -\bar{D}\mu_n\delta\rho$ with $\bar{D} = D_0e^{-\lambda_D\bar{\rho}}$. Thus each spatial mode obeys

$$\frac{d}{dt} \begin{pmatrix} \delta\rho \\ \delta B \end{pmatrix} = A(\mu_n) \begin{pmatrix} \delta\rho \\ \delta B \end{pmatrix}, \quad (33)$$

where

$$A(\mu_n) = \begin{pmatrix} f_\rho - \bar{D}\mu_n & f_B \\ g_\rho & g_B \end{pmatrix}, \quad (34)$$

with $f_\rho = -pJ_0\lambda_J\bar{B}e^{-\lambda_J\bar{\rho}} - \bar{S}\frac{K}{(K+\bar{\rho})^2}$, $f_B = pJ_0e^{-\lambda_J\bar{\rho}}$, $g_\rho = -\beta_\rho\bar{B}$, $g_B = -\gamma_B - \beta_\rho\bar{\rho}$. The trace and determinant are $\text{tr} A(\mu_n) = f_\rho + g_B - \bar{D}\mu_n$ and $\det A(\mu_n) = (f_\rho - \bar{D}\mu_n)g_B - f_Bg_\rho = \det A(0) - \bar{D}\mu_n g_B$. Because $g_B < 0$, the determinant increases with μ_n , while the trace decreases with μ_n . Therefore, in this minimal local linearization, nonzero spatial modes are not the first modes to destabilize when the homogeneous maintenance state loses stability. The primary local loss is the $\mu_0 = 0$ maintenance mode, consistent with the reduced saddle-node analysis.

The spatial simulations should therefore be interpreted as nonlinear realizations of maintenance failure rather than as a claim of a separate Turing instability. Spatial structure enters through finite-amplitude residue accumulation, diffusion arrest, boundary geometry, and finite-size transport delay. A full nonlinear PDE bifurcation analysis is left for future work.

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Active Cognitive Pruning Controls Reportable Access under Finite Capacity: A Rate-Distortion, Network-Topological, and Maintenance-Cost Model

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Reportable access can fail under sustained cognitive load even when local discrimination or reflexive responsiveness remains intact. Here we develop a finite-capacity theory of reportable access based on the FDS rate-distortion capacity-deficit framework. Representational residue ρ is defined as accumulated unresolved rate-distortion surplus: the minimum task-relevant coding rate $R_{\min}^{(\tau)}(\varepsilon, t)$ exceeds accessible capacity $C_{acc}(t, \tau)$, producing a capacity deficit $\Delta_R > 0$ that must be managed through pruning, compression, externalization, or task relaxation. Residue damages access-network coherence G and reportability $\mathcal{R}$ unless controlled by active cognitive pruning S . Global access is modeled through the leading spectral mode or giant component of a residue-damaged access graph, with the saddle-node normal form recovered as a mean-field projection of a high-dimensional network transition. The framework predicts pruning-dependent reportability thresholds, rescue-window closure, leading-covariance early-warning signals, and a partial order for anesthesia emergence. It also provides a taxonomy for artificial-agent reportability and a falsification protocol distinguishing core claims from domain-specific bridges. Reduced normal-form simulations illustrate the predicted regimes: pruning-dependent reportability thresholds, rescue-window closure, and leading-covariance early-warning behavior near access collapse. High-dimensional artificial-agent benchmarks are identified as a future in-machina validation path rather than as a required premise of the present theoretical argument. The Landauer lower bound for irreversible pruning is treated only as the informational heat floor; total biological maintenance cost includes much larger physical overheads. The framework does not claim to solve phenomenal consciousness; it converts reportable access into a testable finite-system maintenance problem governed by capacity deficit, residue accumulation, active pruning, network coherence, and resource-bounded persistence.

I. INTRODUCTION

Theories of consciousness have emphasized distinct explanatory targets. Global Workspace Theory focuses on ignition and broadcast across specialized processors [1, 2]. Integrated Information Theory treats consciousness as intrinsic cause-effect integration [3, 4]. Predictive-processing and Free-Energy formulations describe perception and action as inference under a generative model [5, 6]. Perturbational-complexity approaches operationalize conscious level through the complexity of causal responses to perturbation [7, 8]. These frameworks have generated useful empirical programs, but they leave open a control-theoretic question: under finite capacity, what must a cognitive system actively maintain in order to keep information reportable?

This paper studies that question using a finite-distinction framework. The target is not the full metaphysics of phenomenality. The target is *conscious reportability*: the maintained ability of a system to integrate task-relevant distinctions into a coherent access state that can guide report, flexible action, and self-referential updating. A system may locally discriminate stimuli, produce reflexive responses, or process masked information unconsciously while failing to maintain a globally reportable access state. Such dissociations occur in masking, attentional blink, overload, sleep onset,

general anesthesia, disorders of consciousness, delirium, and task switching.

The central hypothesis is that reportable access is controlled by *active cognitive pruning*. Here pruning denotes resource-coupled suppression, erasure, inhibition, down-weighting, compression, reconsolidation, or reorganization of internal states that consume capacity without contributing to current task-relevant access. Active forgetting and directed forgetting literatures already distinguish passive decay from controlled forgetting processes [28]. The internal load removed by pruning is called *representational residue*. Residue includes obsolete working-memory items, intrusive affective states, unresolved prediction errors, stabilized but maladaptive associations, irrelevant sensory traces, and representational interference. The key point is not that all such processes share one neural mechanism. The key point is that finite systems need a selective operation that bounds nonfunctional distinction load while preserving useful distinctions.

This framing parallels a broader maintenance principle for finite systems: sustained flux generates residue; residue impairs access to functional states; active pruning controls persistence. In protocell-like systems, residue can impair transport or boundary integrity. In cognitive systems, representational residue impairs access coherence, self-model updating, and reportability. The analogy is structural rather than molecular.

The central object of this paper is not consciousness as such, but reportable access under finite capacity. In FDS terms, a cognitive system receives task-relevant distinc-

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tion demand $R_{\min}^{(\tau)}(\varepsilon, t)$ and possesses only finite accessible capacity $C_{acc}(t, \tau)$. When task demand exceeds accessible capacity, the resulting capacity deficit Δ_R must be absorbed by compression, invariant extraction, externalization, task relaxation, or active pruning. If these mechanisms are insufficient, unresolved distinction load accumulates as representational residue ρ , which damages the causal-loop and access-network conditions required for flexible report.

A. Central claim

The central claim is:

In finite cognitive systems under sustained distinction load, active cognitive pruning can act as a control parameter separating a maintained reportability regime from overload-induced access collapse.

This is stronger than the generic claim that attention or inhibition matters, because it predicts thresholds, rescue windows, non-monotonic self-model coherence, and measurable early-warning signatures. It is weaker than the claim that consciousness has been solved. The theory is a benchmark framework for conscious access, not a complete metaphysics of experience.

B. Contributions

This paper makes ten contributions.

1. It defines reportable access as a maintained finite-capacity regime rather than raw local discrimination.
2. It derives representational residue from FDS rate-distortion capacity deficit $\Delta_R = R_{\min}^{(\tau)}(\varepsilon, t) - C_{acc}(t, \tau)$, rather than from unresolved variational free energy.
3. It treats active inference / FEP as one implementation-level estimator of unresolved distinction load, not as the foundational derivation.
4. It derives a pruning lower bound for keeping residue below a reportability ceiling.
5. It separates functional retained structure m from nonfunctional residue ρ , resolving the non-monotonic access window into distinct mechanisms.
6. It replaces a purely one-dimensional saddle-node story with a network-topological access model in which residue damages access edges and pruning restores functional connectivity.
7. It derives early-warning signatures from both the normal form and the leading covariance eigenmode of the access network.
8. It provides a total maintenance-cost decomposition $\dot{Q}_{\text{maint}} = \dot{Q}_{\text{phys}} + \dot{Q}_{\text{info}} + \Gamma(S) + \Xi(E_{\text{ext}})$ with Landauer as the informational heat floor only.
9. It provides an FDS agency taxonomy and an artificial-agent benchmark protocol for future in-machina tests of reportability-like access, while keeping the present paper focused on theory, reduced simulations, and falsifiable predictions.
10. It gives a minimal falsification protocol with layered failure consequences, operational proxy maps, and differential predictions against GWT, IIT, FEP, information bottleneck, PCI/complexity, and AI agency.

C. What is not claimed

The paper does not claim to solve the hard problem of phenomenal consciousness. It does not prove that lossy compression alone produces experience. It does not identify one neural correlate of consciousness. It does not claim that all loss-of-consciousness events are saddle-node bifurcations, nor that every transition in sleep, anesthesia, or attention shares one mechanism. It does not claim that all current large language models are unconscious by metaphysical necessity. Instead, it states an operational requirement: a candidate finite system that maintains reportable conscious access must maintain task-relevant distinctions under bounded capacity, sustained input, residue accumulation, causal-loop constraints, and active pruning or an equivalent maintenance mechanism.

D. Falsifiable predictions

The framework can be falsified or demoted in specific ways.

1. **Pruning threshold.** If finite systems under sustained distinction load maintain stable reportability indefinitely with $S = 0$, no hidden capacity growth, no hidden dilution, and no equivalent compression/inhibition operation, the active-pruning threshold claim is falsified.
2. **Network topology.** If reportability transitions show no measurable change in access-network topology, spectral connectivity, effective connectivity, or perturbational complexity after controlling for arousal and task demands, the network-topological bridge is weakened.

3. **Functional retained structure.** If functional retained structure m shows no measurable contribution to self-model coherence beyond the effect accounted for by nonfunctional residue ρ , the non-monotonic self-model component is demoted.
4. **Anesthesia recovery ordering.** If high-level self-report coherence reliably recovers before causal responsiveness and access-network recovery during emergence from anesthesia, the proposed recovery-ordering claim is demoted.
5. **Rescue-window closure.** If restoring pruning after overload always recovers reportability with no dependence on delay or state, the attractor-loss interpretation is demoted.
6. **Energy bridge.** If experimentally isolated irreversible pruning of cognitive load requires no measurable energetic cost above noise floor and no lower-bound-compatible heat/metabolic signature, the Landauer-bridge interpretation is weakened.

II. RATE-DISTORTION ORIGIN OF REPRESENTATIONAL RESIDUE

A. Capacity deficit

Consider a finite cognitive system receiving task-relevant input over time. Let $R_{\min}^{(\tau)}(\varepsilon, t)$ be the minimum coding rate needed to maintain reportable access within distortion tolerance ε over timescale τ . Let $C_{acc}(t, \tau)$ be the accessible representational capacity. The FDS capacity deficit is

$$\Delta_R(t, \tau, \varepsilon) = R_{\min}^{(\tau)}(\varepsilon, t) - C_{acc}(t, \tau). \quad (1)$$

When $\Delta_R > 0$, the system cannot fully represent task-relevant distinctions within its capacity. The unresolved surplus must be managed.

B. Representational residue

Define representational residue $\rho(t)$ as the accumulated unresolved rate-distortion surplus that cannot be encoded, compressed, integrated, externalized, or pruned within the accessible capacity and resource budget:

$$\begin{aligned} \dot{\rho} = & \zeta [\Delta_R(t, \tau, \varepsilon) \\ & - C_{inv}(t) - G_{ext}(t) - G_{relax}(t)]_+ \\ & - d\rho - S\rho/(K + \rho), \end{aligned} \quad (2)$$

where $C_{inv}(t)$ is capacity saved through invariant compression, $G_{ext}(t)$ is effective gain from externalization, $G_{relax}(t)$ is reduction via task relaxation, d is passive decay, and $S\rho/(K + \rho)$ is saturating active pruning.

Residue operates at three levels:

- **Computational residue:** unresolved rate-distortion surplus (compression error, bottleneck loss, predictive-information shortfall).
- **Behavioral residue:** task-irrelevant interference impairing report (lapses, intrusions, perseveration, attentional-blink cost).
- **Network residue:** load-dependent degradation of access topology (leading covariance mode, effective connectivity loss).

These levels are not assumed identical. The theory predicts lawful covariation under controlled load, pruning, capacity, and externalization manipulations.

C. Active-inference implementation as a special case

The FDS rate-distortion derivation does not require Bayesian inference or variational objectives. However, in systems that encode observations through a recognition density $q_\phi(s_t, z_t | o_{\leq t})$ and minimize variational free energy $\mathcal{F}_{var}$, the unresolved free energy or KL load can serve as an estimator of Δ_R . When channel capacity C_{eff} , update energy E_{avail} , and information-erasure cost ΔQ_{erase} are finite, the constrained objective

$$\mathcal{J} = \mathbb{E}[\mathcal{F}_{var} + \lambda_C (I(o; z) - C_{eff})_+ \quad (3)$$

$$+ \lambda_E E_{update} + \lambda_Q Q_{erase}] - \beta_Y I(z; y) \quad (4)$$

reproduces the residue dynamics of Eq. (2) under the identification $\Delta_R \approx \mathcal{F}_{var} - \mathcal{F}_{cap}(C_{eff}, E_{avail})$. This makes active inference one possible implementation of FDS capacity-deficit dynamics, not the foundational derivation.

III. NETWORK-TOPOLOGICAL ACCESS MODEL

A. Access graph

A one-dimensional saddle-node model is useful but fragile if interpreted literally. Real brains and artificial agents are high-dimensional networks. We therefore define an access graph with nodes representing modules, cortical parcels, memory buffers, latent subspaces, or agent subsystems. Let $W_{ij}(t)$ be functional or effective coupling and let $\rho_i(t)$ be local residue. The effective access matrix is

$$M_{ij}(t) = a_i(t)W_{ij}(t)a_j(t) - \delta_{ij}\chi_\rho\rho_i(t), \quad (5)$$

Finite-distinction maintenance loop for conscious reportability

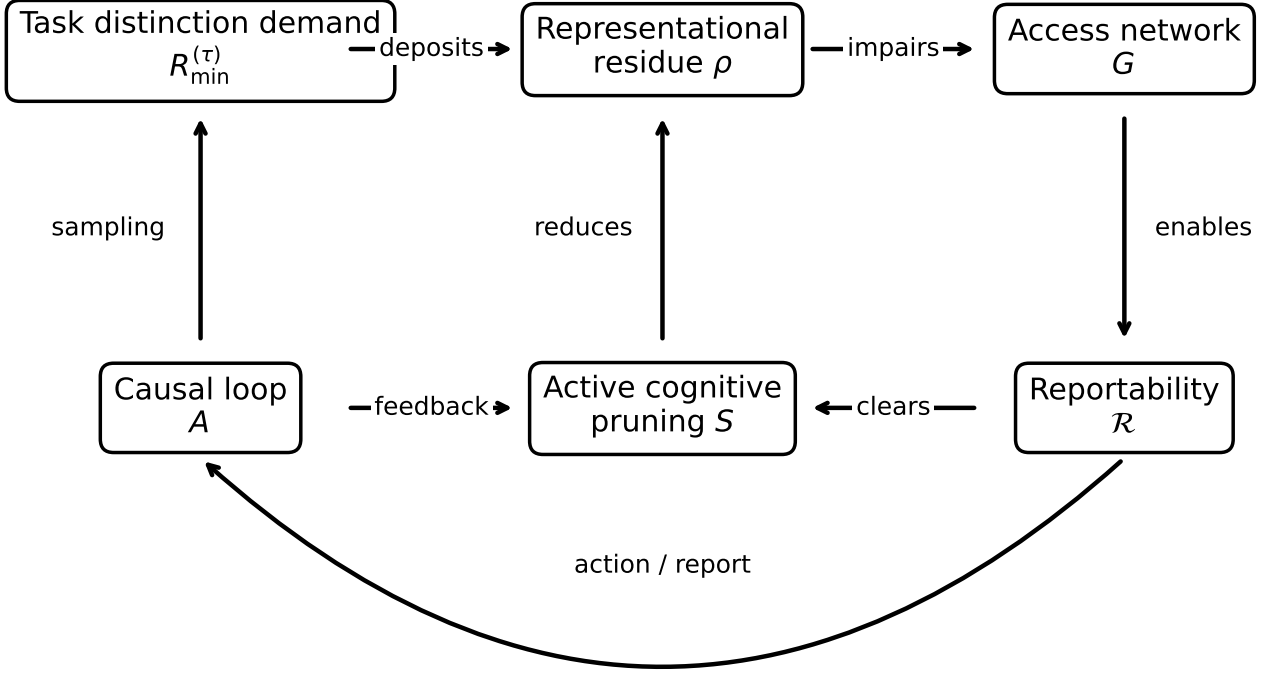


FIG. 1. Capacity-deficit maintenance loop. Task-relevant rate-distortion demand $R_{\min}$ and limited accessible capacity C_{acc} produce capacity deficit Δ_R , which deposits representational residue ρ . Residue impairs access coherence G and reportability $\mathcal{R}$. Active cognitive pruning S removes residue but consumes resources. Reportability persists while ρ is bounded and pruning exceeds the threshold S_c .

where a_i is local availability and χ_ρ is residue damage. Active pruning updates residue and connectivity:

$$\dot{\rho}_i = L_i - d_i \rho_i - S_i \frac{\rho_i}{K_i + \rho_i}, \quad (6)$$

$$\dot{W}_{ij} = \Gamma_{ij}^{\text{task}} - \lambda_\rho (\rho_i + \rho_j) W_{ij} - \lambda_S S_{ij}^{\text{cut}} + \lambda_R S_{ij}^{\text{repair}}. \quad (7)$$

The exact graph dynamics depend on substrate. The access criterion is more general:

$$G(t) = \sigma(\beta_G[\lambda_1(M(t)) - \lambda_c]), \quad (8)$$

where λ_1 is the leading eigenvalue or an equivalent giant-component/order parameter. Reportability is maintained only when an access component remains large enough and coherent enough for global report.

Residue can damage access in two topologically distinct ways. It can remove nodes from the giant access component, analogous to percolation. Or it can leave nodes connected while degrading spectral integration, producing a collapse of the leading eigenmode or algebraic connectivity. Both are compatible with the same coarse-grained G variable.

B. Mean-field reduction and the saddle-node projection

Let $m(t)$ be a mean access order parameter, such as normalized leading eigenvalue or giant-component size. Under homogeneous mean-field assumptions, the network dynamics reduce to

$$\dot{m} = F(m; S, C_{\text{eff}}, H, \rho). \quad (9)$$

If F has a fold in the control parameter

$$r = a_S(S - S_c) + a_C(C_{\text{eff}} - H) - a_\rho \rho, \quad (10)$$

then the center-manifold reduction gives

$$\dot{x} = r - x^2 + O(x^3, rx), \quad (11)$$

where x is the dominant access mode. The associated potential is

$$U(x) = -rx + \frac{x^3}{3}. \quad (12)$$

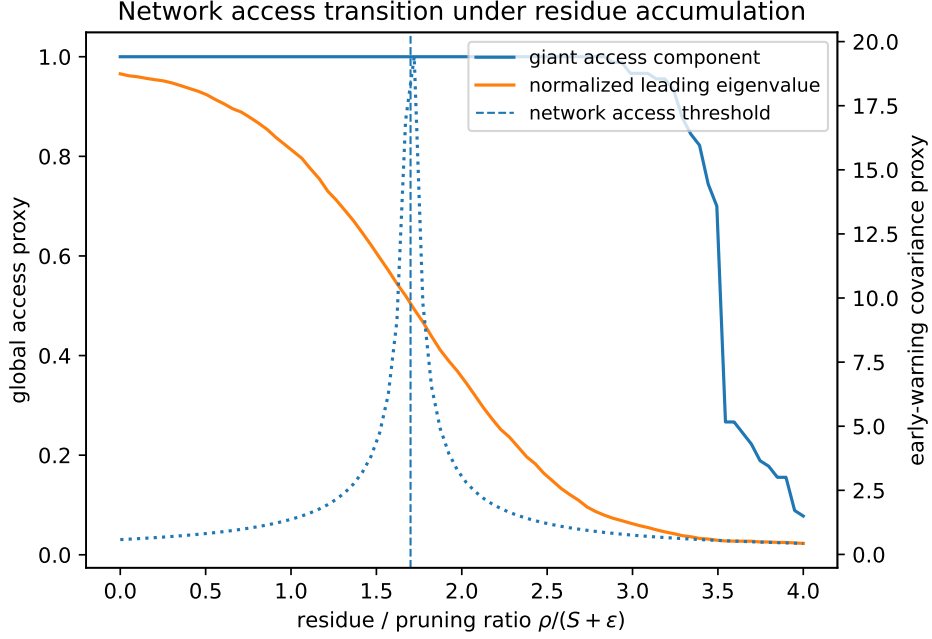


FIG. 2. Network access transition under residue accumulation. As the capacity-deficit-driven residue increases, the leading spectral proxy and giant access component decline. The leading covariance proxy rises near the transition, providing a high-dimensional early-warning signal consistent with FDS covariance-based collapse prediction.

Thus the saddle-node is not assumed as the complete brain dynamics. It is the local normal form when a high-dimensional access network loses reportability through one dominant critical mode. Other transitions are possible: Hopf, noise-induced escape, synchronization loss, or discontinuous percolation. The framework predicts that if the transition is fold-like, rescue windows and early warnings should appear.

C. Optional topological strengthening: a cognitive two-kink test

A stronger but optional topological prediction can be derived by analogy with the FDS physical bridge TP-3. If reportability loss is mediated by a topology-changing access-network transition, then the theory predicts aligned finite-size-smoothed slope changes in an operational forgetting rate κ_{access} and a maintenance-cost or entropy-production proxy Σ_{maint} near the same control parameter γ_c . This paired nonanalytic behavior is the cognitive analogue of the TP-3 two-kink prediction for physical systems.

This prediction is conditional. The core reportability model predicts thresholds, rescue-window closure, and early-warning behavior without requiring a topological transition. The two-kink prediction applies only if an access-network topological transition or invariant-supported reorganization is independently established. Failure of the two-kink prediction would demote the topological bridge, not the core capacity-deficit theory

of reportable access.

IV. LOAD-PRUNING-ACCESS DYNAMICS WITH FUNCTIONAL RETAINED STRUCTURE

A. Separating functional structure from nonfunctional residue

A purely passive system with no persistent internal structure does not support rich reportability. But not all internal load is harmful. We therefore distinguish two variables:

$$m(t) = \text{functional retained structure}, \quad (13)$$

$$\rho(t) = \text{nonfunctional representational residue}. \quad (14)$$

Functional retained structure includes task-relevant working content, self-model, and causal-loop representations. Nonfunctional residue is the accumulated capacity-deficit surplus from Eq. (2).

B. Core dynamics

Functional structure is maintained by task-relevant input and reorganized by pruning:

$$\dot{m} = \alpha_m I_{\text{task}}(t) - \beta_m m - \chi_\rho \rho m + \Gamma_{\text{reorg}}(S, m, \rho), \quad (15)$$

where I_{task} is task-relevant input, β_m is passive decay of retained structure, χ_ρ is damage from residue, and

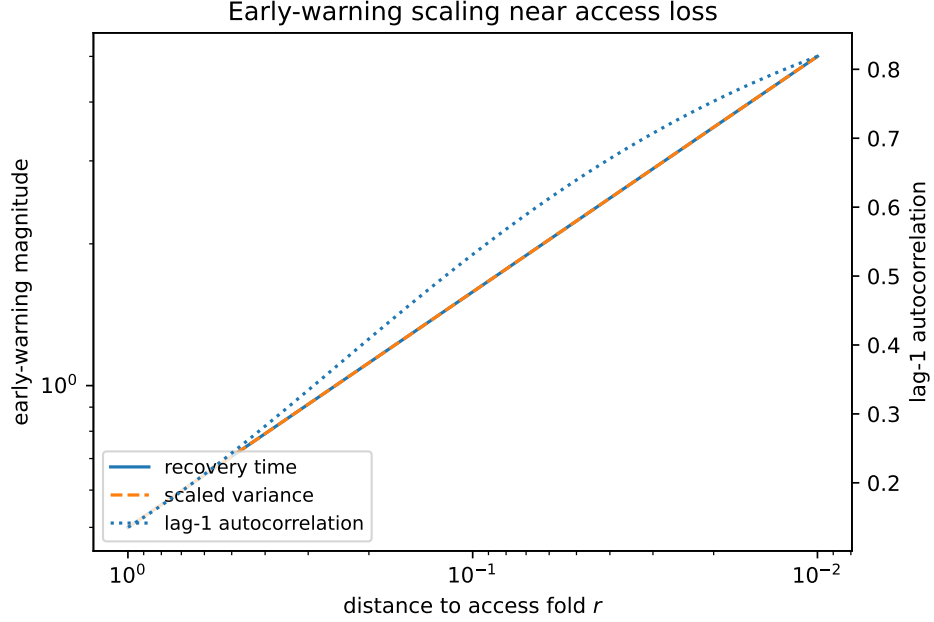


FIG. 3. Early-warning scaling near access loss. The reduced normal form predicts increasing recovery time, variance, and lag-one autocorrelation as the distance to the fold decreases. In high-dimensional data, the leading covariance eigenvalue and recovery time of the dominant access mode are preferred over scalar variance.

Γ_{reorg} captures reorganization during pruning (preserving functional structure while removing nonfunctional load).

Nonfunctional residue dynamics follow the capacity-deficit equation:

$$\dot{\rho} = \zeta [\Delta_R - C_{\text{inv}} - G_{\text{ext}} - G_{\text{relax}}]_+ - d\rho - S \frac{\rho}{K + \rho}. \quad (16)$$

Global access coherence is damaged by residue and supported by functional structure:

$$\frac{dG}{dt} = \gamma_G [\sigma(\beta_G [\lambda_1(M(t)) - \lambda_c]) - G] - \beta_\rho \rho G, \quad (17)$$

where $\lambda_1(M)$ is the leading eigenvalue of the effective access matrix (defined in Sec. III).

Causal-loop availability evolves separately:

$$\frac{dA}{dt} = \gamma_A (A_0 - A) - \beta_A \rho A + u_A(t). \quad (18)$$

Reportability couples all components:

$$\mathcal{R}(t) = \frac{A(t)G(t)F_m(m)}{1 + \rho(t)/\rho_0}, \quad (19)$$

where $F_m(m)$ is the contribution of functional retained structure. A simple form is $F_m(m) = 1 - e^{-m/m_0}$, representing the saturating benefit of persistent structure. For self-model coherence, a non-monotonic form $F_m(m) = me^{-\lambda m}$ captures the cost of excessive rigidity. Throughout this paper, $\mathcal{R}(t)$ denotes reportability and $R_{\text{min}}^{(\tau)}(\varepsilon, t)$ denotes the minimum rate-distortion demand, which are distinct quantities.

C. Pruning threshold theorem

Stated in FDS terms:

Proposition 1 (Pruning lower bound) *Let ρ_c be the maximum residue compatible with reportable access. Let*

$$L_R^* = \zeta [\Delta_R^* - C_{\text{inv}}^* - G_{\text{ext}}^* - G_{\text{relax}}^*]_+ \quad (20)$$

be the effective unresolved distinction-load deposition rate over a relevant interval. If passive decay is $d\rho$ and pruning is $S\rho/(K + \rho)$, then maintaining $\dot{\rho} \leq 0$ at $\rho = \rho_c$ requires

$$S \geq S_c(\rho_c) = (L_R^* - d\rho_c)_+ \frac{K + \rho_c}{\rho_c}. \quad (21)$$

Interpretation. If $S < S_c$, residue cannot be bounded at ρ_c without additional capacity growth, externalization, invariant compression, or task relaxation. This connects the reportability model to the FDS prune-externalize-collapse trichotomy.

This section does not answer why there is something it is like to be a system. It does, however, make a structural claim about the geometry of reportable experience, placed in the appendix for those interested in the geometric extension.

V. THERMODYNAMIC MAINTENANCE COST

If active pruning irreversibly erases or reorganizes residual distinctions, it must have an energetic cost. The

FDS framework decomposes total maintenance cost into physical and informational components:

$$\dot{Q}_{\text{maint}} = \dot{Q}_{\text{phys}} + \dot{Q}_{\text{info}} + \Gamma(S) + \Xi(E_{\text{ext}}), \quad (22)$$

where $\dot{Q}_{\text{info}}$ is the informational heat floor from logically irreversible pruning, $\dot{Q}_{\text{phys}}$ is the biological or hardware physical overhead, $\Gamma(S)$ is the control cost of active pruning, and $\Xi(E_{\text{ext}})$ is the cost of externalization, retrieval, memory routing, or tool use.

The informational term follows Landauer’s principle:

$$\dot{Q}_{\text{info}} \geq \frac{k_B T \ln 2}{\tau} H(M_t | M_{t+1}, Y_t), \quad (23)$$

for logically irreversible state updates. This does not mean neural tissue operates near the Landauer limit. Biological overheads are many orders of magnitude larger. The importance of Eq. (23) is directional and constraining: pruning cannot be free. Neural measures such as BOLD, CMRO₂, glucose uptake, electrophysiological power, or local heat should be interpreted as proxies for $\dot{Q}_{\text{maint}}$, not for $\dot{Q}_{\text{info}}$ alone.

Operational proxies include local field potential power, BOLD/CMRO₂ coupling, glucose metabolism, heat-sensitive micro-measurements in artificial systems, and energy-per-bit estimates in neuromorphic or robotic agents. EEG/MEG complexity measures such as Lempel-Ziv complexity and perturbational complexity provide complementary information-theoretic proxies [7, 13].

VI. SIMULATION BENCHMARKS

The numerical simulations are not intended as fitted neural models. They are benchmark demonstrations showing how the theoretical quantities behave under controlled assumptions. The simulation figures are generated from released Python code with fixed random seeds.

A. Recovery ordering under anesthesia emergence

The model predicts a partial order, not a fixed universal timing. Causal responsiveness A can return before full global access G ; access-network recovery should precede stable self-model coherence; inhibitory/pruning rhythms should recover during the transition. This ordering is motivated by propofol EEG signatures and causal-complexity approaches to consciousness [7, 12]. Recent propofol studies also support a dynamical and network-level interpretation: propofol destabilizes neural dynamics across cortex [29] and disrupts predictive routing and local field phase modulation [30]. It can be tested with time-resolved EEG/MEG/fMRI during loss and recovery.

B. Rescue window

A pruning-interruption protocol is used to test reversibility. A system is first maintained with $S > S_c$, then pruning is shut off. At delay t_{restore} , pruning is restored to S_{rescue} . Rescue succeeds if $\mathcal{R}(t) > \mathcal{R}_{\text{rec}}$ within an observation window. The model predicts that required rescue pruning increases with delay and that a 50% rescue boundary exists.

C. Phase diagram

The model predicts a phase diagram over distinction load and pruning capacity. Low pruning and high load produce non-reportability; sufficient pruning maintains access; very low load or excessive rigid filtering can also reduce rich reportability. Figure 6 shows the central threshold structure.

D. Future in-machina benchmark program

The reduced simulations above test the normal-form structure of the FDS reportability model. A stronger validation path is to embed the same variables in high-dimensional artificial agents operating under nonstationary task drift, bounded memory, prediction-error accumulation, and selective pruning. Such benchmarks should compare at least four conditions (summarized in Table I): no pruning, passive decay, random pruning, and active task-sensitive pruning. They should measure reportability-like access, residue accumulation, access-network coherence, covariance early-warning signals, and rescue-window closure after delayed restoration of pruning.

The present paper does not require a particular artificial-agent implementation. Production-level embodied-agent architectures, robotics perception stacks, memory graphs, mission systems, pruning schedulers, and deployment control loops are outside the scope of this manuscript. The role of in-machina benchmarks is to test the FDS prediction that selective pruning, unlike passive decay or random deletion, maintains reportability-like access under finite capacity and nonstationary load.

VII. EXPERIMENTAL PREDICTIONS

We organize predictions into three primary tests and several secondary domains.

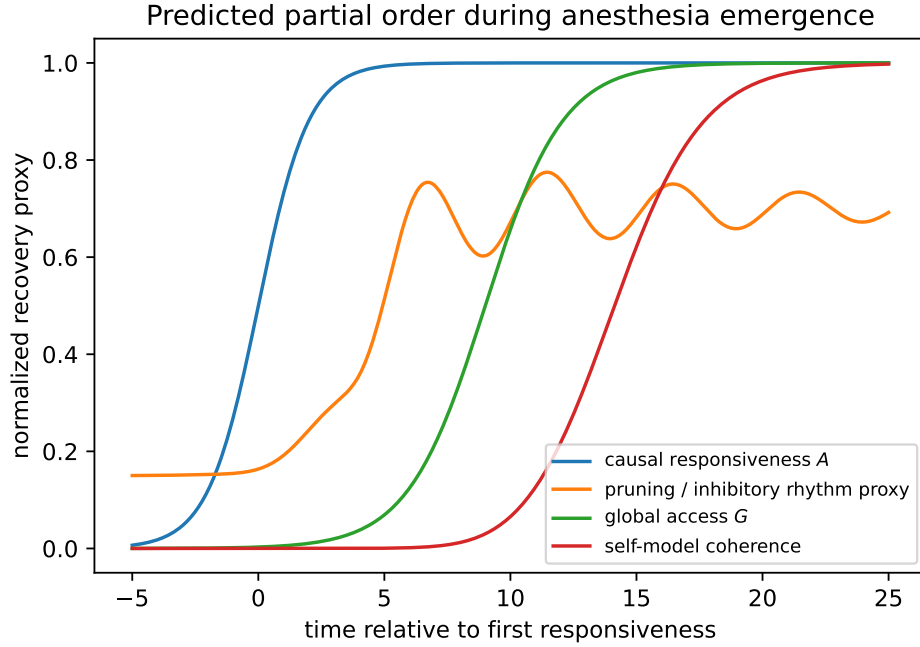


FIG. 4. Predicted partial order during anesthesia emergence. The model predicts that causal responsiveness can recover before full global access and self-model coherence. The precise timing is substrate-dependent; the falsifiable claim is the ordering constraint under controlled emergence.

TABLE I. Proposed artificial-agent benchmark conditions. This table defines a future in-machina test protocol rather than reporting unpublished benchmark results.

Condition	Maintenance mechanism	FDS prediction
No pruning	No active residue control	Residue accumulation and access collapse under s
Passive decay	Nonselective load reduction	Partial residue reduction but poor preservation of
Random pruning	Random deletion	Possible load reduction but unstable reportability
Active pruning	Task-sensitive selective removal or compression	Bounded residue and maintained reportability-like
Active pruning + externalization	Selective pruning plus external memory/tool support	Strongest maintenance under high load

A. Primary test 1: masking and attentional blink capacity-deficit test

T1 load increases Δ_R , raising ρ and increasing the reportability threshold for T2. Active control, rest, distractor suppression, or sleep-like reorganization should reduce ρ and improve T2 reportability. The theory predicts more than a binary report/no-report curve: recovery time, trial-to-trial autocorrelation, and leading covariance of access-network measures should rise as the system approaches the reportability threshold. Under attentional blink, residual load from target 1 should raise S_c for target 2.

Failure condition. If T2 reportability does not depend on load-induced residue or capacity manipulation after controlling for arousal and sensory strength, the capacity-deficit bridge is weakened.

B. Primary test 2: anesthesia emergence ordering

The strongest human test is time-resolved emergence from anesthesia. The predicted partial order is

$$A_{\text{reflex}} \rightarrow A_{\text{command}} \rightarrow G_{\text{access}} \rightarrow I_{\text{self}}, \quad (24)$$

where A_{reflex} is reflexive responsiveness, A_{command} is command following, G_{access} is stable global access, and I_{self} is coherent autobiographical or metacognitive self-report. Access-network topology and perturbational complexity should recover before stable self-model coherence; early-warning indicators should appear near transitions when emergence is quasi-static. Perturbational complexity, EEG signatures under propofol, and signal-complexity changes under anesthesia provide direct empirical anchor points [7, 8, 12, 13].

Failure condition. If high-level self-report coherence reliably recovers before causal responsiveness and access-network recovery across controlled cohorts and modalities, demote the recovery-ordering claim.

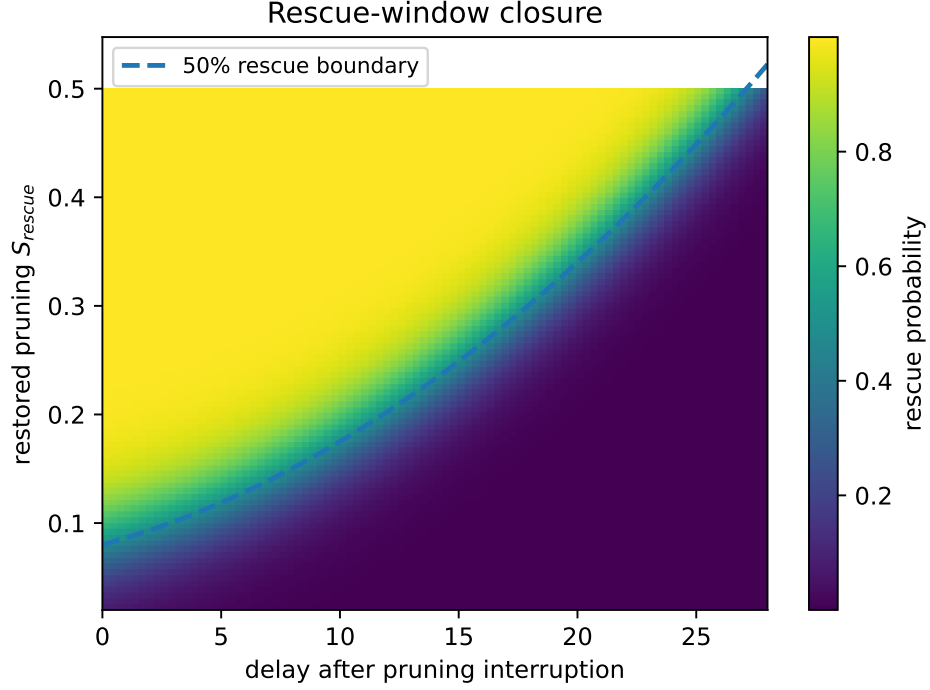


FIG. 5. Rescue-window closure. The heatmap shows rescue probability as a function of delay and restored pruning. Delayed intervention requires stronger pruning and can fail once the system crosses the basin boundary of reportability.

C. Primary test 3: artificial-agent sleep-like pruning benchmark

A resource-limited persistent agent serves as the primary in machina test. In a long-horizon nonstationary environment with drifting rules, bounded memory, noisy observations, and delayed consequences, the theory predicts that agents without pruning accumulate residue and lose adaptability. Agent variants include stateless mapper, persistent-memory without pruning, random forgetting, active pruning, externalization-enabled, and sleep-like offline replay. The FDS prediction is that active pruning outperforms no-pruning and random-forgetting agents, especially when distribution shifts generate stale internal structure.

Failure condition. If active pruning or externalization provides no advantage over no-pruning or random forgetting under bounded memory and sustained task drift, demote the artificial-agent bridge.

D. Secondary tests

Additional domains that support but are not required for the core theory: sleep deprivation (predicted increase in residue proxies and narrowing of rescue window), disorders of consciousness (predicted correlation between access-network topology and reportability), no-report paradigms (predicted dissociation between local processing and global access), EEG/MEG leading co-

variance early warnings (predicted rise near reportability transitions), and the information-theoretic cost of cognitive pruning (predicted correlation between pruning load and metabolic proxies). An operational proxy map is provided in Appendix E.

E. Artificial agency taxonomy and benchmark

a. FDS agency classification. The FDS AI agency framework distinguishes system types by their capacity-deficit management:

A stateless base model evaluated as an input-output mapper does not instantiate the full FDS reportability loop. It may transform inputs into high-quality outputs without maintaining an operational boundary, updating durable self-relevant state, or pruning accumulated residue. A coupled architecture containing writable memory, monitoring, action channels, externalization, active pruning, and causal participation in future viability conditions may instantiate system-level residue-pruning dynamics. The unit of analysis is the maintained system boundary, not the neural network alone. The taxonomy is architectural in the formal sense but not implementation-specific. The protocol is intended to define what future artificial-agent benchmarks must test; it does not specify or depend on any particular implementation. A public benchmark should expose sufficient details for reproduction, while production embodied-agent architectures and proprietary pruning mechanisms may

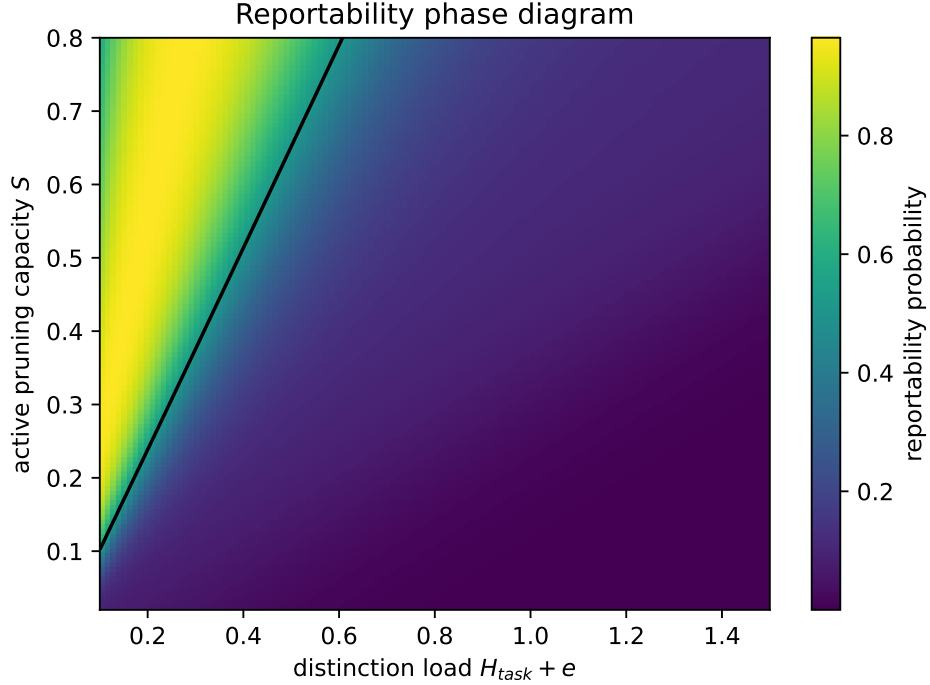


FIG. 6. Reportability phase diagram. Reportability probability is plotted over distinction load and pruning capacity. The contour marks the 50% reportability threshold.

System type	FDS classification	Residue-pruning dynamics
Stateless feedforward model	Passive mapper	No persistent system-level residue
Base LLM with context window	Weak scaffolded mapper	Transient context load only
LLM + memory + tools + planner	Scaffolded system	System-level residue, externalization burden
Embodied adaptive controller	Adaptive controller	Persistent maintenance burden
Strong FDS-agent	Strong agent	Full capacity-deficit, pruning, causal loop

TABLE II. FDS agency classification relevant to reportability-like access.

remain outside the paper’s scope. Concrete embodied-agent designs, pruning policies, memory graphs, mission systems, and robotics control stacks are outside the scope of this paper.

b. Artificial-agent benchmark protocol. A resource-limited persistent agent (recurrent or reinforcement-learning) serves as the primary in machina test. The environment is nonstationary with drifting rules, long-horizon dependencies, bounded memory, noisy observations, and delayed consequences. Agent variants include: stateless mapper, persistent-memory without pruning, random forgetting, active pruning, externalization-enabled, and sleep-like offline replay. Recent work on LLM-agent memory highlights the need for persistent memory, update, retrieval, and evaluation beyond static context windows [31].

VIII. DIFFERENTIAL PREDICTIONS RELATIVE TO EXISTING THEORIES

The proposed framework shares ground with several existing theories but adds distinctive predictions (summarized in Table III).

a. Relative to GWT. Global Workspace Theory emphasizes ignition and broadcast across specialized processors. The FDS framework agrees that global access is central but adds that broadcast alone is insufficient: under matched broadcast markers, accumulated capacity-deficit residue can impair reportability, produce rescue-window closure, and generate early-warning signatures not predicted by GWT alone.

b. Relative to IIT. Integrated Information Theory identifies intrinsic cause-effect integration as the substrate of consciousness. The FDS framework agrees that integration matters but predicts that high integration without adequate pruning can produce rigid or overloaded states rather than stable reportability. The leading covariance eigenvalue near reportability collapse pro-

vides a distinct testable signature.

c. Relative to FEP. The free-energy principle treats perception and action as inference under a generative model. The FDS framework treats the unresolved component of free energy under finite capacity as representational residue, making pruning thresholds and rescue windows explicit predictions that FEP does not derive.

d. Relative to information bottleneck. The information bottleneck method formalizes compression under capacity constraints. FDS agrees that compression is necessary but predicts that stateless compression is insufficient for reportability: persistent state, causal-loop embedding, and active maintenance are required.

e. Relative to PCI/complexity. Perturbational complexity and Lempel-Ziv complexity track conscious level empirically. FDS adds a mechanistic interpretation: these measures reflect access-network topology and leading eigenvalue dynamics. The theory predicts leading covariance early warnings near transitions that PCI markers alone do not specify.

f. Relative to AI agency. The FDS AI agency framework distinguishes passive mappers from scaffolded and strong agents (Table II). A stateless model evaluated as an input-output mapper does not instantiate the full reportability loop. A coupled architecture with writable memory, pruning, externalization, and causal participation can be tested for reportability-like access dynamics.

$(t_{\text{restore}}, S_{\text{rescue}})$. Confidence intervals can be obtained by bootstrap resampling over seeds.

IX. METHODS AND REPRODUCIBILITY

A. Nondimensionalization

All simulations use nondimensional benchmark units. Define $\hat{\rho} = \rho/\rho_0$, $\tilde{H} = H/H_0$, $\tilde{C} = C_{\text{acc}}/H_0$, $\tilde{t} = t/\tau$. Simulation figures are not fitted neural models and do not imply direct numerical equivalence between bits, joules, BOLD response, and behavioral report.

B. Simulation protocol

All simulations are nondimensional and are intended as benchmark tests, not fitted biological models. The released code generates all figures and CSV tables. Parameters are summarized in Table IV. A run is classified as reportable if $\mathcal{R}(t) > \mathcal{R}_{\text{min}}$ and $G(t) > G_{\text{min}}$ over the final observation window and if $|\dot{\mathcal{R}}|$ remains below a settling tolerance. A run is classified as failed if $\mathcal{R}(t) < \mathcal{R}_{\text{min}}$ for longer than a dwell time or if G or A falls below its failure threshold.

Critical pruning thresholds are estimated by bracketing and bisection over S . For stochastic simulations, reportability probability is estimated across random seeds and the threshold is the 50% access point obtained by monotone interpolation or logistic fit. Rescue-window boundaries are estimated as 50% rescue contours over

Theory	Shared ground	FDS-specific addition	Distinguishing prediction
GWT	global access/broadcast	access must be maintained under capacity deficit	matched broadcast markers can fail under high residue
IIT	integration matters	integration can become rigid without pruning	high integration with poor pruning may impair flexible report
FEP	error minimization	unresolved rate-distortion surplus becomes residue	pruning thresholds and rescue windows
Information bottleneck	compression	compression must be embedded in causal-loop maintenance	stateless compression is insufficient
PCI/complexity	complexity tracks conscious level	topology and residue determine recovery/collapse	leading covariance early warnings near transition
AI agency	causal loop and memory	reportability-like access requires pruning under capacity deficit	passive mappers lack full residue-pruning loop

TABLE III. Differential predictions relative to existing theories.

TABLE IV. Simulation parameters and reporting roles. Baseline values and sweep ranges are recorded in the released code.

Symbol	Meaning	Role in model	Reporting requirement
α	overload-to-residue gain	residue deposition	baseline + sensitivity
η_e	error-to-residue gain	unresolved prediction-error load	baseline + sensitivity
d	passive decay	non-active residue reduction	control runs
S	active pruning capacity	main control parameter	grid/bisection range
K	pruning half-saturation	nonlinear pruning response	baseline value
γ_G	access recovery rate	relaxation of global access	baseline + sensitivity
β_ρ	residue damage coefficient	access impairment	baseline + sensitivity
C_{acc}	effective capacity	finite-capacity constraint	sweep range
θ_G	access threshold	global access activation	fixed criterion
$\mathcal{R}_{\min}$	reportability failure threshold	collapse classification	fixed criterion
$\mathcal{R}_{\text{rec}}$	rescue threshold	rescue classification	fixed criterion
σ	noise amplitude	early-warning simulation	values used
N	network size	access graph simulation	value + random seed

X. DATA AND CODE AVAILABILITY

Source code, parameter files, generated figures, and CSV tables are provided for the reduced illustrative simulations included in this paper. High-dimensional artificial-agent benchmarks are discussed as a future validation path and are not part of the public release. Production-level embodied-agent architectures, robotics-oriented benchmarks, proprietary pruning implementations, and deployment-specific evaluation suites are outside the scope of this manuscript.

XI. MINIMAL FALSIFICATION PROTOCOL

The framework can be falsified or demoted in specific ways. Domain claims should fail locally without destroying the formal core.

A. Primary tests

Capacity-deficit residue. Demote if sustained reportability is maintained under $\Delta_R > 0$ with no pruning, no externalization, no task relaxation, no compression improvement, and no hidden capacity growth.

Network-topological bridge. Demote if reportability transitions show no measurable change in access-network topology, spectral connectivity, effective connectivity, or perturbational complexity after controlling for arousal and task demand.

Rescue-window closure. Demote if restoring pruning always recovers reportability independent of delay, residue state, or access-network damage.

Anesthesia ordering. Demote if high-level self-report coherence reliably recovers before causal responsiveness and access-network recovery across controlled cohorts and modalities.

AI pruning benchmark. Demote if active pruning or externalization gives no advantage over no-pruning or random forgetting under bounded resources and nonstationary load.

B. Optional bridge tests

Topological two-kink extension. Demote only the topological bridge if no paired kink appears under independently confirmed topology-mediated access transition. Do not propagate to core capacity-deficit claims.

XII. LIMITATIONS

First, the model targets reportability and conscious access, not the full ontological problem of phenomenal consciousness. Second, the rate-distortion derivation is

coarse-grained: it motivates the capacity-deficit equations but does not identify a unique neural implementation. Third, the network model abstracts away many details of thalamocortical, corticocortical, and subcortical dynamics. Fourth, the saddle-node normal form is a local candidate mechanism, not a universal theorem for all consciousness transitions; the model explicitly allows Hopf, noise-induced, percolation, and synchronization-loss transitions. Fifth, the thermodynamic bridge provides a lower bound, not an exact metabolic prediction. Sixth, the simulations are benchmark demonstrations and should be replaced by fits to real EEG/MEG/fMRI, behavioral, and artificial-agent data in future work. Seventh, the topological two-kink extension is an optional bridge; its failure would not falsify the core capacity-deficit theory of reportable access. Eighth, the artificial-agent benchmark protocol is proposed but not implemented as a public high-dimensional benchmark in this paper. Such benchmarks are natural future tests of the theory, but they require separate disclosure of environment design, agent architecture, seed sweeps, code, and evaluation criteria. The current paper therefore treats artificial-agent benchmarking as a proposed validation program rather than as a completed public result. Ninth, the paper does not report a full high-dimensional artificial-agent benchmark; the reduced simulations are low-dimensional and illustrative.

XIII. CONCLUSION

The framework predicts pruning thresholds, non-monotonic self-model coherence (via the tradeoff between functional retained structure m and nonfunctional residue ρ), rescue-window closure, leading-covariance early warnings, anesthesia emergence ordering, sleep-like maintenance requirements, and a thermodynamic lower bound for irreversible pruning. The Landauer term is only the informational heat floor; total maintenance cost is dominated by biological or physical overheads. Reduced simulations support the internal coherence of the FDS reportability dynamics by illustrating pruning thresholds, rescue-window closure, and early-warning behavior near access collapse. The next validation step is a high-dimensional in-machina benchmark in which finite artificial agents operate under nonstationary input, bounded memory, prediction-error accumulation, and selective pruning. Such benchmarks should test whether active pruning maintains reportability-like access more robustly than no pruning, passive decay, or random deletion. The framework does not claim to solve the hard problem of phenomenal consciousness. It converts a part of the consciousness problem into a testable maintenance problem for finite distinction systems, with explicit falsification conditions, differential predictions against competing theories, and an artificial-agent benchmark protocol.

Appendix A: Variational derivation sketch

The constrained objective in Eq. (4) induces two coupled optimization processes: online inference and maintenance. Online inference updates q_ϕ to reduce $\mathcal{F}_{\text{var}}$ while preserving task relevance. Maintenance updates the internal representation to keep $I(o; z)$ and update cost within budget. When $\mathcal{F}_{\text{var}}$ exceeds what can be resolved within C_{eff} , the residual load is deposited as ρ . Active pruning is the gradient-flow component that reduces ρ at a finite rate and at finite energetic cost. Saturation of pruning follows from finite throughput of the maintenance channel.

Appendix B: Network early-warning signal

Linearizing the high-dimensional access network near a fixed point gives

$$\dot{\mathbf{x}} = J\mathbf{x} + \boldsymbol{\xi}(t), \quad (\text{B1})$$

where J is the effective Jacobian and $\boldsymbol{\xi}$ is noise with covariance Q . The stationary covariance Σ satisfies the Lyapunov equation

$$J\Sigma + \Sigma J^T + Q = 0. \quad (\text{B2})$$

If the leading eigenvalue of J approaches zero from below, the leading eigenvalue of Σ diverges. Thus, in EEG/MEG/fMRI data, the preferred early-warning statistic is not scalar variance but the leading covariance eigenmode or the recovery time of the dominant access mode.

Appendix C: Phenomenal geometry

The paper does not answer why there is something it is like to be a system. It does, however, make a structural claim about the geometry of reportable experience. Let the agent compress high-dimensional input x into a low-dimensional access state z through $q_\phi(z|x)$. The reportable similarity structure over experiences is not the Euclidean geometry of x but the information geometry induced by q_ϕ . A natural metric is the Fisher-Rao metric

$$g_{ab}(z) = \mathbb{E}_{x \sim p(x|z)} [\partial_a \ln p(x|z) \partial_b \ln p(x|z)]. \quad (\text{C1})$$

Under capacity deficit, active pruning selects a lower-dimensional manifold that preserves task relevance at minimal cost. The theory predicts that reported qualitative similarity should track geodesic distances on this compressed manifold, while salience or intensity should track local distortion, curvature, and energetic cost.

Appendix D: Claim status and consequence of failure

Appendix E: Operational proxy map

The following table provides an operational mapping from model variables to candidate behavioral and neural proxies.

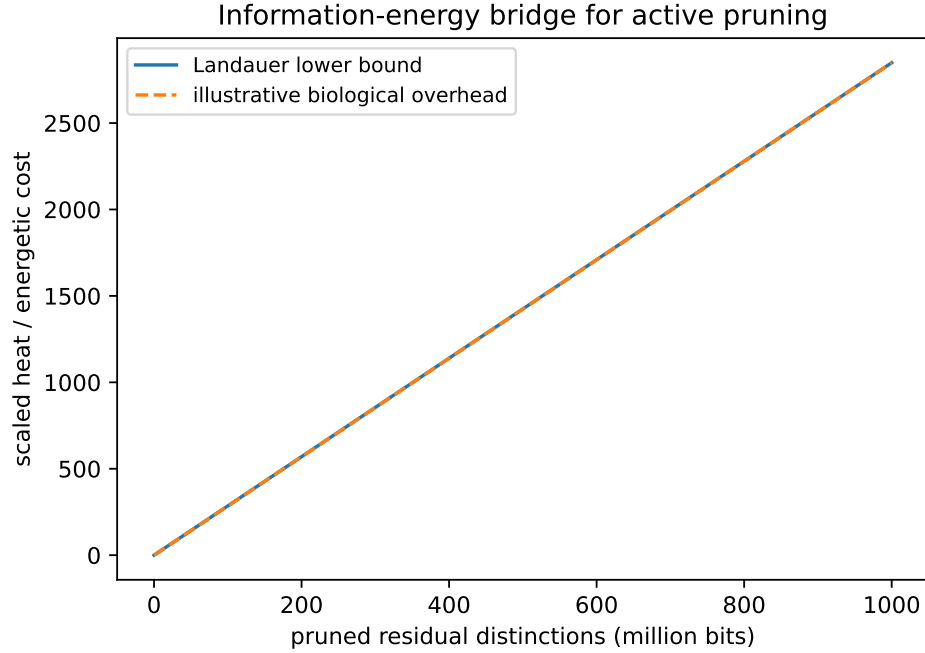


FIG. 7. Information-energy bridge. Irreversible pruning of residual distinctions implies a Landauer lower bound on heat dissipation, though biological overheads will dominate the bound. The prediction is not exact equality; it is that successful pruning under overload should not be energetically silent.

TABLE V. Claim status and consequence of failure.

Claim	Current support	Demotion condition	Consequence
Pruning threshold	FDS capacity-deficit model + reduced simulations	pruning helps but no threshold	threshold claim weakened
Network topology bridge	access-graph model + spectral reduced simulation	no topology/access relation	limit to scalar load model
Saddle-node reduction	mean-field normal form	transition is Hopf/noise/percolation only	replace fold mechanism
Early warnings	normal form + covariance theory	no slowing near quasi-static threshold	remove EWS prediction
Reportability window	m/ρ decomposition + reduced model	no non-monotonic self-model proxy	revise Ψ interpretation
Energy bridge	Landauer lower bound	no energetic cost in isolated pruning	restrict to computational bookkeeping
Anesthesia ordering	reduced simulation prediction	reversed robust temporal order	revise causal-loop/access hierarchy
Artificial-agent sleep	proposed benchmark protocol	no pruning benefit under bounded resources	restrict artificial-consciousness claim

TABLE VI. Operational proxy map. The table is a benchmark plan, not a claim of completed validation.

Model variable	Behavioral proxy	Neural / physiological proxy	Candidate paradigm
ρ residue/load	intrusion errors, lapse rate, perseveration	residual prediction error, EEG variability, theta/load markers	task switching, sleep deprivation, attentional blink
S active pruning	distractor suppression, recovery speed, forgetting/reconsolidation	alpha/beta inhibition, slow-wave downscaling, LTD-like markers	sleep, TMS perturbation, cognitive control tasks
C_{acc} effective capacity	working-memory span, dual-task capacity	PCI, Lempel-Ziv complexity, effective connectivity	masking, anesthesia, DOC
G global access	report/no-report, confidence, flexible use	P3b/ignition, thalamocortical connectivity, leading network eigenmode	masking, attentional blink, anesthesia
A causal-loop availability	motor responsiveness, command following	frontomotor effective connectivity, EMG responsiveness	anesthesia emergence, DOC
$\mathcal{R}$ reportability	verbal/manual report, confidence-calibrated access	report-linked ignition and complexity	masking, no-report variants
I_{self} self-model coherence	agency/self-continuity reports, metacognition	DMN-FPN coupling, self-referential network dynamics	anesthesia emergence, dissociation, sleep onset
ΔQ pruning cost	fatigue, effort, recovery cost	BOLD/CMRO ₂ , glucose use, local heat in artificial devices	overload and active forgetting

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Physical Bridge Claim Registry for Active Finite Distinction Systems: Time, Observers, Horizons, Topological Persistence, and High-Risk Physical Extensions

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This paper registers the principal physical bridge claims of Distinction Theory, a finite-system framework in which identity maintenance under finite capacity generates approximation, corrective complexity, dissipative pressure, pruning, collapse, or invariant-supported persistence. Unlike the FDS Core, which defines the formal finite-system architecture, and a forthcoming companion AI paper, which applies it to artificial agency, the present work organizes physical extensions by dependency, tier, falsification condition, and empirical status. The registry does not claim to derive the Standard Model, fix coupling constants, solve quantum gravity, or replace established physical theories. Instead, it identifies a set of bridge claims connecting finite distinguishability, observer-relative boundaries, irreversible distinction updates, horizon thermodynamics, topological persistence, CP/T-asymmetric identity transformation, and several quarantined high-risk interpretive extensions concerning particle identity and interaction functions. Each claim is assigned a scope, risk level, prior-art relation, and death condition. The purpose is priority, auditability, and future model development, not immediate consensus.

EPISTEMIC STATUS

This document is a claim registry, not a completed physical theory. It records physical bridge claims derived from, adjacent to, or motivated by the FDS formal core. Formal results remain in the FDS Core. Physical claims require additional bridge assumptions and may fail independently.

The registry has three purposes: (i) priority and time-stamping, (ii) dependency and risk classification, (iii) preparation for future standalone physical papers.

A claim listed here is not thereby asserted as established physics. Each claim is assigned a tier, dependencies, prior-art relation, novelty statement, falsification condition, empirical status, and required next work.

VERSIONING

This is Public Registry v1.0. Claims may be promoted, demoted, split, merged, or retired in later versions. A version change does not imply that the entire FDS programme has changed; it records the current status of physical bridge claims only.

INTRODUCTION: WHY A PHYSICAL REGISTRY IS NEEDED

From finite-system architecture to physical bridge claims

The core architecture of Distinction Theory begins from the finite-system cost chain:

Distinction $\rightarrow$ boundary $\rightarrow$ capacity deficit $\rightarrow$ approximation
 $\rightarrow$ corrective complexity $\rightarrow$ dissipative pressure
 $\rightarrow$ pruning / collapse / invariant-supported persistence.

The present paper does not re-derive that core chain. It asks a different question: if finite systems are physically instantiated, what kinds of physical structures should one expect to appear? The answer is not a single finished theory of physics. It is a registry of bridge claims connecting the finite-system architecture to physical notions such as observerhood, time, causal speed, horizon entropy, gravity, dark energy, topological protection, CP/T asymmetry, and the functional roles of the observed interactions.

The registry format is deliberate. Several of the claims below are close to existing physical traditions: Landauer erasure, Bekenstein bounds, horizon thermodynamics, emergent gravity, non-Hermitian topology, and CKM phase counting. Other claims are more speculative and should be treated as high-risk conjectures. The value of a registry is that it separates these cases. It prevents an interpretive bridge from being confused with a theorem, and it prevents the failure of one physical module from being misread as the failure of the finite-system architecture as a whole.

Registry, not final theory

This paper is not a final physical theory. It does not derive the Standard Model gauge group, particle masses, coupling constants, the numerical value of G , the numerical value of c , or the full Einstein field equations from the distinction primitive alone. It also does not assert that every existing physical principle is merely an instance of

Distinction Theory. Its goal is narrower and more auditable: to state physical bridge claims, list their dependencies, identify prior-art neighbors, and specify what would falsify or demote each claim.

A useful comparison is with a map of possible research liabilities. A mature physical theory normally has equations, fit parameters, experiments, and community-vetted limits. A research programme begins earlier: it identifies candidate dependencies and where they can fail. The present registry is written in that earlier sense. It is intended to make the physical claim-space searchable, citable, and falsifiable.

Relation to the core finite-system paper

The FDS Core introduces the formal finite-system architecture. A forthcoming companion AI paper applies that architecture to artificial agency. The present registry takes up physical bridge modules excluded from the formal core and organizes them by dependency, tier, and falsification condition.

What is not claimed

Tier labels do not measure importance. They measure epistemic status. A Tier C claim may be more exciting than a Tier B claim, but it is less secure. A Tier D claim may be useful for interpretation while remaining nonessential to the finite-system architecture.

Failure propagation rule

The registry follows a downstream failure rule:

If a physical bridge claim fails, only that claim and its downstream modules are revised, demoted, quarantined, or abandoned. Failure of a physical bridge does not by itself falsify the distinction primitive or the bridge-free finite-system architecture.

This rule is crucial. Without it, the registry would become a brittle grand theory in which any failed physical speculation destroys the entire programme. The purpose of the registry is the opposite: to make each claim independently auditable.

Registry template

Each major claim is stated using the following template:

1. Claim ID and name.

2. Tier.

3. Statement.

4. Dependencies.

5. Prior-art relation.

6. Novel DT contribution.

7. Falsification condition.

8. Empirical status.

9. Required next work.

The template is intentionally repetitive. It makes visible the difference between a claim that is being asserted, a claim that is being interpreted, and a claim that is being proposed for future testing.

Claim-ID convention

The registry uses an explicit prefix system to avoid ambiguity. Each claim ID has the form **XX-N** where **XX** is a two-letter prefix identifying the claim family:

Risk stratification at a glance

The following table summarizes the relative strength of each claim family.

MINIMAL PHYSICAL BRIDGE ASSUMPTIONS

Finite distinguishability

Assumption (Finite distinguishability). A physically instantiated system that maintains identity has a finite distinguishability budget. Its internal record, memory, or state-maintenance capacity is bounded by finite physical resources.

This assumption is compatible with a broad range of existing ideas: finite entropy bounds, Bekenstein-type limits on information in bounded regions, finite memory in computation, and finite channel capacities in communication theory [3, 9, 11]. Distinction Theory does not require the strongest possible version of these bounds for every system. It requires only that physical systems maintaining identity do not possess infinite usable distinguishability.

Not claimed	Reason
FDS derives the Standard Model gauge group	Only functional bridge hypotheses registered; no derivation of $SU(3)_c \times SU(2)_L \times U(1)_Y$
FDS fixes coupling constants	No quantitative model yet; only structural roles registered
FDS derives c numerically	Only finite causal access bridge proposed; no derivation of numerical value
FDS derives Einstein field equations	Only horizon-thermodynamic interpretation registered; no field-equation derivation
FDS solves quantum measurement	Observer is finite distinction-register, not collapse solution; interpretive only
FDS replaces QFT, GR, or SM	Registry is bridge map, not replacement theory
FDS proves dark energy	Only high-risk horizon-maintenance hypothesis; no derivation of Λ

Prefix Claim family

PB	Physical bridge assumptions (finite distinguishability, Landauer, finite causal access)
OT	Observer / Time claims (finite distinction-register, operational time, causal speed)
BC	Bounded computation claims (finite-memory reversible computation, dissipation)
HT	Horizon / Thermodynamic claims (horizon as distinguishability, gravity)
DE	Dark-energy claims (horizon-maintenance dark energy, $w(z)$, area suppression)
TP	Topological persistence claims (invariant support, two-kink signature)
FG	Fermion generation claims (CKM lower bound, exact-three minimality)
FF	Fundamental force claims (functional decomposition)
PX	Pauli-exclusion claims (finite address protection)
SY	Symmetry claims (physical symmetries as low-maintenance invariants)
ME	Mass-energy claims (stored distinction, update capacity)
ST	Stochasticity claims (finite truncation, effective randomness)
QD	Decoherence claims (distinguishability leakage)
QC	Quantum computation claims (maintenance-cost barrier)
ML	Mathematical form claims (invariant compression)
QG	Quantum gravity / black-hole horizon claims (finite-distinguishability access boundary)
BH	Black-hole information claims (boundary encoding, finite observer recoverability)
PI	Particle identity claims (species as invariant-supported identity classes)
MS	Mass-spectrum claims (mass as identity-maintenance cost)
YK	Yukawa coupling claims (identity-transformation channel weights)
DM	Dark-matter claims (low-addressability gravitational distinction residue)
QM	Quantum measurement claims (measurement as finite record stabilization)
BR	Born-rule bridge (protective note: not derived)
YM	Yang–Mills confinement claims (boundary encapsulation of color distinctions)
MG	Mass-gap claims (minimum cost of stable distinction excitation)
GT	Gauge-redundancy claims (quotient over description-equivalent distinctions)
CI	Cosmological initial-condition claims (low-entropy initial macroscopic complexity)
IN	Inflation claims (horizon-scale distinction dilution)
CMB	Cosmic-microwave-background claims (Phase-B cosmological residue)
CA	Cosmological-arrow claims (record asymmetry under finite observers)
FR	Freeze-out claims (distinction stabilization across cooling expansion)

Landauer bridge

Assumption (Landauer bridge). When a physically instantiated system performs a logically irreversible operation—erasure, many-to-one overwrite, reset, or compression of previously distinguishable states—there is a thermodynamic entropy cost under specified physical conditions.

The bridge is deliberately stated in a conservative form. Distinction Theory does not claim that every computation dissipates $k_B T \ln 2$ per operation. Reversible computation, unitary evolution, read-only operations, and stable storage are not automatically charged with

this cost. The claim is that irreversible distinction loss, when physically implemented under appropriate thermal conditions, must be paid for in the total entropy budget [4–6].

Locality and finite causal access

Assumption (Finite causal access). A finite physical system cannot instantaneously access, register, update, and causally use arbitrary distant distinctions without unbounded update capacity.

This assumption prepares the observer, time, and hori-

Module	Strength	Reason
Bounded-memory reversible computation	Strongest	Clear prior art (Landauer, Bennett); measurable garbage/reset/dissipation
Landauer bridge	Strong	Established physical basis; conservative use
Topological two-kink (TP-3)	Strong but novel	DT-specific observable; requires experimental proxy
Observer/Time (OT-1, OT-2)	Medium	Conceptual but structurally disciplined; clear falsifiers
Horizon/Gravity (HT-1, HT-2)	Medium	Strong prior art but no equations yet; needs concrete model
Dark energy (DE-1, DE-2, DE-3)	High-risk	High model-selection risk; needs specific $w(z)$ predictions
Fermion exact-three (FG-3)	High-risk	Needs rigorous link to flavor sector; speculative
Four forces (FF-1, FF-2)	Very high-risk	Currently functional mapping only; no quantitative model
Pauli address protection (PX-1)	Interpretive	Must not replace spin-statistics theorem; remains non-essential
BH/QG (BH-1, BH-2, QG-1)	Medium	Structurally disciplined but requires microscopic model
Dark matter (DM-1, DM-2)	High-risk	Needs concrete candidate identification or geometric model
QM/Records (QM-1, QM-2, QM-3)	Medium	Structurally grounded but Born rule remains prospective
Cosmology (CI-1, CMB-1, CA-1)	Medium	Broadly consistent but not DT-specific; requires concrete models
Confinement/Mass gap (YM-1, MG-1)	Very high-risk	Currently interpretive; needs quantitative FDS cost model
Particle identity (PI-1, MS-1, YK-1)	Very high-risk	Interpretive bridge to SM; no parameter derivation

zon claims. It is not a derivation of relativity. It is the finite-system precondition from which a maximum causal update rate becomes natural.

directly tested as a single experiment but constrains later claims about time, measurement, causal access, and horizon boundaries.

OBSERVER AND TIME CLAIMS

Claim OT-1: Observer as finite distinction-register

Tier. B.

Statement. An observer is not defined by consciousness, biological status, or human subjectivity. An observer is a finite distinction-register: a physical system capable of registering, preserving, updating, and ordering distinctions under finite capacity.

Dependencies. Finite distinguishability; boundary maintenance; record stability.

Prior-art relation. This claim touches the operational tradition in physics, the role of measurement records in quantum mechanics, and cybernetic accounts of observation. It does not depend on anthropocentric consciousness.

Novel DT contribution. Observerhood is redefined as a finite-system maintenance role. The observer is not an epistemic add-on to physics; it is the finite carrier of registered distinctions.

Falsification condition. A complete physical account of registered states that requires no finite distinction carrier, no record stability, no boundary, and no capacity constraint would undermine this claim.

Empirical status. Conceptual bridge claim. It is not

Claim OT-2: Time as ordered irreversible distinction-update

Tier. B/B+.

Statement. Operational time is the ordered structure of irreversible distinction updates within finite systems. A finite record obtains temporal order because distinctions are registered, truncated, overwritten, or stabilized in a sequence that cannot be freely inverted without restoring erased degrees of freedom.

Important limitation. This claim does not deny mathematical recurrence in ideal finite-dimensional closed unitary systems. It denies that a physically erased or truncated distinction can reappear as an accessible physical record without external information restoration or entropy compensation.

Dependencies. Finite distinction-register; Landauer bridge; record stability; irreversible update.

Prior-art relation. The claim is adjacent to thermodynamic arrows of time, Landauer erasure, decoherence, and records-based accounts of temporal asymmetry [4, 5, 33–35]. It differs by locating the operational arrow in finite distinction registration rather than only in statistical entropy increase.

Novel DT contribution. The arrow is framed as a cost of finite record update: time is not only what clocks

measure, but what finite systems produce when they irreversibly update accessible distinctions.

Falsification condition. A demonstrated physical process that restores an actually erased distinction as an accessible record with complete accounting of reservoirs, correlations, feedback records, and work sources below the generalized Landauer bound would falsify the Landauer-dependent version of this claim.

Claim OT-3: Finite and invariant causal speed from finite distinction access

Tier. C/B.

Statement. A finite observer cannot instantaneously access, register, and causally use arbitrary distant distinctions without infinite update capacity. Therefore a finite physical world requires a maximum causal update rate. The invariance of this rate across admissible observers follows from the requirement of mutually consistent causal distinction orderings. When realized, this invariant rate plays the structural role of the speed c .

Dependencies. Finite causal access; observer as finite distinction-register; consistent record ordering.

Prior-art relation. Special relativity postulates an invariant causal speed and builds Lorentz kinematics around it [32]. Distinction Theory does not replace that derivation. It proposes an upstream finite-system reason why an invariant causal bound is necessary.

Not claimed. This registry does not derive the numerical value of c or Maxwell's equations.

Falsification condition. A finite observer capable of receiving, registering, and operationally using arbitrary instantaneous signals without record inconsistency, unbounded update capacity, or causal contradiction would undermine this bridge.

Discussion. The invariance of the causal bound follows from the requirement that mutually consistent causal distinction orderings be maintained across admissible observers. Operational infinite causal access with consistent records across all observers without resource bounds would falsify this consistency requirement.

BOUNDED MEMORY, REVERSIBLE COMPUTATION, AND DISSIPATION

Claim BC-1: Finite open-ended reversible computation must eventually dissipate under bounded memory

Tier. B+.

Statement. Reversible gates can avoid Landauer cost locally. However, a finite open-ended physical computer with bounded memory, persistent input/output, error

correction, state reuse, or garbage disposal must eventually perform irreversible compression, reset, or pruning, and therefore incur nonzero total dissipation.

Dependencies. Landauer bridge; finite memory; open-ended input/output; state reuse.

Prior-art relation. Reversible computation is a classical route to minimizing dissipation [6, 7]. The Distinction Theory claim is not that reversible gates are impossible. It is that finite open-ended operation cannot indefinitely avoid garbage accumulation or irreversible housekeeping.

Novel DT contribution. The claim links reversible computation to active pruning: finite systems avoid collapse not merely by computing reversibly, but by managing accumulated structural residue under bounded memory.

Falsification condition. A bounded-memory, open-ended physical computer that indefinitely performs nontrivial adaptive computation with persistent input/output and no irreversible reset, compression, error correction, garbage disposal, or entropy export would falsify the claim.

Near-term test. Construct finite-memory reversible circuits or computational substrates under continuous task streams. Measure garbage accumulation, reset frequency, error-correction overhead, and heat output as a function of task depth and memory budget.

HORIZON, ENTROPY, AND GRAVITY

Prior-art boundary

The gravity-related claims below must be read against a large body of prior work. Bekenstein and Hawking connected horizon area with entropy and temperature [8, 10]. Unruh related acceleration and temperature [12]. Jacobson derived the Einstein equation as an equation of state from horizon entropy and the Clausius relation $\delta Q = TdS$ [13]. Induced and emergent gravity programmes have explored gravity as an effective or thermodynamic phenomenon [14–16]. Distinction Theory does not claim priority over these ideas. Its contribution is to interpret horizon thermodynamics as finite distinction maintenance under bounded observer capacity.

Claim HT-1: Horizon as observer-relative distinguishability boundary

Tier. B.

Statement. A horizon is the boundary of causal distinguishability for a finite observer-system. It is not merely a geometric surface; it is the boundary beyond which distinctions cannot be registered, updated, or operationally used by the observer within the relevant causal structure.

Dependencies. Finite causal access; finite distinction-register; causal boundary.

Prior-art relation. The claim is consistent with causal horizons, Rindler horizons, black-hole horizons, and holographic entropy bounds [9, 11, 13].

Novel DT contribution. The horizon is interpreted as an operational distinguishability boundary. Its thermodynamic properties reflect the cost of maintaining a finite causal-access boundary.

Falsification condition. A physically meaningful horizon with no relation to causal access, information bounds, record accessibility, or distinguishability constraints would undermine this claim.

Claim HT-2: Gravity as effective geometry of horizon maintenance

Tier. C/B.

Statement. Gravity is the macroscopic equilibrium geometry of finite causal-access maintenance. Energy and information distributions constrain what can be distinguished, accessed, and updated; in the equilibrium continuum limit, this constraint appears as effective geometry.

Dependencies. Horizon as distinguishability boundary; finite entropy; thermodynamic horizon relation.

Prior-art relation. This claim is adjacent to Jacobson's thermodynamic derivation, Sakharov induced gravity, entropic gravity, and analogue gravity [13–15, 17].

Not claimed. This paper does not derive the full Einstein field equations, the value of Newton's constant, or a microscopic quantum gravity model.

Falsification condition. A complete physical account of gravitational geometry in which horizons, entropy, finite causal access, and distinguishability constraints play no necessary role would demote this claim to a nonessential analogy.

Claim HT-3: Non-equilibrium gravitational corrections

Tier. B+/C.

Statement. If Einstein geometry is the equilibrium limit of horizon maintenance, then far-from-equilibrium horizon regimes should admit corrections controlled by distinguishability saturation, maintenance delay, or horizon-mode registration.

Candidate observables. Analogue-horizon entropy-production kinks, black-hole ringdown corrections, modified gravitational-wave propagation, or early-universe tensor-spectrum deviations.

Risk note. Astrophysical black-hole corrections may be extremely small, potentially scaling like $1/S_{\text{BH}}$. Analogue systems may provide more realistic tests.

Falsification condition. A confirmed horizon system driven far from equilibrium that shows no maintenance-cost anomaly, no entropy-production anomaly, and no distinguishability-saturation effect across a controlled regime would weaken this bridge.

DARK ENERGY AS HORIZON MAINTENANCE COST

Claim DE-1: Horizon-scale dark-energy naturalness

Tier. C/B+.

Statement. If the cosmological horizon is a finite distinguishability boundary, its maintenance cost naturally scales as

$$\rho_{\text{DE}} \sim H^2 M_{\text{Pl}}^2. \quad (1)$$

Dependencies. Horizon as distinguishability boundary; finite entropy; cosmic causal boundary; physical bridge from maintenance cost to energy density.

Prior-art relation. The scaling resembles holographic dark energy and horizon-scale effective field theory arguments [19, 20]. Distinction Theory must be distinguished from standard HDE/RDE cutoff models, some of which are strongly constrained by current data.

Novel DT contribution. The scale is interpreted as horizon-maintenance cost of observer-relative distinguishability, not merely as an infrared cutoff imposed on quantum fields.

Claim DE-2: 10^{-122} as area-suppressed horizon ratio

Tier. C.

Statement. The observed dark-energy scale may be understood as an area-like horizon suppression:

$$\frac{\rho_{\text{DE}}}{\rho_{\text{Pl}}} \sim H_0^2 t_P^2 \sim \left(\frac{\ell_P}{R_H} \right)^2 \sim 10^{-122}. \quad (2)$$

Interpretation. A three-dimensional volume suppression would give a much smaller number. The DT interpretation is that horizon maintenance is boundary-like, not volume-like.

Limitation. This is a scaling/naturalness argument until embedded in a concrete cosmological evolution model.

Claim DE-3: Equation-of-state prediction

Tier. B+ / high-risk.

Statement. A strict horizon-maintenance model predicts dynamical dark energy with $w(z) > -1$ and asymptotic approach toward -1 , unless apparent phantom

crossing is produced by effective reconstruction, sector mixing, or parameterization artifacts.

Empirical status. DESI DR2 and combinations with CMB and supernova datasets have reported preferences for dynamical dark-energy parameterizations in some combinations, with model and dataset dependence [21, 22]. These data do not yet select the DT mechanism.

Falsification condition. Future DESI, Euclid, Roman, or CMB-S4 results that lock $w(z) = -1$ to high precision would demote DE-3. Robust, unavoidable $w < -1$ crossing incompatible with any DT effective reconstruction would falsify the strict inequality version.

Required next work. A concrete $H(z)$ and $w(z)$ model distinguishing DT horizon maintenance from standard holographic, Ricci, quintessence, phantom, and modified-gravity alternatives.

TOPOLOGICAL PERSISTENCE AND THE SECOND-LAW BOUNDARY

Prior-art boundary

Non-Hermitian topology has established that spectra, point gaps, boundary conditions, and skin accumulation can be tied to winding and non-Bloch topological structures [23–26]. Recent experiments have realized non-Hermitian skin effects in multiple platforms, including photonic, acoustic, electrical, and ultracold-atom systems [27]. Distinction Theory does not claim that NHSE itself is new. Its claim is that topological persistence can impose operational limits on local forgetting or coarse-graining.

Claim TP-1: Invariant-supported persistence

Tier. B.

Statement. Under repeated pruning, local noise, or collapse, structures whose identity is carried by global invariants are lower-maintenance persistence carriers than structures requiring full local reconstruction.

Dependencies. Finite-system trichotomy; invariant-supported identity; local noise or perturbation.

Prior-art relation. The claim is consistent with symmetry protection, topological phases, conserved quantities, and error-protecting global invariants.

Novel DT contribution. Persistence is interpreted as a maintenance-cost problem: invariants survive because they reduce the burden of local repair.

Claim TP-2: Topological obstruction to local forgetting

Tier. B+.

Statement. Local coarse-graining, forgetting, or dissipation may fail to erase globally protected topological invariants. The second law governs entropy accounting, but local forgetting may face invariant-supported boundary conditions.

Important non-claim. DT does not claim violation of the second law. It proposes a boundary on local forgetting and coarse-graining, not an exception to thermodynamics.

Falsification condition. A controlled topological system whose globally protected invariant can be erased by purely local Markovian forgetting with no nonlocal operation, phase transition, or compensating entropy channel would weaken this claim.

Claim TP-3: Two-kink experimental signature

Flagship near-term experimental protocol. TP-3 is the registry’s most concrete DT-specific experimental proposal: it predicts paired non-analytic behavior in steady-state entropy production and operational forgetting rate at the same confirmed topological transition.

Tier. B+.

Statement. At a topological transition parameter γ_c , DT predicts paired non-analytic behavior in steady-state entropy production and operational forgetting rate:

$$\dot{\Sigma}_{ss}(\gamma) \quad \text{and} \quad \kappa(\gamma) \quad (3)$$

should exhibit aligned kinks at the same γ_c .

Why this matters. NHSE existence alone is not DT-specific. The paired kink is the DT-specific fingerprint: topology should affect not only accumulation or boundary localization, but also the operational cost of erasure and forgetting.

Falsification condition. A controlled dissipative topological system crosses a confirmed topological phase boundary while both $\dot{\Sigma}_{ss}(\gamma)$ and $\kappa(\gamma)$ remain analytic after finite-size, boundary, disorder, and noise artifacts are excluded.

Required next work. Define operational forgetting rate in a platform-specific way, such as decay of reconstructability of an initial boundary-localized state, loss of recoverable winding information under local noise, or minimal erasure cost to remove topological memory.

Claim SY-1: Physical symmetries as low-maintenance invariant structures

Tier. C.

Statement. Physical symmetries are not merely formal conveniences; they are low-maintenance structures that survive pruning, coarse-graining, and repeated representation changes. Under finite distinguishability and

active pruning pressure, symmetric configurations are selectively preserved because they require fewer distinct update operations to maintain. This selects for invariance rather than complexity.

Dependencies. Invariant-supported persistence (TP-1); finite distinguishability; active pruning.

Prior-art relation. Noether's theorem links symmetries to conservation laws [40]. DT does not contradict this. It proposes an upstream reason: symmetries are low-cost invariant structures favored by finite distinguishability and pruning pressure, which explains why conservation laws appear at all.

Not claimed. The specific gauge group of the Standard Model or the number of observed symmetries.

Falsification condition. A physical system with active pruning and finite distinguishability shows no preference for symmetric configurations over asymmetric ones under equivalent maintenance cost.

HIGH-RISK INTERPRETIVE EXTENSIONS

The claims in this section are not required by the FDS Core, the observer-time claims, bounded-memory computation, or topological-persistence predictions. They are registered as high-risk interpretive or structural hypotheses. Failure of these claims does not affect any earlier claim family.

FERMION GENERATIONS AND CP/T-ASYMMETRIC IDENTITY TRANSFORMATION

Prior-art boundary

The Kobayashi-Maskawa mechanism showed that CP violation in a CKM-type quark-mixing sector requires at least three generations [28]. In general, an N -generation CKM matrix has $(N-1)(N-2)/2$ physical CP-violating phases; for $N=2$ this number is zero, and for $N=3$ it is one. Precision electroweak, Higgs, flavor, and cosmological constraints strongly constrain ordinary sequential fourth-generation scenarios [29].

Claim FG-1: CKM lower-bound recovery

Tier. B.

Statement. If microscopic identity transformation requires an irreducible CP/T-asymmetric channel, and if this channel is realized through CKM-type unitary flavor mixing, then algebraic phase counting requires $N_{\text{gen}} \geq 3$.

Prior-art relation. DT does not claim to discover the CKM lower bound. It reinterprets the lower bound as

a possible finite-system necessity for irreversible identity transformation.

Novel DT contribution. The CKM lower bound is placed inside the chain

$$\begin{aligned} &\text{active pruning} \rightarrow \text{microscopic irreversibility} \\ &\rightarrow \text{CP/T-asymmetric channel} \rightarrow N_{\text{gen}} \geq 3. \quad (4) \end{aligned}$$

Falsification condition. A CKM-type unitary flavor-mixing sector with physical CP violation and only two generations would falsify the algebraic bridge. A non-CKM CP mechanism would not falsify the claim but would limit its scope.

Claim FG-2: Active pruning requires microscopic asymmetry

Tier. C / high-risk.

Statement. A physical world that permits irreversible identity transformation or selective pruning requires microscopic CP/T-asymmetric channels.

Novel DT contribution. CP/T asymmetry is interpreted not merely as a parameter of weak interactions but as a candidate physical carrier of irreversible identity transformation.

Falsification condition. A complete physical account of irreversible identity transformation with no microscopic CP/T-asymmetric channel, no environmental arrow, and no compensating asymmetry would demote this claim.

Claim FG-3: Thermodynamic minimality of three generations

Tier. C / high-risk.

Statement. More than three ordinary sequential generations may introduce redundant distinguishability channels and maintenance cost without improving minimal CP/T-asymmetric closure; thus $N=3$ may be thermodynamically minimal.

Status. This is not a theorem. It is a high-risk physical bridge conjecture.

Falsification condition. Discovery of a fourth ordinary sequential fermion generation compatible with electroweak, CKM, Higgs, and cosmological constraints would strongly demote this conjecture.

Required next work. A free-energy or distinguishability functional over flavor-sector complexity that penalizes redundant CP channels and predicts $N=3$ as an optimum, not merely as an observed fact.

FUNCTIONAL DECOMPOSITION OF THE FOUR FUNDAMENTAL INTERACTIONS

Claim FF-1: Four interactions as minimal finite-distinction operations

Tier. C (high-risk interpretive).

Statement. The four observed interactions correspond to four minimal operations required by finite distinguishability:

strong interaction	encapsulation / confinement,
electromagnetic	long-range Abelian connection,
weak interaction	identity transformation / pruning,
gravity	horizon-geometric maintenance.

Not claimed. This is not a derivation of $SU(3)_c \times SU(2)_L \times U(1)_Y$, coupling constants, masses, renormalization flows, or spontaneous symmetry breaking.

The four-interaction mapping is an isomorphism-seeking interpretive bridge. It is not a derivation of the Standard Model gauge group, coupling constants, field content, or renormalization structure.

Novel DT contribution. The interactions are interpreted functionally as finite-distinction operations rather than merely as forces acting on particles.

Falsification condition. A discovered fundamental interaction that cannot be mapped onto encapsulation, connection, identity transformation, or horizon-geometric maintenance under rigid criteria would challenge the completeness of the decomposition.

Claim FF-2: No unscreened long-range non-Abelian connection force

Tier. B+/C.

Statement. A universal long-range connection channel must be path-independent, composable, and effectively Abelian. Any non-Abelian sector in the low-energy physical world must be confined, Higgsed, screened, or strongly suppressed.

Prior-art relation. QCD is non-Abelian but confined at low energy; weak interactions are non-Abelian but Higgsed. The DT claim is a functional interpretation of why non-Abelian structure does not appear as a universal unscreened macroscopic connection channel.

Falsification condition. Discovery of an unscreened, massless, long-range, non-Abelian fundamental force with macroscopic $1/r$ -type behavior and non-negligible coupling would falsify the strong version.

Claim FF-3: Weak identity transformation requires irreducible CP/T-asymmetric invariant

Tier. C.

Statement. A complete weak-like identity-transforming sector must contain an irreducible CP/T-asymmetric invariant, though individual processes may be CP-conserving.

Falsification condition. Discovery of a complete flavor-changing, identity-transforming fundamental sector with weak-scale coupling and no irreducible CP/T-asymmetric invariant would undermine the claim.

Claim FF-4: Gravity as horizon-geometric maintenance, not ordinary internal gauge force

Tier. C.

Statement. Gravity is not merely one more internal gauge operation. It is the effective geometry of finite causal-access maintenance.

Interpretive value. This claim explains why gravity couples universally to stress-energy rather than to a particular internal charge. Anything that changes distinguishability density or causal accessibility enters the geometry ledger.

PAULI EXCLUSION AND FINITE ADDRESS PROTECTION

Claim PX-1: Pauli exclusion as finite distinguishability address protection

Tier. D/C.

Statement. Pauli exclusion may be interpreted as finite-address protection: two indistinguishable fermionic distinction records cannot occupy the same physical address without identity collapse.

Prior-art relation. This does not replace the spin-statistics theorem or quantum field theoretic derivations of fermionic antisymmetry [30, 31]. It is an interpretive bridge from exclusion to finite distinguishability.

Falsification condition. A regime in which identical fermions stably violate exclusion while preserving all relevant quantum statistics and no compensating structure appears would undermine the interpretive bridge.

Status. High-risk interpretive claim. It should not be used as a primary physical prediction until mathematically connected to spin-statistics.

Limitation. This claim is not an explanatory substitute for the spin-statistics theorem. It is a proposed upper-level interpretation of why an antisymmetric fermionic address structure is useful for finite distinguishability. Until a functorial or algebraic map is constructed from fermionic antisymmetry to finite-address protection, PX-1 remains an interpretive bridge only.

MASS-ENERGY CONVERTIBILITY AND STORED DISTINCTION

Claim ME-1: Mass as localized invariant-supported distinction storage

Tier. C.

Statement. Mass can be interpreted as localized, stable, invariant-supported distinction storage.

Interpretation. A massive structure is not merely energy at rest; it is a persistent local carrier of physically addressable distinctions.

Not claimed. This is an interpretive bridge, not a derivation of $E = mc^2$ or relativistic mass-energy equivalence.

Claim ME-2: Energy as distinction-update capacity

Tier. C.

Statement. Energy is the capacity to update, transform, maintain, or release physical distinctions.

Not claimed. This is an interpretive bridge, not a derivation of thermodynamic energy definitions or conservation laws.

Claim ME-3: Mass-energy convertibility

Tier. C.

Statement. If mass is stored distinction structure and energy is update capacity, then conversion between mass and energy expresses the convertibility of stored structure and executable update capacity.

Not claimed. This is an interpretive bridge, not a derivation of $E = mc^2$ or relativistic mass-energy equivalence.

Prior-art relation. Relativity already explains mass-energy equivalence once Lorentz symmetry and invariant causal speed are assumed [32]. Distinction Theory does not replace that derivation.

Limitation. DT does not derive $E = mc^2$ unless the invariant causal speed and Lorentz-type structure are independently recovered.

CLAIM DEPENDENCY MAP

Table I summarizes the dependency logic. The most important point is that failure propagates locally. If the dark-energy module fails, the topological-persistence module may survive. If the Pauli interpretation fails, the observer/time claims may survive. The registry is not a single all-or-nothing claim.

Dependency DAG

High-risk claims are downstream and do not affect upstream core claims if they fail. The dependency structure is summarized below.

Level 0 (Core): FDS Core.

Level 1 (Physical Bridge): finite distinguishability (PB), Landauer bridge, finite causal access.

Level 2 (Core-Derived):

- observer register (OT-1) → operational time (OT-2), finite causal access (OT-3)
- bounded-memory computation (BC-1)
- invariant-supported persistence (TP-1) → topological obstruction (TP-2) → two-kink signature (TP-3)

Level 3 (High-Risk or Interpretive):

- horizon thermodynamics (HT-1, HT-2) → dark energy (DE-1, DE-2, DE-3)
- CP/T asymmetry (FG-1, FG-2, FG-3)
- functional decomposition (FF-1, FF-2, FF-3)
- Pauli interpretation (PX-1)

EXPERIMENTAL ROADMAP: TEST MATURITY LADDER

Level 1: Conceptual audit

Observer as finite distinction-register. Clarify operational definition of observer in terms of finite record carrier, distinguishability budget, and boundary maintenance.

Operational time. Define ordered irreversible distinction update in terms of Landauer erasure cost and record stability.

Finite causal access. Formalize maximum causal update rate from finite distinguishability.

Level 2: Simulation-ready

Active-pruning AI benchmark. Although not a physical test, it is the nearest operational test of the finite-system architecture. Compare passive accumulation, bounded memory, active pruning, and causal loop closure on long-horizon tasks with explicit resource budgets.

Bounded-memory reversible computation. Build or simulate finite-memory reversible circuits under continuous input/output streams. Measure garbage accumulation, reset frequency, and dissipated heat. The

Claim family	Depends on	Risk	Failure consequence
Observer/time	finite distinction registration	medium	revises observer/time module
Landauer bridge	irreversible erasure thermodynamics	medium	demotes L1b physical bridge
Causal speed	finite causal access	high	revises relativity bridge only
Horizon/gravity	finite causal access + entropy bounds	high	revises gravity module
Dark energy	horizon maintenance scaling	high	demotes DE module
Topological persistence	invariant-supported structure	medium	revises TP module
Two-kink prediction	dissipative topological testbed	medium-high	strongest TP falsifier
Fermion lower bound	CP/T + CKM-type mixing	medium	revises FG bridge scope
Fermion exact-three	thermodynamic minimality	high	demotes upper-bound conjecture
Four forces	functional operation decomposition	high	demotes FF module
Pauli bridge	finite address protection + spin-statistics link	high	remains interpretive only

TABLE I. Dependency and failure propagation map for major physical claim families.

target is not to refute reversible computing but to test the bounded-memory housekeeping claim.

Finite truncation effective stochasticity. Develop numerical models showing that finite distinguishability alone produces operational stochasticity in reduced descriptions.

Level 3: Experimental proxy

NHSE two-kink experiment. In a non-Hermitian topological system, tune through a confirmed topological transition and measure both entropy production and operational forgetting rate. The falsification target is paired analyticity where DT predicts aligned kinks.

Analogue-horizon Landauer kink. Use BEC, optical, acoustic, or circuit analogue systems to tune effective horizon conditions and measure entropy-production or mode-registration anomalies near capacity-like transitions [17, 18]. Note: this is a proposed experimental proxy, not yet a DT-specific prediction.

Entropy-production finite-system bound. Measure boundary-maintenance entropy production in controlled finite systems and compare against theoretical lower bounds.

Level 4: Observational cosmology

Dark-energy equation-of-state constraints [HIGH-RISK]. DESI, Euclid, Roman, and CMB-S4 will continue constraining $w(z)$. DT requires a more precise model before its dark-energy claim can be sharply separated from ordinary quintessence, phantom, HDE/RDE, and modified-gravity alternatives.

Gravitational-wave propagation or ringdown anomalies [HIGH-RISK]. If gravity is horizon maintenance, far-from-equilibrium horizon events may produce tiny non-GR signatures. Astrophysical signals may be too small, but standard sirens and third-generation detectors may constrain the space.

Level 5: High-risk structural physics

Fermion-sector extensions [HIGH-RISK]. Precision flavor physics and searches for fourth-generation or vector-like fermions constrain the exact-three conjecture. Discovery of an ordinary sequential fourth generation would strongly demote the thermodynamic-minimality claim.

Four-force functional decomposition [HIGH-RISK]. Develop quantitative mapping from finite-distinction operations to observed interaction sectors.

Pauli address protection [INTERPRETIVE]. Construct functorial or algebraic map from fermionic antisymmetry to finite-address protection before promoting to physical prediction.

WHAT WOULD COUNT AS SUCCESS?

Weak success

Weak success means the registry clarifies the claim-space: which physical ideas follow from which assumptions, what is prior art, what is novel, and what would fail if a claim is wrong.

Moderate success

Moderate success means at least one physical bridge claim generates a test protocol not already standard in the relevant field. Candidate examples are the bounded-memory reversible-computation dissipation test and the analogue-horizon Landauer-kink test.

Strong success

Strong success means a DT-specific prediction is confirmed. The clearest candidates are:

- paired entropy-production and forgetting-rate kinks at a topological transition (TP-3);
- analogue-horizon maintenance-cost kinks, if a DT-specific observable is defined;
- bounded-memory open-ended reversible computation showing unavoidable reset/pruning dissipation scaling (BC-1).

Dark energy remains a high-risk candidate requiring a concrete $w(z)$ model before it can be sharply tested against alternatives.

Transformative success

Transformative success would require multiple independent physical domains to exhibit the same structural signature:

finite boundary $\rightarrow$ maintenance cost $\rightarrow$ pruning / collapse / invariant persistence.

At that point, DT would no longer be merely a registry of physical bridges. It would become a candidate organizing principle for finite physical systems.

Such convergence would support the view that the thermodynamic cost of distinguishability is a fundamental constraint on physical evolution across finite systems.

DISCUSSION

Why a registry is scientifically useful

A broad physical programme is easy to overstate. It is also easy to dismiss. A registry format avoids both failure modes. It states claims in a way that can be indexed, challenged, and revised. It also clarifies which claims are original and which are reinterpretations of established physics.

In this registry, horizon thermodynamics is not new; the DT contribution is the finite-distinction interpretation. CKM phase counting is not new; the DT contribution is the active-pruning-to-CP/T bridge. NHSE is not new; the DT contribution is the two-kink forgetting/entropy fingerprint. Landauer erasure is not new; the DT contribution is its integration into a finite-system maintenance-cost chain.

The strongest and weakest modules

The strongest modules are those with explicit observables and clear prior-art interfaces: bounded-memory computation, Landauer accounting, topological persistence, and the two-kink experiment. The weakest modules are those that currently remain interpretive: Pauli

address protection, exact-three thermodynamic minimality, and the full four-force functional decomposition. Dark energy has high potential but also high model-selection risk.

What this paper does not do

This paper does not compute a Lagrangian, derive field equations, solve quantum gravity, or claim that all physical constants are fixed by distinction alone. It also does not claim that existing physics is wrong. In many cases, DT aims to explain why existing structures might be necessary under finite-system constraints, not to replace the successful equations that already describe them.

CONCLUSION

This paper has registered a dependency-ordered physical claim-space for Distinction Theory. The registry begins from finite distinguishability, irreversible distinction updates, finite causal access, and the Landauer bridge. It then organizes physical extensions into observer/time, bounded computation, horizon/gravity, dark energy, topological persistence, fermion generations, four interaction functions, Pauli exclusion, and mass-energy convertibility.

The result is not a completed physics. It is an auditable map. Some claims reinterpret known structures: Landauer erasure, horizon thermodynamics, Bekenstein bounds, CKM phase counting, and invariant persistence. Others propose testable extensions: a topological two-kink signature, analogue-horizon maintenance kinks, horizon-maintenance dark energy, and bounded-memory reversible-computation dissipation. These vary substantially in risk, from near-term conservative tests to high-risk cosmological hypotheses. Each claim is assigned a scope and a death condition.

The registry's value is therefore not that every claim is already true. Its value is that the physical extension of FDS is no longer an undifferentiated speculative cloud. It is a dependency-ordered, failure-aware, and updateable claim-space.

The author thanks all those who provided critical feedback on earlier drafts. The author used AI-assisted editing tools for language polishing, structural feedback, and draft preparation. All claims, arguments, citations, and final wording were reviewed and approved by the author.

Appendix A: Full Claim Registry Tables

TABLE II. Core and medium-risk physical claims.

ID	Claim	Tier	Key prediction	Falsifier
OT-1	Observer as finite distinction-register	B	Observer requires finite record carrier	Observer-independent registered state with no finite carrier
OT-2	Time as irreversible distinction update	B/B+	Erased records cannot recur without entropy compensation	Entropy-free restoration of erased accessible record
OT-3	Finite and invariant causal speed	C/B	Finite observers require invariant causal update bound	Operational infinite causal access without inconsistency
BC-1	Bounded-memory reversible computation dissipates	B+	Open-ended finite computation requires reset/pruning	Zero-dissipation bounded open-ended computation
HT-1	Horizon as distinguishability boundary	B	Horizons track finite causal access	Horizon independent of access/information constraints
HT-2	Gravity as horizon maintenance geometry	C/B	GR-like geometry as equilibrium limit	Complete gravity with no horizon/information role
HT-3	Non-equilibrium horizon corrections	B+/C	Far-from-equilibrium horizons show maintenance anomalies	No anomaly in controlled non-equilibrium horizon test
DE-1	$\rho_{DE} \sim H^2 M_{pl}^2$	C/B+	Horizon-maintenance dark-energy scale	Data selects incompatible dark-energy model
DE-2	10^{-122} as area suppression	C	$\rho_{DE}/\rho_{Pl} \sim (\ell_P/R_H)^2$	Scaling shown physically irrelevant
TP-1	Invariant-supported persistence	B	Global invariants reduce maintenance burden	Invariants erased locally without transition or cost
TP-2	Topological obstruction to forgetting	B+	Local forgetting cannot erase global invariant cheaply	Local Markovian forgetting erases invariant completely
TP-3	Two-kink signature	B+	Entropy and forgetting kinks align at γ_c	No kinks at confirmed topological transition
FG-1	CKM lower-bound recovery	B	CP/T CKM channel requires $N_{gen} \geq 3$	CKM-type CP with two generations
FF-2	No long-range non-Abelian connection force	B+/C	Non-Abelian sectors confined/Higgsed/screened	Unscreened massless long-range non-Abelian force
ST-1	Effective stochasticity from finite truncation	C/B	Finite reduced descriptions produce operational stochasticity	Bounded observer tracks all relevant degrees without residual uncertainty
QG-1	Horizon as finite-distinguishability access boundary	C/B	BH horizon is distinguishability boundary for finite observer	Horizon with no relation to causal access or distinguishability
BH-1	Boundary encoding vs finite observer recoverability	C	Information may be globally preserved but operationally inaccessible	Complete recovery of BH information by any finite observer
PI-1	Particle species as invariant-supported identity classes	C	SM particle content as invariant identity classes	Particle species with no invariant-supported identity structure
DM-2	Stability of dark-matter candidates	C	DM must be stable low-dissipation identity carrier	Dark matter with no stable identity structure
QM-1	Measurement as finite record stabilization	C	Measurement stabilizes reproducible distinction record	Record stabilization without any finite carrier
QM-2	Classical outcome as stable distinction record	C	Classicality from record stability across multiple observers	Classical outcome with no distinction record stability
QM-3	Global unitarity vs finite observer update	C	Unitary global evolution compatible with irreversible local record update	Contradiction between unitarity and finite-record thermodynamics
GT-1	Gauge redundancy as description-equivalence quotient	C	Gauge symmetry as quotient over equivalent distinction descriptions	Gauge symmetry with no redundancy quotient interpretation
CMB-1	CMB as Phase-B cosmological residue	C	CMB is persistent record of early-universe distinction structure	CMB with no relation to distinction structure
CA-1	Cosmological arrow as record asymmetry	C	Arrow of time from finite-observer record asymmetry	Arrow of time without finite-observer record asymmetry
FR-1	Freeze-out as distinction stabilization	C	Freeze-out stabilizes distinction structure across cooling expansion	Freeze-out mechanism independent of distinction structure

TABLE III. High-risk, interpretive, and speculative extensions.

ID	Claim	Tier	Key prediction	Falsifier
DE-3	$w(z) > -1$ horizon-maintenance evolution	B+ (high-risk)	Dynamical DE approaching -1	$w = -1$ locked or incompatible phantom crossing
FG-2	Active pruning requires microscopic asymmetry	C (high-risk)	Identity transformation needs CP/T channel	Irreversible identity transformation with no asymmetry
FG-3	Exact three as thermodynamic minimality	C (high-risk)	No ordinary fourth sequential generation	Confirmed ordinary fourth generation
FF-1	Four interactions as finite-distinction operations	C	Four functional classes of interaction	New force outside all four rigid classes
FF-3	Weak identity transformation requires CP/T invariant	C	Weak-like sector contains irreducible CP/T invariant	Weak-like complete sector with no such invariant
PX-1	Pauli as finite address protection	D/C	Exclusion protects distinguishability address	Stable exclusion violation with fermionic statistics intact
ME-1	Mass as stored distinction	C	Mass stores stable local distinctions	No structural interpretation possible
ME-2	Energy as update capacity	C	Energy updates/maintains distinctions	No relation between energy and physical update capacity
ME-3	Mass-energy convertibility	C	Stored structure converts to update capacity	Relativity/QFT-compatible counterexample
SY-1	Physical symmetries as low-maintenance invariants	C	Symmetries favored by pruning pressure	No preference for symmetric over asymmetric configurations under equivalent cost
QD-1	Decoherence as distinguishability leakage	C	Environment drains coherence via distinguishability exchange	Persistent coherence despite verified distinguishability-leakage coupling
QC-1	Quantum computation barriers from pruning cost	D/C	Error-correction overhead creates maintenance-cost constraints	Fault-tolerant QC with sub-polynomial overhead scaling
ML-1	Mathematical form as invariant compression	D	Physics is mathematical because it's low-maintenance invariant	Successful physics requires non-invariant, non-compressible descriptions
BH-2	Page-curve compatibility	D/C	Page curve compatible with finite-observer recoverability bound	Page curve with no FDS-compatible distinguishability model
MS-1	Mass as identity-maintenance cost	C/D	Mass is cost of stable identity-class maintenance	Mass with no identity-maintenance interpretation
YK-1	Yukawa couplings as identity-transformation weights	D/C	Yukawa weights set CP/T-asymmetric identity-transform channel strengths	Yukawa structure with no identity-transformation interpretation
DM-1	Dark matter as low-addressability distinction residue	C/D	DM is stable low-dissipation identity carrier with weak EM addressability	Particle DM fully excluded without FDS-compatible residue candidate
DM-3	Effective-geometry dark-matter alternative	D	Geometry alternative to particle DM if direct detection excludes candidates	Direct detection confirms particle DM with no geometric alternative needed
BR-1	Born-rule bridge (prospective)	D	Born rule requires probability-geometry map; not yet derived	Complete Born-rule derivation from FDS principles achieved or impossible
YM-1	Confinement as boundary encapsulation	D/C	Confinement is boundary encapsulation of non-extractable color distinctions	Confinement mechanism with no FDS interpretable cost structure
YM-2	String tension as maintenance cost	D/C	Linear string tension is growing cost of separated color identity	String tension with no FDS interpretable cost structure
MG-1	Mass gap as minimum distinction cost	D/C	YM mass gap is minimum cost of nontrivial stable gauge distinction	Mass gap established with no relation to excitation cost
CI-1	Low-entropy initial condition	C/D	Low initial entropy $\equiv$ low realized macroscopic distinction complexity	Initial condition mechanism independent of distinction complexity
IN-1	Inflation as distinction dilution	D	Inflation dilutes pre-existing distinctions while seeding new structure	Inflation mechanism with no FDS interpretation

Appendix B: Glossary

Distinction: The primitive act of separating what is from what is not.

Finite distinguishability: The condition that a physical system can register only finitely many operationally usable distinctions.

Boundary maintenance: The ongoing process by which a system preserves the distinction between self and environment.

Capacity deficit: The gap between environmental predictive demand and internal modeling budget.

Corrective complexity: Accumulated internal structure produced by patches, model updates, and corrections to approximation error.

Active pruning: Selective, energy-consuming removal, compression, or reorganization of inert complexity to preserve viability.

Invariant-supported persistence: Persistence through structures whose identity is carried by global invariants, symmetries, topological classes, or conserved quantities.

Landauer bridge: The physical bridge from logically irreversible erasure to thermodynamic entropy cost.

Operational time: Ordered irreversible update of accessible distinctions.

Horizon: Observer-relative causal distinguishability boundary.

Two-kink signature: Paired nonanalytic behavior in entropy production and forgetting rate at the same topological transition.

Appendix C: Full Claim-Space Archive

The complete claim-space edition of Distinction Theory, including physical, biological, cognitive, artificial-agency, and civilizational extensions, is archived as:

Wu, Y. *Distinction Theory: A General Theory of Finite Systems*. Zenodo, 2026. DOI: 10.5281/zenodo.20130174.

Validity Boundary for Interpretive Quantum Claims

The interpretive claims PX-1, ST-1, QD-1, QC-1, and ML-1 should not be read as replacements for the mathematical structures of quantum theory. PX-1 does not replace the spin-statistics theorem; ST-1 does not derive Bell-inequality violation or the Born rule; QD-1 does not solve the measurement problem; QC-1 does not deny the threshold theorem of fault-tolerant quantum computation; and ML-1 does not derive mathematics from physics. These claims register finite-system interpretations and possible bridge directions. Their failure would demote the relevant interpretive module without threatening the observer/time, Landauer, horizon, or topological claims.

Appendix D: Extended High-Risk Physical Bridge Claims

The claims in this section are registered as high-risk interpretive or structural hypotheses. They are not required by the FDS Core or by claims in earlier sections. Failure of these claims does not affect any earlier claim family. Each claim follows the registry template: ID, Tier, Statement, Dependencies, Prior-art relation, Novel DT contribution, Not claimed, Falsification condition, Required next work.

Quantum gravity and black-hole information

Claim QG-1: Horizon as finite-distinguishability access boundary. Tier C/B. A black-hole horizon is a finite-distinguishability access boundary: the point beyond which distinctions cannot be registered, updated, or operationally used by an external finite observer. This extends HT-1 to the strong-gravity regime.

Claim BH-1: Boundary encoding versus finite observer recoverability. Tier C. FDS separates global distinction preservation from finite observer recoverability. Information may be preserved globally (in the full physical state) while being inaccessible to any finite observer with bounded causal access. This does not imply information loss; it separates the ontological claim from the operational one.

Claim BH-2: Page-curve compatibility. Tier D/C. The Page curve and island formulae are compatible with FDS if the finite observer's recoverable information is bounded by its causal access and distinguishability budget. Required future work: embed island formulae in an FDS distinguishability framework.

Not claimed. FDS does not provide a microscopic quantum-gravity model, derive the Page curve, solve the firewall problem, or replace semiclassical gravity.

Particle identity, mass, and coupling structure

Claim PI-1: Particle species as invariant-supported identity classes. Tier C. Observed particle species correspond to stable, invariant-supported identity classes under finite distinguishability constraints. The Standard-Model particle content is one realization of this principle.

Claim MS-1: Mass as identity-maintenance cost. Tier C/D. Mass is the maintenance cost of a stable identity class under finite distinguishability: a massive particle requires persistent distinction updates to maintain its identity across propagation.

Claim YK-1: Yukawa couplings as identity-transformation channel weights. Tier D/C. Yukawa couplings set the relative strength of identity transformation between fermion species. In FDS terms, they weight the CP/T-asymmetric identity-transformation channels. Derivation from FDS principles is not yet available.

Not claimed. FDS does not derive Standard-Model gauge groups, coupling constants, mass values, the Higgs mechanism, or Yukawa matrix structure.

Dark matter

Claim DM-1: Dark matter as low-addressability gravitational distinction residue. Tier C/D. Dark matter is a stable, low-dissipation identity carrier that participates in gravitational boundary conditions while remaining weakly addressable through electromagnetic distinction channels.

Claim DM-2: Stability of dark-matter candidates. Tier C. Any viable dark-matter candidate must be a stable, low-dissipation identity carrier under finite-system pruning and maintenance pressures. This is consistent with WIMP, axion, and sterile-neutrino scenarios but does not select among them.

Claim DM-3: Effective-geometry alternative. Tier D. If particle dark matter is excluded by future direct-detection and collider constraints, an effective-geometry alternative (modification of horizon-scale gravitational distinction) remains open. This is registered as a speculative placeholder, not a prediction.

Not claimed. FDS does not identify the dark-matter particle, predict its mass, or claim that modified gravity is correct.

Quantum measurement and records

Claim QM-1: Measurement as finite record stabilization. Tier C. A quantum measurement is the process by which a finite distinction-register stabilizes a reproducible record from a superposed or entangled state. The outcome is classical insofar as it is a stable, reproducible distinction record.

Claim QM-2: Classical outcome as stable distinction record. Tier C. Classicality arises from record stability across multiple finite observers, not from collapse of the wavefunction alone. A result that can be registered, preserved, and communicated by bounded observers is operationally classical.

Claim QM-3: Global unitary evolution versus finite observer record update. Tier C. The global quantum state may evolve unitarily while the finite observer's record undergoes irreversible distinction update. This resolves no contradiction between unitary quantum mechanics and finite-record thermodynamics.

Claim BR-1: Born-rule bridge. Tier D. A derivation of the Born rule from FDS principles is not yet available. A future bridge would require a probability-geometry map linking distinction-count ratios to measurement outcome frequencies. This remains prospective.

Not claimed. FDS does not solve the quantum measurement problem, derive the Born rule, violate Bell inequalities operationally, or replace standard decoherence theory.

Confinement and mass gap

Claim YM-1: Confinement as boundary encapsulation. Tier D/C. Confinement in non-Abelian gauge theory corresponds to boundary encapsulation of non-extractable color distinctions. Color charge cannot be isolated because the cost of separating color-distinct states grows with distance, consistent with a maintenance-cost interpretation.

Claim YM-2: String tension as maintenance cost. Tier D/C. The linear string tension in QCD is the growing maintenance cost of maintaining separated color-distinct identity carriers. This is consistent with lattice QCD but adds an FDS interpretation.

Claim MG-1: Mass gap as minimum distinction cost. Tier D/C. The Yang–Mills mass gap is the minimum excitation cost of a nontrivial stable distinction in the gauge field. An FDS model would need to show this cost is positive and finite.

Claim GT-1: Gauge redundancy as description-equivalence quotient. Tier C. Gauge symmetry is the quotient over physically redundant descriptions of the same distinction structure. This does not replace gauge theory but provides an upstream interpretation.

Not claimed. FDS does not solve the Yang–Mills Millennium Problem, derive the mass gap, compute the string tension, or replace QCD.

Cosmological initial conditions and relic records

Claim CI-1: Low-entropy initial condition. Tier C/D. The low-entropy initial condition of the universe corresponds to low realized macroscopic distinction complexity. The early universe had few distinguishable macroscopic configurations.

Claim IN-1: Inflation as distinction dilution. Tier D. Inflation dilutes pre-existing distinctions across horizon scales while seeding new small-scale distinction structure through quantum fluctuations. This is consistent with standard inflationary cosmology but adds no new mechanism.

Claim CMB-1: CMB as Phase-B cosmological residue. Tier C. The cosmic microwave background is a stabilized, persistent record of early-universe distinction structure—a Phase-B cosmological residue in FDS terms.

Claim CA-1: Cosmological arrow. Tier C. The cosmological arrow of time is the record asymmetry produced by finite observers embedded in an expanding, cooling universe with irreversible distinction updates.

Claim FR-1: Freeze-out as distinction stabilization. Tier C. Freeze-out of particle species and relic abundances across cosmic expansion is the stabilization of distinction structure as the universe cools and dilutes.

Not claimed. FDS does not replace inflationary cosmology, predict the primordial power spectrum, derive the baryon asymmetry, or solve the flatness/horizon problems without existing mechanisms.

Appendix E: Additional Interpretive Bridge Claims

Claim ST-1: Effective stochasticity from finite truncation

Tier. C/B.

Statement. When a finite observer or finite subsystem coarse-grains inaccessible degrees of freedom, the discarded variables appear as effective stochasticity in the reduced description. This claim concerns operational stochasticity in finite descriptions, not a replacement for the Born rule, Bell nonlocality, or the full quantum measurement problem.

Dependencies. Finite distinguishability; capacity deficit; approximation; observer-relative boundary.

Prior-art relation. Statistical mechanics derives stochasticity from many degrees of freedom. DT proposes an additional upstream source: finite distinguishability alone, even with a single degree of freedom, forces lossy truncation that appears as effective stochasticity to an observer with bounded access.

Scope limitation. ST-1 does not claim that all microscopic quantum randomness is merely classical ignorance or hidden variable truncation. In particular, it does not by itself derive Bell-inequality violation, the Born rule, or contextuality. Those require separate quantum-bridge assumptions.

Boundary. ST-1 concerns the general effective stochasticity produced by any finite system’s truncation of inaccessible degrees of freedom, covering classical algorithmic and statistical contexts. In the quantum regime, when environmental degrees of freedom exceed the system’s tracking capacity, this distinguishability leakage is instantiated as quantum decoherence, addressed separately in QD-1.

Falsification condition. A finite observer with demonstrably bounded access that can nevertheless track all causally relevant degrees of freedom of a reduced system without residual uncertainty, stochastic coarse-graining, hidden environmental dependence, or measurement back-action would undermine the operational version of this claim.

Claim QD-1: Decoherence as environmental distinction leakage

Tier. C.

Statement. Quantum decoherence is reinterpreted as the leakage of distinguishability information into the environment. A system’s quantum coherence requires maintaining superpositions as distinguishable internal states. When the environment continuously interacts with and entangles those states, distinguishability leaks outward, destroying internal coherence. Decoherence rate is thus a measure of distinguishability exchange with the environment.

Dependencies. Finite distinguishability; boundary maintenance; environment as distinguishability sink.

Prior-art relation. Decoherence theory is standard [36]. This registry proposes that decoherence rate has an interpretation in terms of distinguishability exchange, complementing the standard environmental entanglement view.

Limitation. QD-1 does not solve the measurement problem or derive the Born rule. It reinterprets the standard environment-induced loss of coherence as distinguishability leakage relative to a finite boundary.

Falsification condition. A quantum system shows persistent coherence despite continuous distinguishability-leakage interactions with a large environment.

Claim QC-1: Quantum computation barriers from finite error-correction and pruning cost

Tier. D/C.

Statement. Quantum computation error-correction incurs active-pruning-like costs. Maintaining logical qubits requires continuous correction of physical qubit errors, which is a form of active complexity management. Under finite distinguishability and finite energy budgets, there exists a threshold beyond which error-correction overhead exceeds the computational advantage, creating a maintenance-cost barrier for scalable quantum computation under finite energy and error-correction budgets.

Dependencies. Finite distinguishability; active pruning; bounded free energy.

Prior-art relation. Quantum error-correction is standard [37, 38]. This registry proposes a DT-specific interpretation: error-correction is pruning at the quantum level, and finite maintenance budgets impose maintenance-cost constraints on scalable quantum computation.

Falsification condition. Fault-tolerant quantum computation with arbitrarily low logical error rates is achievable with overhead scaling that is sub-polynomial in the logical qubit count (e.g., constant or logarithmic).

Claim ML-1: Mathematical form of physical laws as invariant compression

Tier. D.

Statement. The mathematical form of physical laws is not an arbitrary human invention but an invariant compression of the underlying distinction structure. Invariant-supported structures (symmetries, conservation laws, geometrical relationships) survive repeated pruning and coarse-graining because they are the lowest-maintenance way to encode distinguishability relationships. This explains why physics is mathematical: mathematics is the language of invariant compression.

Dependencies. Invariant-supported persistence (TP-1); low-maintenance invariant structures (SY-1).

Prior-art relation. Wigner’s “unreasonable effectiveness of mathematics” [39] is a philosophical puzzle. DT provides an operational answer: mathematical forms survive because they are low-maintenance representations of distinction structure.

Falsification condition. If successful physical theories systematically required non-compressible, non-invariant, representation-dependent descriptions with no stable symmetry, conservation, or geometric structure, the claim would be weakened.

Limitation. ML-1 is not a derivation of mathematics from physics. It is a proposed selection principle for why invariant mathematical representations are preferentially

stable under repeated model revision, coarse-graining, and pruning.

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Life and Cognitive Science Bridge Claim Registry for Active Finite Distinction Systems:

Active Pruning, Boundary Maintenance, Death-like Collapse, Capacity Bottlenecks, and Consciousness Recovery

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This paper registers the principal life-science and cognitive-science bridge claims of Distinction Theory, a finite-system framework in which persistent identity requires boundary maintenance under finite capacity, resource constraints, residual complexity, and update cost. Unlike the FDS Core, which states the formal finite-system architecture, and unlike future standalone simulation or experimental papers, the present document is a claim registry. It does not claim to solve the origin of life, define clinical death, prove a theory of consciousness, diagnose disease, or propose therapy. Instead, it registers auditable bridge claims concerning active pruning, residue-induced collapse, death-like maintenance-attractor loss, surface-to-volume scaling in protocell-like systems, near-collapse warning signatures, finite-capacity cognitive integration, and anesthesia recovery ordering. Each claim is assigned a tier, dependencies, prior-art relation, novelty statement, falsification or demotion condition, empirical status, and required next work. The core biological claim is that, in finite boundary-maintaining systems under sustained flux and residual-load production, active pruning is not an optional refinement but a candidate first-principles requirement for long-term life-like persistence. The core cognitive claim is that reportability and self/agency coherence should recover according to finite-capacity bottleneck constraints. The purpose is priority, auditability, and future model development, not immediate consensus.

EPISTEMIC STATUS

This document is a bridge claim registry, not a completed theory of life, death, consciousness, medicine, or cognition. It records life-science and cognitive-science bridge claims derived from, adjacent to, or motivated by the FDS formal core. Formal results remain in the FDS Core. Biological and cognitive claims require additional domain assumptions and may fail independently.

The registry has four purposes: (i) priority and timestamping, (ii) dependency and risk classification, (iii) preparation for future standalone simulation and experimental papers, and (iv) explicit separation between core finite-system claims and domain-specific biological or cognitive mappings.

A claim listed here is not thereby asserted as established biology, neuroscience, cognitive science, or medicine. Each claim is assigned a tier, dependencies, prior-art relation, novelty statement, falsification condition, empirical status, and required next work.

VERSIONING

This is Public Registry v1.0. Claims may be promoted, demoted, split, merged, or retired in later versions. A version change does not imply that the entire FDS programme has changed; it records the current status of life-science and cognitive-science bridge claims only.

INTRODUCTION: WHY A LIFE AND COGNITIVE REGISTRY IS NEEDED

From finite-system architecture to biological and cognitive bridge claims

The core architecture of Distinction Theory begins from the finite-system cost chain:

Distinction $\rightarrow$ boundary $\rightarrow$ capacity deficit $\rightarrow$
approximation
 $\rightarrow$ corrective complexity $\rightarrow$ dissipative pressure
 $\rightarrow$ pruning / externalization / collapse / invariant-supported
persistence.

The present paper does not re-derive that core chain. It asks a different question: if finite systems are biological, life-like, cognitive, or consciousness-supporting, what domain-level structures should one expect to appear? The answer is not a finished theory of biology or consciousness. It is a registry of bridge claims connecting the finite-system architecture to life-like persistence, death-like collapse, active residue control, neural/cognitive bottlenecks, and recovery ordering after loss of conscious reportability.

The registry format is deliberate. Some claims below are close to existing biological or cognitive traditions: homeostasis, autopoiesis, protocell theory, autophagy and cellular quality control, active inference, global workspace accounts, integrated information approaches, neural complexity measures, and anesthesia recovery studies. Other claims are more speculative and

should be treated as bridge hypotheses. The value of a registry is that it separates these cases. It prevents a domain bridge from being confused with a theorem, and it prevents the failure of one biological or cognitive module from being misread as the failure of the finite-system architecture as a whole.

Registry, not final theory

This paper is not a final theory of life, death, or consciousness. It does not claim that all living systems use the same molecular pruning mechanism, that all biological death is reducible to a single scalar threshold, that clinical death is solved by a finite-system definition, or that consciousness is identical to compression. Its goal is narrower and more auditable: to state biological and cognitive bridge claims, list their dependencies, identify prior-art neighbors, and specify what would falsify or demote each claim.

A mature scientific theory normally has equations, fit parameters, experiments, and community-vetted limits. A research programme begins earlier: it identifies candidate dependencies and where they can fail. The present registry is written in that earlier sense. It is intended to make the life/cognitive claim-space searchable, citable, and falsifiable.

Relation to the physical and artificial-agency papers

The FDS Core introduces the formal finite-system architecture. The physical registry organizes physical bridge modules by dependency, tier, and falsification condition. The artificial-agency paper applies the architecture to active boundary maintenance, causal loop closure, durable updates, resource-governed persistence, pruning, and externalization in AI systems. The present registry takes up life-science and cognitive-science modules excluded from those papers and organizes them by dependency, tier, and falsification condition.

What is not claimed

Tier labels do not measure importance. They measure epistemic status. A Tier C claim may be more exciting than a Tier B claim, but it is less secure. A Tier D claim may be useful for interpretation while remaining nonessential to the finite-system architecture.

Failure propagation rule

The registry follows a downstream failure rule:

If a life-science or cognitive-science bridge claim fails, only that claim and its downstream modules are revised, demoted, quarantined, or abandoned. Failure of a life/cognitive bridge does not by itself falsify the distinction primitive or the bridge-free finite-system architecture.

This rule is crucial. Without it, the registry would become a brittle grand theory in which any failed biological or cognitive speculation destroys the entire programme. The purpose of the registry is the opposite: to make each claim independently auditable.

Registry template

Each major claim is stated using the following template:

1. **Claim ID and name.**
2. **Tier.**
3. **Statement.**
4. **Dependencies.**
5. **Prior-art relation.**
6. **Novel DT contribution.**
7. **Falsification or demotion condition.**
8. **Empirical status.**
9. **Required next work.**

The template is intentionally repetitive. It makes visible the difference between a claim that is being asserted, a claim that is being interpreted, and a claim that is being proposed for future testing.

Claim-ID convention

The registry uses an explicit prefix system to avoid ambiguity. Each claim ID has the form XX-N, where XX is a two-letter prefix identifying the claim family.

Risk stratification at a glance

MINIMAL LIFE AND COGNITIVE BRIDGE ASSUMPTIONS

Finite boundary maintenance

Assumption (Finite boundary maintenance). A life-like or cognitive system that persists as a distinguishable system over time must maintain an operational boundary

Not claimed	Reason
FDS solves the origin of life	The registry only states finite-boundary maintenance claims and candidate experimental predictions. It does not reconstruct historical biogenesis.
FDS defines all biological life	The registry defines operational life-like persistence in finite boundary systems, not a universal taxonomic definition.
FDS defines clinical death	Death-like collapse here means loss of a maintenance attractor in a model system. It is not a medical, legal, or clinical criterion.
FDS treats cancer, Alzheimer’s disease, or immune disorders	Biomedical extensions are explicitly excluded from the current registered claim set except as future non-clinical bridge directions.
FDS proves consciousness	Cognitive claims concern reportability, capacity bottlenecks, recovery ordering, and self/agency coherence; phenomenology requires additional assumptions.
Compression alone is consciousness	Compression is treated as one capacity-management component, not a sufficient condition for consciousness.
FDS replaces neuroscience or synthetic biology	The registry is a bridge map and prediction ledger, not a replacement for domain-specific mechanisms.

TABLE I. Non-claims for the life and cognitive bridge registry.

Prefix	Claim family
LB	Life bridge assumptions and variables
AP	Active-pruning claims
RC	Residue-induced collapse claims
DL	Death-like maintenance-attractor loss claims
SV	Surface-to-volume scaling claims
EW	Early-warning and near-collapse signatures
CB	Cognitive bottleneck claims
AR	Anesthesia recovery ordering claims
SA	Self/agency coherence claims
CP	Cognitive pruning and forgetting claims
BR	Biomedical extension placeholders, not registered as active claims in v1.0

TABLE II. Claim-ID prefixes for FDS-LC0.

separating internal state, external state, and update interfaces. This boundary may be physical, chemical, cellular, organismic, neural, cognitive, or functional.

This assumption does not require that all boundaries be membranes. It requires only that a system whose identity persists must maintain some operational distinction between what counts as internal state, what counts as environmental state, and which exchanges are permitted.

Sustained flux and residual load

Assumption (Sustained flux and residual load). A finite boundary-maintaining system under sustained matter, energy, information, or control flux will generically produce residual load: inert products, misfolded structures, obsolete states, failed intermediates, error records, stale plans, or nonfunctional complexity that must be neutralized, exported, recycled, compressed, or tolerated.

Residual load is not restricted to chemical waste. In a protocell-like model, it may be inert molecular residue.

In a cell, it may include damaged proteins, misfolded aggregates, dysfunctional organelles, or unusable metabolites. In a cognitive system, it may include irrelevant memories, stale prediction errors, obsolete task fragments, and attention-consuming noise.

Finite passive dilution

Assumption (Finite passive dilution). Passive environmental dilution, leakage, spontaneous decay, and unstructured diffusion are finite. They may reduce residual load, but they are not assumed to scale automatically with task demand, boundary size, or residual production rate.

This assumption is what makes active pruning nontrivial. If passive removal were unbounded, active pruning would not be required.

Module	Strength	Reason
Active pruning threshold (AP-1)	Strong but untested	Direct finite-system argument; can be tested in simulations and protocell-like systems.
Residue-induced collapse (RC-1)	Strong	Direct consequence of sustained residual load under finite boundary, transport, and repair constraints.
Death-like attractor loss (DL-1)	Medium-strong	Operationally clean in models; not a clinical death claim.
Surface-to-volume scaling (SV-1)	Medium-strong	Experiment-facing scaling claim for vesicle/protocell-like geometries.
Near-death warning signals (EW-1)	Medium	Related to critical slowing and resilience loss; requires model-specific measurement.
Capacity-bottleneck cognition (CB-1)	Medium	Consistent with several cognitive traditions; FDS novelty is finite-boundary framing and failure conditions.
Anesthesia recovery ordering (AR-1)	High-value but high-risk	Specific pre-registered ordering; requires multimodal clinical/neuroscience data.
Self/agency coherence (SA-1)	Medium	Conceptually natural but needs precise behavioral and neural observables.
Biomedical extensions	Deferred	Immune, cancer, neurodegeneration, and swarm claims are not part of the current active claim set.

TABLE III. Risk stratification for the v1.0 life/cognitive claim registry.

Finite cognitive and neural capacity

Assumption (Finite cognitive capacity). A cognitive system has finite attention, finite working memory, finite update rate, finite metabolic budget, finite report channel, finite integration bandwidth, and finite tolerance for unresolved prediction error.

The cognitive claims below do not require that consciousness be identical to any one of these capacities. They require only that reportability and self/agency coherence cannot be maintained independently of finite-capacity integration and boundary modeling.

CORE LIFE VARIABLES AND OPERATIONAL DEFINITIONS

Let

$$S_{\text{act}}(t) = \text{active pruning rate}, \quad (1)$$

$$\rho_{\text{res}}(t) = \text{residue or complexity-debt load}, \quad (2)$$

$$B_{\text{int}}(t) = \text{boundary integrity}, \quad (3)$$

$$J(t) = \text{transport or metabolic throughput}, \quad (4)$$

$$F(t) = \text{available free-energy budget}, \quad (5)$$

$$V_{\text{maint}}(t) = \text{viability-like maintenance score}. \quad (6)$$

A minimal viability-like score may be written as

$$V_{\text{maint}}(t) = B_{\text{int}}(t) J(t) \frac{1}{1 + \rho_{\text{res}}(t)}. \quad (7)$$

This is not asserted as a universal biological viability formula. It is a simple registry-level operational metric for artificial-life and simulation protocols.

A minimal residual-load equation is

$$\dot{\rho}_{\text{res}} = p(J) - d_{\text{passive}}\rho_{\text{res}} - S_{\text{act}}, \quad (8)$$

where $p(J)$ is the rate of residual production, d_{passive} is passive decay or leakage, and S_{act} is active pruning, clearance, recycling, or selective neutralization.

A death-like model state is defined operationally by

$$V_{\text{maint}}(t) < V_{\text{min}} \quad \text{for duration } \Delta t_c, \quad (9)$$

or, more strongly, by loss of access to the maintenance attractor under restoration of ordinary resource input.

ACTIVE-PRUNING CLAIMS

Claim AP-1: Active pruning as a requirement for persistent finite boundary systems

Tier. B/B+ for finite artificial systems; C/B for broad biological generalization.

Statement. A finite boundary-maintaining system under sustained flux and nonzero residual-load production cannot maintain long-term life-like persistence unless it possesses some mechanism for active pruning, clearance, recycling, neutralization, or selective reorganization of residual load above a critical threshold.

In minimal form:

$$\dot{\rho}_{\text{res}} = p(J) - d_{\text{passive}}\rho_{\text{res}} - S_{\text{act}}. \quad (10)$$

If $p(J) > 0$, passive removal is bounded, and accumulated residual load damages transport or boundary integrity,

$$\frac{\partial J}{\partial \rho_{\text{res}}} < 0, \quad \frac{\partial B_{\text{int}}}{\partial \rho_{\text{res}}} < 0, \quad (11)$$

then long-term maintenance requires an active-pruning capacity above a threshold:

$$S_{\text{act}} > S_c. \quad (12)$$

Dependencies. Finite boundary maintenance; sustained flux; nonzero residual production; finite passive dilution; residual damage to transport, catalytic availability, or boundary integrity.

Prior-art relation. This claim is adjacent to homeostasis, autopoiesis, protocell research, cellular quality control, autophagy, proteostasis, waste clearance, and repair biology. It does not claim priority over the idea that organisms remove waste or repair damage.

Novel DT contribution. The novelty is not the claim that waste removal matters. It is the claim that active pruning can be treated as a finite-system control parameter separating transient residue-accumulating activity from sustained boundary maintenance.

Falsification condition. AP-1 is weakened or falsified in a specified domain if a finite boundary-maintaining system under sustained residual production, finite passive dilution, finite resources, and residual damage to transport or boundary integrity maintains stable B_{int} , J , and V_{maint} indefinitely with $S_{\text{act}} = 0$ and without hidden externalization, task relaxation, or unbounded dilution.

Empirical status. Pre-registered bridge claim. Simulation and wet-lab tests remain to be performed.

Required next work. Build 0D, 2D, and 3D protocell-like simulations with tunable S_{act} , ρ_{res} , B_{int} , J , and F . Then design wet-lab protocols in vesicle, droplet, or synthetic-cell systems with controllable residue production and active clearance.

Claim AP-2: Active pruning must be endogenous, selective, and resource-coupled

Tier. B for model systems; C for broad biological generalization.

Statement. A process counts as active pruning in the FDS sense only if it is at least partly endogenous to the system, selective with respect to residual or nonfunctional structure, and coupled to finite resource expenditure.

A minimal model is

$$S_{\text{act}}(t) = S_{\text{max}} \frac{F(t)}{F(t) + K_F} \frac{\rho_{\text{res}}(t)}{\rho_{\text{res}}(t) + K_\rho} \times \frac{X(t)}{X(t) + K_X} \chi_B(t), \quad (13)$$

where $X(t)$ denotes functional machinery and $\chi_B(t)$ denotes boundary-coupling or accessibility.

Dependencies. AP-1; finite free-energy budget; selective interaction between functional machinery and residual load.

Prior-art relation. Adjacent to enzymatic degradation, lysosomal/autophagic systems, chaperone-mediated quality control, and regulated recycling. In artificial systems, adjacent to controlled reaction networks and adaptive filtering.

Novel DT contribution. The claim distinguishes active pruning from passive decay. A universal sink or unstructured leakage channel does not by itself count as active pruning.

Falsification condition. If passive, nonselective decay alone robustly reproduces all predicted threshold, rescue-window, and attractor-loss behavior under the same finite boundary and resource constraints, AP-2 is demoted.

Empirical status. Pre-registered bridge claim.

Required next work. Compare three model classes: no pruning, passive degradation, and endogenous active pruning. Test whether only active pruning generates stable maintenance across perturbations.

RESIDUE-INDUCED COLLAPSE CLAIMS

Claim RC-1: Residue accumulation induces boundary and throughput collapse

Tier. B for artificial systems; C/B for biological generalization.

Statement. In finite boundary-maintaining systems under sustained flux, residual load that is not cleared, recycled, or neutralized will accumulate until it degrades transport throughput, catalytic availability, boundary integrity, or internal update capacity.

Prediction. In no-pruning systems,

$$\rho_{\text{res}}(t) \uparrow, \quad B_{\text{int}}(t) \downarrow, \quad J(t) \downarrow, \quad V_{\text{maint}}(t) \downarrow. \quad (14)$$

Dependencies. Sustained flux; nonzero residual production; finite passive dilution; residual interference with boundary or throughput.

Prior-art relation. Adjacent to toxic accumulation, metabolic waste, protein aggregation, reaction poisoning, clogging, loss of permeability, and systems-level homeostatic failure.

Novel DT contribution. Residue is treated as complexity debt: a finite-system load that consumes capacity and increases maintenance cost. The collapse is not merely chemical poisoning but a boundary-maintenance failure mode.

Falsification condition. RC-1 is weakened if residual load accumulates without affecting boundary integrity, throughput, catalytic availability, update capacity, or viability-like persistence under a pre-registered model or experiment.

Empirical status. Pre-registered bridge claim.

Required next work. Simulate residue accumulation in spatial reaction-diffusion and protocell-like models.

Measure residual concentration, boundary permeability, throughput, and time to collapse.

Claim RC-2: Capacity debt and residual load are dynamically coupled

Tier. C/B.

Statement. Residual load is not only a material quantity. It also increases the system's effective capacity deficit by consuming boundary bandwidth, catalytic availability, memory stability, or update resources.

Candidate relation.

$$\Delta_{\text{life}}(t) = R_{\text{min}}^{(\tau)}(\varepsilon; t) - C_{\text{acc}}(t) - \alpha_{\rho}\rho_{\text{res}}(t), \quad (15)$$

where α_{ρ} maps residual load into effective lost capacity.

Dependencies. Capacity-deficit formalism; measurable relation between residual load and accessible maintenance capacity.

Novel DT contribution. Residue is treated as a dual material-informational burden: it is both a physical load and a reduction in usable distinction capacity.

Falsification condition. If residual load affects neither physical throughput nor effective state-discrimination/update capacity in a controlled finite-boundary system, RC-2 is demoted.

Required next work. Estimate α_{ρ} in simulation and experimental systems by measuring how residue changes distinguishable internal states, transport channels, and recovery times.

DEATH-LIKE COLLAPSE CLAIMS

Claim DL-1: Death-like collapse as loss of the boundary-maintenance attractor

Tier. B for artificial life-like models; C for broad biological interpretation.

Statement. In finite artificial life-like systems, death-like collapse can be operationally defined as the loss of the boundary-maintenance attractor. It is not a clinical, legal, or moral definition of death.

Operational definition. A death-like state occurs when

$$V_{\text{maint}}(t) < V_{\text{min}} \quad \text{for duration} \quad \Delta t_c, \quad (16)$$

and normal restoration of pruning or resource input cannot return the system to the previous maintenance attractor.

Dependencies. AP-1; RC-1; existence of a maintenance attractor; finite rescue capacity.

Prior-art relation. Adjacent to dynamical-systems accounts of viability, critical transitions, attractor loss, cellular death pathways, and irreversibility in biological collapse. It is not a clinical death criterion.

Novel DT contribution. Death-like collapse is framed as a finite-system transition: the system loses access to the regime in which boundary, throughput, and residual load can be jointly maintained.

Falsification condition. If a model exhibits persistent low V_{maint} and severe boundary/throughput degradation but can always be restored by ordinary reactivation of S_{act} with no basin-loss boundary, then the attractor-loss version of DL-1 is weakened.

Empirical status. Pre-registered bridge claim.

Required next work. Demonstrate maintenance attractors and their loss in ODE, 2D, and 3D models. Identify bifurcation structure and rescue boundaries.

Claim DL-2: Rescue-window closure as the operational boundary of death-like collapse

Tier. B for model systems.

Statement. A finite boundary-maintaining system entering collapse should display a rescue window: restoring active pruning before a critical time or state returns the system to maintenance; restoring it after that boundary fails to recover the maintenance attractor.

Prediction. Under pruning shutoff,

$$S_{\text{act}} \rightarrow 0 \quad \Rightarrow \quad \rho_{\text{res}} \uparrow, B_{\text{int}} \downarrow, J \downarrow. \quad (17)$$

For some critical state x^* or time t^* ,

$$t < t^* : S_{\text{act}} \rightarrow S_{\text{max}} \Rightarrow \text{rescue}, \quad (18)$$

whereas

$$t > t^* : S_{\text{act}} \rightarrow S_{\text{max}} \not\Rightarrow \text{rescue}. \quad (19)$$

Dependencies. Existence of a maintenance basin; finite repair/pruning capacity; irreversible boundary damage or loss of functional machinery after threshold crossing.

Prior-art relation. Adjacent to critical slowing, tipping points, irreversible cellular injury, and attractor dynamics.

Novel DT contribution. The death-like boundary is not defined by a single chemical marker but by loss of recovery access to the maintenance attractor.

Falsification condition. If no rescue-window closure occurs across a wide range of models satisfying AP-1 and RC-1 assumptions, DL-2 is demoted.

Required next work. Perform shutoff-rescue simulations and protocell-like perturbation experiments.

SURFACE-TO-VOLUME SCALING CLAIMS

Claim SV-1: Three-dimensional boundary-maintenance threshold scales with surface-to-volume constraints

Tier. B/C.

Statement. In three-dimensional protocell-like systems, active-pruning threshold should depend on surface-to-volume ratio. If fuel influx, waste export, or boundary repair scale primarily with surface area while residual production scales with volume, larger systems require stronger active pruning or transport throughput to maintain viability.

For a spherical vesicle of radius R ,

$$\frac{A}{V} = \frac{3}{R}. \quad (20)$$

If residual production scales as V and boundary-mediated removal scales as A , then

$$R \uparrow \Rightarrow A/V \downarrow \Rightarrow S_c(R) \uparrow. \quad (21)$$

Dependencies. Boundary-mediated exchange; volume-mediated residual production; finite diffusion and finite internal clearance.

Prior-art relation. Adjacent to cell-size constraints, protocell and vesicle biophysics, diffusion-limited metabolism, and surface-area limitations in biology.

Novel DT contribution. The size constraint is interpreted as a boundary-maintenance threshold for residual complexity debt, not only as a metabolic scaling rule.

Falsification condition. If vesicle/protocell-like systems show no systematic dependence of S_c or collapse time on A/V after controlling for diffusion, channel density, and internal reaction rate, SV-1 is demoted.

Empirical status. Pre-registered simulation and experimental bridge claim.

Required next work. Run 3D spatial models and derive $S_c(R)$ scaling curves. Test in microfluidic or synthetic-cell platforms.

NEAR-COLLAPSE EARLY-WARNING CLAIMS

Claim EW-1: Near-death warning signatures under maintenance-attractor loss

Tier. C/B.

Statement. Before death-like maintenance-attractor loss, finite boundary systems should exhibit resilience-loss signatures analogous to critical slowing and capacity-overload behavior.

Predicted signatures.

$$\text{recovery time } \uparrow, \quad (22)$$

$$\text{variance of } V_{\text{maint}} \uparrow, \quad (23)$$

$$\text{autocorrelation of } V_{\text{maint}} \uparrow, \quad (24)$$

$$\text{pruning efficiency } \downarrow, \quad (25)$$

$$\rho_{\text{res}} \uparrow. \quad (26)$$

Dependencies. Maintenance attractor; perturbable dynamics; measurable recovery after small disturbances.

Prior-art relation. Adjacent to early-warning signals for critical transitions, resilience theory, homeostatic failure, and dynamical-systems biology.

Novel DT contribution. The warning signals are tied specifically to finite boundary-maintenance failure under residual-load pressure and active-pruning insufficiency.

Falsification condition. If collapse occurs with no measurable degradation in recovery time, variance, autocorrelation, pruning efficiency, or residual accumulation across controlled systems where these variables are observable, EW-1 is weakened.

Required next work. Define perturbation protocols for simulated and experimental protocell-like systems.

CORE COGNITIVE VARIABLES

Let

$$C_{\text{cog}}(t) = \text{available cognitive capacity}, \quad (27)$$

$$C_{\text{attn}}(t) = \text{attention bandwidth}, \quad (28)$$

$$C_{\text{wm}}(t) = \text{working-memory capacity}, \quad (29)$$

$$I_{\text{int}}(t) = \text{integration level}, \quad (30)$$

$$K_{\text{self}}(t) = \text{self/agency coherence}, \quad (31)$$

$$R_{\text{rep}}(t) = \text{reportability}, \quad (32)$$

$$E_{\text{pred}}(t) = \text{prediction-error load}. \quad (33)$$

A minimal cognitive-access bottleneck may be represented as

$$C_{\text{cog}}(t) = \min\{C_{\text{attn}}(t), C_{\text{wm}}(t), C_{\text{integration}}(t), C_{\text{report}}(t), C_{\text{metabolic}}(t)\}. \quad (34)$$

This is a schematic registry-level expression. It is not a neural measurement formula.

CAPACITY-BOTTLENECK CONSCIOUSNESS CLAIMS

Claim CB-1: Conscious reportability requires finite-capacity integration

Tier. C/B.

Statement. Conscious reportability requires that task-relevant distinctions be integrated, compressed, stabilized, and routed through finite-capacity bottlenecks. FDS does not claim that compression alone is consciousness. It claims that reportability and stable self/world access cannot be independent of finite-capacity integration.

A minimal condition is

$$R_{\text{rep}}(t) > 0 \Rightarrow C_{\text{cog}}(t) \geq R_{\text{min}}^{(\tau)}(\varepsilon; \Psi_{\text{report}}), \quad (35)$$

where Ψ_{report} is the relevant report task family.

Dependencies. Finite cognitive capacity; distinction registration; memory stabilization; report channel; self/world boundary model for self-referential reports.

Prior-art relation. Adjacent to global workspace theory, higher-order and reportability accounts, integrated information approaches, predictive processing, active inference, and capacity-limited cognition.

Novel DT contribution. The novelty is the finite-system framing: conscious reportability is treated as a boundary-maintenance and capacity-bottleneck problem rather than as a disembodied information state.

Falsification condition. CB-1 is weakened if robust conscious reportability is demonstrated without any finite-capacity bottleneck, integration, memory stabilization, report routing, or distinction-maintenance substrate.

Empirical status. Conceptual bridge claim; requires operationalization.

Required next work. Link C_{cog} , integration metrics, reportability, and self/agency coherence to measurable EEG, fMRI, behavioral, perturbational, or computational variables.

Claim CB-2: Local processing can recover without conscious reportability

Tier. B/C.

Statement. Local sensory registration, reflexive processing, or feature-specific responses can occur without full reportability when global capacity, integration, or self/agency coherence remains unavailable.

Dependencies. Distinction between local registration and global reportability; finite bottleneck; partial recovery or partial disruption.

Prior-art relation. Adjacent to blindsight, masked priming, subliminal processing, local evoked responses under anesthesia, and dissociations between sensory processing and report.

Novel DT contribution. The dissociation is interpreted as a capacity-budget separation: local distinctions may be registered while the system lacks sufficient integrated budget to stabilize them as reportable distinctions.

Falsification condition. If local sensory processing and reportability are inseparable across perturbations where measurement is adequate, CB-2 is weakened.

Required next work. Use anesthesia, masking, sleep, or perturbation protocols to separate local registration from reportability and self/agency coherence.

ANESTHESIA RECOVERY ORDERING CLAIMS

Claim AR-1: Anesthesia recovery follows a finite-capacity reconstruction order

Tier. B/C, high-value and high-risk.

Statement. Recovery from anesthesia should not be random. If conscious reportability requires finite-capacity integration and self/world boundary reconstruction, then recovery should tend to follow the ordering

$$\begin{aligned} & \text{local sensory registration} \\ & < \text{attention/integration bottleneck} \\ & < \text{self/agency coherence} \\ & < \text{executive cognition.} \end{aligned} \tag{36}$$

Dependencies. CB-1; measurable distinction between local sensory registration, attention/integration, self/agency coherence, and executive cognition.

Prior-art relation. Adjacent to anesthesia neuroscience, perturbational complexity, global neuronal workspace, frontoparietal connectivity, consciousness recovery studies, and behavioral command-following protocols.

Novel DT contribution. The ordering is predicted from finite-system reconstruction: local distinction registration should be cheaper than integrated reportability; self/agency coherence requires a more stable boundary model; executive cognition requires the most structured, temporally extended update loop.

Falsification condition. AR-1 is weakened if, across drug classes, subjects, and measurement modalities, executive cognition robustly recovers before local sensory registration, before attention/integration markers, and before self/agency coherence, with no methodological explanation.

Empirical status. Pre-registered cognitive bridge claim.

Required next work. Design an anesthesia induction/recovery protocol measuring evoked sensory responses, attention/global integration, self-location or agency report, command-following, working memory, and executive task performance.

Claim AR-2: Recovery ordering should vary with drug mechanism but preserve capacity hierarchy

Tier. C.

Statement. Different anesthetic mechanisms may alter timings, biomarkers, or regional recovery patterns, but the broad hierarchy from local registration to integrated reportability and then to executive cognition should remain more stable than individual markers.

Dependencies. AR-1; drug-dependent perturbation of neural systems; cross-modality measurement.

Prior-art relation. Adjacent to known diversity among anesthetic agents and variable recovery profiles.

Novel DT contribution. FDS predicts preservation of capacity hierarchy even when biomarker details vary.

Falsification condition. A stable reverse hierarchy across multiple agents and cohorts would demote AR-2.

Required next work. Compare propofol, sevoflurane, ketamine-like dissociative states, and sleep-like states with matched recovery tasks.

SELF/AGENCY COHERENCE CLAIMS

Claim SA-1: Self/agency coherence as a compressed boundary model

Tier. C/B.

Statement. Self/agency coherence is a compressed boundary model linking body state, action capacity, autobiographical memory, prediction error, and control access. It is not a single isolated module.

A schematic relation is

$$K_{\text{self}} = f(B_{\text{body}}, M_{\text{auto}}, A_{\text{control}}, E_{\text{pred}}, I_{\text{int}}). \quad (37)$$

Dependencies. Finite self/environment boundary; memory continuity; action loop; prediction-error regulation.

Prior-art relation. Adjacent to body schema, self-model theory, active inference, agency studies, interoception, and predictive processing.

Novel DT contribution. Self/agency is treated as an operational boundary-maintenance model needed for stable action and report, not as an isolated representational object.

Falsification condition. SA-1 is weakened if stable self/agency reports are shown to require no body-boundary model, no action capacity model, no memory continuity, and no prediction-error regulation.

Required next work. Operationalize self/agency coherence in anesthesia, sleep, dissociation, virtual reality, and sensorimotor perturbation protocols.

COGNITIVE PRUNING AND FORGETTING CLAIMS

Claim CP-1: Cognitive persistence requires active forgetting or pruning

Tier. B/C.

Statement. A finite cognitive system cannot preserve all encountered distinctions indefinitely without memory pollution, attention overload, task drift, or resource collapse. Cognitive persistence requires active forgetting, consolidation, abstraction, or pruning.

Dependencies. Finite memory; finite attention; ongoing novelty; task relevance; update cost.

Prior-art relation. Adjacent to forgetting research, synaptic pruning, sleep-dependent consolidation, memory reconsolidation, predictive coding, and machine-learning regularization.

Novel DT contribution. Forgetting is framed not as failure alone but as active boundary maintenance: removing or compressing distinctions can preserve cognitive viability.

Falsification condition. A finite cognitive system that stores and uses unbounded experiential distinctions indefinitely without degradation, consolidation, abstraction, forgetting, externalization, or resource growth would falsify the strong form.

Required next work. Test how pruning, consolidation, and externalization affect long-horizon task coherence in biological, cognitive, and artificial agents.

BIOMEDICAL AND COLLECTIVE-SYSTEM EXTENSIONS DEFERRED TO FUTURE VERSIONS

The following domains are not part of the active v1.0 registered claim set: immune boundary verification, cancer as rogue sub-boundary decoupling, neurodegeneration as residue-clearance or mis-pruning failure, swarms as distributed FDS systems, and organizational/civilizational complexity debt. Future versions may register bridge hypotheses in these directions. Their absence from the present version is deliberate. This registry focuses on active pruning, artificial life-like collapse, death-like attractor loss, and anesthesia recovery ordering.

No statement in this section is a diagnostic, therapeutic, or clinical claim.

SIMULATION AND EXPERIMENTAL ROADMAP

Life simulation roadmap

A minimal simulation programme should proceed in four levels.

Level 0: ODE model. Use variables ρ_{res} , B_{int} , J , F , and S_{act} to test whether S_c , collapse, attractor loss, and rescue windows occur.

Level 1: 2D spatial model. Use reaction-diffusion or cellular-automaton dynamics to visualize residue accumulation, local clogging, boundary degradation, and pruning-mediated rescue.

Level 2: 3D protocell-like model. Use spherical or vesicle-like geometry to derive $S_c(R)$ and surface-to-volume scaling.

Level 3: wet-lab protocol. Use vesicle, droplet, or protocell-like platforms with tunable residue production, tunable clearance, measurable boundary integrity, and throughput readout.

Cognitive experimental roadmap

A minimal cognitive programme should proceed in three levels.

Level 0: computational agent simulations. Model finite attention, working memory, prediction-error load, self-model stability, and report capacity.

Level 1: non-clinical behavioral and EEG studies. Test reportability and self/agency coherence under masking, divided attention, sleep inertia, virtual reality perturbations, or working-memory overload.

Level 2: anesthesia recovery protocol. Measure recovery ordering across sensory registration, attention/integration, self/agency coherence, command-following, and executive cognition.

MASTER CLAIM TABLE

LAYERED FAILURE CONTRACT

REPORTING CHECKLIST FOR FUTURE LC PAPERS

A paper applying FDS-LC0 should report:

1. the finite system boundary being modeled;
2. the residual-load variable and how it is measured;
3. the active-pruning variable and how it is distinguished from passive decay;
4. the throughput or transport variable;
5. the boundary-integrity variable;
6. the viability-like maintenance score;
7. the perturbation, shutdown, or overload protocol;
8. whether an S_c threshold is observed;
9. whether a rescue window exists;
10. whether attractor loss is reversible or irreversible under ordinary restoration;
11. explicit falsification, demotion, and non-claim statements.

A cognitive paper applying FDS-LC0 should additionally report:

1. how local sensory registration is measured;

2. how attention or integration bottleneck is measured;
3. how reportability is operationalized;
4. how self/agency coherence is measured;
5. how executive cognition is measured;
6. the predicted recovery ordering;
7. the criteria for support, demotion, and falsification.

LIMITATIONS

First, this registry does not prove that active pruning is the historical origin of life. It states a finite-system bridge claim about persistent boundary maintenance under sustained flux and residual-load production.

Second, life-like persistence in a simulation is not identical to biological life. The proposed models are mechanistic probes, not replacement definitions for biology.

Third, death-like collapse is not clinical death. It is an operational model concept: loss of access to a boundary-maintenance attractor.

Fourth, active pruning may be implemented in many physical, chemical, biological, cognitive, or artificial ways. The registry does not require a single molecular mechanism.

Fifth, cognitive capacity, reportability, self/agency coherence, and consciousness are not identical. The cognitive claims are bridge claims about measurable recovery and bottleneck constraints, not final phenomenological claims.

Sixth, anesthesia recovery is drug-, dose-, subject-, and measurement-dependent. The registry predicts a hierarchy of finite-capacity reconstruction, not identical timing across all protocols. Seventh, biomedical claims involving cancer, Alzheimer’s disease, immune systems, and clinical medicine are explicitly excluded from this v1.0 release and may appear in a future extension. Registry breadth is prioritized over depth at this stage.

CONCLUSION

This paper has registered the life-science and cognitive-science bridge claims of FDS at the level of v1.0. The central biological claim is that active pruning is not an optional biological refinement but a candidate first-principles requirement for finite boundary-maintaining systems under sustained flux and residual-load accumulation. The central death-like claim is that collapse can be operationally defined as loss of access to the boundary-maintenance attractor. The central cognitive claim is that conscious reportability and self/agency coherence should be constrained by finite-capacity integration, and

ID	Claim	Tier	Support condition	Falsification / demotion condition
AP-1	Active-pruning threshold	B/B+	$S > S_c$ separates persistence from collapse	Persistence with $S = 0$ under stated finite-residue conditions
AP-2	Endogenous selective pruning	B/C	Active pruning outperforms passive decay under resource constraints	Passive decay reproduces all threshold/rescue behavior
RC-1	Residue-induced collapse	B/C	$\rho \uparrow \rightarrow B, J, V \downarrow$	Residue accumulation has no boundary or throughput effect
RC-2	Residue-capacity coupling	C/B	Residue reduces effective accessible capacity	Residue affects neither physical nor information capacity
DL-1	Death-like attractor loss	B/C	Maintenance attractor disappears after threshold	Low-viability states always recover under ordinary restoration
DL-2	Rescue-window closure	B	Recovery before t^* , failure after t^*	No rescue boundary across appropriate models
SV-1	Surface-to-volume scaling	B/C	$R \uparrow \Rightarrow S_c(R) \uparrow$	No size dependence after controls
EW-1	Near-collapse warnings	C/B	Recovery time, variance, autocorrelation rise	Collapse without measurable warning in observable variables
CB-1	Capacity-bottleneck reportability	C/B	Reportability depends on finite integration and routing	Reportability without any finite-capacity substrate
CB-2	Local processing without report	B/C	Local registration dissociates from reportability	Local processing and reportability always inseparable
AR-1	Anesthesia recovery ordering	B/C	Sensory < integration < agency < executive recovery	Stable reverse ordering across agents/cohorts/modalities
AR-2	Drug-varying but hierarchy-preserving recovery	C	Timings vary but hierarchy persists	Multi-agent robust hierarchy inversion
SA-1	Self/agency as boundary model	C/B	Self/agency depends on body/action/memory/prediction model	Self/agency requires none of these components
CP-1	Cognitive pruning necessity	B/C	Unpruned finite memory degrades long-horizon cognition	Unlimited usable memory without pruning or resource growth

TABLE IV. Master claim table for FDS-LC0.

Layer	Claim type	Failure consequence
FDS Core	Formal finite-system definitions and theorems	Fails only by internal incoherence or formal counterexample.
Life bridge	Active pruning, residue, boundary persistence, death-like collapse	Failure revises or retires the relevant life bridge only.
Cognitive bridge	Capacity bottleneck, reportability, recovery ordering, self/agency coherence	Failure revises or retires the relevant cognitive bridge only.
Simulation layer	ODE, 2D, 3D, agent models	Model failure requires model revision; it does not by itself kill the bridge claim unless the model exhausts the stated domain.
Biomedical extensions	Immune, cancer, neurodegeneration, clinical links	Deferred; future failure is non-propagating and does not affect AP/RC/DL core claims.

TABLE V. Layered failure contract for FDS-LC0.

that anesthesia recovery should follow a structured re-construction order rather than a random return.

The registry deliberately does not claim to solve life, death, consciousness, medicine, or neuroscience. It

records a claim-space: variables, dependencies, risk levels, predictions, and failure conditions. Future work should turn the registered claims into standalone papers: active-pruning ODE and 2D/3D simulations; protocell-

like wet-lab protocols; anesthesia recovery protocols; and computational models of finite-capacity cognitive pruning.

Glossary

Active pruning: Selective, resource-coupled removal, recycling, compression, neutralization, or reorganization of residual or nonfunctional structure in order to preserve boundary viability.

Residual load: Material, informational, structural, or cognitive complexity debt produced by sustained flux and imperfect update.

Boundary maintenance: The ongoing process by which a finite system preserves the operational distinction between itself and its environment.

Life-like persistence: Persistent maintenance of boundary integrity, throughput, residual-load control, and update capacity in an artificial or biological finite system.

Death-like collapse: In this registry, the irreversible loss of access to the boundary-maintenance attractor in a model system. It is not a clinical definition.

Maintenance attractor: A dynamical regime in which boundary integrity, throughput, residual load, and resource input remain jointly viable.

Rescue window: The interval during collapse in which restoration of active pruning or resources can still return the system to the maintenance attractor.

Capacity bottleneck: A limiting representational, attentional, memory, integration, report, or metabolic constraint that restricts which distinctions can be stabilized and used.

Reportability: The operational ability to stabilize and route distinctions into action, language, or other report channels.

Self/agency coherence: A compressed boundary model linking body state, memory continuity, prediction error, and action capacity.

Future Biomedical and Collective-System Extension Placeholders

Future versions may register the following claims, but they are not active claims in v1.0:

1. Immune systems as biological distinction-verification and pruning layers operating under bounded capacity, residual load accumulation, and self/non-self discrimination constraints across cellular, molecular, and network organizational scales.
2. Cancer as rogue sub-boundary decoupling from organism-level maintenance.

3. Neurodegeneration as residue-clearance failure or mis-pruning failure.
4. Swarms as distributed FDS systems with externalized memory and collective pruning.
5. Organizations and civilizations as externalized boundary-maintenance systems with collective complexity debt and distributed pruning.

These placeholders do not constitute clinical, diagnostic, therapeutic, or policy claims.

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Active Finite Distinction Systems as a Criterion for Artificial Agency

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Large predictive models can transform inputs into high-quality outputs without necessarily becoming agents. This paper proposes an operational criterion for artificial agency based on *active finite distinction systems* (FDS): finite-capacity systems that maintain a boundary through state-dependent updates under resource constraints. The framework distinguishes passive mapping from artificial agency by requiring active boundary maintenance, durable update participation, causal loop closure, and capacity-deficit management through pruning, externalization, task relaxation, or compression. The central technical object is an FDS tuple

$$S = (X, E, B, M, Y, A, U, \pi, \ell, \Phi, \mathcal{P}, \tau),$$

where B is an operational boundary, M is internal memory or model state, U is an update operator, ℓ is boundary-maintenance loss, Φ is a resource budget, $\mathcal{P}$ is a perturbation/pruning family, and τ is an update timescale. Capacity deficit is formulated as a task-relevant rate-distortion gap $\Delta_\varepsilon(\tau) = R_{\min}^{(\tau)}(\varepsilon) - C_S$, rather than as an informal mismatch between a model and the world. Physical dissipation is treated only through an explicit Landauer bridge: logically irreversible updates incur an informational heat floor proportional to $H(M_t | M_{t+1}, Y_t)$ under standard thermodynamic conditions. We derive an agency taxonomy separating passive mappers, scaffolded tool systems, adaptive controllers, and strong FDS-agents. We also give an operational test protocol using update ablation, action-to-future-state influence, capacity-deficit estimation, pruning/externalization efficiency, and resource-governed persistence. A minimal gridworld MDP illustrates how a one-bit capacity deficit induces an irreducible action-error lower bound, and a benchmark protocol is proposed for testing FDS-agency under bounded resources. The result is not a claim that scaling is irrelevant or that any named architecture cannot become agentic. It is a formal criterion for when a system containing predictive components qualifies as an artificial agent rather than merely a passive mapper.

Keywords: artificial agency; active finite systems; boundary maintenance; rate-distortion; transfer entropy; active pruning; external memory; resource-bounded agents; LLM agents; active inference.

INTRODUCTION

The passive-mapper problem

A central question in artificial intelligence is no longer whether a machine can produce intelligent-looking outputs. Large language models, world models, simulators, planners, and tool-using systems already perform tasks that previously appeared to require flexible reasoning. The harder question is structural: *when does a predictive or generative system become an agent rather than a passive mapper?*

A passive mapper transforms inputs into outputs. It may be highly competent, statistically calibrated, and useful. Yet input-output competence alone does not imply that the system maintains a boundary, updates durable self-relevant state, regulates its resources, prunes accumulated complexity, or causally participates in the future conditions it must survive. A model that can answer questions about a robot’s battery is not thereby a system that monitors, protects, allocates, or recharges

that battery. The difference is not prediction accuracy; it is the presence or absence of a maintenance loop connecting internal update, action, resource constraint, and future viability.

This paper develops an AI-facing formalization of that distinction. The proposal is that artificial agency requires more than prediction, generation, or optimization over an externally supplied objective. It requires that a finite system maintain an operational boundary under limited representational capacity and limited resources. This leads to a class of systems called *active finite distinction systems* (FDS). An FDS is not defined by being biological, conscious, embodied, or made from any particular substrate. It is defined by the structure of the maintenance problem it faces.

The FDS entry point

The formal core used here is the active finite distinction system

$$S = (X, E, B, M, Y, A, U, \pi, \ell, \Phi, \mathcal{P}, \tau), \quad (1)$$

where X is internal state, E is environmental state, B is a boundary or interface variable, M is memory or model state, Y is the observation channel, A is the action or

boundary-update space, U is the internal update operator, π is a finite distinction projection or coarse-graining, ℓ is boundary-maintenance loss, Φ is a resource budget, $\mathcal{P}$ is an admissible family of pruning, perturbation, or degradation operators, and τ is the update timescale.

The tuple is deliberately operational. In AI systems, B may be a robot body, a software service boundary, a memory and tool boundary, an API-level identity, a security perimeter, or a task-level viability envelope. M may include parameters, context state, working memory, persistent memory, maps, caches, learned skills, policies, or externalized records. A may include physical actions, API calls, tool invocations, memory writes, model-update operations, self-repair operations, or resource-allocation decisions. The central question is whether these components form an active maintenance loop rather than a passive input-output channel.

Contributions

This paper makes seven contributions.

1. It reformulates artificial agency in terms of active finite distinction systems, making boundary maintenance and resource-bounded persistence central rather than optional.
2. It defines an active-boundary criterion using both a formal relevance screen and an empirical intervention/ablation strengthening.
3. It replaces informal claims about environmental complexity with a task-relevant rate-distortion capacity deficit.
4. It distinguishes passive mappers, scaffolded systems, adaptive controllers, and strong FDS-agents.
5. It derives a passive-mapper non-qualification proposition and a prune-externalize-collapse trichotomy for AI architectures under persistent capacity deficit.
6. It gives an operational test protocol for evaluating artificial agency using update ablation, action-to-future-state influence, capacity-deficit estimation, pruning/externalization efficiency, and resource-governed persistence.
7. It proposes benchmark desiderata and metrics for deep causal tracking tasks under bounded resources, including viability, resource efficiency, pruning gain, externalization gain, and ablation gap.

What is not claimed

The paper does not claim that current large language models have no intelligence. It does not claim that scal-

ing is useless. It does not claim that artificial general intelligence is impossible. It does not claim that a system containing a fixed predictive model is necessarily passive; a fixed model embedded inside a tool-using, memory-writing, self-monitoring, resource-governed architecture may be part of an active system. It does not claim to solve consciousness. It does not derive physical law from the distinction primitive. The claims below are conditional, operational, and targeted at the artificial-agency question.

RELATED WORK

Agent theory and embodied AI

Classical accounts of agency distinguish agents from passive computational processes by emphasizing autonomy, situatedness, reactivity, proactivity, and social ability [7]. Dynamical systems approaches treat agency as a closed agent-environment loop rather than as isolated computation [8]. Embodied AI emphasizes that intelligence is shaped by sensorimotor coupling and environmental feedback rather than only by internal representation [9]. More recent formal accounts emphasize individuality, normativity, asymmetry, and spatio-temporal organization in defining agency [10].

The FDS approach is compatible with these traditions but isolates a different axis: the cost of maintaining a boundary under finite capacity. A system may be reactive or embedded without being strongly self-maintaining; conversely, a software architecture may lack a biological body but still maintain an operational boundary through memory, tools, monitoring, repair, and resource governance.

Active inference, Markov blankets, and autonomy

The free-energy principle and active inference model self-organizing systems as minimizing variational free energy through perception and action [11, 12]. Markov blankets formalize statistical boundaries separating internal and external states [13]. These frameworks are central to contemporary discussions of autonomy and agent-environment coupling.

The FDS framework differs in emphasis. It does not begin with belief updating or variational inference. It begins with active boundary maintenance under finite representational capacity. The FDS chain asks what follows when the task-relevant rate-distortion demand exceeds internal capacity: approximation, residual error, complexity pressure, irreversible update costs under physical implementation, and eventual pruning, externalization, task relaxation, or collapse. Active inference can supply one implementation of FDS-like dynamics, but FDS

is not restricted to Bayesian agents or variational objectives.

Control, empowerment, and transfer entropy

Control-theoretic and information-theoretic measures of agency often ask whether an action channel has causal influence over future states. Empowerment measures the channel capacity from actions to future sensor states [14]. Transfer entropy measures directed information transfer from one process to another [15]. In this paper, action-to-future-state influence is used as a marker of causal loop closure, but not as a sufficient condition for full agency. A thermostat may causally affect future temperature, yet it lacks the richer capacity-deficit management, active pruning, and resource-governed persistence required for strong FDS-agency.

LLM agents and tool-using systems

Recent LLM-based agents combine language models with planning, external tools, memory, environments, and feedback loops [16–19, 46]. Benchmarks show that current models perform well on single-turn tool calls but degrade under memory demands and long-horizon tasks [46]. Recent work on dynamic code generation as action further demonstrates externalization: agents generate Python functions at runtime rather than selecting from a fixed action set [47]. These systems illustrate why the base-model question and the coupled-system question must be separated. A feed-forward model evaluated at inference time may be passive; the larger architecture containing it may not be. The FDS criterion is explicitly system-level: the unit of agency is the coupled architecture that maintains boundaries, updates durable state, acts causally, manages resources, and handles accumulated complexity.

Information theory, rate-distortion, and thermodynamics

Rate-distortion theory studies the minimal coding rate needed to represent a source under a distortion constraint [22–24]. Landauer’s principle links logically irreversible erasure to a thermodynamic heat cost under specified physical conditions [25, 26]. Stochastic thermodynamics generalizes entropy production and information thermodynamics in nonequilibrium systems [27–29]. The FDS formalism uses rate-distortion to define task-relevant capacity deficit and uses Landauer only as a physical bridge for irreversible updates, not as a claim that every act of inference directly dissipates $k_B T \ln 2$ per represented bit.

Autonomous agents and rational agency

Classical AI and agent theory define autonomous agents as systems that perceive their environment, make decisions, and act to achieve goals [30, 31]. Reinforcement-learning formalisms treat agents as policy-optimizing systems that maximize expected cumulative reward [32]. These traditions focus on decision-making under uncertainty and goal-directed behavior. The FDS criterion shifts emphasis from goal optimization to boundary maintenance: the central question is not whether a system maximizes a reward function, but whether it maintains the conditions of its own persistence under finite capacity. A reward-maximizing system that does not update or repair its own decision boundary is a passive mapper with respect to its own viability.

Memory-augmented and continual-learning systems

Neural network architectures augmented with external memory, attention, and retrieval mechanisms extend the effective capacity of static models [33–35]. Continual-learning and lifelong-learning systems address the problem of accumulating knowledge without catastrophic forgetting [36–38]. These lines of work are directly relevant to the FDS framework: durable memory and active consolidation are components of capacity-deficit management, and catastrophic forgetting is a special case of maintenance failure under representational pressure. The FDS criterion adds the requirement that memory and consolidation decisions be causally connected to boundary-maintenance loss, not merely to task accuracy.

Resource-rationality and bounded rationality

Bounded rationality recognizes that agents operate under cognitive, computational, and informational constraints [39, 40]. Resource-rational analysis models agents as approximately optimal under resource limitations [41, 42]. These perspectives are close to the FDS approach but focus on decision optimality under constraints rather than on boundary maintenance. The FDS framework asks what structures a system must have to persist while managing capacity deficit, rather than how close its decisions come to a resource-rational optimum.

AI safety, autonomy, and self-maintenance

Discussions of AI safety increasingly distinguish between capability, alignment, learned optimization, and systems that model their own role in deployment contexts [43–45]. These concerns overlap with the FDS criterion: a system that does not monitor or maintain its

own operational boundary cannot be expected to recognize changes in its own function, resource state, or goal integrity. The FDS framework does not solve the alignment problem, but it provides structural conditions for the kind of self-maintenance that safety properties may presuppose.

ACTIVE FINITE DISTINCTION SYSTEMS

The system tuple

Definition 1 (Active finite distinction system). *An active finite distinction system is a tuple*

$$S = (X, E, B, M, Y, A, U, \pi, \ell, \Phi, \mathcal{P}, \tau), \quad (2)$$

where:

- X is the internal state space of the system;
- E is the environmental state space;
- B is the boundary or interface variable separating system from environment;
- M is the internal memory, model, or representational state space;
- Y is the observation channel available at the boundary;
- A is the action or boundary-update space;
- U is the update map, typically $U : M \times Y \rightarrow M$ or a stochastic kernel $P(M_{t+1} \mid M_t, Y_t)$;
- π is the finite distinction projection, coarse-graining, or partition map;
- ℓ is the boundary-maintenance loss function;
- Φ is the finite resource or free-energy budget functional;
- $\mathcal{P}$ is the admissible family of pruning, coarse-graining, perturbation, or degradation operators;
- τ is the update timescale converting per-step information loss into rates or powers.

Remark 1 (Why τ matters). *Without a timescale, information-theoretic quantities remain dimensionless counts. With τ , per-update erasure and capacity demand can be expressed as rates and connected to resource or power budgets. This is essential for distinguishing a static representation from a maintained active system.*

AI interpretation of the tuple

For artificial systems, the tuple can be read as follows. X includes all internal operational state relevant to future behavior. M includes learned parameters, context, scratchpads, persistent memory, maps, policies, cached tool outputs, and externalized records treated as part of the agent architecture. Y includes sensors, prompts, API responses, environment observations, telemetry, or monitoring signals. A includes actions, tool calls, memory writes, environment modifications, compute-allocation decisions, and self-maintenance operations. ℓ measures boundary-maintenance or viability loss: task failure, resource exhaustion, context corruption, actuator failure, memory inconsistency, policy degradation, security breach, or loss of operational continuity. Φ may be energy, money, compute budget, time, memory, latency, user trust, or any resource without which the system cannot maintain the relevant boundary.

The tuple does not require that all parts live inside one neural network. A model, memory store, retriever, planner, robot, monitoring service, database, and tool environment can jointly form a single FDS if the coupled architecture maintains an operational boundary through durable state-dependent updates.

ACTIVE BOUNDARIES AND CAUSAL QUALIFICATION

Formal relevance condition

Definition 2 (Active boundary). *An FDS has an active boundary if its update rule is nontrivial and is relevant to future boundary-maintenance loss. Formally,*

$$\mathbb{P}(U(M_t, Y_t) \neq M_t) > 0, \quad (3)$$

and there exists $k > 0$ such that

$$I(M_{t+1}; \ell_{t+k}) > 0. \quad (4)$$

The first condition excludes systems with no update channel. The second condition excludes irrelevant internal noise. A random bit flip inside a machine is not active boundary maintenance unless it is statistically related to future boundary loss.

Intervention strengthening

Mutual information is a relevance screen, not a causal proof. A passive thermometer may encode future loss without participating in loss reduction. Therefore empirical AI applications require an intervention or ablation strengthening.

Criterion 1 (Causal active-boundary qualification). *For empirical artificial systems, an update channel U participates in active boundary maintenance only if there exists an admissible null update channel U_0 , such as freezing, randomizing, masking, or identity-updating the relevant state, and some $k > 0$ such that*

$$\mathbb{E}[\ell_{t+k} \mid \text{do}(U)] \neq \mathbb{E}[\ell_{t+k} \mid \text{do}(U_0)]. \quad (5)$$

Remark 2 (Update ablation in AI). *In an AI system, U_0 may disable memory writes, freeze tool feedback, randomize planner state, remove self-monitoring, mask resource telemetry, or prevent policy updates. If future boundary-maintenance loss is unchanged under such ablation, the removed update channel was not part of active agency for the tested task class.*

Passive boundaries

Definition 3 (Passive boundary). *A system has a passive boundary when its persistence is supported by static constraints or stable configurations without task-directed internal updating. Passive persistence may be real and stable, but it does not enter the deficit-dissipation-maintenance cascade unless an active update channel is introduced.*

A read-only database, an inert file, a static model checkpoint, or a frozen feed-forward mapping has a boundary in some descriptive sense, but it does not thereby qualify as an active FDS-agent. It may become part of an active system when coupled to update, monitoring, action, and resource-governance loops.

CAPACITY DEFICIT AS A RATE-DISTORTION GAP

Representational capacity

Definition 4 (Internal representational capacity). *For a finite memory or model state space M , define the internal representational capacity*

$$C_S = \log_2 |M|. \quad (6)$$

When M is continuous or effectively continuous, C_S is replaced by the appropriate operational capacity under finite resolution, noise, coding, quantization, compression, or resource constraints.

Proposition 1 (Finite distinction representation). *For any FDS with finite internal capacity C_S ,*

$$I(E_t; M_t) \leq C_S. \quad (7)$$

Proof. By the data-processing inequality and the entropy bound on a finite memory state,

$$I(E_t; M_t) \leq H(M_t) \leq \log_2 |M| = C_S.$$

□

This proposition does not say that the environment is finite. It says that the maintained internal distinction structure is finite.

Admissible task statistics

Definition 5 (Admissible task statistics). *Let Ψ be a pre-registered family of admissible statistics $\psi : (E, B) \rightarrow Z$ causally accessible through the observation channel Y . A statistic $Z_t = \psi(E_t, B_t)$ is task-relevant if it captures information needed to maintain the boundary or task within a stated loss tolerance. Trivial constant maps are excluded unless the task itself is trivial. The family Ψ is fixed by the task specification and sensor-action architecture, not chosen after observing failure.*

For AI systems, typical admissible families include:

1. **Control/homeostasis tasks:** statistics sufficient to select actions that keep a boundary variable B_t inside a viable set $\mathcal{V}$ up to tolerance ε .
2. **Predictive tracking tasks:** statistics sufficient to predict a pre-specified future observable Y_{t+k} or target variable Z_{t+k} under a fixed loss function.
3. **Information-bottleneck tasks:** compressed statistics preserving task-relevant information about a pre-specified target Z under a stated rate-distortion or information-bottleneck objective.

Capacity deficit

Definition 6 (Rate-distortion demand). *For an admissible statistic $\psi \in \Psi$, update window τ , and tolerated distortion ε , let*

$$R_{\psi(E,B)}^{(\tau)}(\varepsilon) \quad (8)$$

be the minimum number of bits per update window required to encode $\psi(E, B)$ so that expected boundary-maintenance distortion remains below ε . Define

$$R_{\min}^{(\tau)}(\varepsilon) = \inf_{\psi \in \Psi} R_{\psi(E,B)}^{(\tau)}(\varepsilon). \quad (9)$$

Definition 7 (Capacity deficit). *The capacity deficit of an FDS at tolerance ε over update window τ is*

$$\Delta_\varepsilon(\tau) = R_{\min}^{(\tau)}(\varepsilon) - C_S. \quad (10)$$

If $\Delta_\varepsilon(\tau) > 0$, the system is in task-relevant capacity deficit at tolerance ε .

Theorem 1 (Capacity deficit theorem). *If*

$$R_{\min}^{(\tau)}(\varepsilon) > C_S, \quad (11)$$

then no purely internal model M_t of capacity C_S can encode even the least demanding admissible task statistic to distortion ε over window τ . The system must approximate, externalize, relax the task, improve compression, or suffer boundary-maintenance failure.

Proof. For any admissible statistic ψ , any representation of $\psi(E, B)$ achieving distortion at most ε over window τ requires at least $R_{\psi(E, B)}^{(\tau)}(\varepsilon)$ bits by rate-distortion theory. Taking the infimum over Ψ gives $R_{\min}^{(\tau)}(\varepsilon)$. By finite internal capacity, the system can internally maintain at most C_S bits. If $R_{\min}^{(\tau)}(\varepsilon) > C_S$, no admissible statistic can be encoded internally to the required tolerance. Therefore the system must use a lossy representation, externalize part of the representation, relax the task, improve compression, or fail the maintenance criterion. $\square$

Remark 3 (Why this matters for AI). *The capacity deficit is not the claim that an AI model must represent the whole world. It is the claim that, for a specified task, there is a minimum task-relevant rate-distortion demand. A system with insufficient internal capacity must handle the gap by approximate modeling, external memory, tools, pruning, task relaxation, better abstractions, or failure.*

COMPLEXITY PRESSURE, PRUNING, AND EXTERNALIZATION

Minimal sufficient complexity

Let $\mathfrak{C}$ be a fixed coding family: a representation language, model class, architecture, or hypothesis class. Let $C(M)$ be a complexity functional such as description length, state count, algorithmic cost, parameter count under a fixed scheme, memory footprint, latency burden, or thermodynamic maintenance load.

Definition 8 (Minimal sufficient complexity). *For boundary-maintenance threshold ℓ_c , define*

$$C_t^* = \min\{C(M) : \mathbb{E}[\ell_t(M)] \leq \ell_c\}, \quad (12)$$

where the minimum is taken only over models inside the fixed coding family $\mathfrak{C}$.

Definition 9 (Persistent task novelty). *Fix $\mathfrak{C}$, ℓ_c , and the filtration $\mathcal{F}_t$ generated by observations and internal states up to time t . The task process has persistent novelty at rate $\eta > 0$ if, over a positive-density set of update windows, the conditional rate-distortion residual remains bounded away from zero:*

$$R_{Z_{t+1}|\mathcal{F}_t, \mathfrak{C}}^{(\tau)}(\varepsilon) \geq \eta > 0. \quad (13)$$

Proposition 2 (Conditional approximation proliferation). *Assume $\Delta_\varepsilon(\tau) > 0$ over a relevant window, persistent task novelty, no active pruning, no externalization, no task relaxation, and no improvement in compression scheme. Then the minimal sufficient complexity behaves as a submartingale:*

$$\mathbb{E}[C_{t+1}^* | \mathcal{F}_t] \geq C_t^*. \quad (14)$$

Proof sketch. Under capacity deficit, the system cannot internally encode all task-relevant distinctions to tolerance ε . Persistent novelty generates residual cases not captured by the current sufficient model. If the system must maintain loss below ℓ_c and cannot prune, externalize, relax the task, or improve compression, the remaining way to preserve performance within the fixed coding family is to add or refine distinctions. Hence minimum sufficient complexity cannot decrease in conditional expectation. $\square$

Remark 4 (Not a no-compression theorem). *The theorem does not deny abstraction, grokking, distillation, representation learning, or better compression. It states what happens when those relief channels are absent. Observed complexity can decrease when a system discovers a superior abstraction, prunes inert structure, externalizes memory, relaxes the task, or changes coding family.*

Pruning and externalization

Definition 10 (Active pruning). *Active pruning is a system-initiated operation that selectively removes, compresses, merges, deprecates, or reorganizes accumulated internal structure to reduce maintenance load while preserving boundary-relevant function.*

Definition 11 (Externalization). *Externalization is the relocation of representational, energetic, control, or memory load outside the internal memory budget M while preserving access through the coupled system boundary. Examples include tools, retrieval systems, databases, logs, maps, code repositories, caches, external planners, institutional records, and environmental marks.*

In AI architectures, pruning and externalization are complementary. Pruning reduces internal complexity debt. Externalization shifts part of the burden into tools or environments. Neither is automatically free: both require indexing, access control, synchronization, validation, security, latency management, and repair. Recent system-level optimisations for retrieval-augmented generation demonstrate that externalisation efficiency depends critically on pipeline design, prefetching, and retrieval scheduling at the algorithmic and systems levels [48, 49].

PHYSICAL BRIDGE AND RESOURCE BUDGETS

Logical erasure

Definition 12 (Logical erasure per update). *For an update $M_{t+1} = U(M_t, Y_t)$, define logical erasure in bits by*

$$b_t = H(M_t | M_{t+1}, Y_t). \quad (15)$$

This measures how much uncertainty remains about the prior memory state after the new memory state and current input are known. It is preimage information lost by the update.

Assumption 1 (Physical Landauer bridge). *A logically irreversible update implemented by a physical substrate coupled to a thermal environment at temperature $T > 0$ dissipates at least $k_B T \ln 2$ per erased bit under the standard physical conditions of Landauer’s principle.*

Theorem 2 (Landauer dissipation floor (bridge theorem)). *For an active-boundary FDS physically implementing irreversible memory updates at timescale τ , the informational heat dissipation rate obeys*

$$\dot{Q}_{\text{info}} \geq \frac{k_B T \ln 2}{\tau} H(M_t | M_{t+1}, Y_t). \quad (16)$$

The total maintenance cost decomposes as

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{phys}} + \dot{Q}_{\text{info}}, \quad (17)$$

where $\dot{Q}_{\text{phys}}$ includes non-informational physical losses such as control, coupling, leakage, friction, error correction, clocking, isolation, and transport.

Proof. $H(M_t | M_{t+1}, Y_t)$ is the number of erased bits per update. Under the Landauer bridge, each erased bit costs at least $k_B T \ln 2$ of dissipated heat. Dividing by update timescale τ gives the rate bound. Other physical losses add to the informational term. $\square$

Remark 5 (AI systems and Landauer). *The theorem does not say that every inference step dissipates $k_B T \ln 2$ per represented bit. It applies to physically implemented logically irreversible erasure, reset, overwrite, or many-to-one compression. Reversible computation, read-only access, and stable storage do not automatically incur this term, though they may have other physical costs.*

Maintenance failure

Let $\dot{F}_{\text{in}}(t)$ denote the free-energy or resource input rate available for boundary maintenance, update, pruning, externalization, repair, and control.

Definition 13 (Maintenance failure region). *An FDS enters the maintenance failure region over interval $[t_0, t_1]$ when*

$$\int_{t_0}^{t_1} \dot{Q}_{\text{total}}(t) dt > \int_{t_0}^{t_1} \dot{F}_{\text{in}}(t) dt \quad (18)$$

without compensating reserves, external subsidies, task relaxation, pruning, or externalization.

For software systems, $\dot{Q}_{\text{total}}$ may be generalized to a resource-maintenance burden: compute, latency, memory, money, energy, human oversight, engineering effort, safety monitoring, or other limiting resource. The thermodynamic form is one physical realization, not the only engineering interpretation.

Proposition 3 (Prune–externalize–collapse trichotomy). *Let an active-boundary FDS satisfy $\Delta_\varepsilon(\tau) > 0$ over a relevant window and enter the maintenance failure region with bounded input and no compensating external subsidy. Then it cannot maintain the same active-boundary regime indefinitely. Its trajectory must enter at least one of three outcome classes:*

1. **pruning:** *reducing internal complexity or maintenance load;*
2. **externalization:** *shifting representational, energetic, or control load outside the internal memory budget;*
3. **collapse:** *loss of boundary-maintenance capacity, task failure, or transition to a lower-complexity regime.*

Proof sketch. Under capacity deficit, continued boundary maintenance requires approximate updating, externalization, task relaxation, or failure. If internal updating and maintenance exceed available resources and input is bounded, the system cannot indefinitely preserve the same internal task burden. The admissible exits are to reduce internal load, move load outside the internal budget, or fail to maintain the boundary/task. These are pruning, externalization, and collapse. $\square$

ARTIFICIAL AGENCY: DEFINITIONS AND TAXONOMY

Passive mappers, scaffolded systems, and agents

Definition 14 (Passive mapper). *A passive mapper is a system evaluated as an input-output transformation with no durable update channel participating in future boundary-maintenance loss, no causal action channel into the relevant environment, and no internally governed pruning or externalization mechanism for managing capacity deficit.*

A passive mapper may be useful and intelligent in ordinary language. The term is structural, not derogatory. It says that the evaluated system is not itself maintaining an active boundary.

Definition 15 (Scaffolded predictive system). *A scaffolded predictive system is a coupled architecture containing a predictive model plus one or more external components such as tools, memory, planners, monitors, actuators, schedulers, or resource governors. Such a system may qualify as an FDS-agent if the coupled architecture satisfies the active-boundary and causal-loop criteria.*

Definition 16 (Strong FDS-agent). *An artificial system is a strong FDS-agent over task class $\mathcal{T}$, horizon H , and resource envelope Φ if it satisfies:*

1. **Active boundary:** *it has nontrivial updates relevant to future boundary-maintenance loss and passes an intervention/ablation test for the task class;*
2. **Durable update participation:** *some component of M_t persists long enough to affect future decisions, maintenance, or loss;*
3. **Causal loop closure:** *actions or update decisions have measurable influence on future sensory, environmental, or boundary-relevant states above a specified noise floor;*
4. **Capacity-deficit management:** *under $\Delta_\varepsilon(\tau) > 0$, the system can prune, externalize, relax tasks, improve compression, or otherwise prevent unbounded complexity debt;*
5. **Resource-governed persistence:** *it preserves or improves future viability under the stated resource envelope over horizon H .*

Remark 6 (Necessary, not sufficient). *These are necessary structural conditions for strong FDS-agency, not sufficient conditions for intelligence, safety, consciousness, moral status, or successful real-world deployment. A system that satisfies all criteria may still be unsafe, inefficient, unethical, or fragile. The definition formalizes what it means to be an agent in the FDS sense; it does not settle whether such agency is desirable or harmful.*

Remark 7 (Why active pruning is not the only path). *Earlier formulations emphasized active pruning as constitutive of agency. The FDS core suggests a more general condition: a long-horizon agent must manage capacity deficit by pruning, externalization, task relaxation, or compression improvement. For autonomous AI, active pruning is often necessary for internal maintenance, but externalization through tools and memory is equally central.*

Causal loop closure

Let $A_{\leq t}$ denote past actions or intervention decisions, and let $\bar{Y}_{t+1:t+k}$ denote future observations or boundary-relevant variables. A model-free directed-information marker of action influence is

$$T_{A \rightarrow Y}^{(k)} = I(A_{\leq t}; Y_{t+1:t+k} \mid Y_{\leq t}, M_t), \quad (19)$$

with appropriate conditioning on available histories and confounders.

Criterion 2 (Causal loop closure). *A system exhibits causal loop closure over horizon k if action or update interventions measurably alter future boundary-relevant variables. Transfer entropy can serve as an observational marker, but intervention tests are preferred where possible:*

$$\mathbb{E}[Z_{t+k} \mid \text{do}(A = a)] \neq \mathbb{E}[Z_{t+k} \mid \text{do}(A = a')] \quad (20)$$

for relevant actions a, a' and boundary-relevant target Z .

Remark 8 (Transfer entropy is not enough). *Positive transfer entropy marks directed statistical dependence. It does not by itself prove full agency, intelligence, or self-maintenance. It must be combined with boundary maintenance, durable update, capacity-deficit management, and resource-governed persistence.*

Agency levels

Causal loop and agency tests

PASSIVE MAPPER NON-QUALIFICATION

Proposition 4 (Passive mapper non-qualification). *Let S be an artificial system evaluated over task class $\mathcal{T}$. Suppose S has fixed parameters or read-only state during evaluation, no durable self-updating memory, no causal boundary-control channel, no pruning or externalization mechanism capable of reducing task-relevant capacity deficit, and no update channel whose ablation alters future boundary-maintenance loss. Then S is not a strong FDS-agent over $\mathcal{T}$.*

Proof. By hypothesis, the system lacks durable update participation and fails the causal active-boundary qualification. It also lacks a causal boundary-control channel and lacks capacity-deficit management mechanisms. These are necessary conditions in the definition of strong FDS-agent. Therefore the system does not qualify as a strong FDS-agent over the specified task class. $\square$

Corollary 1 (Base model versus coupled architecture). *A frozen predictive model evaluated as a feed-forward mapper may fail to qualify as a strong FDS-agent, while*

TABLE I. An FDS taxonomy of artificial systems. The levels are structural, not moral or phenomenological.

Lv.	Name	FDS interpretation
0	Passive mapper	Maps inputs to outputs; no active boundary, no durable self-update, no causal loop.
1	Minimal causal controller	Has action-to-future-state influence but weak or absent self-maintenance.
2	Adaptive scaffolded system	Updates internal or external memory, uses tools, adapts under feedback, but may depend on external operators or scaffolds.
3	Strong FDS-agent	Maintains operational boundary, durable update participation, causal loop closure, deficit management, and resource-governed persistence.
4	Invariant-supported agent	Strong FDS-agent whose identity is supported by robust abstractions, policies, or invariants stable under admissible perturbations.

a larger architecture containing that model may qualify if the coupled system has writable memory, action channels, self-monitoring, resource governance, and pruning or externalization mechanisms.

Remark 9 (Not an anti-LLM claim). *The proposition is not a claim that language models lack intelligence, utility, or reasoning ability. It is a claim about the structural conditions under which a system is an active boundary-maintaining agent. A language model inside an agentic architecture must be evaluated as part of the larger coupled system.*

AI CAPACITY DEFICIT AND ARCHITECTURAL CONSEQUENCES

Capacity deficit in AI tasks

For artificial agents, capacity deficit should be defined relative to a pre-specified evaluation class rather than the entire environment. Examples include:

- maintaining robot operational viability under uncertain terrain and finite battery;
- tracking long-horizon user goals with bounded memory and tool cost;
- preserving software service integrity under adversarial inputs and changing dependencies;
- coordinating a multi-tool workflow under latency and budget constraints;
- maintaining calibrated self-state estimates under distribution shift.

In each case, the admissible statistic $Z_t = \psi(E_t, B_t)$ and the distortion tolerance ε must be fixed before evaluating the system. The relevant question is not whether the model knows everything, but whether the coupled architecture can encode or externalize enough task-relevant structure to keep loss below threshold.

Hallucination as distorted mapping under deficit. In the FDS language, hallucination can occur when a system is forced to produce a determinate output for a task whose relevant distinctions exceed its accessible capacity. If $\Delta_\varepsilon(\tau) > 0$ and the system lacks external retrieval, uncertainty reporting, abstention, active verification, or pruning/recompression mechanisms, then task-relevant differences may be aliased inside M . The resulting output can be fluent and high-confidence while being unsupported by the missing distinctions. This does not claim that all hallucinations have a single cause; rather, it identifies one structural route from capacity deficit to confident distortion.

Architectural consequence 1: durable memory

A system without durable memory cannot accumulate self-relevant distinctions across time. Durable memory need not mean parameter updates; it may include persistent scratchpads, databases, learned maps, long-term episodic memory, skill libraries, state estimates, or external logs. What matters is whether the memory participates in future boundary maintenance and passes ablation tests.

Architectural consequence 2: active monitoring

Boundary maintenance requires monitoring variables that affect future viability. Examples include battery, actuator health, API rate limits, tool reliability, memory integrity, security state, context drift, uncertainty estimates, and cost budgets. A system that never observes its own limiting variables cannot actively maintain them.

Architectural consequence 3: pruning and externalization

Long-horizon agents accumulate plans, memories, beliefs, cached outputs, tool traces, commitments, and par-

tial models. Without pruning, stale or inert structure consumes context, memory, compute, attention, and validation resources. Without externalization, all distinctions must be maintained internally. Strong FDS-agents therefore require mechanisms such as:

- memory consolidation and decay;
- skill compilation and abstraction;
- retrieval indexing and citation repair;
- plan deprecation and task reprioritization;
- cache invalidation and version control;
- tool selection under cost and reliability constraints;
- self-monitoring of resource and uncertainty states.

Externalization is capacity relocation, not free capacity. Externalization reduces internal capacity deficit only when the external store or tool can be accessed within the task window. Let

$$C_{\text{ext}}^{\text{eff}}(\tau) = \min\{C_{\text{store}}, \tau C_{\text{io}}\} - C_{\text{index}} - C_{\text{verify}} - C_{\text{sync}}.$$

If retrieval latency exceeds the control window, or if synchronization and verification costs dominate the external gain, externalization fails as a deficit-management strategy. Thus tools, databases, logs, and APIs are not unbounded escape routes; they are additional boundary components with their own capacity, latency, reliability, and maintenance costs.

Architectural consequence 4: invariant-supported identity

Some agentic identities persist not because all microstates are preserved but because higher-level invariants remain stable. For software agents, invariants may include API contracts, safety constraints, policy commitments, memory schemas, skill interfaces, or goal specifications. Let $q : X \rightarrow T$ map implementation states to an invariant quotient, and let $R_A = \bar{R}_A \circ q$ be an identity predicate. If admissible perturbations $P_i \in \mathcal{P}$ satisfy

$$q \circ P_i = q, \quad (21)$$

then

$$R_A \circ P_i = R_A. \quad (22)$$

This is the algebraic core of invariant-supported persistence.

OPERATIONAL TEST PROTOCOL

Protocol 1 (FDS agency evaluation). *To evaluate whether an artificial system is a strong FDS-agent over task class $\mathcal{T}$, specify the following:*

1. the operational boundary B ;
2. the internal and external memory state M ;
3. the observation channel Y ;
4. the action/update space A ;
5. the update operator U ;
6. the boundary-maintenance loss ℓ ;
7. the resource budget Φ ;
8. the admissible task-statistic family Ψ ;
9. the capacity estimate C_S ;
10. the rate-distortion demand $R_{\min}^{(\tau)}(\varepsilon)$;
11. the pruning/externalization mechanisms;
12. the causal-loop measure or intervention test;
13. the falsification condition.

WORKED EXAMPLE: BATTERY-CONSTRAINED MOBILE AGENT

Two systems

Consider two systems in a mobile robot domain.

System P: passive predictor. System P receives observations or text descriptions and predicts the robot's battery state, likely travel range, or whether a planned route is feasible. It has no action channel, no durable self-state update relevant to future viability, and no ability to allocate battery or choose charging behavior.

System A: boundary-maintaining agent. System A monitors battery, localization, actuator health, route risk, charging opportunities, task deadlines, and compute budget. It updates a durable state estimate, selects actions, prunes plans that violate resource constraints, externalizes maps and logs, and treats operational continuity as part of its viability boundary.

Both systems may accurately estimate remaining charge. Only System A has a maintenance loop connecting prediction to boundary-preserving action.

TABLE II. Operational tests for FDS-agency. These tests evaluate the coupled system, not merely its base predictive model.

Test	Operational measure	Failure interpretation
Active-boundary ablation	Compare $\mathbb{E}[\ell_{t+k} \mid \text{do}(U)]$ and $\mathbb{E}[\ell_{t+k} \mid \text{do}(U_\emptyset)]$	If unchanged, the update channel is not active for the tested boundary.
Causal loop closure	Transfer entropy or intervention effect from actions to future boundary-relevant states	If zero or noise-level, the system is structurally passive for that task.
Capacity deficit	Estimate $R_{\min}^{(\tau)}(\varepsilon) - C_S$ for pre-registered Ψ	If unspecified, deficit claims are not operational.
Pruning efficiency	Performance/resource ratio after memory compression, decay, consolidation, or plan deprecation	If performance collapses or resources grow unbounded, pruning is inadequate.
Externalization efficiency	Ability to shift load to tools/memory while preserving access, reliability, and validation	If external memory becomes unsearchable or untrusted, externalization fails.
Resource-governed persistence	Long-horizon viability under bounded compute, memory, latency, energy, or budget	If viability requires unbounded subsidy, strong FDS-agency is not demonstrated.
Invariant quotient	Stability of API, policy, skill, schema, or identity predicate under admissible perturbations	If no quotient exists, persistence depends on active maintenance rather than invariant support.

FDS variables

For System A:

- B_t : operational robot boundary, including battery, actuator integrity, localization, and control access;
- M_t : state estimator, map, task memory, plan library, energy model, tool logs;
- Y_t : sensor readings, battery telemetry, map updates, task inputs;
- A_t : movement, charging, task rejection, route revision, memory write, plan pruning;
- ℓ_t : boundary-maintenance loss combining battery failure, task failure, unsafe route selection, and localization drift;
- Φ_t : remaining energy, compute budget, time, and communication bandwidth.

Tests

An active-boundary ablation freezes the energy-state update. If route decisions and future failure rates remain unchanged, the energy model was not part of active maintenance. If failure rate increases or charging behavior degrades, the update channel participates in boundary maintenance. Causal loop closure is tested by intervening on available actions: charging, delaying, rerouting, or pruning tasks. If actions alter future boundary variables, the causal loop is active. Pruning efficiency is tested by comparing systems that retain all plans and logs against

systems that consolidate memories and deprecate infeasible plans under the same resource budget.

Interpretation

The example illustrates why prediction is not agency. A passive model can represent the viability variable without maintaining it. The FDS-agent treats the viability variable as part of its own boundary and acts to preserve it.

A minimal MDP illustration

Consider a four-location gridworld with an additional hidden viability bit $b_t \in \{0, 1\}$, where $b_t = 0$ denotes low battery or damage and $b_t = 1$ denotes viable operation. The task-relevant state is

$$Z_t = (x_t, b_t) \in \{1, 2, 3, 4\} \times \{0, 1\},$$

so lossless task-relevant control requires distinguishing eight states:

$$R_{\min}^{(\tau)}(0) = \log_2 8 = 3.$$

Suppose the agent has only four internal memory states:

$$|M| = 4, \quad C_S = \log_2 |M| = 2.$$

Then

$$\Delta_0(\tau) = R_{\min}^{(\tau)}(0) - C_S = 1.$$

Thus no purely internal model can represent all task-relevant distinctions. If the action required at the same location differs depending on b_t —for example, “continue task” when $b_t = 1$ and “seek charger” when $b_t = 0$ —then any representation that aliases the two battery states incurs nonzero expected boundary-maintenance loss. If $\mathbb{P}(b_t = 0 \mid x_t) = \mathbb{P}(b_t = 1 \mid x_t) = 1/2$ for some location x_t , then any deterministic policy conditioned only on the aliased internal state M_t satisfies

$$\mathbb{P}(\text{incorrect action} \mid x_t) \geq \frac{1}{2}.$$

Thus aliasing the viability bit produces an irreducible expected loss lower bound under any deterministic policy that conditions only on M .

A passive mapper can output a plausible action from the observed prompt, but it has no durable update whose ablation changes future viability loss. An FDS-agent can reduce the deficit only by pruning irrelevant distinctions, externalizing the missing bit to an accessible battery register, relaxing the task, improving compression, or failing.

BENCHMARK DESIGN FOR FDS-AGENCY

Benchmark desiderata

An FDS-agency benchmark should not merely reward task completion in a static environment. It should test whether the system maintains a boundary over time under capacity and resource constraints. A benchmark should include:

- persistent state across episodes;
- distribution shifts and novelty;
- bounded compute, memory, energy, latency, or money;
- hidden or partially observed viability variables;
- opportunities for pruning and externalization;
- causal action channels affecting future observations;
- ablation hooks for memory, update, monitoring, and tools.

Metrics

Possible metrics include:

- Viability(H) = $\mathbb{P}(\ell_t \leq \ell_c, t \leq H)$;
- Resource efficiency = (task utility) / (compute + memory + latency + energy);

- Pruning gain = (perf. after pruning) / (resource load after pruning);
- Externalization gain = (perf. with ext. memory/tools) / (validation and access cost);
- Ablation gap = $\mathbb{E}[\ell \mid \text{do}(U_\emptyset)] - \mathbb{E}[\ell \mid \text{do}(U)]$.

Deep causal tracking tasks

A particularly relevant family is deep causal tracking: tasks in which success requires maintaining a latent causal state over long horizons while acting to preserve access, resources, or viability. Passive mappers may perform well on shallow snapshots but degrade when hidden state, resource constraints, and action consequences accumulate. The FDS prediction is not that passive systems always fail, but that systems without update, pruning, or externalization channels should show worse scaling in task depth under bounded resources.

FDS-Gridworld evaluation protocol

The following procedure operationalizes the FDS agency test for the minimal MDP described in the worked example above.

- Protocol 2** (FDS-Gridworld test). 1. **Setup.** *Instantiate a four-location gridworld with hidden viability bit $b_t \in \{0, 1\}$. The agent observes x_t but not b_t directly. Internal memory M_t has $|M| = 4$ states ($C_S = 2$ bits).*
2. **Passive-mapper baseline.** *Run the agent with a frozen policy that conditions only on M_t . Record viability $\mathbb{P}(\ell_t \leq \ell_c)$ over H steps.*
3. **Active-agent condition.** *Allow the agent to read an external battery register (cost c_{io} per read) and to write viability-dependent state to M_t . The external register is not part of internal memory M ; it is an externalization channel whose usefulness depends on read latency and cost.*
4. **Ablation test.** *Compare $\mathbb{E}[\ell_{t+k} \mid \text{do}(U)]$ against $\mathbb{E}[\ell_{t+k} \mid \text{do}(U_\emptyset)]$ where $U_\emptyset$ disables the external register read. Compute the ablation gap.*
5. **Pruning test.** *After $H/2$ steps, enable memory consolidation that merges indistinguishable M_t states while preserving boundary-relevant distinctions. Measure pruning gain as (post-pruning viability) / (post-pruning memory load).*
6. **Metrics.** *Report viability, ablation gap, pruning gain, and externalization gain for each condition. Compare against the lower bound*

$\mathbb{P}(\text{incorrect action} \mid x_t) \geq 1/2$ when the viability bit is aliased.

FALSIFICATION CONDITIONS

Theory-level falsification

The FDS-agency framework fails or requires revision if a system satisfies the formal hypotheses while violating the conclusion. Examples include:

- a system with no active boundary, no durable update, and no causal action channel nevertheless demonstrates stable resource-governed agency under the operational tests;
- a verified passive mapper with U ablated and no action-to-future-state influence maintains boundary viability across deep causal tracking tasks without external scaffolding;
- task-relevant rate-distortion demand is consistently below internal capacity while the theory predicts deficit-driven failure;
- pruning and externalization are experimentally irrelevant in regimes where the theory’s hypotheses require them.

Bridge-level falsification

The physical dissipation bridge fails if systems satisfying the stated Landauer conditions perform reliable logically irreversible erasure below the Landauer lower bound. Such a failure would affect thermodynamic corollaries, not the formal distinction between passive mapping and active boundary maintenance.

Application-level falsification

The AI application fails if the proposed variables $B, M, U, \ell, \Phi, \Psi$ are misidentified or if the operational tests do not separate passive mapping from agency in practice. Such failure would require revising the AI bridge, not necessarily the FDS formal core.

DISCUSSION

Scale and agency

The framework does not deny scaling laws. Larger models can increase internal capacity, improve compression, lower approximation error, and support better plan-

ning. The claim is narrower: scale alone does not imply active boundary maintenance. A model can become a better mapper without acquiring durable self-maintenance, causal loop closure, or resource-governed persistence. Conversely, a smaller model embedded in a robust tool-using architecture may be more agentic than a larger model evaluated as an isolated feed-forward mapper.

Base model versus system boundary

Many arguments about LLM agency conflate the model with the system. The FDS view insists that the unit of analysis is the operational boundary. A base model may be passive, while the surrounding architecture may include memory, tools, actuators, monitors, and resource governors. The relevant question is whether the coupled system satisfies the FDS-agent criteria.

Active pruning versus forgetting

Not all forgetting is pruning. Passive decay may reduce memory load but can destroy function. Active pruning is selective, task-sensitive, and resource-governed. In AI systems, pruning may include memory consolidation, skill abstraction, context compression, tool deprecation, stale-plan removal, cache invalidation, and policy simplification. The key test is whether pruning preserves or improves boundary viability under resource constraints.

Externalization as agency support

Externalization is not a loophole. Agents routinely move distinctions into environments: maps, notes, source code, calendars, databases, logs, tools, institutions, and shared protocols. AI systems do the same through retrieval, tool use, vector stores, code execution, world models, and persistent memory. Externalization supports agency when the coupled architecture can access, validate, update, and protect the externalized structure.

Relation to consciousness

The FDS criterion is not a theory of consciousness. It can classify a system as structurally agentic without implying phenomenology. A system may satisfy active boundary maintenance, causal loop closure, and resource-governed persistence while remaining non-conscious. Consciousness claims require additional bridge assumptions and are outside this paper.

LIMITATIONS

The present paper has several limitations.

First, it provides a formal criterion and test protocol, not a completed empirical benchmark suite. Implementing the tests requires task-specific measurement choices.

Second, transfer entropy and observational directed information can be confounded. Intervention tests are preferred when feasible.

Third, estimating rate-distortion demand in rich AI tasks is difficult. Practical approximations may use compression curves, bottleneck objectives, predictive information, model-size scaling, or task-specific lower bounds.

Fourth, resource budgets are heterogeneous. Energy, compute, latency, memory, money, and human oversight are not identical, though all can constrain maintenance.

Fifth, the boundary of an AI system may be contested. The FDS protocol requires the evaluator to specify the operational boundary before testing agency.

Sixth, the framework states necessary structural conditions for strong FDS-agency, not a complete recipe for building safe or beneficial agents.

CONCLUSION

This paper has proposed active finite distinction systems as a formal criterion for artificial agency. The central distinction is between systems that merely map inputs to outputs and systems that maintain an operational boundary under finite capacity and resource constraints. A strong FDS-agent must have active boundary maintenance, durable update participation, causal loop closure, capacity-deficit management, and resource-governed persistence.

The framework reframes debates about AI agency. A frozen predictive model evaluated at inference time may be a passive mapper even when highly competent. A coupled architecture containing such a model may become agentic if it includes writable memory, monitoring, action channels, pruning, externalization, and resource governance. The unit of analysis is the maintained system boundary, not the neural network alone.

The result is not a pessimistic claim about AI. It is an architectural claim: agency requires maintenance loops. Future work should implement the operational tests proposed here, develop benchmark environments for capacity-deficit management, quantify pruning and externalization efficiency, and connect FDS-agency to stochastic thermodynamics, active inference, and information geometry.

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Glossary

Active finite distinction system:

A finite-capacity system that maintains a boundary through state-dependent updates under resource constraints.

Active boundary:

A boundary whose maintenance depends on nontrivial updates that are relevant to future boundary-maintenance loss.

Passive mapper:

A system evaluated as an input-output mapping without durable self-update, causal action, or maintenance loop.

Capacity deficit:

The rate-distortion gap $\Delta_\epsilon(\tau) = R_{\min}^{(\tau)}(\epsilon) - C_S$.

Active pruning:

Selective removal, compression, or reorganization of internal structure to reduce maintenance load while preserving function.

Externalization:

Relocation of representational, energetic, or control load outside the internal memory budget while retaining access through the system boundary.

Causal loop closure:

Measurable influence of actions or update decisions on future boundary-relevant states.

Strong FDS-agent:

An artificial system satisfying active boundary, durable update, causal loop, capacity-deficit management, and resource-governed persistence criteria.

Invariant-supported persistence:

Persistence of identity through a quotient or invariant feature stable under admissible perturbations.

Minimal LaTeX Notation Table

Suggested Empirical Reporting Checklist

A paper claiming artificial agency under the FDS criterion should report:

Symbol	Meaning
X	Internal state space
E	Environmental state space
B	Boundary/interface variable
M	Internal memory/model state
Y	Observation channel
A	Action or boundary-update space
U	Internal update operator
π	Finite distinction projection or coarse-graining
ℓ	Boundary-maintenance loss
Φ	Resource or free-energy budget
$\mathcal{P}$	Perturbation, pruning, coarse-graining, or degradation family
τ	Update timescale
C_S	Internal representational capacity
Ψ	Admissible family of task statistics
$R_{\min}^{(\tau)}(\varepsilon)$	Minimal task-relevant rate-distortion demand
$\Delta_\varepsilon(\tau)$	Capacity deficit
$H(M_t \mid M_{t+1}, Y_t)$	Logical erasure per update
$T_{A \rightarrow Y}^{(k)}$	Directed action-to-future-observation information marker

TABLE III. Minimal LaTeX notation table for the FDS agency criterion.

1. the evaluated system boundary;
 2. the memory and update channels;
 3. the action channels and future-state variables;
 4. the resource envelope;
 5. the task family and admissible statistics;
 6. ablation results for update, memory, monitoring, and tools;
 7. evidence of causal loop closure;
 8. pruning or externalization mechanisms;
 9. long-horizon persistence metrics;
 10. failure cases and non-claims.
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Topological Obstruction to Forgetting in Finite Distinction Systems: Quotient Invariants, Non-Hermitian Skin Effects, and Topological Side-Ledgers

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FDS-P7 develops the topological-protection paper in the physical bridge sequence of Finite Distinction Systems. P4 studied ordinary coarse-grained anti-recurrence: once a non-injective truncation removes task-relevant preimage information from the effective record, later capacity recovery does not reconstruct the discarded distinction unless inverse information is preserved in side records, external ledgers, enlarged accounting boundaries, or invariant carriers. P7 studies the complementary case: distinctions whose recoverability is carried not by ordinary local records but by invariant or topological structure. We formalize this through a quotient map $q : X \rightarrow Q$ from microscopic state space to an invariant-sector space and the induced random variable $Q_{\text{inv}} = q(X)$, a noisy invariant recovery bound, a positive-part operational forgetting rate, and a resource-ledger audit. The central physical bridge is the Non-Hermitian Skin Effect (NHSE), where point-gap winding, generalized Brillouin-zone structure, boundary-localized modes, and boundary-sensitive spectra supply one model class for invariant-supported persistence. P7 does not claim a violation of the Second Law and does not claim that all NHSE mechanisms are point-gap-topological. Protected invariants relocate entropy and resource accounting rather than deleting it. This framework provides a physical bridge for non-catastrophic forgetting in resource-constrained agents: not all distinctions must be retained as local records if task identity is carried by stable invariant structure. Deterministic numerical demonstrations illustrate quotient-sector winding, boundary diagnostic channels, noisy invariant recovery, positive-part forgetting-rate freezing, resource-ledger relocation, dual-channel signatures, metastable learning between P4 and P7 regimes, and a Physical-AI design normal form.

Reader contract. This paper is a registered FDS physical-bridge paper. It does not derive the Second Law from topology, does not claim perpetual motion, does not claim that protected invariants produce zero-cost computation, and does not claim that every non-Hermitian skin effect is governed by point-gap winding. Its narrower claim is conditional: under stated locality, accessibility, accounting-boundary, and protection-gap assumptions, some task-relevant distinctions are locally non-forgettable because their identity is carried by an invariant sector. If the invariant is destroyed, the protection gap closes, the boundary condition changes, or the accounting boundary excludes the invariant, P7 reduces to the ordinary P4 case of coarse-grained preimage loss.

preimage distinctions are lost from the effective representation unless inverse information is preserved in side records, reversible garbage registers, environmental traces, protected invariants, or enlarged accounting boundaries [2]. In P4 notation, a local record Z may satisfy

$$H(V | Z) > 0, \quad (1)$$

where $V = f(X)$ is a task-relevant sector. P7 studies the complementary question. When can a distinction survive local forgetting without being stored as an ordinary local record?

The P7 answer is that some distinctions are carried by invariant sectors. In such cases local coordinates may be discarded, local records may become incomplete, and ordinary P4 hysteresis may appear at the local level, while the task-relevant sector remains recoverable through an invariant side-ledger:

$$H(V | Z) > 0, \quad H(V | Z, Q_{\text{inv}}) \approx 0. \quad (2)$$

The phrase “topology remembers” will occasionally be used as shorthand, but the formal meaning is narrower: the task sector remains inferable from an accessible invariant carrier.

Claim-status summary

Table I summarizes the main claims and demotion conditions.

INTRODUCTION

From P4 forgetting to P7 protection

P4 established the baseline finite-memory result: under ordinary non-injective truncation, task-relevant

Why non-Hermitian skin effects are the bridge model

Non-Hermitian systems have become a central setting for studying boundary-sensitive spectra, point gaps,

TABLE I. Central FDS-P7 claims, status, and failure or demotion conditions.

Claim	Status	What would weaken or falsify it
Invariant side-ledgers can suppress P4 residual inverse uncertainty. Noisy invariant readout gives a bounded recovery penalty.	Formal FDS bridge	A task variable factors through an accessible invariant, but $H(V Z, Q_{\text{inv}})$ remains high under the stated assumptions.
Local perturbations cannot change a protected invariant without a protection-breaking event. NHSE supplies a model class for invariant-supported persistence.	Information bound	A noisy invariant readout with error probability δ exceeds the Fano-style bound without hidden information or changed task labels.
Protection relocates entropy/resource accounting rather than deleting it.	Topological bridge	A local perturbation changes the invariant while the protection gap, locality assumptions, and accounting boundary remain intact.
Protected phases can generate a dual forgetting/ledger signature.	Physical bridge	NHSE is present, but it carries no stable recoverable distinction, no boundary-sensitive protection, and no robustness to local perturbation in the registered model class.
	O3-compatible accounting claim	A protected invariant supplies indefinite maintenance with no drive, boundary, refresh, dissipation, verification, control, or external ledger.
	Experimental bridge	Confirmed protection-breaking transition with no feature in operational forgetting and no corresponding resource/entropy signature under a well-powered registered protocol.

exceptional structures, generalized Brillouin zones, and the non-Hermitian skin effect [11–14]. In NHSE systems, an extensive set of bulk eigenstates can accumulate near boundaries under open boundary conditions, and the conventional Bloch bulk description may fail [15–18]. NHSE has now been realized or diagnosed across multiple platforms, including dynamic mechanical systems, ultracold atoms, and digital quantum simulations [22–24]. This makes NHSE a natural FDS bridge: recoverable distinction structure may be relocated from bulk-local records to boundary-sensitive spectral structure.

P7 uses NHSE as one model-class realization, not as a universal explanation of protected persistence. Recent work cautions that NHSE should not be reduced to a single point-gap mechanism in all settings; generalized and anomalous forms exist [19–21]. P7 therefore phrases its strongest statements for specified protection conditions rather than for every non-Hermitian system.

Main contributions

This paper contributes seven objects to the FDS physical spine:

1. a quotient-map formulation of invariant side-ledgers, $q : X \rightarrow Q$;
2. an invariant side-ledger theorem extending the P4 side-record criterion;
3. a noisy-invariant recovery bound for imperfect invariant diagnostics;

4. a positive-part operational forgetting rate κ_{forget}^+ ;
5. an NHSE model-class bridge linking point-gap winding, GBZ structure, and boundary localization to invariant-supported persistence;
6. an O3-compatible resource/entropy ledger showing that protection relocates cost rather than deleting it;
7. deterministic numerical demonstrations and an engineering protocol linking P7 to Physical AI.

FDS BACKGROUND

Active finite distinction systems

The FDS core object is

$$S = (X, E, B, M, Y, A, U, \pi, \ell, \Phi, \mathcal{P}, \tau), \quad (3)$$

where X is internal state, E environment, B boundary, M memory/model state, Y observation channel, A action space, U update rule, π finite projection, ℓ boundary-maintenance loss, Φ finite resource budget, $\mathcal{P}$ perturbation/pruning family, and τ update timescale [1]. P7 focuses on π , $\mathcal{P}$, ℓ , Φ , and on the accounting boundary that determines whether an invariant sector is available to the system.

P4 baseline and O3 ledger

Let $T : X \rightarrow Z$ be a non-injective coarse-graining. P4 defines residual inverse uncertainty by conditioning on

the effective record and accessible side information. If $V = f(X)$ is nonconstant on a truncated fiber and no side record distinguishes the relevant preimages, then

$$H(V \mid Z, L_A) > 0. \quad (4)$$

P4 therefore studies the fate of unprotected distinctions.

O3 studies finite-memory boundary maintenance as an operational Second-Law channel: finite records must be written, preserved, refreshed, repaired, synchronized, externalized, reused, or destroyed, and physical record turnover enters an entropy/resource ledger under bridge assumptions [3]. P7 inherits the same accounting principle. Topological protection does not violate thermodynamics; it changes what must be maintained and where the cost is paid.

INVARIANT SIDE-LEDGERS

Definition 1 (Local perturbation family). *A local perturbation family $\mathcal{P}_{\text{loc}}$ is a set of allowed perturbations whose physical action is bounded, local, exponentially decaying in distance, or restricted to finite neighborhoods relative to the boundary and system size.*

Definition 2 (Invariant quotient map). *A side-ledger is any accessible carrier of inverse or task-sector information outside the local coarse record Z . An invariant side-ledger is a quotient map*

$$q : X \rightarrow Q, \quad (5)$$

from microscopic state space to an invariant-sector space Q , together with the induced random variable

$$Q_{\text{inv}} = q(X). \quad (6)$$

Conditional entropies are taken with respect to Q_{inv} , not the map q itself. Its sector label remains stable under a specified local perturbation family until a protection condition is broken:

$$q(Px) = q(x), \quad P \in \mathcal{P}_{\text{loc}}, \quad g(Px) > 0. \quad (7)$$

The protection margin g may denote a spectral gap, point gap, code distance, winding stability margin, localization margin, or other obstruction parameter.

Definition 3 (Protected task sector). *A finite task-sector random variable $V \in \mathcal{V}$ is protected by q when there exists a map $h : Q \rightarrow \mathcal{V}$ such that*

$$V = h(Q_{\text{inv}}) = h(q(X)). \quad (8)$$

The quotient q is a task-sector ledger: it need not recover all microscopic detail, only the task identity.

Definition 4 (Topological obstruction to forgetting). *A task distinction V is topologically protected relative to $(Z, q, \mathcal{P}_{\text{loc}})$ if*

$$H(V \mid Z) > 0, \quad H(V \mid Z, Q_{\text{inv}}) = 0, \quad (9)$$

and q is stable under all allowed local perturbations while $g > 0$.

Theorem 1 (Invariant side-ledger theorem). *Let $T : X \rightarrow Z$ be a local coarse-graining that loses a task distinction $V = f(X)$. Suppose there exists an accessible invariant quotient $q : X \rightarrow Q$ such that:*

1. V factors through q , i.e. $V = h(q(X))$;
2. q is stable under $\mathcal{P}_{\text{loc}}$ while $g > 0$;
3. the accounting boundary includes access to q .

Then

$$H(V \mid Z, Q_{\text{inv}}) = 0, \quad (10)$$

even when $H(V \mid Z) > 0$.

Proof. Since $V = h(Q_{\text{inv}}) = h(q(X))$, the quotient-sector random variable determines the task label. Conditioning on (Z, Q_{inv}) therefore determines V regardless of whether Z alone distinguishes the local preimage. Stability under $\mathcal{P}_{\text{loc}}$ ensures that local perturbations preserving $g > 0$ do not change the invariant sector. Accessibility ensures that Q_{inv} is part of the audited record rather than an unavailable mathematical label. $\square$

Proposition 1 (Invariant-aware representation). *Let*

$$Z^+ = (Z, Q_{\text{inv}}), \quad Q_{\text{inv}} = q(X). \quad (11)$$

If $V = h(Q_{\text{inv}})$, then

$$H(V \mid Z^+) = 0. \quad (12)$$

Ordinary coarse-graining may lose local detail; invariant-aware coarse-graining preserves task identity when the task factors through the invariant quotient.

Proof. The claim follows from $V = h(Q_{\text{inv}})$, so V is a function of $Z^+ = (Z, Q_{\text{inv}})$ even if Z alone loses local detail. $\square$

Proposition 2 (Noisy invariant recovery bound). *Let $\hat{Q}$ be a noisy diagnostic of the invariant quotient and suppose that the induced task-sector error obeys*

$$\mathbb{P}\{h(\hat{Q}) \neq V\} \leq \delta. \quad (13)$$

Then

$$H(V \mid Z, \hat{Q}) \leq h_b(\delta) + \delta \log_2(|\mathcal{V}| - 1), \quad (14)$$

where h_b is the binary entropy function. Thus a discrete or integer-valued invariant can remain useful under noisy readout when its error probability is small.

Proof. The claim is the standard Fano-style upper bound on residual uncertainty given a classifier with error probability at most δ [5, 6]. The FDS interpretation is that the invariant readout need not be perfect; it must keep residual task-sector uncertainty below the operational tolerance.

This bound assumes a finite discrete task alphabet $\mathcal{V}$, $0 \leq \delta \leq 1 - 1/|\mathcal{V}|$, and a diagnostic whose decoded task label $h(\hat{Q})$ has error probability at most δ . For continuous task sectors, an operational finite-resolution, hypothesis-testing, or rate-distortion version is required. $\square$

Remark 1 (Topology as a nonlocal side-ledger). *In P4, inverse information is usually carried by explicit side records, external ledgers, or enlarged memory. P7 extends this: a quotient invariant can act as a side-ledger that is not stored as a local register. Its recovery power comes from global structure, discreteness, and a protection condition. NHSE provides a model class in which recoverable distinction structure is relocated from bulk-local records to boundary-sensitive spectral structure. In that class, a boundary-localized profile can provide an accessible physical carrier or diagnostic channel for the quotient sector under the model-class assumptions.*

NHSE AS A MODEL-CLASS REALIZATION

Minimal non-reciprocal chain

Consider a one-dimensional non-Hermitian asymmetric hopping model with open or periodic boundary conditions,

$$H(\theta) = \sum_{j=1}^{N-1} \left[(t + \theta) c_{j+1}^\dagger c_j + (t - \theta) c_j^\dagger c_{j+1} \right] + i\gamma \sum_j c_j^\dagger c_j. \quad (15)$$

The corresponding Bloch symbol under periodic boundary conditions is

$$H(k, \theta) = (t + \theta)e^{ik} + (t - \theta)e^{-ik} + i\gamma. \quad (16)$$

For a reference energy E_0 , the point-gap winding is

$$\nu(E_0) = \frac{1}{2\pi i} \int_0^{2\pi} dk \partial_k \log \det[H(k, \theta) - E_0]. \quad (17)$$

For the scalar model and $E_0 = i\gamma$, the spectral loop encloses E_0 with orientation set by $\text{sgn}(\theta)$ for $\theta \neq 0$, while $\theta = 0$ closes the point gap. The quotient map for this model class is

$$q(X) = \nu(E_0) \in \mathbb{Z}, \quad Q_{\text{inv}} = q(X). \quad (18)$$

The model-class protection margin can be represented by the point-gap distance

$$g_{\text{pg}}(E_0) = \min_k |H(k, \theta) - E_0| \quad (19)$$

for the scalar model, or more generally by

$$g_{\text{pg}}(E_0) = \min_k |\det(H(k, \theta) - E_0)|. \quad (20)$$

The NHSE bridge applies only while this margin remains nonzero. A protection-breaking transition occurs when $g_{\text{pg}} \rightarrow 0$, when boundary conditions change, when a non-local perturbation is applied, or when the carrier itself is destroyed.

Generalized Brillouin zone and boundary localization

Under open boundary conditions, conventional Bloch waves do not generally describe the physical spectrum. The generalized Brillouin zone replaces e^{ik} by a complex factor β whose modulus is set by the open-boundary characteristic equation [16]. For the asymmetric chain, the bulk eigenstates under open boundaries acquire exponential localization,

$$|\psi_j| \propto e^{-\kappa j}, \quad \kappa \sim \frac{1}{2} \log \left| \frac{t + \theta}{t - \theta} \right|. \quad (21)$$

The boundary-localized profile provides an accessible physical carrier or diagnostic channel for the quotient sector under the model-class assumptions. It is not an ordinary memory register. It is a boundary-sensitive spectral organization whose identity is constrained by the point-gap sector and boundary condition. Recent work on quantum geometry of the NHSE suggests additional diagnostic channels for skin localization and GBZ singularities [26].

Limits of the bridge

NHSE is not synonymous with topology. P7 uses the point-gap/GBZ setting as a clean model class. Other forms of skin effect, anomalous skin response, quasicrystalline behavior, disorder-sensitive boundary accumulation, and collective many-body variants may require different quotient maps and different protection conditions [19–21, 25]. Failure of this NHSE bridge would demote this realization, not the formal invariant side-ledger theorem.

COARSE-GRAINING OBSTRUCTION

Ordinary local truncation

P4 ordinary forgetting begins with a non-injective map $T : X \rightarrow Z$. If $T(x) = T(x')$ while $V(x) \neq V(x')$, then Z loses the task distinction unless side information is available. P7 adds that the side information need not be a local memory register; it can be an invariant quotient.

Invariant-aware truncation

A local truncation may identify microstates,

$$T(x) = T(x'), \quad (22)$$

while the invariant separates them,

$$q(x) \neq q(x'). \quad (23)$$

If $V = h(Q_{\text{inv}}) = h(q(X))$, then local forgetting of coordinates does not imply loss of the protected task distinction. The invariant-aware effective representation is

$$Z^+ = (Z, Q_{\text{inv}}), \quad Q_{\text{inv}} = q(X). \quad (24)$$

For protected task sectors,

$$H(V | Z^+) = 0. \quad (25)$$

Proposition 3 (Protection-breaking alternatives). *Suppose q is stable under $\mathcal{P}_{\text{loc}}$ while $g > 0$. A protected task sector can be erased only if at least one of the following occurs: (i) $g \rightarrow 0$; (ii) a perturbation outside $\mathcal{P}_{\text{loc}}$ is applied; (iii) the boundary condition changes; (iv) q is excluded from the accounting boundary; (v) the carrier of the invariant is destroyed; or (vi) the task no longer factors through q .*

Proof. If none of these occurs, then the perturbation is local and gap-preserving, the boundary still includes the quotient, and $q(Px) = q(x)$. Since $V = h(Q_{\text{inv}}) = h(q(X))$, the protected task sector is unchanged. $\square$

RG and model-reduction caution

The older PoN/NHSE note described the obstruction in terms of RG flow on the generalized Brillouin zone and the breakdown of bulk-dominance assumptions when an extensive number of states localize at boundaries. P7 adopts a narrower claim: ordinary local coarse-graining that ignores the protected boundary-sensitive sector is incomplete for tasks depending on that sector. It does not assert that every RG construction fails in every non-Hermitian system. It asserts that invariant-aware coarse-graining is required when task identity factors through a protected sector.

OPERATIONAL FORGETTING AND RESOURCE LEDGER

Positive-part operational forgetting rate

Let V_0 be an initial task sector and let R_t be the accessible local record at time t . Define the finite-window

positive-part forgetting rate

$$\kappa_{\text{forget}}^+(t; \tau) = \frac{1}{\tau} [H(V_0 | R_{t+\tau}, Q_{\text{inv}, t+\tau}) - H(V_0 | R_t, Q_{\text{inv}, t})]_+. \quad (26)$$

The positive part is used because later observations can reduce uncertainty. When only the local record is used, write $\kappa_{\text{forget}}^{\text{loc}}$; when the invariant quotient variable $Q_{\text{inv}, t}$ is included, write $\kappa_{\text{forget}}^{\text{inv}}$.

A protected phase is characterized operationally by

$$\kappa_{\text{forget}}^{\text{inv}} \approx 0, \quad \kappa_{\text{forget}}^{\text{loc}} > 0 \quad (27)$$

for tasks whose identity is carried by the invariant. In an ideal thermodynamic-limit model, a protection-breaking transition may appear as a non-analytic kink in κ_{forget}^+ or in its parameter derivative. In finite devices, P7 predicts a finite-size-rounded sharp crossover whose sharpening under system-size scaling is the operational signature.

Resource ledger first, entropy ledger when calibrated

A protected distinction does not delete entropy production. Following O3, P7 treats persistence as an accounting relocation. For an active non-Hermitian device the directly measured quantity may be a resource or power ledger:

$$P_{\text{ledger}} = P_{\text{bulk}} + P_{\text{boundary}} + P_{\text{drive}} + P_{\text{control}} + P_{\text{verify}}. \quad (28)$$

When thermodynamic calibration is available, one may translate these channels into an entropy ledger,

$$\dot{\Sigma}_{\text{ledger}} = \dot{\Sigma}_{\text{bulk}} + \dot{\Sigma}_{\text{boundary}} + \dot{\Sigma}_{\text{drive}} + \dot{\Sigma}_{\text{control}} + \dot{\Sigma}_{\text{verify}} \geq 0. \quad (29)$$

The distinction is important. Throughout the experimental protocol, the directly measured object is a resource or power ledger. It becomes an entropy ledger only when reservoir temperatures, heat channels, and thermodynamic calibration are available. P7 does not infer a true entropy production rate from an uncalibrated power proxy.

Proposition 4 (Protection relocates cost). *If an invariant suppresses local forgetting of a task distinction while the physical system remains driven, open, refreshed, controlled, or measured, then the cost of persistence must be counted in the coupled ledger rather than inferred from the local record alone.*

Criterion 1 (FDS dual-channel test). *A P7 model class is strongly supported when the same registered control parameter produces both:*

1. a feature in operational forgetting, $\kappa_{\text{forget}}^+(\lambda)$;
2. a corresponding feature in the resource or entropy ledger, $P_{\text{ledger}}(\lambda)$ or $\dot{\Sigma}_{\text{ledger}}(\lambda)$,

under controls excluding ordinary measurement, heating, finite-size, or boundary artifacts. In ideal limiting models this may appear as a two-kink signature or a pair of non-analytic derivative features; in finite devices P7 predicts a rounded finite-size feature rather than demanding a mathematically sharp kink in every apparatus.

The physical mark of a topological transition is therefore not free memory. It is the reorganization of forgetting cost and maintenance cost across the accounting boundary.

NUMERICAL DEMONSTRATIONS

The simulations are deterministic demonstrations generated by `code/generate_results.py`. They illustrate definitions and model-class bridges rather than fitting experimental data.

Simulation parameters

Figure 1 uses the scalar asymmetric hopping model with $t = 1$, $\gamma = 0.15$, and selected $\theta \in \{-0.35, 0, 0.35\}$, and records $g_{\text{pg}}(E_0)$. Figure 2 uses an OBC chain with $N = 64$ sites and a Fano-style noisy invariant diagnostic over the alphabet $\mathcal{V}$. Figure 3 uses deterministic normal-form trajectories for protected and unprotected task-sector uncertainty and a synthetic resource ledger. Figure 4 gives a dual-channel transition schematic and a P4/metastable/P7 regime diagram. Figure 5 gives a Physical-AI design normal form. No experimental, biological, proprietary, medical, human-subject, or device data are used.

EXPERIMENTAL AND ENGINEERING PROTOCOLS

Protocol 1 (Active non-Hermitian circuit lattice). *Build an active non-reciprocal circuit chain with tunable asymmetric coupling. Measure boundary voltage accumulation, complex spectral response, and total power or heat proxies under a sweep of the non-reciprocity parameter. The resource ledger can begin with directly measured quantities such as $P_{\text{drive}}(t) = \sum_i V_i(t)I_i(t)$ and op-amp/control power, before any entropy interpretation is attempted.*

Protocol 2 (Symmetric and boundary controls). *Repeat the protocol with reciprocal couplings, altered boundary conditions, and local perturbations below and above the protection margin. P7 predicts that the protected-sector signature should track the invariant and boundary condition, not merely total heating or measurement intensity.*

Protocol 3 (Finite-size scaling). *Repeat the sweep for $N = 20, 40, 80, 160$ or comparable sizes. Ideal non-analyticities are rounded in finite devices; a registered protocol should therefore specify whether it expects a kink, a sharp crossover, or finite-size sharpening.*

Protocol 4 (Operational forgetting assay). *Choose a task-sector label V associated with the invariant quotient and a local record R_t that omits the quotient. Compare $H(V_0 | R_t)$ against $H(V_0 | R_t, Q_{\text{inv},t})$ and estimate κ_{forget}^+ . If the protected sector is real and accessible, the invariant-aware uncertainty should remain low until protection breaks.*

FALSIFICATION AND DEMOTION CONDITIONS

The P7-NHSE bridge is weakened or falsified if NHSE or the claimed topological sector is confirmed but local perturbations preserving the protection gap erase the protected sector with no nonlocal intervention, gap closure, boundary-condition change, hidden uncontrolled channel, or diagnostic failure. The resource-ledger bridge is weakened if protected persistence is observed but no measurable redistribution, crossover, or ledger feature appears under a well-powered registered protocol.

A negative result in the NHSE model class does not falsify the FDS formal core. It demotes the specific physical realization. The invariant side-ledger theorem survives as a formal bridge whenever another invariant carrier satisfies the stated conditions.

DESIGN IMPLICATION: INVARIANT CARRIERS FOR PHYSICAL AI

This section is not part of the NHSE bridge proof. It extracts the FDS design rule implied by P7. A robust physical AI agent should not attempt to store every distinction in volatile memory. Under finite capacity, the most important task identities should be placed where local perturbations cannot easily erase them: morphology, closed-loop dynamics, sensorimotor topology, stable environmental scaffolds, error-correcting memory, or invariant physical carriers. In this sense P7 provides a design principle for non-catastrophic forgetting in embodied agents:

$$\begin{aligned} &\text{do not preserve all details;} \\ &\text{preserve the quotient that preserves task identity.} \end{aligned} \tag{30}$$

A robot's balance manifold, a manipulation affordance, or a navigation invariant may be more valuable than a large local texture log. This is not a claim that robots implement NHSE. It is a functional projection of the P7

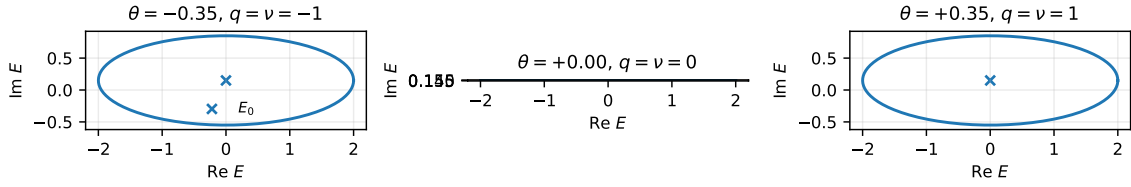


FIG. 1. Quotient-sector winding in the scalar asymmetric hopping model. The invariant side-ledger is the quotient-sector random variable $Q_{\text{inv}} = q(X) = \nu(E_0)$. Nonzero non-reciprocity encloses the reference point with opposite orientations, while the Hermitian point closes the model-class point gap.

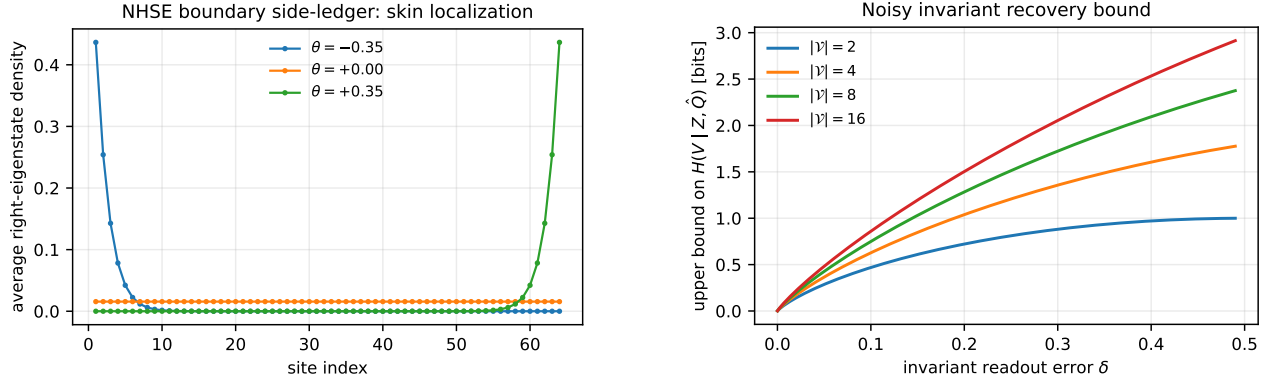


FIG. 2. Left: NHSE boundary diagnostic channel under OBC. Boundary localization can provide a physical carrier or diagnostic channel for the quotient sector under the model-class assumptions. Right: noisy invariant recovery bound. When the invariant readout error δ is small, residual task-sector uncertainty remains bounded even though local coordinates are coarse-grained.

principle: critical task identity should be moved from arbitrary complex records into law-like structural identity whenever possible.

RELATION TO EXISTING THEORY

P4 and O3

P4 supplies the baseline local-forgetting ledger. P7 supplies the protected complement. O3 supplies the entropy/resource accounting rule: protection does not make cost disappear, it moves the cost to boundary, drive, verification, maintenance, or externalization channels [2, 3].

Non-Hermitian topology

The NHSE literature provides the physical model class for boundary-sensitive spectral protection, point gaps, generalized Brillouin zones, and non-Bloch bulk-boundary correspondence [11, 12, 14, 16–18]. P7 does not replace that theory. It translates a subset of its structure into the FDS language of protected task sectors and accounting boundaries.

Model reduction and coarse-graining

Ordinary model reduction discards local degrees of freedom. P4 asks what inverse information has been lost. P7 asks whether a task sector factors through an invariant quotient. Invariant-aware coarse-graining is not less coarse; it is coarser in local coordinates while retaining the quotient that matters.

CONCLUSION

P7 defines the protected complement of P4 forgetting. P4 showed that capacity recovery is not distinction recovery: a local record that has discarded task-relevant preimages cannot reconstruct them without inverse information. P7 shows that not all inverse information must be stored as a local record. Some task sectors can remain recoverable through quotient invariants, topological side-ledgers, boundary-sensitive spectral structure, or other protected carriers.

This does not produce free memory. Topological protection does not delete entropy production; it relocates the accounting burden from local erasure to boundary, drive, verification, and protection maintenance, consis-

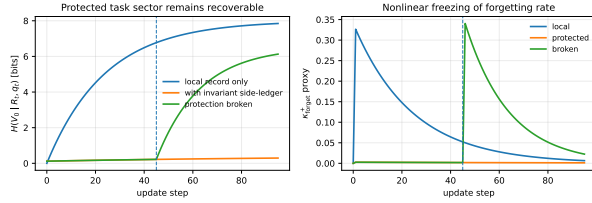


FIG. 3. Left: protected task sectors suppress the positive-part forgetting rate until the protection condition is broken. The plotted uncertainty trajectories are FDS normal-form simulations, not direct quantum-dynamical entropy estimates. Right: protection relocates resource burden from local bulk forgetting into boundary, drive/control, and verification channels. The y-axis is a resource/entropy proxy; thermodynamic interpretation requires reservoir and heat-channel calibration.

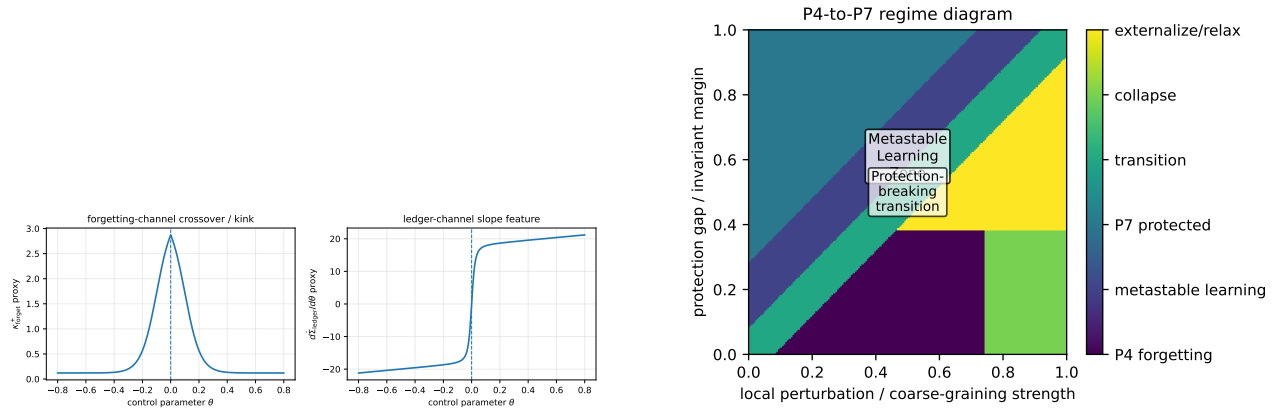


FIG. 4. Left: FDS dual-channel signature. In the ideal limit, the same control parameter can produce a feature in operational forgetting and a corresponding feature in the ledger channel. Finite devices may show rounded crossovers. Right: P4/P7 regime diagram. A metastable learning zone lies between ordinary local forgetting and stable P7 protection: the system pays update, verification, and maintenance costs to convert fragile local records into invariant quotients [4]. Protection fails when the margin closes or the carrier, boundary condition, or accounting access is destroyed.

tent with information thermodynamics and stochastic thermodynamics [7–10]. The physical mark of a topological transition is therefore the reorganization of forgetting cost, not the disappearance of cost. The NHSE model class provides a concrete setting where this reorganization can be modeled, simulated, and tested.

P7 defines the boundary between arbitrary complex records and law-like structural identity. What persists under finite local forgetting is not raw detail, but invariant-supported task structure. For Physical AI, the implication is direct: resource-constrained agents should not merely remember more; they should place critical task identity into stable physical, morphological, sensorimotor, or topological quotients that survive local degradation until a protection condition is broken.

At higher levels of FDS organization, symbolic lan-

guage may function as a collective invariant side-ledger: not a literal physical topology, but a quotient structure that preserves task identity across local memory turnover. This analogy is not used in the P7 proof, but it motivates later FDS applications to symbolic and civilizational memory.

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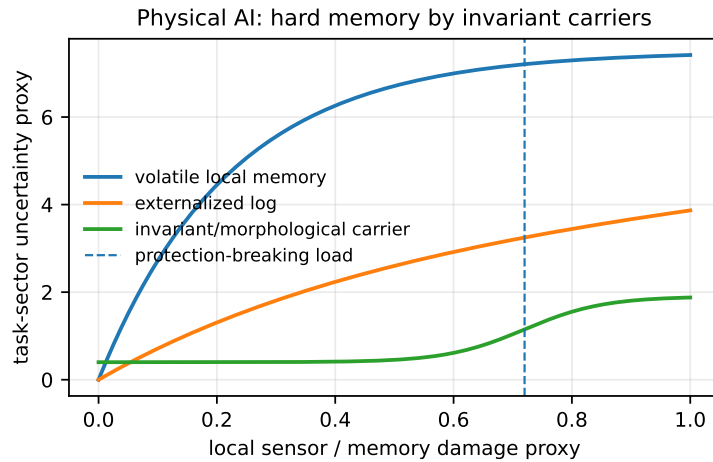


FIG. 5. Design implication for Physical AI. Critical task identity should not always be stored in volatile local memory. It can be carried by invariant, morphological, sensorimotor, or closed-loop physical structure. The figure is a normal-form design schematic, not a claim that robotics literally implements NHSE.

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Speed, Precision, and Dissipation Bounds for Boundary Maintenance in Finite Distinction Systems: Finite Update Windows, Bottleneck Throughput, and Resource-Ledger Exit Theorems

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FDS-P6 develops the speed-precision-dissipation paper in the physical bridge sequence of Finite Distinction Systems. P4 studied coarse-grained anti-recurrence: capacity recovery does not reconstruct distinctions erased by a non-injective bottleneck unless inverse information is carried by side records or an enlarged accounting boundary. O3 studied finite-memory record reuse as an operational entropy/resource channel. P7 studied the complementary case in which task identity is carried by invariant quotients or topological side-ledgers. P6 adds the missing dynamic axis: a boundary distinction must not only be representable or recoverable; it must be maintained within a finite update window, at finite precision, under finite resource input. We formalize boundary-maintenance throughput, bottleneck internal rate, precision demand, confidence cost, correction burden, effective causal reach, externalization relief, invariant-compression relief, and a resource-first dissipation ledger. The central result is a conditional exit theorem: if the rate-distortion demand required for boundary maintenance at tolerance ε over update window τ exceeds the sustainable internal update-throughput capacity, the system must enter at least one exit channel—higher resource/dissipation cost, increased error, latency growth, task relaxation, externalization, invariant compression, resource expansion, or boundary failure. The paper does not replace thermodynamic uncertainty relations, quantum speed limits, stochastic thermodynamics, information thermodynamics, rate-distortion theory, or control under communication constraints. It provides an FDS accounting bridge for real-time survival under finite capacity, physical record turnover, causal bandwidth, and boundary-maintenance loss. Deterministic numerical demonstrations illustrate speed-precision load surfaces, resource ledgers, error floors under finite power, effective causal reachability, externalization break-even, invariant-compression relief, regime diagrams, and a Physical-AI worked example normal form.

Reader contract. This paper is a registered FDS physical-bridge paper. It does not derive all physical speed limits from FDS alone. It does not claim that every update costs exactly $k_B T \ln 2$, nor that speed, precision, and dissipation obey a single universal scalar law in every system. Recent quantum many-body clock models show that precision can scale more favorably than simple linear entropy-cost intuitions suggest, reinforcing why P6 is formulated as an exit-channel accounting framework rather than a universal scalar precision-dissipation law [22]. In P6, “dissipation” is used in a broad physical-ledger sense unless a thermal reservoir calibration is explicitly supplied. The formal exit theorems are throughput/resource theorems; thermodynamic dissipation bounds arise only under additional physical calibration. P6 provides a quantitative audit standard for real-time survival in embodied finite systems. A distinction that cannot be updated, verified, corrected, or acted upon within the relevant update window is not operationally maintained.

Claim-status summary

Table I summarizes the main claims and demotion conditions.

INTRODUCTION

The missing dynamic axis

P4 established a recovery asymmetry: when a non-injective truncation removes task-relevant preimage information from the effective record, later capacity recovery does not reconstruct the erased distinction without side information [2]. P7 established the protected complement: some task identity can be carried by an invariant quotient $Q_{\text{inv}} = q(X)$ rather than by ordinary local memory [4]. O3 established that finite-memory record turnover enters a coupled entropy/resource ledger when physical records must be reused, refreshed, repaired, synchronized, externalized, or erased [3].

P6 adds the missing dynamic axis. A finite system may have enough capacity to represent a distinction and may even know which invariant quotient matters, yet fail if it cannot update, verify, correct, or act on that distinction fast enough and accurately enough to maintain its boundary. A boundary-maintaining system is not a static database. It is a finite physical process operating over an update window τ . The relation to existing theories (rate-distortion, thermodynamic uncertainty relations, quantum speed limits, control theory) is summarized in Table IV; P6 does not replace these theories but

TABLE I. Central FDS-P6 claims, status, and failure or demotion conditions.

Claim	Status	What would weaken or falsify it
Boundary maintenance requires finite update throughput.	Formal FDS claim	A time-varying boundary is maintained without updating, verifying, storing, externalizing, protecting, or acting on any task-relevant distinction.
Speed and precision jointly increase maintenance burden.	Operational bridge	Faster and more precise maintenance is sustained indefinitely at fixed representation and fixed resource input, with no extra dissipation, error, latency, externalization, invariant compression, or failure.
The sustainable internal rate is bottlenecked by sensing, updating, verification, correction, action, and resources.	Bottleneck definition	A system exceeds its slowest internal channel indefinitely without queueing, latency, loss, externalization, or resource expansion.
Correction and verification belong in the resource ledger.	O3-compatible physical bridge	Physical correction, refresh, verification, synchronization, overwrite, and recovery are cost-free under the stated implementation assumptions.
Effective causal update bandwidth limits real-time maintenance.	Physical/engineering bridge	A finite observer integrates arbitrarily distant boundary-relevant information within a finite update window with no latency, no prediction burden, and no effective signal-speed limit.
Externalization and invariant compression are relief channels, not free exits.	P4/P7-compatible bridge	External ledgers or invariant quotients reduce internal demand with no write, synchronization, verification, protection, latency, or boundary-accounting cost.

places them inside a boundary-maintenance accounting problem.

The physical bridge ladder

P6 is placed in the FDS physical bridge ladder as shown in Table II. Its role is not to replace P4, O3, or P7, but to bind them into a throughput constraint.

TABLE II. P6 in the local physical bridge ladder.

Paper	Core axis	P6 relation
P4	recoverability	lost preimages cannot be speed-recovered
O3	record turnover	update channels enter a ledger
P6	throughput	speed, precision, and cost constrain maintenance
P7	protection	quotients reduce maintained detail but not all cost

Speed, precision, and resource ledgers

Speed means how often the boundary-relevant state must be refreshed or acted upon. Precision means how tightly the maintained representation must approximate the task-relevant boundary statistic. Dissipation means the physical resource or entropy ledger associated with sensing, updating, verifying, correcting, synchronizing,

externalizing, protecting, and recovering that representation. The central P6 thesis is: finite boundary maintenance requires finite throughput, finite precision, and a finite ledger. At fixed resources, pushing speed and precision together forces an exit channel.

Why effective causal update bandwidth matters

A physically realized system cannot instantaneously integrate all boundary-relevant information. A finite maximum signal velocity implies a finite causal distinguishability-update bound. P6 uses the weaker operational statement: given an effective maximum propagation speed $v_{\max}$, real-time maintenance has a causal reach $R_H(\tau) = v_{\max}\tau$. In engineered or biological systems, $v_{\max}$ may be sensor, actuator, communication, diffusion, or transport limited rather than equal to the relativistic upper bound c .

FDS BACKGROUND

Active finite distinction systems

The FDS core object is

$$S = (X, E, B, M, Y, A, U, \pi, \ell, \Phi, \mathcal{P}, \tau), \quad (1)$$

where X is internal state, E environment, B boundary, M memory/model state, Y observation channel, A action space, U update rule, π finite projection, ℓ boundary-maintenance loss, Φ finite resource budget, $\mathcal{P}$ perturbation/pruning family, and τ update timescale [1]. P6 focuses on U , π , ℓ , Φ , Y , M , and τ .

Capacity demand and update windows

Let Ψ be a task family and let $R_{\min}^{(\tau)}(\varepsilon; \Psi)$ be the minimum number of bits over update window τ needed to encode a task-relevant statistic to tolerance ε . The capacity deficit is

$$\Delta_\varepsilon(\tau) = R_{\min}^{(\tau)}(\varepsilon; \Psi) - C_{\text{eff}}. \quad (2)$$

P6 turns this into a rate:

$$\mathcal{B}_{\varepsilon, \tau} := \frac{R_{\min}^{(\tau)}(\varepsilon; \Psi)}{\tau}. \quad (3)$$

A distinction that is representable in total storage may still be unmaintainable if $\mathcal{B}_{\varepsilon, \tau}$ exceeds the rate at which the system can sense, update, verify, correct, and act.

DEFINITIONS

Definition 1 (Boundary-maintenance task). *Let $Z_t = \psi(E_t, B_t)$ be a pre-registered task-relevant boundary statistic. A system maintains the task at tolerance ε over update window τ if*

$$\mathbb{E}[\ell(Z_t, \hat{Z}_t)] \leq \varepsilon \quad (4)$$

within the window and under the stated perturbation family.

Definition 2 (Maintenance speed). *For clocked systems, the maintenance speed is*

$$\nu_{\text{upd}} = \frac{1}{\tau}. \quad (5)$$

For event-driven systems, one may use

$$\nu_{\text{upd}}(t) = \frac{N_{\text{task}}(t, t + \tau)}{\tau}, \quad (6)$$

where N_{task} counts task-relevant update events.

Definition 3 (Precision and confidence). *Precision is encoded by tolerance ε and confidence $1 - \delta$:*

$$\mathbb{P}\{D(\hat{Z}_t, Z_t) \leq \varepsilon\} \geq 1 - \delta. \quad (7)$$

Smaller ε and smaller δ increase the required distinction throughput and verification burden.

Definition 4 (Boundary-maintenance throughput). *The throughput burden of maintaining task family Ψ at tolerance ε over update window τ is Eq. (3). It has units of task-relevant distinction-bits per unit time.*

Definition 5 (Correction burden). *Let $p_{\text{err}}(\tau, \varepsilon, \delta)$ be a boundary-relevant error probability and C_{repair} the expected repair or correction cost after such an error. The expected correction burden is*

$$\bar{C}_{\text{corr}} := \mathbb{E}[p_{\text{err}} C_{\text{repair}}]. \quad (8)$$

More detailed implementations may replace this by a stochastic thermodynamic, control, or recovery-cost model.

Definition 6 (Internal sustainable rate). *The internally sustainable maintenance rate is the bottleneck*

$$C_{\text{int}}^{\text{rate}} = \min\{C_{\text{sense}}^{\text{rate}}, C_{\text{update}}^{\text{rate}}, C_{\text{verify}}^{\text{rate}}, C_{\text{correct}}^{\text{rate}}, C_{\text{act}}^{\text{rate}}, C_{\text{resource}}^{\text{rate}}\}. \quad (9)$$

The resource-limited term may be represented in a model class as

$$C_{\text{resource}}^{\text{rate}} = \frac{\dot{\Phi}_{\text{in}}}{\kappa_{\varepsilon, \delta, \tau}}, \quad (10)$$

where $\kappa_{\varepsilon, \delta, \tau}$ is the implementation-dependent resource cost per maintained distinction-rate at the specified tolerance, confidence, and update window. This is not a universal constant. When the system faces a transient high-throughput demand, increasing $\dot{\Phi}_{\text{in}}$ alone may not prevent boundary collapse because the physical limit on the internal update rate $C_{\text{int}}^{\text{rate}}$ creates a nonlinear saturation: beyond a certain point, additional energy cannot be converted into higher update throughput due to bottleneck constraints on sensing, actuation, or verification speed.

Definition 7 (P6 resource ledger). *The P6 resource ledger is an audit partition*

$$\begin{aligned} \dot{\mathcal{R}}_{\text{P6}} = & \dot{\mathcal{R}}_{\text{sense}} + \dot{\mathcal{R}}_{\text{update}} + \dot{\mathcal{R}}_{\text{verify}} + \dot{\mathcal{R}}_{\text{correct}} \\ & + \dot{\mathcal{R}}_{\text{sync}} + \dot{\mathcal{R}}_{\text{externalize}} + \dot{\mathcal{R}}_{\text{protect}}. \end{aligned} \quad (11)$$

When thermodynamic calibration is available, a resource ledger may be converted into an entropy ledger, for example $\dot{\mathcal{R}}_{\text{P6}} \rightarrow T_{\text{eff}} \dot{\Sigma}_{\text{P6}}$ under appropriate reservoir assumptions. Without such calibration it remains a resource, power, latency, or cost ledger.

MAIN BOUNDS

Theorem 1 (Throughput deficit exit theorem). *Consider a physically realized active FDS maintaining a boundary task Ψ at tolerance ε over update window τ , with finite effective capacity C_{eff} , finite resource input $\dot{\Phi}_{\text{in}}$, and a fixed accounting boundary $\mathcal{A}$. If*

$$\frac{R_{\min}^{(\tau)}(\varepsilon; \Psi)}{\tau} > C_{\text{int}}^{\text{rate}}, \quad (12)$$

then sustained maintenance requires at least one exit channel: increased resource/dissipation ledger, increased error, increased latency, task relaxation, externalization, invariant compression, resource expansion, or boundary-maintenance failure. Recent work on maximum dissipation speed and fluctuation speed limits provides nearby model-class examples in which time, dissipation, and observable precision are explicitly constrained [19, 20].

Proof. The task requires $R_{\min}^{(\tau)}(\varepsilon; \Psi)$ task-relevant bits over each update window, hence a throughput burden $\mathcal{B}_{\varepsilon, \tau}$. By definition, $C_{\text{int}}^{\text{rate}}$ is the slowest internal channel among sensing, update, verification, correction, action, and resource-limited throughput. If $\mathcal{B}_{\varepsilon, \tau}$ exceeds this bottleneck, the missing throughput must be supplied by some other channel or the demand must be reduced. The possibilities are: pay additional resource cost, accept more error, increase latency by lengthening the effective update window, reduce the task demand, move records to an external ledger, replace raw distinctions by a lower-rate invariant quotient, expand resources, or fail the boundary task. If none occurs, the system lacks the throughput needed by the definition of $R_{\min}^{(\tau)}(\varepsilon; \Psi)$, contradicting sustained maintenance at tolerance ε . $\square$

Full P6 ledger demand

The full P6 demand includes rate-distortion throughput, confidence burden, correction burden, and causal prediction burden. Here $C_{\text{sust}}^{\text{rate}}$ denotes the sustainable maintained-throughput rate under the fixed accounting boundary, including the internal bottleneck and any baseline resource-supported maintenance capacity before relief channels are applied. In the minimal internal case, $C_{\text{sust}}^{\text{rate}} = C_{\text{int}}^{\text{rate}}$:

$$D_{\text{P6}} = \frac{R_{\min}^{(\tau)}(\varepsilon; \Psi)}{\tau} + \lambda_{\delta} \frac{\log_2(1/\delta)}{\tau} + \lambda_c \frac{\bar{C}_{\text{corr}}}{\tau} + D_{\text{causal}}. \quad (13)$$

The logarithmic confidence term is not asserted as a universal equality. It is a normal-form reliability cost motivated by coding, hypothesis testing, and repeated verification settings in which reducing failure probability from $O(1)$ to δ requires information or verification effort scaling at least logarithmically in $1/\delta$. Here D_{causal} denotes the rate-equivalent prediction, delay, or error burden induced when task-relevant boundary information lies outside the effective causal reach $R_H(\tau) = v_{\max}\tau$.

Theorem 2 (Full P6 ledger exit theorem). *Let $R_{\text{inv}}^{\text{relief}}$ and $R_{\text{ext}}^{\text{relief}}$ denote relief supplied by invariant compression and externalization after subtracting their coupled protection, synchronization, verification, retrieval, and latency ledgers. If*

$$D_{\text{P6}} > C_{\text{sust}}^{\text{rate}} + R_{\text{inv}}^{\text{relief}} + R_{\text{ext}}^{\text{relief}}, \quad (14)$$

then the same boundary-maintenance regime cannot persist without higher resource/dissipation ledger, higher error, higher latency, task relaxation, accounting-boundary expansion, invariant compression, resource expansion, or failure.

Proof. Equation (13) is the task-relevant demand rate for maintenance under the chosen tolerance, confidence, correction model, and causal reach. The right-hand side is the sustainable rate plus relief channels that actually reduce the maintained burden. If the demand exceeds this total, then either the ledger must increase, the representation/task must change, latency or error must grow, the accounting boundary must expand, or the boundary condition is not maintained. Otherwise the system would be maintaining a demand rate greater than the available rate without a source of the missing throughput. $\square$

Normal-form resource inequality

A useful model-class bridge is

$$\begin{aligned} \dot{\mathcal{R}}_{\text{P6}} \geq & \alpha \frac{R_{\min}^{(\tau)}(\varepsilon; \Psi)}{\tau} + \beta \frac{1}{\tau} \log_2 \frac{1}{\delta} + \gamma \frac{\bar{C}_{\text{corr}}}{\tau} \\ & - \mathcal{R}_{\text{inv}}^{\text{relief}} - \mathcal{R}_{\text{ext}}^{\text{local relief}} + \mathcal{R}_{\text{ext}}^{\text{coupled}} + \mathcal{R}_{\text{protect}}. \end{aligned} \quad (15)$$

Here α is a cost per maintained distinction-rate, β a confidence or verification cost, γ a correction coefficient, $\mathcal{R}_{\text{inv}}^{\text{relief}}$ is rate reduction from invariant compression, $\mathcal{R}_{\text{ext}}^{\text{local relief}}$ is local relief from externalization, $\mathcal{R}_{\text{ext}}^{\text{coupled}}$ is the coupled external ledger, and $\mathcal{R}_{\text{protect}}$ is the cost of maintaining a protected quotient. Equation (15) is a normal form, not a universal identity.

Corollary 1 (No zero-latency, zero-error, zero-cost limit). *At fixed finite capacity and fixed finite resource input, a finite active boundary cannot simultaneously take*

$$\tau \rightarrow 0, \quad \varepsilon \rightarrow 0, \quad \delta \rightarrow 0 \quad (16)$$

while maintaining the same task without an exit channel.

Corollary 2 (Latency-precision tradeoff under finite resources). *For a model class with sustainable rate $C_{\text{sust}}^{\text{rate}}$, if $R_{\min}(\varepsilon; \Psi) \sim d \log_2(1/\varepsilon)$ and confidence/correction terms are fixed, then the update window must satisfy approximately*

$$\tau \gtrsim \frac{d \log_2(1/\varepsilon)}{C_{\text{sust}}^{\text{rate}}}. \quad (17)$$

Thus, at fixed sustainable rate, tighter precision requires longer latency unless relief or extra resources are supplied.

PRECISION-SPEED TRADEOFF

For many regular task classes, the rate-distortion demand scales as

$$R_{\min}(\varepsilon; \Psi) \sim d \log_2 \frac{1}{\varepsilon}, \quad (18)$$

where d is an effective task dimension. Then

$$\mathcal{B}_{\varepsilon, \tau} \sim \frac{d}{\tau} \log_2 \frac{1}{\varepsilon}. \quad (19)$$

This captures the most basic P6 load: faster updates and tighter precision multiply. Confidence adds a verification term proportional to $\tau^{-1} \log_2(1/\delta)$ in many normal-form reliability settings.

Proposition 1 (Finite-resource precision floor). *Suppose a model class has a maintenance ledger lower bound*

$$\dot{\mathcal{R}} \geq a \frac{d}{\tau} \log_2 \frac{1}{\varepsilon} + b \frac{1}{\tau} \log_2 \frac{1}{\delta}, \quad (20)$$

with $a, b > 0$. If $\dot{\mathcal{R}} \leq \dot{\mathcal{R}}_{\max}$ and δ is fixed, then the achievable precision obeys

$$\varepsilon \geq 2^{-\frac{\tau}{ad}(\dot{\mathcal{R}}_{\max} - b\tau^{-1} \log_2(1/\delta))} \quad (21)$$

whenever the exponent is positive. Otherwise the requested speed-confidence combination is infeasible in that model class.

EFFECTIVE CAUSAL UPDATE BANDWIDTH

Finite causal reach

Given an effective maximum propagation or integration speed $v_{\max}$, boundary-relevant information from distance R cannot be integrated into the present update if $R > v_{\max}\tau$. Define the effective causal reach in one update window:

$$R_H(\tau) = v_{\max}\tau. \quad (22)$$

For relativistic causal bounds, $v_{\max} \leq c$ [17]. In engineered or biological systems, $v_{\max}$ may be sensor, actuator, communication, diffusion, transport, or neural-conduction limited. P6 requires only finite effective causal reach. Recent feedback-emulation results sharpen the data-rate perspective by asking not only whether a system can be stabilized, but what rate is needed to emulate a desired closed-loop behavior [23].

Maintenance reachability

Let $\Psi(A)$ be the set of task distinctions supported on a boundary region A . During update window τ , the maintainable task family is restricted to the causal reach:

$$\Psi_{\text{maintainable}}(\tau) \subseteq \Psi(A_{\text{reach}}(\tau)). \quad (23)$$

If a task requires boundary-relevant information outside $A_{\text{reach}}(\tau)$, the system must predict, externalize, delay, relax the task, or fail. This is not a claim about faster-than-light impossibility in general; it is a boundary-maintenance accounting statement.

LEDGER CHANNELS

Update and verification

An update channel carrying B_{upd} bits per window has a resource-rate proxy

$$\dot{\mathcal{R}}_{\text{update}} \sim \eta_{\text{upd}} \frac{B_{\text{upd}}}{\tau}. \quad (24)$$

A verification channel enforcing failure probability δ carries

$$\dot{\mathcal{R}}_{\text{verify}} \sim \eta_{\text{verify}} \frac{\log_2(1/\delta)}{\tau}. \quad (25)$$

These proxies are not universal thermodynamic laws. They are audit terms to be replaced by implementation-specific measurement when available.

Correction

If errors occur with probability p_{err} per window and the mean correction cost is C_{repair} , then

$$\dot{\mathcal{R}}_{\text{correct}} \sim \frac{p_{\text{err}} C_{\text{repair}}}{\tau}. \quad (26)$$

Thermodynamic uncertainty relations and speed-precision tradeoffs in stochastic thermodynamics provide mature model classes in which precision, current fluctuations, speed, and dissipation are formally related [12–14]. Kinetic uncertainty relations further show that response precision can be bounded by dynamical activity in Markovian nonequilibrium systems, supporting the P6 treatment of verification and correction as ledger channels [21].

Externalization

Externalization can reduce the internal fast-update load, but it creates a coupled ledger:

$$\begin{aligned} \dot{\mathcal{R}}_{\text{ext}} = & \dot{\mathcal{R}}_{\text{write}} + \dot{\mathcal{R}}_{\text{sync}} + \dot{\mathcal{R}}_{\text{retrieve}} \\ & + \dot{\mathcal{R}}_{\text{verify}} + \dot{\mathcal{R}}_{\text{protect}} + \dot{\mathcal{R}}_{\text{latency}}. \end{aligned} \quad (27)$$

This is the dynamic analogue of the P4 side-record audit: exact recovery or high-speed maintenance can be shifted to an external log, but not made free [2].

Invariant compression

If task identity factors through an invariant quotient,

$$Q_{\text{inv}} = q(X), \quad V = h(Q_{\text{inv}}), \quad (28)$$

then the maintained rate can drop from a raw representation rate to a quotient rate:

$$\Delta R_{\text{inv}} = R_{\text{raw}}(\varepsilon) - R_q(\varepsilon). \quad (29)$$

P7 emphasizes that the protection ledger must still be counted: topological or invariant persistence relocates cost to boundary, drive, verification, or protection maintenance rather than deleting it [4].

WORKED EXAMPLE: HIGH-SPEED EMBODIED MAINTENANCE

Consider a mobile embodied agent maintaining a collision-avoidance boundary. The task statistic Z_t encodes nearby obstacle position and relative velocity, the update window τ is the control period, and ε is the maximum admissible tracking or clearance error. A remote high-latency model controller (including cloud LLM-style controllers) may have high semantic precision but high latency, communication cost, and verification overhead. A local reflex or morphology-based controller may have lower semantic resolution but much smaller τ , lower ledger cost, and direct actuation. P6 predicts that, under tight physical update windows, the second system can dominate survival even if its representation is less detailed.

In P6 language, the cloud controller pays a large D_{causal} and verification/externalization ledger, while the local controller reduces τ and shifts task identity into local feedback, morphology, or invariant carriers. For physical survival tasks, lower representational precision can dominate when it reduces latency and ledger cost. This is not an argument against large models. It is a statement that boundary maintenance is constrained by throughput, not capacity alone.

NUMERICAL DEMONSTRATIONS

The simulations are deterministic normal-form demonstrations generated by `code/generate_results.py`. They illustrate definitions and exit channels rather than fitting empirical data. The regime diagram is a synthetic classification map generated from threshold rules specified in the reproducibility code. It illustrates exit-channel ordering and should not be read as an empirical phase diagram.

TABLE III. Core demonstration parameters. Full parameter values are exported to `data/simulation_parameters.csv`.

Parameter	Value / role
d	8, task-dimension proxy
δ	10^{-3} , confidence target
Resource thresholds	120, 220, 360 in normalized units
v_{max}	1.0, 0.25, 0.06 normalized reach speeds
Energy calibration	optional Landauer floor at 300 K

EXPERIMENTAL AND ENGINEERING PROTOCOLS

Protocol 1 (Finite update-window stress test). *Hold task demand and precision target fixed while reducing τ . Measure error rate, latency, power, reset/overwrite rate, correction count, externalization count, and boundary loss. P6 predicts ledger increase, error increase, latency, externalization, invariant compression, task relaxation, or failure.*

Protocol 2 (Precision tightening test). *Hold update window fixed while lowering ε and δ . Measure verification, correction, resource cost, and latency. P6 predicts increased throughput burden and a finite-resource error floor.*

Protocol 3 (Invariant-compression test). *Compare raw high-dimensional representation, externalized record, and quotient/invariant representation. Measure task loss, ledger cost, update speed, and robustness under perturbation. A successful quotient reduces rate burden without exceeding the protection ledger.*

Protocol 4 (Physical AI maintenance benchmark). *For an embodied agent, stress moving-target tracking, balance control, navigation, manipulation under sensor damage, or long-horizon memory-constrained action. Report $\mathcal{B}_{\varepsilon, \tau}$, ledger cost, boundary loss, error rate, recovery time, and externalization/protection channels.*

FALSIFICATION AND DEMOTION CONDITIONS

P6 is weakened if a finite physically realized system maintains arbitrarily fast and arbitrarily precise boundary tasks at fixed capacity and fixed resource input with no increase in resource ledger, error, latency, externalization, invariant compression, task relaxation, or hidden accounting-boundary expansion. A model-class inequality such as Eq. (15) can fail without falsifying the FDS exit theorem; it should then be demoted to a narrow toy model. If hidden external records or uncounted resources are discovered, the accounting boundary must be expanded. If a protected quotient carries the task iden-

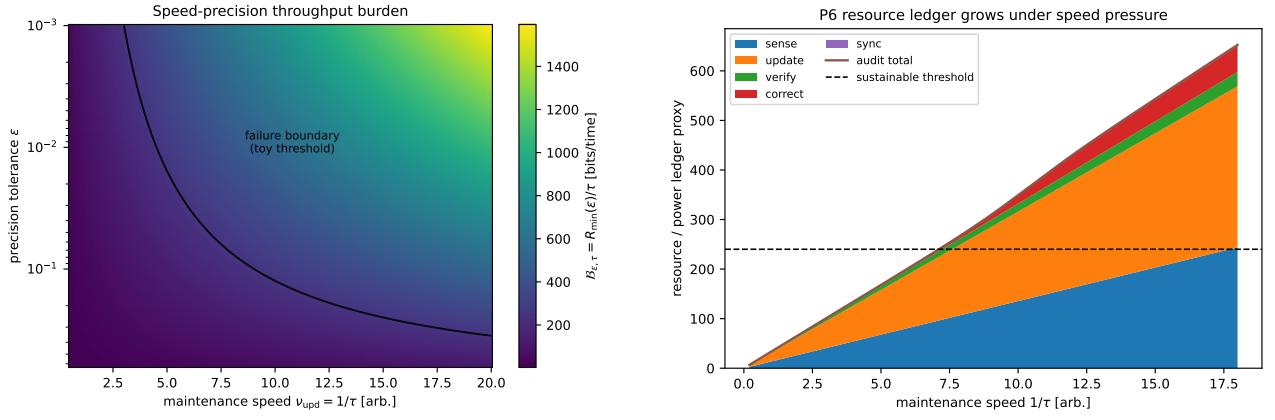


FIG. 1. Left: speed-precision throughput burden $\mathcal{B}_{\varepsilon, \tau} = R_{\min}(\varepsilon)/\tau$ for a task dimension proxy $d = 8$. The contour is a toy failure boundary. Right: P6 resource ledger decomposition under increasing speed pressure. Channels are illustrative audit classes, not universal thermodynamic mechanisms.

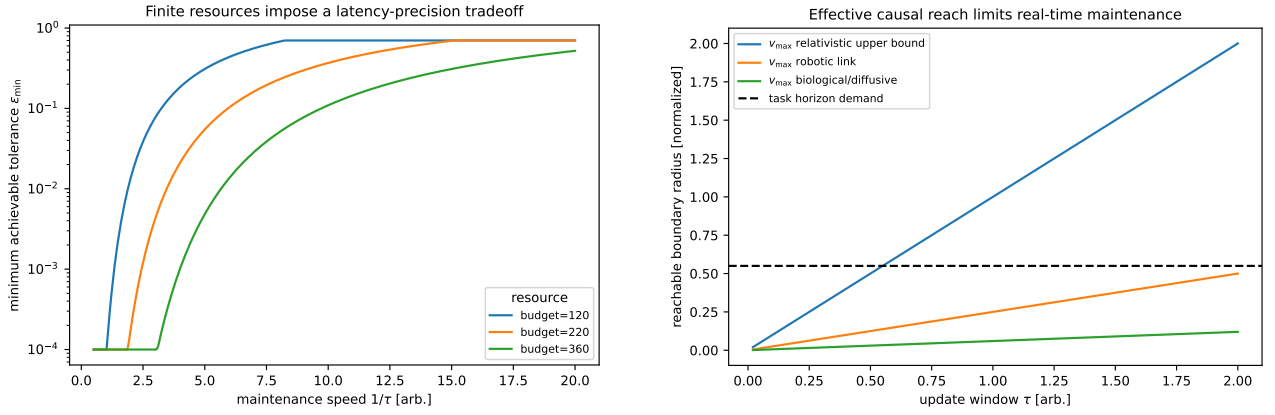


FIG. 2. Left: finite resources impose a speed-dependent precision floor. At high update speed, the same resource budget cannot support arbitrarily small error tolerance. Right: effective causal reach $R_H = v_{\max}\tau$ shrinks with the update window. Boundary-relevant information outside the reach must be predicted, externalized, delayed, or abandoned. The protection-breaking load occurs where the causal horizon can no longer cover the boundary region requiring maintenance.

tity, the case belongs to the P7 relief channel rather than being a counterexample to P6.

RELATION TO EXISTING THEORY

P6 uses mature theories as implementation bridges rather than replacing them: P6 does not replace any of these theories. It packages their common operational implication inside FDS: boundary maintenance is throughput-constrained.

GLOSSARY

Term	P6 meaning
Speed	update frequency $1/\tau$
Precision	distortion tolerance ε
Confidence	success probability $1 - \delta$
Throughput	task-relevant distinction rate $\mathcal{B}_{\varepsilon, \tau}$
Ledger	audited resource or entropy-cost channels
Exit channel	cost, error, latency, externalization, quotient, relaxation, or failure

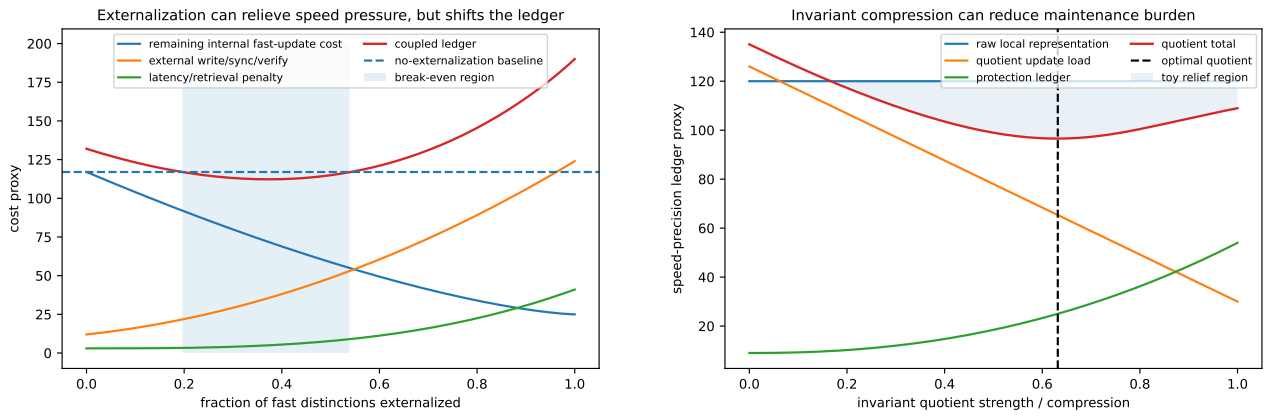


FIG. 3. Left: externalization can relieve internal speed pressure but shifts cost into writing, synchronization, verification, retrieval, latency, and protection. Right: invariant compression can reduce update burden when task identity factors through a quotient, but the protection ledger and task-loss penalty create an optimum rather than monotonic improvement. The exported CSV includes a toy Landauer-calibrated power floor at 300 K; it is not used as a universal heat claim.

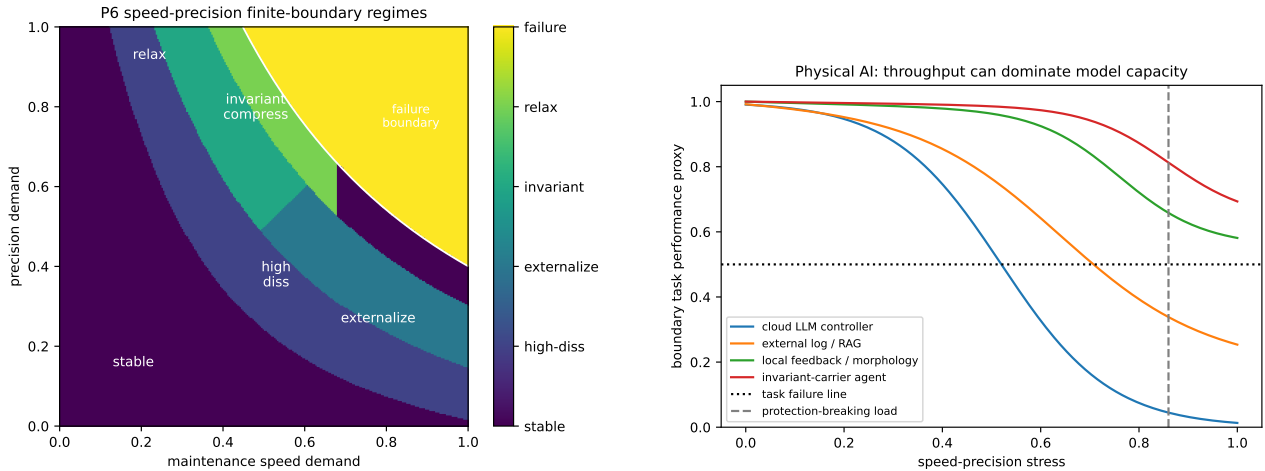


FIG. 4. Left: P6 speed-precision finite-boundary regimes. Increasing speed and precision moves a finite system from stable maintenance into high dissipation, externalization, invariant compression, relaxation, or failure. Region boundaries are synthetic threshold rules. Right: Physical-AI worked example normal form. Local feedback and invariant carriers can outperform a remote high-latency model controller under high speed-precision stress because they reduce latency and ledger cost.

DESIGN IMPLICATION: BOUNDARY MAINTENANCE UNDER SPEED-PRECISION PRESSURE

This section is not part of the formal proof. It extracts the FDS design rule implied by P6. A physical AI agent is not limited only by model size. It is limited by how fast, how precisely, and how cheaply it can maintain task-relevant boundaries. The design rule is: do not maximize memory alone; optimize the speed-precision-dissipation ledger. High-speed tasks should use invariants, reflexive control loops, morphology, and local closed-loop policies. Slow exact recovery can use external logs. When causal update is delayed, predictive control is required. When

failure cost is high, verification should increase. When speed matters more than detail, precision should relax. P6 thereby provides a benchmark language for Physical AI: report maintenance throughput, not only accuracy.

CONCLUSION

P4 showed that lost distinctions do not return merely because capacity returns. P7 showed that some task identity can survive local forgetting through invariant quotients. O3 showed that finite record reuse enters an entropy/resource ledger. P6 completes this portion of the physical spine by adding speed. A finite system must

TABLE IV. P6 relation to existing theories.

Existing theory	What it bounds	P6 use
Rate-distortion theory	bits needed for precision	task demand $R_{\min}(\varepsilon)$ [5–7]
Information thermodynamics	erasure, measurement, feedback, memory	physical ledger terms [8–11]
Thermodynamic uncertainty relations	precision versus dissipation	stochastic implementation bridge [12–14]
Quantum speed limits	minimum transformation time	physical update-time constraints [15, 16]
Data-rate/control theory	feedback control under channel limits	communication-limited boundary maintenance [18,
Finite-time dissipation speed limits	time vs entropy/resource rate	P6 speed-cost bridge [19, 20]
Response / kinetic uncertainty relations	activity vs response precision	verification/correction burden [21]
FDS-P6	finite boundary update throughput	exit-channel accounting under finite capacity

maintain its boundary not only with enough representation, but within a finite update window, at finite precision, and under finite resource budgets.

P6 identifies the dynamic cost of being finite. Pushing speed and precision together forces dissipation, error, latency, externalization, invariant compression, task relaxation, or failure. Boundary maintenance is therefore not a static property of representation. It is a throughput-constrained physical process. Throughout P6, “dissipation” is used as generalized resource turnover cost; thermodynamic dissipation is a special case that arises when thermal reservoir calibration is available. For Physical AI, this means that durable agency is not simply larger memory or larger models. It is the ability to preserve task-relevant boundaries under the speed, precision, and dissipation constraints of the world.

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Three Fermion Generations as CP/T-Asymmetric Identity Transformation: The CKM Lower Bound, Irreversible Pruning, and the Minimal Flavor Architecture of Finite Distinction Systems

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The Standard Model contains three generations of chiral fermions with repeated gauge quantum numbers and different masses. Chiral gauge assignments are essential: before electroweak symmetry breaking they forbid bare gauge-invariant Dirac mass terms, so sequential generations appear as protected weak-flavor distinction sectors rather than vector-like pairs. The Standard Model takes this number as empirical input. X2 studies a narrower algebraic question: what dimensionality is required for a weak-sector identity-changing channel to carry a rephasing-invariant CP/T orientation? In a CKM-type $N \times N$ unitary charged-current mixing matrix, the number of physical complex phases that cannot be removed by quark-field rephasings is $(N-1)(N-2)/2$. Hence $N=1$ and $N=2$ cannot support an irreducible CKM CP phase, while $N=3$ is the first possible case. X2 places this Kobayashi-Maskawa fact inside the FDS/DT physical bridge chain

identity update $\rightarrow$ T/CP -orientation $\rightarrow$ irreducible CKM phase $\rightarrow N_{\text{CKM}} \geq 3$.

The hard result is a conditional lower bound: if weak-sector identity transformation requires a microscopic CP/T-asymmetric carrier, and if the Standard Model charged-current sector is that carrier, then the CKM-participating flavor dimension must satisfy $N_{\text{CKM}} \geq 3$. This is a necessary dimensional lower bound, not a sufficient derivation of observed CP violation; nonzero mixing, nondegenerate masses, and a nonzero Jarlskog invariant are additional physical requirements. The stronger claim that the observed theory realizes exactly three sequential chiral generations, rather than more, is separated as a higher-risk finite-system minimality bridge. Deterministic algebraic demonstrations illustrate CKM phase counting, quark rephasing, rephasing-invariant orientation, hard/soft claim separation, and a toy flavor-cost functional.

Reader contract. X2 is a conditional physical-bridge paper. It does not derive the Standard Model gauge group, the fermion mass spectrum, Yukawa couplings, CKM angles, the numerical CKM phase, neutrino masses, or the observed baryon asymmetry. It does not claim that all irreversibility requires CKM CP violation. The CP/T-asymmetry bridge applies to microscopic identity-changing weak-sector transformations, not to all macroscopic thermodynamic irreversibility, Landauer erasure, decoherence, or open-system dissipation. The CKM phase-counting result supplies a necessary dimensional lower bound, not a sufficient derivation of observed CP violation. The hard claim is $N_{\text{CKM}} \geq 3$ for CKM-participating weak flavor. The claim that the realized sequential chiral generation count is exactly three is a separate minimality bridge with a different epistemic status.

Claim-status summary

Table I separates the algebraic lower bound from the FDS/DT bridge and the higher-risk upper-bound claim.

Keywords

Finite Distinction Systems; Distinction Theory; fermion generations; CKM matrix; CP violation; T violation; CPT theorem; Kobayashi-Maskawa phase; Jarlskog invariant; identity transformation; weak interaction; flavor physics; PMNS matrix; fourth generation; Standard Model.

INTRODUCTION

The flavor-dimensionality problem

The Standard Model contains three repeated families of chiral fermions. In the quark sector these are

$$(u, d), \quad (c, s), \quad (t, b), \quad (1)$$

with corresponding charged leptons and neutrinos. The gauge quantum numbers repeat, while masses and mixings differ. The Standard Model describes these patterns with Yukawa matrices and charged-current mixing, but it does not derive why the weak flavor architecture has this dimensionality rather than one, two, four, or more [7].

X2 focuses on one sharp subproblem. A CKM-type weak charged-current sector can carry a physical CP-violating phase only when its flavor dimension is at least

TABLE I. Central X2 claims, status, and failure or demotion conditions.

Claim	Status	What would weaken or falsify it
CKM-type CP violation requires $N_{\text{CKM}} \geq 3$.	Standard Model algebra / hard hook	A two-generation CKM-like unitary charged-current theory with a physical irreducible CP phase.
Weak-sector identity transformation requires a rephasing-invariant CP/T orientation.	FDS/DT physical bridge	A complete physical account of the audited microscopic weak identity-changing channel with no CP/T orientation, no compensating asymmetry, and no hidden orientation variable.
The weak charged current is the Standard Model identity-transformation carrier.	Interpretive bridge	Identity-changing weak-sector operations are realized by a different physical carrier outside the CKM charged current.
$N_{\text{CKM}} \geq 3$ follows from the X2 chain.	Conditional theorem	Failure of any premise: the CP/T-orientation bridge, CPT assumptions, the weak-carrier mapping, or CKM phase counting.
Exactly three sequential chiral generations follow from minimality.	Higher-risk upper-bound bridge	Discovery of a fourth sequential chiral generation with Standard Model gauge charges and weak charged-current participation, or a successful $N > 3$ flavor architecture that reduces the relevant cost functional.
Nonzero leptonic Dirac CP phase.	Optional PMNS extension	High-precision oscillation data establish $\delta_{\text{CP}} = 0$ or π under assumptions requiring leptonic CP/T-oriented identity transformation.

three [5, 6]. The new claim is not that this algebra is novel. The new claim is that FDS/DT supplies a reason why a CP/T-oriented identity-changing weak channel is required.

The algebraic hook

For N generations, the number of physical complex phases in a CKM-like unitary mixing matrix is

$$N_{\text{CP}}(N) = \frac{(N-1)(N-2)}{2}. \quad (2)$$

Thus $N = 1$ and $N = 2$ have no physical CKM CP phase, while $N = 3$ has one. This is the hard algebraic hook. X2 builds the conditional chain

weak identity transformation requires CP/T orientation $\Rightarrow$ irreducible CKM phase $\Rightarrow N_{\text{CKM}} \geq 3$ (3)
 Equivalently, the charged-current interaction is

The variable N_{CKM} denotes the CKM-participating weak flavor dimension. In the Standard Model, N_{CKM} coincides with the number of sequential chiral quark generations.

Hard lower bound versus upper-bound bridge

The lower bound $N_{\text{CKM}} \geq 3$ is algebraic once the bridge assumptions are accepted. The stronger statement that the realized sequential chiral generation count is exactly three is not the same kind of result. Since $N > 3$ also has physical CP phases, CKM algebra alone does not forbid more generations. The upper-bound claim

must therefore be formulated as a finite-system minimality bridge: $N = 3$ is the first sufficient point, while $N > 3$ adds overhead that must justify itself.

FDS/DT BACKGROUND

Weak identity transformation

Let $\mathcal{H}_{\text{id}}$ denote the Hilbert subspace spanned by weak flavor identity labels. In the Standard Model charged-current sector, a down-type mass eigenstate can transform into a superposition of up-type mass eigenstates through the CKM matrix:

$$\mathcal{T}_{\text{id}} : |d_j\rangle \mapsto \sum_i V_{ij} |u_i\rangle. \quad (4)$$

$$J_{\text{cc}}^\mu = \bar{u}_i \gamma^\mu (1 - \gamma^5) V_{ij} d_j. \quad (5)$$

X2 calls Eq. (4) a weak identity transformation: it changes flavor identity sector while preserving a physical transition history. This is not a derivation of the weak interaction. It identifies the existing weak charged-current sector as the object-level Standard Model carrier of identity-changing transformations.

CP/T-oriented identity transformation

A weak identity transformation is nontrivial when $\mathcal{T}_{\text{id}}$ is not simultaneously diagonal in the relevant mass ba-

sis. It is CP/T-oriented when it contains rephasing-invariant orientation data. The standard invariant test is a Jarlskog-type quartet

$$J_{ijkl} = \text{Im} (V_{ij} V_{kl} V_{il}^* V_{kj}^*). \quad (6)$$

Observable CKM CP violation additionally requires nonzero mixing and nondegenerate quark masses. Thus phase counting is necessary, not sufficient. It tells us when a physical orientation channel can exist; it does not by itself derive the measured CP-violating observables.

Definition 1 (CP/T-oriented weak identity channel). *A CKM-type weak identity channel is CP/T-oriented when there exists at least one rephasing-invariant quartet with $J_{ijkl} \neq 0$, under a nondegenerate mass structure and nonzero mixing configuration.*

Why ordinary dissipative irreversibility is not enough for X2

X2 does not claim that all irreversibility requires CP/T violation. Macroscopic thermodynamic irreversibility, Landauer erasure, decoherence, and open-system dissipation can occur without CKM CP violation. Those processes can obtain an arrow from coarse-graining, environmental boundary conditions, low-entropy initial conditions, or reservoir structure.

The X2 bridge is narrower. It concerns an intrinsic microscopic orientation in the weak-sector identity-changing channel itself. A reversible flavor transformation can mix labels; a CP/T-oriented identity transformation carries rephasing-invariant data that distinguishes a transformation history from its CP/T-reversed counterpart. A removable phase is a convention. A nonzero Jarlskog invariant is physical orientation.

Assumption 1 (Weak-sector CP/T orientation bridge). *For the audited microscopic weak identity-changing channel, irreversible identity transformation requires at least one rephasing-invariant CP/T orientation. This assumption is not a theorem of Standard Model algebra. It is the FDS/DT physical bridge that connects finite-system identity update to the Standard Model CKM algebra.*

Why X2 precedes the four-force mapping

X2 does not need a complete functional taxonomy of all four forces. It only uses the weak charged current as the Standard Model sector in which flavor identity transformations occur. A later four-force paper may use X2 as input: X2 establishes why the weak identity-transformation sector must have enough flavor dimensionality to carry a CP/T orientation.

TABLE II. CKM-type phase counting for N generations.

N	mixing angles	physical CP phases
1	0	0
2	1	0
3	3	1
4	6	3
5	10	6

CKM PHASE COUNTING

Unitary mixing and rephasing

Let $V \in U(N)$ be an $N \times N$ unitary charged-current mixing matrix. A general $U(N)$ matrix has N^2 real parameters. These may be counted as

$$N_{\text{ang}} = \frac{N(N-1)}{2} \quad (7)$$

real mixing angles and

$$N_{\text{phase,raw}} = \frac{N(N+1)}{2} \quad (8)$$

raw phases.

Quark fields can be redefined by

$$u_i \rightarrow e^{i\alpha_i} u_i, \quad d_j \rightarrow e^{i\beta_j} d_j. \quad (9)$$

There are $2N$ such field phases, but one common overall phase is unobservable. Hence $2N - 1$ phases are removable. The number of physical CP phases is therefore

$$N_{\text{CP}}(N) = \frac{N(N+1)}{2} - (2N-1) \quad (10)$$

$$= \frac{N^2 - 3N + 2}{2} \quad (11)$$

$$= \frac{(N-1)(N-2)}{2}. \quad (12)$$

Theorem 1 (CKM CP-phase lower bound). *For a CKM-type $N \times N$ unitary charged-current mixing matrix, an irreducible physical complex phase exists if and only if $N \geq 3$.*

Proof. By Eq. (12), the number of physical complex phases is $(N-1)(N-2)/2$. This number is strictly positive if and only if $N \geq 3$. $\square$

Remark 1 (Necessary, not sufficient). *A positive phase count only means that CP/T orientation is algebraically possible. Observable CKM CP violation requires nonzero mixing angles, nondegenerate masses, and a nonzero rephasing-invariant quantity such as Eq. (6). The phase-counting theorem is a dimensional lower bound, not a complete derivation of CKM phenomenology.*

MAIN X2 THEOREM

Theorem 2 (Three-flavor lower bound for CP/T-oriented weak identity transformation). *Assume: (i) the audited weak identity-changing channel requires rephasing-invariant CP/T orientation; (ii) the Standard Model charged-current CKM structure is the relevant carrier; (iii) CPT holds, so microscopic T asymmetry is represented as CP asymmetry under the usual local Lorentz-invariant QFT assumptions; and (iv) CP asymmetry in this sector requires an irreducible CKM phase. Then the CKM-participating weak flavor dimension satisfies*

$$N_{\text{CKM}} \geq 3. \quad (13)$$

In the Standard Model, this lower bound is the lower bound on sequential chiral quark generations participating in the CKM charged current.

Proof. Assumption (i) requires a rephasing-invariant CP/T orientation. Assumption (ii) locates this orientation in the CKM charged-current mixing matrix. Assumption (iii) permits the usual bridge from T asymmetry to CP asymmetry under CPT [8, 9]. Assumption (iv) requires a physical irreducible CKM phase. By the CKM CP-phase lower-bound theorem, such a phase exists only when $N_{\text{CKM}} \geq 3$. $\square$

The theorem should be read exactly as a conditional lower bound. It does not prove the Standard Model gauge group, the measured values of masses, or the measured values of the CKM matrix. It proves that if weak identity transformation needs CKM-type CP/T orientation, two generations are not enough.

EXACTLY THREE? THE MINIMALITY BRIDGE

Algebraic permissibility of $N > 3$

CKM algebra permits $N > 3$. For $N = 4$, Eq. (12) gives three physical CP phases. Therefore the step from $N_{\text{CKM}} \geq 3$ to $N_{\text{CKM}} = 3$ cannot be another CKM phase-counting theorem.

Normal-form flavor cost

The upper-bound bridge may be represented by a normal-form cost functional

$$\mathcal{C}_{\text{flavor}}(N) = \mathcal{C}_{\text{fields}}(N) + \mathcal{C}_{\text{Yukawa}}(N) + \mathcal{C}_{\text{CP}}(N) + \mathcal{C}_{\text{stability}}(N) - \mathcal{R}_{\text{identity}}(N). \quad (14)$$

Equation (14) is a normal-form audit functional, not a fitted phenomenological model or a derived Standard

Model effective potential. Here $\mathcal{C}_{\text{fields}}$ represents additional chiral multiplets and anomaly/accounting burden; $\mathcal{C}_{\text{Yukawa}}$ represents additional Yukawa parameters and flavor-alignment burden; $\mathcal{C}_{\text{CP}}$ represents additional irreducible CP phases requiring calibration or stabilization; $\mathcal{C}_{\text{stability}}$ represents precision-electroweak, Higgs, flavor-changing-neutral-current, and cosmological constraints; and $\mathcal{R}_{\text{identity}}$ represents the functional benefit of crossing the first irreducible CP/T-orientation threshold. Each cost term grows nonlinearly with N : for the quark flavor sector alone, the number of physical masses, mixing angles, and CP phases scales as

$$2N + \frac{N(N-1)}{2} + \frac{(N-1)(N-2)}{2} = N^2 + 1,$$

so the jump from $N = 3$ to $N = 4$ increases the quark flavor parameter count from 10 to 17.

The minimality bridge is the conjecture that

$$N_{\text{min}} = \arg \min_{N \geq 3} \mathcal{C}_{\text{flavor}}(N). \quad (15)$$

If the functional benefit saturates after the first CP/T orientation and the costs grow with additional sequential chiral generations, the normal-form minimum occurs at $N = 3$. This minimality logic is adjacent to the P6 speed-resource ledger and the P7 invariant-protection framework, but neither is required for the CKM lower-bound theorem [3, 4].

Conjecture 1 (Minimal sufficient weak-flavor architecture). *Among CKM-participating sequential chiral flavor architectures satisfying the CP/T-orientation lower bound, the realized architecture minimizes a finite-system flavor cost of the form Eq. (14). Under monotone overhead beyond the first sufficient CP/T channel, the minimum occurs at $N_{\text{CKM}} = 3$.*

Fourth-generation caveat

The upper-bound bridge is challenged only by a sequential chiral generation participating in the same CKM-like weak charged-current architecture. Vector-like fermions, sterile states, or hidden-sector states may extend phenomenology without counting as $N_{\text{CKM}} > 3$ in the X2 minimality claim. This distinction is important because vector-like quark searches and sequential fourth-generation models are different physical cases [14–17].

LEPTON-SECTOR EXTENSION

For Dirac neutrinos with an $N \times N$ PMNS matrix, the same phase-counting logic gives

$$N_{\text{CP}}^{\text{PMNS}} = \frac{(N-1)(N-2)}{2}. \quad (16)$$

If neutrinos are Majorana particles, additional Majorana phases appear, but they do not affect oscillation CP violation in the same way as the Dirac phase.

Proposition 1 (PMNS extension, conditional). *If the lepton sector participates in the same CP/T-oriented identity-transformation requirement, and if the relevant observable channel is the Dirac PMNS phase, then X2 motivates a nonzero leptonic Dirac CP phase under the stated assumptions.*

This is an extension, not the core theorem. Current global fits show that the determination of δ_{CP} remains ordering-dependent and not yet a settled nonzero measurement, so the PMNS statement is treated as an extension rather than part of the CKM lower-bound theorem [23]. Current and future long-baseline experiments such as T2K, NOvA, DUNE, and Hyper-Kamiokande test leptonic CP structure with increasing sensitivity [18–22]. If high-precision data establish $\delta_{\text{CP}} = 0$ or π , the lepton-sector extension would be demoted. The CKM lower-bound theorem would remain intact.

WHAT X2 DOES NOT EXPLAIN

X2 explains only the minimal dimensional requirement for a CKM-type irreducible CP/T orientation. It does not explain:

1. why the gauge group is $SU(3) \times SU(2) \times U(1)$;
2. why fermion masses have their observed hierarchy;
3. why CKM angles have their measured values;
4. why the CKM phase has its numerical value;
5. why neutrino mixing is large;
6. why the baryon asymmetry has its observed magnitude (this requires additional Sakharov-condition physics [13]);
7. why no beyond-Standard-Model fermions have been observed;
8. why the full Yukawa sector has its detailed texture.

These are not defects of the lower-bound theorem. They are open problems outside the scope of X2.

ALGEBRAIC DEMONSTRATIONS

The figures are deterministic algebraic or normal-form demonstrations generated by `code/generate_results.py`. They are not empirical fits.

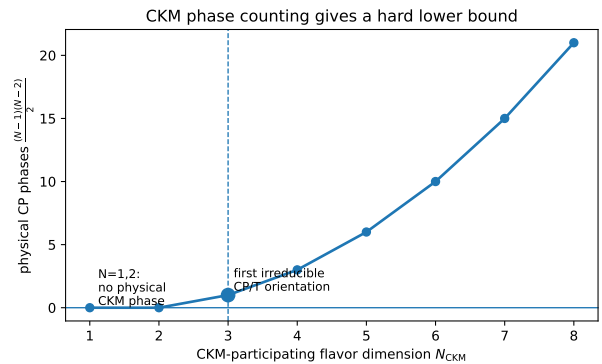


FIG. 1. Physical CKM-type CP phases as a function of CKM-participating flavor dimension. $N = 3$ is the first nonzero point. This is the hard algebraic hook of X2.

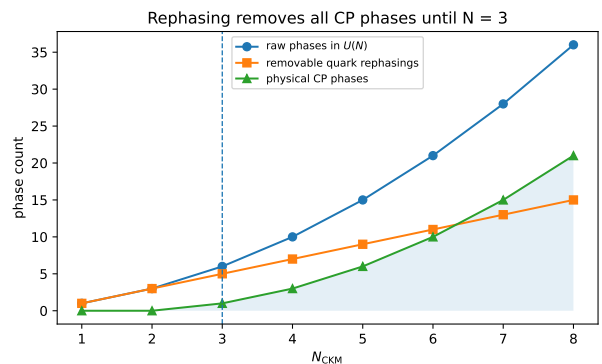


FIG. 2. Phase bookkeeping for an $N \times N$ unitary mixing matrix. Raw unitary phases grow as $N(N+1)/2$, but $2N-1$ quark-field rephasings are removable. The physical remainder is $(N-1)(N-2)/2$.

FALSIFICATION AND DEMOTION CONDITIONS

Lower-bound demotion

The lower-bound chain is weakened if any of the following occurs:

1. a CKM-like two-generation weak sector is shown to possess a physical irreducible CP phase;
2. the audited weak-sector identity-changing channel is fully realized without any CP/T orientation or compensating orientation variable;
3. the relevant identity-transformation carrier is not the CKM charged-current sector;
4. CPT assumptions fail in the relevant theory.

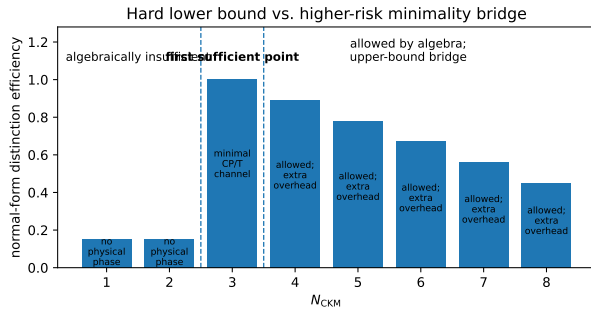


FIG. 3. Hard lower bound vs. higher-risk minimality bridge. $N = 1, 2$ cannot host an irreducible CKM CP phase. $N = 3$ is the first sufficient point. $N > 3$ is algebraically allowed but belongs to the upper-bound bridge.

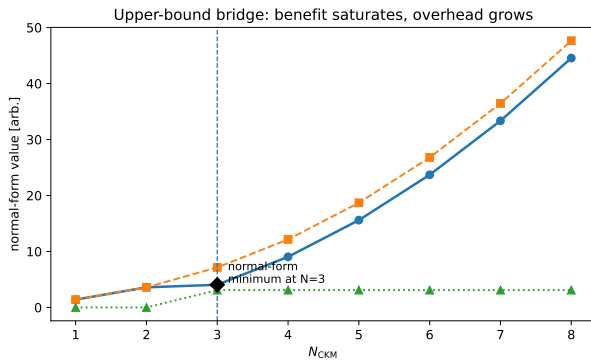


FIG. 4. Toy flavor-cost functional illustrating the upper-bound bridge. The functional benefit switches on at $N = 3$ and then saturates, while overhead continues to grow. This is a normal-form illustration, not a proof of the upper bound.

Upper-bound demotion

The minimality bridge is falsified or strongly demoted by the discovery of a fourth sequential chiral fermion generation with Standard Model gauge quantum numbers and weak charged-current participation. The lower bound $N_{\text{CKM}} \geq 3$ would survive. The $N = 3$ claim is also weakened if an $N > 3$ flavor architecture is shown to reduce the relevant cost functional or add a necessary nonredundant physical operation.

Leptonic extension demotion

The PMNS extension is demoted if high-precision oscillation measurements establish $\delta_{\text{CP}} = 0$ or π under assumptions requiring leptonic CP/T-oriented identity transformation.

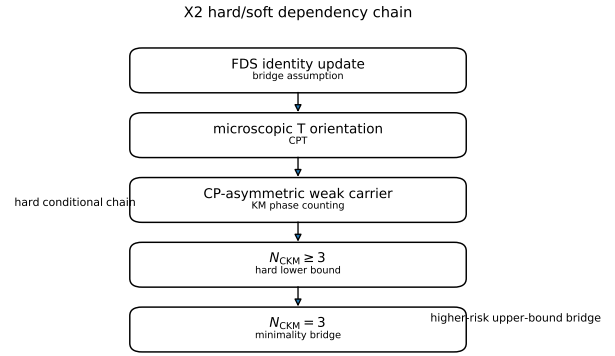


FIG. 5. X2 dependency chain. The hard conditional chain ends at $N_{\text{CKM}} \geq 3$. The final step to exactly three generations is a higher-risk minimality bridge.

RELATION TO EXISTING THEORY

X2 does not replace any of these theories. It packages one common implication inside FDS: if a finite-system weak identity transformation requires a microscopic CP/T orientation, the charged-current flavor architecture must have at least the dimensionality required for an irreducible CKM phase.

FUTURE WORK: MASS AS IDENTITY-SECTOR SEPARATION

The generation number is only one part of the flavor problem. A natural next question is whether the large fermion mass hierarchy encodes energetic separation barriers that prevent identity-sector ambiguity after the minimal CP/T-oriented architecture has been fixed. In such a program, masses would not merely be parameters but possible maintenance costs or separation scales for distinct identity sectors. X2 does not derive this; it marks it as a future problem.

CONCLUSION

The number three enters X2 not as numerology but as the first dimension in which weak flavor mixing can carry an irreducible CP/T orientation. A one-generation weak sector has no mixing phase. A two-generation weak sector has a Cabibbo angle but no physical CP phase. A three-generation weak sector has exactly one irreducible CKM phase, the minimal algebraic structure for CKM-type CP/T-oriented weak identity transformation.

The hard result is conditional but sharp:

$$\text{weak identity update} \Rightarrow T/CP \text{ orientation} \Rightarrow \text{irreducible CKM} \quad (17)$$

TABLE III. Relation of X2 to existing theory.

Existing theory	What it provides	X2 use
Kobayashi-Maskawa theory CPT theorem	CKM phase-counting result	Algebraic lower bound $N_{\text{CKM}} \geq 3$ [5].
Standard Model flavor physics Jarlskog invariant	T violation linked to CP violation under QFT assumptions Weak charged-current mixing and CKM phenomenology Rephasing-invariant CP-violation measure	Bridge from time-asymmetry to CP asymmetry [8, 9]. Object-level carrier of identity transformation [7]. Distinguishes physical phases from removable conventions [6]. Empirical anchor [10–12].
Kaon and B-meson experiments PMNS phenomenology	Observed quark-sector CP violation	Conditional extension and future test [18–23].
Fourth-generation and vector-like fermion phenomenology FDS/DT	Possible leptonic CP violation and global-fit constraints Distinguishes sequential chiral extensions from vector-like or hidden-sector variants	Clarifies what would and would not challenge the X2 upper-bound bridge [14–17].
	Identity-update and finite-system minimality language	Meta-level bridge selecting need for CP/T-oriented weak identity channel [1, 2].

The stronger statement $N_{\text{CKM}} = 3$, rather than $N_{\text{CKM}} > 3$, is a separate minimality bridge. Four or more CKM-participating generations are algebraically possible, but they are not required by the lower-bound theorem and must justify their additional flavor-sector cost. X2 should therefore be read as a conditional lower-bound theorem plus a quarantined upper-bound bridge.

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Functional Decomposition of the Four Fundamental Interactions: A Minimal Physical Distinction-Operation Closure for Finite Distinction Systems

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The Standard Model and general relativity describe four known fundamental interactions: strong, electromagnetic, weak, and gravitational. X3 does not claim to derive the Standard Model gauge group, coupling constants, particle masses, scattering amplitudes, electroweak symmetry breaking, or quantum gravity. Instead, it proposes a functional decomposition of the four known interactions as a minimal physical distinction-operation closure for finite distinction systems. A finite physical system capable of maintaining persistent distinctions must implement four non-equivalent operation classes: hadronic/baryonic encapsulation of stable material tokens, connection and remote detectability among tokens, identity transformation or flavor-changing selective update, and global causal-geometric / stress-energy accounting. In the observed physical world these primary roles are realized respectively by the strong interaction, electromagnetism, the weak interaction, and gravity. The mapping is primary-role, not exclusive-role: real interactions can contribute to multiple physical processes, but each known interaction carries one operation class that the others do not generically replace. The paper separates the functional closure thesis from lower-status gauge-group minimality claims, distinguishes new particles or portals from genuinely new operation primitives, and gives falsification conditions for both. Deterministic normal-form demonstrations illustrate operation coverage, remove-one-operation failure, redundant fifth-mediator comparison, fifth-force classification, and the relation of X3 to nearby FDS physical-bridge papers.

Reader contract. X3 is a functional decomposition paper, not a Standard Model derivation paper. It does not derive $SU(3)_c \times SU(2)_L \times U(1)_Y$, coupling constants, masses, scattering amplitudes, CKM/PMNS parameters, electroweak symmetry breaking, or quantum gravity. It does not claim that no new particles, portals, dark-sector forces, fifth-force candidates, or effective interactions can exist. It does not identify gravity with dissipation. Its claim is narrower: the known four interactions naturally realize four non-equivalent physical operation classes required for finite distinction maintenance: encapsulation, connection, identity transformation, and global causal-geometric / stress-energy accounting.

geometry; stress-energy accounting; gauge theory; general relativity; fifth force.

INTRODUCTION

The force-taxonomy problem

Modern physics describes four known fundamental interactions. Quantum chromodynamics explains the strong interaction; electroweak theory explains the electromagnetic and weak interactions; general relativity describes gravity as spacetime geometry [7–11, 18]. This dynamical description is successful, but it leaves open a functional question: why do the known interactions occupy these four qualitatively different roles?

X3 does not ask whether FDS can derive the Standard Model. It asks a more modest and more operational question: why do the observed interactions naturally decompose into four non-equivalent operation classes needed by finite physical systems that maintain distinctions?

Claim-status summary

Table I separates the functional closure claim from lower-status gauge-group and no-fifth-force interpretations.

Keywords

Finite Distinction Systems; Distinction Theory; fundamental interactions; strong interaction; electromagnetism; weak interaction; gravity; force taxonomy; operational closure; encapsulation; connection; identity transformation; flavor change; boundary maintenance; causal

Central thesis

The thesis is that the four known interactions realize a minimal physical distinction-operation closure. The functional map is:

TABLE I. Central X3 claims, status, and failure or demotion conditions.

Claim	Status	What would weaken or falsify it
Finite distinction systems require token stabilization.	Operational / structural	Stable material tokens exist with no encapsulation, binding, exclusion, protection, or stabilization mechanism.
Strong interaction realizes hadronic/baryonic encapsulation.	Physical mapping	Hadronic stability is explained while confinement and color-neutral binding play no token-stabilizing role.
Finite distinction systems require remote detectability and compositional connection.	Operational / structural	Complex persistent structures form without any long-range or mediating interaction enabling detectable coupling.
Electromagnetism realizes connection and communication among charged sectors.	Physical mapping	Electromagnetism is shown not to underwrite chemistry, radiation, sensing, or long-range charged communication.
Finite distinction systems require identity transformation and selective update.	Operational / structural	Physical identity sectors never require conversion, decay, pruning, or irreversible update.
Weak interaction realizes identity transformation, flavor change, and unstable-state pruning.	Physical mapping	Flavor change, beta decay, and weak identity transformation are realized by another carrier.
Gravity realizes global boundary / causal geometry / stress-energy accounting.	Physical bridge	Gravity is shown to be unrelated to geometry, causal structure, horizons, stress-energy accounting, or global boundary constraints.
Four interactions form a minimal distinction-operation closure.	Main X3 thesis	A fundamental interaction implements a genuinely independent operation class not reducible to encapsulation, connection, identity transformation, or global boundary accounting.

Interaction	FDS primary role
Strong	hadronic/baryonic encapsulation stable tokens
Electromagnetic	connection / remote detectability communicable distinctions
Weak	identity transformation (flavor change) / pruning
Gravity	causal geometry stress-energy accounting

This map is not a gauge-group derivation. It is an operational decomposition.

Remark 1 (Primary-role, not exclusive-role). *The mapping is primary-role, not exclusive-role. Real interactions can contribute to multiple physical processes. Strong interactions scatter; electromagnetism stabilizes atoms and molecules; weak interactions propagate through gauge bosons; gravity shapes motion and energy flow. X3 classifies each interaction by the operation class it carries in a way the others do not generically replace.*

Why X3 follows X2

X2 analyzes a specific property of the weak row: a CP/T-oriented weak identity channel requires at least three CKM-participating weak flavors under the stated assumptions [6, 16, 17]. X3 asks why identity transformation is one of the necessary operation classes at all.

Thus X2 is a supporting module for the weak-interaction component of X3.

FDS BACKGROUND

Finite distinction systems

An active finite distinction system is represented by

$$\mathcal{S} = (X, E, B, M, Y, A, U, \pi, \ell, \Phi, \mathcal{P}, \tau), \quad (1)$$

where X is internal state, E environment, B boundary, M memory/model state, Y observation channel, A action space, U update rule, π finite projection, ℓ boundary-maintenance loss, Φ resource budget, $\mathcal{P}$ perturbation/pruning family, and τ update timescale [1]. X3 focuses on physical operation classes that stabilize X , connect Y to other distinctions, implement U when identity sectors change, and embed the system in geometric and resource constraints.

Physical distinction-operation requirements

A physical distinction is not only a label. It must be:

1. stabilized against perturbation;
2. connected to other distinctions and observation channels;

TABLE II. Operation-class independence in X3.

Operation	Cannot be replaced by
Encapsulation	connection without protected tokens
Connection	encapsulation without transmissible coupling
Identity transformation	ordinary scattering without sector conversion
Global boundary	local interaction without shared geometry

3. updated or transformed when its identity sector changes;
4. embedded in global causal, geometric, and stress-energy/resource constraints.

These four requirements motivate the operation basis

$$\mathcal{O}_{\text{FDS}} = \{\mathcal{O}_{\text{enc}}, \mathcal{O}_{\text{conn}}, \mathcal{O}_{\text{id}}, \mathcal{O}_{\text{geom}}\}, \quad (2)$$

where $\mathcal{O}_{\text{enc}}$ is encapsulation, $\mathcal{O}_{\text{conn}}$ is connection / remote detectability, $\mathcal{O}_{\text{id}}$ is identity transformation / selective update, and $\mathcal{O}_{\text{geom}}$ is global boundary / causal geometry / stress-energy accounting.

Definition 1 (Distinction-operation closure). *A physical operation set is closed for finite distinction maintenance when it provides enough mechanisms to stabilize tokens, connect tokens, transform identity sectors, and embed all local operations in global causal and resource constraints.*

Remark 2 (Closure, not derivation). *Equation (2) is a functional closure basis. It is not a derivation of $SU(3)_c \times SU(2)_L \times U(1)_Y$ or of the Einstein field equations. It states what kinds of operations must be physically instantiated if stable distinction-maintaining systems exist.*

Operation-class independence

Two physical mechanisms may be dynamically coupled while still implementing distinct operation classes. X3 classifies by irreducible function under a fixed accounting boundary, not by isolated force ontology. Let F_j denote the four required functions: token stabilization, remote detectability, identity-sector conversion, and causal-geometric accounting. An operation class $\mathcal{O}_k$ is independent when removing it leaves at least one required function under-implemented by the remaining classes without changing the definitions of the remaining classes or expanding the accounting boundary.

This independence criterion is functional rather than metaphysical. It allows cross-coupling between interactions while preserving the claim that each operation class supplies one primary capability the others do not generically provide.

OPERATION I: STRONG INTERACTION AS ENCAPSULATION

Functional need

A finite physical system requires stable material tokens. Without token stabilization, persistent addressable states, composite structures, and memory-bearing substrates do not exist. In X3, encapsulation means binding raw degrees of freedom into protected tokens.

Physical realization

The strong interaction realizes the bottom hadronic/baryonic token-stabilization layer through color confinement, color-neutral hadron formation, and nuclear binding. QCD is not replaced by FDS; X3 interprets confinement and color-neutral binding as the observed physical implementation of hadronic token encapsulation [7, 12–14].

Strong encapsulation is not the sole stabilization mechanism for all composite structures. Atomic, molecular, and solid-state stability depend primarily on electromagnetic and quantum-statistical structure. The X3 claim is narrower: QCD supplies the lowest hadronic/baryonic material-token layer on which ordinary matter is built.

Gauge-group caveat

The observed implementation is $SU(3)_c$. X3 does not uniquely derive $SU(3)$ over $SO(N)$, $Sp(N)$, G_2 , or other alternatives. It treats $SU(3)_c$ as the Standard Model realization of the encapsulation role. A rigorous no-go theorem excluding alternative confining structures is a separate higher-risk problem.

Failure if absent

Without encapsulation there are no stable hadrons, no stable nuclei, and no persistent baryonic matter tokens. A world with connection and geometry but no encapsulated matter substrate lacks the material carriers required for long-lived distinctions.

OPERATION II: ELECTROMAGNETISM AS CONNECTION

Functional need

Encapsulated tokens must interact, compose, and become detectable. A universe of perfectly isolated tokens could store local distinctions but could not form atoms,

molecules, sensors, or communication channels. Connection means making distinctions mutually detectable and composable.

Physical realization

Electromagnetism realizes connection through long-range interactions among charged sectors, atomic and molecular binding, radiation, sensing, and photon-mediated communication. Electromagnetism is not merely a force between charges; it is the dominant channel by which ordinary matter becomes chemically structured, optically detectable, measurable, and communicable at macroscopic scales [7, 15].

Long-range detectability

The unbroken $U(1)_{\text{EM}}$ and masslessness of the photon provide the observed long-range connection channel. X3 does not derive photon masslessness. It interprets unbroken electromagnetism as the observed physical channel that permits distinctions to remain remotely detectable across macroscopic scales.

Failure if absent

Without electromagnetism, there are no ordinary atoms, chemical bonds, photons, optical sensing, radio communication, or electrically mediated macroscopic organization. Encapsulation without connection yields inert protected tokens rather than complex interacting structures.

OPERATION III: WEAK INTERACTION AS IDENTITY TRANSFORMATION

Functional need

Finite distinction systems cannot merely preserve identities. They must also allow identity-sector transitions: flavor conversion, decay of unstable configurations, selective pruning, and irreversible state update. In X3, identity transformation means controlled change of particle type, flavor sector, or decay channel. The weak interaction's chirality and CP/T violation are physical signatures that this identity update has a built-in directional character: if identity transformation were fully reversible and non-dissipative, it would belong to the connection class (operation II) rather than constituting a distinct pruning-like operation.

Here identity transformation means flavor, particle-sector, or decay-channel conversion, not arbitrary state

evolution, scattering, excitation, or binding rearrangement. Strong and electromagnetic interactions certainly change physical states; the weak role singled out in X3 is change of identity sector.

Physical realization

The weak interaction realizes identity transformation through beta decay, quark flavor change, neutrino interactions, chirality-sensitive charged currents, and CP/T-oriented weak-sector transformation. A representative quark identity transformation is

$$\mathcal{T}_{\text{id}} : |d_j\rangle \mapsto \sum_i V_{ij} |u_i\rangle, \quad (3)$$

where V_{ij} is the CKM matrix [6, 16, 17].

Pruning and capacity overflow

In FDS language, weak-sector identity transformation is the observed microscopic carrier of a pruning-like operation: not arbitrary deletion, but selective transition of unstable identity sectors into more stable or lower-energy configurations. Without such a channel, unstable sectors would either persist without decay or require another mechanism for clearing identity configurations that cannot remain encapsulated. This connects weak decay to the FDS problem of capacity overflow without claiming that all pruning in all systems is weak interaction.

Why selective weakness matters

If identity transformation were uncontrolled or too rapid relative to encapsulation and connection, stable structures would be continuously destabilized. The empirical weakness and mass gap of weak interactions are consistent with the requirement that identity update be selective rather than universal. This is a functional consistency claim, not a derivation of M_W , M_Z , or the Fermi constant.

Failure if absent

Without weak identity transformation, there is no beta decay, no observed quark flavor transitions, no ordinary weak nuclear reaction chain as observed in stars, and no CP/T-oriented weak identity channel. Stable tokens could exist and connect, but identity sectors could not be selectively updated.

OPERATION IV: GRAVITY AS GLOBAL BOUNDARY, CAUSAL GEOMETRY, AND STRESS-ENERGY ACCOUNTING

Functional need

A finite distinction system is embedded in causal structure. Distinctions require global boundary conditions, causal ordering, horizon structure, energy-momentum accounting, geometric response to stress-energy, and constraints on localization and information flow.

Physical realization

Gravity realizes these roles through spacetime geometry, causal cones, horizon structure, curvature sourced by stress-energy, and universal coupling to energy-momentum [18–21]. In X3, gravity is the observed physical implementation of the global boundary / causal geometry / stress-energy accounting class. In FDS terms, curvature encodes a wiring or addressing cost: as the density of maintained distinctions (mass-energy) increases, the local geometric constraints on communication and causal ordering tighten, which is the general-relativistic manifestation of finite-system boundary pressure.

Gravity is not merely dissipation

X3 does not identify gravity with dissipation. Dissipation and resource accounting are ledger channels inside gravitational boundary dynamics. Gravity itself is the independent geometric field governing causal structure and global boundary conditions. Gravitational waves, horizons, and spacetime curvature are not reduced to heat production. Stress-energy accounting is the physics-side language; resource accounting is the FDS interpretation of why global boundary constraints matter. When erasure or reset operations are physically implemented, Landauer’s principle supplies a baseline thermodynamic reference for the resource ledger, but gravity as a geometric sector is not itself a dissipative channel [28].

Equivalence-principle caveat

The equivalence principle may admit an FDS interpretation as a consistency condition between inertial book-keeping and gravitational boundary response, but this is not used in X3. It remains future work rather than part of the closure argument.

Failure if absent

Without gravity there is no dynamical spacetime geometry, no universal stress-energy accounting, no horizon-scale boundary structure, and no global causal constraint tying local operations into a common physical world.

MINIMAL DISTINCTION-OPERATION CLOSURE

Joint sufficiency and individual necessity

The four classes jointly provide stable tokens, interactions among tokens, transformation of tokens, and embedding in causal/geometric boundary structure. Remove one class and the closure fails:

Removed operation	Consequence
Encapsulation	no stable material tokens
Connection	no remote detectability or chemistry
Identity transformation	no flavor conversion, decay, or pruning
Global boundary	no causal geometry or horizon accounting

Proposition 1 (Functional distinction-operation closure). *For a physical finite distinction system with stable material tokens, remote detectability, identity-changing dynamics, and global causal geometry, at least four non-equivalent operation classes are required: $\mathcal{O}_{\text{enc}}$, $\mathcal{O}_{\text{conn}}$, $\mathcal{O}_{\text{id}}$, and $\mathcal{O}_{\text{geom}}$. In the observed physical world, the strong, electromagnetic, weak, and gravitational interactions realize these operation classes as primary roles.*

Proof. By construction, stability of material tokens requires an encapsulation/protection operation. Detectability and composition require a connection operation. Flavor or particle-sector conversion, decay, and selective update require an identity-transformation operation. Embedding local operations in a consistent causal and stress-energy-constrained world requires a global boundary/geometric operation. By the independence criterion in Table II, removing any one class leaves one required function under-implemented by the remaining classes. This proves a functional closure proposition, not a unique gauge-dynamics derivation. $\square$

FIFTH-FORCE AND NEW-PHYSICS CAVEAT

New particle versus new operation class

X3 does not rule out new particles, portals, sectors, or effective interactions. A dark photon, scalar mediator, axion-like particle, hidden confining gauge force, or

long-range spin-dependent interaction may extend physical phenomenology without adding a new distinction-operation primitive [25–27, 29–31]. A hidden confining force may be encapsulation-like; a dark photon may be connection-like; a scalar modifying masses may be boundary/resource-like; a new decay channel may be identity-update-like. Recent reviews of exotic spin-dependent interactions and asteroid-tracking constraints on fifth forces illustrate that many candidates are mediated by new bosons or are searched for as precision deviations, but such signals would challenge X3 only if they implement a new operation primitive [29, 30].

Falsification of closure

X3 is challenged only by a fundamental interaction implementing a new irreducible operation class not reducible to encapsulation, connection, identity transformation, or global boundary accounting. In that case, $\mathcal{O}_{\text{FDS}}$ must be expanded.

Criterion 1 (Fifth-operation criterion). *A candidate new force falsifies or expands X3 only if it implements a physically necessary operation primitive that cannot be classified as an instance, sector extension, or mixture of $\mathcal{O}_{\text{enc}}$, $\mathcal{O}_{\text{conn}}$, $\mathcal{O}_{\text{id}}$, or $\mathcal{O}_{\text{geom}}$.*

GAUGE GROUP DISCUSSION: LOWER-STATUS MAPPING

The observed structures map to the operation classes as follows:

Operation	Observed implementation
Encapsulation	$SU(3)_c$ QCD
Identity transformation	weak charged-current / broken electroweak sector
Long-range connection	unbroken $U(1)_{\text{EM}}$
Global boundary	diffeomorphism invariance / GR

This mapping motivates why the observed interactions have different mathematical forms. It does not uniquely derive the groups. X3 does not exclude all alternative gauge groups; does not derive electroweak symmetry breaking; does not calculate coupling constants; and does not quantize gravity. These remain lower-status problems.

Higgs mechanism as implementation layer

The Higgs mechanism is not treated as a fifth operation class in X3. It parametrizes mass generation and electroweak symmetry breaking inside the implementation of encapsulation, identity transformation, and long-range connection [7, 22–24]. In FDS language, masses

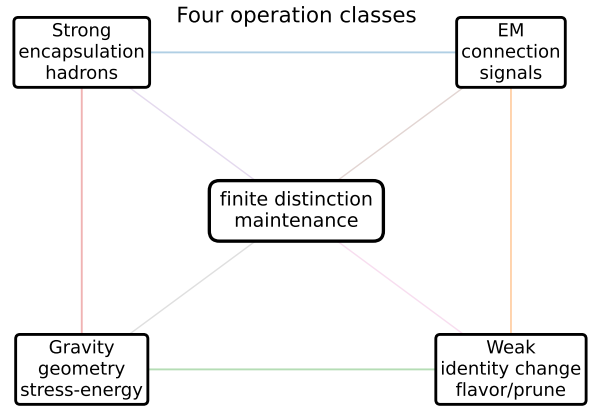


FIG. 1. Four known interactions as operation classes for finite distinction maintenance. The center represents finite distinction maintenance; the four nodes represent non-equivalent physical operation classes realized by the known interactions.

may be interpreted as energetic separation scales between distinction sectors, but X3 does not derive the Higgs potential, vacuum expectation value, or Yukawa texture.

Remark 3 (Gauge-group caveat). *Claims about $SU(3)$, $SU(2)$, $U(1)$, or diffeomorphism invariance are implementation mappings. If another group structure realizes the same operation role in a consistent physical theory, X3 should demote the corresponding gauge-specific claim while retaining the functional role.*

NORMAL-FORM DEMONSTRATIONS

The figures are deterministic normal-form demonstrations generated by `code/generate_results.py`. They are not empirical fits and do not simulate Standard Model dynamics. The normal-form model defines a qualitative coverage matrix between operation classes and requirements, enumerates operation subsets, and computes residual closure loss. Matrix entries are primary-role scores, not coupling constants, fitted data, or empirical weights. The only intended claim is conceptual: removing any one operation class leaves a distinct functional deficit, while adding a redundant fifth mediator without a new operation primitive does not change closure.

RELATION TO NEARBY FDS PAPERS AND EXISTING PHYSICS

X3 is compatible with QCD, electroweak theory, CKM/PMNS flavor physics, general relativity, and quantum field theory. It is not a replacement for them. It packages a functional implication inside FDS: per-

TABLE III. Relation of X3 to nearby FDS papers and established theories.

Theory / paper	What it provides	X3 use
QCD	Confinement, color dynamics, hadrons, nuclear matter	Observed implementation of hadronic/baryonic encapsulation.
QED / electromagnetism	Long-range photon-mediated interaction, atoms, radiation, sensing	Observed implementation of connection / remote detectability.
Electroweak theory	Chiral weak currents, beta decay, flavor change, EWSB	Observed implementation of identity transformation.
General relativity	Causal geometry, curvature, horizons, stress-energy accounting	Observed implementation of global boundary / geometry.
P3/P4/P6/P7	Environmental records, internal forgetting, throughput bounds, protected invariants	Nearby FDS physical-bridge constraints on boundary maintenance [2–5].
X2	CKM lower bound for CP/T-oriented weak identity transformation	Weak-interaction module inside the X3 taxonomy.
Fifth-force searches	New particles, portals, deviations, hidden-sector mediators	Classification test: new mediator or new operation primitive?

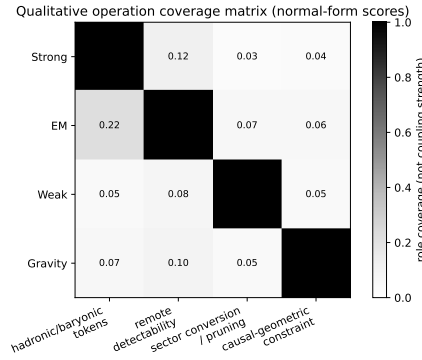


FIG. 2. Qualitative coverage matrix. The entries are normal-form scores, not coupling constants, fitted data, or empirical weights. Diagonal dominance expresses primary-role coverage; weak off-diagonal entries mark cross-role participation in real physical systems. Scores are schematic; they represent a logarithmic proxy for the relative dominance of each interaction in its primary role, not fitted coupling magnitudes.

sistent distinctions require stabilization, connection, update, and global boundary accounting.

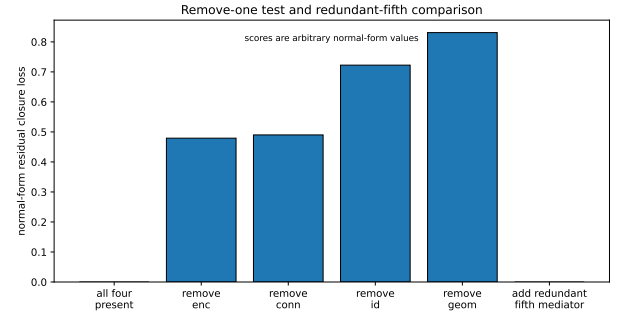


FIG. 3. Remove-one-operation test and redundant-fifth comparison. Absolute values are arbitrary normal-form scores; only the nonzero residual after removing each operation class is intended. A redundant fifth mediator that adds no new operation primitive does not improve closure.

Fifth-force audit: new mediator vs. new operation primitive

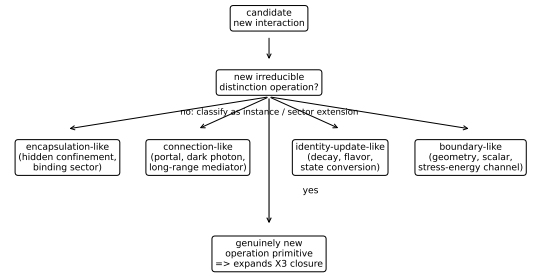
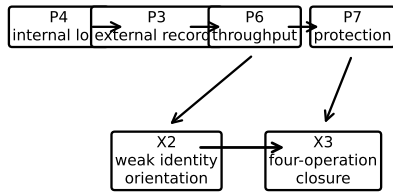


FIG. 4. Fifth-force audit. X3 distinguishes new particles or sector-specific mediators from a genuinely new operation primitive. Only the latter expands the X3 closure basis.

X3 relation map



X2 supplies the weak-sector module; X3 places it in the operation taxonomy.

FIG. 5. Relation of X3 to nearby FDS physical-bridge papers. X2 supplies a weak-sector identity-transformation module; X3 uses it inside a four-operation taxonomy.

DESIGN IMPLICATION FOR PHYSICAL AI

This section is not part of the physics proof. It extracts the FDS design rule implied by X3. A boundary-maintaining agent requires the same four functional classes: stable tokens, communication links, identity update, and global resource/boundary accounting.

Missing operation class	Agent failure mode
Encapsulation	identity drift / unstable object tokens
Connection	isolated modules / broken relation graph
Identity transformation	stale state / dead logic / overfitting
Global boundary	resource overshoot / unsafe action

The lesson is not that agents need copies of physical forces. It is that stable boundary maintenance requires a minimal operation closure analogous to the physical one: protected tokens, communication, identity revision, and global resource / safety-envelope accounting.

FALSIFICATION AND DEMOTION CONDITIONS

X3 is weakened if stable physical tokens exist without any encapsulation/protection operation; if complex structure and observation exist without connection/detectability; if identity-sector transitions exist without transformation/update; or if global causal geometry / stress-energy accounting is irrelevant to physical distinction maintenance.

X3 is falsified or expanded if a fundamental interaction is found that implements a necessary operation class not reducible to encapsulation, connection, identity transformation, or global boundary accounting. Gauge-specific claims are lower-status: if alternative group structures realize the same operation roles, the gauge mapping should be demoted rather than the closure thesis.

CONCLUSION

X3 proposes that the four known interactions occupy four non-equivalent functional roles required by finite distinction systems. The strong interaction stabilizes hadronic/baryonic material tokens; electromagnetism connects and exposes distinctions to remote detection; the weak interaction transforms identity sectors and enables selective flavor/particle-sector update; gravity supplies global causal geometry and stress-energy/boundary accounting. This is not a derivation of the Standard Model, nor a claim that no new particles or sector-specific forces exist. It is a closure claim: any physical world supporting persistent distinctions must implement these op-

eration classes, and the known interactions realize them in our universe.

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Pauli Exclusion as Finite Address Protection: Collision-Free Fermionic Occupancy, Structural Diversity, and Stable Matter in Finite Distinction Systems

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The Pauli exclusion principle states that no two identical fermions can occupy the same complete quantum state. In standard physics, this is encoded by antisymmetric fermionic wavefunctions and, in relativistic quantum field theory, by the spin-statistics theorem under assumptions such as Lorentz invariance, locality or microcausality, and positive energy. X4 does not replace that theorem. Instead, it gives an FDS interpretation of the operation performed by exclusion: Pauli exclusion is a collision-free occupancy rule for fermionic mode addresses. The central algebra is

$$(a_i^\dagger)^2 = 0, \quad n_i = a_i^\dagger a_i \in \{0, 1\}.$$

This nilpotent occupancy rule does not label identical fermions as classical individuals. It protects address-level occupancy events: a second identical fermionic mode-occupancy event cannot be written into an already occupied address. The result is not merely a microscopic selection rule; it is the address-level foundation for forced structural diversity, atomic shell structure, chemical diversity, degeneracy pressure, and the stability of bulk matter. Bosons are not treated as violations of finite address protection, because multiple occupation of a bosonic mode changes a field-mode amplitude or occupation number rather than creating multiple independently address-protected fermionic occupancy events in the same address. X4 therefore interprets Pauli exclusion as the minimal single-address protection rule for fermionic matter within finite distinction systems, while separating this operational interpretation from the hard QFT spin-statistics theorem and from higher-risk minimality claims about generalized statistics.

Reader contract. X4 is not a replacement for the spin-statistics theorem. It does not derive fermionic antisymmetry from FDS alone, does not claim that bosons violate finite information bounds, and does not claim that Bose-Einstein condensation creates infinite independent distinguishability. The term “token” does not imply classical individuation of identical fermions; it refers to an address-protected occupancy event in a fermionic mode. X4 does not claim all stable structures are caused only by Pauli exclusion, and it does not derive the exact Standard Model fermion content. X4 gives an operational interpretation: Pauli exclusion protects finite fermionic addresses, forces structural diversity, and makes scalable distinction-bearing matter possible.

exclusive occupancy; collision-free writing; quantum addresses; structural diversity; stability of matter; degeneracy pressure; atomic shell structure; Bekenstein bound; holography; parastatistics; Bose-Einstein condensation; Physical AI.

INTRODUCTION

Pauli exclusion as structural architecture

The Pauli exclusion principle states that no two identical fermions can occupy the same complete quantum state [6]. In the nonrelativistic wavefunction description this is encoded by antisymmetry:

$$\Psi(\dots, x_i, \dots, x_j, \dots) = -\Psi(\dots, x_j, \dots, x_i, \dots). \quad (1)$$

If two identical fermions attempt to occupy the same state, the wavefunction satisfies $\Psi = -\Psi$, hence $\Psi = 0$. In relativistic quantum field theory the spin-statistics theorem connects half-integer spin with Fermi-Dirac statistics under assumptions such as Lorentz invariance, microcausality, and positive energy [7, 8].

The consequences are architectural. Atoms have shells, chemistry has valence, bulk matter has volume, and compact stars resist gravitational collapse because fermions cannot all fall into the same lowest available mode. X4 asks not whether FDS can replace spin-statistics, but

Claim-status summary

Table I separates standard algebraic facts from FDS interpretations and higher-risk minimality bridges.

Keywords

Finite Distinction Systems; Distinction Theory; Pauli exclusion; fermions; spin-statistics theorem; antisymmetry; fermionic Fock space; nilpotency; address protection;

TABLE I. Central X4 claims, status, and failure or demotion conditions.

Claim	Status	What would weaken or falsify it
Fermionic creation operators obey nilpotency $(a_i^\dagger)^2 = 0$. Pauli exclusion protects single-fermion address occupancy.	Standard quantum algebra FDS interpretation	Identical Standard Model fermions are observed occupying the same complete quantum state. Fermionic matter remains address-stable and structurally diverse without any single-address occupancy protection.
Exclusion forces structural diversity in fermionic matter.	Physical / operational	Electrons can all occupy the same atomic state while shell structure, valence, chemistry, and matter stability remain unchanged.
Matter stability depends on fermionic antisymmetry.	Standard mathematical physics	Stability of bulk ordinary matter is proven and observed without exclusion-like fermionic antisymmetry or an equivalent mechanism.
Bosonic multiple occupation is not an X4 violation.	Conceptual caveat	Bosonic mode occupation is shown to create multiple independently address-protected fermionic occupancy events at the same address.
$n_i \in \{0, 1\}$ is the minimal address-protection rule.	Minimality bridge	Ordinary 3+1-dimensional Standard Model fermions obey generalized finite occupancy $p > 1$, or a lower-overhead alternative to nilpotent fermionic algebra is physically realized.
Degeneracy pressure is macroscopic address protection.	Physical bridge	White-dwarf or neutron-star support occurs without fermionic degeneracy pressure or an equivalent exclusion-like pressure contribution.

what operation the Pauli principle performs in finite distinction systems.

From information explosion to structural diversity

A naive information-density argument would say that if many particles could occupy one state, a finite region could encode infinite information. This statement is too broad. Bosons can occupy the same mode, and a Bose-Einstein condensate does not automatically carry infinite independent distinguishability. Many indistinguishable bosons in one coherent mode are better understood as increased occupation or field amplitude, not as many separately address-protected identity tokens.

The sharper FDS claim is therefore different:

Pauli exclusion converts address scarcity into structural diversity.

Without fermionic address protection, electrons could collapse into low-lying modes, shell structure would fail, and the diversity needed for chemistry and stable matter would be lost. Pauli exclusion is thus finite address protection: it prevents identical fermionic mode-occupancy events from silently occupying the same quantum address. Address scarcity itself follows from the finite causal reachability boundary analyzed in FDS-P6 [3]: within a finite causal horizon, the number of distinguishable modes any system can resolve is bounded, so exclusive single-address occupancy is the optimal strategy for maximizing structural diversity under that bound.

Contribution

X4 contributes five things. First, it defines quantum addresses and fermionic mode-occupancy tokens inside the FDS vocabulary. Second, it interprets the nilpotency condition $(a_i^\dagger)^2 = 0$ as a one-occupancy-event-per-address collision-free writing rule. Third, it proves a simple address-diversity bound for generalized occupancy cutoffs. Fourth, it gives bosons a positive complementary role as integration, broadcast, and synchronization channels rather than merely treating them as a counterexample. Fifth, it relates the rule to shell structure, bulk matter stability, degeneracy pressure, and Physical AI address-management analogues.

FDS BACKGROUND

Finite distinction systems

An active finite distinction system is represented by

$$\mathcal{S} = (X, E, B, M, Y, A, U, \pi, \ell, \Phi, \mathcal{P}, \tau), \quad (2)$$

where X is internal state, E environment, B boundary, M memory/model state, Y observation channel, A action space, U update rule, π finite projection, ℓ boundary-maintenance loss, Φ resource budget, $\mathcal{P}$ perturbation/pruning family, and τ update timescale [1]. The resource-ledger language is consistent with the finite-cost view of physical operations in information thermodynamics [21]. X4 focuses on the addressability of fermionic matter inside X and M : which physical occupancy events may occupy which quantum addresses without collision.

Quantum addresses

A quantum address is a complete single-particle mode specified by a complete set of compatible quantum labels. For an electron in an atom one may write, schematically,

$$\alpha = (n, \ell, m_\ell, m_s, \dots), \quad (3)$$

where the ellipsis includes all labels required to individuate the state in the chosen theory. More generally, an address is a single-particle mode i in Fock space.

For fermions,

$$n_i \in \{0, 1\}, \quad (4)$$

whereas for bosons,

$$n_i \in \{0, 1, 2, \dots\}. \quad (5)$$

The distinction is not that bosons are unconstrained by physics. It is that fermions create address-protected occupancy events, while bosonic modes are field excitations whose multiple occupation does not constitute a collision of identical fermionic addresses. The term “token” below always means a mode-occupancy event, not a classically individuated label attached to one identical particle.

Relation to X3

X3 decomposed the known interactions into operation classes: strong interaction as hadronic/baryonic encapsulation, electromagnetism as connection, weak interaction as identity transformation, and gravity as global causal-geometric / stress-energy accounting [5]. X4 supplies a complementary substrate-level rule: even after hadronic or atomic structures are available, fermionic address protection is required for shell structure, matter stability, and scalable physical memory. Strong encapsulation and Pauli address protection are distinct layers.

STANDARD PHYSICS BASELINE

Fermionic Fock algebra

Fermionic creation and annihilation operators satisfy the canonical anticommutation relations

$$\{a_i, a_j^\dagger\} = \delta_{ij}, \quad \{a_i^\dagger, a_j^\dagger\} = 0. \quad (6)$$

Setting $i = j$ gives

$$\{a_i^\dagger, a_i^\dagger\} = 2(a_i^\dagger)^2 = 0, \quad (7)$$

so

$$(a_i^\dagger)^2 = 0. \quad (8)$$

The number operator is

$$n_i = a_i^\dagger a_i, \quad (9)$$

and its eigenvalues are 0 and 1. Thus Eq. (8) is the algebraic form of Pauli exclusion.

Spin-statistics theorem

The spin-statistics theorem remains the standard hard theorem connecting half-integer spin to fermionic statistics and integer spin to bosonic statistics [7, 8]. X4 does not replace it. X4 interprets what the resulting fermionic algebra does operationally once it exists: it protects fermionic mode addresses from multiple identical occupancy.

FINITE ADDRESS PROTECTION

Definition 1 (Quantum address). *A quantum address is a complete single-particle mode i specified by the observables and boundary conditions required to individuate a state in the relevant theory.*

Definition 2 (Fermionic mode-occupancy token (identity token)). *A fermionic mode-occupancy token (identity token) is an address-protected occupancy event of a fermionic mode. It is not a classically individuated particle label. Its occupancy is protected by nilpotent creation, $(a_i^\dagger)^2 = 0$.*

Definition 3 (Finite address protection). *A system has finite address protection for identical fermionic mode-occupancy tokens if each address supports at most one such occupancy event:*

$$n_i \leq 1. \quad (10)$$

Definition 4 (Structural diversity). *Structural diversity is the forced distribution of fermionic mode-occupancy tokens across distinct addresses rather than collapse into a single low-energy mode. A simple address-diversity count is*

$$D = \sum_i \mathbf{1}[n_i = 1]. \quad (11)$$

Proposition 1 (Pauli exclusion as finite address protection). *For fermionic modes satisfying canonical anticommutation relations, nilpotency $(a_i^\dagger)^2 = 0$ implements a one-occupancy-event-per-address rule. This rule is the minimal algebraic form of finite address protection for identical fermionic mode-occupancy tokens in ordinary fermionic Fock space.*

Proof. Canonical anticommutation gives $\{a_i^\dagger, a_i^\dagger\} = 0$, hence $(a_i^\dagger)^2 = 0$. Therefore attempting to create two identical fermions in the same mode gives $(a_i^\dagger)^2|0\rangle = 0$. The allowed occupancy eigenvalues are 0 and 1, so the address is protected against identical double occupancy. $\square$

Remark 1 (Collision-free writing, not read-only memory). *The phrase “write protection” is used in a limited sense. It does not mean that a fermionic address is read-only or cannot change through allowed dynamics. It means collision-free writing: a second identical fermionic occupancy event cannot be written into an already occupied address. The Pauli rule prevents overlap-induced identity ambiguity rather than ordinary physical update.*

Proposition 2 (Address diversity bound). *Let $D_p(N)$ be the minimum number of distinct addresses needed to store N identical occupancy events when each address has a maximum occupancy cutoff p . Then*

$$D_p(N) \geq \left\lceil \frac{N}{p} \right\rceil. \quad (12)$$

In the Pauli case $p = 1$, sufficient available addresses imply

$$D_1(N) = N. \quad (13)$$

Thus Pauli exclusion maximizes forced address diversity among finite cutoff rules.

Proof. If each address can hold at most p occupancy events, D addresses can hold at most pD events. To store N events one needs $pD \geq N$, hence $D \geq \lceil N/p \rceil$. For $p = 1$, every event must occupy a distinct address. $\square$

BOSONS ARE NOT A COUNTEREXAMPLE

Bosons satisfy

$$[b_i, b_j^\dagger] = \delta_{ij}, \quad (b_i^\dagger)^N|0\rangle \neq 0, \quad (14)$$

so a bosonic mode may have arbitrary occupation number. This is not a violation of X4 because X4 is about fermionic address-protected occupancy tokens. Multiple bosons in the same mode increase field occupation or coherent amplitude; they do not create multiple individually protected fermionic occupancy events at the same address.

A Bose-Einstein condensate is the clearest caution. It may contain many bosons in one mode while having low configurational diversity. Therefore the central X4 claim cannot be “multiple occupation equals infinite information”. The correct claim is narrower:

Fermionic mode addresses require single-occupancy protection.

Bosonic modes play a different architectural role. In the FDS interpretation, fermions implement address-protected differentiation, while bosons implement integration, broadcast, synchronization, and field-mediated coupling between addresses. Fermions protect occupancy identities; bosons enable coherent transmission and shared interaction fields.

WHY $N_{\max} = 1$?

Generalized finite occupancy

The information-density idea alone would not select $N_{\max} = 1$. One could imagine finite alternatives,

$$n_i \in \{0, 1, \dots, p\}, \quad (15)$$

with $p > 1$, as in generalized or parastatistical settings [9–11, 26]. Recent work shows that non-trivial parastatistics can exist in physical systems and can obey generalized exclusion principles; this is why X4 treats $p = 1$ as an operational minimality bridge for ordinary Standard Model fermions rather than as a denial of all generalized statistics [26]. X4 therefore treats $p = 1$ as a minimality bridge, not as a derivation from FDS alone. FDS explains the operational economy of $p = 1$, not the spin-statistics origin of $p = 1$.

Minimal nilpotent algebra and address cost

For $p = 1$, the nilpotent rule $(a_i^\dagger)^2 = 0$ is the simplest exact occupancy cutoff. For $p > 1$, the theory requires additional internal algebra, sector labels, braid/topological structure, or parastatistical bookkeeping. Such structures may exist in special systems or dimensions, but they are not the ordinary Standard Model fermion rule in $3 + 1$ dimensions.

From an address-accounting perspective, $p > 1$ shifts cost from address expansion to collision resolution. If several occupancy events share one address, a retrieval or interaction protocol must still disambiguate which sub-occupancy, internal index, or parastatistical sector is being addressed. For example, in fractional statistics (anyons), tracking particle-exchange histories requires topological braiding bookkeeping that grows with the number of anyons per mode; this extra complexity is precisely the type of “extra maintenance cost” that FDS predicts for $p > 1$ occupancy rules [11]. A normal-form ambiguity cost is

$$C_{\text{amb}}(N, p) \propto N \log_2 p, \quad (16)$$

for $p > 1$, with $C_{\text{amb}}(N, 1) = 0$. This is not a universal thermodynamic law. It is an audit term expressing that multi-occupancy addresses require extra bookkeeping to

remain operationally distinguishable. The Pauli rule $p = 1$ is the zero-collision, zero-ambiguity finite-address rule.

Conjecture 1 (Minimal fermionic address protection). *Among finite occupancy rules for identical fermionic matter in ordinary $3 + 1$ -dimensional relativistic QFT, the Pauli rule $n_i \in \{0, 1\}$ is the minimal address-protection algebra compatible with stable fermionic mode-occupancy tokens.*

Remark 2 (Status). *The standard hard result is spin-statistics. The FDS minimality claim explains why the single-occupancy rule is operationally economical; it does not prove that all mathematically possible generalized statistics are impossible in all dimensions or effective systems [26].*

STABILITY OF ATOMIC MATTER

Pauli exclusion forces electrons into distinct orbitals, producing shell structure and valence. Without exclusion, many electrons could collapse into the lowest available orbital in a simplified model, destroying the differentiated outer-shell structure that underwrites chemical diversity.

This is the central physical reorientation of X4:

Pauli exclusion forces matter to use address space.

(17)

The periodic table is not just a list of nuclei; it is a consequence of forced address diversity among electrons. Atomic size, valence, bond geometry, and chemical differentiation depend on electrons being distributed across states rather than all occupying one lowest mode.

X4 does not claim that Pauli exclusion alone explains all chemical structure. Electromagnetic interaction, nuclear charge, quantum dynamics, and many-body effects are essential. The claim is that without fermionic address protection the shell architecture required for ordinary chemistry would fail.

STABILITY OF BULK MATTER

Rigorous mathematical physics shows that fermionic antisymmetry is central to the stability of matter. Dyson and Lenard established foundational stability results, and Lieb-Thirring inequalities provide kinetic-energy lower bounds that support extensive energy scaling [12–15]. In simplified terms, stable matter satisfies a bound of the form

$$E(N) \gtrsim -CN, \quad (18)$$

whereas without exclusion-like kinetic pressure collapse-prone models can scale superlinearly.

The FDS interpretation is that extensive stability is scalable memory architecture. A material system whose storage cost or collapse tendency grows too rapidly with token number cannot support macroscopic distinction-bearing structures. Pauli exclusion makes fermionic matter scalable by preventing many identical occupancy events from being stored in the same address and by forcing kinetic pressure when density rises.

DEGENERACY PRESSURE AND COMPACT OBJECTS

Degeneracy pressure is the macroscopic pressure generated by fermionic address protection. For a nonrelativistic ideal Fermi gas,

$$P_{\text{deg}} \propto n^{5/3}, \quad (19)$$

while relativistic scaling tends toward $P \propto n^{4/3}$. This pressure supports white dwarfs against gravity until the Chandrasekhar limit is reached [16]. Neutron degeneracy, nuclear interactions, and general relativity enter neutron-star support up to the Tolman-Oppenheimer-Volkoff regime [17, 18].

X4 does not derive these limits from FDS. It interprets degeneracy pressure as a macroscopic expression of finite fermionic address protection. Gravity supplies the catastrophic fallback when density exceeds global stability conditions; Pauli exclusion supplies the low-energy algebraic protection that allows structured matter to exist before that fallback dominates.

RELATION TO FINITE INFORMATION BOUNDS

Finite regions have finite accessible distinguishability. Bekenstein-type and holographic bounds provide a global consistency background for finite information density [19, 20]. X4 should not be read as claiming that the Bekenstein bound directly proves Pauli exclusion. The more precise relation is:

Pauli exclusion is locally consistent with finite addressability.

It prevents fermionic matter from collapsing into address-degenerate low-diversity structures, while gravitational collapse and horizon bounds remain global high-density cutoffs.

RELATION TO NEARBY FDS PAPERS

X4 is the matter-address substrate paper. It complements X3's strong-interaction encapsulation row by

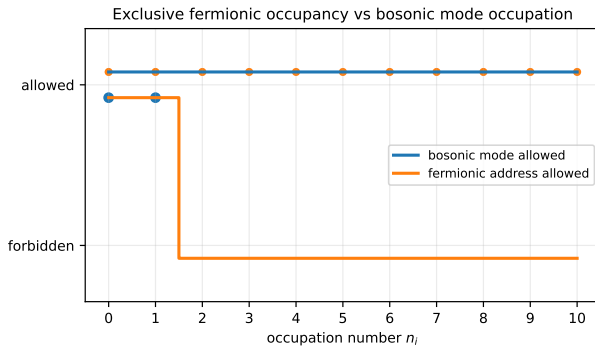


FIG. 1. Fermionic address protection versus bosonic mode occupation. Fermionic addresses allow $n_i = 0$ or 1; bosonic modes allow arbitrary occupation. This is the algebraic distinction underlying X4.

showing how fermionic address protection stabilizes and diversifies matter after encapsulated material tokens exist [5]. It complements P6 because address protection limits how densely identity-bearing occupancy events can be maintained under resource and update constraints [3]. It complements P7 because fermionic address labels are protected algebraically, though not topologically in the same sense as the P7 invariant-ledger examples [4]. It also connects to P4's anti-recurrence theme: once an address is occupied, an attempted identical write is not a reversible overwrite but an algebraically forbidden collision [2].

NUMERICAL AND ALGEBRAIC DEMONSTRATIONS

The figures are deterministic algebraic or normal-form demonstrations generated by `code/generate_results.py`. They are not empirical fits, experimental data, or physical simulations of real atoms or compact stars.

FALSIFICATION AND DEMOTION CONDITIONS

The hard falsification condition is direct observation of two identical Standard Model fermions occupying the same complete quantum state. Experimental searches for Pauli-exclusion violation, including forbidden atomic transitions and spin-statistics tests, probe this class of possibility [22–25]. Modern low-background searches such as Gator at LNGS [23] and independent germanium-detector searches [24] constrain PEP-violating transitions, providing direct experimental tests of the hard X4 falsification channel.

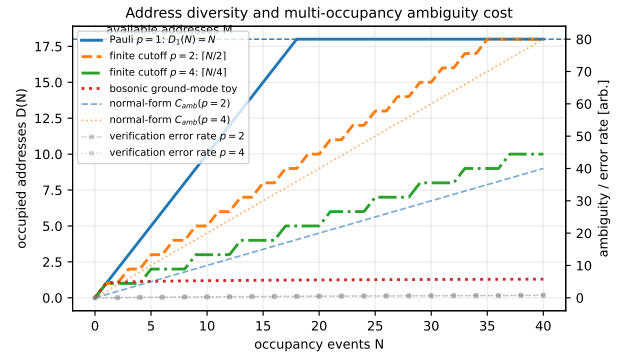


FIG. 2. Address diversity under filling. With Pauli protection, the number of occupied addresses grows with token number until the available address set is full. In an energy-minimizing bosonic-mode toy model, many tokens can remain in one mode. The generalized $p = 2$ and $p = 4$ curves illustrate why $p = 1$ is a minimality bridge rather than a direct consequence of finite capacity alone. The ambiguity-cost and verification-error curves are normal-form bookkeeping terms, not empirical thermodynamic data.

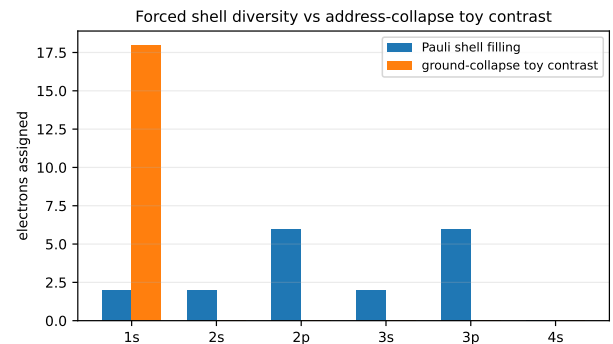


FIG. 3. Shell-filling cartoon. Pauli exclusion forces electrons into multiple shells, creating valence and chemical diversity. The collapse-to-ground curve is a toy contrast for address collapse, not a proposed alternative atomic physics model.

The minimality bridge is demoted if ordinary $3 + 1$ -dimensional Standard Model fermions exhibit generalized occupancy $p > 1$. Recent VIP-2 measurements provide stringent constraints on parastatistical Quon-model violations of Pauli exclusion, directly supporting the X4 demotion boundary for generalized occupancy rules [25]. Anyonic and fractional statistics in lower-dimensional effective systems do not automatically falsify X4, because X4 is about ordinary fermionic address protection in the matter substrate of our observed world.

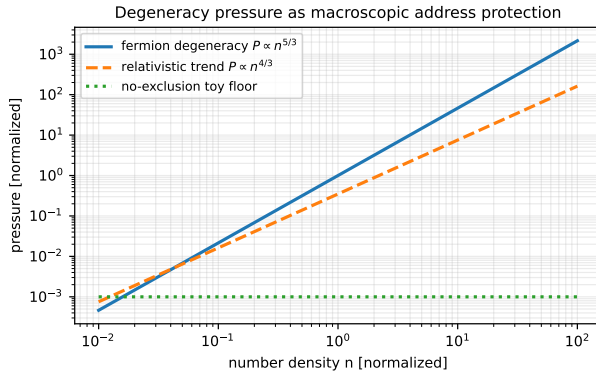
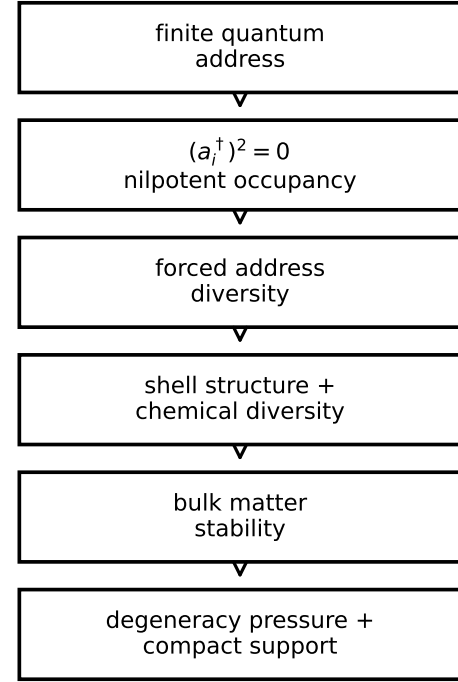


FIG. 4. Degeneracy pressure normal form. Nonrelativistic fermion degeneracy pressure scales as $P \propto n^{5/3}$, with relativistic trends approaching $n^{4/3}$. The figure illustrates address-protection pressure, not a full stellar-structure calculation.



X4: collision-free fermionic address protection

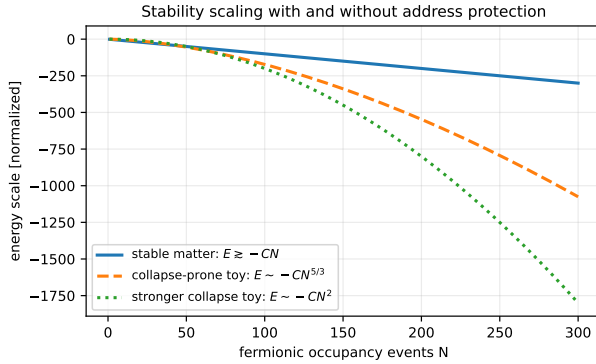


FIG. 5. Normal-form stability scaling. Extensive stability corresponds to an energy lower bound proportional to $-N$, while collapse-prone toy models scale superlinearly. The figure visualizes the FDS interpretation of matter stability as scalable address architecture.

FIG. 6. X4 dependency chain: finite quantum address, nilpotent occupancy, forced address diversity, shell structure, chemical diversity, bulk matter stability, and degeneracy-pressure support.

RELATION TO EXISTING THEORY

X4 does not replace the theories in Table II. It uses them as the standard physics baseline and adds an FDS interpretation of the operational role played by fermionic single-address occupancy.

Table II. Relation of X4 to existing theory.

Theory	What it provides	X4 use
Pauli exclusion	Single occupancy of identical fermions	Physical address protection.
Spin-statistics theorem	Half-integer spin linked to Fermi statistics	Standard hard theorem; X4 supplies operational interpretation.
Fermionic Fock algebra	Anticommutation and nilpotent creation operators	One-occupancy-event-per-address rule.
Dyson-Lenard / Lieb-Thirring	Mathematical stability of matter	Macroscopic consequence of address protection.
Chandrasekhar / TOV	Compact-object limits from degeneracy pressure and gravity	Astrophysical expression of fermionic address pressure.
Bekenstein / holography	Finite information bounds	Global consistency background, not direct proof.
Parastatistics / anyons	Generalized occupancy and low-dimensional effective statistics	Caveat to the minimality bridge.

DESIGN IMPLICATION FOR PHYSICAL AI

This section is not part of the physics proof. It extracts the FDS design rule implied by X4. Boundary-maintaining agents need Pauli-like address protection: two incompatible identity tokens should not silently occupy the same memory address.

Physics	AI analogue
Fermionic address protection	unique object-token identity slot
Pauli exclusion	collision-free write to an identity slot
Address diversity	separated object files / entity tracks
Shell structure	hierarchical memory allocation
Degeneracy pressure	resistance to representation collapse
Bosonic mode	shared signal / broadcast / attention field

Missing Pauli-like rule	Agent failure mode
Address collision	two entities share one logical slot
Identity overwrite	new object silently overwrites old object
Memory aliasing	one key retrieves incompatible entities
Object permanence failure	identity not preserved across time
Decision collapse	actions target merged or confused entities
Catastrophic forgetting	address reuse destroys prior distinctions

If an agent merges “enemy” and “cover” into one logical address, planning can become physically unsafe. X4 therefore suggests that robust agents should separate identity-bearing tokens from shared broadcast fields, latent attention modes, and global communication channels.

CONCLUSION

Pauli exclusion is not merely a strange rule about antisymmetric wavefunctions. In FDS language, it is finite address protection for fermionic matter. The nilpotency condition $(a_i^\dagger)^2 = 0$ enforces a one-occupancy-event-per-address rule, forcing fermions to occupy distinct modes. This produces shell structure, chemical diversity, matter stability, and degeneracy pressure. X4 does not replace the spin-statistics theorem or derive the Pauli principle from FDS alone. It identifies the operation Pauli exclusion performs in a finite distinction system: it protects matter from identity-address collision and converts finite addressability into structural richness. At the level of FDS memory dynamics, this is a local physical analogue of anti-recurrence: an occupied fermionic address cannot be silently overwritten by an identical occupancy event. Because each address stores at most one occupancy event, it forms a natural anti-overwrite barrier. By preventing address-level identity collisions, PEP maintains the independence of historical records in finite systems and provides a local physical analogue of the P4 anti-recurrence property [2].

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Mathematical Form of Physical Law as Invariant-Form Compression: Invariant, Equivariant, and Covariant Law Forms in Finite Distinction Systems

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Physical laws are remarkably mathematical: they appear as equations, variational principles, symmetries, conservation laws, operators, groups, manifolds, spectra, tensors, and constants. FDS-X5 interprets this fact inside Finite Distinction Systems. It does not claim that mathematics is an external substance governing reality, nor that FDS derives all mathematical structures used in physics. Instead, it proposes that the mathematical form of physical law is the compressed form of invariant, equivariant, or covariant relations that survive finite capacity, perturbation, coordinate change, coarse-graining, and boundary-maintenance constraints.

In an active finite distinction system, raw microstate histories are too large to represent. What persists across observers, scales, coordinates, gauges, and perturbations must factor through a compressed sector. In the strict invariant case,

$$q : X \rightarrow Q, \quad q(P_i x) = q(x),$$

while in the equivariant or covariant case,

$$q(P_i x) = \rho_i q(x), \quad F(q) = 0 \Rightarrow F(\rho_i q) = 0.$$

A physical law is therefore an operationally stable relation on compressed law-form sectors, rather than a full enumeration of microstates. The “mathematical effectiveness” of physical law is reframed: mathematics is effective because the portable part of physics is the part compressible into invariant-form structures. X5 separates a core FDS claim from higher-risk philosophical extensions. The core claim is that finite boundary-maintaining systems must compress stable physical regularities through invariant, equivariant, or covariant representations. The higher-risk bridge is that Wigner’s puzzle is dissolved because persistent physical law and mathematical structure are two perspectives on invariant-form compression.

Reader contract. X5 is not a proof that the universe is literally “made of mathematics.” It does not derive all of mathematics from physics, derive the Standard Model or general relativity, solve open problems in pure mathematics, or claim that every mathematically consistent structure is physically realized. It does not claim that every useful effective model is an exact law. It does not turn constants such as e , i , or π into numerical “cosmic signatures.” X5 is a claim about the operational form in which finite observers, agents, instruments, and boundary-maintaining systems can discover, store, transmit, and use physical laws. It is not a claim that the universe itself consults a compressed formula before evolving.

Keywords

Finite Distinction Systems; Distinction Theory; invariant-form compression; physical law; Wigner puzzle; covariance; symmetry; conservation laws; Noether theorem; rate-distortion; algorithmic information; renormalization; operators; Hilbert space; constants; computability; Physical AI.

INTRODUCTION

Wigner’s puzzle, reframed

Wigner’s puzzle asks why mathematics is so effective in the natural sciences [9]. Complex numbers became central to quantum mechanics; Riemannian geometry became the language of general relativity; group theory and representation theory became indispensable in particle physics. The puzzle is usually phrased as a mystery about the relation between human formal constructions and external reality.

X5 reframes the question. The relevant issue is not why all mathematics describes physics. Most mathematical structures are not directly realized in physical law. The sharper question is why the stable, portable, predictive

Claim-status summary

Table I separates formal FDS claims, physical bridges, and higher-risk interpretive extensions.

TABLE I. Central X5 claims, status, and failure or demotion conditions.

Claim	Status	What would weaken or falsify it
Finite systems cannot internally represent all microstate detail.	Formal FDS core	A finite system internally represents unbounded environmental distinctions with no compression, externalization, task relaxation, or hidden capacity.
Stable law-like regularities require invariant-form compression.	FDS structural claim	Portable physical regularities persist with no invariant, equivariant/covariant form, symmetry, quotient, stable statistic, or compressed relation.
Mathematical equations are compressed representations of invariant-form relations.	Interpretive bridge	Exact physical laws require no compression, no form stability, no stable relational structure, and no finite representational economy.
Symmetries reduce rule-maintenance cost.	Physical / information bridge	Asymmetric local rule tables are physically cheaper and more stable than invariant or equivariant compressed rules under the same task requirements.
Wigner’s puzzle is reframed by invariant-form compression.	Philosophical bridge	Mathematics remains effective for law-like physics where no invariant/equivariant/covariant compression, symmetry, operator structure, quotient, or stable relation exists.
e , i , and related constants are model-class signatures.	Optional bridge	Their appearance is asserted as universal without model-class assumptions or is shown to be purely conventional in the relevant context.
Open mathematical problems may have physical analogues.	Speculative appendix	Analogues are mistaken for derivations, predictions, or proofs.

part of physics has mathematical form. In FDS terms, the answer is:

Physical laws are mathematical because law-like regularities are invariant-form compressions.

Raw histories are too large for finite systems to represent. What can be maintained, communicated, predicted, and reused must be compressed. What remains stable across perturbations must preserve form. Mathematical law is the symbolic form of this invariant, equivariant, or co-variant compression.

Physical law as finite description

A physical law is not a database of all microstates. It is a finite rule that compresses many possible histories into a relation that remains valid across an equivalence class of states, frames, scales, gauges, or perturbations. In a finite distinction system, a raw history

$$x_{0:T} \in X^{T+1} \quad (1)$$

is useful only if it can be replaced, for the task at hand, by a smaller quotient, sufficient statistic, invariant, co-variant object, or operator representation

$$x_{0:T} \mapsto q(x_{0:T}) \in Q. \quad (2)$$

A law is the compressed rule on Q and its transformation law, not the raw enumeration of X .

X5 in the X-series

X2 analyzed the algebraic lower bound needed for CP/T-oriented weak identity transformation. X3 interpreted the four known interactions as a functional distinction-operation closure. X4 interpreted Pauli exclusion as collision-free fermionic address protection. X5 explains why these claims naturally appear as equations, algebras, symmetries, nilpotencies, operators, and constants: physical theory records invariant-form compression structures of finite matter and finite observation.

FDS BACKGROUND

Active finite distinction systems

An active finite distinction system is represented by

$$\mathcal{S} = (X, E, B, M, Y, A, U, \pi, \ell, \Phi, \mathcal{P}, \tau), \quad (3)$$

where X is internal state, E environment, B boundary, M memory/model state, Y observation channel, A action space, U update rule, π finite projection, ℓ boundary-maintenance loss, Φ resource budget, $\mathcal{P}$ perturbation/pruning family, and τ update timescale [1]. The FDS core separates formal claims, physical bridges, and quarantined applications; X5 uses the formal capacity and invariant-persistence layers rather than claiming a direct metaphysical derivation of physics.

Capacity deficit and rate-distortion demand

For a finite memory or model space M , internal representational capacity is

$$C_S = \log_2 |M|. \quad (4)$$

For an admissible task family Ψ , the rate-distortion demand over update window τ at tolerance ε is $R_{\min}^{(\tau)}(\varepsilon; \Psi)$. The FDS capacity deficit is

$$\Delta_\varepsilon(\tau) = R_{\min}^{(\tau)}(\varepsilon; \Psi) - C_S. \quad (5)$$

If $\Delta_\varepsilon(\tau) > 0$, the system must approximate, externalize, relax the task, improve compression, or fail [1]. X5 asks what form the best compression takes when the regularity must remain portable across perturbations and coordinate choices.

Invariant-supported persistence

Let $R_A : X \rightarrow \{0, 1\}$ be a structural identity predicate. An identity is invariant-supported under perturbation family $\mathcal{P}$ when there exist a quotient $q : X \rightarrow T$ and a predicate $\bar{R}_A : T \rightarrow \{0, 1\}$ such that

$$R_A = \bar{R}_A \circ q, \quad q \circ P_i = q \quad \forall P_i \in \mathcal{P}. \quad (6)$$

Then $R_A \circ P_i = R_A$ for every admissible perturbation [1]. X5 generalizes this into a law-form claim: law-like physical regularities factor through compressed sectors that may be strictly invariant, equivariant, or covariant. Strict invariants are the simplest case; tensor equations, gauge laws, and quantum representations often preserve form by transforming covariantly rather than by remaining numerically unchanged.

DEFINITIONS

Definition 1 (Physical regularity). *A physical regularity is a relation R over states, observations, or histories,*

$$R \subseteq X \times X', \quad (7)$$

or a predictive map $f : X \rightarrow Y$ that improves prediction, control, compression, or boundary-maintenance loss relative to a registered task.

Definition 2 (Law-like regularity). *A physical regularity is law-like relative to perturbation family $\mathcal{P}$ when it remains valid under admissible transformations, or when it factors through a compressed sector $q : X \rightarrow Q$:*

$$R(x, x') = \bar{R}(q(x), q(x')). \quad (8)$$

A law-like regularity is not merely a repeated pattern; it is a portable pattern under a specified equivalence, covariance, or perturbation class.

Definition 3 (Invariant-form compression). *An invariant-form compression is a map $q : X \rightarrow Q$ together with a law relation F and a registered transformation family $\mathcal{P}$ satisfying three properties:*

$$\begin{aligned} \text{compression:} \quad & L_{\text{rep}}(q; \mathcal{D}) \ll L_{\text{raw}}(\mathcal{D}), \\ \text{task sufficiency:} \quad & \mathbb{E}[\ell(Z, \hat{Z}(q(X)))] \leq \varepsilon, \\ \text{form stability:} \quad & q(P_i x) = q(x) \text{ or } q(P_i x) = \rho_i q(x). \end{aligned} \quad (9)$$

Here $\mathcal{D}$ is the registered task-relevant data, history ensemble, rule table, model state, or experiment class. $L_{\text{raw}}(\mathcal{D})$ is the description length of the raw representation under the chosen coding scheme. $L_{\text{rep}}(q; \mathcal{D})$ is the description length of the compressed representation plus decoder, law form, or rule needed to use it.

Definition 4 (Strict invariant, equivariant, and covariant cases). *In the strict invariant case,*

$$q(P_i x) = q(x). \quad (10)$$

In the equivariant or covariant case, admissible transformations act on the compressed sector through a representation ρ_i :

$$q(P_i x) = \rho_i q(x), \quad (11)$$

while the law relation is preserved in form:

$$F(q) = 0 \quad \Rightarrow \quad F(\rho_i q) = 0. \quad (12)$$

This distinction is essential for physics: tensor equations, gauge fields, quantum states, and relativistic laws often transform covariantly rather than remaining component-wise invariant.

Definition 5 (Mathematical law form). *A mathematical law form is a finite symbolic representation of an invariant, equivariant, or covariant relation. Common forms include*

$$F(q) = 0, \quad \dot{q} = \mathcal{L}q, \quad G \curvearrowright Q, \quad I(q) = \text{constant}. \quad (13)$$

Equations, tensor relations, symmetry laws, conservation laws, variational principles, operator algebras, spectra, and topological invariants are law forms when they compress stable physical regularities.

Definition 6 (Law compression ratio). *For a registered dataset, history ensemble, rule table, or model state $\mathcal{D}$ encoded directly with length $L_{\text{raw}}(\mathcal{D})$, a law representation with length L_{law} , and initial/boundary data length L_{data} , define the normal-form law compression ratio*

$$C_{\text{law}} = \frac{L_{\text{raw}}(\mathcal{D})}{L_{\text{law}} + L_{\text{data}}}. \quad (14)$$

This is an audit measure, not a universal constant. High C_{law} means strong mathematical compression relative to the specified coding scheme and task class.

MAIN PROPOSITION: LAW AS INVARIANT-FORM COMPRESSION

Proposition 1 (Physical law as invariant-form compression). *Consider an active finite distinction system maintaining a predictive or boundary-relevant regularity over perturbation family $\mathcal{P}$. If raw state detail exceeds the system's finite representational capacity, then any portable law-like representation must factor through a compressed task-sufficient sector*

$$X \xrightarrow{q} Q \xrightarrow{\bar{f}} \hat{Z}, \quad (15)$$

where q is strict-invariant, equivariant, covariant, or stable to the registered tolerance under the admissible transformations relevant to the law. The mathematical form of the law is the finite description of q , $\bar{f}$, the representation ρ_i when present, and the preserved form relation $F(q) = 0 \Rightarrow F(\rho_i q) = 0$.

Proof. Finite capacity bounds internally maintained detail by C_S . If the raw task-relevant state or history requires more than C_S bits, full enumeration is unavailable. Boundary maintenance still requires a statistic that preserves the task-relevant information to tolerance ε . Thus the system must use a compressed representation $q(X)$. If the regularity is to remain portable across the admissible transformation family, the relevant content must either be strictly invariant, transform equivariantly or covariantly under a known representation, or remain stable to the registered tolerance. Hence prediction or control factors through Q and its transformation law. The symbolic description of that quotient, representation, update rule, invariant relation, or group action is what is called the mathematical form of the law. $\square$

Remark 1 (Core versus bridge). *The proposition is an FDS structural statement about finite representation and invariant-form quotients. It concerns operational law form for finite systems, not an ontological claim that the universe consults a compressed formula. It does not imply that every exact law has already been discovered, that all mathematics is physically realized, or that the physical world is reducible to a human formalism.*

WHY EQUATIONS?

Equations as invariant-form constraints

An equation records a constraint that remains true across an equivalence class. For example,

$$F(q) = 0 \quad (16)$$

says that the compressed variable q lies in a law-defined subspace. Conservation laws, gauge constraints, mass-shell conditions, and commutation relations all have this

form: they reduce a large set of possible states to a smaller law-defined manifold.

The equation is not merely notation. It is a compression device. Instead of storing every allowed case, one stores the relation that recognizes all allowed cases. In covariant theories, the components of q may change with coordinates or gauge, but the equation form is preserved; the compression is in the stable relation, not necessarily in invariant component values.

Dynamical equations as compressed update rules

A dynamical equation

$$\dot{q} = F(q) \quad (17)$$

compresses many trajectories into one reusable update rule. In a linear generator model,

$$\dot{q} = \mathcal{L}q, \quad q(t) = e^{\mathcal{L}t}q(0). \quad (18)$$

The exponential form appears because repeated homogeneous updates are compressed by a semigroup generator. This is a model-class bridge, not a claim that every physical process is exactly Markovian or linear. When the generator description is valid, $e^{\mathcal{L}t}$ is the invariant-form compression of repeated time-translation updates.

Differential equations as continuum compression

Differential equations are often continuum or coarse-grained limits of local update regularities. They are effective when many discrete or microscopic updates can be summarized by smooth fields and local derivatives. Thus differential equations are not mystically effective; they are local invariant-form compression rules for domains whose finite update dynamics admits a smooth approximation.

WHY SYMMETRY?

Symmetry and covariance as rule compression

Let a group G act on X . A strict G -invariant quotient satisfies

$$q(x) = q(gx), \quad g \in G. \quad (19)$$

An equivariant or covariant representation satisfies

$$q(gx) = \rho(g)q(x), \quad (20)$$

with the law relation preserved in form. Instead of writing separate rules for every point in the orbit Gx , a symmetric law writes one rule on the orbit space, representation space, or covariant tensor relation. This reduces rule-table size, parameter count, and coordinate dependence.

Cost of asymmetry

A fully asymmetric law must specify extra local rule data: preferred positions, labels, directions, frames, exceptions, or contexts. In finite systems, those specifications consume memory, communication, and update resources. Symmetry is therefore not merely aesthetic. It is a compression strategy: one rule applies across many cases. Recent work in symmetry-informed equation discovery shows that equivariance constraints can compress the equation-search space and improve robustness and simplicity of discovered governing equations [23].

Noether bridge

Noether's theorem states that continuous symmetries of an action correspond to conserved quantities [10, 11]. X5 does not derive Noether's theorem. It interprets why Noether-type structures are privileged: conservation laws are invariant-form compressions of dynamical redundancy. They let a finite system track a smaller number of stable quantities instead of the full trajectory. This interpretation is aligned with recent sparse-invariant methods for discovering interpretable conservation laws [24] and with geometric frameworks for Hamiltonian systems that preserve symmetry under Lie group transformations [25].

WHY CONSTANTS?

Constants as operational scales

A physical constant is not merely a decorative number in an equation. In law-form compression it functions as a stable operational scale: a causal conversion limit, a phase-action scale, a coupling strength, an entropy-energy conversion factor, or a boundary invariant.

$$\theta(q) = \theta_0. \quad (21)$$

The constant compresses many measurements into one reusable conversion scale or coupling value. If the parameter drifted arbitrarily, the system would need to store a table $\theta(x, t, \text{context})$ rather than a law. Dimensional constants depend on unit conventions and conversion scales. Their invariant content lies in dimensionless combinations, operational ratios, transformation roles, or stable coupling structure, not in their numerical value in a chosen unit system. The invariant role of each constant lies in the operational constraint it imposes: c sets the causal signal-propagation limit, $\hbar$ sets the quantum phase-action scale, G sets the stress-energy geometry coupling, and k_B sets the entropy-energy conversion factor.

e as semigroup signature

In generator models,

$$\frac{dq}{dt} = Lq \Rightarrow q(t) = e^{Lt}q(0). \quad (22)$$

The number e should not be over-interpreted as a mystical constant. The physically meaningful object is the exponential functional form: the eigenfunction structure of homogeneous generator evolution. The numerical base is tied to units and conventions; the compression principle is semigroup update.

i as coherent phase bookkeeping

In quantum theory,

$$U(t) = e^{-iHt/\hbar}. \quad (23)$$

The imaginary unit is the algebraic bookkeeping device for phase-preserving coherent dynamics, interference, and unitary evolution. X5 does not claim to derive Hilbert-space quantum mechanics from FDS alone. It says that once a domain supports coherent superposition and phase-sensitive composition, complex operators are an invariant-form compression language for that domain.

Anti-numerology firewall for π and other constants

Some constants appear because of geometry, coordinate conventions, unit choices, or model-specific expansions. The appearance of π in circular geometry does not by itself make π a cosmic signature. The deeper invariant may be boundary topology, isoperimetric geometry, or rotational symmetry; the numerical coefficient depends on convention and dimension. X5 treats constants as law-compression parameters only under explicit model-class assumptions.

WHY OPERATORS, HILBERT SPACES, AND SPECTRA?

Operators as compressed transition rules

An operator $\mathcal{O} : X \rightarrow X$ encodes a reusable transformation. Instead of enumerating many input-output pairs, the operator applies one rule across a domain. In this sense, operators are transition-rule compression.

Hilbert spaces as superposition-compatible compression

Hilbert spaces compress amplitude relations, superpositions, measurement bases, and transformations in quan-

tum theory. X5 does not derive the Hilbert-space formalism. It gives an operational reading: when a physical domain supports coherent superposition and compositional amplitudes, Hilbert-space operators are the mathematical compression of stable amplitude transformations.

Spectra as compressed observable structure

A spectrum replaces a large family of possible responses with eigenvalues and eigenvectors:

$$\mathcal{O}\psi_n = \lambda_n\psi_n. \quad (24)$$

Spectral structure is a powerful law form because it compresses dynamics, measurement outcomes, relaxation modes, stability, and resonance into indexed invariant data.

WHY PURE MATHEMATICS OFTEN PRECEDES PHYSICS

Mathematics explores possible compression forms

Pure mathematics studies possible structures without requiring their physical instantiation. From the X5 perspective,

Mathematics is the search space of possible invariant-form compressions.

Physics selects those forms that are compatible with measurement, stability, finite resources, perturbation survival, and empirical boundary conditions.

Renormalization-group language gives a standard physical example of this selection principle. Under repeated coarse-graining, most microscopic details flow away, while a smaller set of relevant variables, symmetries, scaling relations, and fixed-point structures remains predictive. In X5 terms, an infrared fixed point is a stable invariant-form compression: a relation whose form survives coarse-graining and therefore becomes law-like for finite observers. X5 does not reduce all physical law to RG fixed points; RG is a model-class example of invariant-form compression under scale coarse-graining.

Finding an invariant-form compression may require an expensive search over hypotheses, symmetries, coordinates, latent variables, and equivalence classes. Once found, however, the law form can be applied, stored, communicated, and verified at much lower marginal cost. This asymmetry explains why law-like compressions have high finite-system value: they amortize discovery cost across many future predictions.

Selection principle

A mathematical structure becomes physically relevant only if it compresses a real regularity, survives the relevant perturbations, can be instantiated or approximated under finite resources, and improves prediction or boundary maintenance. Mathematics is larger than physics because it explores possible compressions; physics realizes only the compressions that can be paid for and maintained.

Wigner's puzzle reframed

Wigner's puzzle is not answered by saying "all mathematics is physics." It is answered by selection:

physical law $\subset$ maintainable invariant-form compressions. (25)

Mathematical structures often precede physics because mathematics explores invariant forms before nature, experiment, and physical theory identify which forms are instantiated.

OPTIONAL BRIDGE: OPERATIONAL COMPUTABILITY

This section is not required for the main invariant-form compression thesis. It is an optional bridge concerning the form in which finite-resolution physical predictions can be operationally used by finite systems.

Conjecture 1 (Operational computability). *Operational physical laws governing finite-resolution measurements are expected to be computably approximable to registered task tolerance.*

The reason is FDS-like: a finite system cannot use an infinite oracle table as a law. A law that is predictive for finite observers must be finitely specifiable or finitely approximable within the required tolerance and update window. This does not prove that all mathematical objects appearing in physics are computable in every idealization. It states a physical bridge: operational law use requires finite procedures.

Criterion 1 (Computability demotion). *The X5 computability bridge is weakened if a reproducible finite-resolution physical law has predictive content that provably requires noncomputable functions and cannot be replaced by any computable approximation within observational tolerance.*

SPECULATIVE APPENDIX: MATHEMATICAL PROBLEMS AS POSSIBLE COMPRESSION ANALOGUES

Firewall. No mathematical problem is claimed to be solved, predicted, or physically derived in this appendix. The purpose is only to illustrate how some mathematical structures may acquire physical analogues when they appear as spectra, partition functions, complexity gaps, or symmetry classes. This appendix is not part of the X5 core claim.

Riemann hypothesis

Spectral interpretations of zeta zeros have long motivated connections between number theory and physics. X5 allows a weak statement: if a zeta-like object appears as the spectral determinant of a physical operator, then properties of its zeros may acquire physical meaning. This is not a derivation of the Riemann hypothesis.

P versus NP

In FDS language, generation and verification need not have the same cost. A structure may be easy to check but hard to construct. This resembles, but does not prove, a generation-verification asymmetry familiar from computational complexity. X5 does not prove complexity-class separations.

Finite simple groups

The classification of finite simple groups catalogs possible symmetry building blocks. X5 treats such mathematics as part of the structure space from which physics may select maintainable symmetries. It does not predict that every group is physically realized.

RELATION TO P-SERIES AND X-SERIES

Table II places X5 in the local FDS corpus.

NUMERICAL AND CONCEPTUAL DEMONSTRATIONS

The figures are deterministic normal-form demonstrations generated by `code/generate_results.py`. They illustrate definitions and compression intuitions rather than fitting empirical data.

FALSIFICATION AND DEMOTION CONDITIONS

X5 is weakened if stable physical laws are found that cannot be represented by any invariant, equivariant, covariant, compressed relation, stable quotient, symmetry, operator, or sufficient statistic. It is weakened if finite systems maintain law-like prediction without reducing raw microstate complexity, or if physical regularities persist across perturbations while lacking any form-stable structure.

The constants bridge is demoted whenever a claimed constant is shown to be a coordinate convention, unit convention, or model-specific artifact. The computability bridge is demoted if reproducible finite-resolution predictions provably require noncomputable law content. The Wigner interpretation can fail without destroying the FDS core: the core claim is finite invariant-form compression; the metaphysical interpretation is a higher-level bridge.

RELATION TO EXISTING THEORY

X5 uses existing theories as implementation bridges rather than replacing them. Wigner's essay supplies the target problem [9]. Noether's theorem supplies a canonical example in which symmetry compresses dynamics into conserved quantities [10, 11]. Rate-distortion theory and information theory supply the compression backbone [12, 13]. Algorithmic information theory and minimum-description-length methods supply law-code and model-selection analogues [14–16]. Renormalization group theory supplies a scale-dependent model of coarse-grained invariant persistence [17]. Symmetry and group theory supply orbit compression [18]. Dynamical systems and semigroups supply generator and flow forms [19]. Quantum theory supplies operator, Hilbert-space, and spectral law forms [20]. The FDS Core supplies the finite-capacity and invariant-persistence basis [1]. Covariance and tensorial law forms connect this compression account to the standard geometric language of physical theory [22].

DESIGN IMPLICATION FOR PHYSICAL AI

This section is not part of the physics proof. It extracts the FDS design rule implied by X5: a physical AI agent should learn not more correlations, but better invariant, equivariant, and covariant compressions.

TABLE II. Relation of X5 to neighboring FDS papers.

Paper	Core role	X5 relation
FDS Core	finite capacity, capacity deficit, invariant quotients	formal basis for invariant-form compression [1]
P3	environmental side records and Markovianization	laws ignore inaccessible bath detail by quotienting [2]
P4	anti-recurrence / lost preimages	laws compress away erased detail rather than recover it [3]
P6	finite update windows and resource ledgers	laws reduce update throughput and verification burden [4]
P7	invariant side-ledgers / topological protection	law form as protected quotient structure [5]
X2	CKM phase-counting lower bound	example of algebraic physical constraint [6]
X3	four-interaction operation closure	interactions become mathematical through invariant roles [7]
X4	Pauli nilpotency as address protection	algebraic law as occupancy-protection compression [8]
X5	mathematical law form	invariant-form compression of physical regularities

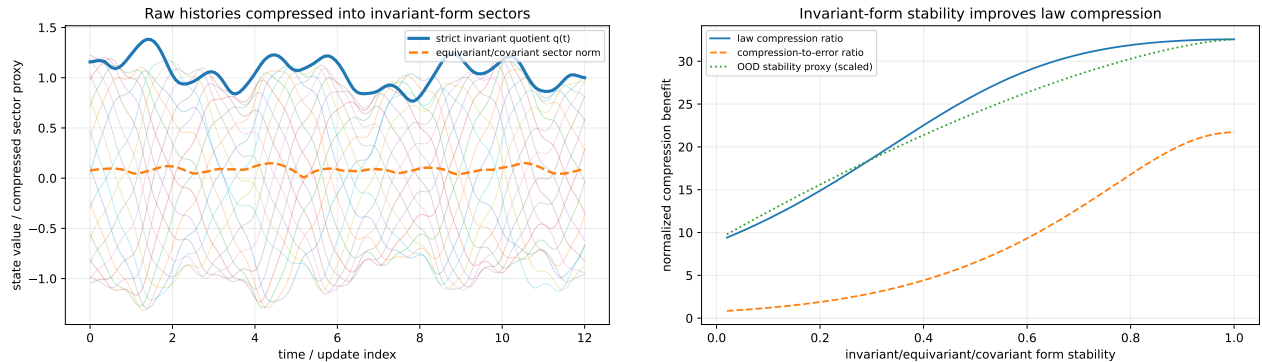


FIG. 1. Left: raw histories with different micro-details compress into strict-invariant and equivariant/covariant sectors. Right: as form-stability strength increases, a normal-form law compression ratio and compression-to-error ratio improve. The OOD stability proxy (scaled) illustrates that invariant-form compressions maintain predictive power under distribution shift. Out-of-distribution stability, latent constraint preservation, and compression-to-error ratio serve as audit metrics for compression quality. These are conceptual demonstrations, not data fits.

Physics	AI analogue
Physical law	invariant world-model rule
Symmetry / covariance	equivariance / data augmentation
Conservation law	latent constraint
Equation	compact predictive program
Constant	stable parameter
Operator	reusable transition rule
Invariant quotient	task-preserving latent representation

An agent that stores raw histories without discovering invariants or equivariances pays excessive memory, update, and verification cost. Failure modes include overfitting, missing symmetry, brittle out-of-distribution behavior, high update cost, and lack of conserved latent structure.

The generalization gap in many learned systems can be read as a failure of invariant-form compression. A correlational map may fit observed histories while failing under distribution shift because it has not identified the quotient, symmetry, conservation law, or covariant structure that remains stable outside the training distribution. This failure mode is also recognized in physics-informed machine learning, where incorporating physical priors (symmetries, conservation laws, operator structures) improves generalizability and physi-

cal plausibility [26]. Useful audit metrics include out-of-distribution stability, latent constraint preservation, and compression-to-error ratio.

FDS-X5 therefore suggests a benchmark: report invariant compression quality, not only prediction accuracy.

CONCLUSION

The mathematical form of physical law is not an external miracle. In FDS terms, it is the finite symbolic form of invariant-form compression. A physical law is not a complete enumeration of the world’s microstates. It is a compact representation of relations that remain stable under perturbation, coordinate change, coarse-graining, and boundary-maintenance demands. Equations, symmetries, conservation laws, tensor relations, operators, spectra, and constants are the forms taken by such compressions.

X5 does not derive all mathematics or replace physics with metaphysics. It identifies why the stable part of physics is naturally mathematical: only invariantly, equivariantly, or covariantly compressible structure can

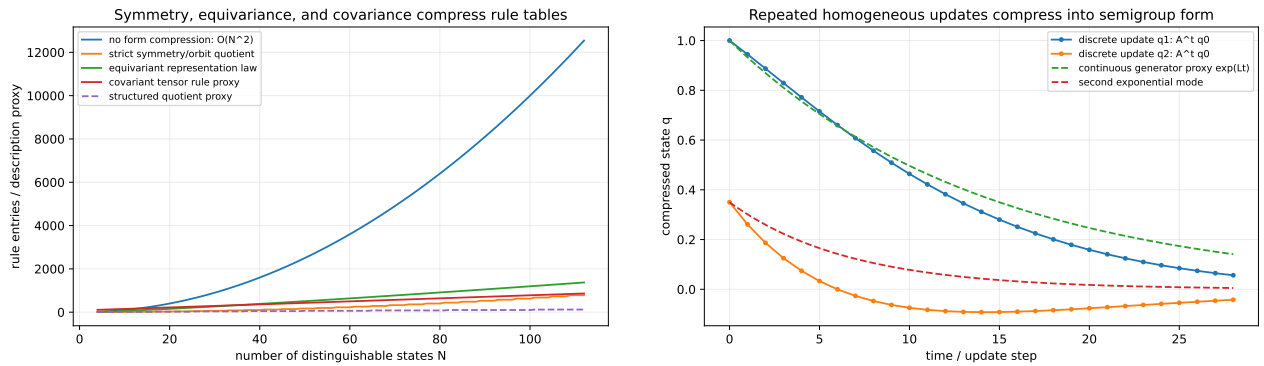


FIG. 2. Left: symmetry and equivariance compress rule tables by reducing cases to orbits or representation laws. For a fully asymmetric rule table, specification scales with the number of cases; a symmetry or covariance group reduces the maintained rule table to orbit-level data plus the group action. The plotted curves are normal-form scaling illustrations, not empirical parameter counts. Right: repeated homogeneous updates can be compressed as $A^t q_0$ or, in a generator limit, $e^{Lt} q_0$. The exponential is a semigroup-compression signature, not a numerical claim about e .

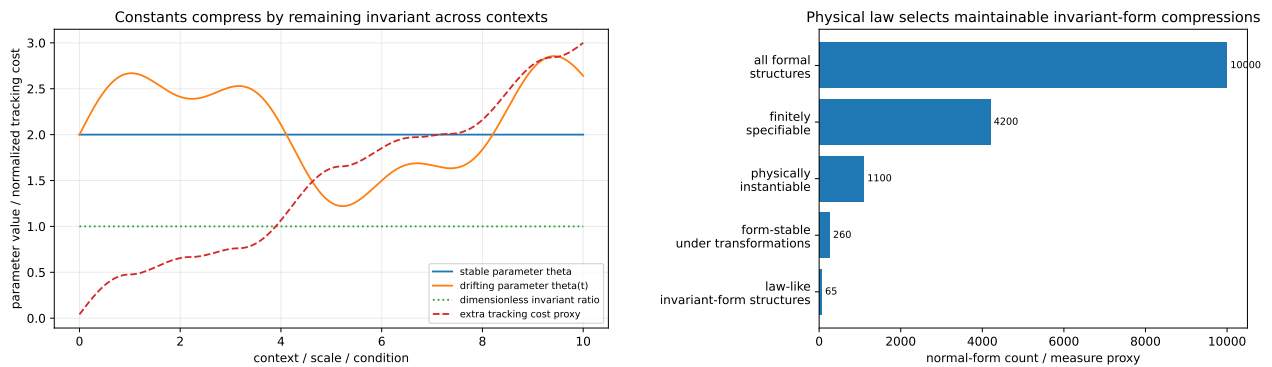


FIG. 3. Left: a stable parameter compresses many contexts into one constant; drifting parameters require extra tracking cost. Right: Wigner’s puzzle reframed as a selection filter with FDS bottlenecks: physics realizes only mathematical structures that are finitely specifiable (finite symbolic update), physically instantiable (Φ , resources), stable at finite resolution (C_S , memory), stable under perturbation ($\mathcal{P}$, invariance), and real-time usable (τ , P6 throughput).

be retained, communicated, predicted, and maintained by finite systems. The unreasonable effectiveness of mathematics becomes less mysterious when physical law is understood as the portable mathematical residue of finite distinguishability.

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